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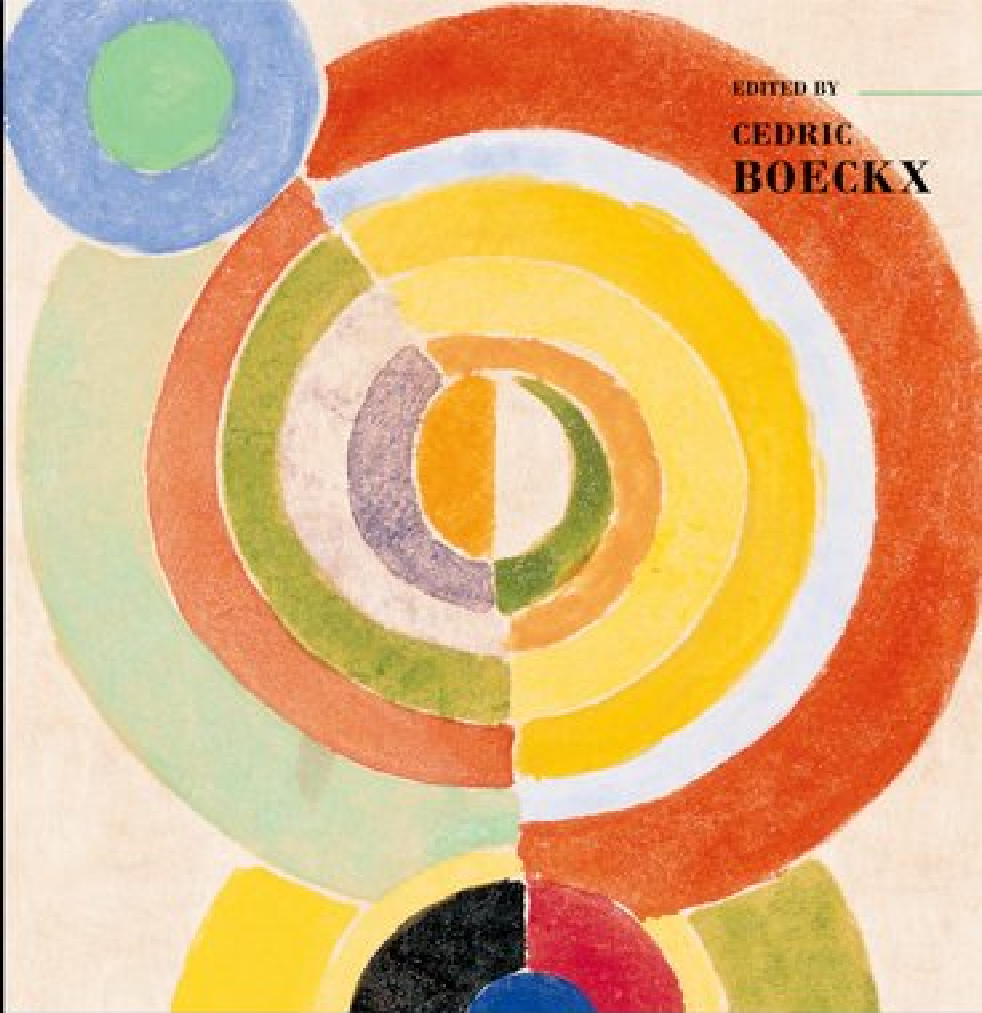

The Oxford Handbook *of* LINGUISTIC MINIMALISM

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The Oxford Handbook of Linguistic Minimalism
Edited by Cedric Boeckx

Print Publication Date: Mar 2011 Subject: Linguistics
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The Contributors

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The Contributors

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Robert Freidin is Professor of Linguistics in the Council of the Humanities at Princeton University. Starting with his 1971 Ph.D. dissertation, he has been concerned with the foundations of syntactic theory and with the central concepts of syntactic analysis and their evolution, pursuing the minimalist quest for an optimally simple theory of syntax. His work focuses on the syntactic cycle, case and binding, and the English verbal morphology system, and utilizes the history of syntactic theory as a tool for explicating and evaluating current theoretical proposals. A collection of the full range of this work is published in *Generative Grammar: Theory and its History* (Routledge, 2007). He is also the author of *Foundations of Generative Syntax* (MIT Press, 1992) and *Syntactic Analysis: A Minimalist Approach to Basic Concepts* (CUP, in press). He is the editor of *Principles and Parameters in Comparative Grammar* (MIT Press, 1991), and *Current Issues in Comparative Grammar* (Kluwer, 1996), and co-editor with Howard Lasnik of the six-volume collection *Syntax: Critical Concepts in Linguistics* (Routledge, 2006), and with Carlos P. Otero and Maria Luisa Zubizarreta of *Foundational Issues in Linguistic Theory: Essays in Honor of Jean-Roger Vergnaud* (MIT Press, 2008).

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David Pesetsky is Ferrari P. Ward Professor of Linguistics at the Massachusetts Institute of Technology. He has worked on a number of topics in syntactic theory, including the relation between argument structure and syntax, as well as the cross-linguistic typology of *wh*-constructions. In an ongoing collaboration with Esther Torrego, he has helped develop a new proposal concerning the nature of case, and has recently also investigated case morphology in Russian. He is the author of two books, *Zero Syntax* (MIT Press, 1995) and *Phrasal Movement and its Kin* (MIT Press, 2000), and is currently engaged in a collaborative investigation (with Jonah Katz) of the syntax of tonal music.

Paul M. Pietroski is Professor of Philosophy and Professor of Linguistics at the University of Maryland. He is the author of *Causing Actions* (OUP, 2000) and *Events and Semantic Architecture* (OUP, 2005), along with many papers in semantics, related areas of philosophy, and (in collaboration with colleagues) psycholinguistic studies of semantic competence and acquisition.

Gillian Ramchand's research concerns the relationship between syntactic and semantics representations of natural language. She has worked in areas of tense, aspect, predication, and argument structure on languages as diverse as Bengali, Scottish Gaelic, and English. She has published articles in *Natural Language Semantics, Linguistic Inquiry*, and *Lingua* as well as a number of edited volumes. She is the author of two books *Aspect and Predication* (OUP, 1997) and *Verb Meaning and the Lexicon* (CUP, 2008), where she argues for a syntactic implementation of an event structural view of verbal meaning and participant relations. She is currently Professor of Linguistics at the University of Tromso, Norway and Senior Researcher at the Center for Advanced Study in Theoretical Linguistics (CASTL) there. Before moving to Norway in 2003, she was lecturer in General Linguistics at **(p. xii)** the University of Oxford. She holds a Ph.D. in Linguistics from Stanford University, and Bachelor's degrees in Mathematics and Philosophy from the Massachusetts Institute of Technology.

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List of Abbreviations and Symbols

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List of Abbreviations and Symbols

A	adjective
ABC	Agree-Based Construal
ABS	absolutive
Acc	accusative
ACD	Antecedent-Contained Deletion
AFL	abstract families of languages
Agr	Agree
AP	articulatory-perceptual
AP	adjective phrase
ASP	attention span parameter
Asp	Aspect
ATB	across the board
Aux	auxiliary

BCC	Borer-Chomsky Conjecture
BPS	bare phrase structure
BS	base set
C	complementizer
CBC	Chain-Based Construal
CBT	canonical binding theory
CCL	Constraint on Chain Links
CDE	Condition on Domain Exclusivity
CED	Condition on Extraction Domains
CHL	computational system for human language
CI or C-I	conceptual-intentional
CMG	conflated minimalist grammars
CNP	conjunctive participle particle
CNPC	complex NP constraint
COMP	complement
COSH	Constraint on Sharing
CP	complementizer phrase
CP	compounding parameter
D	determiner
(p. xvi) Dat	dative
DbP	Derivation by Phase

Dem	demonstrative
DO	direct object
DP	determiner phrase
DS	deep structure, D-Structure
ECM	Exceptional Case Marking
ECP	Empty Category Principle
EF	edge feature
EGKK	*A Derivational Approach to Syntactic Relations* (Epstein et al. 1998)
EM	External Merge
EPP	Extended Projection Principle
ERG	ergative
EST	Extended Standard Theory
F	feature
FI	full interpretation
FL	language faculty
FLB	Faculty of Language—Broad Sense
FLN	Faculty of Language—Narrow Sense
FOC	focus
FOFC	Final-over-Final Constraint
FP	F Parameter
FSA	Finite State Automaton

FUT	future
GB	Government and Binding Theory
Gen	genitive
GEN	gender
GPSG	Generalized Phrase Structure Grammar
H	head
HAB	habitual
HC	Head Constraint
HFL	human faculty of language
HMC	Head Movement Constraint
HPSG	Head-Driven Phrase Structure Grammar
I	inflection
IC	Immediate Constituent (analysis)
IC	Interface Condition
iff	if and only if
IM	Internal Merge
INFL	inflection
(p. xvii) initP	causing projection
INSTR	instrumental
IO	indirect object
IP	inflection phrase

IRR	irrealis
L	language
L_2	second language
LA	lexical array
LCA	Linear Correspondence Axiom
Lex	lexicon
LF	logical form
LI	lexical item
LOT	language of thought
LRC	Last Resort Condition
MD	movement-driven
MG	minimalist grammar
MLC	Minimal Link Condition
Mod	modifier
MP	minimalist program
MSO	Multiple Spell-Out
MSP	Minimal Structure Principle
MWF	multiple w/2-fronting
N	noun
NCI	negative-concord items
Nom	nominative

NP	noun phrase
NS	narrow syntax
NSP	null subject parameter
NSR	nuclear stress rule
NTC	Non-Tampering Condition
Num	number
O	object
OT	Optimality Theory
P	preposition
P&P	Principles and Parameters (Theory)
PCMG	phase-based conflated minimalist grammar
PD	Push Down
PDA	Push Down Automaton
PDbP	phonological derivation by phase
PDH	Procedural Deficit Hypothesis
PF	phonetic form
(p. xviii) PHC	phase-head-complement
PHON	phonological component
PIC	Phase Impenetrability Condition
PL	plural
PL	pair list

PLD	primary linguistic data
PM	phrase marker
PMG	phase-based minimalist grammar
PP	prepositional phrase
PRO	phonetically null argument
ProcP	process projection
PSG	phrase structure grammar
QR	quantifier raising
Quant	quantifier
ResP	result projection
RM	Relativized Minimality
RMG	relativized minimalist grammar
RNR	right node raising
S_0	initial state
SC	Serbo-Croatian
SEM	semantic component
SF	semantic structure
SG	singular
SHA	Spec-Head agreement
SLI	specific language impairment
SLQZ	San Lucas Quiavini Zapotec

SM	sensorimotor
SMC	Shortest Move Constraint
SMT	Strong Minimalist Thesis
SO	syntactic object
Spec	specifier
SS, S-Structure	surface structure
SSOs	scope-shifting operations
SUT	Strong Uniformity Thesis
SYN	Syntax
T	tense
TD	target-driven
TOP	topic
TP	tense phrase
TPM	*A theory of phrase markers and the extended base* (Chametzky 1996)
(p. xix) UG	universal grammar
UTAH	Uniformity of Theta Assignment Hypothesis
v	'little v'
V	verb
V_2	verb second
VCN	(virtual) conceptual necessity
VP	verb phrase

XP	phrasal category
*	unacceptable
π	the sound of the linguistic expression
λ	the ‘meaning’ of the linguistic expression

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Dedication

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Dedication

(p. xx) *In memoriam:*

Kenneth L. Hale (1934–2001)

Tanya Reinhart (1943–2007)

Carol Chomsky (1930–2008)

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Overview

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Overview

Linguistic Minimalism refers to a research program—not a specific theory—that grew out of results obtained within a specific research tradition (the Extended Standard Theory/Principles and Parameters approach in generative grammar), although the intuitions or research guidelines at the heart of minimalism are largely independent of the specific implementations that provided the immediate historical context for their emergence. As such, minimalist concerns ought to be of interest to a wide range of researchers in the language sciences, and cognitive sciences more generally.

A minimalist program for linguistic theory was first formulated by Noam Chomsky almost twenty years ago. In the intervening years, minimalism has developed into a very rich research tradition touching on many aspects of our species-specific language faculty. It quickly established itself as a major, if healthily controversial, approach to linguistic phenomena, so much so that I felt the time ripe for a handbook (the first of its kind) exclusively devoted to minimalist ideas applied to core aspects of human language. The primary goal of the handbook is to provide an essential source of reference for both graduate students and more seasoned scholars and prepare them for the somewhat bewildering panorama of ideas that can be found in the primary literature.

The present work is not a textbook, nor is it a book that is meant to be read from cover to cover. But in preparing the final draft and in circulating the chapters among my students, I was pleasantly surprised by the depth and breadth of discussion and analysis found in each of the contributions included here. I feel that as a whole the present volume offers an authoritative survey of what linguistic minimalism is today, how it emerged, and how far it has advanced our knowledge of the human language faculty. Quite a few contributors did not hesitate to highlight the many gaps that remain to be filled, and the many limitations—some contingent, others perhaps more inherent to the enterprise—of a minimalist approach to linguistic phenomena. Students coming to linguistic minimalism should regard these—at times severe—limitations as challenges to tackle, if indeed they feel (as I hope they do) that looking at the human language faculty through minimalist lenses can yield valuable insights that would otherwise remain hidden. I, for one, would be delighted if this volume served as the source of many a dissertation advancing the minimalist agenda.

(p. xxii) To maximize the resourcefulness of the material contained in this work, I thought it necessary to devote some space to clarifying both the form (organization) and content of this handbook. This is the goal of the present overview.

I began by pointing out that linguistic minimalism is a research program, not a theory. That is to say, the pursuit of the minimalist program, beginning with its original formulation in the early 1990s, is meant to construct a certain theoretical space within which specific theories of the various components of the human language faculty can be elaborated. Crucially, minimalism does not intend to offer a final product. Rather, like the best cookbooks, it aims at identifying a few key ingredients that in the hand of creative minds may shed light on the true nature of our language faculty. Because of this open-ended character of the minimalist program, there is not a single 'best' way to introduce the various ideas that practitioners take to be central. As a result, there is not a single 'best' way to order the contributions that make up this volume. It is all of them, taken as a whole, that constitute the most fertile ground for minimalist investigations. Needless to say, this rendered my task an an editor particularly difficult. It often seemed to me that the richness of each contribution would be best reflected if this handbook could have existed in loose-leaf form, made up of loosely associated, easily reorganizable parts. Although in the end the very linear structure of the book's table of contents prevailed, I urge the reader to bear in mind at all times that this book can be read in multiple ways.

To make matters worse (for the editor), linguistic minimalism departs in important ways from a central assumption of The Extended Standard Theory/Principles and Parameters. Instead of viewing the internal organization of the language faculty as modular (an argument structure module, a phrase structure module, a locality module, etc.), the minimalist program seeks to identify very general computational properties that cut across the traditional modules. This means that in practice it becomes virtually impossible to decompose the language faculty into quasi-independent sub-components or areas. This means that for the editor there is no easy way to keep the chapters of a handbook on minimalism separate, let alone order them sequentially: all of them should come first. Ideally, for the reader to benefit the most from the chapters, and to truly grasp the nature of minimalist inquiry, all the chapters should be read in parallel.

I have used the following guiding idea in arranging the material contained here: experienced scholars should be able to find their way around much more easily than students. To help the latter, I have ordered the chapters according to both anticipated familiarity and difficulty. Given that linguistic minimalism first emerged in the context of syntactic theory, I had placed the chapters touching on issues of (narrow) syntax first. More advanced readers who maybe wondering about possible extensions of minimalist questions to other fields are encouraged to skip ahead. I have tried as far as possible to place chapters comparing the relative merits of two possible processes after those chapters in which the processes being compared were first introduced independently.

(p. xxiii) I stress that these were organizational guidelines. In many chapters the very same concept or process is introduced but articulated somewhat differently, and put to somewhat different uses. I have resisted the editorial temptation to eliminate all but one of the passages in which a particular concept was introduced because this very variety of subtle nuances and perspectives is one of the central elements of linguistic minimalism, and I wanted the reader to develop a taste for it. For this very reason I have let many authors start their chapters by giving their own take on what minimalism is. The reader confronted with this multitude of perspectives will thus gain first-hand experience of the very nature of a research program. As a final example of what it is to pursue a program, I commissioned several chapters dealing with roughly the same set of phenomena, but approaching them from very different perspectives. The reader will find in the following pages quite a bit of controversy regarding the nature of anaphora, Last Resort, Merge, the mapping from syntax to semantics, and so on. I hope that the reader will be tempted to formulate his or her own synthesis on the basis of these conflicting views.

Having warned the reader, let me now briefly describe the content of the various chapters that make up this handbook. Chapter 1, by Robert Freidin and Howard Lasnik, discusses the historical roots of linguistic minimalism, and seemed to me to be the best point of entry. Chapter 2, by David Adger and Peter Svenonius, focuses on the nature of the most basic building block in all current minimalist analyses: the feature. As far as I have been able to determine, all minimalist analyses currently available begin with lexical items as bundles of features (in some cases, these bundles consist of single features), combine and recombine these essential elements via processes such as Merge or Agree, and use these features as units for interpretation. This provides sufficient justification to place this chapter on features very early in the book. Chapter 3, by David Pesetsky and Esther Torrego, deals with a particular feature, viz. Case, which played an essential role in the development of the Principles and Parameters (P&P) approach (and minimalism), and which continues to be involved in the formulation of many minimalist investigations.

Chapters 4, 5, and 6, by Naoki Fukui, Jan-Wouter Zwart, and Barbara Citko, respectively, address issues pertaining to phrase structure, and the mechanisms behind them. Fukui examines the nature of Merge (the standard structure-building operation in virtually all minimalist studies), and issues of dominance (projection), long-distance dependencies/movements (can movement be reduced to Merge?), and the minimalist reformulation of the well-known X-bar theory in terms of bare phrase structure. Zwart is concerned with how structures formed by Merge map onto linear order—the issue of linearization that Richard Kayne made central with his 1994 monograph. Finally, Citko examines the possibility of multi dominance, an option that some have argued follows from the very nature of Merge.

Chapters 7 (Jairo Nunes), 8 (Norvin Richards), and 9 (Ian Roberts) all concentrate on movement dependencies. Nunes examines the implications of reanalyzing movement in terms of a copy (and Merge) operation. Richards looks at 'A-bar' (p. xxiv) (operator-variable) dependencies, and Roberts discusses head dependencies. The material in Roberts's chapter, in particular, shows how a central construct in previous frameworks can have its status threatened by certain minimalist ideas, and also how various alternatives can be used to save the phenomena.

Because the theme of locality has traditionally been closely associated with movement, it made sense to me to place chapters dealing with locality right after Chapter 9. Within the P&P tradition, and within minimalism as well, a distinction is often made between intervention-based locality and domain-based locality. Luigi Rizzi tackles the former (in terms of relativized minimality) in Chapter 10, and Juan Uriagereka tackles the latter (by clarifying the concept of cycle, or phase, in minimalist parlance) in Chapter 11. In Chapter 12, Kleanthes K. Grohmann discusses a new area of research within locality (and minimalism)—that of anti-locality, which deals with the minimal distance that dependencies must span to be licit (as opposed to the traditional question of the maximal distance that a dependency can cover).

The locality section seemed to me to be the perfect place to bring up the longstanding debate between derivational and representational approaches to grammatical phenomena: Should constraints (on, say, movement) be seen as emerging from how the syntactic computation takes place, step by step, or should deviant results be generated and then filtered out because of what the output of the syntactic computation is, not because of how that output came about? Samuel D. Epstein, Hisatsugu Kitahara, and T. Daniel Seely examine the nature of derivations in Chapter 13, and Robert Chametzky scrutinizes representations in Chapter 14.

The next group of chapters deals with why syntactic processes apply when they do. In Chapter 15, Željko Bošković looks at how operations can be seen as subject to a Last Resort operation, and in Chapter 16, Shigeru Miyagawa analyzes instances of optional movement, and how optionality can be motivated in a minimalist context, which appears to favor the idea that all but one option should be available. Miyagawa's discussion crucially involves the notion of interpretation and the systems external to narrow syntax that further manipulate linguistic expressions. In Chapter 17, Eric Reuland looks at the division of labor between syntax and the interpretive systems by focusing on patterns of anaphoric dependencies. The very same patterns are given an alternative approach, suggesting a different division of labor, by Alex Drummond, Dave Kush, and Norbert Hornstein in Chapter 18.

Chapters 19 through 22 further delve into interpretive matters, beginning with argument structure (Heidi Harley, Chapter 19), moving to the syntactic representations of events (Gillian Ramchand, Chapter 20), the relation between words and concepts (Paul M. Pietroski, Chapter 21), and culminating with the relation between language and thought (Wolfram Hinzen, Chapter 22). This set of chapters, two of which are written by trained philosophers with excellent knowledge of the linguistic literature, illustrates how minimalist concerns extend beyond the narrow realm of syntax, and how specific articulations of minimalist guidelines may **(p. xxv)** inform questions traditionally addressed by non-linguists. This is the sort of interdisciplinary research that linguistic minimalism, with its emphasis on interfaces, promotes.

Chapters 23 (Ágel J. Gallego) and 24 (Charles Yang and Tom Roeper) revisit traditional concerns in generative grammar—patterns of variation (‘parameters’) and the acquisition of specific linguistic systems—and ask to what extent minimalist guidelines can shed light on these issues. It is fair to say that these are areas of research that have not figured as prominently within linguistic minimalism as they did in previous frameworks. The reader will no doubt notice that many basic questions remain open, and one may anticipate that they will figure prominently on the minimalist research agenda in the coming years.

Chapters 25 (Bridget Samuels) and 26 (Víctor Longa, Guillermo Lorenzo, and Juan Uriagereka) extend the research horizon by applying minimalist concerns to the domain of morphophonology and language evolution, respectively. One can only hope that these forays into new territories will promote work that applies minimalist thinking to other domains such as language processing, neurolinguistics, or other cognitive realms (music, number, and moral systems come to mind, as these have already been studied from a generative perspective, albeit by a minority of researchers).

The final chapter (27) in some sense brings us back where we started. Ed Stabler raises computational concerns that were very prominent in the early stages of generative grammar, and re-examines basic units and operations of minimalist syntax in a formally explicit context.

Let me close this overview by noting that I honestly believe that the material that follows constitutes a representative sample of current minimalism. No doubt, some will feel that specific issues discussed here as part of a chapter ought to have received a more comprehensive treatment as a separate chapter. But I think that all the key concepts within linguistic minimalism today have found their way into the chapters of the handbook. This is thanks to the excellent researchers who found the time to contribute to this project. I want to take this opportunity to thank them all for making this handbook possible. I would also like to thank Noam Chomsky, Michael Brody, Alec Marantz, Danny Fox, Hagit Borer, Bob Berwick, and Richard Kayne, who unfortunately could not contribute to the volume, but who nevertheless offered me advice at various stages along the way. Their works have shaped linguistic minimalism in significant ways, and I hope that the following pages reflect this.

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This volume is dedicated to the memory of three wonderful linguists and remarkable individuals: Kenneth L. Hale, Tanya Reinhart, and Carol Chomsky.

C.B.

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Some Roots of Minimalism in Generative Grammar

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[–] Abstract and Keywords

While the perspective of a minimalist program for linguistic theory (MP) constitutes a significant departure from previous versions of linguistic theory, many of the fundamental ideas under investigation within it can be found in some form in earlier work on generative grammar. This article shows how MP is deeply rooted in the work on generative grammar of the past half-century. It has validated some significant early ideas, while at the same time leading us to new ones and also to new horizons.

Keywords: minimalist program, linguistic theory, generative grammar

WHILE the perspective under a minimalist program for linguistic theory (MP) constitutes a significant departure from previous versions of linguistic theory, many of the fundamental ideas under investigation within it can be found in some form in earlier work on generative grammar. This is what we will be examining in what follows.[1]

The MP arises from a more general research program into the nature of language that takes linguistic knowledge in the mind of the speaker as the object of inquiry. This linguistic knowledge is modeled in terms of a computational system that operates on elements from the lexicon of a language to form representations of linguistic expressions for the sound and meaning of each linguistic expression formed. The lexicon and computational system constitute a grammar of a language, serving as a theory of the I-language, where 'I' stands for internal, individual and intensional. The form of I-languages is largely determined by the grammatical mechanisms (p. 2) available for constructing linguistic expressions and the general constraints on their operation and output. These mechanisms and constraints apply generally across I-languages and therefore belong to the general theory of I-language, what is called universal grammar and thought to characterize the initial state of the language faculty in all speakers.

The MP is extremely easy to state, but nonetheless less easy to explicate. As a research program in linguistic theory, it addresses two fundamental questions about I-languages—i.e. about the computational system plus lexicon that models each I-language.

(1)
- **a.** To what extent is human language a 'perfect' system?
- **b.** To what extent is the computational system for human language optimal?

These questions are interrelated to the extent that an optimal computational system might be considered a prerequisite for a language system that is perfect, though not necessarily the converse. Part of the task of the MP is to render precise interpretations for the terms 'perfect' and 'optimal' as they apply to human language.

These core questions of the MP are raised within a milieu of background assumptions based on decades of empirical and theoretical research on linguistic structure. It is this work that motivates the formulation of a MP. More precisely, these assumptions and the work on which they are based suggest that we can expect the answers to the two core questions to be: 'to a significant extent.'

In the initial discussion of the MP, Chomsky (1993) enumerates a set of 'minimalist' assumptions. The first constitutes the starting point of modern generative grammar—namely that there is a language faculty (FL) in the mind/brain, a cognitive system that interacts with other cognitive systems. It is the FL that accounts for the fact that humans acquire natural languages but other biological organisms do not. FL contains a computational system for human language (henceforth C_{HL}) whose initial state S_0 contains invariant principles and parameters (options for variation restricted by hypothesis to functional elements of the lexicon, e.g. the categories C and T). The selection S of parameters determines a language. In effect, language acquisition can be characterized as two main tasks: acquiring a lexicon and fixing the values of parameters. The language acquired is a generative procedure that determines an infinite set of linguistic expressions given as a pair of representations (π, λ). π represents the 'sound' of the expression, its phonetic form (PF), which interfaces with the articulatory-perceptual (AP) components. λ represents the 'meaning' of the linguistic expression that interfaces with the conceptual-intentional (CI) components.[2] (π, λ) are interpreted at the interfaces as (p. 3) 'instructions' to the performance systems. These assumptions have been standard for the past three decades, some of them obviously longer.[3]

Chomsky (1993) articulates a further assumption that could have been considered prior to the MP: there is no variation in overt syntax or the Logical Form (LF) component.[4] This is identified as 'a narrow conjecture', which, given the considerable potential empirical evidence to the contrary regarding overt syntax, it seems to be. However, if we put aside the kind of arbitrariness that is endemic to the lexicon (e.g. the phonetic labels of lexical items (what Chomsky calls 'Saussurean arbitrariness') and other idiosyncratic properties[5]), then the remaining variation might reduce to parameters and the mapping to PF, as Chomsky wants to claim. This leads to a rather spectacular follow-up conjecture—namely, 'there is only one computational system and one lexicon [apart from Saussurean arbitrariness and idiosyncratic lexical properties]' (1993: 3). This is certainly in line with the direction and results of work in the Principles and Parameters framework, though whether it is correct remains an open question.[6]

The remaining assumptions Chomsky 1993 articulates are in essence unique to the MP. The fundamental assumption from which the rest more or less follow is that the theory of grammar must meet a criterion of conceptual necessity. To a large extent this appears to be a version of Ockham's razor (the law of parsimony), which is essentially a methodological consideration that far predates modern generative grammar. However, the sense of this criterion goes beyond methodology (p. 4) by considering an initial substantive hypothesis: language constitutes 'an optimal way to link sound and meaning' to the extent that FL needs to satisfy only the interface conditions imposed by those components that connect it to the general cognitive system (what Chomsky calls the Strong Minimalist Thesis, SMT: (Chomsky 2008a: 135).[7] From this it follows that the interface levels PF and LF are the only linguistic levels. A linguistic representation of sound that connects with the sensorimotor components of the brain is a minimal requirement for a theory of language. The same is true of a linguistic representation that interfaces with the conceptual/intentional components of human cognitive systems.[8] Beyond PF and LF, the MP discards other proposed linguistic levels, specifically D-Structure and S-Structure, and attempts to reanalyze any empirical evidence that might be used to motivate them.

The MP as spelled out in the two questions in (1) along with the set of background assumptions discussed above constitutes a program of research in linguistic theory. As Chomsky noted over a decade ago, 'it is a program, not a theory, even less so than the Principles and Parameters approach. There are minimalist questions, but no specific minimalist answers' (1998: 119–20).[9] This assessment still holds. However, as a guide for research on linguistic theory the MP continues to have an important heuristic and therapeutic effect in limiting the hypothesis space for linguistic analyses.[10]

Central to the MP is the notion of a language as a lexicon plus a computational system that together generate representations for linguistic expressions. This notion originates in Chomsky (1965), where the lexicon is analyzed as an entity separate from the system of grammatical rules, i.e. a set of phrase structure rules and transformations. This proposal is motivated methodologically on the grounds that it allows for a simplification of grammatical rules, specifically the elimination of context-sensitive phrase structure rules from the phrase structure rule (p. 5) component.[11] Context sensitivity is thus located in the lexicon, which becomes the locus of idiosyncrasy in languages, eventually to include parametric variation as well. The constraint that the phrase structure rule component of a grammar may only contain context-free rules was one of the first major steps in limiting the descriptive power of syntactic theory.[12] Such steps demonstrate how the simplification of grammars is achieved by placing limitations on the descriptive power of grammatical rules.

The separation of the lexicon from the computational system ultimately leads to the Inclusiveness Condition, whereby syntactic representations are restricted to the features of lexical items. As discussed in Chomsky (1995b: 228),

> A ‘perfect language’ should meet the condition of inclusiveness: any structure formed by the computation (in particular, PF and LF) is constituted of elements already present in the lexical items selected for [the numeration] N; no new objects are added in the course of computation apart from rearrangements of lexical properties (in particular, no indices, bar-levels in the sense of X-bar theory, etc.).

In this way the lexicon places severe restrictions on the computational system itself, as well as providing a significant criterion for determining the perfection of the language system.

(p. 6) How optimal the computational system for human language turns out to be depends significantly on the form and function of grammatical operations which generate linguistic representations. It is generally assumed that optimality can be measured in terms of the simplicity of the system, the simplest system being the most optimal. The evolution of singulary transformations in syntactic theory provides a striking example of how the formulation of the computational system has moved toward greater and greater simplicity.

Consider, for example, one of the earliest formulations of a wh-movement transformation, as given in *Syntactic Structures* (Chomsky 1957: 112). The rule is divided into two parts, one that moves an NP to the left edge of the string under analysis and another that adjoins an element *wh-* to the fronted NP, which is then converted into an interrogative pronoun. The rule is stipulated to be optional but conditional on the application of another rule that moves a finite auxiliary in front of a subject NP. The first part of the rule is formulated as (2).

(2)

T_{w1}:	Structural analysis:	X − NP − Y
	Structural change:	$X_1 - X_2 - X_3 \rightarrow X_2 - X_1 - X_3$

Taking X − NP − Y to analyze a clause, the transformation moves NP, the only constant term identified in the structural description of the rule, to clause-initial position.

From the outset (e.g. Chomsky 1955) it was understood that transformations like (2) would produce deviant constructions in English and therefore that something further was required for a grammar of the language.[13] For example, Chomsky (1964a) notes that (3a) cannot be derived from the structure (3b), whereas (3c) can.

(3)

a. *what did he know someone who has (of yours)?
b. he knew [$_{NP}$someone who has [$_{NP}$something] (of yours)]
c. who who has something (of yours) did he know?

However, (2) applied to the NP *something* in (3b) will yield the deviant (3a). When (2) applies to the larger NP *someone who has something (of yours)*, the result (3c) is not deviant. Chomsky's solution is to propose a general constraint on the operation of transformations that prohibits ambiguity in their application. In (3b) the transformation (2) can target either the NP *something* or the larger NP *someone who has something (of yours)*, which contains it. Chomsky proposes ‘a general requirement that the dominating, rather than the dominated, element must always be selected in such a case’ (1964a: 931). He goes on to suggest that when appropriately formalized, such a constraint ‘might then be proposed as a hypothetical linguistic universal’.[14]

(p. 7) Having general constraints on the application of transformations makes it possible to retain an optimally simple formulation of transformational rules—in this case, the rule for *wh*-movement.

The history of the passive transformation for English demonstrates the same impetus towards a minimal formulation of rules in a more spectacular way. Consider the transformational rule for passives in *Syntactic Structures* (4).

(4)

Passive—optional:	
Structural analysis:	NP − Aux − V − NP
Structural change:	$X_1 - X_2 - X_3 - X_4 \rightarrow X_4 - X_2 + be + en - X_3 - by + X_1$

As formulated in (4), the passive transformation performs four separate sub-operations. It inserts both the English passive auxiliary *be+en* and the English preposition by, thus making it a language specific transformation. In addition it reorders the underlying subject NP to follow the passive *by* and also reorders the NP immediately following V to the position immediately preceding the phrase Aux. By the mid 1970s (Chomsky 1976), this rule had been reduced to (5).

(5) Move NP

The two lexical insertion operations in (4) are no longer part of a grammatical rule that derived passive constructions. The sub-operation that reordered the underlying subject NP to follow the passive *by* no longer plays a role in the derivation of passive constructions.[15] And, perhaps the most interesting development, the sub-operation that effects the preposing of the underlying NP following V in passive constructions under the formulation in (5) now generalizes to interclausal movements involved in the derivation of raising constructions. Thus passive constructions as in (6a) and raising constructions (as in (6b–c)) can now be accounted for as instances of the same phenomenon, NP-movement. (The symbol *t* merely indicates the position from which the NP in boldface was moved by rule (5)and α designates a clause boundary.)

(6)

a. [$_{\alpha}$ **the student** was praised *t* by the professor]

b. [$_{\alpha}$ **the professor** is expected [$_{\alpha}$ *t* to praise her students]]

c. [$_{\alpha}$ **the professor** seems [$_{\alpha}$ *t* to have lost his lecture notes]]

In this way, the theory of transformations abandons the formulation of construction-specific rules like (4) in favor of more general and more abstract formulations of transformations.

(p. 8) A formulation like (5) was made possible by extending the approach developed for *wh*-movement (as discussed above) to NP movement. General conditions on the application of transformations and on the representations produced determine the behavior of the grammatical operation (e.g. movement) specified in the rule; therefore, this operation need not be further constrained by stipulating a specific syntactic context via a structural description in which the rule can or must apply. In effect, the rule in *Syntactic Structures* for *wh*-movement (2) now extends to non-*wh* NP movement. The movement of the constant term NP, the second term in the structural description of (2), crosses a variable, the first term of (2). The A-over-A Principle (see note 14 above) and the Ross island constraints (Ross 1967a) are the first proposals for general constraints on such variables. Both proposals focus almost exclusively on the overt movement of *wh*-phrases. Chomsky (1973) extends this approach to incorporate non-*wh* NP movement.

The development from the passive transformation formulated as (4) to a more general movement rule like (5)—and ultimately to the most general formulation as ‘Move α’ in Chomsky (1981a)—constitutes a shift from a language-particular and construction-specific rule to one that must be part of universal grammar, since the grammar of every language that manifests some type of movement will incorporate this rule.

Rules like (2) and (5) involve a single grammatical operation, in contrast to the original passive transformation (4), which contains several. Restricting all grammatical transformations to a single transformational operation places a significant limitation on the notion ‘possible rule of grammar’, and hence on the class of possible grammars. Such restrictions contribute to the simplicity of grammars and presumably reduce computational complexity.

There is of course a more precise analysis of transformational operations that has been in the background since the outset of modern generative grammar (Chomsky 1955), where grammatical transformations consist of one or more elementary transformational operations. In the case of a movement transformation like (2) or alternatively (5), this analysis raises the question of what elementary operations are involved. Under the optimally simple formulation, the answer would be a single elementary operation, which results from extending the restriction on grammatical transformations to the level of elementary operations. In effect, grammatical transformations cannot compound elementary operations.[16] It therefore follows that a movement transformation cannot include a deletion, where the deletion operation accounts for the fact that at PF a moved constituent is pronounced in the position to which it finally moves (e.g. the phrases in boldface in (6)) and not in the position from which it initially moves (e.g. the syntactic object designated as *t* in (6)).

(p. 9) Limiting grammatical transformations to single elementary transformational operations has another important consequence for the analysis of movement phenomena. The constituent that appears to be moved must be a copy of the constituent that is targeted by the operation. It is the placement of the copy that constitutes the effect of movement. However, the constituent targeted by the operation Copy remains in its original place. It does not move. This insight into the nature of displacement phenomena has been a central part of syntactic theory since the advent of trace theory in the mid 1970s (Fiengo 1974, Chomsky 1973).[17]

The elementary operation that effects movement by positioning a copy in a phrase-marker has had two distinct formulations, one as a substitution operation, which is structure-preserving, and the other as a form of adjunction operation,[18] which is structure-building. The latter prevails in minimalist analysis for a number of reasons. On the basis of economy of derivations, the adjunction analysis is preferable because it requires only one step, whereas the substitution analysis requires two. Under substitution, there must be a constituent that is replaced by the moving constituent, so the replaced constituent must be created first and then replaced. Under adjunction, the adjunction of the moved constituent to a root creates the landing site in the process of adjoining to the root. Furthermore, the substitution analysis unnecessarily violates the Extension Condition (hence cyclicity). It also violates the Inclusiveness Condition unless the lexicon contains items that consist of solely syntactic category features (i.e. without phonetic, morphological, or semantic features), which seems unlikely.

The adjunction analysis is also preferable for an even more important reason. Taking the adjunction operation to be Merge, which has both a grouping function and a labeling function,[19] we can use it to create phrase structure and thereby (p. 10) dispense with phrase structure rules.[20] Instead of treating transformations as a grammatical mechanism that must be tacked onto a core phrase structure grammar for empirical reasons (see Chomsky 1957, Postal 1964), transformations emerge as the only core grammatical mechanism.[21] In effect, all syntax is transformational in nature.[22]

With Merge as the operation that constructs phrase structure and also accounts for displacement phenomena (‘movement’), it is technically impossible to construct canonical D-structure, which in previous theory is a level of representation that results from the application of phrase structure rules plus lexical insertion but crucially no movement transformations. For example, the D-structure of (6a) would be along the lines of (7), where **NP** indicates an ‘empty’ category—i.e. one without phonetic, morphological, or semantic features.

(7) $[_{TP}$ **NP** $[_{T}$ was $[_{VP}$ $[_{V}$ praised $[_{NP}$ the student$]$ $[_{PP}$ by $[_{NP}$ the professor$]]]]]]$

Merge cannot produce (7) first of all because the lexicon does not contain a lexical item **NP** and also because this operation is completely unnecessary given that merging a copy of *the student* with the T-phrase *was praised the student by the professor* creates the target structure (6a) without having to posit additional grammatical machinery (i.e. a separate substitution operation) or the questionable lexical item **NP.**

(p. 11) The first minimalist argument offered against D-structure is, however, purely conceptual. In a major minimalist development, Chomsky (1993: 169) argues that the interface levels LF and PF are the **only** levels of representation:[23]

> [UG] must specify the interface levels (A-P, C-I), the elements that constitute these levels, and the computations by which they are constructed. A particularly simple design for language would take the (conceptually necessary) interface levels to be the only levels. That assumption will be part of the ‘minimalist’ program I would like to explore here.

This elimination of D-structure is, in interesting respects, a return to the roots of transformational generative grammar.

In the earliest work in transformational grammar (Chomsky 1955/1975a), there is no level of D-structure. A phrase structure component constructs P-markers for simple sentences. Generalized transformations combine these single clause structures into multiple clause structures. The 'recursive' component of the grammar is thus part of the transformational component, in particular the generalized transformations, which combined pairs of P-markers either by coordinating the pair or by subordinating one to the other through embedding. Chomsky (1965) rejects this model in favor of one with recursion in the phrase structure rule component. The output of this component and the lexical insertion transformations is 'deep structure'. Chomsky's major arguments for this innovation are that it results in a simpler overall theory, and at the same time it explains the absence of certain kinds of derivations for which there appeared to be no empirical motivation. Chomsky's second point is based on the observation in Fillmore (1963) that while there is extensive ordering among singulary transformations (see note 28 below), 'there are no known cases of ordering among generalized transformations although such ordering is permitted by the theory of Transformation-markers' (Chomsky 1965: 133). Further, while there are many cases of singulary transformations that must apply to a constituent sentence before it is embedded, or that must apply to a 'matrix' sentence after another sentence is embedded in it, 'there are no really convincing cases of singulary transformations that must apply to a matrix sentence before a sentence transform is embedded in it' (1965: 133). In other words, the earlier theory allows for a class of grammars that do not appear to exist.

As for the argument from simplicity, Chomsky claimed that the theory of transformational grammar is simplified by this change, since the grammatical machinery of 'generalized transformations' and 'transformation-markers' (T-markers) are eliminated entirely. The P-markers in the revised theory, what Chomsky designates as 'generalized P-markers', contain all of the information of those in the *LSLT* version (Chomsky 1955/1975a), but they also indicate explicitly how the clauses are embedded in one another—information that had been provided by the embedding transformations and T-markers.

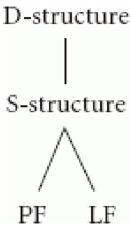
(p. 12) This extension of the theory of phrase structure rules to include recursion, which makes generalized transformations redundant, also has consequences for the theory of singulary transformations. As indicated above, in the *Aspects* theory, as in the *LSLT* theory, there is extensive ordering among singulary transformations. In both frameworks, the set of singulary transformations was seen as a linear sequence: an ordered list. Given the *Aspects* modification, this list of rules applies 'cyclically', first operating on the most deeply embedded clause, then on the next most deeply embedded, and so on, working 'up the tree' until they apply on the root clause, the entire generalized P-marker. Thus, singulary transformations apply to constituent clauses 'before' they are embedded and to matrix clauses 'after' embedding has taken place. 'The ordering possibilities that are permitted by the theory of Transformation-markers but apparently never put to use are now excluded in principle' (Chomsky 1965: 135).

Since minimalism returns to generalized transformations, in fact giving them even more prominence since **all** structure building is done by them instead of by PS rules, we must reconsider the *Aspects* argument against them. Recall that Chomsky argues that the *Aspects* model, lacking generalized transformations, excluded certain undesired interactions between generalized and singulary transformations (basically, anti-cyclic derivations). However, on closer inspection, it was not actually elimination of generalized transformations that had this limiting effect. Rather, it was the constraint that transformations operate bottom-up, starting on the most deeply embedded clause and proceeding cyclically up the tree. Chomsky (1993) observes that a condition with the same effect can be imposed on the operation of generalized transformations and their interaction with singulary transformations. Chomsky imposes the condition that operations before Spell-Out must extend their target (the Extension Condition[24]), and observes that this yields a version of the strict cycle. This guarantees the same sort of monotonic derivations as those permitted by Chomsky (1965).

The one remaining *Aspects* argument against generalized transformations can also be straightforwardly addressed. Chomsky had argued that eliminating generalized transformations yields a simplified theory, with one class of complex operations jettisoned in favor of an expanded role for a component that was (p. 13) independently necessary: the phrase structure rule component. This was a very good argument. But since then, the transformational component has been dramatically restricted in its descriptive power. In place of the virtually unlimited number of highly specific transformations available under the theories of the 1950s and early 1960s, we have instead a tiny number of very general operations: Merge (the generalized transformation, expanded in its role so that it creates even simple clausal structures, in fact all basic structure), Copy, Delete. The complex apparent results come not from complex transformations, but from the interactions of very simple ones with each other, and with very general constraints on the operation of transformations and on the ultimate derived outputs. The 1965 argument can then be reversed on itself: Eliminate phrase structure rules, in the culmination of a program initiated in Stowell (1981).

Given the severely restricted inventory of elementary operations available, recent work in minimalism suggests that ‘single-cycle’ derivations would be the ideal for a computational architecture of human language. Under the single-cycle view, there is a single-cyclic derivation, beginning with a selection of items from the lexicon, which builds the structure, successively transforms it, and periodically (at phases) sends information to the phonological and semantic interfaces. In such a derivation, syntactic, phonological, and semantic rules are interleaved in their application to linguistic structures. This stands in marked contrast to the principles and parameters ‘Y-model’ in (8).

(8)



In this model, a transformational cycle links an internal level D-structure to another internal level S-structure. A subsequent phonological cycle connects S-structure to the interface level PF. Parallel to this latter cycle, a (‘covert’) transformational cycle relates S-structure to LF. This three-cycle model depends on the existence of internal levels of representation (i.e. beyond interface levels), which are prohibited under minimalist assumptions. The argument for a single-cycle derivation is thus conceptual. As Chomsky notes, ‘This computational architecture, if sustainable, seems to be about as good as possible. S-structure and LF are no longer formulable as levels, hence disappear along with D-Structure, and computations are reduced to a single cycle’ (2005: 18).[25]

(p. 14) The modern source of the single cycle concept is the Multiple Spell-Out proposal (Uriagereka 1996, 1999), which Uriagereka (1999: 276) suggests approximates the cyclic Spell-Out proposal in Chomsky (2000a). Under Multiple Spell-Out, the operation of Spell-Out is itself cyclic, applying at several points during the derivation rather than just one. This makes it impossible to separate a derivation into an overt vs. a covert part, as is possible in a derivation that has only one point where Spell-Out applies. The computational architecture that results from cyclic Spell-Out is reminiscent of a proposal in Bresnan (1971), which both Uriagereka and Chomsky cite as an early (if only partial) antecedent. Bresnan argues on empirical grounds that at least one class of phonological rules—those assigning stress patterns to sentences, and in particular the Nuclear Stress Rule (NSR) of Chomsky and Halle (1968)—must apply after the syntactic transformations in each cyclic domain. Therefore the NSR must be part of the syntactic cycle rather than part of a separate phonological cycle.

Bresnan notes that ‘the stress patterns of certain syntactically complex constructions appear to violate the general prediction made by the NSR’ (1971: 258). Among several examples she discusses, there is the following contrast, first noted in Newman 1946:

(9)

a. George has **plans** to leave
b. George has plans to **leave**

The words in boldface receive the heaviest stress in each sentence; the difference in stress pattern correlates with two distinct interpretations. In (9a) *to leave* is interpreted as an infinitival relative clause, where *plans* is interpreted as the object of the transitive verb *leave*, as well as the object of *has*. In contrast, *leave* in (9b) is interpreted as an intransitive verb and the infinitival phrase *to leave* functions as a complement of *plans*. So George is planning to leave in (9b), whereas in (9a) he possesses plans which he intends to leave at some unspecified location. If the NSR applies in a separate phonological cycle after the syntactic cycle, only one of these patterns (9b) is derivable. Bresnan demonstrated that in all the cases she discusses both patterns are in fact ‘predictable without any special modifications in that rule, given one assumption: THE NUCLEAR STRESS RULE IS ORDERED AFTER ALL THE SYNTACTIC TRANSFORMATIONS ON EACH TRANSFORMATIONAL CYCLE’ (Bresnan 1971: 259, capitalization original).

The Chomsky and Halle NSR assigns primary stress to the rightmost primary stressed (i.e. by the prior operation of word stress rules) item in its cyclic domain of operation. On successive cycles, primary stressed items can be reassigned primary stress by the NSR, if they are rightmost in those successive domains. Finally, by convention, each time primary stress is assigned, all other items in the domain have their stress weakened by 1. For Bresnan, the derivation leading to (9a) would be something like (10).

(p. 15) **(10)** [s George has [$_{NP}$ plans [$_S$ PRO to leave plans]]

1	1	1	1	1	
1	1	1	2	1	Cycle 1: NSR
				ϕ	Cycle 2: Syntax
		1	3		Cycle 2: NSR
2	2	1	4		Cycle 3: NSR

For (9)b, on the other hand, intransitive *leave* is always rightmost, hence, it receives and retains primary stress throughout the derivation. Bresnan's basic point is that if we waited until the end of the syntactic derivation to begin applying the NSR, the relevant distinction between (9a) and (9b) would be lost.

On the semantic side, Jackendoff (1969, 1972) proposed that the rules assigning coreference relations to pronouns, reflexives, and null subjects of non-finite complements apply, just like Bresnan's version of the NSR, at the end of each syntactic cycle. Jackendoff's representation of coreference relations is specified in a 'table of coreference'. This is a list of pairs of referential expressions in an utterance, marked as coreferential if certain conditions, primarily structural, are satisfied. What is important from our perspective here is that entries in the table are determined at the end of each syntactic cycle. There were already existing arguments for the cyclic nature of the rules establishing anaphoric relations. Especially influential were those presented by Ross (1967b). But for Ross, the relevant rules were all transformations (e.g. a pronominalization transformation deriving a pronoun from an NP identical to its antecedent), rather than 'interpretive' rules applying to base-generated pronouns, so the issue of separate cycles vs. single cycle did not arise. For Jackendoff's interpretive approach, however, the question is of substantial significance, as is Jackendoff's cyclic answer.

Among the phenomena of interest to Jackendoff is one of the paradigms used by Postal (1970) to argue against Ross's cyclic ordering of pronominalization:

(11)
a. Who that Mary knew do you think she visited
b. Who that she knew do you think Mary visited

Both examples allow coreference between *Mary* and *she*. But, Postal argues, if (transformational) pronominalization were cyclic, only (11b) would be derivable because of the obligatory character of the rule. At the relevant point in the derivation of (11a) we would have:

(12) [$_{S1}$ You think [$_{S2}$ Mary visited [$_{NP}$ who that Mary knew]]]

On the S_2 cycle, pronominalization would obligatorily apply forwards and downwards, rendering (11a) ungenerable. For Jackendoff, this problem for cyclic application doesn't obtain, since pronouns are present in the base. Thus the structure underlying (11a) would be (13) rather than (12).

(p. 16) **(13)** [$_{S1}$ You think [$_{S2}$ she visited [$_{NP}$ who that Mary knew]]]

His interpretive pronominalization rule would be inapplicable on S_2 because *she* both precedes and commands *Mary*. Then on the S_1 cycle after *wh*-movement relocates the interrogative pronoun with its relative clause to the front of the root clause, Jackendoff's interpretive rule could apply—optionally because the preferential interpretation of the pronoun in (4a) is optional—thus yielding (14).

(14) [$_{S1}$ [$_{NP}$ who that Mary knew] do you think [$_{S2}$ she visited]]

Bresnan's argument is mainly empirical, though she hints at a more general conceptual justification: ‘we see that the stress-ordering hypothesis provides a kind of “naturalness condition” on syntactic derivations: the formal properties of surface structures cannot diverge too greatly from those of deep structures without destroying the relationship between syntax and prosodic stress’ (1971: 272). Jackendoff's argument is, in contrast, essentially technical. Note that what we discussed above merely makes it *possible* for interpretive pronominalization to be in the transformational cycle. For Jackendoff, what ultimately makes it *necessary* is that the rule should partially collapse with the rules interpreting reflexive pronouns (‘reflexivization’) and null complement subjects (‘control’), which are assumed to be cyclic (like their transformational predecessors). Lasnik (1972) presents a theoretical argument. Assuming that the only possibilities for scope assignment are that it takes place internally to the syntactic derivation or else ‘post-cyclically’ (essentially, at surface structure), Lasnik reasons that ‘strict cyclicity’ in Chomsky's (1973) sense demands the former, once again implicating the single-cycle model:

> Briefly, strict cyclicity requires 1) that no cyclic transformation apply so as to involve only material entirely within a previously cycled domain; and 2) that a transformation only involving material in an embedded cyclic domain be cyclic. Requirement 2 is simply a way of saying that whether or not a rule is cyclic should depend solely on its domain of application. It excludes the possibility of calling passive, for example, a post-cyclic transformation, thereby allowing it to escape from requirement 1 by a notational trick. This convention, if extended to interpretive rules, would require that a rule assigning not a scope be S cyclic, since in the examples I have discussed scope relations would be the same even if the sentences were deeply embedded.[26] (Lasnik 1972: 69–70)

As we have just discussed, cyclic computation and its principles, including the Extension Condition and the No Tampering Condition (see note 24 above), Multiple (cyclic) Spell-Out and derivation by phase (Chomsky 2001, 2008a), significantly (p. 17) restrict the function of the computational system. In so doing, they move the theory of grammar closer to a MP goal of optimal and efficient computation. Within the MP, these concepts fall under the broader notions of economy (of derivation and representation) and the associated notion of simplicity, both of which contribute to the optimality of the system of grammar and thereby support the notion that language may be a perfect system. These notions, like others we explore here, have their roots in earlier, sometimes much earlier, work in generative grammar.

Significantly, even the earliest investigations sharply distinguished between simplicity as a general criterion for evaluating scientific theories and the empirical notion of simplicity embedded within a particular theory of language. Thus, Chomsky (1955/1975a), in Chapter 4, ‘Simplicity and the form of grammars’, states:

> Note that when we speak of the simplicity of linguistic theory, we are using ‘simplicity’ in the still vague sense in which simplicity is an ideal for any science, whereas when we speak of the simplicity of grammars, we are using it in a sense which we hope to make as precise as the definition of ‘phoneme’ or ‘morpheme.’ The simplicity of linguistic theory is a notion to be analyzed in the general study of philosophy of science; the simplicity of grammars is a notion defined within linguistic theory.’ (1955/1975a: 119)

A substantial portion of Chomsky's early work was devoted to investigating this latter concept, in particular his MA thesis at the University of Pennsylvania (Chomsky 1951/1979), which offers several relevant comments.

> … one of the considerations involved in setting up linguistic elements in a particular way, and consequently, in determining what are in fact the grammatical sentences, will be the total simplicity of the grammar in which these elements appear.[…] the grammar must be designed in such a way as to be the most efficient, economical, and elegant device generating just these sentences. (1979: 3)

Note this very early allusion to economy of derivation. Chomsky's conception of grammar as a device for ‘synthesizing utterances’ rather than one which merely states regularities in a language in terms of various levels of analysis—sharply departing from the prevailing approach of his American structuralist teachers (cf. Harris 1951)[27]—naturally leads to these general issues of the economy and efficiency of linguistic computation.

In the 1950s, rule ordering emerged as a fundamental mechanism in grammars, in both phonological and syntactic components.[28] Some of Chomsky's early discussions therefore focused on simplicity implications of ordering statements. (p. 18) While rule ordering is no longer regarded as a language-particular property of grammars, the discussion of simplicity in certain respects foreshadows more recent concerns:

> … the simplicity of the system is at what might be called a 'relative maximum' with this ordering of statements [i.e. phonological rules]. It is not excluded that some complicated set of interchanges of the statements might give a simpler grammar, or in fact, that a total recasting in different terms might be more elegant. Thus this investigation is limited in that only one 'dimension' of simplicity is considered, viz., ordering. Actually a complete demonstration would have to show that the total simplicity is greatest with just the given ordering, segmentation, classification, etc. (1979: 5)

Chomsky here formulated a concrete (if still incomplete) notion of simplicity, one that played an important role in his work through the 1960s: 'As a first approximation to the notion of simplicity, we will here consider shortness of grammar as a measure of simplicity ….' (1979: 5). 'Shortness' in this context refers to the formulation of rules under conventional notations like bracketing (see below). This is the core of the definition of 'simplicity' in Chomsky and Halle (1968).

Two comments are in order here. First, in this early work the major role of this theory-internal notion of simplicity was to guide the linguist toward the appropriate grammar of the language under analysis, given data and an articulated theory of grammar, a kind of evaluation procedure (see Chomsky 1957: ch.6). This task of the linguist seems strikingly similar to the task of the language learner. This latter was surely a subtext, but it didn't become the major text until Chomsky (1965), with the introduction of the 'evaluation metric' as a central portion of the language acquisition device. We acknowledge that neither of these is quite the same as the notion of simplicity found in minimalist analysis, but there is, it seems to us, a family resemblance. Second, while this notion of simplicity (shortness of grammars) sounds a lot like the general aesthetic criterion for scientific theory construction, conceptually it is very different. It is part of a theory of language, and is subject to empirical investigation. Notice that size of a grammar in terms of number of symbols depends on choice of notation, also an empirical matter, as notational conventions must be chosen in order to capture actual generalizations. Chomsky (1951/1979) emphasized this issue, asserting:

> [we] will use such notations as will permit similar statements to be coalesced. To keep this notion of simplicity from reducing to an absurdity, the notations must be fixed in advance, and must be chosen to be neutral to any particular grammar, except with respect to the considerations they are chosen to reflect. (1979: 5–6)

The following again makes it clear to what extent the relevant notion is theory-internal:

> Given the fixed notation, the criteria of simplicity governing the ordering of statements are as follows: that the shorter grammar is the simpler, and that among equally short grammars, the simplest is that in which the average length of derivation of sentences is least. (1979: 5–6)

(p. 19) While shortness of grammar can easily be confused with a general aesthetic notion, length of derivation is straightforwardly internal to the theory of grammar. Chomsky (1955) offers further elaboration on simplicity, and on notation:

> we discuss the possibility of defining simplicity of grammar within linguistic theory. We can approach such a conception by providing notations for grammatical description which convert considerations of simplicity into considerations of length by permitting coalescence of similar grammatical statements. This favors grammars that contain generalizations. We have a generalization when we can replace a set of statements, each about one element, by a single statement about the whole set of elements. More generally, we have a partial generalization when we have a set of similar (not identical) statements about distinct elements. By devising notations that permit coalescence of similar statements, we can measure the amount of generalization in a grammar by length. Other features of simplicity can also be measured in a natural way in terms of length. For this approach to be significant, we must develop a fixed set of notations in linguistic theory, and a fixed form for grammatical statement. The definition of these notations (essentially, the construction of a 'language of grammar') constitutes the basic part of the definition of simplicity. (1955/1975a: 67)

A classic example of such notational conventions in phonology is representation of phonemes in terms of binary distinctive features, whereby, for instance, a phonological rule affecting p, t, and k is simpler than one affecting just two of these three. In syntax, one famous case (slightly altered here for ease of exposition) is the device of parentheses, whereby (15)

(15) S → Tense (Modal) (Perfect) (Progressive)

is evaluated as no more costly than the single statement with all four elements, and dramatically less costly than the corresponding eight statements that (15) 'expands' to.

(Chomsky 1958/1962) elaborates further on the empirical nature of the relevant notion of simplicity, and on how it can be investigated:

> The problem of giving a general definition of simplicity of grammar is much like that of evaluating a physical constant. That is, we know in many cases what the results of grammatical description should be (e.g., we know which sentences are structurally ambiguous, and should, correspondingly, have dual representations on some level in a successful grammar), and we can attempt to define simplicity in general terms in such a way as to force us to choose the correct solution in particular cases. (p. 242)

Since the computation of simplicity is dependent on the notation chosen, there is the potential danger of circularity. Chomsky addresses that issue as well:

> There is no vicious circularity here. In the same way, we will try to define every other notion of general linguistic theory so that, in certain crucial and clear cases, we arrive at desired results, just as a scientist in any other field will try to construct his theories so that they predict correctly in particular cases. As long as the definitions given of simplicity, phoneme, etc., are quite general, this will not be reduced to a triviality. (p. 242)

(p. 20) Simplicity, in its early instantiation as a grammar evaluating metric for the linguist or the learner, has a certain 'global' character, in that crucially it evaluates the grammar as a whole:

> Notice that simplicity is a systematic measure; the only ultimate criterion in evaluation is the simplicity of the whole system. In discussing particular cases, we can only indicate how one or another decision will affect the over-all complexity. Such validation can only be tentative, since by simplifying one part of the grammar we may complicate other parts. It is when we find that simplification of one part of the grammar leads to corresponding simplification of other parts that we feel that we are really on the right track. (Chomsky 1957: 55–6)

The reasoning is valid, but given the unlimited number of grammars available, there would seem to be substantial, perhaps intractable, computational complexity involved, parallel to that involved in the 'global' economy computations of early minimalism. The former concern ultimately led to the Principles and Parameters approach, with its severe limitation on the class of possible grammars; the latter led to 'local economy', as in Collins (1997), where at each derivational step, only the most economical next step is permitted. Notice that the early discussions of simplicity focus on the formulation of grammatical rules—which is not the focus of the MP.

The core ideas for the current perspective on economy, focusing as they do on derivations and representations, predate the initial formulation of the MP in Chomsky (1993), having already appeared in Chomsky (1986b, 1991). Economy of derivation, in the form of a 'last resort' character of operations, occurs in Chomsky (1986b), crystallizing earlier observations:

> … movement is a kind of 'last resort.' An NP is moved only when this is required, either because it is a wh-phrase that must appear in an operator position (at least at LF), or in order to escape a violation of some principle: the Case filter, as in the case of passive and raising.… (1986b: 143)

In essence, this begins to reverse the leading idea in GB that all movement is optional, with the bad results of failing to move, or of moving when not appropriate, being filtered out. This GB position had something in common with the earliest transformational work, where many of the transformations were designated as optional. However, just slightly later Chomsky reconsiders, suggesting that optionality is to be avoided, returning to an even earlier view in Chomsky (1958/1962) that 'an obvious decision is to consider minimization of the optional part of the grammar to be the major factor in reducing complexity'. That is, a desideratum is that syntactic derivations be deterministic.

Chomsky (1965) pursues this thought still further:

> … it has been shown that many of the optional singulary transformations of [Chomsky 1955] […] must be reformulated as obligatory transformations, whose applicability to a string is determined by presence or absence of a certain marker in the string. (1965: 132)

(p. 21) In the course of this discussion, following Katz and Postal (1964), Chomsky further suggests the germ of the minimalist Inclusiveness Condition, which restricts derivations from introducing new structure not already part of the lexical items selected for a derivation: ‘transformations cannot introduce meaning-bearing elements’ (1965: 132). This relates naturally to the point at issue, since many of the earlier optional transformations did introduce ‘meaning-bearing elements’ such as negation or the *wh*-morpheme. In the *Aspects* model, the constraint is on post-D-structure insertion of such elements. Under minimalist analysis, the constraint is that the numeration, the selection of items from the lexicon that begins the derivation, contains all of the material that will ever appear.

In contrast to economy conditions on derivations, which are hinted at in some of the earliest work in transformational grammar, the topic of economy conditions on representations does not come up until a few years before the advent of the MP. Chomsky (1986b) introduces the fundamental concept with a principle constraining representations, which he calls Full Interpretation, such that representations at the interface levels LF and PF must consist entirely of elements that are interpretable by the systems external to the FL that these levels interface with.

> there is a principle of full interpretation (FI) that requires that every element of PF and LF, taken to be the interface of syntax (in the broad sense) with systems of language use, must receive an appropriate interpretation—must be licensed in the sense indicated. None can simply be disregarded. At the level of PF, each phonetic element must be licensed by some physical interpretation. The word *book*, for example, has the phonetic representation [buk]. It could not be represented [fburk], where we simply disregard [f] and [r] […] Similarly, we cannot have sentences of the form (88), interpreted respectively as ‘I was in England last year,’ ‘John was here yesterday,’ ‘John saw Bill,’ and ‘everyone was here,’ simply disregarding the unlicensed bracketed elements *the man, walked, who*, and *every*:
>
> **(88)**
>
> **(i)** I was in England last year [the man]
> **(ii)** John was here yesterday [walked]
> **(iii)** [who] John saw Bill
> **(iv)** [every] everyone was here
>
> This is not a logically necessary property of all possible languages; for example, FI is not observed in standard notations for quantification theory that permit vacuous quantifiers in well-formed expressions, as in (89i), which is assigned the same interpretation as (89ii):
>
> **(89)**
>
> **(i)** $(\forall x)\ (2+2 = 4)$
> **(ii)** $2+2 = 4$
>
> But FI is a property of natural language.’ (1986b: 98–9)

FI gives special significance to the levels of LF and PF: ‘The levels P[F] and L[F] constitute the interface of the language faculty with other cognitive systems, and correspondingly, the licensing conditions at P[F] and L[F] are, in a sense, “external” (1986b: 100)’.

We have seen economy and simplicity playing a role in theorizing from the very outset of the generative program. What is perhaps new in the minimalist program (p. 22) is the centrality that those notions have assumed in current theorizing under the MP and the new perspective on the FL they have revealed.

As part of this new perspective, one of the most ambitious projects within the MP is the attempt to move beyond explanatory adequacy. It is worth recalling that ‘explanatory adequacy’ is in fact a technical term in the theory of grammar. Chomsky (1965) defines it as follows:

> To the extent that a linguistic theory succeeds in selecting a descriptively adequate grammar on the basis of primary linguistic data, we can say that it meets the condition of *explanatory adequacy*. That is, to this

> extent, it offers an explanation for the intuition of the native speaker on the basis of an empirical hypothesis concerning the innate predisposition of the child to develop a certain kind of theory to deal with the evidence presented to him. Any such hypothesis can be falsified (all too easily, in actual fact) by showing that it fails to provide a descriptively adequate grammar for primary linguistic data from some other language—evidently the child is not predisposed to learn one language rather than another. It is supported when it does provide an adequate explanation for some aspect of linguistic structure, an account of the way in which such knowledge might have been obtained. (1965: 25–6)

Explanatory adequacy concerns linguistic theories that meet a condition of descriptive adequacy, which itself concerns an accurate account of the intuitions of the native speaker.[29] A theory that meets the condition of explanatory adequacy provides an explanation for these intuitions by addressing the question of how they could have been acquired by the native speaker on the basis of (impoverished) primary language data (PLD).[30]

(p. 23) Before the MP, this explanation has (by hypothesis) taken the form of innate constraints on the computational system for human language, thus considered to be part of UG. UG constraints on the function and output of the computational system (i.e. on the application of grammatical mechanisms and on the representations they create in conjunction with a lexicon) are, again by hypothesis, part of the initial state (S_0) of the language faculty (FL). S_0 is one of three factors whose interaction yields the properties of the I-language attained. According to Chomsky, 'assuming that the faculty of language has the general properties of other biological systems, we should, therefore, be seeking three factors that enter into the growth of language in the individual' (2005: 6). The other two factors involve individual linguistic experience (essentially PLD) and 'principles not specific to the faculty of language' (2005: 6). This third factor has also been characterized as 'general properties of organic systems' (Chomsky 2004a: 105).

The third factor itself subdivides into distinct parts, including general principles of data analysis not unique to language acquisition[31] and 'principles of structural architecture and developmental constraints that enter into canalization, organic form, and action over a wide range, including principles of efficient computation' (Chomsky 2005: 6). Among the latter would presumably be the interface constraints (ICs), what Chomsky (1995b) refers to as 'bare output conditions' (p. 221), imposed by the cognitive components that interact with the FL. Being imposed from outside, ICs are not part of the FL itself.

The one proposal we have that seems to fit the properties of an IC is the one condition on the economy of representations, Full Interpretation (FI: see Chomsky 1986b, 1991), which as noted above predates the formulation of the MP.[32] Given the intuitive content of FI as a prohibition against superfluous symbols in representations, a kind of legibility requirement, it follows immediately that phonetic features cannot occur in representations that interface with the C-I components, and that semantic features cannot occur in representations that interface with the SM components. In effect, Spell-Out must be part of the computational system. Furthermore, it appears that some UG constraints might be redundant given FI. Consider for example the θ-Criterion (Chomsky 1981a), which prohibits an argument that is assigned no θ-role (cf. Functional Relatedness of Freidin 1978)and which also prohibits multiple assignments of a single θ-role of a single predicate (Freidin 1975: n. 20).[33] The former prohibition rules out constructions like (16a), while the latter blocks constructions like (16b).

(p. 24) **(16)**

a. *John seems that Mary is happy.

b. *John [$_{VP}$ gave Mary [$_{NP}$ abook] [$_{PP}$ to Bill]]

If the derivation of (16b) satisfies the constraint on the unique assignment of θ-roles, then it too violates the constraint on argument relatedness. Now, if an argument without a θ-role is uninterpretable and hence superfluous at the C-I interface, then the constructions in (16) fall out under FI and we can eliminate these portions the θ-Criterion on the grounds that they are redundant. The same kind of analysis can be given to the Case Filter (Vergnaud 2008 [1977], Chomsky 1980a) under the hypothesis that Case features, whether unlicensed or unvalued, are uninterpretable at one or both of the two interfaces. This has the effect of eliminating UG constraints, part of S_0, which is genetically determined, in favor of ICs that exist outside FL. As a result, there is a significant reduction in the domain-specificity of FL.[34]

This reduction is consistent with Chomsky's skepticism about a purely genetic account of evolution.[35]

> It does seem very hard to believe that the specific character of organisms can be accounted for purely in terms of random mutation and selectional controls. I would imagine that biology of 100 years from now is going to deal with evolution of organisms the way it now deals with evolution of amino acids, assuming that there is just a fairly small space of physically possible systems that can realize complicated structures. (Chomsky 1982b: 23)

As an alternative to the second factor (i.e. genetic) approach, Chomsky goes on to mention the third factor approach exemplified by 'D'Arcy Thompson's attempt to show that many properties of organisms, like symmetry, for example, do not really have anything to do with a specific selection but just with the ways in which things can exist in the physical world' (Chomsky 1982b: 23).[36] It is worth noting that Chomsky had been entertaining the third factor approach from almost the outset, as illustrated in the following quote from the first chapter of Chomsky (1965), written seven years earlier (see again Chomsky 1982b: 62).

> (p. 25) It is clear why the view that all knowledge derives solely from the senses by elementary operations of association and 'generalization' should have had much appeal in the context of eighteenth-century struggles for scientific naturalism. However, there is surely no reason today for taking seriously a position that attributes a complex human achievement entirely to months (or at most years) of experience, rather than to millions of years of evolution or to principles of neural organization that may be even more deeply grounded in physical law … (1965: 58–9)

Here Chomsky offers an evolutionary explanation equally plausible as the third factor explanation, which 'may be even more deeply grounded in physical law'.[37] In 2004 he noted that this reference to principles of neural organization and physical law was essentially a 'throwaway' 'because it didn't seem you could do anything with it' (Chomsky 2004b: 162). With the MP, this may no longer be the case; rather, the plausibility of the second factor account of language design, one that relies heavily on genetic endowment, may be in question.[38]

A third factor account of language design also opens a new and radical approach to questions of economy and simplicity.

> Within the biolinguistic framework, methodological considerations of simplicity, elegance, etc., can often be reframed as empirical theses concerning organic systems generally. For example, Morris Halle's classical argument against postulating a linguistic level of structuralist phonemics was that it required unmotivated redundancy of rules, taken to be a violation of natural methodological assumptions. Similarly conclusions about ordering and cyclicity of phonological and syntactic rule systems from the 1950s were justified on the methodological grounds that they reduce descriptive complexity and eliminate stipulations. In such cases, the issues can be recast as metaphysical rather than epistemological: Is that (p. 26) how the world works? The issues can then be subjected to comparative analysis and related to principles of biology more generally, and perhaps even more fundamental principles about the natural world; clearly a step forward, if feasible. Such options become open, in principle at least, if the inquiry is taken to be the study of a real object, a biological organ, comparable to the visual or immune systems, the systems of motor organization and planning, and many other subcomponents of the organism that interact to yield the full complexity of thought and action, abstracted for special investigation because of their apparent internal integrity and special properties. From the earliest days there have been efforts to explore closer links between general biology and the biolinguistic perspective. Insofar as methodological arguments in linguistics can be reframed as empirical ones about general operative principles, the analogies may become more substantive. (Chomsky 2007: 1)

Under a third factor approach to language design, methodological considerations of simplicity and economy might be recast as empirical hypotheses about how the world is. It's an exciting prospect, though whether such speculations will be fruitful remains to be determined.

As we have demonstrated in part, the MP, while in important respects a new approach to linguistic theory, is deeply rooted in the work on generative grammar of the past half century. Perhaps unsurprisingly, it has validated some significant early ideas while at the same time leading us to new ones and also to new horizons.

Notes:

(1) We would like to thank Noam Chomsky, Terje Lohndal, and Carlos Otero for helpful comments on this work.

(2) We take λ to refer to that part of the representation of 'meaning' that is generated by the computational system of FL and interfaces with CI. In this regard it is useful to consider the model discussed in Chomsky (1975b), where 'meaning' is constructed in two parts. First there is a mapping from S-Structure to logical form (LF) by rules of semantic interpretation involving 'bound anaphora, scope, thematic relations, etc.' Chomsky goes on to say that 'the logical forms so generated are subject to further interpretation by other semantic rules (SR-2) interacting with other cognitive structures, giving fuller representation of meaning' (1975b: 105). If the representation of 'meaning' involves cognitive modules beyond FL, as seems plausible, then its derivation would involve processes on the other side of the CI interface. This raises a question about what rules of semantic interpretation (broadly construed) might plausibly apply on the other side of the interface (i.e. not on the FL side). λ is determined by only those that apply before the CI interface is reached. We could then equate λ with LF, though the equation is no more than terminological. On this view, LF is no more and no less than the linguistic representation produced by a derivation in narrow syntax (i.e. excluding phonology) that interfaces with CI.

More recently the existence of LF as a linguistic level of representation has been challenged (see esp. Chomsky 2004a). See also the discussion below on single cycle syntax for further comment.

(3) Much of this comes from the Principles and Parameters framework (see esp. Chomsky 1981a, 1981b).

(4) We take the 'LF component' to be that part of the derivation in narrow syntax to CI that occurs after Spell-Out, the covert part of the derivation (i.e. without any PF effects). It is generally assumed that there is no variation in the LF component because the language learner receives no overt evidence.

(5) For example, *want* cannot occur with a finite clause complement, whereas the corresponding verb in French (*vouloir*) can, and furthermore the French verb cannot occur in an ECM construction while its English counterpart can.

(6) An answer largely depends on how linguistic variation is handled. For important discussions see Borer (1984), Baker (2001), Kayne's work on microcomparative syntax (e.g. Kayne 2008), and Manzini and Savoia's extensive work on Romance dialects (e.g. Manzini and Savoia 2008 and the work cited there), to name but a minuscule fraction of the literature on comparative syntax of the past three decades.

(7) Chomsky (2008a) distinguishes the two perspectives as methodological vs. substantive. See also Martin & Uriagereka 2000, who characterize them as methodological minimalism vs. ontological minimalism. In the past few years the formulation of the SMT has sharpened in focus to (i).

((i)) recursion + interfaces = language

Taking Merge as the sole recursive grammatical operation, the entire burden of constraining its operation and output now falls on interface conditions. For further discussion see our comments on third factor explanations below. See also Chomsky (2005, 2007, 2010) for discussion of the asymmetry between the two interfaces.

(8) It is worth noting that an architectural model of the FL containing two interface levels may not be minimal. See Chomsky (2004a) for a proposal in which there is no level of LF in the technical sense because syntactic derivations interface with the conceptual components of the cognitive system involved in full semantic interpretation at multiple points. See also the discussion below on single-cycle syntax.

(9) See also Chomsky (2007: 4).

(10) And, in particular, by prohibiting analyses that merely mirror the complexity of the data.

(11) See e.g. the phrase structure rules in Chomsky (1958/1962). In addition to context-free rules like number 3 in §VII,

((i))

$$\mathrm{VP}_1 \longrightarrow \mathrm{V} \left(\begin{Bmatrix} \mathrm{NP} \\ \mathrm{Pred} \end{Bmatrix} \right)$$

there are context-sensitive rules like number 4.

((ii))

$$V \longrightarrow \left\{ \begin{array}{l} \left\{ \begin{array}{c} V_s \\ \textit{become} \end{array} \right\} \text{ in env.\_\_\_Pred} \\ V_t \text{ in env.\_\_\_NP} \\ V_i \text{ in env.\_\_\_ } \left\{ \begin{array}{c} \# \\ \text{Adv} \end{array} \right\} \end{array} \right\}$$

(12) In the late 1960s, the formulation of the X-bar theory of phrase structure (Chomsky 1970) places further constraints on the computational system. At the time it was formulated, X-bar theory could have led to the abandonment of even context-free phrase structure rules, but didn't. The initial formulation utilized context-free phrase structure rule schema involving category variables, as in (i) (Chomsky 1970: 210).

((i)) $X'' \rightarrow [\text{Spec}, X']X'$

where X ranges over the major syntactic categories N, V, and A. Phrase structure rules remained the only grammatical device for generating phrase structure until the advent of Merge in the mid-1990s.

The MP goes further by eliminating phrase structure rules altogether, as will be discussed below.

(13) See §95.3 for discussion.

(14) The original proposal, which became known as the A-over-A Principle, was formulated as follows: 'if the phrase X of category A is embedded within a larger phrase ZXW which is also of category A, then no rule applying to the category A applies to X (but only to ZXW)' (Chomsky 1964a: 931).

Ross (1967a) replaces this with his set of island constraints, for (3a) in particular the Complex NP Constraint (CNPC). In Chomsky (1973), the CNPC is subsumed under the more general Subjacency Condition (see also Chomsky 1977b). More recent proposals involve a constraint on derivations whereby certain parts of a derivation called phases become inaccessible to other parts during the course of the derivation (see Chomsky 2001, 2008a).

(15) For a more detailed discussion of this history, see Freidin (1994).

(16) This constraint appears in Chomsky (1980a) and remains a background assumption of current minimalist analysis.

(17) In early trace theory, the unpronounced traces were considered to be special 'empty' categories designated as *t* (for trace) or for example $[_{NP}$ e] (for a NP trace). Under minimalism, such special symbols are prohibited by the Inclusiveness Condition because it is assumed that they do not exist in the lexicon—i.e. as features of lexical items. Recall that the Inclusiveness Condition is postulated as a criterion for determining whether language constitutes a perfect system (Chomsky 1995b: 225).

(18) The substitution analysis, which predates the minimalist program by almost three decades (see Chomsky 1965), is used in early minimalism (see Chomsky 1993). The adjunction analysis via Merge, which replaces it, first appears in Chomsky (1995a). Note that we are using the term 'adjunction' as synonymous with 'merger', the result of applying Merge. Note also that in Chomsky (2004a) Merge is distinguished from another operation called 'adjunction' where the former is assumed to be a symmetrical operation, 'yielding syntactic objects that are sets, all binary', whereas the latter is an asymmetric operation that forms an ordered pair from two syntactic objects. According to Chomsky, 'set-merge and pair-merge are descendents of substitution and adjunction in earlier theories' (2004a: 117).

(19) The labeling function of Merge, under which a head assigns its syntactic category label on a phrase created by merging it with another syntactic object (lexical item or phrase), instantiates the property of phrasal projection from X-bar theory (Chomsky 1970).

(20) This is a rational reconstruction of these theoretical considerations. In the actual history of syntactic theory, the elimination of phrase structure rules was motivated instead by the recognition that certain properties stipulated in such rules were derivable from UG constraints that had independent motivation. For example, the fact that VP

containing NP and CP complements would have a linear order V–NP–CP or CP–NP–V followed from the Case Filter, given that Case licensing (or marking or valuation) requires adjacency between a NP and the 'Case-assigning' element. See Stowell (1981) for an extensive detailed analysis.

(21) Consider the definition of transformation in Chomsky (1955/1975a): 'In the abstract development of the level of transformational analysis, we construct a set of 'grammatical transformations' each of which converts a string with phrase structure into a string with derived phrase structure' (p. 72). Merge is just the generalized form of a generalized (as opposed to singulary) transformation, which operates on pairs of such strings. The earliest formulations of generalized transformation were restricted to pairs of strings representing clauses. Merge drops this restriction and instead allows the two objects merged to be any syntactic object—i.e. single lexical items or strings constructed of lexical items. Hence Merge is a form of transformation.

(22) In the regard, it is worth recalling Chomsky's comment in Chapter 7 of *Syntactic Structures*: 'I think it is fair to say that a significant number of the basic criteria for determining constituent structure are actually transformational. The general principle is this: if we have a transformation that simplifies the grammar and leads from sentences to sentences in a large number of cases (i.e., a transformation under which the set of grammatical sentences is very nearly closed), then we attempt to assign constituent structure to sentences in such a way that this transformation always leads to grammatical sentences, thus simplifying the grammar even further' (1957: 83). The evolution of the MP strongly supports this view, but also strengthens it: the basic criteria for determining constituent structure are transformational.

(23) Postal (1972) offers a somewhat similar argument against D-structure.

(24) See Chomsky (1993: 22–3) for discussion. More recently, Chomsky (2008a) has proposed a No Tampering Condition for Merge whereby Merge cannot affect the internal structure of the two syntactic objects it combines. This has the same effect as the Extension Condition with respect to strict cyclicity. We will not comment further on the relationship between the two conditions. One last comment: this No Tampering Condition should not be confused with the proposal in Chomsky (2004a), where lexical items are taken to be 'atoms' in a derivation, so that their internal parts (features) cannot be operated on by syntactic rules. No Tampering in this case means that there can be no feature movement as in previous analyses (e.g. Chomsky 1995b: ch. 4). Note further that each of these conditions would prohibit the kind of generalized transformations proposed in the earliest work on transformational grammar.

(25) Chomsky specifically mentions LF here, but the reasoning directly carries over to PF.

(26) The facts at issue involved transformations affecting scope of negation, as in (i).

((i))
 (a.) Often, I don't attend class.
 (b.) I don't often attend class.
 (c.) I don't attend class often.

Lasnik argues that the process determining scope must at least wait until after the application of syntactic transformations on each cycle—an empirical argument. He then argues, entirely on conceptual grounds, that the process should not be post-cyclic, as spelled out in this quotation.

(27) Harris mentions this notion at the end of Harris (1951), but as discussed in Freidin (1994) Harris's interpretation is quite different from Chomsky's, and furthermore, Harris does not pursue it.

(28) A standard example in the syntactic component was the stipulated ordering of the passive transformation prior to the subject—verb agreement transformation, to guarantee that agreement is with the derived subject rather than the underlying one. This is a case of extrinsic ordering, where the rules could technically apply in either order but where one order would produce a deviant result. In the case of intrinsic ordering, one rule of a pair could not apply at all if the other had not applied first.

(29) Note that the term 'descriptive adequacy' is likewise a technical term. Chomsky (1965) explicates it as follows: 'A grammar can be regarded as a theory of a language; it is *descriptively adequate* to the extent that it correctly describes the intrinsic competence of the idealized native speaker. The structural descriptions assigned to

sentences by the grammar, the distinctions that it makes between well-formed and deviant, and so on, must, for descriptive adequacy, correspond to the linguistic intuition of the native speaker (whether or not he may be immediately aware of this) in a substantial and significant class of crucial cases. A linguistic theory must contain a definition of 'grammar,' that is, a specification of the class of potential grammars. We may, correspondingly, say that a *linguistic theory is descriptively adequate* if it makes a descriptively adequate grammar available for each natural language.'

(30) Chomsky points out that a grammar may be justified on two distinct levels, one involving descriptive adequacy and the other, explanatory adequacy: 'To summarize briefly, there are two respects in which one can speak of "justifying a generative grammar." On one level (that of descriptive adequacy), the grammar is justified to the extent that it correctly describes its object, namely the linguistic intuition—the tacit competence—of the native speaker. In this sense, the grammar is justified on *external* grounds, on grounds of correspondence to linguistic fact. On a much deeper and hence much more rarely attainable level (that of explanatory adequacy), a grammar is justified to the extent that it is a *principled* descriptively adequate system, in that the linguistic theory with which it is associated selects this grammar over others, given primary linguistic data with which all are compatible. In this sense, the grammar is justified on *internal* grounds, on grounds of its relation to a linguistic theory that constitutes an explanatory hypothesis about the form of language as such. The problem of internal justification—of explanatory adequacy—is essentially the problem of constructing a theory of language acquisition, an account of the specific innate abilities that make this achievement possible'. (Chomsky 1965: 26–7)

(31) See Yang (2002) for discussion.

(32) See Freidin (1997) for discussion.

(33) The θ-Criterion also prohibits the non-assignment of a θ-role of a predicate and also the multiple assignment of distinct θ-roles to a single argument or its chain (Functional Uniqueness of Freidin 1978). The former rules out (i) on the analysis (ii.a) and the latter rules out (i) on the analysis (ii.b) where *t* constitutes a trace of *Mary* and thus the chain {*Mary, t*} is assigned two θ-roles by *mentioned.*

((i)) *Mary mentioned.

((ii))

(a.) [$_{TP}$ Mary T [$_{VP}$ mentioned]]

(b.) [$_{TP}$ MaryT [$_{VP}$ mentioned *t*]]

Given the θ-role assignments in (ii.b), (i) should have the interpretation of 'Mary mentioned herself', which of course it doesn't.

(34) See Chomsky (2004b: 162–4) for further discussion.

(35) For Chomsky's most recent views on the evolution of language, see Chomsky (2010).

(36) Chomsky (2004b), an updated version of Chomsky (1982b), includes a reference to A. M. Turing's 1952 paper 'The chemical basis of morphogenesis' in addition to referring to Thompson (1942 [1917]). Turing's abstract states the third factor approach as follows: 'The purpose of this paper is to discuss a possible mechanism by which the genes of a zygote may determine the anatomical structure of the resulting organism. The theory does not make any new hypotheses; it merely suggests that certain well-known physical laws are sufficient to account for many of the facts.'

(37) For additional discussion, see Freidin & Vergnaud (2001). Chomsky (2004b) notes that in general biology the research program illustrated in Thompson (1942 [1917]) and Turing (1952) is 'basically regarded as a failed research program, or maybe premature' (p. 162). In this regard, consider for example Sydney Brenner's discussion of the icosahedral shape of the head of a virus: 'If one takes the extreme view that one would like to compute an entire organism from its genome, one has to first understand what one might call the 'principle of construction'. For example, one can look at the head of a virus and one can see that it's a perfect icosahedron. We know that's genetically determined because it's inherited. What we want to know is how is the equation for an icosahedron written in the DNA? We know how such an equation can be written on the back of a package of Corn Flakes. It says, 'Cut here, bend there, glue here and you can fold it all up into this icosahedron.' Similarly, viral icosahedrons are made out of molecules of proteins packing together in a special way. So if we were to unravel all of the

structure we would find that the ‘equation’ for a viral icosahedron is written in little bits and pieces in the genome—in a little sequence of amino acids here, and another bit there. But we could not disentangle this *a priori* unless we understood the ‘principle of construction’. (Brenner 2001: 118–19)

The alternative is easy enough to imagine. Suppose that the genome determines the size of the head of the virus and that it is covered with a protein sheath. It could well be that at that magnitude, such a structure will assume the shape of an icosahedron because of the structure of the protein molecules that compose it, and that an icosahedron requires minimal energy in much the same way as soap bubbles form—the kind of explanation that Thompson might have offered, grounded ultimately in physical law.

(38) See Chomsky's discussion of approaching UG from below (Chomsky 2007).

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Features in Minimalist Syntax

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Abstract and Keywords

This article outlines a number of major issues concerning features in minimalist syntax. The purpose is to delineate the core conceptual issues that the notion of feature raises within minimalist approaches to (transformational) grammar. The article begins by pointing out the different perspectives taken on the notion of feature by minimalist and unification-based approaches. It then clarifies the notions of category and feature, taking category to have essentially a positional definition, while feature is defined as a property of a category that sub-classifies it. The article distinguishes two further properties of features: the potential of organizing features into feature classes (where the property of being in a class holds of all tokens of a feature), and the possibility of features having what we call second-order properties, which may vary from one token of the feature to another. With this in hand, two kinds of features in minimalist syntax are distinguished: those that play a role primarily at the interfaces with sounds and meaning, and those whose function is primarily syntax internal. The article then explores the way in which features create configurations that can be compositionally interpreted, highlighting their role in dependency formation, in constituent construction, and in displacement effects. Finally, it considers the role that features play at the interfaces of sound and meaning.

Keywords: minimalist approach, transformational grammar, category, features, sound, meaning

2.1 Introduction

This chapter outlines a number of major issues concerning features in minimalist syntax. Our purpose is neither to survey the field nor to provide a particular theory of features in minimalism. It is rather to delineate what we take to be the core conceptual issues that the notion of feature raises within minimalist approaches to (transformational) grammar.

We begin by pointing out the different perspectives taken on the notion of feature by minimalist and unification-based approaches. We then clarify the notions of category and feature, taking category to have essentially a positional definition, while feature is defined as a property of a category that sub classifies it. We then distinguish two further properties of features: the potential of organizing features into feature classes (where the property of being in a class holds of all tokens (p. 28) of a feature), and the possibility of features having what we call second-order properties, which may vary from one token of the feature to another. With this in hand, we distinguish two kinds of features in minimalist syntax: those which play a role primarily at the interfaces with sounds and meaning, and those whose function is primarily syntax-internal. We then explore the way that features create configurations which can be compositionally interpreted, highlighting their role in dependency formation, in constituent construction, and in displacement effects. Finally, we consider the role that features play at the interfaces with sound and meaning.

2.1.1 Some ontological clarifications

An important issue that needs to be clarified when discussing features is their theoretical status, since they are used rather differently in different frameworks in formal linguistics. In certain unification-based frameworks, such as HPSG, features are used as part of a description language for grammatical theory:

> Intuitively, a feature structure is just an information-bearing object that describes or represents another thing by specifying *values* for various *attributes* of the described thing; we think of the feature structure as providing partial information about the thing described. (Pollard and Sag 1987: 28)

From this perspective, a rich feature theory is a reasonable thing to posit, since the feature theory does not constrain the objects of the linguistic theory, but merely describes them (see King 1994 for discussion). Something else is required to constrain the linguistic objects themselves (for example the type hierarchy in HPSG).

The alternative view assumed in minimalist work is that the features are properties of syntactic atoms and hence are directly objects of the theory: a feature [plural] for example is used analogously to chemists' use of H for the real-world thing hydrogen. From this perspective, it is crucial to say what the possible feature structures are such that the properties of the features allow them to enter into relationships with other features, analogously to saying what the properties of atoms are such that they can enter into relationships with other atoms.

From this viewpoint, the constraints on the feature theory are substantive and amount to constraining the theory itself, something which is not true when features are seen as a description language. This means that entertaining alternative hypotheses about feature structures is tantamount to entertaining alternative theories. The minimalist framework can be seen as a set of guidelines which constrain the general hypothesis space within which these various theories can be entertained. Of course, it is important to be explicit about what the theories themselves are so as to be able to evaluate their empirical adequacy and their theoretical parsimony.

(p. 29) 2.1.2 Some terminological clarifications

Generative grammar traditionally makes a distinction between the notions of category and feature, a distinction which stems from the immediate constituent analysis advocated by the American Structuralists (Bloomfield 1933, Wells 1947), formally implemented as a phrase structure grammar (PSG) (Chomsky 1957) (see Chomsky 1965: 75ff. on the inadequacy of immediate constituent systems in dealing with certain cross-classification issues in syntax). The categories used in PSGs represent classes of elements that (i) occur in complementary distribution and (ii) are restricted to certain positions within constructions (e.g. Bloomfield 1926): hence we have the categories of Noun (e.g. *cat, sincerity* …), Complementizer (*that, if, Ø* …), Adjective (*cold, final, Scottish*), etc. These categories often need to be subclassified (animate count Noun (*cat*), question complementizer (*if*), gradable Adjective (*cold*), etc.). In the structuralist and generativist tradition, this subclassification is not correlated with positions in constructions. From this perspective, categories are essentially defined syntagmatically, while subcategories are paradigmatic. The distinction has its roots in the Aristotelian conceptions of substance and form, which Greek and medieval grammarians used to understand parts of speech (categories) and inflections (featural specifications) (see e.g. Lyons 1968).

Following an observation by G. H. Matthews, Chomsky (1965: 79–80) notes that extending PSGs by adding further production rules to capture subcategories (as was done in Chomsky 1957) loses the possibility of capturing generalizations that cross-cut the subcategories. Take, for example, a set of of PS rules like those in (1) (Chomsky 1965: 80):

(1) N → Proper
N → Common
Proper → Pr-Human
Proper → Pr-non-Human
Common → C-Human
Common → C-non-Human

In such a system, the symbols ‘Pr-Human’ and ‘C-Human’ are unrelated as they are atomic category labels. However, this obviously poses problems when we want to capture generalizations about, for example, ‘human’ nouns. To solve this problem, Chomsky proposes an extension of PSGs which allows the categories to bear distinctive features (an idea imported from work in phonology: Jakobson et al. 1951). The distinctive features used in *Aspects* are mainly binary, but other possibilities are also considered (see, for example the discussion on pp. 170ff., and associated notes). We take up the question of what kinds of values features can have in more detail below. An important consequence of the introduction of features is that the extended phrase structure system no longer models the kind of taxonomic theory of linguistic structure defended by the American Structuralists, since the features add an extra cross-classificatory dimension.

(p. 30) In *Aspects*, Chomsky suggests that there may be no category–feature distinction at all, the parts of speech N, V, A, P being simply another set of features alongside Animate, Past, etc. (Chomsky 1965: 207ff). He gives lexical entries such as those in (2) in which category labels have no special status.

(2)

a. *sincerity*, [+N, −Count, +Abstract]
b. *boy*, [+N, −Count, +Common, +Animate, +Human]

But in practice, the Aristotelian distinction was not abandoned; thoughout the Extended Standard Theory and Government–Binding periods, various kinds of rules were commonly formulated to single out the features N, V, A, and P, for example X-bar theory and government. When COMP and INFL and D were eventually integrated as functional categories, it was by treating them like N, V, A, and P, subject to the same rules of projection and so on, and distinct from features like Past and Wh. In fact, the original arguments for treating COMP and INFL as projections went back to positional arguments in work by den Besten (1981), Platzack (1983), Emonds (1978), and others, in an essentially structuralist sense: the verb appeared in more than one place, therefore an additional category was needed. Similar arguments were raised for other functional categories such as Agr (Pollock 1989) and D (e.g. Taraldsen 1991).

The category/feature distinction is also commonly assumed within minimalism, although it is little discussed. Despite the elimination of a base component, the syntagmatic ordering of expressions in constituent structure must still be captured somehow, and one might take only a subset of features to be relevant to that ordering, in which case the notion of category is still present (e.g. if both C and T can 'have' φ features, but it is their C-ness or T-ness and not their φ features which determine their first merge position; cf. Chomsky and Lasnik 1993: 526). In minimalist grammars, this syntagmatic ordering is typically dealt with via a hierarchy (or sequence) of functional categories (an extended projection, in Grimshaw's 1991 sense; a functional sequence, in Starke's 2004 terms—we will adopt Starke's term in what follows). To the extent that this hierarchy implies a special class of features which make syntactic atoms visible to the constituent-forming operations of language (e.g. a special class of category features, but also the Edge features of Chomsky 2008a), the notion of category in the sense discussed here is still present.

There is one branch of minimalist work, namely cartography (e.g. Cinque 1994, 1999, Rizzi 1997) which actively investigates the hypothesis that the number of features which are 'categorial' in that they project phrase structure is quite large. This work reopens the possibility suggested by Chomsky in *Aspects* of eliminating the category/feature distinction, a possibility which arises perennially (for example, Kayne 2005a, b discusses the possibility that each syntactic node bears at most one syntactically active feature). However, if every feature projects, then the cross-classification problems originally noted by Matthews and Chomsky must be addressed, something which is yet to be done.

(p. 31) 2.2 Structures of feature systems

The question of how features are structured itself splits into two: (i) what is the internal structure of a feature, and (ii) how do features come together to make larger structures? Our purpose in this section is to lay out the range of possibilities for what a theory of features might be under the basic assumption that the features are to be interpreted as primitives of the system. We begin with what we see as the simplest system that can be entertained and then investigate systems of increasing complexity, evaluating them on the basis of various minimalist assumptions. Thus in what follows we do not assume from the outset the category/feature distinction, but revisit it eventually.

2.2.1 Privativity

We begin with what we call PRIVATIVE features. We will assume that syntax involves a set s={a, b, c, …} of syntactic atoms, as well as a set S of operations and constraints affecting those atoms and structures built from them, S={A, B, C, …}. Elements of s are distinguished from each other only insofar as they are affected differently by one or more elements of S.

A feature is by definition a property that distinguishes some elements from others. A privative feature is a feature with no other properties beyond its distinctiveness from other features; and a privative feature system is one in which all features are privative. Since the feature has no properties, two linguistic structures will be different from each other with respect to a privative feature just in case the feature is present in one and absent in the other. In this system features are simply defined as a list:

(3) Features:
a. Syntax builds structure through recursive application of Merge.
b. The smallest element on which Merge operates is a syntactic atom.
c. A syntactically relevant property of a syntactic atom which is not shared by all syntactic atoms and which is not derivable from some other property is a feature.

(4) Privative features:
The inventory of features in a language is a set F = {*α, β, γ*, …}.

For example, one could represent the difference between *cat* and *cats* or *this* and *these* as:
(5)
a. *cat* [N]; *cats* [N, plural]
b. *this* [Dem]; *these* [Dem, plural]

(p. 32) Here we have two privative features, [N] and [plural], and we can construct two different linguistic objects by maintaining the presence of [N]/[Dem] and allowing the presence of [plural] to vary (we remain neutral here on whether the difference is at the level of syntactic atom or syntactic structure).

The notion of privativity is directly connected to the richness of the set S of syntactic operations. If we allow operations to be defined so that they can invoke specific features, then such a system is tantamount to allowing the privative features to have a complex property: their job is to index the relevant operation, so that their presence triggers the operation and their absence does not (leaving open the question of whether operations need to be triggered to apply).

For example, taking the lexical items from (5), we have:

(6)
a. *this cats
b. these cats

If we index the feature [plural] to a rule that copies [plural] from N to D, we could capture this pattern.

However, even such a powerful system will require some organization of the features, since syntactic operations are typically more general than would be expected if they were triggered by individual features. So the aforementioned copying rule will apply not just to [plural] but also to [feminine] in a language where D agrees in gender and number, while it may not apply to, say, [count]. Given such cases are ubiquitous, it is necessary to organize features in some way.

One way to organize features is in terms of a functional sequence; the functional sequence C–T–*v*–V is sometimes interpreted as a constraint on external merge, and can be understood to encode co-occurrence restrictions from top to bottom (e.g. if C is present then so is T, though not vice versa).

Another kind of organization of privative features is a 'geometry': for example, person, number, and gender features can be grouped under a single node for *φ* which is relevant to agreement (e.g. Harley and Ritter 2002, Béjar 2004, drawing on feature theory in phonology; see Clements 1985). Such a geometry is normally understood to encode distributional implications from bottom to top, e.g. [speaker] and [addressee] both imply [person], so if [person] dominates them in the geometry, then a rule can be formulated that refers to both by invoking the [person] feature; and if 3rd person is the absence of speaker and addressee features, then the [person] feature covers all persons.[1] If φ dominates a node which includes number and gender features but which excludes person features (Harley and Ritter's 2002 (p. 33) INDIVIDUATION), then participial and adjectival agreement can be understood as referring to this node (cf. Chomsky's 2000a use of the notion 'φ-completeness').

One theoretical issue here is to what extent the geometry must be stipulated: does such a system require us to posit a syntax-external module to organize the features, or is it possible to derive the properties of the geometry from the syntax of the structure or the semantics of the features (Harbour 2007)?

Returning to the relation between features and operations, within minimalist approaches, the rule systems are assumed to be extremely general. In fact, the sole rules are (i) the operation that forms constituents (Merge, whether internal or external) and (ii) the operation that creates syntactic dependencies (Agree, including a matching requirement on features). Given this, together with the Inclusiveness principle (Chomsky 2000a: 113), which forbids the introduction of features not present in the 'numeration' of syntactic atoms drawn from the lexicon, features cannot be privative in the strict sense defined above. A feature entering into a syntactic dependency must have at least the property of saying what kind of element it enters into a dependency with, since this information cannot be stated in the general rule.

For example, we might treat our pattern above by endowing *cats* with [plural] and *these* with [agrees-with: plural], where the notation [agrees-with] simply specifies that the dependency-forming operation Agree applies to it. The information about what elements Agree applies to is stated here not as part of Agree, but effectively as a property of the feature [plural]. If the feature [plural] were privative, some of that information would have to be specified in the operation, leading to an expanded set of syntactic operations.

In addition, as noted above, syntactic operations apply to classes of features, rather than to individual features. For example, the dependency-forming operation of Agree may apply, in a particular language, to the group of features [person], [number], [gender] and not to, say, [past], [present], and [future]. But our definition of privativity disallows reference to feature classes, since such classes are effectively further properties of features (beyond presence/absence).

The other general syntactic operations behave similarly: they cannot make reference to classes of features, in a privative system, and so cannot pick out those features that should be involved in selection or movement from those that should not. For these reasons, a purely privative system is inadequate for human language, which displays syntactic dependencies. Either we need to abandon Inclusiveness, or we need to increase the descriptive capacity of the feature theory to something more powerful than a strictly privative system.

One response to this argument might be to take agreement effects to arise because of a movement operation; the formulation of 'checking domains' in Chomsky (1993) essentially ensured that all feature-checking occurred under movement or external merge, and Chomsky's (1995b: 262) notion of feature movement provides a way of partially unifying agreement and movement. For example, if the (p. 34) feature [plural] has the (unique) property that it always moves to D, and spells out in both places (at the bottom of its chain as *-s* and at the top as some part of *these* and *those*), that would not make a system non-privative. However, this approach would still entail a rather specific rule which ensures that the feature moves to D, rather than elsewhere. Furthermore, the privative account fails to extend to other cases. The unification of movement and agreement is only partial: an agreement chain spells out in more than one place, unlike the chain formed by phrasal movement, and an agreement target is never syntactically complex, unlike the landing site for movement, which can be. Thus, even if agreement is modelled as feature movement, we still need to distinguish two circumstances—something which is (apparently) not possible in a strictly privative feature system.

An alternative response might be to deny that agreement dependencies are fundamentally syntactic, and to say that interface constraints play a role in enforcing agreement, as in Dowty and Jacobson's (1989) analysis of agreement as a semantic phenomenon. However, there are cases of irreducibly syntactic agreement, for example the appearance of case and gender agreement morphology on various dependents of a noun phrase (see Svenonius 2007 for further discussion of this issue). These are furthermore not reducible to a morphological component, as they are constrained by syntactic conditions of locality. Hence, at our current level of understanding, mainstream minimalist work is correct in assuming that there is some kind of syntactic agreement, and the syntactic feature system of natural language cannot be entirely privative.

2.2.2 Feature classes

Every feature system so far proposed for natural languages assumes at least tacitly that features are organized in some way, for example if the categories N, V, A, P, C, T, and D are visible to Merge but other features are not (category features, or Edge features), or if N, V, and A assign thematic roles but other features do not (lexical features), or if C, T, and V are organized in one functional hierarchy and P, D, and N in another (different extended projections), or if only the features NOM, ACC, DAT and GEN satisfy the Case Filter (case features), or if the features PLURAL, PARTICIPANT, and SPEAKER are copied in an agreement process (φ features) but certain other features are not.

Any property or rule which applies to the members of a subset of features defines a CLASS of features; such a system is descriptively distinct from, and more powerful than, a system which arranges privative features in an implicational geometry.

For the sake of discussion, we can state this quasi-formally as follows, where F is understood as defined in (3) and (4) above.

(p. 35) **(7)** Feature class:
A feature class is a subset O of F, where the members of O share some syntactically relevant property

For example, N, V, A, P, C, T, and D can be members of a feature class CATEGORY; NOM, ACC, DAT and GEN can be members of a feature class CASE; and SPEAKER can be a member of the feature class PERSON. Feature classes here have an entirely extensional definition.

To see that such a system is descriptively more powerful than a standard rooted directed acyclic graph geometry of features, consider that any such graph can be represented as a set of sets of features, where each node in the graph corresponds to the set of all nodes that it (reflexively) dominates; the reverse, however, is not true, e.g. a feature system which consists of the classes {A, B} and {B, C}. To take an example from the literature, in Chomsky (2000a), C and *v* are phase heads, while *v* and V are θ-assigners.

Below we discuss a different conception of attribute-value matrices, one which is even more descriptively powerful in that it allows the notion of 'valuation' to be stated.

A system with classes of features is not privative in our terms, since features have the property of belonging to a class which is itself 'active' in the grammar (i.e. there are rules or principles which refer to the class). When, in the literature, a lexical entry is represented something like: *me* [D, ACC, SPEAKER], the notation is silent as to whether the feature system is privative or not; in practice, feature classes are almost always assumed.

2.2.3 Second-order features

In our description of feature classes, in the previous section, we discussed cases in which a given feature either belongs or does not belong to a given class, e.g. DATIVE is a CASE feature. Any such class can be stated as a property of features, and in practice, in the literature, the term 'feature' is variably applied to what we are calling features and what we are calling feature classes (for example, DATIVE is a feature if datives behave differently from other DPs, but one might also state the Case Filter as requiring that a DP must bear a CASE feature).

We can think, then, of feature classes as being predicates that hold of features, where the predicate is true for all tokens of the feature in the language (or, indeed, in UG). Thus, acc is always a CASE feature and never a TENSE feature. This gives us a very simple feature-typing system.

However, in practice we often need to associate a particular property with a particular feature only some of the time—for example, we talk of C with and without EPP. To recognize this asymmetry, we introduce the term SECOND-ORDER FEATURE for a property a feature in F can have in some instances but not in (p. 36) others. Unlike feature class, second-order features are properties that hold of tokens of features rather than features *qua* types. Thus, EPP is a second-order feature, rather than a feature class, if some first-order features can have or not have EPP.[2]

(8) Second-order feature:
a. A feature in F is a first-order feature.
b. A property which syntactically distinguishes some instances of a first-order feature α from other

instances of α is a second-order feature.

This means that rather than a non-syntactic bundle of [C, EPP] or a class EPP defined to include C, we can have an association of a first-order feature, C, with a second-order feature, EPP.

The most common example of a second-order feature in linguistic literature outside minimalism is the minus sign, normally interpreted as negation.[3] In such a binary system, the union of features [A] and [B] is [A, B], and the union of [A] and [−B] is [A, −B], but the union of features [A] and [−A] is impossible or empty (but see Harbour 2007, who takes this combination to be equivalent to uninterpretability). In the absence of negation, such incompatibilities must be ruled out by other factors. In practice, the minus sign is not widely used in minimalist syntax (as opposed to morphology); we discuss the related use of uninterpretability below.

Second-order features are used in minimalism to capture dependency relations. This idea has multiple incarnations in various instantations of minimalism: strong versus weak features (Chomsky 1993); interpretable versus uninterpretable features (Chomsky 1995b); features with the ‘EPP’ property and features which lack this property (Chomsky 2000); valued versus unvalued features (Chomsky 2001); etc.[4] Each of these properties itself can be construed as a feature associated with another feature.

Although the content of the second-order features has had various incarnations in different implementations of minimalism, the core notion of second-order feature has remained constant (although largely unrecognized).

There are two possibilities for interpretation of second-order features: they have interpretations at the interfaces or they have interpretations via operations within the syntactic system itself. Both possibilities have been considered. For example, Chomsky (1993) proposes that strength should be thought of as uninterpretability at the SM interface, while Chomsky (1995) takes a strong feature to be one that must be checked as soon as possible after it has been Merged. The first hypothesis (p. 37) connects the property to the external systems, while the second maintains that it is internal to the syntax proper, a distinction we discuss further in §2.3.

A further bifurcation in the notion of strength has to do with whether the strong feature can be satisfied by movement into the specifier of the category bearing the feature or to the category itself. For example, Chomsky (1993) proposes that the T head may bear strong features that cause its specifier to be filled by a DP, as well as strong features that cause the T^0 category to be filled with a verb.

The EPP PROPERTY of a feature replaces the notion of strength in Chomsky (2001). It is entirely formal, simply requiring that some syntactic unit be Merged as the specifier of the category whose feature bears this property. It is more general than strength, as it does not specify any properties of the element to be Merged, and hence it is also less restrictive.

Strength and EPP are essentially ways of ordering and localizing the elements that bear matching features; but more fundamental than this is the notion of feature matching itself. That is, what property of a feature entails that it must match another feature? The property of (UN)INTERPRETABILITY (Chomsky 1995b: 277ff.) is used as the driving force behind the establishment of syntactic dependency in many minimalist systems. The idea is that uninterpretability forces feature matching, and any uninterpretable feature which has been matched is deleted. Feature matching is constrained by the structure of the derivation: uninterpretable features can be thought of as triggering a search for their sister (and whatever it dominates), the search terminating when a matching feature is found, or when some other barrier to the search is encountered (e.g. a phase boundary).

This interpretable/uninterpretable asymmetry in feature–feature relations is rather natural in a derivational system, since the uninterpretable features are those that drive the derivation, while the interpretable ones are those that are used, in the final representation, to connect with the semantic systems (or the phonological ones). Brody (1997) points out that an alternative ‘bare’ checking theory is more natural in a representational system: in such a system, features are interpreted where they can be, and the interpretation of matching features, if those features are in an appropriate syntactic relation (a chain respecting c-command and locality), collapses to the relevant chain-position. Frampton and Gutmann (2000, 2001) develop a (derivational) model of agreement as feature sharing which has this property as well, as do Adger and Ramchand (2005), in their interface principle Interpret Once under Agree. On these models, there are uninterpretable instances of features, but arguably no syntactic features which never have an interpretation (though Chomsky has suggested that Case is such a feature; see e.g.

Chomsky 1995b: 278–9; see also Svenonius 2007 for discussion).

If matching and movement always co-occurred, then we could reduce everything to a single property which would simply create a local configuration between two matching features. However, this seems to be empirically incorrect, as we see feature matching (apparently) without overt movement:

(p. 38) **(9)**

a. *There seems to be many men in the garden.
b. *There seem to be a man in the garden.

Of course, if elements of a movement chain can be phonologically realized in either the lowest or highest chain position (Groat and O'Neil 1996), then one might take movement to always occur when feature matching takes place, but in some cases the moved element is spelled out in situ. For evidence that in fact we need to distinguish the feature matching case from the covert movement case, see Pesetsky (2000). An alternative takes there to be only overt movement, whose impact on linear order is disguised by other movements (Kayne 1998).

2.2.4 Valuation

The second-order feature of VALUATION is more complicated than that of interpretability, at least as it is usually used, in that a feature is not merely valued or unvalued, but rather it is valued *as something*, so valuation is strictly speaking a function rather than a property.

Recall that a simple system of feature classes can be represented in terms of sets: the name of the set is the feature class, and rules can generalize over all of its members. That representation does not straightforwardly lead to a notion of 'unvalued' feature.

Chomsky (2001) replaces the notion that uninterpretability drives feature-checking with the idea that unvalued features do so. We can state valuation in the following way: one class of features (the attributes) have the fixed second-order feature that they can take another class of features as their values. Formally stated:

(10) Valued feature:

a. A valued feature is an ordered pair ⟨Att, Val⟩ where
b. Att is drawn from the set of attributes, {A, B, C, D, E, …}
c. and Val is drawn from the set of values, {a, b, c, …}.

In our terms, an attribute is a feature class in the sense of (7), and a value is a feature in the sense of (3). If an attribute can lack a value, non-valuation can be characterized as a kind of second-order feature (actually a property of a class of features, rather than a property of a feature).

In this kind of system, the matching property that seems so important for capturing syntactic dependencies is built directly into the nature of the second-order feature: identity of attribute is the precondition for matching, and the valued and unvalued features unify (specifically, in Adger forthcoming, an unvalued feature is one which has the empty set as its value; in feature checking, the empty set is replaced by non-empty values).

Once we allow such second-order features, we can ask a number of questions about them, with the answers to these questions defining different theories of (p. 39) features. Are there multiple second-order features in the system (e.g. both strength and interpretability) or is there only one? Relatedly, can individual features have multiple second-order features or just one? Do only a subset of the features have second-order features, or is this option open to all features (i.e. are features organized into classes with respect to the second-order features they bear, over and above the kinds of feature classes discussed above)? If there are feature classes, are they TYPED, so that some features have some subset of the second-order features and some others have some different subset? Even within this fairly minimal setup, there are many options.

For example, Adger (2003) has a rather rich system, where a feature can be interpretable or uninterpretable, valued or unvalued, and weak or strong. The first property is used to establish syntactic dependencies (essentially agreement without movement), the second to capture the particular morphological category associated with agreement, and the third to ensure locality between the two features (i.e. to trigger movement). The distinction between the first two properties is maintained because case features are taken to be uninterpretable even when they have a value. Pesetsky and Torrego (2007) also argue that there is a difference between a feature's interpretability and its status as being valued or unvalued.

There are further options within a system that structures features into attributes and values. Can both attributes and values themselves have second-order features? Can values be drawn from the set of attributes, allowing recursion into the feature structure, as proposed by Kay (1979) and adopted into HPSG (Pollard and Sag 1994)? Can values be structured syntactic objects, as in GPSG-style SLASH features (Gazdar et al. 1985)? The general minimalist answer, insofar as the question is addressed, would be that this kind of complexity should be handled in the syntax rather than in the structure of the lexical items. That is, one would like to adopt the No Complex Values hypothesis of Adger (forthcoming).

A further question is whether there are other second-order features than the ones just discussed. For example, do certain features have the property that they can 'percolate' or be transmitted in ways other than standard projection (Chomsky 2007)? Clearly, the more restricted the options are, the simpler the resulting theory.

2.3 The interaction of features with syntax

It is not clear how many systems syntax interfaces with, but there are at least two, one concerned with perception and expression (SM, or Sensorimotor) and the other with meaning; this latter might be a semantic module interfacing with other (p. 40) systems of thought, or it might be the systems of thought directly (the C-I systems, Conceptual-intentional). Following Svenonius (2007), we call features which play a role in both syntactic processes and phonological or semantic interpretation INTERFACE FEATURES; features which play a role only in syntax, we call SYNTAX-INTERNAL FEATURES. In these terms, we can ask whether the features visible to syntax are all interface features, or whether there are any syntax-internal features: that is, features which only play a role in conditioning the application of purely syntactic operations.

Within minimalist approaches to syntax, the syntactic operations are few and very general. We have already discussed the operation of feature matching (usually called Agree). In addition to Agree, there are two other core operations: Merge and Move. The function of Merge is to create larger syntactic units out of smaller ones, Merging two independent elements A and B to form C, which has A and B as immediate constituents; Move does the same thing, except that it draws B from within A. On this definition of Move, it is simply a variant of Merge (so we have External Merge and Internal Merge, in Chomsky's 2004a terms; see also Starke 2001, Gartner 2002). We consider these in turn.

2.3.1 Merge, External and Internal

Observationally, at least three instances of Merge can be distinguished: extended projections, complements, and specifiers.[5] First, there is the merge of the extended projection, which follows a functional sequence of categorial features (C over T over *v* over V, for example). This is normally construed in terms of heads merging with complements (Brody 2000a being an exception, since non-morphological dependents in an extended projection are represented as specifiers). If the functional sequence is stated over categorial features, then this does not require additional selectional features to be posited.

Second, there is the Merge of a selecting lexical category with an internal argument, for example, an adjective, noun, or verb with a subcategorized complement. Since the categories involved vary considerably from lexical item to lexical item (e.g. different verbs may take finite or non-finite complements, DP or PP complements, or no complement at all), this Merge falls under the descriptive heading of subcategorization or c-selection.

There are two ways to think about how features are relevant to c-selection. A widespread view is that there is little or no c-selection, and that complementation is determined by non-syntactic factors (see Borer 2005 for such a view). An alternative is to use the technology of features to implement c-selection, for example taking a verb which selects an object to bear a feature which implements this (p. 41) requirement (e.g. Chomsky 1965, Emonds 2000). Crucially, such features will have to be subject to a locality constraint on how they are checked, since c-selection is always local (Baltin 1989). Svenonius (1994) captures this by tying c-selection to head-movement, while Adger (2003) suggests that sub categorization features are always strong, and therefore always require local checking. Hallman (2004) invokes checking domains for selection defined in terms of mutual c-command.

A third instance of Merge is the merge of an argument into a specifier position. The conditions under which this takes place appear to be somewhat different from those governing complements, and hence the features involved may be different. For one thing, the specifier is apparently not merged until after the complement has been merged, judging from the fact that the specifier appears to asymmetrically c-command the complement. Additionally, there do not seem to be c-selectional relations between a verb and its subject (Chomsky 1965); that is, a verb does not subcategorize for the syntactic category of its subject. It is by now commonly assumed that subjects are not introduced by the verb, but rather by some functional category (e.g. Chomsky 1995b, Kratzer 1996); if a distinction between functional and lexical categories is maintained, then this asymmetry can be made to follow from the restriction of subcategorization features to lexical categories.

Internal Merge (Move) only takes place to specifier/head positions, and never to complement position, and so complement-style c-selectional features are irrelevant to this case. It is possible that Internal Merge can be unified with the third instance of External Merge mentioned above, i.e. Merge of an argument into a specifier.

This is where the second-order features of strength/EPP come into play. As discussed above, these impose a requirement that some category containing the lower of the matched features is moved to some position local to the higher of the matched features. However, strength and EPP here behave differently, since an EPP feature is satisfied if any category is merged in the specifier, while strength requires movement of a projection of the matching category. The looser EPP formulation appears to be required for Icelandic *Stylistic Fronting* constructions (Holmberg 2000b), where any category can satisfy the requirement (the closest appropriate category moves). However, most cases of movement seem to target more specific features.

The question of whether Internal Merge can be unified fully with the introduction of arguments into specifiers is then partly a question of whether there are cases in which a probe (that is, a strong, uninterpretable, or unvalued feature driving movement) is specified as requiring Internal versus External Merge. In the canonical cases of the EPP, it can be satisfied by either. If natural language agreement is going to be unified with movement, then at least that will have to be done by somehow requiring Internal Merge. McCloskey (2002) argues in some detail that Irish complementizers distinguish whether a *pro* specifier is internally or externally merged.

An alternative motivation for movement has been proposed by Starke (2004). Starke points out that something must motivate the order of the merge of (p. 42) functional projections, and proposes that that is also what motivates movement to specifiers. The mechanism he suggests is the functional sequence, in effect using the functional sequence to replace the traditional spec-head relation. In Starke's system (see also Brody 2000a), the moved element satisfies whatever constraint would license Merge of an X^0 category in that position. This has the interesting consequence of collapsing the notions of specifier and head, at least for functional heads and their specifiers.

2.3.2 Agree

In minimalist papers from the 1990s, feature checking was generally held to be possible only in a very local configuration, called a checking domain (cf. Chomsky 1993); a distant goal would have to move into a local relationship with a probe in order to check its features. However, since features are assumed to drive movement, it seems that a relation must be established *prior* to the movement taking place; hence in Chomsky (2000a, 2001), and subsequent papers, the checking mechanism, called AGREE, is assumed to be able to create a relationship between features at a distance. Phases are part of the theory of locality that constrains the Agree relation (though see Hornstein 2009 for arguments that Agree should be constrained to checking domains and distinguished from movement).

As already stated, Agree as it is usually conceived is a syntax-internal operation, without a specific interpretation at an interface, and so the second-order feature of valuation could well be classified as a syntax-internal feature. On the other hand, if Agree is modelled as the matching of pairs of interpretable and uninterpretable features, and if interpretability is understood as interpretability at the interface, then the second-order feature of (un)interpretability is an interface feature.

2.3.3 Licensing

Various natural language phenomena are discussed under the rubric of 'licensing', for example anaphors, polarity items, deletion sites, traces, and various other elements are sometimes said to need licensing or to be licensed by certain configurations or elements. The general assumption underlying most work in the minimalist program is that these phenomena are either to be subsumed under the kinds of featural relations we have discussed here—as when Kayne (2002) casts binding theory in terms of movement, or when Zeijlstra (2004) analyzes negative concord in terms of the checking of uninterpretable negative features—or else are to be handled by different modules; the minimalist reluctance to posit additional syntactic modules (such as a binding module or a theta module) means that phenomena which are not reducible to known syntactic mechanisms are handled, outside of the syntax proper, (p. 43) by the semantic component, which is usually thought of as strictly interpretive (cf. Chomsky and Lasnik 1993 on binding as an interpretive procedure; phenomena like polarity licensing are regularly analyzed in semantic terms, cf. e.g. Giannakidou 1997 and references there).

2.4 Features and the interfaces

In this section, we discuss the connection of the formal features which are visible to syntax to the systems that syntax interfaces with.

2.4.1 Sensorimotor

A given feature might consistently spell out in one way or another, for example the plural feature in nouns in English consistently spells out as some allomorph of *-s*, except in listed idiomatic morphological forms (such as *sheep*). This simply reflects the usual Saussurean arbitrary pairing of phonological content with syntactic and/or semantic content, and does not show anything different from the fact that *dog* has a pronunciation as well as a meaning.

In Chomsky (1995b), following *Aspects*, this arbitrary pairing is listed in a lexicon that associates phonological and syntactic features. In a 'lexicalist' model, the lexicon is the input to the computation. Alternatively, there is a distinction between the input symbols and the vocabulary items which replace them, a notion known as "Late Insertion", with complex morphological structures being built up syntactically and then associated with phonological forms via a computation which is part of the Spell-Out operation (McCawley 1968, Halle and Marantz 1993). This approach can be extended to non-concatenative meaning—sound linkages in a straightforward fashion. For example, if in a language like Igbo a possessor is marked with a high tone, we can assume that this represents the insertion of a morpheme whose phonological content is autosegmental. In none of these cases do we want to say that a phonology—syntax interface feature is at play.

Sometimes a non-segmental phonological property might be directly associated with a syntactic feature; for example, the **L* H H%** contour associated with yes/no questions in English (Pierrehumbert 1980). If the pairing of phonological information and syntactic content is arbitrary, and the intonational contour could in principle have been associated with some other feature, then this can properly be thought of as another example of lexical insertion, no different in principle from the situation with *dog*.

More interestingly, features might represent instructions to the SM system in ways distinct from the sound-content pairing of the sign, in which case they play a (p. 44) role in both modules and are potentially true interface features. One way in which features might connect to the SM (sensorimotor) systems is if they correspond to instructions to the spell-out procedure, for example in specifying that a head be spelled out to the right of its complement (for a head-final structure), or in specifying that a chain be spelled out at its foot (for LF movement). This is one way of characterizing the traditional overt vs. covert movement dichotomy: see Groat and O'Neil (1996). If the system is so configured that only the highest link in a chain is spelled out, then all movement would be overt, but this seems to be empirically incorrect (at least to the extent that scope phenomena are handled in the syntax, rather than via independent semantic mechanisms; cf. e.g. Huang 1982, Fox 1999). It would seem then that some property of links in a chain is necessary to ensure the parameterization of overt vs. covert movement.

Connected to the issue of overtness is the question, in derivational versions of minimalism, of whether operations take place after Spell-Out. Chomsky (1995b) suggested that LF movement could be modeled as the movement of features, while overt movement would be the pied-piping of additional material along with the attracted features.

The overt vs. covert parameter could then be a function either of the probe or of the goal, for example, the probe could be specified to attract additional features, in the case of overt movement, or the goals could be specified as having features which percolate. Each case would involve a second-order feature which could be modeled as syntax-internal.

Pesetsky (2000) argues that phrasal movement at LF must be distinguished from feature movement; if so, then an additional distinction needs to be drawn. Here again the notion of strength has been appealed to, in conjunction with a general principle of chain linearization. For example, Nunes (2004) argues that feature checking determines the linearization of a chain: one copy checks features, while deletion of the other removes the unchecked features from the derivation.

In a strictly antisymmetric system of linearization with no LF movement, such as has been advocated by Kayne (1994, 1998), there would be no such features, and hence perhaps no features relating syntax to linearization, the linearization algorithms being sensitive only to non-featural information.

Brody (2000a, b) has proposed a 'mirror theory' which replaces head movement while capturing various mirror effects observed in morphology. In that theory, a functional sequence spells out as a morphological word at a position designated by a diacritic '*'; for example, French verb movement is represented by T*–*v*–V, while English would be T–*v**–V. In our terms, Brody's * is a second-order feature (parametrically fixed) which is an interface feature. It is present in the syntax (as it is associated with syntactic features like *v*, not with vocabulary items), but simply instructs the phonological component where to spell out a word.

Outside of linearization, there are other possible syntax—phonology interface features. For example, scrambled elements are often phonologically destressed. If a single feature marked an element for movement and destressing, it would (p. 45) be a syntax—phonology interface feature (see, e.g. Neeleman and Reinhart 1998). Alternatively, if sentential prosody is assigned by an algorithm sensitive only to structure, as has been suggested for example by Cinque (1993), then such features may not be needed.

It is sometimes assumed that a feature might flag a constituent for not spelling out, as with Merchant's (2001) E feature, which marks ellipsis sites. The alternative is again lexical, for instance to analyze ellipsis as a kind of zero-pronominal (see, e.g. Lobeck 1995).

An interesting example of a potential interface feature is that observed in focus by stress. In some languages, including English, a word may be focused by stress, as witnessed in the difference between Rooth's (1985) examples (11a) and (11b), where capitals indicate focus stress.

(11)
- **a.** I only claimed that CARL likes herring.
- **b.** I only claimed that Carl likes HERRING.

The interesting thing about this is that the focus must be represented in the syntax as well as in the phonology if one adopts the classic Y-model of syntax, where the only connections between semantics and phonology are through the syntax and the lexicon. However, if one denies this architecture, such focus can also be seen as a phonology—semantics feature.

One syntactic approach is a lexical treatment. There would be a functional head, FOC, which carries the semantic content of focusing whatever it is sister to (cf. Rooth 1985), and which would have an autosegmental phonological content. However, this would suggest that increased pitch and intensity is an autosegmental feature which could in principle be associated with any meaning, for example negation or past tense.

If, on the other hand, the focus feature is not a separate lexical item, then it would have to be a feature, present in syntax, which carries an instruction to the phonology (pronounce loud) and to the semantics (place in focus). Possibly, this semantic instruction does correspond to some syntactic operation (e.g. LF movement), but its insensitivity to syntactic islands (Anderson 1972) suggests not.

In sum, some syntactic features might have interpretations at the SM interface, though there are no uncontroversial examples of this. In the case of linearization, the most likely examples involve different options in the spell-out of chains and of heads. Another class of cases involves non-pronunciation, and another concerns prosodic correspondences with information structure.

2.4.2 Meaning

At the other end of syntax is meaning, and here there is a lack of consensus regarding exactly how many modules there are and where the boundaries lie between (p. 46) them. In *Aspects*, some semantic features were tentatively posited in order to handle the selectional restrictions which render anomalous such sentences as *#The boy frightened sincerity* (*frighten* was taken to require a [+Animate] object). However, it became apparent that most such constraints are strictly semantic in that they are satisfied under paraphrases and not tied to individual lexical items.[6] The consensus since the early 1970s has been that semantic selection is not part of syntax, but rather part of some semantic module or of the conceptual-intentional domain of thought. In general, then, there will be thousands of nominal roots in any language which are featurally indistinct as far as syntax is concerned.

However, the kinds of meanings which distinguish edibles from inedibles and draft animals from mounts are arguably different from those with which formal semantics is concerned, and it is there we find the language of logic, set theory, predicate calculus, and other tools. Notions such as quantification, negation, grad-ability, boundedness, telicity, plurality, and so on are part of this system. Let us use the term semantics for such formal representations, excluding vaguer, prototype-based meanings such as whatever distinguishes camels from reindeer or a joke from an insult (and leaving open the question of whether animacy is one or the other).

The question of how syntactic features relate to meaning can then be posed in the following way: how are first-order formal features of syntax, and their second-order features, related to semantic representations?

Suppose, for example, that we have developed a model in which the only second-order features that a syntactic feature can have are those bearing on Merge, Agree, and Spell-Out, e.g. merge with something of feature X, or spell-out that something at the bottom of its chain. Then evidence that a given semantically interpretable feature is visible to syntax comes from data that shows that that semantically interpretable feature triggers Merge, Agree, or Spell-Out operations.

As an example, contrast a notion like ‘dangerous’ with a feature like *wh* or negation, which are visible to syntax while also clearly corresponding to semantic interpretations. First, we apparently never find a language in which dangerous is a syntactic feature, classing syntactic elements which behave in some consistent way together and correlating at the same time with a semantic interpretation of danger. This suggests that dangerous should be relegated to the conceptual-intentional realm along with whatever distinguishes camels from reindeer.[7]

Negation seems to be different from the feature dangerous. To see in what sense negation can be a syntactic feature, consider the rule of ‘Neg-inversion’ in English: (p. 47) if a negative element appears to the left of the subject, and scopes over the main predication, then the verb moves to second position.[8]

(12)
- **a.** I have at no time betrayed this principle.
- **b.** At no time have I betrayed this principle.
- **c.** *At no time I have betrayed this principle.

(13)
- **a.** I have never betrayed this principle.
- **b.** Never have I betrayed this principle.
- **c.** *Never I have betrayed this principle.

Thus, sentence negation does not require subject—auxiliary inversion, and subject-auxiliary inversion does not require sentence negation (it occurs, for example, in yes/no questions, with *wh*-expressions in interrogatives, with the VP pro-form *so*, and expressions with *only*). However, there is a non-arbitrary connection between sentence-level negation and subject—auxiliary inversion, in that any element which induces sentence negation and occurs to the left of the subject requires subject—auxiliary inversion.

The property of attracting a finite auxiliary is a second-order feature, which is borne by certain features including one which is interpreted as sentence negation; thus we can say that some feature neg has a consistent interpretation in the semantics as well as a consistent effect in the syntax. To the extent that other features behave the same way in the syntax, they are due to a fixed second-order feature, and this might be parametrically fixed if other languages differ on this point.

Note that semantic interpretation alone does not seem to be enough, since for example *no more than three* and *at most three* are semantically equivalent, but only the one containing a morpheme with the negative feature can trigger inversion.

(14)

a. On no more than three occasions in history has there been a land bridge connecting Asia with North America.
b. *On at most three occasions in history has there been a land bridge connecting Asia with North America.
c. On at most three occasions in history, there has been a land bridge connecting Asia with North America.

This suggests that the expression *no more than* bears a formal feature, call it NEG, which is not present in the expression *at most*, despite the logical equivalence.

(p. 48) On the basis of phenomena like NEG inversion, we posit a functional head with the following second-order features: it attracts an XP with a NEG feature to its specifier, and it attracts a finite auxiliary (or has the * feature, in Brody's terms). Semantically, it has the result that the TP is interpreted as a denial. This does not appear to be a universally present functional head; Cinque (1999) concluded after a cross-linguistic survey that NEG was different from modality, tense, and aspect in not being ordered in a universal sequence.

In this particular case, it might well be that the second-order features of this element in English are essentially accidental, a language-specific combination of properties. If so, then it should in principle be possible for a language to have a similar inversion rule for plural, or accusative, or some other syntactic feature.

All we can say is that in English neg is visible to the syntax, since a syntactic head attracts it, and that it has a consistent semantic interpretation, as all elements with the NEG feature have something in common semantically.

2.4.3 Universal correlations

As noted, there are widely differing conceptions of where the interface lies between syntax and meaning. In a syntactically austere model, syntax consists only of such very basic operations as Merge and Agree and some variable aspects of linearization. In this case, syntax interfaces with a semantic module, where syntactic outputs are translated into semantic representations and such phenomena as quantification, telicity, scope, and so on are located. This module interfaces with CI.

In that case, the relationship of second-order features with particular interface features such as NEG or INTERROGATIVE is accidental, and could at least in principle vary from language to language.

An example which suggests that this is on the right track comes from a comparison between English and certain Norwegian dialects: In English, *wh*-questions require a finite auxiliary in second position, a kind of V2 phenomenon, but topicalization does not require V2. In Norwegian, on the other hand, V2 is obligatory whenever any element is topicalized, but the V2 condition is relaxed, in some dialects, for *wh*-questions, where it is optional (see Vangsnes 2005 and references there).

(15)

a. There he has been.
b. Where has he been?

(16)

a.

Der	har	han	vært.
there	has	he	been
'There he has been'			

b.

Kor	han	har	vært?
where	he	has	been
'Where has he been?' (Tromsø Norwegian)			

(p. 49) This suggests that the association of V-movement features with interrogative C but not with declarative C is 'accidental' in English; this is a classic kind of 'parameter' in the literature, where both languages have comparable first-order features and fixed second-order features (or feature classes, cf. (7), (8)), but differently associated.

Of course, in both languages, topics and *wh*-elements move, and it is argued that similar movement occurs covertly in languages like Chinese and Japanese (Huang 1982, Watanabe 1993). Nevertheless, if syntax and semantics are strictly separated, then semantic properties such as being a quantifier should not have syntactic effects. If it turns out that quantifiers undergo movement, then it might be through 'accidental' placement of features, where the wrong arrangements lead to unusable structures.

On this view, being a quantifier in the semantics and taking scope in the syntax are correlated only functionally, in the sense that if an element X is interpreted as a quantifier and yet is not attracted to a scope position by some feature F in the syntax, then a derivation containing X will crash at the semantic interface, because it will be impossible to interpret X in its non-scopal position. Thus, a language will only be able to use X if it also has F.

However, there are many proposals in the literature for a tighter correspondence between syntax and semantics. For example, it is sometimes assumed that T universally carries an EPP feature, and that T maps onto tense semantically. It is quite typically assumed that *v* and V assign thematic roles but that C and T do not, and that C and T dominates *v* and V hierarchically. Similarly, if projections of N need case cross-linguistically while projections of V assign case, and N and V are semantically distinguishable, this is another universal correlation between syntax and semantics.

Work in cartography, in particular Cinque (1994, 1999), posits a hierarchy of functional categories whose labels reflect their semantic content. In the functional hierarchy, for example, epistemic dominates deontic or root modality, tense dominates aspect, perfect dominates progressive, and so on. If each of the categories in the functional sequence otherwise has cross-linguistically variable second-order features relating to Merge, Agree, and Spell-Out, then the only property which is universally connected to semantics is the functional hierarchy itself.

2.5 Conclusion

We have discussed a wide range of different phenomena which bear on the analysis of features in natural languages, and we have outlined what we take to be the different options for a minimalist theory of features. We have suggested a number of (p. 50) distinctions in feature theories: (i) the distinction between categories (which have positional motivation) and features (which have a cross-classificatory motivation); (ii) the distinction between first-order and second-order features and, within this, the distinction between fixed and variable second-order features; (iii) the distinction between interface features and module-internal features.

Second-order features are motivated by the fact that languages have syntactic dependencies, and within a minimalist system which embraces Inclusiveness, first-order features must have some property which implements this dependency. Variable second-order features—i.e. properties which can be borne or not by individual instantiations of features in a given language—are the means of this implementation. Current minimalist theory takes the notion of valuedness as a variable second-order feature which drives the formation of dependencies, as values are copied to the attribute lacking them.

Since fixed second-order features have to do with the creation of dependencies for such relations as Merge, Move, and Agree, much of parametric variation can be thought of as residing in which first-order features are associated with which fixed second-order features in a given language (i.e which dependencies are syntactically active in a language).

Given these distinctions, various theoretical questions arise: can all featural distinctions be reduced to positional distinctions? How many second-order features are there and what is their correct characterization? Can all types of first-order features be associated with second-order features (or, for example, is this restricted to, say, category features)?

A second major issue is the interaction of the notions of first- and second-order features with the notion of interface vs. syntax-internal features. Brody (1997: 143) posits the hypothesis of Radical Interpretability: that syntax never makes use of elements which lack interface properties (either semantic/conceptual content or instructions concerning Spell-Out). The strongest version of this hypothesis would hold that all first-order features have semantic interpretation (such as tense or negation or entity or quantity), and all second-order features are interface properties (such as uninterpretability, or licensing overt Spell-Out).

Pesetsky and Torrego (2001: 364) discuss a slightly weaker notion under the rubric of 'relativized extreme functionalism', namely that all grammatical features which are first-order features, in our terms, have a semantic value (though some instantiations of such features may be uninterpreted).

Given the distinction we draw here, one might take first-order features to be interface features in the sense of having some consistent meaning representation, while the second-order features would be only syntax-internal. Such a view would, we think, be compatible with relativized extreme functionalism, and would lead to the interface between syntax and the interpretive component being extremely minimal. The parametric data discovered in the cartographic tradition could be handled by allowing functional categories to have a simple format, associating **(p. 51)** certain second-order properties with the first-order features which are organized hierarchically by the functional sequence.

It is common in minimalist literature to assume many of the distinctions discussed above without due care in questioning the addition of new first-order or second-order features into the system. It seems to us that a concentration on what might constitute a more minimalist theory of features is necessary, and we hope that by identifying and clarifying the issues here, we have brought such a theory one step closer.

Notes:

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(1) In our sense of the term 'privative', the existence of a sequence or geometry does not make a system non-privative if the only way in which it classes features is in the hierarchy itself. However, if e.g. the nodes [person], [number], and [gender] have a different status from their dependents [speaker], [plural], and [feminine], then that implies that there are classes beyond the geometric structure, and the system is non-privative; see further below.

(2) In fact, we can think of both feature-class and second-order features as different kinds of properties of features: the former are fixed for tokens of the feature, while the latter may vary.

(3) In contrast, the plus sign is often used simply to flag something as a feature, without any assumption being made that there are minus values.

(4) The EPP (Extended Projection Principle, Chomsky 1982a: 10) was originally thought of as a fixed second-order

feature of Infl, but was recast as a variable second-order feature, in our terms, in Chomsky 2000a: 102.

(5) Setting aside adjunction; Kayne (1994) suggests that adjuncts and specifiers can be unified; see Kidwai (2000), Chomsky (2004a) for a different view of adjunction.

(6) Hence you can eat the result of a baking process, but you cannot eat the result of a syntactic transformation, although both objects are headed by the noun *result*; cf. Jackendoff (1972: 18).

(7) One might imagine that noun classifications could be sensitive to such a feature, even if syntactic operations are not. The Australian language Dyirbal was made famous by George Lakoff's bestseller *Women, Fire, and Dangerous Things* for having a gender class in which the concepts in the title of the book were grouped together. Plaster and Polinsky (2007) argue that this is incorrect, and dangerous things is not the basis for a gender class in Dyirbal.

(8) Famous minimal pairs include the following, which show that if the fronted element is not interpreted as scoping over the main predication, it does not trigger inversion.

((17))
(a.) With no job would Kim be happy.
(b.) With no job, Kim would be happy.

The first example can only mean that Kim would not be happy with any job, while the second can only mean that Kim would be happy if unemployed (the observation goes back to Klima 1964, but this example is based on Liberman (1974); see Haegeman and Zanuttini 1991, Rizzi 1996 for syntactic treatments).

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Case

David Pesetsky and Esther Torrego

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[–] Abstract and Keywords

The minimalist program includes the important conjecture that all (or most) properties of syntactic computation in natural language should be understood as arising from either (1) the interactions of independent mental systems; or (2) 'general properties of organic systems'. The study of case morphology and the distribution of nominal expressions in the languages of the world is one of the areas in which generative syntax has made the most-profound advances over previous approaches. A large (and increasing) number of studies have identified patterns and principles of great generality – achievements that also shed light on other topics and puzzles. At the same time, it is fair to say that the phenomenon of case represents one of the more outstanding challenges for the minimalist conjecture. Though case is indeed an area in which complex phenomena can be predicted on the basis of more general principles, these principles themselves look quite specific to syntax and morphology, with little apparent connection to external cognitive systems. The discussion in this article reflects the provisional character of the investigation.

Keywords: minimalist program, syntactic computation, natural language, syntax, morphology

3.1 Introduction

The minimalist program includes the important conjecture that all (or most) properties of syntactic computation in natural language should be understood as arising from either (1) the interactions of independent mental systems, or (2) 'general properties of organic systems' (Chomsky 2004a). The study of case morphology and the distribution of nominal expressions in the languages of the world is one of the areas in which generative syntax has made the most profound advances over previous approaches. A large (and increasing) number of studies have identified patterns and principles of great generality—achievements that also shed light on other topics and puzzles. At the same time, it is fair to say that the phenomenon of case represents one of the more outstanding challenges for the minimalist conjecture. Though case is indeed an area in which complex phenomena can be predicted on the basis of more general principles, these principles themselves look quite specific to syntax and morphology, with little apparent connection to external cognitive systems (not to mention general properties of organic systems). Our discussion in this chapter reflects the provisional character of the investigation.

(p. 53) 3.2 Case theory in Government—Binding syntax

By the late 1970s, it had become clear that the distribution of nominals is governed by special laws cross-linguistically. Though Chomsky and Lasnik (i977) had assembled a structured list of syntactic contexts in which NPs in languages like English are either disallowed or heavily restricted, it remained an open question why these restrictions should hold. Shortly thereafter, Vergnaud (2008[1977]), in a letter addressed to Chomsky and Lasnik, proposed that these restrictions in languages like English might be linked to another special cross-linguistic property specific to nominals: the presence of special morphology whose shape correlates with syntactic position—so-called case morphology. As Vergnaud observed, the distribution of certain types of case morphology on nominals in languages like Latin appears to match the distribution of nominals in languages like English whose case morphology is sparse or non-existent.

In languages like Latin, Russian, Japanese, and many others, a kind of case called accusative (henceforth ACC) is found on the complements of V and sometimes P (cf. (1), (2))—but not on the complements of N and A (cf. (3b) and (4b, d)). A complement of N and A either bears a different type of case affix (e.g. genitive morphology), or else must appear as a PP (cf. (3a, c) and (4a, c)). The following examples from Latin illustrate this pattern, which is found in many other languages as well:

(1) Complement to V (accusative)

[$_{VP}$	scripsit libr-*um*]
	wrote book-ACE

(2) Complement to P (accusative)

[$_{PP}$	ad Hispani-*am*]
	to Spain-ACC

(3) Complement to N (*accusative)

a.

[$_{NP}$	amor libertat-*is*]
	love liberty-GEN
‘love of liberty’	

b.

*[$_{NP}$	amor libertat-*em*]
	love liberty-ACC

c.

[$_{NP}$	amor	[$_{PP}$	in patriam]]
	Love		into country
‘love for one's country’			

(p. 54) **(4)** Complement to A[1] (*accusative)

a.

urbs	[$_{AP}$	nuda praesidi-*o* (Att. 7.13)
city		naked defense-ABL
‘a city deprived of defense’		

b.

*urbs [$_{AP}$ nuda praesidi-um]
city naked defense-ACC

c.

[$_{AP}$	liberi	[$_{PP}$	adelici-*i*s]] (Leg.Agr. 1.27)
	Free		from luxuries
‘free from luxuries’			

d.

*[$_{AP}$ liberi delici-*α*s]
free luxuries-ACC

Observations like these suggest the presence in the grammar of rules of case assignment. A first approximation adequate for the examples above might be

(5) Accusative case in Latin-type languages
a. V and P assign accusative case to an NP complement.
b. N and A do not assign accusative case (to an NP complement).

At first glance, some languages—for example, English—appear to lack the phenomena of (5) entirely. Nonetheless, the distribution of complements in English strongly resembles the generalizations captured in (5). As (6)–(10) make clear, English permits nominal complements in precisely those contexts in which languages like Latin assign ACC, and disallows bare nominal complements where Latin disallows ACC (but allows other cases such as genitive, and allows PP complements):

(6) Facts about the availability of NP complements in English
a. V and P allow an NP complement.
b. N and A do not allow an NP complement.

(7) Complement to V (NP)
[$_{VP}$ wrote the book]
(8) Complement to P (NP)
[$_{PP}$ to Spain]
(9) Complement to N (*NP)
a. [$_{NP}$ love of liberty]
b. *[$_{NP}$ love liberty]
c. [$_{NP}$ love [$_{PP}$ for their country]]

(10) Complement to A (*NP)
a. [$_{AP}$ free from luxuries]

b. *[$_{AP}$ free luxuries]

As Vergnaud noted, in languages with case marking of the sort we have been examining, any nominal that is morphologically capable of showing case morphology *must* do so. This observation could be stated explicitly as the Case Filter in (11).

(11) Case Filter
*[NP –case]

(p. 55) The relevance of the Case Filter to languages like Latin is self-evident. Remarkably, however, Vergnaud suggested that the Case Filter is also true of English, even though English lacks non-zero case morphology (at least outside the domain of personal pronouns). In particular, if we assume that English has an 'abstract' variant of ACC that may be assigned by V and P, a nominal complement to V or P will receive case (even if no overt morphology reflects this fact), and will satisfy the Case Filter. If we also assume that English, unlike Latin, lacks other cases that may be assigned to nominals (such as the genitive case assigned by N in (3a) and the ablative case assigned by A in (4a)), bare nominal complements to N and A will receive no case specification at all, and will be excluded by the Case Filter.

(12) Differences between English and Latin
a. Case morphology in English is phonologically zero
b. English has accusative case, but does not have genitive, dative, ablative, etc. as Latin does.

Phonologically zero case morphology is a phenomenon independently found in languages where the existence of a rich case system is not in doubt. In Russian, for example, though most nouns show overt case morphology much like Latin, there is also a productive class of 'indeclinable' nouns. Most of these are foreign borrowings, which, because they display certain non-Russian phonological properties, cannot receive case morphology. Unlike English, but like Latin, Russian is a language that does have cases such as genitive and dative that may be assigned by N and A. Indeclinable nouns may thus appear in all the same positions as their 'declinable' counterparts that do display case morphology:

(13) Declinable vs. indeclinable nouns

a.	[VP	vidit mašin-*u*]	b.	[PP	v mašin-*u*]			*Declinable*
		sees car-ACC			into car-ACC			
a′.	[VP	vidit kengure]	b′.	[PP	v kenguru]			*declinable*
		sees kangaroo-ACC			into kangaroo-ACC			
c.	[NP	uničtoženic mašin-*y*]	d.	[NP	ljubov’	[PP	k mašin-*e*]]	*declinable*
		destruction car-GEN			love		to car	
c′.	[NP	uničtoženic kengure]	d′.	[NP	ljubov’	[PP	k kenguru]]	*declinable*
		destruction kangaroo-GEN			love		to kangaroo	
e.	[AP	dovolen mašin-*oj*	f.	[AP	serdit	[PP	na mašin-*u*]]	*declinable*
		satisfied car-INSTR			angry		at car	
e′.	[AP	dovolen kenguru]	f′.	[AP	serdit	[PP	na kenguru]]	*indeclinable*
		satisfied kangaroo-INSTR			angry		at kangaroo	

Insofar as indeclinable nouns are concerned, Russian looks like English with respect to (12a), and unlike English with respect to (12b). Since phonologically null case marking is clearly available in Russian, we may conclude that it is not an implausible proposal for English either.

The grammar of case for complements in a language where only accusative case is available to a complement makes two types of distinctions, both of which seem (p. 56) to refer to syntactic category. First, it distinguishes between assigners (V, P) and non-assigners (N, A) of accusative case. Second, it distinguishes between NP, which needs to receive case (by the Case Filter in (11)) and other categories such as PP and CP, which act as if they do not need case. The fact that PPs do not appear to need case has already been demonstrated in (9a, c) and (10a). The examples in (14) make the same point for CP:

(14)
- **a.** Complement to A
  her proof [$_{CP}$ that the world is round]
- **b.** Complement to A
  satisfied [$_{CP}$ that the world is round]

All these distinctions among the various syntactic categories may be attributed to the interaction of the Case Filter with the rule in (15) (a rule that we will improve on as this chapter proceeds; see the discussion of Exceptional Case Marking in section 3.4):[2]

(15) Accusative case assignment

α assigns accusative case to β only if:

i. α is V or P (not N or A); and

ii. β is the complement of α.

For subjects, it appears to be T that assigns case (nominative), and in languages like English, only finite T, as (16a, b) show. As (16c) shows, English allows a prepositional complementizer *for* in certain circumstances to assign case (presumably ACC) to the subject of a non-finite clause:

(16) Only finite T assigns case to its specifier
a. We were happy [that Mary won the prize].
b. *We were happy [ø Mary to win the prize].[3]
c. We would be happy [for Mary to win the prize].

We thus add the rule in (17):[4]
(p. 57) **(17)** Nominative case assignment (English)
Finite T assigns nominative case to its specifier.

After Vergnaud made his proposal, it was soon observed that 'Case Theory' could account not only for the overall distribution of NPs (vs. non-NPs) but also for the obligatoriness of NP-movement in constructions like passive, unaccusative VPs, and raising infinitivals in English and in many other languages. The complement position of a passive (or unaccusative) verb in a language like English could be understood as a position in which ACC was not assigned by V(a special effect of passive morphology). This failure of ACC assignment would have no consequence for a non-NP, but would force movement of an NP into a position where the NP could be assigned case. That is why the italicized NPs in (18) (where the underscore marks their original location) must move, while the PP and CP complements remain in their VP-internal positions:

(18)
a. *The book* was put__[$_{PP}$ under the desk].
b. *Mary* was persuaded__[$_{CP}$ that we should leave tomorrow].
c. *The door* opened__suddenly.

Similarly, the failure of non-finite T to assign case to its specifier motivates obligatory Raising of the subject of an infinitival clause functioning as the complement to an unaccusative verb like *seem:*
(19) *Mary* seemed [__to have written the letter].[5]

Of course, there is another reason why *some* element must move to Spec, TP in (18) and (19)—the requirement that TP needs a specifier, called the Extended Projection Principle (EPP). If a verb takes only a CP complement, for example, an expletive *it* may satisfy the EPP in a language like English—and the CP does not need to move. That is why verbs like *believe* have two passives, as in (20), while a verb that takes an NP complement has only one passive, as seen in (21). Both examples satisfy EPP in (21), but only the first satisfies the Case Filter:

(p. 58) **(20)**
a. [That the world is round] was believed__by the ancient Greeks.
b. It was believed by the ancient Greeks [that the world is round].

(21)
a. The book was put__under the table.
b. *It was put the book under the table.

Furthermore, in many languages, such as Spanish, nominative case appears possible on NPs that are c-commanded by T (and structurally close to T), even without movement. This suggests that nominative case is not necessarily assigned by finite T to its specifier directly. Instead T might assign nominative case 'down', i.e. to a nearby position that it c-commands—and that the EPP property of T in English (but not in all languages) independently moves the NP targeted by nominative case assignment to the specifier position of the assigner. The discovery of the precise relation between nominative case, EPP, and movement is an important research topic, as is the relation between all three of these factors, and verb-subject agreement, which also goes hand in hand with nominative case and movement to Spec, TP cross-linguistically (but not universally).

3.3 Burzio's generalization and *v*

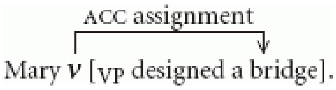
It was also observed by Burzio (1981, 1986), early in the development of Case Theory, that the failure of passive and unaccusative verbs to license ACC case seems systematic, and is linked in some fashion to their failure to take an external argument (at least in the same way as transitive active verbs do; we put aside the question of *by*-phrases in passive).

(22) Burzio's generalization
If a verb licenses accusative case, it has an external argument.

That is, for example, there is no verb *ypen* 'open' that could appear in a frame like (23), where *it* is intended to be expletive (not a referential pronoun):

(23) *It ypened the door. [expletive *it*]
'The door opened.'

Burzio's generalization states, but leaves unexplained, the link between licensing of ACC and external argument. One approach to this problem was developed by Chomsky (1995b: ch. 4), who suggested that the assigner of ACC might not be V after all, but might be a separate head *v* (read as 'little *v*') that is simultaneously responsible for assigning ACC and the external argument thematic role.[6] The category *v* takes VP as its complement, and combines with V by some operation such (p. 59) as head movement to yield the transitive verbs we actually hear in languages like English.



(24)

In support of this idea, Wurmbrand (1998, 2001) provided evidence from 'restructuring' constructions that ACC is indeed not assigned by V itself, but by a higher head. Restructuring constructions are configurations in which a verb belonging to a certain class takes an infinitival complement that is transparent for many processes that are usually blocked by a clause boundary.[7] Wurmbrand showed that these complements are structurally smaller than other infinitivals. For example, they lack independent Tense and are unable to show negation. Chomsky's *v*-hypothesis makes it possible to suggest that the infinitival complement in a restructuring environment might be a bare VP, and that it is *v* (and all higher clausal material) that is absent. If so, we would still hear a verb in the embedded clause, but any NP complement of that verb would have to be dependent on the *v* of the matrix verb (or some other element) for its case. Wurmbrand showed that restructuring constructions in German do indeed behave this way. In particular, when the matrix verb of a restructuring configuration is passivized, it is the embedded object that shows the behavior seen in (18)—that is, it must receive NOM from the matrix T, a construction known as 'long passive'. Nominative case and plural agreement on the verb in (25) make it clear that the underlying object of *reparieren* 'repair' is receiving case from the higher finite T:

(25) Long passive in German

… weil	[der Lastwagen und der Traktor]	zu reparieren	versucht	wurden
since	[*the truck and the tractor*]-NOM	to repair	tried	were
lit. 'since the truck and the tractor were tried to repair'				
(i.e. someone tried to repair them)				

(p. 60) Because negation (or tense) is higher than *v*, the addition of negation (or tense) to the embedded clause forces it to be bigger than *v*P. Consequently, long passive is disallowed, as predicted:

(26) Long passive in German

*…weil	[der Lastwagen und der Traktor]	*nicht*	zu	reparieren
since	[*the truck and the tractor*]-NOM	not	to	repair
Versucht	wurden			
Tried	were			

The discovery of *v* marked a major advance in work on NP-licensing and case more generally, as results like Wurmbrand's make clear. At the same time, opinions vary concerning the functions other than case assignment that the VP-external case-assigner plays. Chomsky linked the case-assigner to the assignment of an external *θ*-role, with distinct flavors of *v* either assigning this role and assigning accusative case (in transitive constructions) or assigning neither (in unaccusative and passive constructions).[8] In a similar vein, Kratzer (1996) has identified the same category as instantiating Voice, and offered an explicit semantics for a proposal similar (though not identical to) Chomsky's. Other researchers have suggested other connections, however. Travis (1992), for example, identified the category as Aspect; Torrego (2002) argued that it is a locative preposition; and Pesetsky and Torrego (2004) suggested that it is a lower instance of Tense. If any of these proposals are correct, the question of Burzio's generalization again arises, and remains something of a mystery.

3.4 Other forms of licensing

Up to now in our presentation, the ability to license an NP has been a property of particular functional categories—finite T and *v* in the versions of case theory just discussed. Accusative and nominative case morphology on nominals reflects the distinctions among these functional elements. In many languages, however, **(p. 61)** members of *lexical* categories—for example, specific verbs—also determine nominal morphology. For example, a particular verb may require dative case on its complement in a language like Latin, Icelandic or Warlpiri—and dative morphology may supplant the otherwise expected accusative (or nominative) morphology on the nominal. Such requirements appear to be linked to argument structure and thematic role assignment: a lexical item may require a special case on only those nominals that it takes as semantic arguments (and assigns a thematic role to).

When such a requirement is found, an immediate question arises: Does the assignment of this idiosyncratic morphology also license the nominal, as accusative or nominative assignment does? In other words, is such morphology on a nominal just ‘paint’, that obscures an underlying nominative or accusative, or does it represent an alternative form of licensing that makes additional case assignment unnecessary? It turns out that both options are realized in the languages of the world.

In Icelandic, for example, it seems that idiosyncratic nominal morphology required by individual verbs is indeed ‘paint’ that covers up a system that is underlyingly English-like in most respects. Such morphology has come to be called *quirky case* (Andrews 1982). (Nominative and accusative case, which are not linked to argument structure or thematic role, are called structural case by contrast.) Consider, for example, the unpredictable assignment of dative morphology by the verb ‘finish’ in (27a) and genitive by ‘visit’ in (27b). When these verbs are passivized, the dative and genitive morphology remain, and (if other circumstances do not intervene) the underlying complement moves to Spec, TP much as in English—as seen in (28a, b) (data from Andrews 1982):

(27)

a.

Ðeir	luku	kirkjunni.
they	finished	the-church.DAT

b.

Við	vitjuðum	Olafs.
we	visited	Olaf.GEN

(28)

a.

Kirkjunni	var	lokið	(af Jóni).
the-church.DAT	was	finished	

b.

Olafs	var	vitjað	(af Jóni).
Olaf.GEN	was	visited	

In sentences whose main verb does not require a particular quirky case, the object of an active verb bears accusative morphology, while the corresponding argument in a passive sentences bears nominative (in a finite clause)—as we expect if passive *v* fails to assign accusative. Furthermore, in an environment where an overt nominal is not licensable by abstract case in a language like English, quirky case morphology is not sufficient to license a nominal in a language like Icelandic. The subject position of the infinitival complement 'to try' provides a relevant example:

(29)

a.

Mér	býður	við	setningafræði.
me.DAT	is.nauseated	at	syntax

(p. 62) **b.**

*Hún	reyndist	mér	bjóða	við	setningafræði.
he	tried	me.DAT	to.be.nauseated	at	syntax

Quirky case in Icelandic thus appears to be irrelevant to the licensing of nominals. It does not constitute an alternative to abstract accusative or nominative assignment, but merely makes the assignment of accusative or nominative morphologically undetectable.

By contrast, some lexically governed nominal morphology does appear to license nominals, with the result that no other licenser such as abstract case is necessary. Such morphology is called *inherent case* (Chomsky 1986b). In Russian, for example, when a complement nominal bears dative or instrumental morphology as a consequence of the requirements of verbs such as 'help' and 'manage' (cf. (30)), and the verb is passivized as in (31), the noun does not raise to Spec, TP with its idiosyncratic case morphology as in Icelandic. Instead, the construction is disallowed, since the complement does not need to move for case reasons and general principles of ECONOMY prevent a licensed nominal from raising to Spec, TP to satisfy EPP:[9]

(30)

a.

Ivan	pomog	studentam.
Ivan	helped	students.DAT.PL

b.

Maša	upravljaet	zavodom
Masha	manage	factory.INSTR.SG

(31)
a.

*Bylo	pomoženo	studentam.	(*any word order, any case pattern)
Was	helped	students.DAT.PL	

b.

*Bylo	upravleno	zavodom.	
was	managed	factory.INSTR.SG	(*any word order, any case pattern)

As in Icelandic, the norm with verbs that do not impose particular morphological requirements on their complements is for the complement of an active transitive verb to show accusative case, and the corresponding argument of a passive to show nominative.

In both English and Icelandic, accusative case sometimes appears to be assigned by a verb that takes a clausal complement across the clause boundary, to the subject of the complement. This pattern, which is somewhat rare cross-linguistically, was dubbed *Exceptional Case Marking* (ECM) by Chomsky (1981). In (32a), the verb *believe* case-marks *him* in ECM fashion, while in (32b) the replacement of the verb *believes* with a related nominal eliminates the possibility of case marking for *him*; and in (32c) passive morphology on *believe* suppresses its ability to assign accusative case—supporting the analysis of (32a) as ECM.[10]

(p. 63) **(32)** Exceptional Case Marking: English
a. Mary believes [Sue to have read the book].
b. *Mary's belief [Sue to have read the book].
c. Sue was believed [_ to have read the book].

Icelandic ECM behaves like its English counterpart, except that quirky case assigned by the lower verb may overwrite the otherwise expected accusative marking of the embedded subject (as expected):
(33) Exceptional Case Marking: Icelandic
a.

Hann	telur	**Jón**	hafa	kysst	Maríu.
He-NOM	believes	**John-ACC**	to-have	kissed	Mary

b.

Hann	telur	**mér**	bjóða	við	setningafræði.
He-NOM	believes	**me-**DAT	to.be.nauseated	at	syntax

Crucially, however, as predicted, the higher verb never imposes a quirky or inherent case on the embedded subject, since though it case-marks this nominal, it does not assign it a thematic role.

The phenomenon of ECM makes it clear that accusative case is not necessarily assigned to the complement of the assigner, contrary to our initial proposal in (15). This point is independently made by the assignment of accusative case by little *v* discussed in section 3.3, if that hypothesis, rather than (15), is the correct one. Both the phenomenon of ECM in early work on case theory and *v* in later work raise questions of locality: how much distance can separate a nominal from its case assigner? We will not discuss this issue in detail here, but it appears that case may be assigned so long as no barrier of a particular sort intervenes, and as long as there is no nominal closer to the assigner than the assignee.[11] In early work, most maximal projections were taken to function as barriers of the relevant sort, and the relevant locality was called 'government'. (An exception to the general rule was made for non-finite TP, precisely so as to permit ECM.) In more recent work, it has been argued that the relevant notion is 'phase', which has the advantage of linking the conditions on case assignment to other properties of language.

(p. 64) 3.5 Ergative patterning and inherent case

So far, we have seen that nominal licensing may involve nominative or accusative case assignment, or assignment of inherent case. Furthermore, the shape of case morphology may be transparent (when the distinction between nominative and accusative is overtly marked) or opaque—as in English (where all nominals behave like Russian indeclinables) or Icelandic (where quirky case may mask nominative and accusative).

In recent work, Legate (2008) has argued (building on work by Woolford 1997, among others) that these distinctions may in fact be all we need to explain case systems that appear at first glance to be quite different from those we have examined so far. If Chomsky (1995b: ch. 4) is correct in suggesting that *v* is responsible for assigning a *θ*-role to the external argument of a clause, does *v* ever assign inherent case to its specifier? Legate argues that in certain languages it does.

Suppose in a given language the following conditions hold, each of which is independently allowable within the system already developed:

1. Abstract case is morphologically zero (or shows the same morphology for both NOM and ACC)—as in English.
2. Inherent case exists, and is not morphologically zero.
3. One of the categories that assigns inherent case is *v* (the category that introduces an external argument), which assigns inherent case to its *θ*-marked specifier.

A language with these three properties would look like a language with a case-marking pattern called 'ergative/absolutive'. In such a language direct objects and subjects of unaccusative and passive verbs bear the zero or invariant case morphology given in point 1 above—but external arguments will bear the special inherent-case morphology required by *v*. The morphology present on external arguments in such a language is called 'ergative'; and the morphology (or absence of morphology) that (on this view) marks nominative and accusative is called 'absolutive'. Example (34a) shows a transitive verb taking an ERG external argument and an ABS direct object, while (34b) shows an intransitive verb whose sole argument is marked ABS.

(34) Warlpiri (Pama-Nyungan: Central Australia)[12]

a.

nyuntulu-**rlu**	ka-npa-ju	ngaju	nya-nyi
you-**ERG**	PRS-2SG-1SG	me-**ABS**	see-NPST
'You see me.'			

(p. 65) **b.**

ngaju	ka-rna	parnka-mi
me-ABS	PRS-1SG	run-NPST
'I am running.'		

If this analysis is correct, then the possibility of an ABS subject as in (34b) should disappear in a non-finite clause (since structural case to the subject is unavailable in non-finite clauses), but the possibility of an ERG subject as in (34a) should remain (since ERG is an instance of inherent case, unaffected by finiteness). Legate shows that this prediction is correct in Warlpiri. Example (35a) shows a non-finite embedded clause with ERG subject, while example (35b) shows that an ABS-marked subject is blocked in a similar environment:

(35)

a. Warlpiri: ERG possible as subject of non-finite clause

Kurdu-lpa	manyu-karri-ja,	[**ngati-nyanu-rlu**
child-PASTIMPF	play-stand-PAST	[**mother-**POSS**-**ERG
karla-nja-rlarni.]		
dig-NONFIN-OBVC]		
'The child was playing, while his mother was digging (for something).' (Laughren 1989: [44a])		

b. Warlpiri: ABS impossible as subject of non-finite clause

Ngarrka-patu-rlu	ka-lu-jana	puluku
man-PAUC-ERG	PRES.IMPERF-3PL.SUBJ-3PL.OBJ	bullock.ABS
turnu-ma-ni …	*… [kurdu parnka-nja-rlarni].	
group-CAUSE-NONPAST	[child.ABS run-**NONFIN**-OBVC]	
'The men are mustering cattle while the children are running.' (Legate 2008: 62)[13]		

Legate also shows that ABS on the direct object does not disappear in non-finite environments (in contrast to subject ABS), thus confirming her approach—since finiteness is not expected to affect ACC case.

(36) Warlpiri: object ABS still possible in non-finite clause

Ngarrka-patu-rlu	ka-lu-jana	puluku	turnu-ma-ni,
man-PAUC-ERG	PRESIMPF-3PL.SUBJ-3PL.OBJ	bullock	muster-NPAST

[karnta-patu-rlu	**miyi**	purra-nja-puru.]
[woman-PAUC-ERG	**food.**ABS	cook-NONFIN-TEMPC]

'The men are mustering cattle while the women are cooking the food.' (Legate 2008: 63)

(p. 66) If Legate's analysis of Warlpiri and similar systems is correct, then what might appear at first glance to be a radically different organization of case marking is actually an expected variation on patterns already attested in other languages.[14]

3.6 Deeper questions about case

In the discussion so far, we have taken for granted that case is a property of languages that display the relevant morphology. The interest of the preceding sections lies mainly in the presentation of a relatively simple theory of case that turns out to have a much wider set of empirical consequences cross-linguistically than one might have thought. We saw first that the distribution of nominals in a language like English that appears to lack case morphology is nonetheless governed by the same laws that regulate nominative and accusative case in languages like Latin and Russian. We then saw (following work by Legate) that the morphological parameter that distinguishes English from more obvious case languages, when crossed with a distinct parameter that distinguishes structural from quirky and inherent case, provides an immediate account of languages otherwise said to show a distinct ergative/absolutive pattern.

A key question, however, has been left unasked so far: why languages should show 'case phenomena' in the first place. This question is particularly urgent in the context of a minimalist program that seeks to attribute syntactic properties that do not arise directly from the action of Merge to properties of the interfaces between syntactic computations and adjacent systems (or else to language-external factors). In this context, then, we should ask a 'why' question about every aspect of case theory. For example:

1. Nature of case: What is case, and why is it a necessary licenser for nominals?
2. Specialness of nominals: Why do only nominals seem to need licensing by case (while clauses, PPs and APs do not)?
3. Assigners vs. non-assigners: Why do nouns and adjectives not license case in the same fashion as verbs and prepositions, at least in languages like Latin and English? Why does finite T differ from infinitival T in its ability to assign case?

It is fair to say that none of these questions has been answered in a satisfying fashion so far. Nonetheless, some progress has been made, perhaps, in achieving less ambitious goals that might serve as first steps towards answering these questions. In (p. 67) particular, there have been several attempts to anchor case theory to more general, possibly syntax-external properties of language. To the extent that the anchor itself must be stipulated, these attempts fall short—but to the extent that the proposed links between case and other aspects of language prove real, these proposals might constitute progress towards answering questions like 1–3 above.

An early attempt of this sort was presented by Chomsky (1981a: 176ff. and 336ff.), developing a suggestion by Aoun (1979). Chomsky speculated that case is a precondition for an NP to receive an interpretation at Logical Form (LF) (the interface of syntax with semantic systems). This came to be called the Visibility Hypothesis. The θ-criterion of Chomsky (1981a) (and his more general later proposal (Chomsky 1986b), the principle of Full Interpretation) disallows any NP that fails to receive such interpretation. Thus, the Case Filter—at least insofar as it has the effect of rendering certain expressions unacceptable—can be said to follow from more general principles that require all components of a syntactic expression to be 'legible' at the interface with semantic components. Of course, the link between case and legibility remains stipulated; so question 1 above, though addressed by this proposal, is not really answered. Furthermore, this proposal leaves unexplained the fact that clausal arguments (and PPs) may be interpreted without receiving case (question 2). Chomsky (1981a: 337ff.) discusses the problem for CPs, but ultimately leaves it unresolved—suggesting only a tentative stipulation that would allow CPs to be visible at LF under conditions different from those that apply to nominals.[15] Question 3 remains entirely open.

More recently, we have attempted to develop Chomsky's proposal in a different fashion (Pesetsky and Torrego 2001; henceforth P&T). As Chomsky (1995b: ch. 4, 2000a) and others have pointed out, features that are morphologically expressed on a particular word are not always interpreted in that position. For example, a finite verb in a language like Latin or English will often bear person or number morphology that is semantically relevant to the subject of the sentence, but not to the verb itself. More specifically, ϕ-features such as person, number and gender appear to have both interpretable and uninterpretable variants, depending on what category of word they appear on.

In standard proposals, case (in particular, nominative and accusative) is strikingly different. As normally described, case features have no semantic interpretation associated with them—no matter where they occur.[16] P&T suggested that this anomaly in the theory might be a sign that case is actually the uninterpretable counterpart of some interpretable feature—a feature, perhaps, to which a different (p. 68) name is normally given. The well-known correlation between tense and nominative case suggests that case might in fact be an uninterpretable instance of tense (T).[17]

If case is actually T on the head of nominals, how might one explain the requirement that a nominal occupy a 'case position' (and the absence of such a requirement for clauses and PPs)? The answer would have to link the nominal's apparent need for case to some more general type of need found with other features. As it happens, a need of this sort is indeed found with some other features.

As discussed by Chomsky (1995b: ch. 4) and many others, the features of lexical items quite generally come from the lexicon in two forms: valued and unvalued. A lexical item with an unvalued feature is an element that 'knows' that it must bear a feature such as number, but does not know in the lexicon whether it will be, for example, singular or plural. The existence of unvalued features can be seen in the phenomenon of agreement. The number and gender features of a past tense verb in Russian, for example, co-vary with the corresponding features of the head noun of a subject nominal. The noun's versions of these features come from the lexicon knowing their values. This is most transparent in the case of gender, which is often an unpredictable property of individual nouns—and also for number, which is occasionally stipulated as plural (as in the case of *šči* 'cabbage soup', which only occurs as a plural). By contrast, an inflected verb does not come from the lexicon valued for gender (nor for number). No verb is intrinsically neuter, for example, nor obligatorily plural. The number and gender features of a past tense verb in Russian are assigned to it by a syntactic process called Agree, in which unvalued features receive their value from a local element that bears valued counterparts of these features. Unvalued features must receive a value, or else the derivation fails: no past tense verb in Russian, for example, lacks number and gender morphology.

P&T suggested that the process otherwise called 'case assignment' is in fact nothing but Agree applying so as to value an otherwise unvalued T-feature on a nominal. When nominative case is assigned to a nominal expression, they argued, what is actually happening is the valuation of an unvalued T-feature on the nominal by a valued counterpart on Tense itself (ultimately on the finite verb). Instances of nominals that violate the case filter (and therefore produce a judgment of deviance) are simply nominals whose unvalued T-feature has remained unvalued, for lack of an appropriately situated local bearer of a valued counterpart.

This proposal connects the difference between nominals (which must appear in a 'case position') and CPs (which have no such need) to an independent difference between these two categories in languages like English. CPs contain T locally inside them, whereas nominals do not.[18] Consequently, all arguments may be assumed to have the same type of T on their head, CPs as well as nominals—unvalued T, in the (p. 69) instances under discussion. The difference between unvalued T on C and unvalued T on the head of a nominal lies in whether or not valuation requires an external instance of valued T. Because T on C need not search external to CP for valuation, a CP may occur in non-case positions as well as case positions, unlike nominals.[19]

P&T extend their proposal to accusative case, by arguing that T-features on categories within the VP system can function in much the same fashion as Tense itself to value T on a nominal. Furthermore, inherent case is straightforwardly distinguishable from structural case within this system as a situation in which valued rather than unvalued T is present on the head of a nominal. Like a CP, such a nominal will have no need of external valuation for its T-feature, and thus will show no Case Filter effects—not, in this instance, because T on the argument receives its value internally, but rather because it is in no need of valuation in the first place. P&T (2001, 2004) argue that PP arguments have this property as well: a preposition bears valued T-features (and in fact shares a significant amount of its syntax with Tense; cf. the use of prepositional vocabulary such as English *to* within the tense system).

Arguments both against and in favor of P&T's proposal arise when case morphology is considered in more detail. On the negative side, consider the fact that if P&T's proposal is correct, we expect to find the shape of case morphology co-varying with tense. For example, we might expect nominative morphology to look different in a past tense sentence and a present tense sentence. Though there are many languages in which case-related phenomena such as ERG-ABS vs. NOM-ACC patterning vary between past and present tenses (or perfective vs. imperfective aspect), there are few if any languages in which case morphology directly mirrors the tense of the sentence. One example offered by P&T (due to Ken Hale, personal communication) is the Australian language Pitta-Pitta, where subject nominals bear future tense morphology when the tense of the sentence is future (Blake and Breen 1971, Hale 1998):

(37) Pitta-Pitta

Ngapiri-ngu	thawa	paya-nha.
father-FUT	kill	bird-ACC
‘Father will kill the bird (with missile thrown).’		

But such examples appear to be vanishingly rare, so a puzzle remains.

Another similar question raised by P&T's proposal, however, may receive a more positive answer. P&T provide no reason why T on nominals should have to be semantically uninterpretable. If they are correct that case is actually T on nominals, and that all arguments must bear T, we expect to find languages in which nominals (p. 70) bear instances of T that are interpretable—i.e. that temporally situate the individual denoted by the nominal, just as interpretable T in an English clause temporally situates an event. In fact, such languages are known. In the Halkomelem (Salish) examples seen in (38), past tense morphology on N places the lifespan of the object denoted by N in the past. Similar phenomena have been discussed for Somali by Lecarme (1996).[20]

(38) Halkomelem Salish interpretable T on N

a.

te-l	má:l-**elh**
DET-1SG.POSS	father-PAST
‘my late father’	

b.

te-l	xéltel-**elh**
DET-1SG.POSS	pencil-PAST
‘my former pencil’	

(Burton 1996: 67)

3.7 What distinguishes nom from acc morphology

If case in a language like Latin or Russian is an instance of unvalued T on nominals, a question immediately arises concerning the nature of the distinction between nominative and accusative morphology. To answer this question, we might adopt Chomsky's (1995b: ch. 4) suggestion that the choice of nominative or accusative morphology on a nominal reflects whether the nominal entered an Agree relation with a feature of T or with a feature of *v*. In the context of P&T's proposal, this would amount to a claim that nominal morphology reflects the identity of the element whose T-features were responsible for valuing T on the nominal.

Chomsky's actual proposal arose in the context of a theory different from the one we have just discussed. Chomsky suggested that case assignment is a consequence of an Agree relation (just as P&T later argued). For Chomsky, however, the crucial Agree relation involves neither T nor case itself, but instead involves ϕ-features. (p. 71) In Chomsky's view, case is valued on a nominal as a stipulated by-product of an Agree relation that actually involves a distinct set of features. In this proposal, the distinction between nominative and accusative morphology on a nominal is keyed to whether ϕ-feature agreement took place with T (yielding subject agreement, where visible) or with *v* (yielding object agreement, in languages in which this is visible).

Common to both approaches is the assumption that NOM and ACC morphology, when distinguishable, reflects agreement or licensing directly.[21] We end this chapter by noting an alternative view, developed by Marantz (1991), that is in principle compatible with any of the proposals concerning case that we have discussed so far: the classical proposal, P&T's proposals, or Chomsky's proposals that treat case as a by-product of ϕ-feature agreement.

Marantz suggests that *no* case morphology reflects licensing directly. The distinction between NOM and ACC morphology is sensitive to syntactic structure, he argues—but does not depend on the source of nominal licensing. In a sense, in this theory, *all* case morphology is quirky, in that it is all unrelated to nominal licensing. In particular, Marantz proposes, for languages like Latin or Icelandic with a NOM-ACC case system, that NOM is assigned under rather familiar structural conditions related to T (though unrelated to any Agree or assignment relation involving T). Crucially, however, the assignment of ACC is different. ACC in this theory is a dependent case, assigned to the lower of two nominals, when the higher nominal bears NOM. If the higher nominal bears a distinct quirky case, NOM, rather than ACC, will be found on the lower nominal. In effect, this builds Burzio's generalization into the morphology, rather than into the syntax of nominal licensing or Agree:[22]

(39) A dependent-case account of NOM and ACC (based on Marantz 1991)

a. NOM is the morphology found on the highest non-case-marked nominal in a clause in which V has entered a relationship with T.

b. ACC case is the morphology found on a nominal within a domain in which a higher nominal has received NOM.

A key argument from nominative—accusative systems in favor of a strictly morphological view of NOM as a dependent case is provided by Icelandic constructions like (40). Here, the verb 'believe' licenses the embedded subject *henni*, which, however, bears dative morphology as a lexical property of the embedded verb 'think'. Crucially, although the embedded verb is non-finite, its complement (a small clause) is marked with NOM. Marantz argues that morphological NOM rather than ACC appears on the complement precisely because the subject of the embedded clause does not bear morphological NOM, as predicted by (39b).

(p. 72) **(40)**

Eg tel	[henni	hafa	alltaf	þótt	[Olafur	leiðinlegur]]
I believe	her-DAT	to-have	always	thought	Olaf-NOM	boring-NOM

Marantz goes on to suggest that an ERG—ABS system is one in which a dependent case is assigned to the *higher* of two nominals, and also presents arguments in favor of this view (which we will not consider here).

Dependent-case approaches to case morphology (and the NOM—ACC distinction) are often presented as alternatives to agreement-based accounts. Bobaljik (2008), for example, argues that verbal agreement is sensitive to a morphological hierarchy based in part on the dependent/non-dependent distinction, and that agreement and case thus both belong to a morphological, 'post-syntactic' component, rather than to syntax proper.[23] More recently, however, Baker and Vinokurova (2008) present arguments from the Turkic language Sakha that morphological case rooted in the dependent/non-dependent distinction coexists with agreement-based case, and both systems may play a role in licensing nominals and determining agreement targets.[24]

3.8 Conclusion

As we have shown, a remarkable series of results and generalizations have grown from Vergnaud's initial suggestion that the grammar of case might be central to the syntax of all languages—not just those with rich case morphology. At the same time, it should be obvious (as we noted at the outset) that these achievements still fall short of the goals set out in Chomsky's minimalist conjectures, and much controversy remains. Furthermore, many fundamental questions, including the reason why case should exist at all, do not yet have substantive answers. In this sense, research on case remains very much work in progress.

Notes:

(1) Examples drawn from corpora searchable at the Perseus Digital Library (http://www.tufts.perseus.edu). 'ABL' stands for 'ablative', a case in Latin.

(2) The discussion so far tells us what sorts of complements are possible for a noun or adjective, in contrast to a verb or preposition. Case Theory does not tell us that a transitive verb with an NP object will probably have a nominal counterpart in English that takes a PP object (or a nominal counterpart in Latin or Russian that takes an object marked with some case other than accusative). The relation between *destroy the car* and *destruction of the car* (and the reason why * *destroy of the car* is an unacceptable VP) must follow from some other property of the grammar.

(3) The example **I am happy* [**that** *Mary to leave the room*] is also impossible, but this results from the fact that the clause-introducer *that* is limited to finite clauses in English.

(4) If the subject of a controlled infinitive is a special null pronominal called PRO, as was argued at length in literature of the 1980s, then an immediate question arises concerning its status under the Case Filter. Though *Mary* in (16b) violates the Case Filter, yielding a judgment of unacceptability, PRO in the same environment is fully acceptable: *We were happy PRO to win the prize*. One early answer exempted PRO from the Case Filter entirely. The formulation of the Case Filter in Chomsky (1980a) referred to N, rather than NP. PRO was assumed to be a bare NP, lacking an N terminal node—a distinction unavailable, of course, under the later theory of Bare Phrase Structure (Chomsky 1995b: ch. 4).

(5) The behavior of PRO in Raising constructions like (19) provides an argument (due to Chomsky and Lasnik 1993) that PRO must move for Case reasons, just like overt NPs—contradicting the widely held assumptions about PRO cited in note 4. For example, in the Raising constriction [*PRO to seem to him to have written the letter*] *would be strange*, PRO is obligatorily disjoint in reference from *him* (and replacement of *him* with *himself* allows coreference), which is expected only if *PRO* is required to raise. This observation led Chomsky and Lasnik to propose a special type of case that could only be assigned to PRO (and only by appropriate instances of infinitival T), which they called 'null case'. The self-evident ad hoc character of this proposal, along with other long-standing issues in the distribution of control, has been one of the impulses behind work that seeks to reanalyze Control constructions in a variety of ways (e.g. as involving movement, as suggested e.g. by Bowers 1973, 1981, Wehrli 1980:115–31, 1981, and Hornstein 1999; cf. also Landau 1999, 2003 and much subsequent literature for alternative views).

(6) Chomsky's proposal united many streams of contemporaneous research. Pollock (1989) had argued for the existence of a head lower than T but higher than VP to which French infinitival verbs (which do not move to T) optionally raise. Kayne (1989a) had argued for the existence of a head (once again lower than T, but higher than VP) to which French direct objects raise—for example, in the course of *wh*-movement, passive, or cliticization—

triggering object agreement. Koopman (1992) argued for a head higher than VP that assigns ACC case on the basis of data from Bambara (a Mande language of West Africa); while Holmberg (1986) argued for a similar head associated with ACC case to which particular nominals raise in Scandinavian, as part of a construction called Object Shift.

Chomsky (1991, 1993; reprinted as chs. 2–3 of Chomsky 1995b) was the first to propose that Pollock, Kayne, Koopman, and Holmberg had all independently discovered the same syntactic head, which he suggested was both the assigner of ACC and trigger for object agreement. This head was dubbed AGR_{O}. The suggestion that this head also assigns the external argument role was brought into the picture later (Chomsky 1995b: ch. 4). Hale and Keyser (1993, 2002) had suggested that a head distinct from V assigns the external argument role, which they called '*v*'. Chomsky proposed that this head too should be identified with the head whose other functions had been discovered by Pollock, Kayne, Koopman, and Holmberg (as part of an account of Burzio's generalization)—hence the *v*-hypothesis in the form presented in the text.

(7) Many of the special properties of restructuring constructions, particularly in Romance languages, were first discovered and investigated by Aissen and Perlmutter (1970) and Rizzi (1978a). See also the bibliography at http://wurmbrand.uconn.edu/Bibliographies/res-bib.html.

(8) The *v*P of an unaccusative or passive clause is transparent to agreement processes, while the *v*P of a transitive clause appears to be opaque. This led Chomsky (2001) to suggest a further correlation, with the maximal projection of case-assigning, external-argument *v* constituting a 'Strong Phase' that is spelled out once fully constructed and thus blocks processes like agreement; and the *v* of unaccusatives and passives heading a 'Weak Phase' that is not spelled out in this fashion.

In more recent work, Chomsky (2008a) has suggested (as a consequence of issues in the theory of PHASES) that V does assign ACC case after all, but 'inherits' this ability from the *v* that selects it (and that a similar process allows T to inherit its own case-assigning ability from the higher functional head C). We will not discuss this proposal further here. Wurmbrand's results probably can be achieved under these ideas as well, since a VP not directly selected by *v* will not inherit the property of ACC assignment—so it will once again have to be a higher V that performs this task.

(9) Such a nominal will also block the raising of lower phrases, an instance of what Chomsky (2000a) calls a 'defective intervention constraint'.

(10) ECM constructions require modifications of the proposal for accusative case assignment in (15), to allow not only complements, but certain other positions c-commanded by V to receive accusative. Much discussion over the past 20 years has been devoted to specifying the precise nature of these structural restrictions. The analysis of constructions like (32a) as involving case-marking across a clause boundary stands in opposition to earlier proposals concerning such constructions, in which the embedded subject moves to form a new object position in the higher VP ('Raising to Object', proposed by Rosenbaum 1967, and strongly defended by Postal 1974). Aspects of the Raising-to-Object analysis have been revived by Lasnik and Saito (1991; see also Branigan 1992, Chomsky 2008a), on the basis of the anaphoric properties of the embedded subject. We will not discuss the details of these proposals here, except to note that the topic remains an area of active discussion.

(11) In some cases, it appears that the recipient of accusative case moves to a position nearer the case assigner. As mentioned in note 10, Lasnik and Saito (1991) argued that in the ECM constructions of (32), the embedded subject moves to the specifier of its case assigner *v* (Agr_{O}, for Lasnik and Saito; see note 6). Their argument involved constructions in which Binding Theory effects showed that the embedded subject of the infinitive c-commands into adverbials that modify the higher clause. In later work, Chomsky (2008a) suggested movement only as far as the specifier of V (rather than *v*), which, he suggested, inherited the property of case assignment from *v*.

(12) These data come from Bittner and Hale (1996), who label absolutive as 'nominative', in keeping with their proposal that the two cases are underlyingly identical.

(13) As Legate discusses (2008: 63), the subject of the non-finite clause may be marked with DAT case, a case that is also found on the subjects of nominals. The direct object in (36) below, however, may not be replaced with DAT.

(14) Legate's discussion extends beyond the facts cited here, and suggests a more general typology of ergative systems. One natural question that arises in the context of our presentation is whether languages exist in which *v*

assigns quirky, rather than inherent, case. Such a language would lose the possibility of ERG as well as ABS subjects in non-finite environments. We are not sure whether this pattern is attested.

(15) Similar problems arise for PRO, if it is assumed that PRO does not need or receive case (see note 4), since PRO bears a θ-role. Chomsky (1981a) suggested some possible solutions to these difficulties. Note that the problem disappears if the proposal discussed in note 5 is adopted.

(16) There have been attempts to attribute a semantic value to the structural cases NOM and ACC, most famously by Jakobson (1984[1936]). For the most part, these efforts have been unsuccessful, since there appear to be no semantic generalizations about the meanings of NOM and ACC that have predictive power.

(17) This suggestion was advanced earlier by Williams (1994: 11) and developed in a different manner by Haeberli (2002).

(18) A noun may take a CP complement or adjunct (a relative clause), and these elements may contain T—but this instance of T is too far from N to help.

(19) P&T support this theory with an extended argument that their proposals concerning case, when combined with a new analysis of the complementizer system of English, can explain certain constraints on *wh*-movement as well as the effects of the Case Filter. We will not discuss this supporting evidence here.

(20) Building on P&T's proposals, Wiltschko (2003) has argued that Halkomelem Salish shows no Case-filter effects for nominals—precisely because, like English CPs (and inherently case-marked nominals cross-linguistically), a Halkomelem nominal does not need to value its T externally, since it is not only interpretable, but lexically valued (like the T on an English finite verb). Matthewson (2005), however, has argued against Wiltschko's analysis. It should also be noted that Tonhauser (2007) argues that what is commonly called nominal tense in languages like these should instead be viewed as an instance of aspect (which would not affect the syntactic proposals discussed in this chapter).

(21) Both solutions may be extended to quirky case by assuming that morphology that reflects licensing directly may be overwritten by quirky-case requirements.

(22) The actual proposal is somewhat more complex, since it takes into account the role of finiteness and consequences of movement (which we have omitted here).

(23) Bobaljik proposes that agreement systems target the highest syntactically accessible arguments in the hierarchy Unmarked case > Dependent case > Lexical/Oblique case, with multiple agreement a possibility, so long as the hierarchy is respected. Preminger (2009) argues, however, that apparent instances of secondary agreement in Basque are actually instances of clitic doubling—raising the possibility that multiple agreement might not exist in general, with potential consequences for the status of the proposed hierarchy (and the support it brings to dependent-case theories).

(24) The possible post-syntactic nature of case morphology is itself a topic of some discussion. Norvin Richards (2007), for example, has offered arguments from the distribution of 'case stacking' (in which multiple case affixes attach to a single element, as in the Australian language Lardil) that case morphology itself is assigned in the course of the syntactic derivation, rather than in a post-syntactic component. Rezac (2008) argues for a similar conclusion specifically in the domain of dependent-case phenomena.

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Merge and Bare Phrase Structure

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[–] Abstract and Keywords

This article examines the nature of Merge (the standard structure-building operation in virtually all minimalist studies), and issues of dominance (projection), long-distance dependencies/movements (can movement be reduced to Merge?), and the minimalist reformulation of the well-known X-bar theory in terms of bare phrase structure. It is organized as follows. Section 4.2 briefly reviews the history of modern linguistics to see how these characteristics have been captured by various different components of grammar. Section 4.3 focuses on the operation Merge, which is assumed in bare phrase structure theory to be the fundamental operation in human language, and discusses its properties and problems. It also explores a few different interpretations of Merge and related operations, and discusses some implications for comparative syntax, particularly Japanese syntax. Section 4.4 summarizes the discussion, trying to figure out the current stage of our understanding of the relevant issues, and speculates on future directions.

Keywords: minimalist program, dominance, long-distance dependencies, X-bar theory, bare phrase structure

4.1 Introduction

Since Aristotle, language has been taken as a system of associating sound—or sign, as recent research has shown—and meaning over an infinite range. One of the most important discoveries in linguistic sciences is that this association is actually not direct, but rather is mediated by 'structure' whose exact nature remains to be clarified by empirical investigations. Modern linguistics has identified certain fundamental properties of the 'structure' of human language and the system that generates it. These properties can be summarized as follows.

(1)
- **a.** hierarchical structure
- **b.** unboundedness/discrete infinity
- **c.** endocentricity/headedness
- **d.** the duality of semantics

(p. 74) There is a fair amount of consensus by now that these are the properties that ought to be captured, in one way or another, in any linguistic theory that aims to explain the nature of human language. The questions are: How much mechanism is needed to account for these properties elegantly, and is it possible to figure out what is behind these properties?

The rest of this contribution is organized as follows. In section 4.2, I briefly review the history of modern linguistics (particularly generative linguistics), to see how these characteristics have been captured by various different components of grammar. In section 4.3, I focus on the operation Merge, which is assumed in bare phrase structure theory to be the fundamental operation in human language, and discuss its properties and problems. In this section

I also explore a few different interpretations of Merge and related operations (if such operations exist), and discuss some implications for comparative syntax, particularly Japanese syntax. In the concluding section (4.4), I summarize the discussion, trying to figure out the current stage of our understanding of the relevant issues, and speculate on future directions. Throughout the discussion, I confine myself to those issues directly related to phrase structure theory, particularly bare phrase structure theory. Accordingly, I cannot pay sufficient attention to various other important problems of minimalism that may in principle be related to the issues at hand. In addition to the other chapters of this volume, the reader is referred to introductory books such as Hornstein et al. (2005) for a more comprehensive discussion on minimalist syntax at large, in which the following discussion is couched.

4.2 A Brief History[1]

Let us consider (1a) ‘hierarchical structure’ first. That linguistic expressions have abstract hierarchic structures, not merely sequences of words and formatives, is one of the fundamental discoveries of modern linguistics. This discovery goes back to pre-generative structural linguistics, particularly in the form of ‘Immediate Constituent (IC)’ analysis (Wells 1947). IC analysis is couched in the ‘procedural’ approach developed in (American) structural linguistics, and as such cannot be carried over to the theory of generative grammar, which explicitly denies the procedural approach (see e.g. the introduction to Chomsky 1975a). However, the insights of IC analysis, along with important concepts drawn from historical phonology—the concept of ‘ordered rewriting rules’ in particular—can be incorporated into the theory of phrase structure grammar. The theory of phrase structure grammar is developed on (p. 75) the basis of Post's combinatorial system (Post 1943), with an important modification regarding the notion of ‘vocabulary’ (the terminal vs. non-terminal distinction), and is a set of rules (phrase structure rules) of the following form, where A is a single symbol and X, Y and Z are strings of symbols (Z non-null; X and Y possibly null):

(2) XAY → XZY

Phrase structure rules express the basic structural facts of the language in the form of the ‘P(hrase)-markers’ they generate, with terminal strings drawn from the lexicon. P-markers generated by phrase structure rules express three kinds of information about a linguistic expression:

(3)

- **a.** the hierarchical grouping of the ‘constituents’ of the structure (Dominance);
- **b.** the ‘type’ of each constituent (Labeling);
- **c.** the left-to-right order (linear order) of the constituents (Precedence).

Thus, the specific kind of hierarchical structure of a linguistic expression (i.e. the labeled hierarchic structure), along with how the elements are stringed (linear order), is explicitly expressed by phrase structure grammar generating a set of P-markers.

In (2), X and Y need to be non-null, when the environment in which A is to be rewritten as Z needs to be specified. This situation arises when a lexical item is inserted into a particular terminal position of a P-marker. This type of ‘lexical insertion’ is abolished in favor of the lexicon with subcategorization features (Chomsky 1965). The separation of the lexicon from the computational system—phrase structure grammar (PSG)—makes it possible to simplify the form of phrase structure rules for human language from the context-sensitive rule (2) to the context-free rule (4).

(4) A → Z

In (4), A is a single non-terminal symbol, and Z is either a non-null string of non-terminal symbols or the designated symbol ‘Δ’ into which a lexical item is to be inserted in accordance with its subcategorization features (see Chomsky 1965 for details).

Thus, context-free phrase structure grammar (coupled with the lexicon) is responsible for expressing the properties of phrase structure of human language, particularly its labeled hierarchic structure with the designated left-to-right linear order. Property (1a) is thereby accounted for.

Let us skip properties (1b) and (1c) for the moment, and consider property (1d) next. This property, the duality of semantics, calls for a device other than phrase structure grammar. The duality of semantics refers to the fact (as has been noticed and studied from various points of view over the years) that generalized predicate—argument structure is realized in the neighborhood of the predicate (within the core (p. 76) part of a clause), whereas all other semantic properties, including discourse-related and scopal properties, involve an 'edge' or a 'peripheral' position of a linguistic expression (generally a sentence). This duality, particularly the latter fact, requires a device that relates two non-sister positions in the structural description of a sentence, i.e. a device that refers back to some earlier—not necessarily immediately preceding—step in the phrase structural derivation. However, reference to constituent structure (i.e. to the past history of a phrase structural derivation) cannot be neatly expressed by context-free phrase structure grammar (see e.g. Chomsky 1957). Thus, a new grammatical device has to be introduced to deal with the duality of semantics, and the notion of 'grammatical transformation' is introduced for this and related purposes.[2]

Human language clearly exhibits the property of discrete infinity (1b), taken to be the most elementary property of the shared language capacity. Language is discrete, as opposed to dense or continuous, roughly in the sense that linguistic expressions are constructed on distinct and separate units (rather than continua), so that there are n word sentences and n+1 (or n−1) word sentences, but there are no n.5 (or n.3, etc.) word sentences (just like natural numbers). And language is infinite, since there is no n (in any human language) such that n is the number of words contained in the longest sentence (so that a sentence with n+1 words is a non-sentence). Most important cases of discrete infinity exhibited by human language are handled by special types of transformations —'generalized transformations'—in the early theory of transformational-generative grammar. These transformations are equipped with the function of embedding a structure (typically a sentence) into another structure of the same type (a sentence). With the abolishment of generalized transformations in the Standard Theory in the 1960s (Chomsky 1965), this function of (self-)embedding is transferred to phrase structure grammar with 'recursive symbols' that appear both in the left-hand and right-hand sides of the phrase structure rules, allowing a kind of non-local recursion.

Toward the end of the 196os, it became apparent that certain important generalizations about the phrase structure of human language, i.e. endocentricity/headedness (1c), cannot be stated in terms of phrase structure rules alone (nor in terms of transformations for that matter). Phrase structure in human language is generally 'endocentric', in the sense that it is constructed based on a certain central element—called the 'head' of a phrase—which determines the essential properties of the phrase, accompanied by other non-central elements, thus forming a larger structure.[3] This is the right intuition, but, as pointed out by Lyons (1968), the theory (p. 77) of phrase structure grammar cannot capture this. Phrase structure rules are too permissive as a theory of phrase structure in human language, in that they over-generate phrase structures that are never actually permitted in human language, i.e. those structures that are not headed ('exocentric' structures). We thus need some other mechanism which correctly captures the endocentricity/headedness of phrase structure that appears to be a fundamental property of human language. X-bar theory is introduced mainly for this purpose.

The basic claims of X-bar theory, as it was introduced in Chomsky (1970), can be summarized as follows.

(5)

a. Every phrase is headed, i.e. has an endocentric structure, with the head X 'projecting' to larger phrases.
b. Heads are not atomic elements; rather, they are feature complexes, consisting of primitive features.
c. Universal Grammar (UG) provides the general X-bar schema of the following sort, which governs the mode of projection of a head:
$X' \rightarrow X \ldots$
$X'' \rightarrow [\text{Spec}, X'] \; X'$

The version of X-bar theory presented in Chomsky (1970) was in a preliminary form, and numerous refinements have been made since then. However, it is also true that all the crucial and fundamental insights of X-bar theory were presented in this study, and have been subjected to few substantive changes since. While claim (5c), the existence of universal X-bar schema, has been subjected to critical scrutiny in recent years, claims (5a) and (5b) have survived almost in their original forms throughout the ensuing development of grammatical theory, and are still assumed in the current framework (but see below for a recent proposal to eliminate the notion of projection). In this way, property (1c), the headedness of phrases in human language, is explicitly captured by X-bar theory.

As the Principles and Parameters approach (P&P) took shape around 1980, indicating the emergence of an explanatory theory of a radically different sort from the earlier traditional rule-based systems, 'rules of grammar' virtually disappeared, replaced by the principles of various modules of UG (Case theory, X-bar theory, etc.). Accordingly, phrase structure rules disappear, for a substantial core system, which is in fact a rather natural move, since phrase structure rules are redundant to a significant extent, recapitulating information that must be stated in the lexicon. On the other hand, transformational rules are not redundant, and thus are ineliminable, although the exact form in which they should be expressed is open to question. It seems that complex transformational rules—which are specific to constructions in particular languages—need not be stipulated, and that over a large range, transformations can be reduced to the simple general schema Move-α (which says 'Move anything anywhere'), given trace theory and the other principles of UG.

Thus, within the P&P framework, we are essentially left with X-bar theory (of some sort; see e.g. Chomsky 1986a) and Move-α. X-bar theory is responsible for (p. 78) properties (1a) (hierarchic structure), (1b) (discrete infinity)[4], and (1c) (endocentricity/headedness), while Move-α is mainly responsible for property (1d) (the duality of semantics). During the mid to late 1980s, attempts were made to integrate the theory of phrase structure (X-bar theory) and the theory of movement (Move-α) by proposing that phrase structures are built 'from the bottom up' by means of a formal operation very similar to adjunction (or substitution, depending on the structural property of the target[5]) employed in a transformational operation (Move-α). The basic claims of one such attempt, which are relevant for our present discussion, can be summarized as follows.[6]

(6)

a. Heads project as they 'discharge' their features (selectional features, agreement features, etc.).
b. Iteration is possible, particularly at the single-bar level.
c. One and the same operation ('Adjunction') is responsible for both structure building and movement.
d. There is no X-bar schema. (Thus, the notion of 'maximal projection' cannot be defined in terms of bar-levels, and ought to be characterized contextually.)
e. Agreement closes projections.

Claim (6a) is based on the intuition that phrases are constructed around their heads, and that heads are driven to project in terms of their inherent features. (6b) is to account for the infinitude of phrase structure composition (but see note 4 above). The simplification of transformations makes it possible to search for a fundamental operation (or a small number of fundamental operations), and (6c) is one possible answer to this important question. Given the idea expressed in (6a), the X-bar schema in the traditional X-bar theory seems eliminable. This is in fact a desirable result, not just because of Occam's razor, but because of the highly specific, rather stipulative nature of the postulated X-bar schema. If the effects of X-bar schema can be shown to derive from more natural, simpler principles, that would be a highly desirable result. The last claim, (6e), is based on the observation that in some type of languages (e.g. Japanese), phrases seem to be never 'closed', in the sense that given a phrase, it is always possible (in syntax) to expand that phrase by combining it with some other element, as long as the combination is licensed by being assigned an appropriate interpretation. This is of course not the case in, say, English. And a hypothesis is put forth that the 'closure' property of phrases seems to be linked (p. 79) to the presence of agreement. Thus, in a language like Japanese, where there is no conventional ϕ-feature agreement, phrases are never closed, while in English-type languages, agreement closes projections of phrases, from which a variety of other differences also follow. See Fukui (1986, 1995). See also Kuroda (1988) for a different approach to the 'agreement parameter'.

It is perhaps worth pointing out here some fundamental differences between Kuroda's and Fukui's approaches to phrase structure and comparative syntax (in particular, the 'agreement parameter' just mentioned). Kuroda's theory, as it is stated in Kuroda (1988), is 'geometric' in nature. This orientation seems to derive from the spirit of modern geometry as laid out in Felix Klein's well-known Erlanger Programm (1872), whose basic idea is that each geometry (Euclidean, affine, projective, etc.), given a certain space, can be characterized by a 'group of transformations,' and that a geometry is really concerned with invariants under this group of transformations (rather than a space itself). Simply put, Kuroda attempts to establish the geometry of human language. For Kuroda, then, the space is (universally) given by the X-bar schema in this case. English and Japanese are exactly the same in this regard (apart from linear order, of course). A group of transformations, particularly an abstract relation 'Agreement,' is also universal. English and Japanese are minimally different with respect tot he enforcement of this operation: Agreement is forced in English, but it is unforced in Japanese. While the very nature of this 'parametric' statement remains unclear, it is clear that Kuroda's approach is (i) geometric (and thus, in part, tends to be 'representational' in the familiar linguistic terminology) and (ii) universalist (in that both the space and the group of transformations—Agreement, in particular—are taken to be universal). Fukui's approach, on the other hand, is couched in a more conventional 'economy/last resort' paradigm (as we just saw above). It attempts to eliminate the X-bar schema, claiming that structures are built from the bottom up, starting out with a head and continuing on as long as syntactic objects are licensed (by feature-checking, for example). There is no superfluous structural position, and there is no superfluous operation. Since Japanese lacks the relevant Agreement-inducing heads (e.g. ?-features) in its lexicon, Agreement simply doesn't occur in the language, although even in Fukui's system, the operations are universally available as they are provided by UG. Thus, despite the widely held view that Kuroda's and Fukui's approaches exhibit many similarities (and they actually do share many insights), their background philosophies are very different, yielding quite a few important empirical differences. As it turns out, Fukui's approach went along with subsequent developments of linguistic theory (leading to bare phrase structure theory, for instance), but Kuroda's geometric approach offers interesting future research topics and is not to be discounted, in my opinion.

Returning to our main discussion, the total and explicit elimination of X-bar theory is carried out by Chomsky's (1995a) 'bare phase structure' theory (see also Kayne 1994 for a different approach). The bare theory is couched within the (p. 80) minimalist program, according to which all the principles and entities of UG must be motivated and justified either by the properties of (at least) two interfaces that language has with the other performance systems—sensorimotor and conceptual-intentional systems—or by 'third factor' conditions (see Chomsky 2005) that are not specific to language and that govern the way language satisfies the conditions imposed by the interfaces. Most of the basic claims of the approach mentioned above are actually incorporated into the bare theory in a greatly refined form, and the fundamental operation is identified as Merge, with respect to which various interesting theoretical and empirical problems arise, as discussed in the following section.

4.3 Merge and Related Issues

4.3.1 Merge

Chomsky (1995a) argues that standard X-bar theory specifies much redundant information, while the only structural information needed is that a 'head' and a 'non-head' combine to create a new unit. He then proposes that a phrase structure is constructed in a bottom-up fashion by a uniform operation called 'Merge,' which combines two[7] elements, say α and β, and projects one of them as the head. Since Merge combines (not 'concatenates') two elements and does not specify the linear order between the two, the resulting object can be more accurately represented as in (7).

(7) K = $\{\gamma, \{\alpha, \beta\}\}$, where $\gamma \in \{\alpha, \beta\}$

(7) states that Merge forms a new syntactic object K by combining two objects α and β, and specifies one of them (γ) as the projecting element (hence the 'head'/'label' of K). Merge applies iteratively to form indefinitely many structures. Note incidentally that in this formulation, the operation Merge—Merge $(\alpha, \beta) = \{\gamma, \{\alpha, \beta\}\}$, where $\gamma \in \{\alpha, \beta\}$—is an *asymmetric* operation, projecting either α or β, the head of the syntactic object that 'projects' becoming the head/label of the complex object formed by Merge.

How are the basic properties of human language listed in (1) captured by the bare theory, particularly by the simple operation Merge? Property (1a), hierarchic (p. 81) structure, can be straightforwardly captured by assuming iterative Merge to be non-associative (i.e., [A # B] # C ≠ A # [B # C], where # denotes Merge).[8] It requires some discussion to see how (and whether) this simple elementary operation captures the other properties listed in (1), to which we now turn.

4.3.2 Labeling

Notice that in the formulation of Merge (7) above, headedness (labeling) is a part of the operation, thereby rendering Merge asymmetric, as we just saw. However, the status of labeling under minimalist assumptions does not seem obvious. It is not entirely clear whether labels are indeed required by either of the two—sensorimotor and conceptual-intentional—interfaces, nor is it likely that labels as such are derived from some third-factor considerations. To sharpen the fundamental operation of Merge, therefore, it is desirable to dissociate the labeling part from Merge itself, and see if labels are predictable by general principles. Under this scenario, Merge is formulated simply as a *symmetric* set-formation operation: Merge $(\alpha, \beta) = \{\alpha, \beta\}$.

How are labels determined, then? Since Chomsky (1995a), there have been various attempts to determine the label optimally (i.e. without reference to idiosyncrasies, without stipulating a special rule, without look-ahead to check eventual convergence of a derivation, etc.). One line of such attempts converges on the elimination of labeling. Collins (2002a) examines various areas of syntax (X-bar theory, selection, Minimal Link Condition, etc.) in which the notion of labels has been used, and proposes that the notion of labels can be totally eliminated from the theory of syntax if we assume the mechanism of 'saturation/discharge'—and the derived notion 'locus'—similar to the one employed in an approach discussed above (see (6a); see also the references cited in note 6). In a series of works (e.g. Chomsky 1995a, b, 2000a, 2001, 2007, 2008a), Chomsky explores various possibilities of predicting labels in major cases, and reaches the conclusion that the notion 'labels' may be illusory with no theoretical status, and that it can be eliminated entirely (along with Collins's notion of locus), by virtue of the third-factors such as minimal search conditions. Thus, in a structure of the form H-XP (where H is a(n) L(exical)I(tem), order irrelevant), a representative case of merger, minimal search conditions trivially determine that H is the element that enters into further computations, since the other element, XP, has no information available with its head too deep down within it. Headedness is simply an epiphenomenon under this approach.

Another line of research takes labeling/headedness to be a fundamental property of human language, and proposes that this property be explicitly expressed by the (p. 82) theory of grammar (see Boeckx 2008a, Hornstein 2009, among many others, and the references cited). For example, Fukui (2005) proposes that in addition to Merge, which is, as Chomsky formulates it, a simple set-formation operation, we need another operation called 'Embed,' which is responsible for expressing the 'local self-embedding' character of linguistic structure. Operating on the workspace—called the Base Set (BS)—created by an application of Merge (i.e., BS = $\{\alpha_1, \ldots, \alpha_n\}$, which is yielded by Merge $(\alpha_1, \ldots, \alpha_n) = \{\alpha_1, \ldots, \alpha_n\}$), Embed picks out one member of the BS, say α_i, and forms a union of α_i and the BS.

(8) Embed $(\alpha_i, \text{BS}) = \alpha_i \cup \text{BS} = \{\alpha_i, \{\alpha_1, \ldots, \alpha_n\}\}$[9]

The α_i, the target of the operation Embed, is the 'head' of the resulting structure. As discussed above, the number n is usually 2 in human language. So, BS is of the form $\{\alpha, \beta\}$, and applying Embed to, say, α, we get the self-embedding structure K = $\{\alpha, \{\alpha, \beta\}\}$, where α is the label/head. By adding Embed to the structure-building component of grammar (the Merge/Embed system), properties (1a) (hierarchical structure) and (1c) (endocentricity/headedness) are naturally accounted for.

Fukui (2005, 2008) argues that more refined analyses become available in many areas of grammatical investigation within the Merge/Embed system than within the standard Merge-alone system. In fact, these works factor out three relevant factors to look at in examining the properties of phrase structure, (i) Merge, (ii) Embed, and (iii) iterativity. By combining these, we have (at least) the following possibilities.[10]

(9)

- **a.** non-iterative Merge without Embed
- **b.** iterative Merge without Embed
- **c.** non-iterative Embed
- **d.** iterative Embed

The last possibility (with all three factors in action) represents unbounded endocentric structures found in the core of human language, as traditional X-bar theory indicates. The other three possibilities have rarely been considered, but there may be cases in which they are attested. Thus, the data on Japanese Broca's aphasics indicate the following basic properties (see Fukui 2004 and references cited there).

(10)

- **a.** Word order is retained in general.
- **b.** Case and sentence-final particles such as *-ga* 'Nom', *-o* 'Acc', *-ka* 'Q' are generally dropped.
- (p. 83) **c.** Postpositions (e.g. *-kara* 'from', *-made* 'until') are generally retained.
- **d.** The maximal number of constituents in a noun phrase is two.[11] That is, only one *-no* 'Gen' is allowed within a single noun phrase, e.g. *ore-no kyoodai* 'my brother,' *Sapporo-no sigai* 'the town of Sapporo',

but * *Taro-no imooto-no yoofuku* ‘Taro's sister's clothes’, **a-no uti-no mon-no mae* ‘in front of the gate of that house’.

Property (10a) could indicate the importance of the ‘head-parameter’ (linear order) in the core part of language, but let's set this aside for the moment. Properties (10b), (10c), and (10d) all point to the conclusion that only sister elements can be related in Japanese Broca's aphasics. While the exact mechanism of case-marking in Japanese is still largely unknown, it is relatively clear that particles such as *-ga* and *-o* (as well as the question particle *-ka*) require ‘non-sister information’ for their licensing, i.e. it is necessary to look at portions of a phrase marker that are not in a sister relation to these elements. Postpositions, on the other hand, require only ‘sister information’ for their licensing, as they generally theta-mark their sisters. (10d) indicates that X and Y are linked within a noun phrase and *-no* can be attached to X (i.e. [X-*no* Y]), but one more application of Merge/Embed is impossible (i.e. *[Z-*no* [X-*no* Y]]). All of these properties can receive a natural account if we assume that iterativity is lost in Japanese aphasics.[12] Note incidentally that the traditional ‘working memory’ account cannot explain why the crucial line is actually drawn where it is (rather than between, say, three and four, etc.).

Note incidentally that the complement vs. non-complement distinction, which plays a major role in various modules of grammar, including the theory of movement, is grounded in a kind of third-factor considerations. Merge can be taken to provide a recursive definition of syntactic objects. Thus, its first application (first-Merge), which—apart from the case of singleton-formation (see note 18)—yields a head-complement configuration, directly corresponds to the base of a recursive definition, while subsequent applications of Merge, which introduce non-complements (specifiers), correspond to the recursion part of a recursive definition. The complement vs. non-complement distinction therefore, is based on the independently motivated base vs. recursion distinction in the characterization of recursive definitions. The concept of recursive definition is certainly not specific to language (UG); thus, to the extent that the complement vs. non-complement distinction is reduced to the nature of recursive definition, the former distinction is justified on a kind of third-factor grounds. This reasoning holds under either interpretation of Merge we're discussing, but the distinction between first-Merge and subsequent applications of Merge is highlighted under the approach taking iterativity as a separate property.

(p. 84) As this brief discussion indicates, it is generally the case that by considering the three factors separately, it becomes possible to address the questions of their respective roles in evolution, acquisition (development), and, as we just saw, (partial) loss of the human language capacity (see also Fujita 2007 for relevant discussion). Thus, it seems that the Merge/Embed system—with the treatment of iterativity as a separate property—offers a certain descriptive promise, although the addition of a new operation like Embed, thereby enriching UG, may run counter to general minimalist guidelines. The plausibility of this type of approach seems to depend in part on the ‘naturalness’ of the additional operation (Embed or whatever), as well as the empirical significance of labels/headedness in the syntax and at the interfaces. The matter is still contested in the literature.

A recent proposal by Narita (2009b, 2010) presents a rather radical view on the notion of labels and related concepts such as projection and feature percolation. He is in agreement with the first type of approach discussed here, i.e. an approach under which the role of label/headedness is being diminished to almost zero. He goes on to take a stronger view that when α and β are merged, forming a new object K, no feature of α/β is transmitted to K. That is, there is no projection, let alone feature percolation. Narita then assumes the following claim made by Chomsky (2008a: 139).

> For an LI to be able to enter into a computation, merging with some SO [Syntactic Object], it must have some property permitting this operation. A property of an LI is called a *feature*, so an LI has a feature that permits it to be merged. Call this feature the *edge feature* (EF) of the LI. If an LI lacks EF, it can only be a full expression in itself: an interjection. When merged with a syntactic object SO, LI forms {LI, SO}; SO is its *complement*. The fact that Merge iterates without limit is a property at least of LIs—and optimally, only of LIs, as I will assume. EF articulates the fact that Merge is unbounded, that language is a recursive infinite system of a particular kind. (emphasis original)

Adopting the concept of EF discussed in this quote, Narita assumes that EF triggers Merge, and that only an LI can utilize its EF to do so. Notice that EF is a feature of an LI, and if there is no projection/feature-percolation, no features of an LI, including its EF, can be transmitted to a larger structure created by merging the LI and a syntactic object SO. Thus, an LI, once merged, can no longer trigger further application of Merge, which is clearly against '[t]he fact that Merge iterates without limit' (see the above quotation). To solve this problem, Narita assumes a multiple transfer model (e.g. Chomsky 2000a, 2008a), according to which syntax interfaces with the performance systems multiple times at well-defined steps of a derivation (called 'phases'). Each application of Transfer strips off the complement of a designated LI, a phase head, thereby rendering the phase head a kind of 'revived' LI. Thus, only phase heads allow more than one application of Merge, and only phase heads can 'move,' with the apparent effect of pied-piping in disguise. In this way, the system proposed by Narita (2009b, 2010) permits a specific type (p. 85) of unbounded Merge. He goes on to explore an impressive array of consequences of this proposal, including a new analysis of freezing effects (Boeckx 2008), an account of an asymmetry in coordinate structures, and so on. The next subsection attempts to reinterpret Narita's proposal by recasting the notion of EF.

4.3.3 The edge feature

Whether an application of Merge is triggered or spontaneous has been a matter of debate. One line of inquiry sees Merge as a free-standing option made available by UG that can be utilized any time in a derivation, and the result is acceptable or unacceptable, depending on whether the merged element is properly licensed (at the interfaces). On this view (see Chomsky 1995a, b), Merge is a cost-free operation, not 'triggered' by anything, and is not subject to Last Resort. There is empirical evidence that this may in fact be true (cf. Saito and Fukui 1998), and thus this possibility is not to be discounted (see also section 4.3.4).

Another view on Merge is that its application is on a par with the other operations in a grammar, in that it should be triggered by some kind of feature-checking (selection and other requirements). This view is motivated by the general (minimalist) guideline according to which there is no superfluous step in a derivation and every operation has the 'reason' for it to apply. The notion of EF seems to have developed from these lines of inquiry as an abstract concept covering various cases that have been treated separately.

The EF of an LI is taken to be a 'feature' (along with the other features) of the LI, and it indicates that the LI can be merged. As a feature, however, EF exhibits some peculiar properties. First, it is a feature associated with virtually every LI, apart from interjections or frozen expressions. Features in the lexicon are generally for the purposes of distinguishing different classes of lexical items. Thus, it is not entirely clear why such a feature should exist at all, if it is universally associated with every lexical item. In fact, EF seems to be equivalent to being a lexical item, at least a part of its definition. Second, EF does not play any direct interpretive role at the interfaces. Thus, assuming some sort of full interpretation to hold at the interfaces (not a trivial assumption, though), EF is surely an 'uninterpretable' feature that is not allowed to exist (survive) at the interfaces. So it must be deleted, but it cannot be deleted when Merge applies, because if EF is always deleted when satisfied (by an application of Merge), then there will never be second-Merge, third-Merge, etc. (i.e. specifiers/non-complements) per given LI. Empirical evidence indicates that specifiers/non-complements do exist. In particular, only this choice permits Internal Merge. When merging X to Y (the asymmetric phrasing here is only for expository purposes), X can be either external to Y or part of Y. In the former case, we have External Merge (responsible for predicate-argument structures), whereas the latter possibility represents Internal Merge, which comes free and which is assumed (p. 86) in the bare theory to be an equivalent to Move/transformations.[13] And Internal Merge is responsible for one of the fundamental properties of human language, i.e. property (1d), 'the duality of semantics'. Thus, if the expressive potential of human language is to be used in full, EF must be undeletable, and undeletability of EF provides the basis for the duality of semantics via Internal Merge. Moreover, unlike other uninterpretable features which get deleted when checked under certain structural conditions, deletion of EF is presumably carried out as part of the operations of Transfer (see Chomsky 2007: 11).[14] In all of these respects, EF is a unique 'feature', quite distinct from all other features generally assumed to be associated with lexical items—which casts serious doubt about EF as a lexical feature. A natural conclusion, then, seems to be that EF is not a lexical feature *per se*, but rather a term describing the general properties of Merge (as it relates to lexical items).

If Merge applies freely without any drive (cf. the first view just discussed), then there is no need for EF anyway. But if an application of Merge indeed requires a driving force, we need to state the conditions, replacing EF, under which Merge applies. It is important to note in this connection that in the overwhelming number of cases of Merge applications, Merge always operates on a head (H or LI) and another syntactic object α, i.e. H-α (order irrelevant).[15] That is, merger almost always occurs between a head and some other syntactic object, and if we look at the actual cases, it is again almost always the case that α is a non-head (typically a phrase). Thus, the situation is such that when Merge applies to two syntactic objects, one of them must be a head, and the other non-head, i.e. there is a certain '(structural) asymmetry' (in some sense to be made precise) between the two objects. This suggests the following generalization concerning an application of Merge.[16]

(11) Merge is driven by asymmetry.

The intuition behind (11) is that syntax is unwilling to tolerate asymmetries, and tries to fix one by applying Merge. Merge in this scenario has the important function of maximizing symmetry. The asymmetry that exists between H (a head) and α (a non-head) is covered and eliminated by an application of Merge yielding a set {H, α}, a symmetric object. Human language has an intrinsic source of asymmetries, namely, the lexicon—lexical items with their selectional properties. As these LIs enter into computation, they will necessarily create asymmetries, triggering Merge.

(p. 87) If a condition like (11) is sustainable, and if Merge applies only when it is driven (this is not obvious; see the discussion above), then there is no merger between two 'symmetric' objects, such as head-head and XP-YP. In fact, this is what the preceding discussion predicts. But there are well-known 'exceptions'. One notable exception is an external merger of an external argument, which has all sorts of exceptional properties, as Chomsky (2007, 2008a) discusses. As briefly discussed above, Narita (2009b, 2010) tries to deal with this case by means of a phase-by-phase multiple Transfer mechanism. The phase head *υ* in the complex {*υ*, {V, Obj}} can eliminate its complement by Transfer ({*υ*, ~~{V, Obj}~~}, so that the revived phase head *υ*, an LI, can utilize its EF again, triggering an application of Merge, which merges an external argument. We are assuming EF is not a feature of a lexical item, so the same problem does not arise. But the insight of Narita's analysis, i.e. that merger of XP and YP is possible only when one of them constitutes a phase (whose head LI can Transfer its complement and thus become a revived head), can be stated as follows.[17]

(12) Transfer creates asymmetry.

Transfer strips off the complement portion of a phase, and by doing so, it makes the whole structure asymmetric and unstable, which triggers an application of Merge (External or Internal). Thus, the external merger of an external argument is properly motivated by the independently established fact that *υ*P is a phase (and *υ* is a phase head). This approach also accounts for the fact that movement (Internal Merge) only occurs at the phase level (basically at *υ*P and CP, in informal terms), since only a phase head can trigger Transfer, which in turn creates asymmetry, thereby driving an application of (Internal) Merge.

As for the head-head merger case, particularly the first step of a derivation, when two LIs are to be merged, condition (11) does not force Merge to apply. Here we might adopt Kayne's (2008) idea that one option for Merge is the direct formation of the singleton set {x}, and that the singleton formation applies only to nouns (see Kayne 2008). Thus, when N and V are to be merged, Merge (singleton formation) first applies to N[18], forming {N}, and this creates an asymmetry (V vs. {N}) that prompts an application of Merge, in accordance with the condition (11), yielding a set {V, {N}} as desired.

In this way, we might be able to account for the conditions under which Merge should apply, without recourse to EF as an actual feature associated with a lexical item.

Our discussion is rather sketchy and inconclusive, and to make the proposal more substantive, it is certainly necessary to sharpen the notion of 'asymmetry' invoked (p. 88) in condition (11). This is hard to do, however. The reason is that there is no general agreement as to what information Merge can obtain when it is to apply to syntactic objects X and Y. It is widely claimed that LIs are 'atoms' for computations. If they are indeed atoms, Merge cannot obtain any information stored inside LIs' bundles of features, including, I suppose, EF, if it is really a feature of an LI. Then how does Merge know X or Y is an LI (with EF)? Bar-levels and other diacritics are excluded in the bare theory as violations of inclusiveness. So they are not available. The only entity that seems to be available to Merge is the 'braces', i.e. the information regarding the layers of sets formed by (prior) applications of Merge (i.e. the concept of nth order). Thus, x and {x} should be distinguishable, if braces are indeed ontological entities (not an innocuous assumption) and Merge is able to 'see' them (ditto). In this way, Merge should be able to detect the existence of asymmetry (or lack thereof), to see whether it should apply. Beyond this, it is hard to determine how much information (of syntactic objects) is accessible to Merge. This casts some doubt on any system in which Merge is 'triggered' by some feature—including EF, if it is indeed a feature of a lexical item. The proposal suggested above is to barely get around this fundamental problem of feature-triggered Merge approaches. Note also that throughout the discussion, it remains open whether Merge can apply when not driven by asymmetry. It may be that Merge is always available and is free to apply without trigger, but it is forced to apply when driven by a condition like (11). As briefly mentioned at the outset of this subsection, this possibility is not to be discounted.

As an alternative view, one might propose that Merge freely and blindly applies to X and Y without caring about what they are, what features they have, etc., and that some other mechanism (e.g. the labeling algorithm of Chomsky 2008a: 145) deals with various properties of the structures constructed by Merge. Although the same 'atom' problems arise for the mechanism at hand (how it distinguishes LI and non-LI, etc.), Merge itself could maintain the simplest possible form under this approach.

In the absence of crucial evidence at the moment, I leave all these possibilities open for future research, and turn to a brief discussion of Japanese syntax in the following subsection, as it appears to show some basic properties of Merge (and licensing mechanisms) in a straightforward way.

4.3.4 Merge and Japanese syntax

It is by now widely known that expressions in Japanese—clauses in particular, but noun phrases show a very similar pattern—exhibit a certain iterativity property at the edge that is not generally observed in languages like English.[19] Thus, given the (p. 89) sentence (13), which looks like a complete sentence, it is possible to keep expanding the sentence indefinitely by adding an expression (a noun phrase with a nominative marker) at the edge, as illustrated in (14).

(13)

Dare-mo	(sono	gakkai-ni)	ko-nakat-ta.
anybody-MO(even)	that	conference	come-not-Past
Lit. 'Anybody did not come (to that conference).'			
'Nobody came (to that conference).'			

(14)

a. Daigakuinsei-ga [dare-mo (sono gakkai-ni) konakatta].
graduate students-Nom
Lit. 'Graduate students, anybody did not come (to that conference).'
'As for the graduate students, none of them came (to that conference).'
b. Seisuuron-ga [daigakuinsei-ga [dare-mo (sono gakkai-ni) konakatta]].
number theory-Nom
Lit. 'Number theory, graduate students, anybody did not come (to that conference).'
'As for number theory, none of the graduate students (in that field) came (to that conference).'
c. Suugakuka-ga [seisuuron-ga [daigakuinsei-ga [dare-mo (sono gakkai-ni) Mathematics department-Nom
konakatta]]].
Lit. 'Mathematics department, number theory, graduate students, anybody did not come (to that conference).'
'As for the mathematics department, in the area of number theory, none of the graduate students (in that field) came (to that conference).'
d. Harvard-ga [suugakuka-ga [seisuuron-ga [daigakuinsei-ga [dare-mo (sono gakkai-ni) konakatta.
Lit. 'Harvard, mathematics department, number theory, graduate students, anybody did not come (to that conference).'
'As for Harvard, none of the graduate students in the mathematics department in the area of number theory came (to that conference).'
And so on.

While the actual acceptability of a sentence depends on how easily one can think of an appropriate relation holding between the added nominative phrase and the rest of the sentence, it is safe to claim that the syntax of Japanese allows indefinite expansions of a sentence at the edge, and submits the resultant structure to interpretation.[20] More generally, given a sentence (S) and a phrasal category (XP), typically a noun phrase or a postpositional phrase—these are of course informal terms only for expository purposes—it is always possible in Japanese to combine them, yielding the structure {XP, S}. It is then necessary to submit the structure to (p. 90) interpretation, to see if an appropriate interpretive relation holds between XP and S. This ‘appropriate interpretive relation’ includes all sorts of semantic/discourse (or even pragmatic) relations such as topic—comment, focus—presupposition, part—whole, predication, ‘aboutness’, which seems to indicate that the situation is such that, as suggested by Wolfram Hinzen, Juan Uriagereka, and others (see Hinzen 2006, Chomsky 2007, and references therein), that syntax carves the path interpretation must blindly follow (cf. Uriagereka 2002: 64), i.e. the output of syntax (the Japanese syntax, in this case) prompts the semantic/pragmatic interface to try its best to come up with an appropriate interpretation. If such a relation holds, then the structure is appropriately interpreted. If not, the resulting expression is judged to be unacceptable or unnatural. Abstracting away from all the important issues concerning case, cartography (which I think provides us with valuable data, calling for minimalist explanations), and others, I maintain that this is basically all that happens at the edge of a sentence (and a noun phrase) in Japanese. In other words, ‘unbounded Merge’ is in full force in the syntax of Japanese.

This basic property can be taken to be responsible for a variety of phenomena, apart from the obvious ‘topic prominence’ of Japanese, listed in random order below.

(15)

a. multiple *ga/no* (see above)
b. scrambling
c. gapless topic constructions
d. gapless relative clauses
e. indirect passives
f. multiple-headed constructions (multiple-headed relative clauses and clefts)

These phenomena have been noted widely in the literature, and have been listed as peculiar properties characterizing the Japanese language—as opposed to, say, English, where these constructions are generally impossible. But they all conform to the pattern just discussed, i.e. they all fall under the schematic structure {XP, S}, where the appropriate interpretation is required to obtain between XP and S. The syntax of Japanese permits the structure quite freely, and sends it off to interpretation. Speakers' acceptability judgments may naturally vary depending on the availability of an appropriate interpretive relation between XP and S, but crucially, the structure does not violate the laws of form (syntax) and thus is not ungrammatical (even though it maybe unnatural or nonsensical).[21]

Interestingly, the iterative mechanism sometimes comes into play, circumventing allegedly deviant island violations. Thus, it has been long noted that subjacency effects with scrambling or relativization in Japanese are rather ‘weak’, in that some of the examples supposedly violating the condition are actually quite acceptable. For example, (16a) and (16b) are acceptable, even though they apparently involve (p. 91) subjacency violations. (The underscored portion indicates a position where the noun phrase supposedly receives a theta-role.)

(16)

a.

Sono-haiyuu-ni	[$_S$ boku-wa	ima	[$_{NP}$[$_S$ sensyuu	Tokyo-de __ atta] hito]-o	sagasite-iru tokoro da].
that-actor-Dat	I-Top	now	last week	-in met person-Acc	be-looking for
Lit. ‘That actor, I'm now in the process of looking for the person who met in Tokyo last week.’					

b.

[$_S$[$_{NP}$[$_S$ __ kawaigatte ita]	inu]-ga	kyonen sindesimatta]	otokonoko
took loving care of	dog-Nom	last year died	boy
Lit. ‘the boy who the dog he was taking good care of died last year’			

Example (16a), involving scrambling from within a relative clause, is not perfect,[22] but as compared to the English counterpart (which is hopeless), it can be arguably regarded as grammatical. And it is quite easy to construct similar examples for the other types of island violations—the subject condition, the adjunct condition, the non-relative complex NP cases, etc. (16b) (adapted from Kuno 1973: 239) is a case of one relative clause embedded in another relative clause, and is plainly grammatical in Japanese. In both cases, the structure in which an XP is merged with an S is created by Merge at the top of the structure, and as long as an appropriate interpretive relation can be established between the XP and the S, the XP is locally licensed where it is merged, effectively nullifying the movement (if any) which violates the subjacency condition.

Thus, unbounded Merge not only provides the basis for the constructions in (15), but also offers a kind of repair strategy for certain island violations, as illustrated by the examples in (16). Note that there is no need to stipulate anything to account for these ‘peculiar’ facts in Japanese. They are direct consequences of unbounded Merge along with necessary interface (interpretive) licensing. Japanese syntax, on this view, exhibits the nature of UG in its purest form. What needs to be accounted for are the cases where such unbounded Merge appears to be prohibited, e.g. the lack of those constructions (15) in, say, English.[23]

(p. 92) In brief, the discussion in this subsection indicates that the syntax of Japanese seems to exhibit how the bare minimum of UG—Merge and interface requirements—works in its barest form. A class of properties of Japanese, which has been treated as peculiar properties of the language, can be reduced simply to unbounded Merge and interpretive licensing conditions at the interface. Properties of Japanese seem to support the view that Merge applies rather freely without any drive, unless a reasonable concept of ‘(a)symmetry’ may come up so that structures in this language can be appropriately characterized as inherently asymmetric (i.e. ‘open’).

4.3.5 Summary

Let us summarize our discussion on Merge and related issues. Merge is a simple set-formation operation. Applying to two SOs, α and β, it forms a set $\{\alpha, \beta\}$. This symmetric view on Merge leaves unaccounted for the notion of labels/headedness, the core insight of traditional X-bar theory as we saw in section 4.2. We discussed this problem in section 4.3.2. One approach to this problem simply states that labeling is not part of the structure-building operation; rather, it is an epiphenomenon and to the extent that it holds, it's predictable by general principles of minimal search. Another approach takes labeling to be an essential part of the structure-building processes of human language, and claims that there is an additional operation of self-embedding (Embed) that directly accounts for headedness/labeling. Furthermore, it is argued that by factoring out combination (Merge), self-embedding (Embed), and their iterativity (recursiveness), more refined analyses become available, which has implications for the issues of the evolution, development, and (partial) loss of the human language faculty. A radical variant of the first approach claims that there is no notion of ‘projection’ in syntax. Merge is assumed to be triggered by EF, but since EF (a feature of an LI) cannot project on this view, Merge cannot apply more than once per LI. This problem is resolved by assuming that Transfer, which applies only at the phase level, has a side effect of reviving the head as an LI. Thus, only phase heads permit specifiers/non-complements, only phase heads can ‘move’, etc.

The notion of EF is critically examined in section 4.3.3 It is pointed out that EF is a unique feature, distinct from all the other ‘conventional’ lexical features. It plays no direct interpretive role at the interfaces, so it is definitely an ‘uninterpretable’ feature. However, unlike other uninterpretable features, it doesn't seem to be deleted when satisfied. All of these properties cast some doubt on the treatment of EF on a par with other lexical features. It is thus suggested that EF is a term describing the general conditions under which Merge applies as it relates to an LI. There are two views on the applicability of Merge. One is that Merge applies freely and blindly, without any need for trigger. Under this view, there is basically no need for EF. (p. 93) The other view is that Merge is driven by some factor such as EF. Based on the observation that Merge almost always operates on a head and another non-head syntactic object, it is suggested that a certain kind of 'structural asymmetry' that holds between the two syntactic objects is actually the driving force for Merge. Basically, Merge applies to reduce asymmetries, maximizing symmetry in syntax. The special role of Transfer, mentioned above, with respect to applications of Merge is attributed to the fact that Transfer creates asymmetry, thereby making it possible for Merge to apply more than once per LI.

In section 4.3.4, certain well-known characteristics of Japanese syntax (cf. (15) and (16)) are discussed in light of general properties of Merge. These characteristics have been noted as peculiar properties of Japanese in the literature. In the present context, however, they can be analyzed as direct consequences of unbounded Merge, backed up with interface mechanisms that attempt to assign appropriate interpretations to the structures generated by syntax. Thus, they are no longer peculiar properties of Japanese, but rather, the situation in which free applications of Merge appear to be prohibited calls for an explanation.

4.4 Concluding Remarks

After going over the historical background briefly (section 4.2), I focused on the core of the bare phrase structure theory, the operation Merge, in section 4.3, to see how the fundamental properties of human language listed at the outset of this contribution can be naturally accounted for by this simple operation. As our discussion in the preceding section indicates, the current situation seems to be as follows. Property (1a), the existence of hierarchical structure, is straightforwardly handled by Merge, if we assume the operation to be iterative and non-associative. The unboundedness/discrete infinity (property (1b)) is expressed by EF, although there is an issue as to whether EF (or structural asymmetry of some sort) triggers Merge or Merge freely applies without any trigger. Either way, unbounded Merge directly accounts for the unboundedness/discrete infinity exhibited by human language (but see note 4 above). The status of property (1c), endocentricity/headedness (labeling), which is the core insight of traditional X-bar theory, is actually rather controversial. It is either a non-property of human language *per se*, determined by a third-factor principle such as minimal search, or it is indeed an essential property of the human language faculty that needs to be captured by UG, by means of an additional operation such as Embed (or its equivalent). Finally, property (1d), the duality of semantics, which is one of the basic motivations for transformations in earlier models of grammar, is also captured by Merge, Internal Merge in this case, as we saw above (but see note 13 above for a potential problem).

(p. 94) In this way, the bare phrase structure theory, with the elementary operation Merge, successfully accounts for all the fundamental properties listed in (1) at the outset of our discussion. This is a remarkable and certainly welcome result, given minimalist goals. On the other hand, this situation may give rise to a fear that there is little left for future research, as far as the core phrase structure theory is concerned. Notice that iterative Merge is the simplest possible mode of recursive generation, and thus, apart from a few interesting issues (e.g. multiple dominance, the role of linear order, labeling), the story is pretty much finished once we have reached this operation. Does this mean there are no 'deep' results to be drawn from the study of phrase structure theory in the future? I don't think so.

I maintain that one promising area of study in this connection is the mathematical study of strong generative capacities of grammars, as they relate to questions of empirical significance (with respect to explanatory adequacy and beyond). Despite its importance for theoretical linguistics, the study of strong generative capacity has been, after the classical study by Chomsky and Schützenberger (1963), set aside for a long time. Kuroda's (1976) topological study of strong generation of phrase structure languages, although it has not been widely recognized, takes a significant step forward, unveiling important mathematical problems embedded in this domain of inquiry. Kuroda introduces a class of topological spaces associated with finite sets of phrases of a phrase-structure (tree) language, and then defines the notion of 'continuous function' from one tree language to another as a continuous function with respect to these topological spaces. Then, grammars that generate tree languages can be classified nicely into structurally homeomorphic types. He argues that the topological method introduced in this fashion provides a better and more appropriate means, both mathematically and linguistically, than the notion of strong equivalence in the traditional sense for investigating the structural similarity of languages and grammars.

Recent discoveries by Kuroda (2008) further confirm, in a rather unexpected way, the conviction that the mathematical studies of strong generation offer a rich area of inquiry with significant empirical import. Kuroda notes some similarities between phrase-structure languages and ζ functions. ζ functions, first explicitly noticed by Euler in the eighteenth century and later developed by a number of mathematicians (Riemann, Dirichlet, Hecke, to name just a few), are a 'major' mathematical object in the sense that they show up at virtually every corner of mathematical structures—particularly in number theory[24], but sometimes even in elementary particle theory in physics—with profound mathematical substance. Kuroda invents a formal procedure for transforming the Euler product representations of certain ζ functions into phrase structure representations (in the extended form); by arithmetizing such phrase structure languages, the values of ζ functions can be calculated, (p. 95) leading to the situation where one can say that a ζ function has a representation as an accumulative sum of a phrase structure language. It is demonstrated in Kuroda (2008) that this procedure holds for Ramanujan's second-degree ζ functions (it trivially holds for first-degree ζ functions such as Riemann's and Dirichlet's), and it can be conjectured (though not explicitly considered in Kuroda 2008) that the procedure can be naturally extended to the cases of higher-degree ζ functions, possibly even including congruence ζ functions.[25] Kuroda's procedure is stated in terms of concatenative systems such as context-free phrase structure grammars, but his results ought to be readily translatable into the Merge-based generative system.

Mathematical studies of phrase structure as represented by these works have not yet attracted much attention in theoretical linguistics. But if the Galilean dictum that 'Nature's great book is written in mathematical language' indeed holds, and if the human language capacity is part of the natural world, as has been assumed throughout the development of biolinguistic inquiries, then the linkage between language and mathematics might actually turn out to be much more substantial than it first appears to be, giving real substance to the claim (see Chomsky 1980b) that the language faculty and the 'number faculty' maybe rooted in the same origin.

Notes:

I have given talks and lectures at various places discussing part of the material presented here, and I thank the audiences on these occasions for their comments and questions. I also would like to thank Cedric Boeckx (the editor of this volume), Teresa Griffith, Terje Lohndal, Hiroki Narita, and Mihoko Zushi for their valuable input.

(1) The discussion in this section is overly simplified, as it serves just to set the stage for the discussion in the following section. See Fukui 2001, for a more comprehensive historical survey.

(2) The original reasons for the need for grammatical transformations are not strictly restricted to the duality of semantics. See Chomsky (1956, 1957) for details. Also, the concept of transformations has important precursors in pre-generative structural linguistics, particularly in Zellig Harris's work (Harris 1957). See Introduction to Chomsky (1975a) for relevant discussion.

(3) The observation goes back to the pre-generative era. See Harris 1951. The endocentric-exocentric distinction is even older (see, for example, Bloomfield 1933), though the intended meaning of the distinction is somewhat different.

(4) By allowing iteration at one or more levels of projection. It is not entirely clear how the 'global' recursion of the type expressed by the use of recursive symbols in phrase structure grammar (see the discussion above) is to be captured by X-bar theory, or in any subsequent theory for that matter.

(5) Care must be taken to interpret the adjunction/substitution dichotomy, as it has carried different contents over the years, depending on the theory of transformations at the time. See Fukui 2001, notes 11 and 12, for relevant discussion.

(6) Fukui 1986, Speas 1986, 1990, Fukui and Speas 1986, among others. The exposition here is based largely on Fukui 1986 and naturally benefits from hindsight.

(7) In general, *n* number of objects. In human language, however, the number seems to be restricted to 2, yielding only binary structures. See Chomsky 2004a: 115 for the relevant discussion regarding why *n* should be 2 under the probe-goal system. The possibility of allowing *n*-ary structures ('nonconfigurational' structures) may still be open, if/when the probe-goal mechanism is not in operation. See Chomsky 2004b: 167–8 for discussion.

(8) As argued in Fukui and Zushi 2003, 2004: 12.

(9) Note the similarity between Embed and the von Neumann version of the successor function $n^+ = n \cup \{n\}$. Merge, on the other hand, is very similar to Zermelo's formulation of the successor function $n^+ = \{n\}$. Although these two versions are generally assumed to be equivalent in regard to the construction of natural numbers, it is not clear whether two (mathematically) 'equivalent' operations always represent and perform the 'identical' operations, in human cognition, in this case.

(10) Note that Embed presupposes Merge.

(11) There is of course no such a restriction in non-aphasic Japanese, so the cited examples are all grammatical. See also the discussion below.

(12) Fukui 2004 puts forth the hypothesis that Embed is also lost in Japanese Broca's aphasics. But the crucial evidence is hard to come by, and the available evidence does not seem compelling.

(13) It is not entirely clear, however, exactly how two (or more) non-sister copies in the case of Internal Merge can be appropriately linked to each other in a way that can be carried out within the bounds of Merge, which is a very local operation.

(14) Problems may arise for Narita's analysis if Transfer (of a complement) always deletes the EF of a phase head.

(15) See Chomsky 2008a: 145. Narita 2009b, 2010 calls this the 'H-α schema.'

(16) Condition (11) shares a spirit similar to (but not quite the same as) the one in the antisymmetric approach initiated by Kayne 1994, or to the proposal by Moro 2000, both mainly for motivating movement.

(17) See also Boeckx 2009d

(18) What motivates this application of Merge remains a problem. The apparent lack of 'complements' in Japanese noun phrases—as opposed to verb phrases in the same language—discussed in Fukui 1986 may constitute empirical evidence for the singleton-formation applying to nouns (but not to, say, verbs).

(19) The relevant literature is too numerous to mention. The reader is particularly referred to the references mentioned in note 6 above. See also papers in such collections/handbooks as Fukui 2003 and Miyagawa and Saito 2008, and sources cited therein.

(20) As mentioned, this conclusion largely carries over to noun phrases.

(21) See Hoji 2009 for an extensive study of methodology to sort out various factors involved in linguistic judgments.

(22) This is actually the hardest kind of scrambling out of an island for salvaging, because of the associated particle *-ni*, taken by the verb *atta* 'met'. If the scrambled phrase is marked by *-o*, interpretive licensing will be much easier, thanks to the hidden possibility of taking the fronted phrase as a reduced form of *-no koto-o* Lit. '*-no*-formal noun-Acc', i.e. as a kind of 'major object' akin to a topic.

(23) Given the vital role of agreement in English-type languages vis-à-vis the apparent lack of such processes in Japanese, there have been attempts to deal with the problem by stipulating that agreement somehow 'closes' the projection (cf. (6e)). While such a stipulative account cannot count as a true explanation in the current minimalist setting, its main insights, i.e. that additional requirements of English (including agreement) render the constructions impossible, should somehow be incorporated into the ultimate explanation.

(24) Thus, according to Goro Shimura, a prominent number theorist, 'Seisuuron, itarutokoro ζ kansuu ari' (Lit. 'Number theory, ζ functions are everywhere'), i.e. 'In number theory, ζ functions show up everywhere' (quoted by Kato 2009: 69).

(25) See Fukui 1998: 209 for a brief remark about how (portions of) the 'Weil Conjectures' (Weil 1949) might be related to human language structure.

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Structure and Order: Asymmetric Merge

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[–] Abstract and Keywords

This article discusses the relation between structure and linear order in the minimalist approach to syntactic theory. The general idea of Kayne that linearization is a function of structural asymmetry among syntactic nodes can be maintained in the bare phrase structure theory of Chomsky, if we take the history of the derivation into account. On its simplest definition, Merge is the same at each step of the derivation, i.e., first Merge should have no special properties. This is achieved if Merge is taken to be an operation transferring one element at a time from a resource to a workspace (the object under construction). Simplifying even more, and adopting a top-down derivational approach, we can take structure to result from an operation that splits off one element at a time from the resource ('Split Merge') until the resource is empty. Either way, sister pairs are not sets but ordered pairs, and the set of elements merged/split off is a nest, which is equivalent to an ordered n-tuple. This allows us to consider structure-to-order conversion as a trivial equivalence relation (where material between slashes is ordered in time, i.e. realized one after the other in sound or gesture).

Keywords: linear order, syntax, grammar, syntactic theory, trivial equivalence relation

5.1 Introduction

In this chapter I understand '(linear) order' (said of linguistic elements α, β) in terms of temporal organization, such that α precedes β (is ordered before β) if and only if the time at which α is realized (in sound or gesture) precedes the time at which β is realized. Order in this sense is traditionally considered to be the domain of syntax (cf. Ries 1927: 141), but in linguistic minimalism, order is not established in narrow syntax but at the interface component dealing with sound (cf. Chomsky 1995b: 334–5).

This assumes a model of grammar where syntax in the narrow sense ('narrow syntax') is a computational system that takes elements from a lexicon (hereafter also called 'resource' or 'numeration') and merges them, creating a structure to be delivered for interpretation at interface components dealing with sound (PF) and meaning (LF). In this model, linear order comes in only 'after syntax', i.e. as a modality-specific realization of a structure that is ordered hierarchically, but not linearly.

(p. 97) Syntax in the minimalist program, then, retreats to its core business of defining the way elements combine to create larger units (cf. Ries 1927: 142), referring many traditional aspects of the theory of syntax (including also inflectional morphology) to more peripheral components. Nevertheless, the question of how order relates to structure has been a formative element of minimalist syntactic theory since Kayne (1992, 1994), in that it prompted the articulation of the 'bare phrase structure' theory (cf. Chomsky 1995b: 249) replacing traditional X-bar theory.

5.2 The Linear Correspondence Axiom

Traditional X-bar theory (Chomsky 1970, Jackendoff 1977) specifies a universal format for the structure of phrases, distinguishing heads, complements, specifiers, and adjuncts as occupying well-defined structural positions in the phrase. The order of elements in the phrase is a function, partly of structure (in that complements appear closer to the head than specifiers and adjuncts) and partly of language-specific properties of the head, taking a complement to its right or left (the 'head parameter' or 'directionality parameter', yielding head-initial and head-final syntax, respectively).

In the bare phrase structure theory, structure is a function of the merger operation combining elements from the lexicon (hence 'Merge'), which creates sets. Merge is recursive, in that the set created by Merge is extended by each next operation Merge, yielding the familiar hierarchical phrase structure organization. We return to the details of Merge below, but the point to be made here is that the operation is autonomous, i.e. not bounded by requirements posed by a theory of phrase structure like X-bar theory.

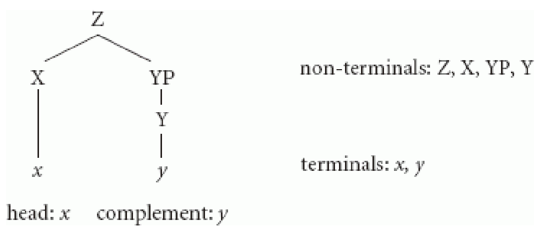
Since the notion 'head' is not given up in the bare phrase structure theory (see Chomsky 1995b: 245), it remains possible to describe linear order in terms of the setting of a head parameter. In bare phrase structure theory, the head determines the category of the output of Merge, and it may be stipulated that it precedes or follows the non-head (cf. Saito and Fukui 1998: 452). But Kayne (1994) proposes that linear order is an automatic reflection of structural organization, through his Linear Correspondence Axiom, leaving no room for a directionality parameter (p. 47).

The Linear Correspondence Axiom (LCA) was formulated with the traditional X-bar theory in mind, and proposes that the linear order of the terminals of a phrase structure Π reflects the asymmetric command relations among non-terminals of Π (the terminals are the actual linguistic items, and the non-terminals nodes in the phrase structure dominating the terminals). Command is the familiar notion of c-command, where α c-commands, β iff, β is (dominated by) the sister of α (and α is **(p. 98)** the sister of β iff α and β are merged together, and α dominates β iff α is the outcome of an operation merge involving [γ dominating] β). In order for the structure-to-order mapping envisioned in the LCA to be possible, non-terminals must be in asymmetric c-command relations, which in turn leads to proposals about structure to that effect (i.e. ensuring asymmetry of c-command relations among nonterminals) which are in part incompatible with the bare phrase structure approach.

The relevance of the LCA for the viability of a directionality parameter is the following. We observe within a language that linear order is antisymmetric, i.e. given two elements *x* and *y* we (generally) don't find both orders *xy* and *yx* (antisymmetric ordering of *x* and *y* means that *xy* and *yx* do not both occur). Kayne (1992) observes that this antisymmetry applies across languages, in that certain phenomena involving directionality (such as movement) do not co-vary with a supposed directionality parameter. For example, movements to the left in a head-initial language like English (*wh*-movement, subject—auxiliary inversion, etc.) are not mirrored as movements to the right in a head-final language like Japanese. Therefore, if the moved element is *x* and its trace is *y*, we do not find both *xy* and *yx* in this domain across languages. Antisymmetry, then, is a characteristic of 'universal grammar' (the language faculty as realized in all languages). If this can be generalized, i.e. if languages show no mirror effects at all, a directionality parameter cannot be part of universal grammar.

Kayne (1994) explicitly argues against a directionality parameter on theoretical and empirical grounds. Theoretically, he derives from the LCA that the specifier and the complement must be on opposite sides of a head (p. 35). For linguistic items *x, y* functioning as a head and its complement to be ordered, the non-terminals *X* and *Y* associated with *x* and *y* must not c-command each other. Let *x* be the head; then *y* cannot be a head, since the complement of a head must be a phrase (Kayne 1994: 8). If *y* is not complex, its non-terminals must include both a head-non-terminal and a phrase-non-terminal. The structure of the head—complement configuration, then, is as in (1):

(1) Head—complement configuration (Kayne 1994: 10)



In (1), an asymmetric c-command relation exists between nonterminals *X* and *Y*, so that the terminals can be ordered as *xy*, but if *YP* were not present, *X* and *Y* would c-command each other, and no ordering of the terminals *x* and *y* would be (p. 99) possible. It is easy to see that a specifier *Q* (dominating terminal *q*) merged to *Z* in (1) would asymmetrically c-command both *X* and *Y*, so that the order of the terminals becomes *qxy*. Hence, the specifier and complement must be on opposite sides of the head, leaving no room for a directionality parameter regulating the head—complement order.

Empirically, the observations lead us to reject the only alternative ordering of the terminals of a specifier—head—complement structure, namely the inverse order *yxq*. This is because the specifier is predominantly on a left branch in both head-initial and head-final languages, witness the distribution of typical specifier occupants such as subjects (preceding predicates) and displaced *wh*-elements (fronted, i.e. moved to the left) across languages.

The combined theoretical and empirical observations lead Kayne to conclude that the linear order of phrases is universally specifier—head—complement. Deviations from the universal order must be the result of movement (where movement occurs when a term of Π is merged with Π), and languages differ not in a directionality parameter setting, but in the amount (and perhaps the type) of movement.

In assessing the LCA, it is important to separate the empirical observations from the theoretical proposal. The empirical observations (essentially of a typological nature) raise important questions relating to the significant *absence* of otherwise expected phenomena (e.g. why there is no verb-second-to-last, Kayne 1992). These questions find a useful answer in a sweeping generalization like the LCA. But perhaps equally important is the observation that few (if any) languages are altogether free from disharmonic word order phenomena. This requires that we define domains where deviations from the unmarked structure-to-order correspondence might originate (see section 5.6).

5.3 Bare phrase structure theory

In the bare phrase structure theory, structure is a function of Merge, i.e. no nodes exist that are not the product of an operation merging two items. Projection levels (head, phrase) are contextually defined, and not given in advance by rewrite rules. It follows that a non-branching complement (such as *y* in (1)) is simultaneously a head and a phrase: it is a head because it does not branch, and it is a phrase because it acts as the complement to another head. Furthermore, the bare phrase structure theory does not distinguish lexical items from nodes in the structure: the items *are* the elements merged, and so the items themselves constitute the structure.

These properties of the bare phrase structure theory are incompatible with Kayne's proposal to root the LCA (i.e. the structure—order correspondence) in (p. 100) asymmetric command relations. *YP* and *Y* in (1) are collapsed in a single node, yielding a symmetric sister pair of *X* and *YP/Y*. While Chomsky (1995b: 340) accepts the main empirical conclusions of Kayne (1994), including the universal head-complement order and the idea that deviations from the universal head—complement order are caused by movement, he proposes to rethink the role of the LCA in the theory of grammar.

In the bare phrase structure theory, the LCA no longer blocks the generation of symmetric structures—it's just not clear how to convert a symmetric structure to an ordered string (a sequence of sounds). If the LCA is a principle of the phonological component, the problem posed by symmetric structures disappears if the phonological component may ignore one of the elements of the symmetric structure. Chomsky therefore proposes that a non-branching complement has to move (cliticize, incorporate) before the structure is turned over to the interfaces (Chomsky 1995b: 337). A trace of a moved category is ignored (or deleted) at the sound interface, obviating any ordering requirements (see Moro 2000 for extensive discussion of this proposal and its consequences).

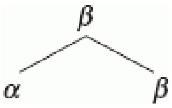
This leads to the strong prediction that any right-branching structure ends in a trace. This raises at least two questions: (1) Is it true that every right-branching structure ends in a trace? and (2) If so, what triggers the movements creating these traces? Note that the movement trigger must be independent of the need to create a structure that is interpretable at the sound interface, as the movement takes place in narrow syntax and must be oblivious of interface requirements such as the LCA (*pace* Moro 2000: 28–9). These questions have not been vigorously pursued in the literature, as far as I am aware.

If structure is a function of Merge, as in the bare phrase structure theory, it becomes possible (and perhaps necessary) to think of structure not in terms of tree configurations, but in terms of sets. Chomsky (1995b: 243) therefore describes the output of Merge of α and β, as the set K = $\{\alpha, \beta\}$. K may be merged again yielding another set containing K. Every phrase, then, is a recursively defined set of sets.

It is easy to see that when Merge yields sets, asymmetric c-command relations can be expressed in terms of set membership properties. Thus, a specifier γ merged with K = $\{\alpha, \beta\}$ yielding L = $\{\gamma\ \{\alpha, \beta\}\}$ and α and β are not elements of the same set, as α and β are elements of K, a co-member of γ in L. Corresponding to this γ c-commands α and β, but not vice versa. However, ordering by set membership yields no result among sisters, i.e. does not derive head—complement linear order (and there are other problems, having to do with the fact that set membership is not a transitive relation).

A separate question is posed by adjuncts. In traditional X′-theory, adjuncts are merged to maximal projections via Chomsky-adjunction (i.e. the node dominating the adjunct after adjunction is identical to the node to which the adjunct is merged; see Chomsky 1986a: 6). As a result, the sister of the adjunct involves two segments, with only the higher segment including the adjunct:

(p. 101) **(2)** Adjunction structure



One interpretation of the configuration in (2) would be to state that α is neither included in nor excluded by β (cf. Chomsky 1986a: 7, 9). If so, linear ordering of adjuncts cannot be a function of the set membership created by merge.

Replacing the familiar tree structure notation of phrase structure with set notation requires a rethinking of the notion projection (i.e. the determination of the features of a whole based on features of its parts). Chomsky (1995b: 244) views K, the set resulting from merger of α and β, as a slightly more complex object $\{\alpha, \{\alpha, \beta\}\}$, where α is the head of K and projects. α, then, is the *label* of K. For adjunction, Chomsky (1995b: 248) proposes that the label reflect the two-segment character of the object construed, and he suggests using to that end the ordered pair $\langle\alpha, \alpha\rangle$ as the label instead of just α. The set notation of (2) would then become $\{\langle\alpha, \alpha\rangle, \{\alpha, \beta\}\}$.

Notice that the label α of the output of Merge $\{\alpha, \beta\}$ is not itself ordered before or after the elements α and β at the sound interface PF (likewise with the adjunction label $\langle\alpha, \alpha\rangle$). This suggests that the label is a mere notational device, needed to express an inherent asymmetry among elements merged (see Collins 2002a). This asymmetry (that one element is the head and the other is not) allows us to think of K as an ordered pair $\langle\alpha, \beta\rangle$, on the understanding that any dissimilarity between α and β in property P renders α and β ordered with respect to P. In this connection, Langendoen (2003: 310) notes that $\{\alpha, \{\alpha, \beta\}\}$ is the set-theoretic definition of the ordered pair $\langle\alpha, \beta\rangle$ (more exactly, the set-theoretic definition of $\langle\alpha, \beta\rangle$ is $\{\{\alpha\}, \{\alpha, \beta\}\}$, cf. Kuratowski 1921: 171).

If the output of Merge is an ordered pair by definition, all that is needed for the structure-to-order conversion is a correspondence rule that says (where material between slashes is ordered in time):

(3) Structure-to-order conversion
$\langle\alpha, \beta\rangle = /\alpha\ \beta/$

On this view, the head—complement distinction in itself is sufficient to bring on the asymmetry required for ordering at the interface. This raises the question (not further addressed here) whether the device of the label (essentially, the property of *projection*) is necessary and sufficient to turn the output of Merge into an ordered pair.

I take (3) to be the ‘silver bullet’ of structure-to-order conversion. If it can be derived that Merge itself yields an ordered pair rather than an unordered set, linear order follows almost trivially.

(p. 102) 5.4 Timing and nesting

If the derivation of a syntactic structure D involves a sequence of steps, then stages of D $d_1, \ldots, d_n$ maybe distinguished, and elements of D maybe differentiated as to their existence in each stage. Assuming a bottom-up derivation, elements merged to D at stage d_i are not part of D at stage d_{i-1}. At each step of the derivation, then, an asymmetry exists between the two sisters being merged, in that one of the two sisters is already part of a derivation to which the other is newly merged (Jaspers 1998: 109).

It follows that the output of Merge is inherently asymmetric, except with first Merge (assuming binary merge, i.e. involving exactly two elements). For most of the derivation, then, we may conclude that Merge yields an ordered pair rather than an unordered set.

What about first Merge? Let us assume that a derivation involves a single resource (lexicon, numeration) and a single target, the object under construction. If so, first Merge is special in that it involves the selection (from a resource) of *two* elements. For each next step, it suffices to select a single element from the resource, the other element involved in the merger being the derivation under construction itself. Now if the target can be held constant (i.e. it is the unique object under construction), then Merge can be simplified as in (4) (Zwart 2004):

(4) Unary Merge
Merge selects a single element from a resource and includes it in the object under construction.

Before first Merge, the object under construction is empty. First Merge, then, simply includes an element from the resource in an empty workspace. At the next step, the workspace is no longer empty, and 'including an element in the object under construction' implies merger with that object.

Fortuny (2008: 18f.) demonstrates that this sequence of steps can be described in set-theoretic terms, where Merge is an operation of set formation taking two sets *A* and *B* and producing the union of *A* and *B*. For Merge to be successive, one of *A* and *B* (say, *B*) must be the output of the immediately preceding operation Merge. At first Merge, where there was no preceding operation, this output is zero, so that in that case *B* is the empty set ϕ. Since the empty set is not phonetically realized, Fortuny derives the result of Chomsky (1995b: 337) and Moro (2000) that one of the elements involved in first Merge must be empty at the sound interface PF.

It now follows that the asymmetry among the elements merged brought on by the derivational history (i.e. of two elements merged, one was already part of the derivation and the other is newly merged) applies to all stages of the derivation, including first Merge. This allows us to think of Merge as yielding an ordered pair, i.e. as being inherently asymmetric. If so, (3) may be taken to hold.

(p. 103) In formulating the structure-to-order rule (3), I assumed that Merge yields an ordered pair automatically, i.e. as a function of the difference between the existing element (the derivation under construction) and the newly merged element. Fortuny (2008) shows that linear order can be derived from the history of the derivation, even if Merge yields a set rather than an ordered pair.

Fortuny (2008: 19) takes a derivation which successively merges the members of are source S = $\{\alpha, \beta, \gamma, \delta, \varepsilon\}$ to yield a derivational record K = $\{\{\alpha\}, \{\alpha, \beta\}, \{\alpha, \beta, \gamma\}, \{\alpha, \beta, \gamma, \delta\}, \{\alpha, \beta, \gamma, \delta, \varepsilon\}\}$, which is the set of sets of elements merged at each stage of the derivation (for notational clarity, the empty set is left out in the member sets of K). K is a *nest* (i.e. every set in K is a subset or superset of another set in K), which is shown by Kuratowski (1921: 164) to provide a linear ordering of the members of S, i.e. the ordered n-tuple $\langle\alpha, \beta, \gamma, \delta, \varepsilon\rangle$. The order can be seen as a function of the number of sets in which elements are included, α being included in all sets of K, β in all but one, etc.

Note that the linear order of the example derivation would be /$\varepsilon\ \delta\ \gamma\ \beta\ \alpha$/, assuming bottom-up derivation and continuing to assume that specifiers are ordered before their associated heads and complements (i.e. if α and β form a head—complement combination, then γ is a specifier or a higher head, and must precede α and based β, on Kayne's observations). On Fortuny's derivation of the structure-to-order conversion, (3) reads as (3′):

(3′) $\langle\alpha, \beta\rangle$ = /$\beta\ \alpha$/

Notice that this timing/nesting approach to structure-to-order conversion has nothing to say on the question of the order of the (most deeply embedded) head and complement: it is not immediately clear whether the first element merged should be a head or its complement. However, since complements are typically transparent (allowing sub-extraction), they cannot be 'lexical' in the sense of section 5.6.1, i.e. they cannot be construed in a separate derivation, and hence they cannot be merged as single items, as specifiers/adjuncts must be (see also Toyoshima 1997). The derivation, then, must start with the construction of the complement, as Fortuny (2008: 20) also assumes.

The general approach discussed here makes no special provisions for adjuncts, and tacitly assumes that adjunction has no special status as far as syntactic structure is concerned (i.e. all Merge = adjunction).

5.5 Order without Merge

The approaches to structure-to-order conversion discussed above share the assumption that Merge is a process transfering elements from a resource to a (p. 104) workspace (the structure under construction). Order can be derived from the circumstance that this transfer process involves a sequence of steps, yielding an ever-increasing structure. This approach to structure-building is questioned in Bobaljik (1995b), who instead proposes that the derivation merely forges relations among the members of the resource. On this approach, the elements of 'transfer' and 'structure building' are just metaphors. (Consequently, Bobaljik makes no distinction between a 'resource' and a 'workspace', using the term 'workspace' for the numeration, which is now not depleted in the course of the derivation, but instead grows.)

Ignoring functional elements, the example *The man hit a ball* might be analysed in Bobaljik's system as involving an initial workspace (5), which is expanded to (7) after the steps in (6):

(5) {the}
{man}
{hit}
{a}
{ball}
(6)
1. relate {the} and {man}
2. relate {a} and {ball}
3. relate {hit} and {{a}, {ball}}
4. relate {{the}, {man}} and {{hit}, {{a}, {ball}}}

(7) {the}
{man}
{hit}
{a}
{ball}
{{the}, {man}}
{{a}, {ball}}
{{hit}, {{a}, {ball}}}
{{{the}, {man}}, {{hit}, {{a}, {ball}}}}

The family of sets in (7) is not a nest, suggesting that the timing/nesting approach tothestructure-to-orderconversionislostinasystemwithoutatransferprocess taking elements from the resource to the structure under construction. However, the timing/nesting approach can be restored if Bobaljik's system is sufficiently sharpened.

I believe that the system proposed in Bobaljik (1995b) is not sufficiently restrictive, in that any (original or created) element may enter into a relation with any other element. Bobaljik (1995b: 56) capitalizes on this property of the system to derive what corresponds to inter-arboreal operations: this occurs when an element *x* previously merged with *y* (yielding *A*) then enters into a second relation with *z* (yielding *B*), after which *B* may enter a relation with *A* (or an element including *A*). Bobaljik (1995b) argues that such operations are needed to derive a movement operation, such as head movement, where the moved element does not merge with the root node of the structure (i.e. it violates the Extension Condition of Chomsky 1993: 22); in Bobaljik's analysis, a verb moving out of VP to T takes a sidestep to (p. 105) merge with T in a separate tree structure, yielding a complex head which is itself merged with the VP, yielding TP.

The system as a whole seems too unrestricted however. What is missing is a sense of direction in the derivation, and that is what precludes a straightforward structure-to-order conversion.

Two other remarks on Bobaljik's system are relevant here. First, the system does not involve as a first step 'merger with nothing/the empty set'. As a result, no ordering between *a* and *ball* results. Second, *the* and *man* in (7) occur in three subsets, as does the verb *hit*, suggesting that *the, man*, and *hit* have equal depth, which would not allow a straightforward transfer from structure to order. But that problem is remedied if *the man* can be construed in a separate derivation and is included in the workspace (5) as a single item (see section 5.6.1). In what follows, I assume that this is the correct approach to complex specifier elements: they are construed in separate derivations and included as a single item in the numeration that feeds the derivation in which they are to appear as specifiers.

What I would like to show here is that the intuitive appeal of Bobaljik's system, namely that the derivation merely establishes relations among elements in a resource, can be maintained without losing Fortuny's result that the structure-to-order conversion is a straightforward rule of the type in (3)/(3′), interpreting an ordered n-tuple as a sequence of elements ordered in time.

The proposal discussed here, developed in Zwart (2009a), takes 'Merge' to be a process that splits the resource into a pair consisting of one item from the resource and the resource's residue ('Split Merge'; see also Fukui and Takano 1998, where a similar operation, 'Demerge', is part of the linearization process). The syntactic position and the grammatical function of the element split off from the resource are contextually defined, as a function of the relation with its sister, the residue of the resource. The residue itself becomes a dependent of the element split off. The derivation proceeds by split-merging the residue of the resource, splitting off one element with each step, until the resource is empty.

On this approach, starting from a resource S = $\{\alpha, \beta, \gamma, \delta, \varepsilon\}$, and splitting off α first, β next, etc., the derivation proceeds as in (8):

(8) Split Merge

STEP	SPLIT	RESOURCE
1.		$\{\alpha, \beta, \gamma, \delta, \varepsilon\}$
2.	α	$\{\beta, \gamma, \delta, \varepsilon\}$
3.	β	$\{\gamma, \delta, \varepsilon\}$
4.	γ	$\{\delta, \varepsilon\}$
5.	δ	$\{\varepsilon\}$
6.	ε	ϕ

The derivational record K can now be defined as a the set of sets of elements split off from the resource at each step, i.e. K = $\{\{\alpha\}, \{\alpha, \beta\}, \{\alpha, \beta, \gamma\}, \{\alpha, \beta, \gamma\}, \{\alpha, \beta, \gamma, \delta,\} \{\alpha, \beta, \gamma, \delta, \varepsilon\}\}$, (p. 106) which is a nest yielding the ordered n-tuple $\langle\alpha, \beta, \gamma, \delta, \varepsilon\rangle$. The structure-to-order conversion then follows straightforwardly from rule (3).

The Split Merge system shares with Bobaljik's system the characteristic that it does not need to involve 'merger with nothing' as a first step, here a straightforward result of the top-down orientation of the derivation. Likewise, it involves no transfer from a resource to a workspace, reducing the importance of the concept of movement significantly (i.e. movement, 'Internal Merge' is now an additional mechanism, no longer modeled on the basic structure-building operation '(External) Merge'; see Zwart 2009a for discussion).

Note that in the Split Merge system, the only constituents are, at each step, (a) the elements split off from the resource, and (b) the state of the resource (the combination of these two elements was defined as a constituent at the preceding step in the derivation). It follows that complex elements split off from the resource (such as specifiers, adjuncts) must be included in the resource as single items, i.e. must be the output of a separate derivation. Therefore, the derivation must be layered, as I believe is inevitable in a restrictive system.

In the minimalist literature, top-down derivations have been explored several times, most notably by Phillips (2003), and, building on Phillips's work, Richards (1999) and Chesi (2007). These proposals involve transfer from the resource to the structure, and the key concept is that Merge expands structure to the right. A difference with the Split Merge approach is that the right branch at each stage of the derivation is a linguistic item (rather than an unordered set), which is then replaced by a newly created branching structure at the next step. Space considerations prevent me from discussing this line of research in more detail.

5.6 Deviations

The pursuit of a regular and automatic structure-to-order conversion was motivated empirically by the word-order asymmetries noted by Kayne (1992, 1994), and theoretically by the desire to keep narrow syntax free from ordering considerations. I have suggested that the output of Merge may be an ordered pair rather than a set, and that the output of the derivation as a whole may be defined as an ordered n-tuple (Fortuny 2008, Zwart 2009a). If so, a straightforward structure-to-order conversion of the type in (3) may be maintained.

However, if structure is created uniformly across languages, and something like (3) applies, how come the surface syntax of languages is riddled with deviations from the expected linear order pattern? It is to this question that we now turn, expecting to make some minor inroads at best.

(p. 107) 5.6.1 Lexical and morphological

To begin with, a distinction must be made between head-final orders that may or may not be brought about by movement. The question of deviating word order is acute only for word orders that cannot be (or are unlikely to be) brought about by movement. This is because movement (a sub-case of Merge) of *x*, a complement of *y*, merges *x* with a phrase dominating *y*, establishing a new hierarchical relation between *x* and *y*. This is what allowed Kayne (1994) to maintain that word-order variation need not be a function of a directionality parameter. So the question of deviating linear order has to abstract away from the effects of movement, and needs to address construction types where a movement analysis is impossible or unmotivated.

(In a Split Merge approach, the status of movement is unclear, but we may assume as a starting point that the moved category is split off first, and that its 'base position' is filled by a contextually interpreted empty category; see Zwart 2009a for some discussion.)

More telling, then, is the observation that even strict head-initial languages like English show some amount of head-finality, for instance in the formation of compounds. English does not generally reorder heads and complements via movement, so a syntactic explanation of the complement—head order is not obvious (even if technically possible).

In the minimalist program, linear order is established at the sound interface, and it needs to be established to what extent deviations from the automatic structure-to-order conversion can be understood in terms of processes particular to the sound interface, i.e. processes that are not syntactic but rather 'morphological' or 'lexical'.

I take 'morphology' to be the inventory of forms expressing the properties of syntactic objects. Syntactic objects are created by merge, but at the sound interface they must be realized in forms which are stored and may have idiosyncratic properties (as is obvious from the example of inflectional paradigms). Syntactic features are instrumental in selecting the most suitable candidate from the set of stored forms (cf. Halle and Marantz 1993: 121–2). On this view, complex forms that we mostly consider to be morphological, such as compounds, are construed in syntax via Merge, but realized at the sound interface after an exchange with the morphological component (see Ackema and Neeleman 2004 for a general discussion of the syntactic nature of derivational morphology).

I take a form to be 'lexical' if it is enlisted in a numeration as a single item. That is, an item is lexical only in the context of a single numeration—derivation pair. Lexical integrity is the property of items of a numeration N that their parts are not manipulated in the course of the derivation built on N. Crucially, 'lexical' is opposed to 'syntactic' only in the context of a single (sub)derivation. Thus, a compound C may be created in derivation D_1, realized at the sound interface at the conclusion of D_1, and listed as a single item in the numeration for the next (p. 108) derivation D_2; C then is syntactic in the context of D_1 and lexical in the context of D_2. Idiosyncratic properties, including potentially deviating linear order, arise at the sound interface between D_1 and D_2. This definition of 'lexical' crucially assumes that derivations are layered, for which see Zwart (2009a).

The model of grammar assumed here identifies the sound interface as the precise point of contact between automatic creative processes (Merge) and stored knowledge (morphology). It remains to be shown that deviating linear order, overriding the automatic structure-to-order conversion, has this lexical/morphological character. If so, we expect constructions with deviating linear order to show idiosyncratic sound/meaning properties and/or reduced productivity.

5.6.2 Typological generalizations

Caballero et al. (2008) studied ordering effects in productive and unproductive noun incorporation in head-final and head-initial languages. They found that NV order is dominant in noun incorporation generally (36 out of 49), but exclusive in unproductive noun incorporation (8 out of 8). Moreover, about half of the VO languages (9 out of 19) has NV order with productive noun incorporation (with OV languages this is 15 out of 17). The survey shows more deviation from the expected structure-to-order conversion (i.e. from head-initial order) in unproductive types and almost exclusive deviation in the direction of head-final order. I take this to support the idea that linear order deviation is a function of morphology, on the assumption that a noun incorporation complex (certainly of the less productive type) is construed in a separate derivation and passes through the sound interface before being enlisted in the numeration for the next derivation as a single lexical item.

Caballero et al. (2008: 398f.) point out that similar results are obtained from a typological survey of synthetic compounds (of the type *skyscraper, witch hunt*) where they find 'morphologically driven departures from syntactic word order' exclusively in VO languages with NV synthetic compounds (in 8 out of 23 languages, with 0 out of 26 OV languages showing VN compounds; cf. Bauer 2001). In this context, Caballero et al. (p. 400) note that clearly lexical incorporations (Type I of Mithun 1984) are NV in 3 out of 6 VO languages, but VN in 0 out of 9 OV languages, and the results are even more striking with grammaticalized noun incorporations where instrumental nouns have become derivational affixes (Caballero et al. 2008: 400–401). Again, it appears that head-finality, certainly in head-initial languages, is brought about by morphology:

> **(9)** Generalization I
> Head-finality in a head-initial language is established at the sound interface.

Conversely, it appears that head-initiality in a head-final language is syntactic, i.e. a function of regular structure to order conversion. This may be concluded from (p. 109) a study of noun phrase coordination in head-final languages (Zwart 2005, 2009b). If we take coordination to be a prototypical syntactic operation (developed out of mere juxtaposition if Mithun 1988 is correct), we expect the pattern to be the inverse from the pattern observed in noun incorporation and the like. That is, we expect head-final languages to show head-initial noun phrase coordination, and we do not expect head-initial languages to show head-final noun phrase coordination.

The terms 'head-initial' and 'head-final' coordination are justified on the hypothesis of Kayne (1994: 12) that coordination structures are regular binary branching structures, headed by the conjunction, where the second conjunct is the complement of the conjunction. Head-initial and head-final coordination, then, are of the types *A* & *B* and *A B* &, respectively. The survey reported in Zwart (2009b: table 3) indicates that 47 out of 57 head-final languages display head-initial coordination, whereas 0 out of 85 head-initial languages show head-final coordination. When we ignore coordination strategies not using a pure coordinating conjunction, but an adposition or some other device, head-final languages show no head-final coordination at all (Zwart 2009b: 1598). These observations suggest that structure-to-order conversion is regular in the syntactic domain:

> **(10)** Generalization II
> Head-initiality in a head-final language is established in narrow syntax.

The generalizations in (9) and (10) predict that if a noun phrase coordination takes on lexical (idiomatic) properties, head-finality will again become possible. Precisely this seems to be the case in Waigali, which has regular head-initial noun phrase coordination (11a), but also the head-final fixed expression (11b) (in Degener's (1998: 166) words, this requires that the two conjuncts form a natural group):

(11) Indo-European, Iranian: Waigali (Degener 1998: 166)

a.

e	meši	ye	e	muša
a	woman	and	a	man
‘a man and a woman’				

b. meši-moša-y
woman-man-and
‘men and women’

I have not found any examples of the reverse (regular head-final coordination and idiomatic head-initial coordination).

5.6.3 The Final-over-Final Constraint

If head-final order is brought about at the sound interface, head-final constructions are lexical items in the sense understood here (single numeration items), and (p. 110) we expect head-final constructions to occur embedded in regular syntactic head-initial structures, but not the reverse (head-initial structures embedded in head-final constructions), or at least not generally. Precisely this generalization has been formulated in Holmberg (2000a: 124) as the Final-over-Final Constraint or FOFC (see also Biberauer et al. 2008):

(12) Final-over-Final Constraint (FOFC)
If α is a head-initial phrase and β is a phrase immediately dominating α, then β must be head-initial. If α is a head-final phrase, and β is a phrase immediately dominating α, then β can be head-initial or head-final.

The FOFC essentially states that head-finality must be lower in the structure than head-initiality. Consider how this might follow.

If a complement β of a head α is complex, i.e. β is not the output of a separate derivation, then the phrase [α β] is linearized as / $\alpha\beta$ / even on the original formulation of the LCA, where Merge is symmetric but the structure asymmetric (given that β is complex), and likewise if Merge yields asymmetric structure regardless of the complexity of β (as suggested in this chapter). It follows that if β is head-initial, it will be dominated by a head-initial phrase, as stated in the first clause of the FOFC.

If a complement β of a head α is the output of a separate derivation (i.e. is not complex in the context of the derivation in which α and β are merged), then the LCA does not apply on its original formulation, given bare phrase structure theory (see section 3). On the bare phrase structure approach, β must move in order to create a pair of α and a trace (which is ignored at Spell-Out, ensuring vacuous linearization); moved β then is manipulated in a syntactic derivation (becoming a specifier or adjunct) where we find only head-initial structure (on both the original and more recent formulations of the LCA). Assuming asymmetric Merge (yielding an ordered pair), there is no need for β to move, but if no movement occurs we expect the head α to precede its complement β (by (3)). In this situation, where β is the output of a separate derivation, passing through the sound interface before being enlisted in the numeration, β may have acquired a deviating (head-final) linear order. But the mechanism does not allow for α to come out as head-final, as α and β are merged in a regular syntactic derivation.

The mechanism does not preclude α (dominating β) being subject to reordering at the sound interface concluding the derivation in which α was created via merger of its head with β. Potentially this might yield an exception to the FOFC, if β itself happens to be head-initial. But it is predicted that this situation will be idiomatic, having the flavor of the exceptional. The empirical motivation of the FOFC, as discussed in Biberauer et al. (2008) and references cited there, however, is squarely based on observations of regular syntax.

I conclude that the system of structure-to-order conversion contemplated in this chapter provides a rationale for the FOFC: since head-final order is a lexical property, head-final phrases will occur at the bottom of the syntactic tree, and (p. 111) head-finality will never be a property of the main projection line, which is a function of Merge, not of the lexicon.

5.6.4 Head-finality in a head-initial language: a closer look

Let us call head-finality as the result of movement 'pseudo-finality'. Kayne (1994) conjectured that all head-finality is pseudo-finality. The discussion in this section acknowledges that some head-finality may be pseudo-finality, but that there is also a large amount of real head-finality which is brought about by morphology at the sound interface. This raises the question how to distinguish the two types of head-finality.

In my earlier work on the syntax of Dutch (Zwart 1993, 1994) I tried to argue that deviations from head-initial syntax were caused by leftward movement of various elements to specifier positions in the functional domain. Some of the movements proposed, in particular the object shift movement affecting the position of definite and specific indefinite objects, were already assumed in the standard literature (cf. Vanden Wyngaerd 1989). Other movements, involving leftward shift of nonspecific indefinite objects, particles, and secondary predicates, were novel and somewhat suspect, in that the elements involved remained adjacent to the verb. One possibility is that the earlier analysis erred in conflating the two types of head-finality, and that the OV order without obligatory object—verb adjacency involves pseudo-finality (caused by movement), whereas the complements left-adjacent to the verb owe their surface position to linearization at the sound interface.

To illustrate the basic facts, Dutch shows an asymmetry between nominal and clausal verb complements, such that nominal complements precede, and clausal complements follow the verb in clause-final position (example sentences involve embedded clauses with the verb in its base position; in main clauses, the verb is realized in the 'verb-second' position, following the first constituent).

(13) Nominal vs. clausal complements in Dutch

a.

dat	Jan	**Die**	**dingen**	niet	beweert
COMP	John	DEM:PL	thing:PL	NEG	claim:3SG
'that John does not claim those things'					

b.

dat	Jan	niet	beweert	**dat**	**het**	**regent**
COMP	John	NEG	claim:3SG	COMP	it	rain:3SG
'that John does not claim that it is raining'						

In (13a), the nominal complement *die dingen* 'those things' precedes the verb (obligatorily), and is separated from it by the negation element *niet* 'not'. In (13b), the clausal complement *dat het regent* 'that it is raining' follows the verb (obligatorily). Assuming that the complement of the verb is generated as a sister to the verb, the position of *die dingen* 'those things' in (13a) must be the result of leftward movement (p. 112) (object shift), and although traditionally the base position of the object is taken to precede the verb (e.g. Koster 1975), a possible extension of that analysis is that the object originates in the position to the right of the verb, occupied by the clausal complement in (13b) (thus Zwart 1994).

Other elements obligatorily preceding the verb (in clause-final position) include verbal particles (14a), secondary predicates (14b), and nonspecific indefinite objects (14c). In most dialects of spoken Dutch, the past participle also precedes the clause-final auxiliary (14d), though many patterns occur across Dutch dialects, especially if larger verb clusters are taken into account (cf. Zwart 1996, Barbiers et al. 2008).

(14) Elements left adjacent to the verb in Dutch

a.

dat	Jan	die	dingen	niet	**op**	schrijft
COMP	John	DEM:PL	thing:PL	NEG	up	write:3SG
‘that John does not write those things down’						

b.

dat	Jan	het	hek	niet	**rood**	verft
COMP	John	the	fence	NEG	red	paint:3SG
‘that John does not paint the fence red’						

c.

dat	Jan	zelden	**een**	**boek**	leest
COMP	John	rarely	a	book	read:3SG
‘that John rarely reads a book’					

d.

dat	Jan	het	boek	niet	**gelezen**	heft
COMP	John	the	book	NEG	read:PART	have:3SG
‘that John didn't read the book’						

The boldface elements in (14) cannot be separated from the finite verb by negation, adverbs, and the like. The only evidence that their placement might be the result of movement offered in Zwart (1994: 400) is that stranded prepositions of adjunct prepositional phrases (PP) may break up the adjacency. Compare (15a), with a full adjunct PP and (15b) with a stranded adjunct preposition.

(15) Stranded preposition breaking up adjacency

a.

dat	Jan	het	hek	**met**	**die**	**kwast**	rood	verft
COMP	John	the	fence	with	DEM	brush	red	paint:3SG
‘that John paints the fence red with that brush’								

b.

de	kwast	waar	Jan	het	hek	(mee)	rood	**(mee)**	verft
the	brush	REL	John	the	fence	with	red	with	paint:3SG
‘the brush that John paints the fence red with’									

The position of the secondary predicate *rood* ‘red’ in (15b), preceding the stranded preposition *mee* ‘with’ (at least optionally), was taken to indicate that the secondary predicate *rood* had been moved to the left, on the assumption that (a) adjunct PPs are outside the verb phrase, and (b) the stranded preposition could not have been (p. 113) lowered. However, the distribution of stranded prepositions is not well understood, and in the present context it might be assumed that the position of the stranded preposition in linear order is the result of linearization at the sound interface, as stranded prepositions are typically light elements (cf. Zwarts 1997). If so, no evidence for leftward movement of the boldface elements in (14) remains.

Could the head-finality in (14) be the effect of linearization at the sound interface? Some suggestion that this might be right is provided by the observation that idiomaticity in Dutch is sensitive to the left—right asymmetry as reflected in head-final vs. head-initial linear order. For example, whereas adjunct and complement PPs can freely appear to the right of the verb in final position, verb-PP idioms require the PP to precede the verb (cf. Veld 1993: 148):

(16) Verb-PP idioms require head-final order

a.

dat	Jan	de	pijp	(aan	Marie)	geeft	(aan	Marie)
COMP	John	the	pipe	to	Mary	give:3SG	to	Mary
‘that John hands the pipe to Mary’								

b.

dat	Jan	de	pijp	(aan	Maarten)	geeft	(*aan	Maarten)
COMP	John	the	pipe	to	Marten	give:3SG	to	Marten
‘that John quits’								
idiom: *de pijp aan Maarten geven* = to quit								

In the model of grammar considered here, idioms are construed in a separate sub-derivation, and pass through the interfaces before being enlisted in the numeration for the next derivation. They are, then, ‘lexical elements’ with special sound-meaning properties acquired when passing through the interface separating two derivation layers (see section 5.6.1). The head-final order of the idiom *de pijp aan Maarten geven* ‘quit’ (and in fact of all verb-PP idioms) may then be the outcome of linearization at PF.

I would like to suggest that all head-final constructions in (14) are construed in separate derivations, and hence are ‘lexical’ in the sense of section 5.6.1 (i.e. they are ‘complex predicates’, cf. Neeleman 1994). First, verb—particle combinations (14a) are invariably highly idiomatic (e.g. *op bellen* [up ring] ‘phone’, *uit vinden* [out find] ‘invent’, *in dikken* [in thick] ‘thicken’, *aan vallen* [on fall] ‘attack’, *voor stellen* [fore put] ‘propose’), and the verb cannot be fronted without pied-piping the particle:

(17) No particle stranding under verb fronting

a.

*Geschreven	heeft	Jan	dat	niet	op
write:PART	have:3SG	John	DEM	NEG	up
intended: 'John did not write that down.'					

b.

Op	geschrevenheeft	Jan	dat	niet
up	write:PART have:3SG	John	DEM	NEG
'John did not write that down.'				

(p. 114) If the verb—particle combination is the output of a separate derivation, it is expected that the verb and the particle cannot be separated in the context of the next derivation in which the combination appears. The verb and the particle may be separated under verb-second, where the finite verb moves to the position following the first constituent in main clause, but this is not an operation of narrow syntax if Chomsky (2001: 37f.) is correct.

The verb and the particle may also be separated by other material belonging to a verb cluster, as in *op heeft geschreven* [up has written] 'has written down', but the logic of the analysis entails that the entire verb cluster is to be the output of a separate derivation, and hence the separation may be the effect of linearization as well. In fact, the combination of a perfective participle (like *geschreven*) and the auxiliary *hebben*, originally a verb of possession, has come to be used to refer to a relative past tense as the outcome of a grammaticalization process. As a result, the verb—particle combination in (14d) also has idiosyncratic sound—meaning properties that might be taken to betray construction in a separate derivation.

The fact that verb clusters in Continental West Germanic dialects display a wide variety of linear orders, both across and within dialects, has been taken to suggest that the derivation of these clusters requires special rules, such as reanalysis (Haegeman and van Riemsdijk 1986), flipping (i.e. inversion of sister nodes, den Besten and Edmondson 1981: 43), and rightward movement of heads ('verb raising') and larger projections of the verb ('verb projection raising') (Evers 1975). Of these, reanalysis is arguably a process not of narrow syntax but of the sound interface (Zwart 2006; cf. Zubizarreta 1985: 286, stating in this context that 'the grammar does allow for mismatches between morphophonology and morphosyntax'). Flipping is clearly not Merge, so it too must be considered a function of linearization at PF. We know that stylistic injunctions against 'German-sounding' patterns play a role in determining auxiliary—participle orders in written and carefully spoken Dutch (Stroop 1970: 252), indicating that structure-to-order conversion is not free from influence by stored knowledge in this domain. The concept of rightward movement has no status assuming bare phrase structure theory, so the directionality aspect of it must come in only at the sound interface.

For secondary predicates (14b), a complex predicate analysis has been proposed and extensively argued for by Neeleman (1994). Semantically, the verb and the secondary predicate form a tight connection. When the secondary predicate is a PP, we again observe that the PP cannot be extraposed, just like PPs that are part of a verbal idiom (cf. (16)).

(18) No extraposition of secondary predicate PPs

a.

dat	Jan	in	de	sloot	sprong
COMP	John	in	the	ditch	jump:PAST.SG
OKadverbial reading: 'that John was jumping [*sc*. up and down] in the ditch'					
OKsecondary predicate reading: 'that John jumped into the ditch'					

(p. 115) **b.**

dat	Jan	sprong	in	de	sloot
COMP	John	jump:PAST.SG	in	the	ditch
OKadverbial reading: 'that John was jumping [*sc*. up and down] in the ditch'					
*secondary predicate reading ('that John jumped into the ditch')					

As the absence of the resultative (secondary predicate) reading of the post-verbal PP in (18b) shows, secondary predicate PPs behave like idioms. Both PP and non-PP secondary predicates readily lend themselves to the formation of idioms with the verb (e.g. *zwart maken* [black make] 'badmouth', *om de tuin leiden* [around the garden lead] 'mislead', etc.).

Nonspecific indefinites (14c) lose their nonspecific interpretation as soon as they are not adjacent to the verb (again, not counting the effect of 'verb second' placement of the finite verb in main clauses). Thus, in (14c′) the leftward shifted indefinite object acquires a generic reading.

(14)

c′	dat	Jan	**een**	**boek**	zelden	leest
	COMP	John	a	book	rarely	read:3SG
'that John rarely READS a book'						
'(i.e. what John rarely does to a book is read it')						

This indicates that the combination of the verb and the adjacent indefinite acquires idiosyncratic sound—meaning properties, which we took to be indicative of construction in a separate sub-derivation.

While firm conclusions cannot be drawn at this point, the observations may be taken to indicate that head-final linear order in Dutch, overriding the automatic structure-to-order conversion, may be a function of the punctuated character of the derivation, where elements construed in one derivation acquire idiosyncratic sound—meaning properties (including deviating linear order) while passing through the interfaces before being enlisted in the numeration for the next derivation layer. It would be interesting to consider the question how many movements (and movement types) may be dispensed with when the potential of linearization at the interface between derivation layers is more fully taken into consideration.

5.6.5 Conclusion: head-finality as a linguistic sign

If I am correct that deviating (head-final) linear order originates at the sound interface separating two derivation layers, the following holds:

(19) Head-finality is a linguistic sign, signaling derivation layering.

The function of reordering at the sound interface might be to brand complex, derived structures as single items for use in a further derivation. Put differently, (p. 116) linear order identifies structures as being either open-ended (head-initial) or sealed off (head-final). Using head-final linearization to signal derivation layering is clearly not obligatory, but the crucial observation is that only head-final orders can be argued to perform such a signal function.

If the relation between linearization and derivation layering suggested in (19) is real, linear order is intimately connected with a fundamental property of the faculty of language, recursion (see Hauser et al. 2002). To be clear, I understand 'recursion' slightly different from what is standard. The operation Merge is standardly taken to be recursive in that the output of Merge may be subject to further operations Merge. However, each sub-derivation may just as well be taken to be iterative rather than recursive, for instance if each step of the derivation transfers a single element from the resource to the workspace (section 5.4), or splits a single element off from the resource (section 5.5). What is unquestionably recursive, though, is the process whereby the output of one derivation functions as a single item in the next derivation, and this is how I understand recursion here (cf. Hofstadter 2007: 83, who identifies the recursive capacity of treating complex concepts as single packets, to be combined with other concepts ad infinitum, as a species-specific property of human cognition). If (19) is correct, linear order signals recursion in this sense.

In this connection, we may understand the memory limitation on center-embedding discussed in Yngve (1961) as reflecting a limit on the number of derivational loops that can be tracked in the context of a single utterance. Inasmuch as the concept of center-embedding refers to the linear order of a branching and a non-branching category, it has no status in the minimalist approach to structure building (there is no 'center'). But a 'left branch element' (an adjunct or a specifier) must be merged with the object under construction (or split off from the resource under the approach of section 5.5) as a single whole, and therefore it has to be the output of a separate derivation.

5.7 Conclusion

In this chapter I have discussed the relation between structure and linear order in the minimalist approach to syntactic theory.

The general idea of Kayne (1994) that linearization is a function of structural asymmetry among syntactic nodes can be maintained in the bare phrase structure theory of Chomsky (1995b), if we take the history of the derivation into account.

On its simplest definition, Merge is the same at each step of the derivation, i.e. first Merge should have no special properties. This is achieved if Merge is taken to be an operation transfering one element at a time from a resource to a (p. 117) workspace (the object under construction). Simplifying even more, and adopting a top-down derivational approach, we can take structure to result from an operation that splits off one element at a time from the resource ('Split Merge') until the resource is empty. Either way, sister pairs are not sets but ordered pairs, and the set of elements merged/split off is a nest, which is equivalent to an ordered n-tuple.

This allows us to consider structure-to-order conversion as the trivial equivalence relation in (3) (where material between slashes is ordered in time, i.e. realized one after the other in sound or gesture).

(3) Structure-to-order conversion
$\langle \alpha, \beta \rangle = /\alpha, \beta/$

Deviations from (3) can be of two types: (i) pseudo-finality, which is the result of movement, or (ii) real finality, which is the outcome of a morphological process at the sound interface PF, where syntactic structures are replaced by linear strings. Movement rearranges the hierarchic relations among elements, and is therefore, on the system considered here, expected to lead to a change in linear order. The only real head-finality, then, is of type (ii), i.e. is essentially lexical.

In this context, I defined 'lexical' in relation to the process of derivation layering. A derivation is layered when an element in its numeration is the output of a previous derivation. The output of a derivation passes through the interfaces before being enlisted in the numeration of a subsequent derivation. I have argued that head-final order may result from a marked structure-to-order conversion at the sound interface, as part of a set of idiosyncratic sound—meaning properties acquired at that particular juncture. I have suggested that head-final order is a sign indicating that a complex element (the output of a derivation) is to be considered as a single item in the context of the next derivation.

Capitalizing on the architecture of layered derivations allows one to redefine the concepts 'syntactic' and 'lexical' in relation to derivations, such that an element may be syntactic in one derivation layer (i.e. created by Merge) and lexical in the next (i.e. treated by Merge as a single item). Since ordering does not exist in narrow syntax, it must be brought in at the sound interface. The layered derivation approach entails that linearization is not a once-only process, but may be interlaced with stages of structure-building in which linear order plays no role (cf. Uriagereka 1999). Linear order, then, may be frozen in 'packets' manipulated by Merge in the context of subsequent derivation layers.

Once this is understood, it becomes possible to reconsider analyses that have been proposed to derive head-final order in narrow syntax. It will be clear that syntactic operations that seem to lack independent motivation are the first suspects. I have tried to argue that some aspects of Dutch head-final order, unsuccessfully described in terms of movement in previous analyses, may indeed be explained as a function of linearization at the interface.

(p. 118) Further research might be directed at those instances of head-final order which are currently derived via the somewhat suspect process of 'roll-up movement' (turning [A [B [C]]] into [[[C]+B]+A] via successive incorporative movements). Inasmuch as the model for roll-up movement was provided by the distinctly morphological process of inflectional affixation (e.g. Baker 1985b, Pollock 1989), a reduction to morphological linearization processes might not be unattractive for at least some of the cases.

What is missing from the present discussion is a more fundamental assessment of the factors entering into linearization at the sound interface. I have suggested that linear order is partly simply given by the morphological inventory, i.e. is stored knowledge of a fixed form—meaning pairing. But in addition, prosodic factors having to do with 'lightness' of elements (clitics, stranded prepositions, discourse particles) (cf. Anderson 1993) and ordering requirements particular to linkers (Maxwell 1984: 253) are arguably involved, as may be other factors, and a more fundamental investigation of these factors and their interaction is needed for a complete understanding of the nature of structure-to-order conversion.

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Multidominance

Barbara Citko

The Oxford Handbook of Linguistic Minimalism
Edited by Cedric Boeckx

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Abstract and Keywords

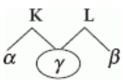
This article examines the status of multidominance in minimalist syntax. It addresses the following questions: How are multidominant structures generated? How are they linearized? What empirical insights do they offer? The article is organized as follows. Section 6.2 introduces a mechanism to generate such structures, relying on minimalist assumptions about Merge and Move. Section 6.3 turns to empirical support for multidominance, surveying a number of unrelated constructions that naturally lend themselves to a multidominant analysis. Section 6.4 considers the issue of linearization, focusing on how different accounts handle it, and pointing out the similarities and differences between them. Section 6.5 offers a brief conclusion and suggests directions for future research.

Keywords: minimalist syntax, multidominant structures, merge, move, linearization

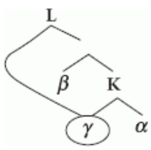
6.1 Introduction

This chapter examines the status of multidominance in minimalist syntax. A multidominant structure is a structure in which a single node has two mothers.[1] The result could be a multi-rooted structure of the kind illustrated in (1a) or the one shown in (1b), in which one of the mothers dominates the others.

(1)



a.



b.

The concept of multidominance is often described as unorthodox, non-standard, or incompatible with basic assumptions about phrase structure.[2] My main goal in this (p. 120) contribution is to show that just the opposite is the case: multidominant structures are quite orthodox, standard, and compatible with current minimalist assumptions about phrase structure and movement. If anything, the lack of such structures, rather than their presence, would be a surprise. This is not to say that they do not raise some non-trivial issues for the grammar. In this chapter, I examine these issues, and the ways in which they have been addressed in minimalist literature. The ones I focus on are the ones that any multidominant account has to address: (i) How are multidominant structures generated? (ii) How are they linearized? and (iii) What empirical insights do they offer? I proceed as follows. In section 6.2, I introduce a mechanism to generate such structures, relying on minimalist assumptions about Merge and Move. In section 6.3, I turn to empirical support for multidominance, surveying a number of unrelated constructions that naturally lend themselves to a multidominant analysis. In section 6.4, I turn to the issue of linearization, focusing on how different accounts handle it, and pointing out the similarities and differences between them. And in section 6.5, I offer a brief conclusion and suggest directions for future research.

Before proceeding any further, a couple of disclaimers are in order. First, given the mimimalist theme of this handbook, I will not discuss how multidominant structures are generated in other frameworks, such as tree adjoining grammars (see Chen-Main 2006) or phrase linking grammars (see Peters and Ritchie 1981). Second, I will not attempt to provide a complete survey of constructions that have been claimed to be multidominant. Third, due to space considerations, I will not be able to make an adequate comparison of multidominant and non-multidominant accounts of the constructions I will discuss here.

6.2 Generation of multidominant structures

This section shows that the existence of multidominance in the grammar is a natural consequence of basic assumptions about phrase structure and movement (p. 121) in recent minimalist theory. In particular, it follows from Chomsky's (2004a) distinction between two kinds of Merge, External Merge and Internal Merge. External Merge is the familiar kind of Merge; it takes two syntactic objects (*i.e.* α and β) and forms one larger object from them, as shown in (2a). Internal Merge differs from External Merge only in that one of these two objects is a sub-part of the other, as shown in (2b).

(2)

a. External Merge of α and β

b. Internal Merge of α and β

In Citko (2005) I argued that the existence of External Merge and Internal Merge predicts the existence of a third kind, combining the properties of these two. This third kind, which I dubbed Parallel Merge, is what is responsible for generating multidominant structures. Parallel Merge structures are a result of a two-step process. First, α merges with γ, as shown in (3a). And, second, β, merges with a subpart of α, as shown in (3b). As a result, γ is shared between α and β.

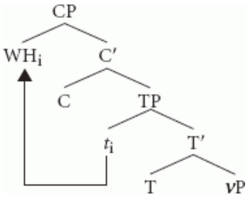
(3)

a. Merge α and γ

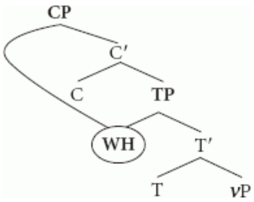
b. (Parallel) Merge β and γ

If the idea that movement is Internal Merge is taken literally, all cases of movement lead to multidominance. On this view, the result of *wh*-movement of the subject, for example, is (4b) not (4a); the *wh*-phrase, instead of being copied and merged with C (becoming its specifier) is simply re-merged with C. As a result, it is immediately dominated by two nodes, TP and CP.

(p. 122) **(4)**



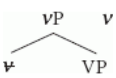
a.



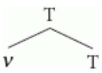
b.

There is, however, a natural distinction between multidominance resulting from Parallel Merge and multidominance resulting from movement. Gračanin-Yüksek (2007) refers to the two types as horizontal sharing and vertical sharing, respectively.[3] The *wh*-phrase in (4b) is vertically shared between CP and TP, because one of its mothers dominates the other. In cases of horizontal sharing, neither dominates the other. My main focus in this chapter is on horizontal sharing, which is sharing resulting from Parallel Merge (not Internal Merge), even though in some fundamental sense the two are the same process. Before moving on to Parallel Merge, however, let me note one further consequence of treating movement as Internal Merge. It involves sideward movement of the kind proposed by Nunes (2001, 2004) (called 'interarborial movement' by Bobaljik and Brown (1997)). If movement is Internal Merge, sideward movement becomes indistinguishable from Parallel Merge. To see why, let us first consider a sideward movement derivation of head movement.[4] First, the υ head is copied. Next, the copy of υ is merged with T. At this point in the derivation, there are two rooted objects (υP and T), hence the intuition that movement proceeds in a sideward fashion. And finally, the T complex is merged with υP.

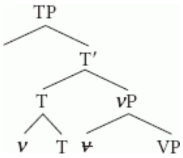
(5)



a.



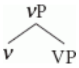
b.



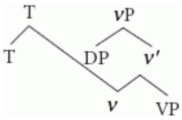
c.

If movement is Internal Merge, (5a–c) reduces to (6a–c). There is no copying involved; *υ* undergoes Parallel Merge with T instead.[5]

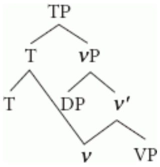
(p. 123) **(6)**



a.



b.



c.

To sum up briefly, I hope this section has removed some of the biggest theoretical obstacles to multidominance, by deriving it from independent assumptions about the properties of Merge. In the next section, I turn to the empirical evidence in favor of multidominance, coming from the fact that it can account for a number of properties of a number of distinct constructions.

6.3 Empirical support

6.3.1 General properties of multidominant structures

Multidominant structures generated by Parallel Merge (or some mechanism like it) have been implicated in an analysis of a number of distinct constructions, a complete discussion of which would lead us too far astray. (7) below lists some that have been claimed to involve multidominance. The list (and the references accompanying it) is by no means complete; its purpose is to show that multidominance is quite a common mechanism, which has been applied to a wide range of unrelated constructions.[6]

(7)

a. across-the-board *wh*-questions (Citko 2005, Goodall 1987, Muadz 1991, de Vries 2009)
b. *wh*-questions with conjoined *wh*-pronouns (Citko forthcoming a, Gračanin-Yüksek 2007)
c. right node raising (Bachrach and Katzir 2009, Larson 2007, McCawley 1982, de Vries 2009, Wilder 1999, 2008)
d. gapping (Citko forthcoming b, Goodall 1987, Muadz 1991)
e. determiner sharing (Citko 2006b)
f. standard free relatives (Citko 2000, van Riemsdijk 1998, 2000, 2006)
g. serial verbs (Hiraiwa and Bodomo 2008)
h. parasitic gaps (Kasai 2008)
(p. 124) **i.** idioms (Svenonius 2005)
j. comparatives (Moltmann 1992)
k. transparent free relatives (van Riemsdijk 1998, 2000, 2006)
l. parentheticals (McCawley 1982, de Vries 2007)
m. *wh*-amalgams (Guimarães 2004)
n. cleft-amalgams (Guimarães 2004, Kluck and Heringa 2008)

I will not be able to do justice to all the arguments that have been made in favor of (or against) treating every construction in (7) as multidominant.[7] What I would like to do instead is address a more general question of whether there is something that they all have in common, which could be called a hallmark of multidominance.

In general, the constructions in (7) can be divided into three groups. The first five (i.e. ATB *wh*-questions, *wh*-questions with conjoined *wh*-pronouns, right node raising, gapping, and determiner sharing) involve coordination, which raises the question of whether coordination is a necessary (or sufficient) condition for multidominance. The answer to both questions is negative for obvious reasons; not all the constructions listed in (7) are coordinate and not all types of coordinate structures are listed in (7). A multidominant approach to coordination in general would be hard to reconcile with the evidence that the conjuncts in a coordinate structure stand in an asymmetric c-command relationship (as argued convincingly e.g. by Johannessen 1998, Munn 1993, and Progovac 1998).

Some of the constructions listed in (7) are constructions that have been analyzed as involving ellipsis. These are gapping and determiner sharing (which is essentially a form of gapping). Again, it is not true that ellipsis is a hallmark of multidominance (as there are many elliptical constructions that are not multidominant).

All of them, however, involve sharing, the intuition that one element is simultaneously in two positions (without moving from one to the other). In many cases, such movement would violate well-established constraints on movement. For example, in ATB questions, movement of the *wh*-pronoun from one conjunct to the other would violate the Coordinate Structure Constraint. In free relatives, movement of the *wh*-pronoun from the specifier of CP to the head position would violate the ban on improper movement. In right node raising, movement would be to the right. This idea of sharing thus seems to be quite a reliable diagnostic of multidominance. I examine its effects in the next two sections, which discuss a subset of the multidominant constructions listed in (7) in more detail. I start with ATB *wh*-questions, summarizing the account of Citko (2005). I follow up with a brief discussion of other multidominant constructions, highlighting their properties that follow most naturally from a multidominant structure.

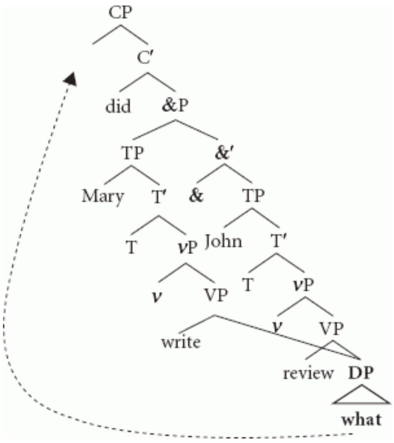
(p. 125) 6.3.2 Case study: ATB *wh*-questions

To see how a multidominant account of ATB *wh*-questions might work, let us consider a derivation of the *wh*-question given in (8a). Its numeration, given in (8b), contains only one *wh*-pronoun. This is the element that is going to be shared between the two conjuncts, as shown in (8c).[8]

(8)

a. What did Mary write and John review?

b. N = {what, Mary, write, John, review, v_2, T_2, C_1, and_1}



c.

The most straightforward support for such structure comes from case and category matching, most visible in languages with rich morphological case-marking, such as Polish. Borsley (1983) and Dyła (1984) note that in questions and relative clauses involving ATB movement, the fronted *wh*-phrase has to satisfy case requirements imposed on it within both conjuncts (see also Franks 1993, 1995 for a discussion of this requirement). This case matching requirement is what accounts for the data in (9a–c); (9a) is a base example showing that ATB movement is possible if the verbs inside the two conjuncts assign identical cases to the moved element. (9b) is ungrammatical because the two verbs (*lubić* 'like' and *nienawidzić* 'hate') assign different cases to their complements (accusative and genitive, respectively). The grammaticality of (9c) follows from case syncretism; the masculine form of the *wh*-pronoun is the same in the accusative and genitive case.

(9)

a.

dziewczyna,	**którą**	Maria lubi t_{ACC}	a	Ewa	uwielbia t_{ACC}
boy	who.ACC	Maria likes	and	Ewa	adore
'the girl who Maria likes and Ewa adores'					

(p. 126) **b.**

*dziewczyna,	**którą/której**	Janek lubi t_{ACC}	a	Jerzy	nienawidzi t_{GEN}
girl	who.ACC/who.GEN	Janek likes	and	Jerzy	hates
'the girl who Janek likes and Jerzy hates'					

c.

chłopiec,	**którego**	Maria lubi t_{ACC}	a	Ewa	nienawidzi t_{GEN}
boy	who.ACC/GEN	Maria likes	and	Ewa	hates
'the boy who Maria likes and Ewa hates' (Franks 1995: 62)					

The case matching requirement follows from the multidominant structure given in (8c) as follows. Since a single *wh*-pronoun is shared between two clauses, it has to satisfy whatever category and case requirements are imposed on it within these two clauses.

The multidominant structure in (8c) can also shed some light on why ATB questions are an exception to the Coordinate Structure Constraint. Ross's (1967a) original formulation of the Coordinate Structure Constraint essentially stipulated this exception.[9] In a multidominant structure, however, movement is never from just a single conjunct (or from one conjunct to the other). Technically speaking, then, it does not violate the Coordinate Structure Constraint.

Another property of ATB questions that follows nicely from a multidominant structure is the lack of multiple ATB *wh*-fronting, illustrated by the ungrammaticality of questions involving the configuration schematized in (10), in which the number of fronted *wh*-elements matches the total number of gaps inside the two conjuncts.

(10) $[_{CP}$ WH$_i$ WH$_j$ $[_{TP}$ … t_i…] and $[_{TP}$ … t_j…]]

In English, the ungrammaticality of (11a), which is an example of the configuration in (10), can be attributed to the fact that its C heads do not allow multiple specifiers (or, to put it in a more atheoretical fashion, to the fact that English does not allow multiple *wh*-fronting), which is what accounts for the ungrammaticality of (11b).

(p. 127) **(11)**

a. * *What$_j$ what$_i$* did John write t_i and Mary review t_j?

b. * *Who$_i$ what$_j$* t_i read t_j?

However, if this were the case, we would expect such questions to be possible in languages that do allow multiple *wh*-fronting, such as Polish. This is not the case, as shown by the contrast between the ungrammatical ATB question in (12a) and the grammatical multiple *wh*-question in (12b):

(12)

a.

**Co$_i$*	*co$_j$*	Jan polecił t_i	a	Maria	przeczytała t_j ?
what	what	Jan recommended	and	Maria	read
'What did Jan recommend and Maria read?'					

b.

Kto$_i$	co$_i$	t_i	czyta	tj?
who	what		reads	
'Who reads what?'				

This generalization might seem falsified in view of the grammaticality of the ATB *wh*-question with multiple fronted *wh*-elements in (13). This is only apparent; what distinguishes it from the ungrammatical case in (12a) above is the fact that both *wh*-phrases are extracted from both clauses simultaneously. As a result, the number of fronted *wh*-elements matches the number of gaps in each conjunct, not the total number of gaps in the structure.

(13)

Co$_j$	*komu$_i$*	Maria	dała t_j t_i	a	Jan zabrał	t_j	t_i ?
what	whom	Maria	gave	and	Jan took.away		
'What did Mary give and Jan take away from whom?'							

In addition to these properties, in Citko (2005) I also discussed the interpretation of ATB questions and the lack of covert ATB movement, and in Citko (2006a), the lack of ATB left branch extraction. The last two are linked to linearization, which I discuss in section 6.4 below. In short, I argue that the multiply dominated element has to move, as it can only be linearized in its landing site. This explains the lack of covert ATB movement and the fact that the nominal cannot be stranded by left branch extraction.

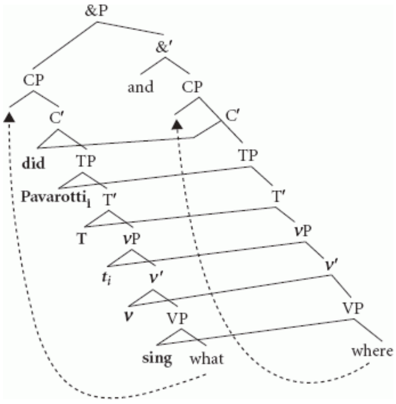
6.3.3 Extensions

In this section, I turn to other constructions listed in (7) and highlight the empirical advantages of analyzing them in a multidominant fashion. The ones I focus on are *wh*-questions with conjoined *wh*-pronouns, right node raising, gapping, determiner sharing, standard free relatives, transparent free relatives, and amalgams. Due to space constraints, I can only offer a bird's-eye view of how a multidominant analysis for these constructions might work; I refer the interested reader to Citko (forthcoming c) for a monograph-length multidominant treatment of a larger subset.

(p. 128) An example of a *wh*-question with coordinated *wh*-pronouns is given in (14a); what is interesting about such questions is the fact that the two coordinated elements (*what* and *where*) are of different categorial status and the result does not violate the Law of the Coordination of Likes. Gračanin-Yüksek (2007), whose structure is given in (14b), analyzes such questions as involving two clauses in which everything *but* the two *wh*-pronouns is shared between the two conjuncts.

(14)

a. What and where did Pavarotti sing?



b.

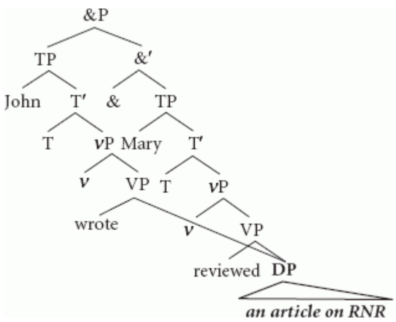
Such a multidominant structure can explain, among other things, why (14a) does not violate the Law of the Coordination of Likes. Since coordination in (14b) is between two CPs, with each one hosting a *wh*-phrase in its specifier, it does not matter that the two *wh*-phrases are of different categories.[10] This structure also captures the fact that the question in (14a) is most naturally paraphrased as (15); again, this follows from the fact that it involves two CPs with one *wh*-phrase in the specifier of each.

(p. 129) **(15)** What did Pavarotti sing and where did Pavarotti sing?

The next construction listed in (7) is right node raising, which is quite commonly analyzed along the lines schematized in (16) (see e.g. Bachrach and Katzir 2009, Johnson 2007, Larson 2007, McCawley 1982, de Vries 2009, Wilder 1999, 2008).

(16)

a. John wrote and Mary reviewed an article on RNR.



b.

The most straightforward argument in favor of such an analysis (and against alternatives that treat right node raising as rightward ATB movement, as argued e.g. by Postal 1998, Ross 1967a, and Sabbagh 2007) comes from the behavior of right node raising with respect to standard movement diagnostics. To illustrate briefly, the grammaticality of the examples in (17a-b) shows that right node raising can escape *wh*-islands and complex NP islands.

(17)

a. John asked when Bill wrote __ and Mary wondered when LI will review __ *Sue's new article on right node raising.*

b. Mary read a book that praised __ and John read an article that criticized __ *ellipsis accounts of RNR*.

Preposition stranding points toward a similar conclusion. Languages that disallow preposition stranding in movement constructions allow it in right node raising, as shown by the following contrast from Irish. (18a) involves Heavy NP Shift, which disallows preposition stranding, and right node raising, which does. The stranded prepositions in both examples are in bold.

(18)

a.

*Bhí	mé	ag éisteacht	le__	inné	clár	mór fada	ar an ráidió	faoin	toghachán.
was	I	listen.prog	with	yesterday	program	great long	on the radio	about-the	election
‘I was listening yesterday to a great long program on the radio about the election.’									

(p. 130) **b.**

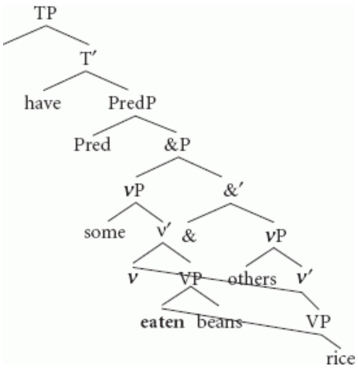
Níl	sé	in aghaidh	an	dlí	A thuilleadh	bheith	ag éisteacht le __	nó	ag breathnú	ar	ráidió	Agus	teilifís	an Iarthair
is-not	it	against	the	law	anymore	be(-fin)	listen.prog with	or	look.prog	on	radio	And	television	the West.gen
‘It is no longer against the law to listen to, or to watch, Western radio and television’. (McCloskey 1986: 184–5)														

The next two constructions listed in (7), gapping and determiner sharing, have traditionally been analyzed as involving ellipsis. An example of each is given in (19a–b).

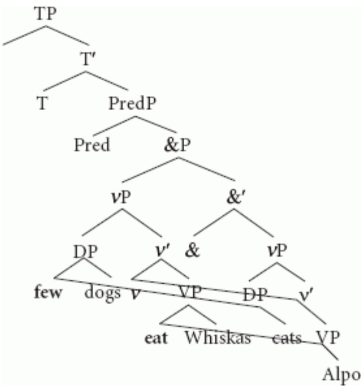
(19)
a. Some have eaten beans and others ~~have eaten~~ rice.
b. Few dogs eat Whiskas or ~~few~~ cats ~~eat~~ Alpo.

The two are obviously related in that determiner sharing (in addition to a gapped determiner in the second conjunct) has a gapped verb. The multidominant structures for both given below build on Johnson's (2000a, 2009) idea that gapping and determiner sharing involve coordination of small conjuncts (*v*Ps not TPs) (see also Lin 2000, 2002 for a small conjunct approach to determiner sharing). In these accounts, the gapped verb undergoes ATB movement to some position above the coordination level, such as the head of PredP.[11] If, as suggested above, ATB movement is a result of multidominance, gapping and determiner sharing receive the representations in (20a–b) instead.

(20)



a.



(p. 131) **b.**

In a multidominant structure, the gapped elements (which are understood to be shared between the two conjuncts) are literally shared between the two conjuncts. This is obvious for the verb and the determiner, both missing from the second conjunct. In Citko (forthcoming b), I show that in gapping constructions the *v* head has to be shared as well, in order to account for the fact that the two conjuncts have to match in voice. This is illustrated in (21a–b) for gapping and in (22a–b) for determiner sharing.[12]

(21)
a. *Some eat beans and rice ~~was eaten~~ by others.
b. *Beans were eaten by some and others ~~ate~~ rice.

(22)
a. *Few dogs eat Whiskas and Alpo ~~is eaten~~ by ~~few~~ cats.
b. *Whiskas is eaten by few dogs or ~~few~~ cats ~~eat~~ Alpo.

Determiner sharing is further evidenced by the fact that the determiner assigns the same case to both subjects. In (23a) both subjects have genitive case, assigned by the quantifier *mało* ‘few’. If the second conjunct subject has nominative case instead, the result is ungrammatical, as shown in

(23b).[13]

(23)

a.

Mało	kotów	je	Alpo	a	~~mało~~	psów	~~je~~	Whiskas.
few	cats.GEN	eat	Alpo	and	few	dogs.GEN	eat	Whiskas
'Few cats eat Alpo and dogs Whiskas.'								

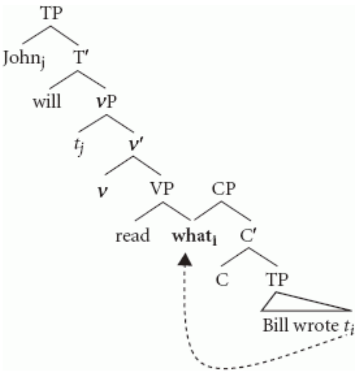
(p. 132) **b.**

*Mało	kotów	je	Alpo	a	~~mało~~	psy	~~jedzą~~	Whiskas.
few	cats.GEN	eat	Alpo	and	few	dogs.NOM	eat	Whiskas
'Few cats eat Alpo and dogs Whiskas.'								

The last set of constructions I consider in this chapter consists of standard free relatives, transparent free relatives, *wh*-amalgams, and cleft-amalgams. What distinguishes them from the ones discussed above is that they are not coordinate; one of the clauses is subordinate or parenthetical to the other. What they have in common with the ones discussed above is the idea that one element is shared between two clauses. In free relatives, for example, the *wh*-pronoun is shared between the matrix and the relative clause, as shown in (24a–b).

(24)

a. John will read what Bill wrote.



b.

Such a structure can naturally account for the presence of matching effects in free relatives, illustrated in (25a–c) with data from German. Since in (25a) the *wh*-pronoun is assigned the same case in both the matrix and the relative clause, the result is grammatical. In (25b) and (25c), on the other hand, the *wh*-pronoun is assigned nominative case in one clause and accusative (or dative) in the other, which leads to ungrammaticality. The matching requirement follows from the fact that the *wh*-phrase is simultaneously in both clauses in a multidominant structure.

(25)

a.

Wer	nicht	stark	ist	muss	klug	sein.
who.NOM	not	strong	is	must	clever	be
'Who is not strong must be clever.'						

b.

*Wen/*wer	Gott	schwach	geschaffen	hat	muss	klug	sein.
whom.ACC/who.NOM	God	weak	created	has	must	clever	be
'Who God has created weak must be clever.'							

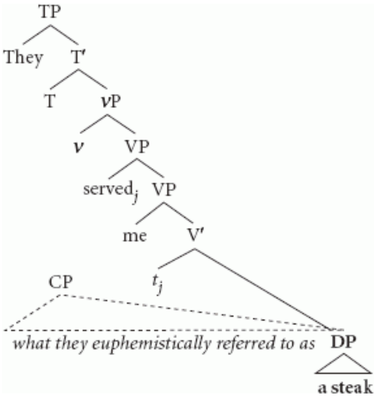
(p. 133) **c.**

*Wem/*wer	Gott	keine	Kraft	geschenkt	hat	muss	klug	sein.
who.DAT/who.NOM	God	no	strength	given	has	must	clever	be
'Who God has given no strength must be clever.' (van Riemsdijk 2006: 344)								

Transparent free relatives differ from standard ones in that the link between the matrix and the relative clause is established by some element other than the *wh*-pronoun. In (26a) below, for example, this element is the DP *a steak*. This leads van Riemsdijk (20o6) to propose the structure in (26b), simplified in irrelevant ways, in which this DP, which he refers to as *a callus*, is shared between the matrix and the relative clause.

(26)

a. They served me *what they euphemistically referred to* as a steak.
(van Riemsdijk 2006: 364)



b.

The relative CP in transparent free relatives is understood to be parenthetical, as it can be omitted without affecting the grammaticality of the sentence, as shown in (27).

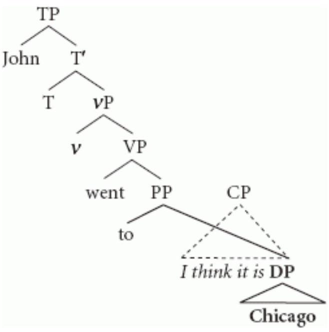
(27) They served me (what they euphemistically referred to as) a steak.

In this respect, transparent free relatives pattern with the so-called *wh*-amalgams and cleft-amalgams, illustrated in (28a–b) below. In both types of amalgams one clause is parenthetical to the other, and furthermore, one phrase serves as the link between the main and the parenthetical clause.

(28)

a. John invited *you will never guess* how many people to his party.
(Guimarães 2004: 46)
(p. 134) **b.** John went to *I think it is* Chicago on Sunday.
(Kluck and Heringa 2008: 2, citing Lakoff 1974)

This suggests a structure in which this 'linking phrase' is shared between the main and the parenthetical clause, such as the one in (29) for the example in (28b) above.



(29)

De Vries (2007) examines a number of predictions stemming from assigning such multidominant structures to parentheticals in general. Since the two clauses are not connected at the root, the straightforward prediction is that there should be no interaction between them.[14]

In transparent free relatives, for example, it is the shared element (not the *wh*-element) that counts for the purposes of coordination. This is evidenced by the contrast in (30a–b). Given the structure in (26b) above, coordination in (30a) is between a DP and an AP (in violation of the Law of the Coordination of Likes), whereas in (30b) it is between two DPs (and the result is grammatical).

(30)

a. *The creature changed from *a frog* and what can only be termed *slimy* into *a prince* and what truly deserves to be called *radiant.*
b. The creature changed from *a frog* and what can only be termed *a slimy abomination* into *a prince* and what truly deserves to be called a

prospective lover. (van Riemsdijk 2006: 365)

The invisibility of amalgams is evidenced by the fact that no quantifier binding is possible from the main clause into the parenthetical clause:

(31) *Everybody$_i$ wants to go *he$_i$ says it's France* this summer.
(Kluck and Heringa 2008: 7)

(p. 135) This concludes our brief survey of multidominant constructions. The upshot of the discussion in this section is that there are certain properties that we can expect from a multidominant structure. These are: matching whose exact nature depends on the nature of the shared element, insensitivity to islandhood (if the shared element does not undergo further movement), and certain types of transparency effects (if the two nodes do not get further integrated into the structure). In the next section, I turn to linearization of multidominant structures.

6.4 Linearization of multidominant structures

This section is devoted to the issue every multidominance account faces: how multidominant structures are mapped onto linear strings. This issue has received a fair amount of attention in the literature, and I will not be able to do justice to all the proposals that have been made to address it. My goal here is to show why the issue is important, outline the types of solutions that have been given to it, and point out the similarities and differences between. The differences are especially important, as different linearization constraints impose different restrictions on what counts as a possible multidominant structure.

Before getting into the details, let me outline some basic assumptions about the relationship between syntactic structure and linear order. First, I assume that the ordering of two elements (either precedence or subsequence) is a relationship that has to satisfy the following three conditions:

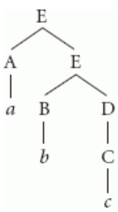
(32)
a. it is transitive (xRy & yRz → xRz)
b. it is total (for all distinct x, y, either xRy or yRx)
c. it is antisymmetric (not (xRy & yRx))

Second, in line with what seems to be a general consensus in minimalist literature, I assume Kayne's (1994) Linear Correspondence Axiom (LCA).[15] The LCA, stated formally in (33a) and informally in (33b), derives precedence from asymmetric c-command.
(33) Linear Correspondence Axiom (Kayne 1994: 6)
a. d(A) is a linear ordering of T.
b. Let X, Y be nonterminals and x, y terminals such that X dominates x and Y dominates y. Then, if X asymmetrically c-commands Y, x precedes y.

(p. 136) T in (33a) above is a set of all terminal nodes, A is a set of ordered pairs <X, Y> such that the first member asymmetrically c-commands the second one, and d(A) (the image under d of A) is the set of terminals that A dominates.[16]

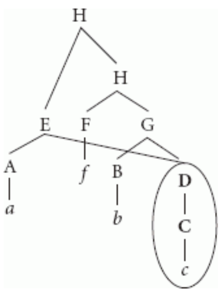
To illustrate, let us consider the structure given in (34a), with its A given in (34b) and its d(A) in (34c).[17] The resulting ordering is transitive, total, and antisymmetric, thus satisfying all the requirements linear ordering has to satisfy.

(34)



a.
b. A = {<A, B>, <A, D>, <A, C>, <B, C>}
c. d(A) = { <a, b>, <b, c>, <a, c>}

With this as background, let us turn to a schematic representation of a multidominant structure, given in (35a). In (35a), D (and whatever it dominates) is multiply dominated (by E and G). The set A for this structure is given in (35b), and d(A) in (35c).
(35)



a.
b. A = {<A, C>, <E, F>, <E, G>, <E, B>, <E, D>, <E, C>, <F, B>, <F, D>, <F, C>, <B, C>}
c. d(A) = {<a, c>, <a, f>, **<c, f>**, **<c, b>**, <c, c>, <a, b>, <f, b>, **<f, c>**, **<b, c >**

(p. 137) The problem arises with ordering the terminals of D, the multiply dominated node. The problematic pairs are in bold. On the one hand *c* should precede *f* and *b* (since C is dominated by E and E asymmetrically c-commands F and B). On the other hand, *c* should follow *f* and *b* (since C is asymmetrically c-commanded by F and B). The set d(A) thus contains <f, c>, <b, c>, <c, f>, and <c, b>. Such linearization violates the antisymmetry requirement on linear order; the same element cannot both precede another element and be preceded by it.[18] This is an important issue that needs to be resolved if we want to maintain the idea that there is a direct mapping between hierarchical structure and linear order, which seems desirable from a minimalist perspective.

In principle, there are four ways to resolve the issue of linearizing multidominant structures.[19] They are given in (36a–d):

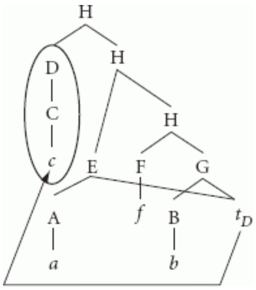
(36)

a. abandon multidominance since it violates the LCA;
b. abandon the LCA since it disallows multidominance;
c. modify multidominance structures to make them compatible with the LCA;
d. modify the LCA to make multidominance compatible with it.

I will not discuss the first two options, since there is good empirical evidence in favor of both multidominance (see section 6.3 above) and the LCA. The only reasonable options seem to be (36c) and (36d). Let me discuss them in turn.

My analysis of ATB *wh*-questions in Citko (2005) is an example of an approach that modifies multidominant structures to make them compatible with the LCA, as it keeps the LCA essentially intact but ‘undoes’ multidominance in the course of the derivation. It relies on a natural assumption that the LCA, being a principle of linearization, needs to apply only at a level relevant for linearization, namely the level of Spell-Out. This is in line with Chomsky's (1995b) suggestion on how to make LCA compatible with bare phrase structure theory, and Moro's (2000) dynamic approach to antisymmetry, which allows symmetric structures as long as they become antisymmetric by the time of Spell-Out. This suggests that multidominance is possible as long as the multiply dominated element moves overtly out of the shared position (crucially, to some non-shared position, in which it can be linearized according to the LCA). Crucially, copies of moved elements (being unpronounced) do not have to be linearized.[20] The structure in (37a) is thus going to be linearized as (37c).

(p. 138) **(37)**



a.
b. A = { <D, E>, <D, A>, <D, F>, <D, G>, <D, B>, <E, F>, <E, G>, <E, B>, <F, B>}
c. d(A) = { <c, a>, <c, f>, <c, b>, <a, f>, <a, b>, <f, b>}

Linking linearization to overt movement, while intuitively very appealing, is quite restrictive in what types of multidominant structures it allows. It works for ATB questions, gapping, and determiner sharing, but it is not going to work for constructions in which the multiply dominated element does not move, such as right node raising or questions with coordinated *wh*-pronouns. This shows that it needs to be augmented with a way to linearize multiply dominated elements in situ, which brings us to the option given in (36d), which is to make multidominant structures compatible with the LCA by modifying the LCA. Such approaches are quite common in the literature on right node raising (see e.g. Bachrach and Katzir 2009, Fox and Pesetsky 2007, Gračanin-Yüksek 2007, Johnson 2007, Wilder 1999, 2008).[21] I will illustrate this general approach with Wilder (1999), since many others (p. 139) build on his basic insight: that multiply dominated nodes are treated in a special way by the LCA.

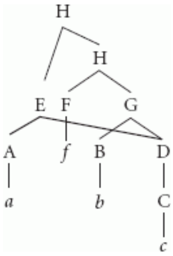
Wilder's basic idea is to allow the LCA to ignore some of the orderings involving multiply dominated nodes. In particular, by introducing the notion of full dominance (and making c-command sensitive to full dominance), he is able to exclude multiply dominated nodes from the set of nodes relevant for linearization. Full dominance, defined below, is stricter than standard dominance.

(38)

a. X fully dominates α iff X dominates α and X does not share α.
b. α is shared by X and Y iff (i) neither of X and Y dominates the other, and (ii) both X and Y dominate α.
c. X c-commands Y only if X does not fully dominate Y.
d. d(A) = the set of terminals fully dominated by A. (Wilder 1999: 590–1)

With these modifications in mind, let us look again at the multidominant structure in (34a) above, repeated below for convenience. The modified version of c-command allows E to c-command D, since E does not fully dominate D. Furthermore, the terminals of D, not being fully dominated by E, are not included in the image of E. Thus c-command by E does not cause the terminals of D to precede anything that E precedes. The resulting ordering is transitive, total, and antisymmetric.

(39)



a.
b. A = { <A, C>, <E, D>, <E, F>, <E, G>, <E, B>, <E, C>, <F, B>, <F, D>, <F, C>, <B, C>}
c. d(A) = {<a, c>, <a, f>, <a, b>, <f, b>, <f, c>, <b, c>}

The big advantage of Wilder's linearization algorithm is the fact that it can capture the sensitivity of right node raising to right edges. This sensitivity has been dubbed the Right Edge Generalization by Abels (2004), the Right Edge Effect by Johnson (2007), the Right Edge Condition by Wilder (1999), and the Right Edge Restriction by Sabbagh (2007), whose formulation is given in (40).

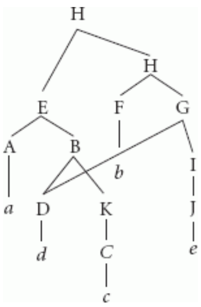
(p. 140) **(40)** Right Edge Restriction (Sabbagh 2007: 356)[22]
In the configuration:
[[$_A$….X…] Conj [$_B$…X…]]
X must be rightmost within A and B before either (i) X can be deleted from A; (ii) X can be rightward ATB-moved; or (iii) X can be multiply dominated by A and B.

The Right Edge Restriction is illustrated below; in (41a) the pivot (in italics) is rightmost in both conjuncts, whereas in the ungrammatical (41b–c) it is rightmost only in one of them.

(41)
a. I [$_{VP}$ invited into my house__] and [$_{VP}$ congratulated *all the winners*]
b. *I [$_{VP}$ gave __ a present] and [$_{VP}$ congratulated *all the winners*]
(Wilder 1999: 587)
c. *[$_{TP}$ I congratulated__] and [$_{TP}$ Mary will give *all the winners* a present]

Let us see how the Right Edge Restriction follows from Wilder's linearization algorithm by considering a structure in which the shared element is not final. (42a) below is an example of such a structure; *d* is followed by *c* within E and by *e* within G. If E and G are two conjuncts in a coordinate structure, this could be a representation of a right node raising construction violating the Right Edge Restriction. The set A for this structure is given in (42b), and its d(A) in (42c). The problematic pairs are given in bold; the set contains both <d, c> (by virtue of standard c-command within E) and <c, d> (by virtue of E commanding G). This is a violation of antisymmetry. If there were no *c* in the first conjunct, the violation (p. 141) would disappear, which is how this linearization algorithm derives the Right Edge Restriction on right node raising.[23]

(42)



a.
b. A = { <A, D>, <A, K>, <A, C>, <D, C>, <E, F>, <E, G>, <E, I>, <E, J>, <F, D>, <F, I>, <F, J>, <D, J>}
c. d(A) = {<a, d>, <a, c>, **<d, c>**, <a, b>, <a, e>, <c, b>, <c, e>, **<c, d>**, <b, d>, <b, e>, <d, e>}

The question that I would like to conclude with is whether movement-related and non-movement-related linearization algorithms discussed above are compatible with each other. I believe the answer to this question has to be yes; the evidence presented in sections 6.3.2 and 6.3.3 suggests that we need the insights of both in order to account for all the properties of ATB *wh*-questions and right node raising, for example.

An obvious linearization issue that arises here concerns the two types of doubly rooted non-coordinate multidominant structures discussed in section 6.3.3: free relatives and amalgams. They pose a problem for either of the linearization algorithms discussed above. I do not believe, however, that this warrants abandoning a multidominant analysis for these constructions. One way to salvage such an analysis would be to say that the two roots undergo re-Merge in a way that makes them linearizable according to the LCA. De Vries (2007), for example, argues that the relationship between the two roots is mediated by a Parenthetical head. In Citko (forthcoming c), I argue that in free relatives, the relative CP undergoes late adjunction to the head (which is the *wh*-phrase in standard free relatives and the pivot in transparent free relatives). I will leave the choice between the two as an open question here.

(p. 142)
6.5 Conclusion

My goal in this chapter was to show that there are no theoretical objections to multidominance and that there is a fair amount of empirical support for it. I showed that the existence of multidominant structures follows from standard assumptions about the nature of Merge and Move, thus removing the theoretical objections that have been levied against multidominance in the past. Furthermore, I showed that multidominance allows us to capture a number of properties of a number of unrelated constructions, thus adding empirical support in favor of its existence. The issue of whether a single linearization algorithm can cover all of them remains somewhat open, and I hope this chapter will stimulate further research into this issue.

Notes:

(1) The term 'multidominance' is sometimes used interchangeably with 'multidimensionality'. Here, I keep the two distinct, reserving the term 'multidimensionality' for constructions consisting of two separate substructures (or planes, in Muadz's 1991 terminology) that are only brought together as part of a linearization process.

(2) Multidominant structures do not meet the standard definition of a tree, given in (i) below. This, by itself, however, cannot be a reason to banish them from the grammar. All this means is that we need a somewhat more relaxed approach to what counts as an admissible syntactic object, such as the one developed by McCawley (1982), Moltmann (1992), or Gärtner (2002).

((i)) A *constituent structure tree* is a mathematical configuration ⟨N, Q, D, P, L⟩, where:

N is a finite set, the set of *nodes*;
Q is a finite set, the set of *labels*;
D is a weak partial order [i.e. it is transitive, reflexive, and antisymmetric] in N × N, the *dominance* relation;
P is a strict partial order [i.e. it is transitive, irreflexive, and asymmetric] in N × N, the *precedence* relation;
L is a function from N into Q, the *labeling junction*.

And such that the following conditions hold:
(a.) (∃x ∈ N) (∀y ∈ N) ⟨x, y⟩ ∈ D (*Single Root Condition*)
(b.) (∀x, y ∈ N) ((⟨x, y⟩ ∈ P v ⟨y, x⟩ ∈ P ↔ (⟨x, y⟩ ∉ D & ⟨y, x⟩ ∉ D)) (*Exclusivity Condition*)
(c.) (∀w, x, y, z ∈ N) ((⟨w, z⟩ ∈ P & ⟨w, y⟩ ∈ D & ⟨x, z⟩ ∈ D) ↔ ⟨y, z⟩ ∈ P) (*Nontangling Condition*) (Partee et al. 1990: 443–4)

(3) De Vries (2007) refers to the two as External Remerge and Internal Remerge, respectively.

(4) This is the derivation Bobaljik and Brown (1997) propose for head movement in general, as it does not violate the Extension Condition.

(5) The representations in (6a–c) ignore directionality.

(6) The references also do not include non-multidominant accounts of these constructions.

(7) The discussion of ATB *wh*-questions is a summary of Citko (2005). See Citko (forthcoming c) for a monograph-length treatment of ATB *wh*-questions, *wh*-questions with conjoined *wh*-pronouns, gapping, right node raising, serial verb constructions, and free relatives.

(8) I leave open the possibility that other elements in ATB questions might be shared as well.

(9) Interestingly, violations of the first part of the Coordinate Structure Constraint (the so-called Conjunct Condition, which prohibits movement of the entire conjunct) do not get ameliorated by ATB movement, as shown by the following contrast:

((i)) *$What_i$ did John read t_i and a book?
((ii)) *$What_i$ did John read t_i and t_i?

This, however, could be reduced to a more general prohibition against null conjuncts, illustrated in (iii–iv):
((iii)) *John read a book and __.
((iv)) *John read __ and a book.

(10) Gračanin-Yüksek (2007) derives a number of other properties of questions with conjoined *wh*-pronouns from this structure, such as the restriction illustrated in (i), which shows that in English such questions are only possible with optionally transitive verbs (see also Whitman 2005 for a discussion of this restriction).

((i)) *What and why did you devour?

Furthermore, Gračanin-Yüksek (2007) proposes a different structure for questions with coordinated *wh*-pronouns in Slavic languages (see also Citko forthcoming a, Gribanova 2009, Scott 2010).

(11) In cases of complex gaps, the verbal complex (vacated by the remnants) undergoes remnant ATB movement to the specifier of PredP.

(12) One could argue that the ungrammaticality of the examples in (21–22) might have to do with the fact that the two verbal forms (the present and the passive participle) are distinct. However, the same judgments obtain if the two forms are identical, as shown in (i–ii) for gapping:

((i)) *Some brought cookies and cakes ~~were brought~~ by others.
((ii)) *Cookies were brought by some and others ~~brought~~ cakes.

(13) I depart from Johnson (2000a) and Lin (2000, 2002), who derive the effects of determiner sharing from generating the determiner above the coordination level.

(14) De Vries (2007) focuses on parentheticals and examines their behavior with respect to diagnostics involving movement, idiom chunks, binding, and negative and positive polarity licensing. In general, he shows that the paratactic constituent does not interact with the main clause with respect to movement and binding diagnostics.

(15) See, however, Citko (forthcoming c) for arguments that the grammar also has to allow symmetric dependencies.

(16) The definitions of c-command and asymmetric c-command the LCA relies on are given below:

((i)) X *c-commands* Yiff X and Y are categories and X excludes Y, and every category that dominates X dominates Y.
((ii)) X *asymmetrically c-commands* Y if X c-commands Y and Y does not c-command X.

(17) Because of the segment/category distinction, the lower E in (34a) does not c-command A.

(18) Note that d(A) contains the pair <c, c> which also raises issues concerning reflexivity.

(19) I focus here on ways to linearize multidominant structures that are compatible with the LCA.

(20) Deletion is another way to 'undo' multidominance.

(21) There are, of course, differences between these approaches, a full consideration of which goes beyond the scope of this chapter. For example, Bachrach and Katzir (2009) work out a linearization algorithm compatible with cyclic Spell-Out. Their basic idea is that only fully dominated constituents can be spelled-out, and a spelled-out constituent cannot be changed in a way (as also argued on independent grounds by Fox and Pesetsky 2004).

Johnson (2007) allows the LCA to generate inconsistent linearization orderings as long they contain a subset that meets the three requirements on linear order (totality, antisymmetry, and transitivity) and do not violate the well-formedness alignment constraints in (i–ii).

((i)) Align the right edge of αP to the right edge of the projection of αP.
((ii)) Align the left/right edge of α to the left/right edge of α's head. (Johnson 2007: 12)

Gračanin-Yüksek (2007) proposes an overarching constraint on multidominant structures, which she dubs Constraint on Sharing (COSH), arguing that its effects can also be derived from a modified version of the LCA. Informally stated, COSH requires the mothers of shared nodes to dominate identical sets of terminal nodes.

(22) Alternative formulations are given below:

((i)) Right Edge Effect (Johnson 2007)
Let α be shared by β and γ through Right Node Raising, and let β precede γ. The right edge of α must align with the right edge of γ, and be able to align with the right edge of β.
((ii)) Right Edge Generalization (Abels 2004: 48)
In a configuration of the form $[_{XP}1$ … Y …] *conj* $[_{XP2}$ … Y …], Y must be the rightmost element within XP_1 and XP_2 before RNR may apply.
((iii)) Right Edge Condition (Wilder 1999: 587)
If α surfaces in the final conjunct (RNR), gap(s) corresponding to α must be at the right edge of their non-final conjuncts.

Wilder's constraint differs in empirical predictions from the others in that it allows the pivot to be non-final in the final conjunct. Wilder cites the following data in support of his formulation:

((iv)) John should fetch __ and give __ *the book* to Mary.

It is not clear, however, that such examples do not simply involve coordination of two verbs (rather than RNR of *the book*). Fox and Pesetsky (2007) also note that Wilder's data are far from being clear in this respect.

(23) Note that Wilder's linearization algorithm thus allows the shared element to be non-final in the final conjunct.

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The Copy Theory

Jairo Nunes

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Abstract and Keywords

This article reviews the general features of the Copy Theory of movement, focusing on some empirical gains prompted by its adoption in the Minimalist Program. It is organized as follows. Section 7.2 presents Chomsky's original conceptual arguments for reinterpreting traces as copies. Section 7.3 discusses some of the questions that the Copy Theory poses to the syntax-phonetic form mapping and presents Nunes' approach to phonetic realization of copies in terms of linearization and economy computations. Section 7.4 provides an overview of the kinds of empirical material that may receive a natural account within the Copy Theory but remain rather mysterious within the Trace Theory. Section 7.5 discusses how the Copy Theory makes different predictions for the current debate on how to analyse obligatory control within minimalism. Some concluding remarks are presented in Section 7.6.

Keywords: movement, minimalist program, Chomsky, traces, copies, syntax-phonetic form

7.1 Introduction

One of the major goals of the generative enterprise has been to capture the so-called 'displacement' property of human languages—the fact that syntactic constituents may be interpretively associated with positions different from the ones where they are phonetically realized. The Government and Binding (GB) model captured this property in terms of the Trace Theory, according to which a movement operation displacing a given element from one structural position to another leaves behind a co-indexed trace. Under this view, traces are conceived of as phonetically unrealized categories that inherit the relevant interpretive properties of the moved element, forming with it a discontinuous object—a (nontrivial) chain. The derivation of a sentence such as (1), for instance, is as represented in (2), where *John* moves from the object to the subject position.

(1) John was arrested.
(2) [John$_i$ was arrested t$_i$]

With the wholesale conceptual evaluation of the GB apparatus that arose with the emergence of the minimalist program (Chomsky 1993, 1995c), the Trace Theory of movement became an obvious topic to be examined under minimalist lenses. Chomsky (1993) argues that the Trace Theory should actually be abandoned in favor of an earlier interpretation of movement as copying. More specifically, he proposes **(p. 144)** that a movement operation leaves behind a copy of the moved element which gets deleted in the phonological component (in case of overt movement). From this perspective, the derivation of (1) proceeds along the lines of (3), where the crossed material represents lack of phonetic realization at PF.

(3)
a. [was arrested John]
b. Copy:

John [was arrested John]
c. Merge:
[John was arrested John]
d. Delete:
[John was arrested ~~John~~]

Although Chomsky's arguments for incorporating the Copy Theory of movement into the minimalist framework were largely architectural in nature, the Copy Theory made it possible to analyze recalcitrant data and opened new avenues to analyze the syntax–PF mapping. By presenting us with a clear case where the simplification of the theoretical apparatus has led to significant gains in empirical coverage, the Copy Theory has become one of the most stable pillars of minimalist theorizing. As a consequence of its solid status within the framework, the Copy Theory has also been employed to empirically distinguish between competing minimalist analyses, as is the case of the PRO-based and movement-based approaches to obligatory control.

This chapter reviews the general features of the Copy Theory of movement, focusing on some empirical gains prompted by its adoption in the Minimalist Program. The chapter is organized as follows. In section 7.2 I present Chomsky's (1993) original conceptual arguments for reinterpreting traces as copies. In section 7.3 I discuss some of the questions that the Copy Theory pose to the syntax–PF mapping and present Nunes' (1995, 1999, 2004) approach to phonetic realization of copies in terms of linearization and economy computations. In section 7.4 I present an overview of the kinds of empirical material that may receive a natural account within the Copy Theory, but remain rather mysterious within the Trace Theory. In section 7.5 I discuss how the Copy Theory makes different predictions for the current debate on how to analyze obligatory control within minimalism. Some concluding remarks are presented in section 7.6.

7.2 Conceptual motivations for the Copy Theory

Sentences such as (4a) and (4b) below are transparent examples of the displacement property of human languages. In (4a) the anaphor requires being interpreted (p. 145) in the object position in order to be c-commanded by the subject, whereas the idiom chunk *the shit* in (4b) must be interpreted in [Spec, vP] in order to form a constituent with *hit the fan*. The representation of these sentences under the Trace Theory given in (5) must therefore be supplemented with extra assumptions in order to ensure that the required configurations obtain at the point where interpretation applies.

(4)
a. Which picture of himself did John see?
b. The shit will hit the fan.

(5)
a. [[which picture of himself]$_i$ did John see t_i]
b. [[the shit]$_i$ will [t_i hit the fan]]

Within GB, two main lines of inquiry were explored to account for facts like these. The interpretation of the anaphor and the idiom was taken to be computed at a level prior to movement (D-Structure) or an operation of reconstruction applying in the LF component was employed to restore the moved material back in its original position. Chomsky (1993) observes that both alternatives are suspect from a minimalist perspective. The first alternative is at odds with one of the major tenets of minimalist downsizing, which is the elimination of non-interface levels of representation such as D-Structure or S-Structure. The second approach circumvents this problem by computing the relevant relations at LF, but at the price of invoking a lowering application of movement (the reconstruction operation).

As Chomsky points out, the Copy Theory remedies both of these problems. If the sentences in (4) are to be associated to the structures in (6) below, the expected interpretation can be computed at LF via the lower copies. Crucially, the lower copies are deleted in the phonological component, but are available for interpretation at LF.

(6)
a. [[which picture of himself] did John see ~~[which picture of himself]~~]

b. [[the shit] will [[~~[the shit]~~ hit [the fan]]]

Another important conceptual advantage of the Copy Theory over the Trace Theory has to do with the Inclusiveness Condition. Chomsky (1995c: 228)has proposed that the mapping from the numeration to LF should be subject to an Inclusiveness Condition requiring that an LF object be built from the features of the lexical items of the numeration. The Inclusiveness Condition can be viewed as a metatheoretical condition that ensures internal coherence within the model. First, it restricts the reference set of derivations that can be compared for economy purposes. If the system could add material that is not present in the numeration in the course of syntactic computations, the role of the numeration in determining the class of comparable derivations would be completely undermined. Thus, given the minimalist assumption that economy matters in the computations from the numeration to LF, something like the Inclusiveness Condition must be enforced in the system. The second important role played by the Inclusiveness Condition is that it ensures that the inventory of syntactic primitives is kept to a minimum, by (p. 146) preventing the syntactic component from creating objects that cannot be defined in terms of the atoms that feed the derivation.

Given this general picture, it is clear that the Trace Theory within GB was ripe for a minimalist reanalysis, as it is flagrantly incompatible with the underpinnings of the Inclusiveness Condition. Traces are not part of the initial array, but are introduced in the course of the computation. Besides, they are taken to be independent grammatical formatives, with their own properties and licensing conditions. In comparison, the Copy Theory provides a much more congenial way for movement operations to comply with the Inclusiveness Condition. First, a copy is not a new grammatical formative; it is either a lexical item or a phrase built from lexical items[1] Second, the copies are built from the material that was present in the numeration. Note that it is not the introduction of objects in the course of the derivation *per se* that is problematic, for both traces and copies are introduced in this way. The difference is that the operation Copy, like the structure-building operation Merge, creates an object by manipulating material that is available in the numeration, thus permitting a simple formulation of the reference set for economy computations. By contrast, under the Trace Theory traces pop up as completely new elements in the computation, thereby requiring that the reference set be further specified with respect to which new elements can or cannot be introduced in the computation out of the blue.

At first sight, this way of satisfying the Inclusiveness Condition may look too costly, as it seems to require the introduction of two operations in the system: Copy and Delete (cf. (3)). Appearances are misleading, though. These operations are in fact independently motivated. Delete, for example, must be invoked in the derivation of ellipsis constructions,[2] regardless of whether it is interpreted as an erasure operation in lexicalist approaches or as a blockage to late insertion in approaches based on Distributed Morphology.[3] As for Copy, standard cases of morphological reduplication provide evidence of its effects elsewhere in the system and, as we will see in section 7.4 below, we may also find unequivocal reflexes of its applications in the syntactic component when more than one chain link gets pronounced. But postponing the presentation of this empirical evidence until section 7.4, it is worth observing at this point that a Copy-like operation must also be independently resorted to in the mapping from the lexicon to the numeration/derivation.[4] After all, when we say we take an item from the lexicon to form a given numeration, we (p. 147) definitely do not mean that the lexicon has lost one item and is now smaller. Rather, we tacitly assume that numerations are formed by *copying* items from the lexicon.

To summarize: in addition to conforming to the Inclusiveness Condition, the Copy Theory considerably simplifies the analysis of reconstruction phenomena. Furthermore, by making it possible to treat reconstruction as an LF phenomenon, the Copy Theory contributes to the attempt to eliminate non-interface levels of representation. Finally, by eliminating traces *qua* grammatical formatives, it reduces the number of theoretical primitives in our inventory: if traces are copies, they are either lexical items or complex objects built from lexical items (see note 1 above).[5]

7.3 The Copy Theory and syntax-PF mapping

In his reanalysis of reconstruction in terms of the Copy Theory, Chomsky (1993) argues that there is more than one possibility for the interpretive systems to read LF objects. He proposes that the ambiguity of a sentence like (7) below, forinstance, is due to the different parts of the *wh*-chain that the interpretive systems may compute. If the whole *wh*-phrase is computed upstairs after deletion of the lower copy, as represented in (8a) with the outlined material annotating lack of interpetation at LF, we obtain the reading under which the anaphor is bound by the matrix (p. 148) subject. By contrast, if only the *wh*-element is computed upstairs after scattered deletion within the *wh*-chain, as in (8b), we get the embedded subject reading for the anaphor.[6]

(7) John$_i$ wonders which picture of himself$_{i/j}$ Bill$_j$ saw

(8)

a. John wonders [$_{CP}$ [which picture of himself] [$_{IP}$ Bill saw [which picture of himself]]]

b. John wonders [$_{CP}$ [which picture of himself] [$_{IP}$ Bill saw [which picture of himself]]]

In addition to these two interpretive alternatives, there also arises the logical possibility that only a lower link gets interpreted after deletion of the higher copy, as is arguably the case of reconstruction of idiom chunks in (9) (cf. (4b)).

(9)[[the shit] will [[the shit] hit the fan]]

On the LF side we thus have some choice in deciding how deletion should take place in nontrivial chains (see note 6). That being so, questions arise with respect to the syntax–PF mapping. Take the structure in (10) below, for instance, which has been formed after the object DP moved to the subject position. One wonders why the only well-formed PF output for (10) is (11a), where only the highest copy is pronounced. The ungrammaticality of (11b), with no deletion, is particularly interesting, as it is the most transparent output with respect to the structure that reaches LF.

(10) [[the student] was arrested [the student]]

(11) *PF outputs*:

a. [[the student] was arrested [~~the student~~]]

b. *[[the student] was arrested [the student]]

c. *[~~[the student]~~ was arrested [the student]]

d. *[[the ~~student~~] was arrested [~~the~~ student]]

(p. 149) At first sight, the Trace Theory fares better in this regard, as it need not be concerned with potentially different phonetic outputs for a given nontrivial chain. However, appearances are again illusory. The issue of phonetic realization of chains is mute within the Trace Theory because traces are *stipulated* to be phonetically null. A truly explanatory account of movement in terms of traces should provide an appropriate answer for why traces are necessarily devoid of phonetic content. In other words, even under the Trace Theory, the fact that only chain heads are phonetically realized needs explanation. Upon close inspection, this conundrum may in fact provide the Copy Theory with the upper hand on this issue. Consider why.

In section 7.2, we discussed conceptual reasons for why the Copy Theory was a better alternative than the Trace Theory, given minimalist considerations regarding the mapping to LF. Recall that the argument was not that the Trace Theory was empirically flawed, but that it required additional assumptions that did not fit snugly within the general architectural features of minimalism. By contrast, the mapping from the syntactic component to PF may offer a deadly empirical argument against the Trace Theory. Suppose, for instance, that some constructions (in some languages) may allow pronunciation of more than one copy, pronunciation of a lower copy, or scattered deletion within a chain in a way analogous to (11b–d). If such cases do exist, we will then have a very powerful argument for choosing the Copy Theory over the Trace Theory. The latter has no room to accommodate facts like these, for it is a defining property of traces that they are phonetically empty.

section 7.4 below is devoted to showing that we do indeed find PF outputs parallel to (11b–d). But before we examine such cases, let us first discuss why the PF output represented in (11a), where only the head of the chain is pronounced, is by far the most common pattern found across languages. It is productive to break the puzzle into two different questions: (i) why is it the case that in general a chain cannot surface with all of its links phonetically realized (cf. (11b))? and (ii) why is it the case that full pronunciation of the highest copy is in general the only grammatical PF output (cf. (11a) vs. (11c–d))?

By combining a convergence requirement with economy considerations, Nunes (1995, 1999, 2004) provides a general answer to these questions that is able to account for both the standard option illustrated in (11a) and exceptional cases that parallel the options in (11b–d). The convergence ingredient is related to linearization at PF.

The gist of the proposal is that copies count as 'the same' for purposes of linearization because they are non-distinct elements (i.e. they relate to the same occurrences of lexical items of the numeration; see note 5 above) and this creates problems. Take the structure in (10), for example. Given that the higher occurrence of *[the student]* asymmetrically c-commands was, Kayne's (1994) Linear Correspondence Axiom (LCA) dictates that both *the* and *student* should precede *was*. Likewise, given that *was* asymmetrically c-commands the lower occurrence of *[the student]*, it should precede *the* and *book* in compliance with the LCA. Given (p. 150) that these occurrences of *[the book]* are non-distinct, we reach acontradiction: *was* should precede and be preceded by *the* and *student*. Similarly, given that the higher occurrence of *[the student]* asymmetrically c-commands the lower one, we obtain the undesirable result that *the*, for instance, should be required to precede itself.

To make the same point in a slightly different way, the fact that a nontrivial chain is a discontinuous object that simultaneously occupies different structural positions in the syntactic structure creates an impasse for linearization. On the one hand, a chain cannot be assigned a single slot in the PF linear sequence resulting from the LCA, for it is associated with more than one structural position; on the other hand, the assignment of multiple slots should create contradictory requirements, preventing the whole structure from being linearized. Thus, the reason why a chain cannot (in general) surface with all of its links phonetically realized (cf. (11b)) under this view is that the structure containing it cannot be linearized.

Nunes (1995, 1999, 2004) argues that deletion comes into play in this scenario as a rescuing strategy to permit the linearization of structures containing chains. More specifically, deletion of the 'repeated' material within chains before linearization ('Chain Reduction', in Nunes' terms) circumvents the problem of linearizing *was* with respect to *the* and *student* in (10). If the material of the chain CH = (*[the student]*, *[the student]*) is deleted in any of the ways depicted in (11a, c,d), the structure in (10) can be linearized without any problems. The question now is why only the deletion sketched in (11a) yields an acceptable sentence.

This is the point where economy plays a crucial role. More specifically, economy considerations should ensure that deletion applies as few times as possible. Applying to the DP chain in (10), Chain Reduction may yield the output in (11d), with two applications of deletion, or the outputs in (11a) and (11c), with a single application targeting the whole DP node. Once there is arguably no convergence problem resulting from these reductions, the three derivations are eligible for economy comparison and the derivation yielding (11d) is excluded for employing more operations of deletion than necessary. What is now missing is an explanation of why the actual reduction of the DP chain in (10) must involve the deletion of the lower copy, rather than the head of the chain (cf. (11a) vs. (11c)), despite the fact that both reductions may employ a single operation of deletion targeting the whole DP node. Obviously, we cannot simply say that lower copies must delete. Conceptually, that would amount to reintroducing traces and empirically, it would be just wrong, as we will see in section 7.4.

The most plausible answer should again be formulated in economy terms.[7] There should be some factor that makes the pronunciation of the highest copy more economical in the general case. One possibility is that such an independent factor is feature checking/valuation (see Nunes (1995, 1999, 2004). If the highest copy always has more features checked/valued than the lower copies, it should be (p. 151) the optimal candidate for phonetic realization. Actual implementation of this idea depends on specific assumptions regarding the inner workings of feature checking/valuation and the relation among copies when one of them undergoes feature checking/valuation.[8]

For concreteness, I will here assume Bošković's (2007) proposal that a given element can only have its uninterpretable features valued if it acts as a probe. Under this view, the derivation of (10) proceeds along the lines of (12) below. Given (12a), T probes the structure and has its *Φ*-features valued by agreeing with *[the student]*, yielding (12b). In order to have its Case-feature valued, the internal argument then moves to [Spec, TP] and from this position, it probes T and values its Case-feature, yielding (12c).

(12)

a. [T $_{\Phi:?}$ be arrested [the student]$_{\text{Case:?}}$]
b. [T $_{\Phi:3SG}$ be arrested [the student]$_{\text{case:?}}$]
c. [[the student]$_{\text{case:NOM}}$T $_{\Phi:3SG}$ be arrested [the student]$_{\text{case:?}}$]

The structure in (12c) arguably causes the derivation to crash at LF, as the lower copy does not have its Case feature valued. Let us then assume that once a given element has its uninterpretable features valued, it is allowed to probe the structure again and value the features of its lower copies (in a domino fashion if more than one copy is involved). If so, before the structure in (12c) is shipped to the conceptual-intentional (C-I) interface, the upper copy values the Case feature of the lower copy, yielding (13) below. This suggestion captures in a derivational fashion Chomsky's (1995c: 381, n. 12) proposal that ‘the features of chain are considered a unit: if one is affected by an operation, all are.’

(13) [[the student]$_{Case:NOM}$ T $_{\Phi:3SG}$ be arrested [the student]$_{Case:NOM}$]

We now have all the ingredients we need. At the point where the structure in (12c) is assembled, probing by the upper copy is not required for PF purposes. If no such probing is forced to apply before Spell-Out, the structure in (12c) is shipped to the phonological component as is and the probing yielding (13) takes place after Spell-Out. In the phonological component, the choice for pronunciation between (11a) and (11c) is then determined by the copy that has its features valued (cf. (12c)), as this information is needed by morphology. Thus, (11a) trumps (11c).

To sum up: the combination of a convergence requirement in terms of linearization and economy considerations regarding the number of applications of deletion provides an account for why a chain (in general) does not surface with all of its links phonetically realized (the structure containing such a chain cannot be linearized) and why scattered deletion constructions are uncommon (they employ an unnecessary number of applications of deletion). Finally, an independent (p. 152) asymmetry among copies due to feature checking/valuation establishes a specific economy metric that favors deletion of lower copies.

For the sake of completeness, let us consider how this proposal accounts for the standard output of remnant movement constructions (see section 7.4.3 below for further discussion). Take the derivation of (14) below, for instance, whose representations under the Trace Theory and the Copy Theory are given in (15). The interesting thing about the representation in (15b) (see Gärtner 1998) is that the leftmost copy of *John* gets deleted despite the fact that it doesn't form a chain with either of the other copies (it neither c-commands nor is c-commanded by the other copies, for instance). This potential problem for the Copy Theory is in fact analogous to the one faced by the Trace theory in accounting for how t_j in (15a) is not c-commanded by its antecedent.

(14) … and elected, John was.

(15)

a. …and [$_{XP}$ [elected t_i]$_k$ [$_{X'}$ X[$_{TP}$ John$_i$ [$_{T'}$ was t_k]]]]

b. …and [$_{XP}$ [elected ~~John~~] [$_{X'}$ X [$_{TP}$ John [$_{T'}$ was ~~[elected John]~~]]]]

Within the Copy Theory, there are two possible approaches to this issue. A more representational answer is offered in Nunes (2003, 2004), building on Chomsky's (1995c: 300) observation that the representation of a chain such as CH = (*a*, *a*) should be seen as a notational abbreviation of CH = ((a, K), (a, L)), where K and L are each the sister of one occurrence of a. In other words, the individual links of a chain must be identified not only in terms of their content, but also in terms of their local structural configuration. Hence, movement of *John* in (15b) first forms the chain CH = ((*John, V*), (*John, elected*)) and movement of the remnant VP then forms the chain CH_2 = ((*[elected John]*, X'), (*[elected John], was*)). Under the assumption that Spell-Out ships the whole structure in (15b) to the phonological component, Chain Reduction inspects CH and instructs the phonological component to delete the occurrence of *John* that is the sister of *elected*. Interestingly, there are two elements in (15b) that satisfy this description: the leftmost and the rightmost copies of *John*. In fact, these two copies are technically identical: they are non-distinct in terms of the initial numeration, they have participated in no checking relations, and their sisters are non-distinct. Assuming that the phonological component blindly scans the structure to carry out the deletion instructed by Chain Reduction, it ends up deleting the two copies that satisfy the instruction, as represented in (15b); Chain Reduction of CH_2 then deletes the lower copy of VP and the sentence in (14)isderived.

Under a more derivational approach (see Bošković and Nunes 2007), linearization/Chain Reduction applies as the phonological component is fed with Spell-Out units. From this perspective, the system spells out TP after the structure in (16a) below is built and Chain Reduction deletes the lower copy of *John*. From this point on, the copy of *John* in the object position will be unavailable to any operation of (p. 153) the phonological component. Hence, movement of VP later on in the derivation, as shown in (16b), will be oblivious of this copy. After the whole structure in (16c) is spelled out, deletion of the lower VP copy then yields the sentence in (14).

(16)

a. [$_{XP}$ X [$_{TP}$ John [$_{T'}$ was [$_{VP}$ elected ~~John~~]]]]

b. [$_{XP}$ [$_{VP}$ elected ~~John~~] [$_{X'}$ X [$_{TP}$ John [$_{T'}$ was [$_{VP}$ elected ~~John~~]]]]]
c. [$_{XP}$ [$_{VP}$ elected ~~John~~] [$_{X'}$ X [$_{TP}$ John [$_{T'}$ was ~~[$_{VP}$ elected John]~~]]]]

This is not the place to decide between these alternatives. For our purposes, it suffices that both of them correctly enforce deletion of traces in standard remnant movement constructions and can also handle the remnant movement constructions involving multiple copies to be discussed in section 7.4.3 below.

7.4 Empirical payoff

As mentioned in section 7.3, the most powerful argument for the Copy Theory should come from the mapping from Spell-Out to PF. If lower copies can somehow be pronounced, we will have a knock-out argument for the Copy Theory, for under the Trace Theory traces are phonetically null by definition.

Below we consider several cases that instantiate the possibility that lower copies can be pronounced.[9]

7.4.1 Phonetic realization of a lower copy

In section 7.3, the preference for pronouncing chain heads was taken to follow from an economy condition. Given the derivation sketched in (17) below, where α moves to value its uninterpretable feature F, the structure in (17b) can be shipped to the phonological component without the additional valuation of the lower copy by the higher one. Such valuation, as sketched in (18), is only required for LF purposes. Once the additional valuation is not required to apply before Spell-Out, (local) economy prevents it from doing so and Spell-Out applies to (17b), yielding the preference for deleting lower copies.

(17)
a. [H [… $_{aF:?}$ …]]
b. [$_{aF:✓}$ H [… $_{aF:?}$ …]]

(18) [$_{aF:✓}$ H [… $_{aF:?}$…]]

(p. 154) Now suppose that in a given derivation, independent convergence requirements of the phonological component ban the pronunciation of the higher copy of α in (17b). In such circumstances, the system will then be forced to trigger valuation in (18) before Spell-Out in order to ensure convergence. Once (18) is shipped to the phonological component, each copy has its features valued and is eligible for pronunciation. But if the higher copy violates well-formedness conditions of the phonological component, it should be deleted and the lower copy should be pronounced instead, as sketched in (19).

(19) [~~$_{aF:✓}$~~ H[… $_{αF:✓}$…]]

The scenario depicted above where a lower copy is pronounced instead of the head of the chain has been increasingly documented in the literature (see note 9 above). Consider the contrast between (20) and (21) below, for instance. (20) illustrates the well-known fact that Romanian is a multiple *wh*-fronting language; hence the unacceptability of the *wh*- in situ in (20b). (21), on the other hand, seems to be an exception to the paradigm illustrated in (20), in that a *wh*-element in situ is allowed.

(20) Romanian
a.

Cine	ce	precede?
Who	what	precedes

b.

*Cine	precede	ce?
who	preced	what
‘Who precedes what?’		

(21) Romanian

a.

*Ce	preced	ce?
what	precede?	what
‘What preceds what’		

b.

Ce	precede	ce?
what	precede	what
‘What precedes what?’		

Bošković (2002b), however, argues that the appearances here are deceiving. The unacceptability of (21a) is related to a restriction in the phonological component prohibiting adjacent occurrences of *ce* ‘what’. That is, from a syntactic point of view, there is no difference between (20) and (21); we have multiple *wh*-fronting in both cases. It just happens that if the higher copy of the moved object of (21) is realized, it will violate this ban on adjacent identical words, which is found in several languages.[10] The phonological system then deletes the higher copy of the object *ce* ‘what’, as sketched in (22) below, allowing the structure both to be linearized and to **(p. 155)** comply with this adjacency restriction. Bošković provides independent evidence for the deletion sketched in(22) by showing that the object in (21b) patterns like moved *wh*-objects in being able to license a parasitic gap, as shown in (23), something that a truly in situ *wh*-object cannot do.

(22)[ce$_{SU}$ [~~ce$_{OB}$~~ [~~cesu~~ precede ce$_{OB}$]]

(23) Romanian

Ce	precede	ce	fara	sa	influenteze?
What	precedes	what	without	SUBF.PRT	influence.3SG
‘What precedes what$_i$ without influencing iti?’					

Another interesting argument for pronunciation of lower copies is provided by Bobaljik's (1995a) account of Holmberg's (1986) generalization (see also Bobaljik 2002). Holmberg (1986) has observed that object shift in Scandinavian can take place in matrix main verb V2 clauses, but not in auxiliary+participle clauses or embedded clauses, which do not involve main verb movement. This can be seen in (24a), where *ekki* ‘not’ is taken to mark the VP boundary.

(24)

a.

Í gær	máluðu	stúdentarnir	husið$_i$	[$_{VP}$ ekki t$_i$]	(Icelandic)
yesterday	painted	the-students	the-house	not	
‘The students didn't paint the house yesterday.’					

b.

*at	Peter	den$_i$	[$_{VP}$ læste t$_i$]	(Danish)
that	Peter	it	read	

c.

at	Peter	[$_{VP}$ læste den]
that	Peter	read it
‘that Peter read it’		

d.

*Hann	hefur	bókina$_i$	[$_{VP}$ lesið t$_i$]	(Icelandic)
he	has	the-book	read	

e.

Hann	hefur	[$_{VP}$ lesið bókina]	
he	has	read	the-book
‘He has read the book.’			

Bobaljik argues that in clauses in which V-movement does not take place, the relevant Infl head (finite or participial) must be adjacent to the verbal head in order for them to undergo morphological merger after Spell-cut. Thus, obligatory overt movement of (specific, non-contrastive definite) objects with standard pronunciation of the head of the chain disrupts the adjacency between Infl and V and yields an ungrammatical result (cf. (24b)/(24d)). Bobaljik proposes that in these circumstances, the head of the object shift chain is deleted and its tail is pronounced, as sketched in (25) (cf. (24c)/(24e)), which allows morphological merger between Infl and the verb, as they are now adjacent.

(p. 156) **(25)**

a. [at [$_{IP}$ Peter I [$_{AgroP}$ ~~den~~ [$_{VP}$ læste den]]]]
b. [hann hefur [$_{PartP}$ Part [$_{AgroP}$ ~~bókina~~ [$_{VP}$ lesið bókina]]]]

The possibility of pronouncing lower copies due to independent requirements of the phonological components can also account for some interesting facts concerning V2 in Northern Norwegian, as argued by Bošković (2001). Rice and Svenonius (1998) have observed that the V2 requirement in Northern Norwegian is stricter than in other Germanic V2 languages in that the material preceding the verb must minimally contain one foot (i.e. two syllables), as illustrated by the contrast in (26). Rice and Svenonius further note that (26b) can be saved by using the *wh*-subject–V order, as shown in (27).

(26) Northern Norwegian

a.

Korsen	kom	ho	hit?
how	came	she	here
'How did she get here?'			

b.

*Kor	kom	du	fra?
where	came	you	from
'Where did you come from?'			

(27) Northern Norwegian:

Kor	du	kom	fra?
where	you	came	from
'Where did you come from?'			

Bošković (2001) argues that a uniform analysis in terms of V-to-C movement in the syntactic component can be maintained for (26) and (27) if it is coupled with the possibility of pronouncing lower copies under PF demands. That is, in order to comply with the stricter prosodic requirements of Northern Norwegian regarding V2 constructions, the head of the of verb chain, which is adjoined to C, is deleted and the lower copy in [Spec, TP] is pronounced instead, as illustrated in (28) (cf. (26b) vs. (27)).

(28) [$_{CP}$ kor ~~kom~~ [$_{IP}$ du kom fra]]

Recall that the preference for pronouncing the head of the chain is ultimately related to an economy condition. if the highest copy does not need to probe the structure to value the features of lower copies before Spell-Out, it doesn't. In the case of (26b)/(27), such probing was required because otherwise the lower copy could not be pronounced. In the case of the derivation of (26a), on the other hand, no such probing before Spell-Out is needed to ensure convergence at PF. It is therefore blocked from applying and the highest copy of the verb must be pronounced, as illustrated by the contrast between (26a) and (29a).

(p. 157) **(29)** Northern Norwegian

a.

*Korsen	ho	kom	hit?
how	she	came	here
'How did she get here?'			

b. *[$_{CP}$ korsen ~~kom~~ [$_{IP}$ ho kom hit]]

Let us finally consider one more argument for lower copy pronunciation triggered by PF considerations, this time based on the relationship between word order and stress assignment in a 'free' word order language such as Serbo-Croatian. Stjepanović (1999, 2003, 2007) offers a variety of arguments showing that S, V, IO, and DO all move out of VP overtly in Serbo-Croatian. However, a focused element must surface as the most embedded element of the sentence, as illustrated in (30).

(30) Serbo-Croatian

a. [Context: Who is Petar introducing to Marija?]

Petar	Mariji	predstavlja	Marka.
Petar	Marija-DAT	introduces	Marko-ACC
‘Petar is introducing Marko to Marija.’			

b. [Context: Who is Petar introducing Marija to?]

Petar	Mariju	predstavlja	Marku.
Petar	Marija-ACC	introduces	Marko-DAT
‘Petar is introducing Marko to Marija.’			

In order to reconcile the evidence showing that the verb and its arguments leave VP with the position of focused elements, Stjepanović argues that the lower copy of a moved focus element may be pronounced instead of the head of the chain so that it surfaces in a position where it can receive focus stress. Under this view, Chain Reduction in the derivation of the sentences in (30b), for instance, proceeds along the lines sketched in (31).

(31) [SV~~IO~~DO [$_{VP}$ ~~SV~~IO~~DO~~]]

To summarize: under the specific implementation of the Copy Theory reviewed here, standard pronunciation of the head of the chain is more economical, as it does not employ probing by the highest copy to value the features of the lower copies before Spell-Out. However, the more economical option does not always lead to a convergent result at PF. In such circumstances, the additional probing is required to apply overtly and a lower copy can be pronounced instead. Notice that by relying on economy, we have an account for why pronunciation of the head of a chain is always preferred, all things being equal, while also being able to account for the output when things are not equal, i.e. when additional convergence requirements of the phonological component block pronunciation of the highest copy and remove this derivational option from the comparison set, allowing pronunciation of a lower (p. 158) copy. As mentioned earlier, by stipulating that traces do not have phonetic content, the Trace Theory is not so fortunate and cannot account for data such as those discussed in this section in a principled manner.

7.4.2 Scattered deletion

Let us now examine what would be necessary for scattered deletion within a chain, as illustrated in (32), to obtain in the phonological component, allowing different parts of different links to be phonetically realized.

(32) [[a $\cancel{\beta}$] H [...[~~a~~, β]...]]

The first requirement necessary for such an output to be derived is that the links that surface have their features valued; otherwise, the derivation would crash. That is, given the derivational steps in (33) below, where the constituent [a, β]moves to have its feature F valued, the higher copy of [a, β] in (33b) must probe the structure and value the feature F of its lower copy, as represented in (34), before Spell-Out.

(33)

a. [H [... [a β]$_{F:?}$...]]

b. [[a β]$_{F:✓}$ H [... [a β] $_{F:?}$...]]

(34) [[a β]$_{F:✓}$ H[... [a β] $_{F:✓}$...]]

As we saw in detail in section 7.4.1, overt probing by a higher copy to value a lower copy is not an economical option. To derive the output in (32) from the structure in (34), the system must still resort to an additional non-economical route, namely, two applications of deletion when just one application targeting the upper or the lower link would suffice to allow the structure to be linearized in accordance with the LCA. In other words, the non-economical alternative in (32) can only be an optimal output if neither alternative employing just one application of deletion converges.

A persuasive example of this possibility is provided by Bošković's (2001) in his analysis of the contrast between Macedonian and Bulgarian with respect to their surface location of clitics, as illustrated in (35) and (36).

(35) Macedonian (Rudin et al. 1999)

a.

Si	mu	(gi)	dal	li	parite?
are	him-DAT	them	given	Q	the-money

b.

*Dal	li	si	mu	(gi)	parite?
given	Q	are	him-DAT	them	the-money
‘Have you given him the money?’					

(36) Bulgarian (Rudin et al. 1999)

a.

*Si	mu	(gi)	dal	li	parite?
are	him-DAT	them	given	Q	the-money

(p. 159) **b.**

Dal	li	si	mu	(gi)	parite?
given	Q	are	him-DAT	them	the-money
‘Have you given him the money?’					

Bošković argues that in both languages the complex head [si+mu+gi+dal] left-adjoins to the interrogative particle *li*, leaving a copy behind, as represented in (37) below. Deletion of the lower copy of [si+mu+gi+dal], as shown in (38), yields a well-formed result in Macedonian (cf. (35a)), because in this language pronominal clitics are proclitic and *li* is enclitic. The unacceptability of (35b), then, follows from the general ban on scattered deletion imposed by economy considerations regarding the number of applications of deletion. In Bulgarian, on the other hand, *li* as well as the pronominal clitics are enclitics; thus, deletion of the lower copy of the complex head does not lead to a convergent result (see (36a)). Bošković proposes that the system then resorts to scattered deletion, as shown in (39), allowing the chain to be linearized while at the same time satisfying the additional requirements of the phonological component.

(37) [[si+mu+gi+dal] +li …[si+mu+gi+dal] …]

(38) Macedonian

[[si+mu+gi+dal]+li… ~~[si 1 mu ι gi ι dal]~~ …]

(39) Bulgarian

[[~~si+mu+gi~~+dal] +li … [si+mu+gi+~~dal~~] …]

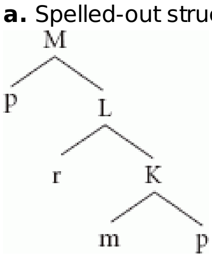
Under the linearization approach reviewed in section 7.3, the fact that constructions involving scattered deletion are rare follows from their having to resort to non-economical derivational routes in the mapping from the syntactic component to PF. But to the extent that they do exist,[11] they provide very convincing arguments for the Copy Theory and against the Trace Theory.

7.4.3 Phonetic realization of multiple copies

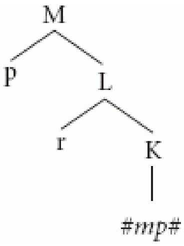
Let us finally examine cases where different links of a given chain are pronounced, but with the same phonetic material.[12] Recall from section 7.3 that according to the linearization approach to deletion of copies, a chain cannot surface with more than one link realized with the same phonetic material because the structure containing (p. 160) it cannot be linearized. This line of thinking predicts that if two given copies somehow manage to not interfere with linearization, they should in principle be able to both surface overtly. Nunes (1999, 2004) argues that under certain conditions, this actually happens. Here is the reasoning. Suppose that after the syntactic structure in (40a) below, with two copies of p, is spelled out, the morphological component fuses (in the sense of Halle and Marantz 1993) the terminals *m* and *p*, yielding the atomic blended terminal *#mp#* (or *#pm#*, for that matter), with no internal structure accessible to further morphological or syntactic computations, as sketched in (40b).

(40)

a. Spelled-out structure M



b. Fusion in the morphological component



The content of *#mp#* in (40b) cannot be directly linearized with respect to *r* or the upper copy of *p* because it is an inaccessible part of #mp#. From an LCA perspective, for instance, the blended material within *#mp#* is not accessible to c-command computations. However, it can be indirectly linearized in (40b) by virtue of being an integral part of *#mp#:* given that the upper copy of *p* asymmetrically c-commands *r* and that *r* asymmetrically c-commands *#mp#*, we should obtain the linear order *p>r>#mp#*. In other words, the material inside *#mp#* gets linearized in a way analogous to how the phoneme /l/ is indirectly linearized in *John loves Mary* due to its being part of the lexical item *loves*. But, crucially, once the lower copy of *p* in (40b) becomes invisible for standard linearization computations, the linearization problems caused by the presence of multiple copies discussed in section 7.3 cease to exist. Thus, the structure in (40b) not only can but must surface with two copies of *p* at PF.

(p. 161) With this in mind, consider verb clefting constructions in Vata, as illustrated in (41) below. Koopman (1984) shows that the two verbal occurrences in (41) cannot be separated by islands, which indicates that they should be related by movement. The problem from the present perspective is that if these occurrences are to be treated as copies, the structure containing them should not be able to be linearized in accordance with the LCA, as discussed in section 7.3. Nunes (2004) proposes that this possibility does not in fact arise because the highest copy of the clefted verb gets morphologically fused, thereby evading the purview of the LCA. More precisely, he analyzes verb-clefting in Vata as involving verb movement to a Focus head, followed by fusion in the morphological component between the moved verb and the Focus head, as represented in (42a). Of the three verbal copies in (42a), the LCA only 'sees' the lower two after the highest copy gets fused with Foc0.[13] The lowest copy is then deleted (cf. (42b)) and the structure is linearized as in (41), with two copies of the verb phonetically realized.

(41) Vata (Koopman 1984)

li	a	*li-da*	zué	saká
eat	we	eat-PAST	yesterday	rice
'We ATE rice yesterday'				

(42)

a. Fusion

$[_{FocP}\ \#[_{Foc}{}^{0}\ V[_{Foc}{}^{0}\ Foc^{0}]]\#[_{TP}\ldots\ [_{T}{}^{0}\ V[_{T}{}^{0}\ T^{0}]]\ [_{VP}\ldots\ V\ldots]]]$

b. Deletion of copies

$[_{FocP}\ \#\ [_{Foc}{}^{0}\ V\ [_{Foc}{}^{0}\ Foc^{0}]]\#\ [_{TP}\ \ldots\ [T^{0}\ V[T^{0}\ T^{0}]]\ [_{VP}\ \ldots\ \cancel{V}\ldots]]]$

Nunes (2004) presents two bits of evidence in favor of this account of verb-clefting in Vata. The first one relates to Koopman's (1984:158) observation that the restricted set of verbs that cannot undergo clefting in Vata have in common the property that they cannot serve as input for morphological processes that apply to other verbs. If these verbs cannot participate in any morphological process, they certainly should not be able to undergo the morphological fusion with Foc^{0} depicted in (42a) and should not be allowed in predicate-clefting constructions. The second piece of evidence is provided by the fact, also observed by Koopman, that the fronted verb in these focus constructions must be morphologically unencumbered; in particular, none of the tense or negative particles that occur with the verb in Infl may appear with the fronted verb, as illustrated in (43) below. This makes sense if these particles render the verb morphologically too complex, thereby preventing the verb from undergoing fusion with the focus head.

(p. 162) **(43)** Vata (Koopman 1984)

a.

(*nav-)*le*	wa	ná-*le*-ka	
NEG	eat	they	NEG-eat-FT
'They will not EAT'			

b.

li(*-wa)	wã	*li-wa*	zué
eat TP	*they*	*eat-TP*	*yesterday*
'They ATE yesterday'			

These restrictions can be interpreted as showing that if the realization of multiple copies is licensed via morphological fusion, it should naturally be very sensitive to morphological information. The first kind of relevant information regards the feature composition of the elements that are to be fused. After all, not any two elements can get fused, but only the ones that satisfy the morphological requirements of one another. In Vata, for instance, the duplication of focused material only affects verbs, and many languages only allow multiple copies of *wh*-elements, as we will see below. This may be viewed as a reflex of the morphological (categorial) restrictions a given head may impose on the copy with which it may fuse. The second kind of information concerns morphological complexity. As a rule, the more morphologically complex a given element is, the less likely it is to undergo fusion and become part of a terminal. Thus, the addition of specific morphemes (which may vary from language to language) may make the resulting element morphologically too heavy' to become reanalyzed as part of a word. This seems to be what is going on in (43), with the addition of INFL particles to the fronted verb. Of course, if a given copy is syntactically complex, i.e. it is phrasal, it is also morphologically complex and not a good candidate to undergo morphological fusion.[14]

This general approach provides a natural account of *wh*-copying constructions found in many languages, as illustrated by German in (44) below. *Wh*-copying constructions are subject to two intriguing constraints. First, although more than one trace may be phonetically realized (cf. (44)), only intermediate traces can be pronounced, as shown by the ungrammaticality of (45), where the tail of the *wh*-chain is realized, as well. The second pervasive characteristic of *wh*-copying constructions is that, roughly speaking, they can only involve simplex, not complex *wh*-phrases, as illustrated by (46).

(p. 163) **(44)** German (Fanselow and Mahajan 2000)

Wen	denkst	Du	*wen*	sie	meint	*wen*	Harald liebt?
who	think	you	*who*	she believes	*who*	Harald loves	
'Who do you think that she believes that Harald loves?'							

(45) German

*Wen	glaubt	Hans	*wen*	Jakob	*wen*	gesehen	hat?
whom	thinks	Hans	*whom* Jakob	*whom*	seen	has	
'Who does Hans think Jakob saw?'							

(46) German (McDaniel 1986)

**Welche*	*Bucher*	glaubst	du	*welche Bücher*	Hans	liest?
which	*book*	think	you	*which book*	Hans	reads
'Which book do you think Hans is reading?'						

Nunes (1999, 2004) argues that this paradigm can be accounted for if long-distance *wh*-movement in languages that allow for *wh*-copying constructions may proceed via head adjunction to C, as illustrated in (47a),[15] and if a[-wh] C fuses with the adjoined *wh*-element in the morphological component, as represented in (47b).

(47)

a. $[_{CP}[_{C}{}^{0}\mathrm{WH}[\mathrm{c}^{0}\mathrm{Q}]]\ldots[_{CP}[_{C}{}^{0}\mathrm{WH}[_{C}{}^{0}\mathrm{C}_{[\_wh]}]]\ [_{TP}\ \ldots\mathrm{WH}\ \ldots]]]$

b. $[_{CP}\ [_{C}{}^{0}\mathrm{WH}\ [_{C}{}^{0}\mathrm{Q}]]\ \ldots[_{CP}\#[_{C}{}^{0}\ \mathrm{WH}\ [_{C}{}^{0}\mathrm{c}_{[\_wh]}]\#]\ [_{TP}\ \ldots\mathrm{WH}\ \ldots]]]$

The *wh*-chain in (47b) has only two links visible to the LCA, as the intermediate *wh*-copy becomes invisible after it undergoes fusion. The two visible copies should then prevent the structure from being linearized unless chain Reduction is employed. Thus, the derivation of (45), for instance, cannot converge because the relevant structure cannot be linearized.[16] Under the assumption that the highest copy in (p. 164) (47b) has more features checked, it should be kept and the lowest copy deleted, as discussed in section 7.3, yielding (48).

(48) $[_{CP}\ [_{C}{}^{0}\ \mathrm{WH}\ [_{C}{}^{0}\ \mathrm{Q}]]\ \ldots.\ [_{CP}\ \#\ [_{C}{}^{0}\ \mathrm{WH}\ [\mathrm{c0}\ \mathrm{C}_{[\_wh]}]\#]\ [_{TP}\ \ldots\ \cancel{\mathrm{WH}}\ldots]]]$

We now have an answer for why the tail of the wh-chain contrasts with intermediate traces with respect to phonetic realization. There is nothing intrinsic to intermediate traces themselves that allows them to be phonetically realized. Rather, morphological requirements of the intermediate C^0 may trigger fusion with the adjoined *wh*-copy, making it invisible for the LCA and, consequently, for deletion. Once the system only 'sees' the highest and the lowest wh-copies in (47b), its linearization as in (48) is no different from the linearization of a standard *wh*-movement construction such as (49), where economy considerations regarding applications of operations before Spell-Out ultimately determine the deletion of the lower *wh*-copy (see section 7.3).

(49)

a. What did John see?

b. [$_{CP}$ what did [$_{IP}$ John see ~~what~~]]

Finally, by having wh-copying be dependent on morphological fusion, we reach a natural explanation for why complex wh-phrases do not license wh-copying (cf. (46)). The more morphologically complex a given element is, the harder it is for it to be fused and reanalyzed as part of a word. Thus, the unacceptability of sentences such as (46) is arguably due to the fact that the wh-phrases cannot undergo fusion with the intermediate C0 due to their morphological complexity. This in turn entails that all the copies of the moved wh-phrase are visible to the LCA, and failure to delete all but one link prevents their structures from being linearized.[17]

Nunes (2003, 2004) shows that the reasoning presented above also accounts for phonetic realization of more than one link in remnant movement constructions. Consider duplication of emphatic focus in Brazilian Sign Language (LSB) in (50) below. Nunes and Quadros (2006, 2008) argue that in constructions such as (50), the focused element moves and adjoins to a Focus head, followed by remnant movement of TP and fusion between Foc and the adjoined element in the morphological component.[18] Under the implementation of the linearization approach to copy deletion proposed in section 7.3, the derivation of (50a), for instance, is as sketched in (51).

(p. 165) **(50)** Brazilian Sign Language

a. I *LOSE* BOOK *LOSE*

'I LOST the book.'

b. [JOHN BUY *WHICH* BOOK YESTERDAY]$_{WH}$[*WHICH*]$_{WH}$

'Which book exactly did John buy yesterday?'

(51)

a. [$_{FocP}$ Foc [$_{TP}$ ILOSE$_{F:?}$ BOOK]]

b. Adjunction to Foc

[$_{FocP}$ [$_{Foc0}$LOSE$_{F:✓}$ [Foc^0 Foc^0]] [$_{TP}$ I LOSE$_{F:?}$ BOOK]]

c. Probing by the higher copy

[$_{FocP}$ [$_{Foc}{}^0$ LOSE$_{F:✓}$ [$_{Foc}{}^0$ Foc^0]] [$_{TP}$ I LOSE$_{F:✓}$BOOK]]

d. Remnant movement of TP

[[$_{TP}$ I LOSE$_{F:✓}$ BOOK] … [$_{FocP}$ [$_{Foc}{}^0$ LOSE$_{F:✓}$ [$_{Foc}{}^0$ Foc^0]] [$_{TP}$ I LOSE$_{F:✓}$ BOOK]]]

e. Spell-Out + fusion

[[$_{TP}$ I LOSE$_{F:✓}$ BOOK] … [$_{FocP}$ #[$_{Foc}{}^0$ LOSE$_{F:✓}$ [$_{Foc}{}^0$ Foc^0]]# [$_{TP}$ I LOSE$_{F:✓}$BOOK]]]

f. Chain Reduction of CH = (TP, TP)

[[$_{TP}$ I LOSE$_{F:✓}$ BOOK] … [$_{FocP}$ #[$_{Foc}{}^0$ LOSE$_{F:✓}$ [$_{Foc}{}^0$ Foc^0]]# ~~[$_{TP}$ I LOSE$_{F:✓}$ BOOK]~~]]

As discussed earlier, after the verb adjoins to Foc in (51b), valuation of the lower copy by the higher one, as in (51c), is not economical and will be resorted to only if triggered by independent requirements. This is indeed the case here. After LOSE and Foc fuse in the morphological component, as in (51e), the fused copy becomes invisible to the LCA and Chain Reduction is not called upon to delete the lower link of the chain CH = (LOSE, LOSE) formed when the verb adjoined to Foc. Thus, in order for the derivation to converge at PF, valuation of the lower copy of LOSE in (51c) must occur before Spell-Out. That being so, the only chain subject to reduction is the TP chain and deletion of its lower link as in (51f) yields the sentence in (50a).

Despite being optional, focus duplication is a pervasive phenomenon in LSB, being able to affect several kinds of constituents. However, there is a major restriction on this construction: the duplicated material cannot be morphologically complex (see Nunes 2003, 2004, Nunes and Quadros 2006, 2008), as illustrated in (52a) below, with a verb that requires agreement morphology (annotated by the indices), and (52b), with a w/2-phrase. Once the phonetic realization of multiple copies is dependent on morphological fusion and fusion is sensitive to morphological complexity, the ungrammaticality of the sentences in (52) can be attributed to the impossibility of fusion involving the moved elements. The presence of multiple copies that are visible to the LCA in the phonological component then prevents the structures underlying these constructions from being linearized (see section 7.3).

(p. 166) **(52)** LSB

a. *JOHN $_{a}$*LOOK*$_{b}$ MARY $_{a}$*LOOK*$_{b}$

‘John LOOKED at Mary’
b. *[JOHN BUY *WHICH BOOK* YESTERDAY]$_{wh}$ [*WHICH BOOK*] wh
‘Which book exactly did John buy yesterday?’

Martins' (2007) analysis of verb duplication in emphatic affirmation constructions in European Portuguese provides further evidence for this approach. In European Portuguese, the denial of a previous statement in a given context may involve verbal duplication at the right edge of the sentence, as illustrated in (53a) and (53b) below. Martins argues that both types of sentences are derived by movement of the verb to the polarity head Σ, followed by verb movement to C and fusion between C and the verb in the morphological component. The difference between them is that (53a) involves ellipsis, whereas (53b) involves remnant movement, as sketched in (54a) and (54b), respectively.

(53) European Portuguese
a.
A:

O	João	não	comprou	um	carro,	pois	não?
The	João	not	bought	a	car	POIS	NEG
‘John didn't buy a car, did he?’							

B:

Comprou	*comprou*.
Bought	bought
‘Yes, he DID.’	

b.
A:

Ele	não	comprou	o	carro.
he	not	bought	the	car
‘He didn't buy the car.’				

B:

Ele	*comprou*	o	carro,	*comprou*.
he	bought	the	car,	bought
‘He did buy the car.’				

(54)
a. [$_{CP}$[$_{C'}$[$_C$ comprou] [$_{\Sigma P}$ [$_{\Sigma'}$ comprou [$_{TP}$ [$_{T'}$ ~~comprou~~
bought bought bought
[$_{vP}$ ~~o João comprou o carro~~]]]]]]]
the João bought the car

b.

[$_{CP}$ [$_{\Sigma P}$ ele comprou o carro] [$_{C'}$ [$_{C}$ comprou]
he bought the car bought
~~[$_{\Sigma P}$ ele [$_{\Sigma'}$ comprou[$_{TP}$ [$_{T'}$ comprou [$_{VP}$ ele comprou o carro]]]]]~~]]
he bought bought he bought the car

Relevant for our current purposes is Martins' documentation of a series of contexts that block verbal duplication, as illustrated in (55), with compound verbs and verbs with stressed prefixes. As Martins argues, the ungrammaticality of sentences such as (55) is to be attributed to the morphological complexity of their verbs, which (p. 167) should block fusion; in turn, once more than one copy of the verb is visible to the phonological component, the whole structure cannot be linearized.

(55) European Portuguese

a.

A:

Ele	não	fotocopiou	o	livro	sem	autorização,
he	not	photocopied	the	book	without	permission,
pois	não?					
POIS	NEG					
‘He didn't copy the book without your permission, did he?’						

B:

??Fotocopiou,	*fotocopiou*.
Photocopied	Photocopied
‘Yes, he DID.’	

b.

A:

O	candidate	não	contra-atacou,	pois	não?
the	candidate	not	counter-attacked,	POIS	NEG
‘The candidate didn't counter-attack, did he?’					

B:

?? *Contra-atacou*,	*contra-atacou*.
counter-attacked	counter-attacked
‘Yes, he DID.’	

Remnant movement constructions thus provide further empirical support for the Copy Theory in that they can also allow more than one chain link to be phonetically realized, provided that linearization and morphological requirements are satisfied.

7.5 The Copy Theory and the debate on obligatory control

In the same way that mapping from Spell-Out to PF can provide compelling evidence for the Copy Theory over the Trace Theory, it can also set up independent grounds for choosing between the two major minimalist approaches to obligatory control which are currently under debate. As far as the mapping from Spell-Out to PF is concerned, the PRO-based approach, be it in terms of null Case (see e.g. Chomsky and Lasnik 1993, Martin 2001) or in terms of Agree (see e.g. Landau 2004), is no different from the GB approach. In other words, once PRO is by definition devoid of phonetic content, its chain will receive no interpretation at PF. By contrast, under the movement approach to obligatory control proposed by Hornstein (2001), obligatorily control is actually a trace (i.e. a copy) of the controller. Under this view, the fact that the controller is the element that surfaces at PF follows from (p. 168) the fact that in general heads of chains are the ones that are realized at PF. But recall that pronunciation of chain heads is just the optimal output when there are no additional requirements on specific chain links. Thus, given the several possibilities discussed in section 7.4 for a chain to be realized at PF, there arises the possibility that control constructions may also display similar unorthodox realizations at PF.

Two such realizations have gained prominence recently. The first involves backward control constructions (see e.g. Polinsky and Potsdam 2002, Boeckx et al. 2010b). As convincingly argued by Polinsky and Potsdam (2002), Tsez, for instance, allows control constructions where the thematic matrix subject is obligatorily null and obligatorily bound by the embedded overt subject, as illustrated in (56).

(56) Tsez

[$\Delta_{1/*2}$	[kidbā$_1$	ziya	bisra]	yoqsi]
	girl.ERG	cow.ABS	feed.INF	Began
‘The girl began to feed the cow.’				

Polinsky and Potsdam present several kinds of evidence pointing to the conclusion that the phonetically realized subject in sentences such as (56) does indeed sit in the embedded clause. For instance, (57) below shows that the case-marking on the overt subject is determined by the dative-assigning verb in the embedded clause, whereas (58) shows that the overt subject cannot precede a matrix adverb. Polinsky and Potsdam then propose that backward control constructions involve movement to a thematic position, as in standard instances of control under Hornstein's (2001) approach, with the difference that a lower copy is pronounced instead, as sketched in (59).

(57) Tsez

kid-ber	babiw-s	xabar	teq-a	7y-oq-si
girl.II-DAT	father-GEN	story.III-INF	hear-INF	begin-PAST.EVID
‘The girl began to hear the father's story.’				

(58) Tsez

a.

ħuł	[kidbā	ziya	bišra]	yoqsi
Yesterday	girl.ERG	cow	feed	began

b.

*kidba	ħuł	[ziya bišra]	yoqsi
girl.ERG	yesterday	cow feed	began
'Yesterday the girl began to feed the cow.'			

(59)
a. [DP$_1$ V [DP$_1$...]]
b. Deletion in the phonological component (forward control)
[DP$_1$ V [~~DP$_1$~~...]]
c. Deletion in the phonological component (backward control)
[~~DP$_1$~~ V [DP$_1$...]]

(p. 169) The second type of unorthodox control constructions discussed recently regards copy-control. Consider the data in (60) from San Lucas Quiaviní Zapotec (SLQZ), discussed by Lee (2003).

(60) SLQZ (Lee 2003)
a.

R-cààa'z	Gye'eihlly	g-auh	Gye'eihlly	bxaady.
HAB-want	Mike	IRR-eat	Mike	grasshopper
'Mike wants to eat grasshopper.'				

b.

B-quìi'lly	bxuuhahz	Gye'eihlly	ch-iia	Gye'eihlly	scweel.
PERF-persuade	priest	Mike	IRR-go	Mike	school
'The priest persuaded Mike to go to school.'					

c.

B-ìi'lly-ga'	Gye'eihlly	zi'cygàa'	nih	cay-uhny	Gye'eihlly
PERF-sing-also	Mike	while	that	PROG-do	Mike
zééiny.					
work					
'Mike sang while he worked.'					

Each of the sentences in (60) shows a bound copy in the embedded subject position. Interestingly, the similarities of these constructions with standard control constructions go beyond translation. They also trigger a sloppy reading under ellipsis, as shown in (61), and the bound copy displays complementarity with a co-referential pronoun, as shown in (62).
(61) SLQZ (Lee 2003)
a.

R-càà'z	Gye'eihlly	g-ahcnèe	Gye'eihlly	Lia	Paamm
HAB-want	Mike	IRR-help	Mike	FEM	Pam
zë'cy	cahgza'	Li'eb.			
Likewise	Felipe				
'Mike wants to help Pam, and so does Felipe (want to help Pam/*want Mike to help Pam).'					

b.

Zi'cygàa'	nih	cay-uhny	Gye'eihlly	zèèiny	b-ìi'lly-ga'
while	that	PROG-do	Mike	work	PERF-sing-also
Gye'eihlly	zë'cy	cahgza'	Li'eb.		
Mike	likewise	Felipe			
'While Mike$_i$ was working, he$_i$ sang, and so did Felipe$_k$ (sing while he$_k$ worked).'					

(62) SLQZ (Felicia Lee, p.c., 2003)

a.

R-caaa'z	Gye'eihlly	g-ahcnèe-ëng	Lia	Paamm.
HAB-want	Mike	IRR-help-3SG.PROX	FEM	Pam
'Mike; wants him$_{k/*l}$ to help Pam'				

b.

Zi'cygàa'	nih	cay-uhny-ëng	zèèiny	b-ìi'lly-ga'
while	that	PROG-do-3SG.PROX	work	PERF-sing-also
Gye'eihlly.				
Mike				
'While hei/*k worked, Mikek sang.'				

(p. 170) Boeckx et al. (2007, 2008) argue that the data in (6o)–(62) are indeed cases of control, i.e. movement to thematic positions, with both the controller and the controllee copies being phonetically realized. More specifically, they propose that these constructions involve morphological fusion of the controllee copy with a null 'self' morpheme available in this language.[19] As we should expect, given the discussion above, if a control chain involves morphologically encumbered copies, fusion will be blocked and phonetic realization of more than one copy leads to an ungrammatical result. That this prediction is correct is illustrated by the copy control constructions in (63a), which involves a quantifier phrase, and in (63b), whose links contain an anaphoric possessor.

(63) SLQZ (Lee 2003)

a.

*Yra'ta'	zhyàa'p	r-cààa'z	g-ahcnèe'	yra'ta'	zhyàa'p	Lia
every	girl	HAB-want	IRR-help	every	girl	FEM
Paamm.						
Pam						
'Every girl wants to help Pam.'						

b.

*R-e'ihpy	Gye'eihlly	behts-ni'	g-a'uh	behts-ni'
HAB-tell	Mike	brother-REFL.POSS	IRR-eat	brother-REFL.POS
bx:àady.				
grasshopper				
'Mike told his brother to eat grasshoppers.'				

Let us re-examine the adjunct copy control case in (60c). As argued by Hornstein (2001), adjunct control involves sideward movement (in the sense of Nunes 2001, 2004) of the embedded subject before the adjunct clause attaches to vP. The fact that sideward movement may also lead to phonetic realization of multiple copies shows that sideward movement is nothing more than one of the instantiations of Copy plus Merge. Interestingly, there are languages which only allow adjunct copy control, which indicates that the relevant head that triggers fusion in these languages is within the adjunct clause. In his detailed study on control structures in Telugu and Assamese, Haddad (2007) shows that adjunct copy control constructions such as (64) and (65) below (CNP stands for conjunctive participle particle) display all the traditional diagnostics of obligatory control and argues that they should also be analyzed in terms of sideward movement and phonetic realization of multiple copies.

(p. 171) **(64)** Telugu (Haddad 2007)

[[Kumar	sinima	cuus-tuu]	[Kumar	popcorn
Kumar.NOM	movie	watch-*CNP*	Kumar.NOM	popcorn
tinnaa-Du]]				
ate-3-M.s				
'While watching a movie, Kumar ate popcorn.'				

(65) Assamese (Haddad 2007)

[[Ram-Or	khong	uth-i]	[Ram-e	mor	ghorto	bhangil-e]]
Ram-GEN	anger	raise-CNP	Ram-NOM	my	house	destroyed-3
'Having got angry, Ram destroyed my house.'						

Given the role of morphological fusion in making possible the phonetic realization of multiple copies, it comes as no surprise that multiple copies are only possible if, in Haddad's (2007: 87) words, the subject 'does not exceed one or two words', as illustrated by the ungrammaticality of (66).

(66) Telugu (Haddad 2007)

*[[Kumar	maryu	Sarita	sinim	cuu-tuu]	[Kumar
Kumar. NOM	and	Sarita.NOM	movie	watch-CNP	Kumar.NOM
maryu	Sarita	popcorn	tinna-ru]]		
and	Sarita.NOM	popcorn ate			
'While Kumar and Sarita were watching a movie, they ate popcorn.'					

To the extent that backward control and copy control constructions are roughly analyzed along the lines suggested above, they provide decisive grounds for choosing between PRO-based and movement-based approaches to control. More specifically, these constructions prove fatal to PRO-based approaches to control, as PRO is taken to be a phonetically null element by definition. In contrast, backward control and copy control are, in fact, expected under a movement-based approach to control under the Copy Theory, given its potential different outputs at PF.

7.6 Concluding remarks

Chomsky's (1993) original arguments for incorporating the Copy Theory into the minimalist framework had to do with interpretation effects and the mapping from the numeration to LF. However, optimizing this mapping by assuming the Copy Theory seems to have an unwelcome consequence, as it appears to require stipulating that in the mapping from Spell-Out to PF, lower copies (the old traces) must be deleted. After all, the null hypothesis regarding the Copy Theory is that if α_1 is a **(p. 172)** copy of a_2, they should have the same status in the computational system. Thus, if the highest can be pronounced, lower copies should in principle be pronounceable, as well.

Upon close inspection, vice can be turned into virtue. As seen in the previous sections, lack of phonetic realization is not an intrinsic property that characterizes traces as grammatical primitives. Traces or parts of traces may be phonetically realized if the pronunciation of the head of the chain causes the derivation to crash at PF. The fact that traces in the general case are not phonetically realized results from the interaction among convergence and economy factors. On the one hand, linearization requirements trigger deletion of 'repeated' material within a chain. On the other, economy considerations regarding the valuation of lower copies by higher ones before Spell-Out and the number of applications of deletion within a chain work in such a way that they render higher copies more PF-palatable than lower ones. Thus, if the phonological component imposes no additional convergence conditions that can affect these optimality computations, the head of a chain will always be the optimal option for phonetic realization. However, these economy considerations may be overruled by convergence requirements in the phonological component—in which case we may have pronunciation of a lower copy, pronunciation of different parts of different copies, and even pronunciation of more than one copy.

In sum, we have seen that rethinking movement operations in terms of the Copy Theory, which was driven by the minimalist search for conceptual elegance, has led to a considerable enlargement of the empirical coverage previously handled.

Notes:

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(1) Due to space limitations, here I will not discuss Chomsky's (1995c) Move-F approach, according to which the syntactic component can also move/copy the set of formal features of a given lexical item (see Aoun and Nunes 2007 for a comparison with the Agree approach). However, it should be pointed out that nothing substantially changes regarding the Inclusiveness Condition if the Move-F approach is correct, as the copied set of formal features is a replica of features present in the numeration.

(2) In fact, when Chomsky (1993) proposed adopting the Copy Theory, he suggested that deletion of traces could be related to deletion in ellipsis constructions (see e.g. Nunes 2004, Saab 2008).

(3) See Saab (2008) for recent illuminating discussion.

(4) See Hornstein (2001) for discussion of this point.

(5) One question that arises in any version of the Copy Theory of movement is how the computational system distinguishes copies from elements that happen to have the same set of features. The derivation of (i) below, for instance, should converge if it starts with the numeration N1 in (iia) (with one instance of *Mary*), but not with the numeration N_2 in (iib) (with two instances).

((i)) [Mary [was [hired Mary]]]
((ii))
(a.) N_1 = {$Mary_1$, was_1, hired1,...}
(b.) N_2 = {Mary2, was1, hired1,...}

Chomsky (1995c: 227) suggests that two lexical items *l* and *l'* should be marked as distinct if they enter the derivation via different applications of Select. Alternatively, Nunes (1995, 2004) suggests that it is in fact the operation Copy that assigns a non-distinctiveness index; in other words, all elements will be taken to be distinct for purposes of the computational system unless they are specified as non-distinct by the Copy operation. It is worth pointing out that both suggestions run afoul of the Inclusiveness Condition, as the added (non-)distinctiveness markings are not part of the numeration. A possibility that is in consonance with the Inclusiveness Condition (see Nunes 2004: 165) is to allow the system to compute (non-)distinctiveness by comparing derivational steps. For instance, if two contiguous derivational steps σ_1 and σ_2 differ in that a new term τ is introduced into the computation, two possibilities arise: if from σ_1 to σ_2 the numeration has been reduced, *τ* is to be interpreted as distinct from all the other syntactic objects available at σ_2; if the numerations of σ1 and σ_2 are the same, *τ* must be a copy of some syntactic object available at σ_1. Whether or not it is desirable that the recognition of copies by the computational system proceeds along these lines remains to be determined.

(6) Based on the obligatory reconstruction in sentences such as (ia) below, Chomsky (1993) proposes that in the case of A'-chains, there is actually an economy preference for minimizing operator restrictions in LF, which normally leads to scattered deletion (cf. (ib)/(8b)). To force reconstruction in (ia) while allowing the upstairs reading of *himself* in (7), Chomsky suggests that either the higher or the lower copy of *himself* undergoes anaphor movement covertly. When the lower copy of *himself* moves, deletion along the lines of (8b), which complies with this preference principle, yields a well formed result. By contrast, if the higher copy undergoes anaphor movement, scattered deletion as in (8b) would 'break' the anaphor chain, causing the derivation to crash. The system is then allowed to employ deletion as in (8a), for only convergent derivations can compete for purposes of economy.

((i))
(a.) *Mary wondered which picture of Tom_i he_i liked.
(b.) *Mary wondered [$_{CP}$ [which picture of Tom_i] he_i liked [which picture of Tom_i]]

(7) See Franks (1998) for the seeds of the economy approach to be explored below.

(8) For relevant discussion and alternatives, see e.g. Nunes (1995, 1999, 2004), Kobele (2006), and Bošković and Nunes(2007).

(9) For additional examples and general discussion, see e.g. Nunes (1999, 2004), Bošković (2001), Bošković and Nunes (2007), Saab (2008), Kandybowicz (2008), Corver and Nunes (2007), and references therein.

(10) See Golston (1995) for a discussion of many such cases, and N. Richards (2006) for some related issues.

(11) For examples of other constructions that are argued to involve scattered deletion, see e.g. Ćavar and Fanselow's (1997) analysis of split constructions in Germanic and Slavic languages and Wilder's (1995) analysis of extraposition.

(12) Due to space limitations, I will not discuss cases where it has been argued that lower copies are realized as (resumptive) pronouns, reflexives, or partial copies (see e.g. Lidz and Idsardi 1997, Pesetsky 1997, 1998, Hornstein 2001, 2007, Grohmann 2003b, Fujii 2007, and Barbiers et al. 2010). For our purposes, suffice it to say that to the extent that these lower copies are rendered distinct from the head of the chain, no linearization problem is at stake.

(13) The point is not that every instance of head movement renders the adjoined element invisible to the LCA, but rather that *fused* elements are not computed by the LCA (cf. (40)).

(14) There are languages that allow a fronted predicate to be duplicated, as illustrated by Yoruba in (i). If (i) does involve non-distinctive copies, they should be somehow prevented from being computed ‘at the same time’ for purposes of linearization. See Aboh (2006) and Kobele (2006)for specific suggestions.

((i)) Yoruba (Kobele 2006)

Rira	adię	ti	Jimọ	ọ	ra	adieẹ
Buying	chicken	TI	Jimo	HTS	buy	chicken

‘the fact/way that Jimo bought a chicken’

(15) For arguments that head adjunction should in general be preferred over movement to specifiers, all things being equal, see Nunes (1998) and Bošković (2001). Suggestive evidence that *wh*-movement in *wh*-copying does indeed involve head adjunction is provided by the fact the *wh*-copying is more restricted than regular wh-movement. In particular, it is subject to negative islands even when arguments are moved, as illustrated in (i), which can be accounted for if *wh*-copying involves head adjunction to comp and if an intervening Neg head blocks such head movement.

((i)) German (Reis 2000)

*Wen	glaubst	du	nicht,	*wen*	sie	liebt?
whom	believe	you	not	*whom*	she	loves

‘Who don't you think that she loves?’

(16) contrary to what may seem to be the case at first glance, movement of the verb from T to Foc in (42a) or

movement of the *wh*-element from one head-adjoined position to another in (47a) is not incompatible with Baker's (1988) account of the general ban on excorporation (if the ban indeed holds). According to Baker, given the head adjunction structure [Y^0 X^0 Y^0], if X^0 moves, the morphological component will receive a head with an adjoined trace, which is taken to be an illicit morphological object. Under the copy Theory, Baker's proposal can be interpreted as saying that deletion of copies cannot take place under an X^0 element. Notice that it is a crucial feature of the analysis reviewed above that the V-copy adjoined to F in (42a) and wh-copy adjoined to the intermediate C^0 in (47a) do *not* delete.

(17) It should be noted that considerable dialectal and idiolectal variation is found among speakers who accept *wh*-copying constructions. From the perspective reviewed here, variation in this regard is not due to syntactic computations proper, but to the degree of morphological complexity a given dialect or idiolect tolerates under fusion. As a rule, the more complex a constituent, the less likely it is to undergo fusion and become invisible to the LCA.

(18) Independent differences aside, the analysis of duplication of focus in LSB to be sketched below can also be extended to the constructions involving focus duplication in American Sign Language originally discussed by Petronio (1993) and Petronio and Lillo-Martin(1997) (see further Nunes 2004, Nunes and Quadros 2006, 2008).

(19) Boeckx et al. (2007, 2008) argue that fusion with this null ‘self’ morpheme is also what underlies the existence of copy-reflexive constructions in San Luca Quiavinì Zapotec such as the ones illustrated in (i).

((i)) SLQZ (Lee 2003)

B-gwa	Gye’eihlly	Gye’eihlly.
PERF-shave	Mike	Mike

‘Mike shaved himself.’

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[–] Abstract and Keywords

The earliest work in minimalism was primarily concerned with A-movement and its effects. Still, the mechanisms proposed in this work and in its sequels have had profound implications for our understanding of A-bar dependencies as well. This article sketches some of these implications, describing some of the history of this part of the field and identifying controversies where they arise. It focuses on properties of *wh*-movement. One of the earliest proposals of the minimalist framework was the elimination of D-structure and S-structure. Each of these is considered in turn. The article then moves on to consider some problems having to do with successive-cyclic *wh*-movement, and concludes with a discussion of implications that the properties of A-bar movement have for the interfaces between syntax, phonology, and semantics.

Keywords: A-bar movement, minimalism, wh-movement, syntax, phonology, semantics

THE earliest work in minimalism (e.g. Chomsky 1993, 1995b) was primarily concerned with A-movement and its effects. Still, the mechanisms proposed in this work and in its sequels have had profound implications for our understanding of A-bar dependencies as well. In this chapter I will try to sketch some of these implications, describing some of the history of this part of the field and identifying controversies where they arise.

Work in generative grammar has identified a large number of kinds of A-bar dependencies, including *wh*-movement, relativization, topicalization, and tough-movement (Chomsky 1977b and much subsequent work):

(1)
- **a.** Who do you think that John saw?
- **b.** the man [who I think that John saw]
- **c.** That man, I don't think that John saw
- **d.** That man is tough to convince people that they should talk to

A-bar dependencies are characterized by comparatively loose restrictions on locality (for instance, the ability to cross tensed clause boundaries in many languages, as in (1)), and inability to repair Condition A violations by creating new binders for anaphors:

(2)
- **a.** *[The picture of himself$_i$]hit John$_i$ on the head.
- **b.** *Who$_i$ did [the picture of himself$_i$] hit__on the head?

I will concentrate here on properties of *wh*-movement, mainly for reasons of space.

(p. 174) One of the earliest proposals of the minimalist framework was the elimination of D-structure and S-structure. Let us consider each of these in turn. We will then move on to consider some problems having to do with successive-cyclic *wh*-movement, and will end with a discussion of implications that the properties of A-bar movement have for the interfaces between syntax, phonology, and semantics.

8.1 Eliminating D-structure: Merge, Move, and the Derivation

Minimalism inherits from its predecessors a derivational model, in which the building of a syntactic tree can involve movement operations, as well as other operations which simply create hierarchical structure. In such a model, the question arises of how movement and non-movement operations are to be ordered. Prior to minimalism, the general assumption was that all movement operations took place after the tree had been entirely built; D-structure was taken to be the level at which the thematic requirements of predicates were satisfied, after which movement transformations could alter the structure of the sentence. This model offers a straightforward answer to the question of how movement and non-movement operations are to be ordered: movement operations invariably follow non-movement ones in the derivation. The elimination of D-structure led to the abandonment of this straightforward answer. In the following sections, we will consider the answers that were offered in its stead.

8.1.1 Late Merger

Minimalist syntactic work now assumes that movement and non-movement tree-building operations (Move and Merge, or, in more recent work, Internal and External Merge) may be freely interspersed, with their ordering determined by general principles. One of these general principles is the cycle, the nature of which has become a fruitful topic of research.

Lebeaux 1994, 2009 offered the first argument for the interspersal of Internal and External Merge, based on properties of reconstruction in A-bar chains. The argument begins with contrasts like the one in (3) (van Riemsdijk and Williams 1981, Freidin 1986, Lebeaux 1988, Chomsky 1995b, Fox 2000):

(3)

- **a.** *Which argument [that John$_i$ is a genius] did he$_i$ believe?
- **b.** Which argument [that John$_i$ made] did he$_i$ believe?

(3a) is apparently subject to obligatory reconstruction, thus violating Condition C of the Binding Theory; (3b), interestingly, avoids the same violation. Lebeaux **(p. 175)** suggested that this contrast reflects a difference in the timing of Merge of the bracketed constituents. On his proposal, the nominal complement clause *that John is a genius* in (3a) is required to Merge early in the derivation, while the relative clause in (3b) has the option of merging later, after *wh*-movement has taken place. Thus, (3a) goes through an early stage of the derivation at which Condition C is violated, prior to *wh*-movement; in (3b), by contrast, there is no point in the derivation at which Condition C is violated, since the relative clause containing the name *John* is not merged until after *wh*-movement has moved the object out of the *c*-command domain of the subject, an operation known as Late Merger. For this type of account to work, we must assume either that Condition C applies throughout the derivation (which was Lebeaux's suggestion) or that movement operations leave behind null copies which are subject to Binding Theory (Chomsky 1995b and much subsequent work).

For Lebeaux, the requirement that the nominal complement clause be merged early was an instance of the Projection Principle, which required all theta assignment to take place at D-structure. His argument showed, however, that D-structure could no longer be the level at which all instances of Merge took place. Chomsky (1993) concluded, on this basis of this argument and others, that D-structure could be done away with entirely. Having eliminated D-structure, of course, we must develop some other account of the contrast in (3), and several options present themselves. For instance, in place of a requirement that all theta assignment take place at the beginning of the derivation, we could posit a requirement that theta assignment take place as early as possible, perhaps as soon as the theta assigner has been introduced into the tree. Chomsky (1993) develops a general principle, to be discussed in the next section, which prevents Merge in embedded positions (his Extension Condition), and explicitly exempts adjuncts from this requirement. Fox (2002) offers a condition on the interpretation of movement structures which requires complements to be present in all the positions in a movement chain; in effect, all the positions in the chain contribute to the interpretation, and if any of these positions have missing argument structure, the result will be uninterpretable.

Theories crucially assuming Late Merger have subsequently been offered in a number of domains, including Fox and Nissenbaum's (1999) work on extraposition, Nissenbaum (2000) on parasitic gaps, Bhatt and Pancheva (2004) on comparatives, and Takahashi and Hulsey (2009) on the A/A′ distinction.

8.1.2 Interspersing Internal and External Merge: the cycle

With D-structure eliminated, the burden of constraining the ordering of Internal and External Merge operations falls mainly on the cycle. See Uriagereka (Chapter 11 below) for a full discussion of the role of the cycle; in this section I will concentrate on its role in our understanding of A-bar dependencies.

The cycle is another concept inherited from pre-minimalist approaches to syntax; it requires that operations which affect lower parts of the tree occur earlier in the **(p. 176)** derivation than operations which affect higher parts of the tree. As Chomsky (1993) points out, for example, the cycle could be used to rule out certain derivations which circumvent locality violations, like the one in (4):

(4)

a. is certain [John to be here]

b. seems [is certain [John to be here]]

c.

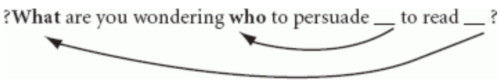
d. John seems [it is certain [__to be here]]

In (4a), the derivation begins by creating a context for raising, constructing a raising infinitive which appears as the complement of the raising predicate *be certain*. Instead of performing raising at this point, however, the derivation continues to merge more material on top of the result of (4a), adding another raising predicate seems. In (4c), *John* undergoes raising into the matrix clause, and finally, in (4d), an expletive is inserted in the embedded clause. The resulting structure has the appearance of a super-raising example, but in fact no movement of *John* past another DP has taken place. If we think that the ill-formedness of super-raising has to do with movement of one DP past another, then the derivation in (4) must be ruled out on some other grounds. The cycle, Chomsky pointed out, allows us to achieve this; Merge of the expletive in (4d) is countercyclic, since it applies to a subset of the existing tree.

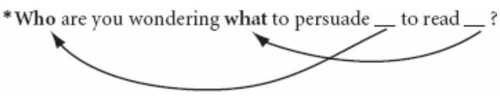
Chomsky (1993) noted that a version of the cycle could come from a condition on Merge operations which required the Merge operation to ‘extend the tree’, creating a new node which dominates a previously undominated node. In (4d) above, the Merge of the expletive fails to extend the tree, since the new node created by the operation is dominated by the material forming the matrix clause. This Extension Condition required the tree to be built from the bottom up, with operations occurring as soon as was structurally possible[1].

One of the early triumphs for the cycle in minimalism was Kitahara's (1994b, 1997) derivation of Pesetsky's (1982) Path Containment Condition. The Path Containment Condition required intersecting movement paths to nest, rather than crossing, and was responsible for contrasts like the one in (5):

(5)



a.



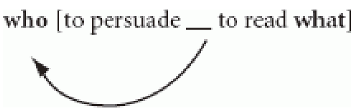
b.

(p. 177) Both of the examples in (5)are *wh*-island violations, but the one in (5b) violates the Path Containment Condition as well, since the paths of *who* and *what* intersect and neither is contained by the other.

Kitahara derived the Path Containment Condition from the cycle, together with a condition requiring that movement paths must be maximally short (in his terminology, the Minimal Link Condition). The derivation of an example like the one in (5a) would run as in (6):

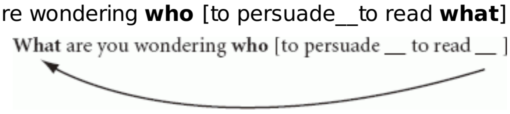
(6)

a. to persuade **who** to read **what**



b.

c. you are wondering **who** [to persuade__to read **what**]



d.

In this derivation, the first instance of *wh*-movement takes place in (6b), and the Minimal Link Condition requires that the higher of the two *wh*-phrases move, since this creates the shortest possible movement path. The matrix clause is then built, and in (6d), the remaining *wh*-phrase moves, incurring a *wh*-island violation.

There are at least two derivations that could yield the Path Containment Condition-violating order in (5b). In one, the Minimal Link Condition is violated in the embedded clause:

(7)

a. to persuade **who** to read **what**

what [to persuade **who** to read __]

b.

c. you are wondering **what** [to persuade__to read **who**]

Who are you wondering **what** [to persuade __ to read __]

d.

Here the Minimal Link Condition is violated in step (7b), in which *what* moves past *who*. We can therefore rely on the Minimal Link Condition to rule out the derivation in (7), in favor of the derivation in (6). Kitahara pointed out another derivation, however, which we must rely on cyclicity to rule out:

(8)

a. to persuade **who** to read **what**

b. you are wondering [to persuade **who** to read **what**]

Who are you wondering [to persuade __ to read **what**]

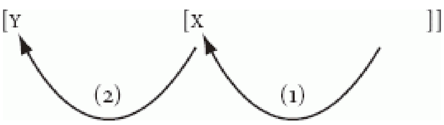
(p. 178) **c.**

Who are you wondering **what** [to persuade __ to read __]

d.

The first *wh*-movement in this derivation takes place in (8c), and since the higher of the two *wh*-phrases is the one to move, the Minimal Link Condition is obeyed.[2] The difficulty with derivation (8) is the violation of cyclicity; the move in (8d) is countercyclic (in Chomsky's 1993 terms, it fails to extend the tree, thus violating the Extension Condition). The offending example (5b) is therefore ruled out; one of its possible derivations is excluded by the Minimal Link Condition and the other by cyclicity. Kitahara's proposal thus derives the Path Containment Condition from more general principles, leading to a deeper explanation of the facts.

Descriptively, cyclicity declares that an operation whose effects are completely contained in a projection, call it X, must take place earlier in the derivation than another operation whose effects are contained in another projection Y, such that X is completely contained in Y:



(9)

Some work has focused on the question of what kinds of nodes can act as X and Y for purposes of the cycle. Chomsky's (1993) Extension Condition enforces the cycle for every syntactic node; once a node X is contained in another node Y, Merge to X can no longer extend the tree, and is blocked.

Chomsky (1995b) offers a distinct version of the cycle for movement, which increases the size of the domains which are relevant for cyclicity. His proposal is couched in terms of a theory which assumes the existence of features responsible for driving overt movement operations, which he called 'strong features'. His proposal is that strong features must trigger movement immediately, as soon as they have been introduced into the structure; names in the literature for the proposal include the virus theory (Uriagereka 1998) and featural cyclicity (N. Richards 1997, 2001). Consider, on this view, Kitahara's cyclicity-violating derivation in (8), repeated as (10):

(10)

a. to persuade **who** to read **what**

b. you are wondering [to persuade **who** to read **what**]

Who are you wondering [to persuade __ to read **what**]

(p. 179) **c.**

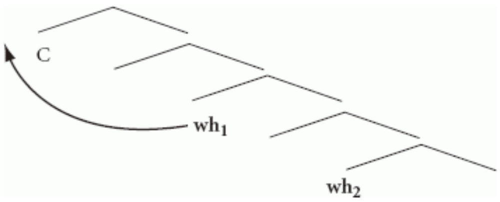
Who are you wondering **what** [to persuade __ to read __]

d.

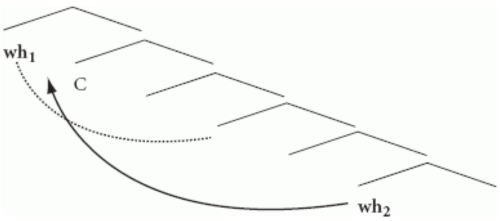
The C of the embedded clause in this derivation bears a strong feature, which ultimately drives *wh*-movement in (10d). However, this C is introduced into the derivation in step (10a). The delay in *wh*-movement between (10a) and (10d) is in violation of 'featural cyclicity'; the strong feature of the embedded C could trigger movement immediately, but does not do so.

Chomsky's (1995b) version of cyclicity enlarges the minimal domains to which cyclicity applies, at least for overt movement. Movement operations need no longer expand the tree, as long as they contribute to the immediate checking of a strong feature. Richards (1997, 2001) makes a proposal about movement to multiple specifiers of a single head which takes advantage of this looser version of cyclicity. If movements to multiple specifiers are not required to expand the tree, and if the Minimal Link Condition requires movement paths to be as short as possible, then we should expect each movement to land in a specifier underneath all the previously created specifiers, a type of movement which has come to be known as 'tucking in':

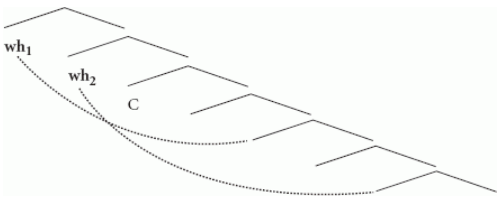
(11)



a.



b.



(p. 180) **c.**

In (11a), the Minimal Link Condition forces the higher of the two *wh*-phrases to undergo *wh*-movement to a specifier of C. In (11b), the second *wh*-phrase must move. If cyclicity is enforced by Featural Cyclicity rather than the Extension Condition, then this movement need not extend the tree; consequently, we should expect the Minimal Link Condition to force this move to land in the lowest possible position, creating a specifier of CP below the existing specifier. The consequence, shown in (11c), is that the paths of *wh*-movement in this kind of case are required to cross, preserving the original hierarchic relation between the two moved phrases.

As Rudin (1988) had previously shown, this is indeed the correct description of multiple *wh*-movement in languages like Bulgarian:

(12)

a.

Koj	Kakvo	kupi?
who	what	bought
‘Who bought what?’		

b.

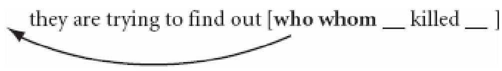
*Kakvo	koj	kupi?
what	who	bought

Combining the reasoning behind ‘tucking in’ with Kitahara's derivation of the Path Containment Condition, we arrive at a new conclusion: languages like Bulgarian ought to exhibit the opposite of the Path Containment Condition, a requirement that multiple-*wh* paths must maximally cross. To see this, consider a Bulgarian equivalent of Kitahara's derivation:

(13)

a.

b. [**who whom**__killed__]



c.

d. Who are they trying to find out [**whom**__killed__]?

This derivation begins in (13a) with the creation of an embedded clause with an interrogative C, and both *wh*-phrases move to specifiers of this C, via the tucking-in derivation just described. This yields the structure in (13b). In (13c), the matrix clause is constructed, and a *wh*-phrase must move to the specifier of the higher C. (p. 181) The Minimal Link Condition requires that this be the higher of the two *wh*-phrases, yielding the result in (13d). Since (13d) is a structure in which the intersecting *wh*-paths cross, we arrive at the prediction that Bulgarian should prefer crossed paths to nested paths, in exactly those contexts in which English has been shown to have the opposite preference. And indeed, this is the correct prediction:

(14)

a.

Koj	se	opitvat	da	razberat	kogo	__	e	ubil__?
Who	SELF	try	to	find.out	whom		AUX	killed
‘Who$_1$ are they trying to find out whom$_2$__$_1$ killed__$_2$?’								

b.

*Kogo	se	opitvat	da	razberat	koj	__	e	ubil__?
Whom	SELF	try	to	find.out	who		AUX	killed
‘Whom$_2$ are they trying to find out who$_1$__$_1$ killed__$_2$?’								

Kitahara's (1994b) logic for deriving the Path Containment Condition, together with Chomsky's (1995b) replacement of the Extension Condition with the less stringent Featural Cyclicity, accurately predicts these Bulgarian facts.

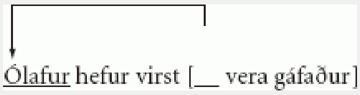
More recent work (Hiraiwa 2005, Chomsky 2008a) has suggested that the size of the relevant domain for the cycle should be increased even further, so that operations may occur in any order even when they are triggered by distinct heads in the tree, as long as those heads are all contained in the same *phase*. We will discuss the nature of phases later, in section

8.4; for the time being, all that is relevant for us is that they are portions of structure larger than a single maximal projection, though still smaller than a complete tree.

One set of facts that militate in favor of this further loosening of the cycle comes from Icelandic, as described by Holmberg and Hróarsdóttir (2003). Icelandic has a process of raising, which is disrupted in (15b) by the presence of an experiencer:

(15)

a.



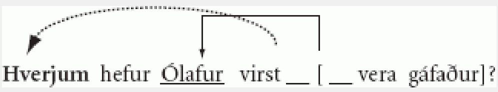
Olaf	has	seemed	to.be intelligent
'Olaf has seemed to be intelligent'			

b.

Olaf	has	seemed **to.her**	to.be intelligent
'Olaf has seemed to her to be intelligent'			

However, Holmberg and Hróarsdóttir note that (15b) can be redeemed by *wh*-extraction of the experiencer:

(p. 182) **(16)**



to.whom	has	Olaf	seemed	to.be intelligent
'**To whom** has Olaf seemed to be intelligent?'				

The contrast between (15b) and (16) is puzzling for a number of reasons, and one of the problems has to do with cyclicity. On either of the first two versions of cyclicity discussed here, *wh*-movement in (16) ought to be required to take place after raising of the subject. If raising takes place after *wh*-movement, it cannot extend the tree, and will thus violate the Extension Condition; moreover, the head which triggers raising would be unable to immediately check off the feature triggering raising, violating Featural Cyclicity. Unless wh-movement can somehow retroactively erase the locality violation created by raising of the subject past the experiencer, then, some more relaxed version of cyclicity would seem to be necessary, so that *wh*-movement need not follow raising in the derivation. Hiraiwa (2005) and Chomsky (2008a) in fact claim that operations within a phase are all simultaneous.

8.2 Eliminating S-structure! syntax and the interfaces

Chomsky (1993) proposed to eliminate not only D-structure but also S-structure. In earlier work, S-structure had been the last point in the narrowly syntactic part of the derivation, before the representation was sent to the phonological and semantic interfaces for interpretation. The model assumed in Chomsky (1993) still had such a point in the derivation but the proposal was that no formal reference would be made to it; syntactic conditions would either hold throughout the derivation or would make reference to one or another interface, but would never refer directly to S-structure. Ultimately, the goal is to derive the properties of syntax, as much as possible, from the need for syntax to provide the interfaces with interpretable material.

This ultimate goal is consistent with at least three imaginable stands on the relationship between syntax and the interfaces. On one type of view, syntactic operations are directly guided by their consequences for the interfaces; the syntax is able to consider the consequences of its actions for the eventual interface representations, and chooses its actions accordingly. On another type of view, we might imagine that the syntax simply performs syntactic operations without regard for the needs of the interfaces, and that the interfaces then act as filters, ruling out some of the representations given to them by the syntax (or, perhaps, (p. 183) altering unacceptable representations postsyntactically in ways that make them interpretable). A third proposal would find a middle ground between these extremes: the syntax does perform operations which improve the suitability of the representation for the interfaces, but it does so in response to objects which are present in the narrow syntax, not because of any direct 'knowledge' of the interface representations. Of course, these proposals are not mutually exclusive: a single theory might posit cases in which the syntax makes direct reference to interface representations, other cases in which interface representations must impose an interpretation on a syntactic operation, and still other cases in which syntactic objects prompt syntactic operations that have the effect of improving the interface representations.

Chomsky (1993) emphasizes the role of the third of these options. He posits features in the syntactic representation which trigger syntactic operations, and these syntactic operations have the effect of making the representation interpretable by the interfaces, but the syntax itself makes no direct reference to properties of the interfaces. Moreover, he suggests that certain kinds of syntactic operations cannot take place unless motivated by these features. More recent work (Bošković 2001, Chomsky 2008a) has explored the second kind of relationship between syntax and the interfaces; the syntax generates structures, perhaps without any 'motivation' in the classic sense, and the interfaces assign an interpretation to the result. Comparatively little work has proposed direct connections between syntax and the interfaces, but there are cases in which it is difficult to see how such direct connections can be avoided; for instance, Fox (2000) argues compellingly that the best description of Quantifier Raising involves a syntactic operation which makes crucial reference to semantic interpretation.

8.2.1 Agree

Chomsky (1993) proposes that syntactic operations are driven by the need to delete features which are uninterpretable at one or another of the interfaces. In Chomsky (1995b), he proposes that this deletion takes place when features on a head (sometimes called the Probe) seek corresponding features on a phrase (sometimes called the Goal); I will refer to this operation (anachronistically) as Agree. Agree is sometimes associated with a movement operation that moves the Goal into a specifier of the Probe; we will explore the conditions under which this takes place in the following sections.

8.2.1.1 The Probe

Ascribing movement to the actions of Probes allows us to describe more straight-forwardly the interaction of potential participants in movement, for instance, in multiple-*wh*-constructions. Consider the facts in (17)–(18):

(p. 184) **(17)**

a. Who bought **what**?
b. ***What** did **who** buy?

(18)

a. What did you give to **whom**?
b. ***Who** did you give **what** to?

The facts in (17)–(18) are traditionally referred to as Superiority effects, and can be described as a preference for short moves over long ones; given a choice between *wh*-phrases to move, it is the higher of the two which must move[3]. We saw the same preference in effect in section 8.1.2, where it was crucial for Kitahara's derivation of the Path Containment Condition. If we make the interrogative C the Probe responsible for triggering *wh*-movement, then this locality condition can be described as a condition on C; the Agree relation initiated by C must be with the closest available *wh*-phrase.

Another case in which it is useful to make reference to Probes has to do with island conditions on *wh*-movement. In English, certain types of island effects block *wh*-movement, but appear to have no effect on *wh*-in situ:

(19)

a. ***What** did the senator deny [the rumor that he wanted to ban__]?
b. Who denied [the rumor that he wanted to ban **what**]?

Wh-movement out of the clausal complement of the noun *rumor* in (19a) is impossible, but *wh*-in situ can appear inside such a relative clause, as in (19b). If we think that *wh*-in situ is related to its scope position via movement (a possibility we will consider further in section 8.2.1.3), then the well-formedness of (19b) is surprising; covert movement of *what* appears to be immune to the island effect that rules out *wh*-extraction in (19a). Huang (1982) proposed that this is a general difference between overt and covert *wh*-movement; covert *wh*-movement is not subject to the same island effects that we find in overt *wh*-movement.

In Richards (1997, 2001) I claimed that the contrast in (19) was actually a contrast between single and multiple *wh*-movement. We find contrasts like (19) in languages like Bulgarian, in which all of the *wh*-movements involved are overt:

(20)

a. ***Koja kniga** otrece senatorat
which book denied the-senator

$\boxed{\text{malvata ce iska da zabrani \_\_}}$

the-rumor that wanted to ban
'Which book did the senator deny [the rumor that he wanted to ban t]?'

(p. 185) **b.**

Koja senator __ otrece
which senator denied
$\boxed{\text{malvata ce iska da zabrani } \textit{Vojna i Mir}}$
the-rumor that wanted to ban war and peace

'Which senator denied

the rumor that he wanted to ban *War and Peace*?'

c.

Koj senator **koja kniga** __ otrece
which senator which book denied
$\boxed{\text{malvata ce iska da zabrani \_\_}}$
the-rumor that wanted to ban

'Which senator denied the rumor that he wanted to ban which book?'

The contrast in (2oa–b) shows that Bulgarian, like English, is unable to extract *wh*-phrases from inside complement clauses of nominals. In (20c), we see another similarity between Bulgarian and English: multiple *wh*-questions in which one *wh*-phrase extracts out of an island are well-formed, as long as another *wh*-phrase extracts from a position not inside the island. Bulgarian differs from English, however, in that the movement operations in question are both overt. Watanabe (1992) noted a similar contrast in Japanese, in which all *wh*-phrases are left in situ:

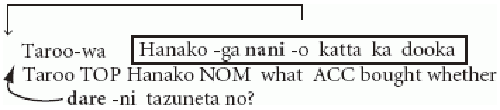
(21)

*Taroo-wa $\boxed{\text{Hanako -ga } \textbf{nani} \text{ -o katta ka dooka}}$ tazuneta no

a.
Taroo TOP Hanako NOM what ACC bought whether asked
'What did Taroo ask [whether Hanako bought]?'

Taroo-wa $\boxed{\text{Hanako -ga kuruma-o katta ka dooka}}$
Taroo TOP Hanako NOM car ACC bought whether
dare-ni tazuneta no?

b.
who DAT asked Q
'Who did Taroo ask [whether Hanako bought a car]?'


Taroo-wa Hanako -ga **nani** -o katta ka dooka
Taroo TOP Hanako NOM what ACC bought whether
dare -ni tazuneta no?

c.
who DAT asked Q
'Who did Taroo ask whether Hanako bought what?'

It seems to be generally true, then, that multiple *wh*-questions are well-formed as long as one of the *wh*-movements obeys island conditions, regardless of whether the *wh*-phrases involved are overtly moved or in situ. We can view this as evidence that these island conditions are in fact conditions on Probes, rather than the movements themselves; the island conditions require each Probe to participate in a **(p. 186)** non-island-violating *wh*-question, and as long as this takes place, other *wh*-questions may freely violate island conditions.

8.2.1.2 The Goal

The preceding section concentrated on the properties of the Probe in the Agree relation. Let us now turn our attention to proposals about the nature of the Goal.

On the view of Agree relations first articulated in Chomsky (1995b), it is the Probe that bears the features which determine whether the Agree relation will trigger movement or not ('strong' features, in Chomsky's (1995b) terminology). As we have already seen, the distribution of feature strength is a matter of cross-linguistic parametrization, with some languages (like English and Bulgarian) having strong features responsible for overt *wh*-movement, and others (like Japanese) lacking such features. The behavior of English multiple-*wh*-questions seems to offer support for this perspective. In this language, there is no straightforward generalization to be made about whether *wh*-phrases undergo movement; some do, and others do not:

(22) What did you give__to **whom**?

On the other hand, there is a straightforward generalization about Probes in ordinary questions in English (leaving echo-questions aside); they always trigger a *wh*-movement. Giving the Probe the responsibility for determining whether *wh*-movement will happen overtly or not seems to capture the English facts well.

Bošković (1999) explores the possibility that in Serbo-Croatian, it can be the Goal that bears the features which force overt movement. He notes, first of all, that all *wh*-phrases in Serbo-Croatian are in fact required to move overtly; in other words, it is perfectly possible to make a generalization about the behavior of Goals that are agreed with by C in this language:

(23)

a.

Ko	je	**koga**	vidio?
Who	AUX	whom	seen
'Who saw whom?'			

b.

Koga	je	**ko**	vidio?
whom	AUX	who	seen

c.

***Ko**	je	vidio	**koga**?
who	AUX	seen	whom

Moreover, Serbo-Croatian is unlike Bulgarian in that the fronted *wh*-phrases may be in any order. Bošković suggests that we can capture this fact by putting the features responsible for driving overt movement on the *wh*-phrases themselves in this language. In the multiple-*wh*-questions we have considered so far, on this view, the feature that forces *wh*-movement to be overt is on the Probe, and this feature must Agree with the closest available Goal. In Serbo-Croatian, on the other hand, Bošković's theory posits features on the *wh*-phrases which force them to overtly (p. 187) move; each of these features is required to Agree with the closest available Probe, but in a sentence like the one in (23), nothing determines which Goal is to be Agreed with first, and consequently, the order of the *wh*-phrases is free.

Another debate in the literature has to do with where exactly the features in a Goal for the Agree operation are housed. A (perhaps largely unexamined) assumption of the literature on *wh*-movement, for example, has assumed that the C Probe is Agreeing with a feature in the *wh*-word itself, perhaps a feature of D when the *wh*-phrase is a DP. Hagstrom (1998) offers arguments that there is a third participant in *wh*-questions, in addition to C and the *wh*-word, which he calls Q. Q is phonologically null in English, but appears overtly in a number of languages, including Sinhala, Shuri Okinawan, and older stages of Japanese[4]:

(24)

a.

Siri	**mokak**	**də**	keruwe?
Siri	what	Q	did-E
‘What did Siri do?’			

b.

oyaa	[**kauru**	liyəpu	potə]	**də**	kieuwe?
You	who	wrote	book	Q	read-E
‘Who did you read [a book that__wrote]?’					

As the Sinhala examples in (24) show, the Q particle *də* appears after the *wh*-phrase, and need not be adjacent to it. Hagstrom develops a semantics for *wh*-questions in which the *wh*-phrase, Q, and the complementizer all play a role.

Cable (2007) argues, on the basis of the properties of Q in Tlingit, that the Agree relation initiated by C is actually with Q, and not with the *wh*-phrase at all. One of his goals is to account for the fact that *wh*-movement often can, and sometimes must, move a phrase which is larger than the *wh*-phrase itself, a phenomenon known in the literature as pied-piping:

(25)

a. [**How** many dogs] did you see?
b. *[**How** many] did you see [__dogs]?
c. ***How** did you see [__many dogs]?

Cable argues that a theory of the possible positions for Q would effectively be a theory of pied-piping; since C Agrees with Q, it is the projection dominating Q which undergoes movement. He develops such a theory, based on the distribution of Q in Tlingit:

(26)

a.

[**X'oon**	keitl	**sá**]	ysiteen?
how.many	dog	Q	you.saw.them
‘How many dogs did you see?’			

b.

*[**X'oon**	**sá**]	keitl	ysiteen?
how.many	Q	dog	you.saw.them

(p. 188) For a different minimalist approach to the problem of pied-piping, see Heck (2004, 2009), who offers an account of the facts in terms of locality restrictions on the Agree relation.

8.2.1.3 Agree and Move

In the domain of A-dependencies, it is often claimed that Agree operations need not trigger any type of movement. Existential constructions offer a standard example of this:

(27)

a. There **is/*are** a book on the table.
b. There ***is/are** books on the table.

In (27), T Agrees with the postverbal DP, but this DP does not move to the specifier of TP.

We have already seen examples of *wh*-phrases which do not appear to undergo movement:

(28)

a. Who bought **what**?
b.

Taroo-wa	**nani-o**		katta	no?
Taroo TOP	what	ACC	bought	INT
'What did Taroo buy?'				

In multiple-*wh*-questions in English, or in any *wh*-question in a language like Japanese, we find *wh*-phrases which undergo no overt movement. The literature offers two classes of approaches to this kind of fact.

On one approach, the existence of *wh*-questions like the one in (28b) demonstrates that the syntax—semantics interface is more complex than we might have hoped. In particular, we need to posit a process for interpretation which will yield the semantics of a *wh*-question when confronted either with a syntactic structure involving *wh*-movement or with a structure in which *wh*-phrases are left in situ. On the other hand, this model allows the relationship between syntax and phonology to be comparatively simple, at least in this domain; *wh*-phrases undergo movement just when that movement is actually detectable from its effects on word order. For approaches along these lines, see Tsai (1994, 1999), Shimoyama (2001, 2008), and references cited there.

An alternative view posits types of movement which do not affect word order; such a view must complicate the relationship between syntax and phonology. On the other hand, the semantics of *wh*-questions can be left relatively simple; all *wh*-questions can have the same syntactic form, as far as interpretation is concerned. Approaches along these lines include Huang (1982), Pesetsky (2000), Richards (1997, 2001, 2008), and references cited there.

Among works which do posit some covert type of movement, a number of mechanisms for covert movement have been proposed. We will return to this issue in section 8.4 below.

(p. 189) 8.3 Successive-cyclic *wh*-movement

Thus far our discussion has largely centered on monoclausal *wh*-constructions, in which the Probe responsible for triggering *wh*-movement has consistently been an interrogative C. We have excellent evidence, however, that *wh*-movement can take place to positions other than the specifier of an interrogative C in the course of the derivation.

One piece of evidence for this conclusion comes from McCloskey's (2ooo) work on the West Ulster dialect of English. West Ulster English allows *wh*-movement to strand floating quantifiers next to the original position of the *wh*-phrase:

(29)

a. What all did you do__after school the day?
b. What did you do **all**__after school the day?
c. *What did you do__after school the day **all**?

In (29a–b), we can see that *all* can appear either next to the *wh*-phrase or in a stranded position next to the site of extraction; (29c) shows that *all* is not capable of appearing in random positions in the clause.

In long-distance *wh*-questions in this dialect, *all* may appear in a number of positions:

(30)
a. What **all** did he say [that he wanted__]?
b. What did he say [that he wanted__**all**]?
c. What did he say [**all** that he wanted__]?

In (30), *all* may be stranded next to the theta-position of the *wh*-phrase, as before, but (30c) shows that *all* can also appear in an intermediate position. We can account for this fact if we say that *wh*-movement is successive-cyclic, stopping in the specifier of the intermediate CP on its way to its eventual destination.

Another argument for successive-cyclic movement comes from Lebeaux's (1994, 2009) work on reconstruction, discussed in section 8.1.1 above. Lebeaux discusses the contrast in (31):

(31)
a. [Which paper that he_i gave to $Bresnan_j$] did every $student_i$ think that she_j would like?
b. *[Which paper that he_i gave to $Bresnan_j$] did she_j think that every $student_i$ would like?

The examples in (31) involve a relative clause modifying the *wh*-phrase which contains both a pronoun that must be bound by a quantifier and a name which must be free. The contrast between the examples, in Lebeaux's theory, is related to the fact that in (31a), but not in (31b), there are positions in the tree where both of these requirements may be satisfied. In (31a), if the relative clause in question is interpreted in the specifier of the embedded CP, then the pronoun *he* can be bound by *every student*, and the name *Bresnan* can be free from being bound by *she*. (p. 190) In (31b), by contrast, there is no position for the relative clause that satisfies both constraints; if the relative clause is c-commanded by *every student*, the pronoun *he* will be bound, but so will the name *Bresnan*. On Lebeaux's account, then, the contrast in (31) demonstrates that the relative clause can undergo Late Merger in a position which is neither the theta-position for the *wh*-phrase nor the position in which it is pronounced. If we allow long-distance *wh*-movement to stop in the specifier of the intervening CP, and if (as we argued in section 8.1.1) a relative clause may be merged into the structure at any point in the derivation, then we can account for the contrast in (31), merging the relative clause to the *wh*-phrase as it stops in its intermediate position.

Fox (2000) pushes this approach further, developing an argument for additional landing sites for *wh*-movement, on the basis of contrasts like the following (Fox 2000: 164):

(32)
a. [Which of the papers that he_i wrote for Mrs $Brown_j$]
did every $student_i$ get her_j to grade?
b. *[Which of the papers that hei wrote for Mrs $Brown_j$]
did she_j get every $student_i$ to revise?

Again, the contrast is to be explained in terms of the timing of Merge; in (32a), there is a section of the tree into which the relative clause can be safely merged, where it will be c-commanded by *every student* but not by *her*, and hence satisfy the needs of both the pronominal variable *he* and the name *Mrs* Brown. In(32b), by contrast, there is no such section of the tree. By the reasoning outlined just above, this leads us to the conclusion that the *wh*-phrase lands in a position between *every student* and *her* in an example like (32), and that the relative clause can be merged while the *wh*-phrase is in this position.

The successive-cyclic nature of *wh*-movement seems to be conclusively proven (for further evidence for this conclusion, see Chung 1998, Nissenbaum 2000, Legate 2003, Abels 2003). From the standpoint of the theory developed so far, the existence of successive-cyclic movement raises at least two puzzles. First, if we were right in section 8.2 to conclude that movement is always triggered by an Agree operation involving some feature, what feature drives the movements to the intervening positions? And second, why should movement be successive-cyclic at all?

There are languages in which successive-cyclic movement is associated with agreement morphology on intervening complementizers or verbs (Kinande, from Schneider-Zioga 2007: 422, and Passamaquoddy, from Bruening 2001: 217):

(33)

a.

Ekihi	**kyo**	Kambale	a-asi	nga-**kyo**	Yosefu	a-kalengekanaya	nga-**kyo**	Mary'	a-kahuka
what.7	7	Kambale	AGR-know	C-7	Joseph	AGR-think	C-7	Mary	AGR-cook
'What did Kambale know that Joseph thinks that Mary is cooking?'									

(p. 191) **b.**

wen-**ihi**	piluwitaham-oc-**ihi**	Piyel	kisi-komutonom-ac-**ihi**?
who-**OBV.PL**	suspect-2-OBV.PL	Piyel	PERF-rob-3-OBV.PL
'Who all do you suspect that Piyel robbed?'			

We might conclude from this that successive-cyclic movement is driven by Probes, much as movement to the specifier of an interrogative complementizer is, and that the Agree relations initiated by these Probes have morphological effects in some languages. A potential problem for this conclusion is that the relevant Probes will have to be absent when no *wh*-phrase is present in the sentence (or will have to satisfy their need to Agree in some other way). One challenge for theories that attempt to account for successive-cyclic movement is the well-formedness of examples like (34):

(34) Who said that he would buy **what**?

Examples like (34) show that we cannot simply require *wh*-phrases to move, in a language like English, whenever they are present. In this example, *what* must remain in situ, even though (on standard minimalist assumptions about bottom-up tree-building) the higher *wh*-phrase *who* is not introduced until much later in the derivation.

A variety of solutions to these problems have been offered in the literature, which space restrictions prevent me from discussing at any length. Many of them involve abandoning, to some degree, the assumption that movement must be driven by Agree relations; movement is either taken to be entirely free, or is allowed to take place without Agree under special circumstances. See Heck and Müller 2000, McCloskey 2002, Boeckx 2007, Bošković 2007, Preminger 2008a, Chomsky 2008a, and references cited there for further discussion.

As for why movement should be successive-cyclic at all, a common answer relates this fact to locality conditions on movement which prevent movement from being too long, thus forcing it to proceed in short steps (Takahashi 1994, Boeckx 2003, and much other work). One version of this kind of theory is the one positing *phases*, to which we will turn in the next section.

8.4 The role of the interfaces

The syntax of A-bar extraction has been one source of data bearing on the nature of the interaction between the syntax and the semantic and phonological interfaces. In particular, researchers have hoped to shed light on the difference between overt and covert movement, and on the locality conditions constraining A-bar movement. I discuss each of these issues briefly in turn.

(p. 192) As we saw in section 8.2.1.3, there are a number of approaches to the problem of *wh*-in situ that have been defended in the literature. On one type of view, *wh*-in situ is simply intepreted in situ; on another, *wh*-in situ undergoes some type of syntactic movement which does not alter word order. Several versions of this second type of approach have been proposed, and in fact we may discover that different options are correct for different instances of *wh*-in situ. One kind of theory would posit movement of elements which are not pronounced; for example, if Agree relations are in fact relations between features on heads, then we might not be surprised to discover cases in which it is the features themselves which undergo movement, rather than any actual phrases or heads. Alternatively, we could see *wh*-in situ as teaching us about the nature of the interfaces; on this view, we must understand *wh*-in situ as undergoing a type of movement which feeds the interpretive component but not the phonological component.

Pesetsky (2000) argues that there are in fact multiple ways for *wh*-in situ to be achieved; in particular, he argues that *wh*-in situ sometimes involves feature movement, and sometimes a type of phrasal movement which does not affect word order. One of his arguments has to do with differences in the status of different types of *wh*-in situ with respect to resolution of Antecedent-Contained Deletion (ACD). He notes, first, that *wh*-in situ is in principle capable of hosting ACD:

(35) I need to know [**which girl** ordered [**which boy** [that Mary also did [__]]]] to congratulate Sarah.

Here the elided VP inside the in situ *wh*-phrase can be interpreted as meaning 'ordered to congratulate Sarah'. The example in (35) contrasts with the one in (36):

(36) *I need to know [**which girl** Sue ordered [**which boy** [that Mary also did [__]]]] to congratulate__.

Here ACD resolution is impossible. Pesetsky claims that the *wh*-insituin (35) undergoes phrasal movement in a manner which does not affect word order but which does affect interpretation. In (36), by contrast, he argues that the *wh*-in situ is related to the interrogative C via feature movement, and that the *wh*-phrase itself remains in situ. The ACD contrast follows; ACD is only possible if the *wh*-phrase moves out of the VP which is to act as an antecedent for the elided VP.

If Pesetsky is right, then we apparently need some mechanism (or set of mechanisms) for allowing syntactic movement of phrases to feed the semantic representation but not the phonological one[5]. At least three such mechanisms have been proposed in the literature. One, sometimes known as 'single-output syntax' (Bobaljik 1995a), simply allows the phonological component to choose whether to pronounce the head or tail of a movement chain; on this view, there is no syntactic (p. 193) difference between overt and covert movement. Another approach posits movement after the narrow syntax is complete and has already been sent to the phonological component; this is sometimes called the 'T model' or 'Y model' (Huang 1982, Richards 1997, 2001). A third approach, sometimes called a 'multiple spell-out model', proposes that syntactic material is periodically sent to the phonological and semantic components each time a syntactic object of a certain size has been constructed, before the syntactic derivation is completely over (Uriagereka 1999, Chomsky 2001, 2007, 2008a). The syntactic objects that are sent to the interfaces once completed are what have come to be known as 'phases'.

Space considerations prevent me from discussing phases here in depth; see Uriagereka (Chapter 11 below) for further discussion. Accounts positing phases typically hold that once a phase has been completely constructed in the syntax, the material in the phase is sent to the interfaces (or, in some theories, just to the phonological interface) after which it becomes inaccessible to the syntactic computation (this is the PIC, or Phase Impenetrability Condition). This property of phases has been of interest to syntacticians working on A-bar dependencies, since it promises to offer us a new account of some of the locality restrictions on A-bar movement.

Such an account would be very welcome. Some of the classic island effects have been successfully subsumed under the Minimal Link Condition and its descendants; for instance, we can hope to deal with *wh*-island effects in this way, and possibly also with relative clause islands:

(37)

a. ***What** are you wondering [**who** Mary persuaded__[to buy__]]?

b. ***What** did you meet a man [**who** Mary persuaded__[to buy__]]?

Moreover, as we saw in section 8.1.2 above, the minimalist approach to *wh*-islands is cross-linguistically better supported than its predecessors; we have an account, for example, of why English prefers nested *wh*-paths to crossed ones, while in Bulgarian the reverse is true.

Still, many of the classic island effects discussed by Ross (1967) and in much subsequent work are no better explained under minimalism than they were in previous theories. For instance, we have no generally accepted account of Huang's (1982) Condition on Extraction Domains (CED), which banned extraction out of subjects and adjuncts:

(38)

a. ***What** do you think that [to buy__] would be a mistake?

b. ***What** are you angry [because Bill bought__]?

Cross-linguistic work has cast doubt on the claim that adjunct and subject island effects should be handled by the same principle (Kitahara 1994a, Stepanov 2007), but our understanding of the facts has not grown much deeper since they were first discovered; certain domains just seem to be opaque to extraction. The advent (p. 194) of phases has given us hope of a deeper explanation: see, in particular, Uriagereka (1999), Rackowski and Richards (2005), and Chomsky (2008a) for discussion[6].

8.5 Conclusions

In this chapter we have considered some properties of A-bar dependencies from a minimalist perspective. We have seen that the minimalist approach to the cycle and to locality principles can account for a variety of subtle facts about reconstruction and Superiority effects in several different languages. A number of topics are still fruitful points of research, including the nature and motivation of successive-cyclic movement and the full array of locality effects.

Notes:

(1) For an alternative approach to the cycle in which trees are built from the top down, see Phillips (1996, 2003), Richards (2003).

(2) If we are concerned about the shorter possible move to the embedded CP complement of *wonder*, we can imagine that this CP layer is countercyclically merged between steps (8c) and (8d).

(3) For careful discussion of how 'length' should be measured for these purposes, see Fitzpatrick (2002). See Chomsky (2008) for a radically different take on these facts.

(4) For more discussion of these particles, see Kishimoto (1992, 2005), Sugahara (1996), Hagstrom (1998), Cable (2007), and references cited there.

(5) See also Sauerland and Elbourne (2002) for an analysis of reconstruction as involving phrasal movement which feeds phonology but not semantics.

(6) See also Miyagawa (2010) for a phase-based account of the distribution of A- and A-bar movement.

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Head Movement and the Minimalist Program

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[–] Abstract and Keywords

This article discusses head dependencies. It begins by recapitulating the essentials of the analysis of head movement as it was largely agreed on in mainstream syntactic theory by the late 1980s. This approach was in essence unaltered in the earlier versions of minimalism. Section 9.3 considers the reasons that led Chomsky to suggest excluding head movement from the core operations of the narrow syntax. Section 9.4 reviews the various alternatives to the earlier conception of narrow-syntactic head movement that have been put forward: phonetic form movement, remnant phrasal-category movement, and 'reprojective movement', focusing on a case study of each alternative. Finally, Section 9.5 considers the conceptual status of head movement in relation to the general goals of the minimalist program.

Keywords: head dependencies, syntactic theory, minimalism, Chomsky, narrow syntax, phonetic form movement, phrasal-category movement, reprojective movement

9.1 Introduction

I will begin by recapitulating the essentials of the analysis of head movement as it was largely agreed on in mainstream syntactic theory by the late 1980s. This approach was in essence unaltered in the earlier versions of minimalism (Chomsky 1993, 1995b, aside from section 4.10). In section 9.3, I consider the reasons which lead Chomsky (2001: 37–8) to suggest excluding head movement from the core operations of the narrow syntax. section 9.4 reviews the various alternatives to the earlier conception of narrow-syntactic head movement which have been put forward: PF movement, remnant phrasal-category movement, and 'reprojective movement', focusing on a case study of each alternative. Finally, in section 9.5, I will consider the conceptual status of head movement in relation to the general goals of the minimalist program.

(p. 196) **9.2 The GB Approach**[1]

Earlier versions of generative grammar often featured head movement operations; see for example Affix Hopping in Chomsky (1957), McCawley's (1971) Tense-attraction rule, Emonds' (1971, 1976) *have/be*-raising and his (1978) verb movement rule for French, den Besten's (1983) analysis of Germanic verb-second, French subject—clitic inversion and English subject—auxiliary inversion. But it was only in the GB period that these ideas were systematized and a series of theoretical postulates were put forward that together provided a clear characterization of head movement, arising primarily from the work of Koopman (1984), Travis (1984), and Baker (1985a, 1988).

The central idea in these approaches is (1):

(1) Head movement is the case of Move-α where α is X°.

In most versions of X′-theory assumed in GB theory, X° was defined as the head of XP. This was the position into which terminals could be substituted; unlike the bare phrase structure notion of minimal X, however, it could have internal structure (beyond simply being a bundle of features of some kind), in part thanks to the possibility of head movement.

As an instance of Move-α, head movement was argued to be subject to the standard well-formedness conditions applying to movement operations and their outputs generally. These conditions were of three main types, not necessarily exclusive in their empirical effects: structure preservation, locality, and the requirement that the trace created by the movement operation meet the relevant well-formedness conditions on traces. Let us look at each of these in turn.

Concerning structure preservation, Chomsky (1986a: 4) posits two general conditions on movement: 'only X° can move to a head position' and 'only a maximal projection can move to a specifier position'. He remarks that these 'would follow from an appropriate form of Emonds' Structure Preservation Hypothesis' (the second given 'the X-bar theoretic assumption that heads cannot be base-generated without a maximal projection so that a bare head cannot appear in the specifier position to receive a moved X° category'). Later, Chomsky (1986a: 73) suggests that only maximal projections may adjoin to maximal projections, ruling out adjunction of heads to maximal projections. This proposal ('a kind of generalization of Emonds' Structure Preserving Hypothesis') follows if 'we were to regard movement of a lexical category as analogous to NP-movement, barring [it] either on the grounds that *t* [the trace of this movement—IGR] is an unlicensed free variable or that there is 'improper movement' with *t* ultimately bound in the domain of the head of its chain' (p. 73). Given the assumptions in Chomsky (1986a), this would violate Principle C of the Binding Theory. If the head moves on to an 'A-position' (p. 197) (i.e. a position adjoined to another head), then, we have improper movement. If it does not, the trace of head movement counts as an unlicensed free variable. This proposal does not, however, rule out head-to-head adjunction, and was not intended to. In fact, the upshot of Chomsky's reasoning is that head movement can only move a head to another head position. It was generally assumed that head movement adjoined the moved head to the host head, forming a structure like (2):

(2) $[_{Y}$ XY]

(But see Rizzi & Roberts 1989 for a more elaborate proposal.) Kayne (1991, 1994) proposed that head adjunction is always left adjunction, as depicted in (2).

Concerning locality, the central condition on head movement was the Head Movement Constraint (HMC), first explicitly formulated in Travis (1984). I give it in the following form:

(3) Head movement of X to Y cannot skip an 'intervening' head Z.
(Roberts 2001: 113)

'Intervention' is understood in terms of asymmetric c-command in the usual way (Z intervenes between Y and X iff Y asymmetrically c-commands both X and Z, while Z asymmetrically c-commands X). The HMC has the effect of forcing head movement to be cyclic, in an obvious sense. Moreover, it has typically been assumed that formation of the complex head in (2) could not be undone by a later step of movement. Hence further movement of Y to a higher head W would form the complex head $[_{W}[_{Y}\ X\ Y]\ W]$. In other words, iterated head movement always involves cyclic 'roll-up', forming a successively more complex head. On the other hand, 'excorporation' of X from Y (or W), or of Y from W (or X), is not allowed (but see Roberts 1991 for the observation that this assumption did not follow from any aspect of the theory of movement assumed at the time, and a discussion of two empirical candidates for excorporation).

Finally, and perhaps most importantly in the GB context, the trace of head movement was subjected to the standard conditions on traces. Indirectly, many of the conditions on head movement were derived in this way, since a structure containing an ill-formed trace would be ruled out. The movement itself was allowed to overgenerate illicit representations, which were filtered out by specific conditions. The most important of these conditions was the Empty Category Principle (ECP), which required all traces to be properly governed. Definitions of proper government varied somewhat in detail, and for present purposes it is simpler to break down the requirements imposed by the ECP into a number of separate cases, bearing in mind that the ECP provided a unified characterisation of this range of cases.

First, one effect of the ECP was that head movement of X to Y out of an XP not contained in the structural complement of Y is impossible. Thus head movement from subjects and adjuncts was impossible. Baker (1988), in particular, showed in (p. 198) detail that the various forms of incorporation he proposed satisfy this condition. Second, 'downward' head movement is not allowed, since a fundamental requirement imposed by the ECP is that an antecedent c-command its trace. This implies that the Affix Hopping, as conceived in Chomsky (1957) and elsewhere, could not be an instance of head movement if this is seen as a core syntactic operation. (Pollock 1989 and Chomsky 1991, among others, sought to avoid this consequence by treating the ECP as holding of LF representations and allowing downward movement in the overt syntax as long as the effects of this were obliterated by the time the ECP applied.) Third, to the extent that, following Rizzi (1990b), the ECP featured some form of relativized minimalism constraint, the HMC itself can be derived from the ECP. Hence the local nature of head movement follows.

The GB conception of head movement, then, was that this was a core syntactic operation raising a head X to an immediately superjacent (governing) head Y where X is contained in Y's immediate structural complement. The effects of this highly articulated and restricted conception were observed in a very wide range of empirical phenomena: noun incorporation, many kinds of morphologically complex causative constructions, applicatives, passives, verb movement within the clause of the French/Romance kind, to C of the Germanic kind and to clause-initial position of the kind found in VSO languages, English subject—auxiliary inversion, French subject—clitic and complex inversion, Italian Aux-to-Comp, inversion of inflected infinitives in European Portuguese, a whole range of phenomena involving movement of the Noun within DP, including Semitic construct states, Balkan and Scandinavian postposed articles, and the relative ordering of nouns in relation to possessors and modifiers of various kinds (see Cinque 1994, Longobardi 1994), clitic movement, and many other phenomena (see Roberts 2001 for overview, illustration, and further references).

9.3 The Minimalist Program

In the early versions of the minimalist program, the GB conception of head movement was by and large retained. The discussion of V movement to T and Agr and related issues in Chomsky (1993: 27–32/1995b: 195–9) introduces checking theory, and makes it clear that V movement, like other forms of movement, obeys the core constraints that this theory imposes. The same is true for the notions of checking domain, internal domain, and complement domain; moreover, head movement plays a role in giving rise to equidistant positions, which is central to capturing the generalization that objects move only when the verb moves (an early version of Holmberg's 1986 generalisation); see the discussion in Chomsky (1993:10–19/1995: (p. 199) 176–86). In Chomsky (1995b: 4.10), the picture changes somewhat, partly as a consequence of the abandonment of Agr as a syntactic category. Here, Chomsky proposes analyzing 'multiple subject constructions' (e.g. Germanic transitive expletive constructions such as *There painted a student the house* or passives like *There have some cakes been baked for the party*) in terms of multiple specifiers of T, since SpecAgrP is not available if Agr is not a functional head. Chomsky argues for an analysis which features the substring *Expletive Subject T* … If V is in T, this is clearly the wrong order, the attested order being *Expletive V Subject*. Taking this order to be a direct reflex of the V2 property of the languages in question, Chomsky suggests (1995b: 368) that 'the V-second property … may belong to the phonological component. If that is the case, the observed [i.e. V2–IGR] order is formed by phonological operations … and may observe the usual constraints (V→C), but need not, as far as we know.' Although V-to-T movement is assumed (Chomsky 1995b: 367), the possibility that V2 orders are derived by something other than syntactic head movement of T to C is at least questioned here.

But it was in Chomsky (2001: 37–8) that a series of arguments of a range of types are presented that, together, lead Chomsky to conclude that 'a substantial core of head-raising processes, excluding incorporation in the sense of Baker (1988), may fall within the phonological component' (p. 37).[2]

First, Chomsky claims that head movement never affects interpretation: 'the semantic effects of head-raising in the core inflectional system are slight or nonexistent, as contrasted with XP-movement' (2001: 37). The core point here is that, while French or Icelandic verbs occupy a different structural position in finite clauses from their English or Mainland Scandinavian counterparts, analysing this in terms of different extents of head movement, as was standard in GB or early minimalism (see in particular Vikner 1995), leads to the expectation that there may be some LF-related differences between verbs—perhaps involving scope or reconstruction effects—in the two classes of languages. Such effects are not found, leading to the suggestion that head movement is confined to the PF part of the grammar.

Second, Chomsky raises the question of the nature of the trigger for head movement. The issue arises when we consider, for example, T in a language such as French, which has consistent DP movement into SpecTP and consistent V movement to T (following Pollock 1989). Hence T must contain the relevant triggers for these movements: (uninterpretable/unvalued) ø-features and an EPP feature to trigger DP movement and, presumably, some form of V features combined with a movement-triggering feature triggering V movement. All other things being equal, the system has to have sufficiently rich featural information to be able to correctly distinguish the two sets of triggers: an XP movement trigger for D and (p. 200) head movement trigger for V. Similarly in V2 languages: T must move to C and XP to SpecCP, but not vice versa. Note that the prediction is not that the inverse properties may not exist as parametric options: perhaps they do. For example, D movement to T can be seen as a form of subject cliticization, while VP movement to SpecTP, satisfying T's EPP feature, was argued for for Niuean by Massam (2000) and others (see section 9.4.2 below). The point is that the movement-triggering mechanism needs to be enriched in such a way that head movement has a special kind of triggering feature. Chomsky suggests that such a complication is not needed if head movement is treated as something outside the core computational system of narrow syntax.

Third, Chomsky points out that the derived structure of head movement, as construed in section 9.2, is countercyclic; in fact, it violates the Extension Condition.[3] The Extension Condition requires that all movement operations extend the root of the structure that they apply to. For example, a standard case of A-movement raising the subject to SpecTP (triggered by T's EPP feature in the system in Chomsky 2001) applies at the point in the derivation after T is combined with its complement vP. T Agrees with the nearest DP that it asymmetrically c-commands, which, in a simple transitive clause, is the DP merged in SpecvP, the external argument. In virtue of this Agree relation and T's EPP feature, this DP is raised, forming SpecTP The formation of SpecTP extends the root at this point in the derivation. It is fairly clear that *wh*-movement to SpecCP, as well as various kinds of adjunct operations, can be seen in the same light, whatever the precise details concerning the triggers for these operations. However, as we have seen, head movement was thought to derive structures such as that in (2), by adjoining one head to another. Such an operation does not involve extension of the root, at least in any obvious way without appeal to a special notion of 'root' (which is imaginable but has not been proposed; the assumed notion of root is that node X such that there is no node Y that irreflexively dominates X).

Fourth, Chomsky makes the related point that, owing the fact that head movement adjoins one head to another, in the derived structure the moved head is unable to c-command its trace/copy. This is true if we maintain the most natural definition of c-command: that it is the transitive closure of sisterhood and containment (this is the natural definition, since both sisterhood and containment can be directly defined in terms of Merge; see Chomsky 2000a: 116). If we adopt a definition of the kind assumed in Kayne (1994:18), which allows an adjoined category to c-command both the category to which it adjoins and out of that category,[4] then the moved head (p. 201) would be able to c-command its trace in a typical head movement configuration such as that shown in (4):

(4) $[_{YP} \ldots [_{Y}\ YX] \ldots [_{XP} \ldots (X) \ldots]]$

But Chomsky suggests that such complications of the definition of c-command are unnecessary and undesirable (they do not 'fall under the notion of c-command derived from Merge', i.e. transitive closure of sisterhood and containment: Chomsky 2000a: 116). If so, then head movement features a major anomaly in relation to other types of movement in that the moved category does not c-command its trace.

Fifth, Chomsky pointed out that head movement was suspect as a core syntactic operation since onward cyclic movement is never successive-cyclic, instead it always involves 'roll-up' (i.e. movement of the entire derived constituent formed by movement, of the type in (2)). We commented on this point in section 9.2: after adjunction of X to Y, forming (2), further movement of Y to a higher head W forms the complex head $[_W[_Y X\ Y]\ W]$. In other words, iterated head movement always forms a successively more complex head. Successive-cyclic head movement, on the other hand, would involve excorporation of X from $[_Y$ X Y], moving X on to form $[_W X\ W]$. As already pointed out, Roberts (1991) observed that nothing prevented this in the GB conception of head movement, and that it was in fact empirically desirable. The general view, however, has remained that this possibility is not found (the empirical cases Roberts adduced can be analyzed in other ways). If so, then an explanation is required. Chomsky says that if head movement were seen as a morphological operation, then this might be why we do not observe excorporation (‘iterability is a general property of operations of narrow syntax, and these alone’ 2000a: 38). But if we treat head movement as syntactic movement, then we have to explain why successive-cyclic movement, so clearly available for phrasal movement (both A and A′-movement) is not available to head movement.

Chomsky's arguments have given rise to various reactions, as we shall see. In general they have been influential, in that many researchers have been led to look for alternatives to the earlier approach to head movement, either by eliminating it altogether, eliminating it from the core computational system of narrow syntax, or radically redefining it. In many cases, new phenomena have been brought to bear on the issues, or at least older data have been reconsidered in a new light. Two points should, however, be made here. First, although Chomsky's arguments naturally lead to a re-evaluation, at least, of the account of head movement sketched in section 9.2, he does not articulate a theoretical principle which would force, either directly or as a deductive consequence, the elimination of head movement from narrow syntax. The question that remains open if we accept Chomsky's conclusions is then: why is head movement not part of narrow syntax? The second point is related: to what extent do these questions bear on the conceptual goals of the minimalist program? To put the question, in a sense, the other way around (albeit tendentiously): could **(p. 202)** this discussion regarding the nature of head movement have been GB-internal? I will return to these points in section 9.5.

9.4 Alternatives to Core Syntactic Head Movement

Three main alternatives to the earlier form of syntactic head movement have been proposed since Chomsky (2001), one of which developed to some extent independently of Chomsky's remarks. These are the PF movement approach, which Chomsky himself advocated, the remnant movement approach, which partly stems from Kayne (1994), and the ‘reprojective’ approach. I will look at each of these in turn, focusing on one case study of how an earlier analysis or family of analyses involving core-syntactic head movement is replaced by the alternative mechanism.

9.4.1 PF movement

To judge from Chomsky's (1995b: 4.10, 2001: 37–8) comments, the alternative he has in mind to syntactic head movement is a PF operation. This becomes clear when we consider that the PF movement alternative is unproblematic in relation to all the arguments Chomsky makes: clearly we do not expect PF movement to have to obey the Extension Condition or the c-command condition (English Affix-Hopping could be a case of PF head movement but cannot be syntactic movement: see section 9.2); we expect it to be triggered quite separately from syntactic XP movement, perhaps to involve morphological ‘roll-up’, as already mentioned, to be subject to special, non-syntactic, locality constraints, and, of course, to lack LF effects. The existence of head movement(-like) operations in PF is frequently assumed: alongside Affix-Hopping one can point to Halpern's (1992) operation of Prosodic Inversion, which switches the positions of a clause-initial enclitic and a potential host, so that the enclitic can ‘lean left’, as required; this operation may underlie many 2nd-position clitic phenomena, and the general approach is characteristic of distributed morphology outlined in Embick and Noyer (2001).[5]

(p. 203) The question becomes, then, one of providing evidence that head movement processes which appear to be syntactic are really PF processes. Here, decisive evidence is somewhat lacking. One interesting argument is made by Boeckx and Stjepanović (2001), who propose that pseudogapping, as in (5), provides evidence for PF verb movement in English:

(5) Although John doesn't eat pizza, he does—pasta.

Starting from Lasnik (1995b), examples of this kind have been taken as evidence for syntactic object shift in English, combined with remnant VP deletion after object shift (i.e. deletion of *[vp eat (pasta)]* in (5)).[6] Boeckx and Stjepanović (2001) observed that Lasnik's original account of why the verb moves when there is no pseudogapping (to derive *John eats pasta*) cannot be maintained in the Agree-based theory of movement of Chomsky (2000a, 2001). Assuming all three operations (object shift, verb movement, and ellipsis) to be intrinsically unordered, the question then becomes why V movement followed by ellipsis is not possible, giving the ungrammatical (6):

(6) *… he eats ~~[VP (eats) pasta]~~.

Boeckx and Stjepanović argue that the question concerns ordering, and point out that object shift must precede both head movement and ellipsis, while the latter two can appear in either order:

(7)

a. Object Shift > ellipsis (head movement bled) → pseudogapping:

.. he does pasta ~~[VP eats (pasta)]~~.

b. Object shift > head movement > ellipsis:

.. he eats pasta ~~[VP (eats) (pasta)]~~.

c. *Head-movement > ellipsis (object shift bled):

*.. he eats ~~[VP (eats) pasta]~~.

They conclude that the right result can be guaranteed if object shift is a syntactic operation, with both ellipsis and V movement taken to be PF processes. Hence ellipsis can either precede or follow PF V movement; in the former case, as in (7a), pseudogapping results, in the latter, VO order results, as in (7b). (7c) is impossible since object shift, as a syntactic operation must precede verb movement.

However, Baltin (2002: 655) observes that the same movement/deletion options apply to non-verbal predicates such as *fond* in (8) and to phrasal categories as in (9):

(8) Although he isn't fond of pizza, he is (fond) of pasta.

(p. 204) **(9)**

a. Although he isn't very fond of pizza, he is (very fond) of pasta.

b. Although he didn't try to persuade Mary, he did (/tried to persuade) Martha.

In (9a) the gapped string is *very fond*, presumably an AP, and (9b) it is *try to persuade*. Baltin further observes that it seems that the *of*-PP has undergone 'object shift' in (8) and (9a), raising questions about Lasnik's initial conclusion. The following examples underscore both points:

(10)

a. Although John isn't easier to please than Mary, he is—than Bill.

b. Although John isn't easier to convince the students to talk to than Mary, he is—than Bill.

Here, *than Bill* must have undergone putative 'object shift', which is surprising, since this category is usually taken to be either a PP or an elliptical CP and the pseudogapped constituent is the complex AP, containing a possibly unbounded A′-dependency.

In fact, it appears that the 'object shift' operation should really be seen as an optional focusing operation, moving an XP to the left edge of vP (see Belletti (2004a) on the idea that the vP, like CP, may have an extended left periphery). This operation seems to be like scrambling in other West Germanic languages, in that it can apply to many XPs, but not readily to small clauses, particles or small-clause predicates (see Johnson 2001: 463 for the same suggestion, and his n. 41 for one or two provisos):

(11)

a. Even though John didn't put Mary down, he did put her up.

b. * … he did up—.

c. Even though John didn't get Mary drunk, he did get her angry.

d. * … he did her angry—.

e. * … he did angry—.

Let us suppose, then, that English has an XP movement operation, a highly restricted residue of scrambling, that moves an XP out of VP to the left edge of the vP phase, subject to that element receiving a special interpretation. This operation is associated with VP deletion, which then applies to the remnant VP, giving pseudogapping. Nothing further needs to be said. In particular, V movement plays no role in accounting for the salient facts of this construction.

In fact, head movement may be relevant in one respect, and this points to exactly the opposite conclusion from that drawn by Boeckx and Stjepanović. In examples where VP is headed by a main verb, V-to-T movement is impossible and *do* is inserted in the standard way, in order to bear T's ø and Tense features. Examples like (8, 9a, 10, 11) can also be seen as involving VP-ellipsis combined with obligatory *be*-raising to T. The ungrammaticality of the corresponding examples without *be* can then be taken to argue that V-to-T movement must apply before VP ellipsis, and (p. 205) hence is a syntactic operation (the ungrammaticality of '*do*-support' here further implies that that operation, too, cannot be a purely PF matter).

The one open question concerns the relation between leftward XP movement and VP ellipsis. The latter can clearly apply without leftward XP movement, but leftward XP movement appears to be conditioned by VP ellipsis, in that *he pasta eats/he does pasta eat* are ungrammatical.[7] This fact seems to be connected to the intrinsic link between VP ellipsis and focus, also manifest in the very well-known fact that the auxiliary cannot be contracted here:

(12)

a. *John is fond of pizza, and Bill's—too.
b. *Although he isn't fond of pizza, he's—of pasta.

A focus feature on v seems required for both VP ellipsis and optional XP movement.

It appears, then, that Boeckx and Stjepanović's argument does not support the postulation of PF head movement. Many other cases of head movement could be treated as PF phenomena, in part for the reasons given by Chomsky as summarized in section 9.3. However, PF head movement must be entirely without LF effects, and a number of arguments showing that some cases of head movement have LF effects have been given, notably by Lechner (2005; see also Cinque 1999: 184, n. 8, Roberts forthcoming: ch. 1, Zwart 2001). Roberts (forthcoming: ch. 1) points to the following paradigm (see also McCloskey 1996: 89, Kayne 2000: 44):

(13)

a. *Which one of them does anybody like?
b. Which one of them doesn't anybody like?
c. *They succeeded in finding out which one of them anybody liked.
d. *They succeeded in finding out which one of them anybody didn't like.
e. They succeeded in finding out which one of them wasn't liked by anybody.

Here it appears that the NPI *anybody* in subject position in (13b) is licensed by the auxiliary raised to C. This argument depends on the standard assumption that NPIs must be c-commanded by their licensers at LF. Movement of the auxiliary in examples like (13b) above affects LF by altering c-command relations involving the moved item, and as such is the head movement analog of raising in (14):

(14)

a. After the meeting, nobody seemed to anybody to be satisfied with the outcome.
b. *After the meeting, it seemed to anybody that nobody was satisfied with the outcome.

Furthermore, Matushansky (2006: 102–4) provides a plausible reason for why it should be the case that verb movement, in particular, often lacks semantic effects: essentially this is because verbs are predicates. To quote Matushansky, 'whether we (p. 206) assume that predicates must reconstruct […] or allow them to be interpreted in their final position, the outcome is the same: predicate movement is not reflected at LF' (p. 103).

There may well be reasons, then, to think that not all head movement takes place at PF. This does not imply that no head movement takes place at PF, of course, although unambiguous evidence to this effect is lacking (and if the suggestion in note 5 above that Internal Merge cannot take place at PF is correct, then it may be that PF head movement is impossible after all).

9.4.2 Remnant phrasal movement

To some degree as a direct response to Chomsky's (2001) arguments, summarized in section 9.3, and to some extent as a consequence of the re-evaluation of the status of clitic pronouns following on from Kayne (1994), a number of authors have proposed remnant movement accounts for some of the phenomena previously handled as head movement, including verb movement of various kinds (see Koopman and Szabolcsi 2000, Nilsen 2003, Müller 2004a, Wiklund and Bentzen 2007, Wiklund et al. 2007, Bentzen 2007, 2009, and several of the contributions in Mahajan 2003; see also the recent treatments of various forms of inversion in French in Kayne and Pollock 2001, Poletto and Pollock 2004, Pollock et al. 2003, Pollock 2006, and several of the papers on verb-initial languages in

Carnie et al. 2005, and, on the syntax of nominals, Shlonsky 2004, Cinque 2005, forthcoming, and the references given there).

These approaches share the central idea that analyses positing head movement relations of the type schematized in (15) should be replaced by analyses of the general type in (16):

(15) … H …[$_{XP}$ Z (H) Y] …
(16) … XP … Z..Y.. [$_{XP}$ (Z) H (Y)] …

Other things being equal, both scenarios convert underlying *-ZHY-* order to surface *-HZY-*. In (15), this is achieved by H movement out of the category XP containing H, Z, and Y prior to movement. In (16), on the other hand, H does not move: instead XP moves, but thanks to presumably independent operations moving Z and Y, the moved XP contains only H; all the other material has been moved out of XP before XP movement takes place. XP is thus a 'remnant category', in that it contains only a subset of the elements it contained at an earlier stage of the derivation (this point should really be stated in terms of the categories realized at PF, since copies/traces are presumably present in core syntax but deleted in PF). Movement of XP in scenarios like that schematized in (16) is thus referred to as 'remnant movement'.

(p. 207) Strictly speaking, the term 'remnant movement' does not denote a form of movement, but rather a (sub-part of) a derivation where, given a complex constituent *[$_{XP}$Y Z]*, both movement of Y or Z from XP and movement of XP itself take place. Derivations of this type are allowed and attested quite independently of the issues surrounding head movement. Typically, this movement is subject to certain constraints, though. In particular, various notions of Freezing and (strict) cyclicity are relevant. Freezing (originally put forward by Ross 1967a, Wexler and Culicover 1980) bans movement out of moved constituents; this forces movement of Y or Z to take place before XP movement in the derivation (if XP is a cyclic domain, then the Strict Cycle has the same effect). Moreover, the Strict Cycle, on many formulations, requires XP to move to a higher position than Y or Z. The Extension Condition, combined with Freezing, will also have this effect. The schema in (16) reflects this order of operations.

Perhaps the best-known independent motivation for remnant movement comes from so-called 'remnant topicalization' in German, as in examples such as the following, discussed by den Besten and Webelhuth (1990):

(17)

a.

Gelesen	hat	er	das	Buch	nicht.
Read	has	he	the	book	not
'He hasn't read the book.'					

b. [$_{VP}$ (das Buch) gelesen] hat er das Buch nicht (VP).

Here, *das Buch* has undergone scrambling, an operation which productively applies to definite DP objects, raising them outside VP to some TP-internal position in German (the exact nature and trigger for scrambling in German and elsewhere is much debated; see Grewendorf and Sternefeld 1990, Corver and van Riemsdijk 1994, Thráinsson 2000). This is followed by VP fronting to the first position in the clause, usually thought to be SpecCP, satisfying the V2 constraint here. This combination of operations is entirely licit, and explains what would otherwise be an anomalous V2 construction, involving just a participle preceding the inflected verb. (18), from Müller (1998), illustrates the interaction of remnant movement with Freezing and the Strict Cycle:

(18)

*Worüber	hat	[$_{DP}$	ein	Buch	(worüber)]	keiner (DP)	gelesen?
what-about	has		a	book		noone	read
‘What did noone read a book about?’							

(18) can be derived by moving the DP *ein Buch worüber* first, and then by subextracting *worüber*; this violates Freezing. Alternatively, if *worüber* is first moved, and then DP, movement of the DP violates the Extension Condition. We see, then, that remnant movement and the constraints relevant to it are motivated.

Müller (2004a) argues that the analysis of V2 constructions which postulates two separate movements, one head movement of the verb and the other XP fronting, originally proposed by den Besten (1983), should be replaced by a single operation (p. 208) of remnant fronting. Specifically, Müller proposes that a vP evacuated of all overt material other than the verb and a single constituent on the left edge undergoes this fronting operation; note that here the remnant category must contain more than just an unmoved head, and so the schema in (16) does not exactly apply. The creation of the appropriate initial domain is achieved by what Müller calls the Edge Domain Pied-piping Condition, which states exactly this: only one maximal constituent, occupying the left edge of vP, can be present in a vP which moves (the definition of Edge Domain is given in (21) below). This analysis is claimed to have certain interesting empirical advantages, and notably to have the theoretical advantage of allowing us to dispense with a well-known case of head movement.

The central innovation in Müller's analysis is the idea that V2 is derived by a single movement operation, remnant vP fronting, rather than by the interaction of movement of the finite verb and movement of an XP. Thus, instead of the standard derived structure for an object-initial V2 clause as in (19) we have (20):

(19)

[$_{CP}$	Das Buch [$_{C'}$	hat-C	[$_{TP}$ Fritz [$_{vP}$ (Fritz) [$_{vP}$ (das Buch)	gelesen] (hat)] (hat)]]]
	the book	has	Fritz	read

(20)

[$_{CP}$ [$_{vP}$	Das Buch (Fritz) (VP)	hat] [$_{C'}$ C	[$_{TP}$ Fritz [$_{T'}$ [$_{vP}$ (das Buch)	gelesen] [$_{T'}$ (vP)T]]]]]
	The book	has	Fritz	read
‘Fritz has read the book.’				

As Müller points out:

> In this approach, the pre-V/2 position is occupied by whatever category happens to be at the left edge of vP earlier in the derivation—this will typically be the subject NP or an adverb, but, after scrambling, it may also be an object NP, a PP, a CP, or a VP (complete or remnant..). (2004a: 182–3)

In addition, there is no reason to postulate head movement; in examples of the kind in (20), *hat* is assumed to have merged directly in v. Where a main verb appears in second position, it has not moved to v, but rather counts as being on the edge of vP owing to the first clause of the definition of Edge Domain, which runs as follows (Müller's (6), p. 184):

(21) EdgeDomain
Acategory α is in the edge domain of a head X iff (a) or (b) holds:
- **a.** α is the highest overt head reflexively c-commanded by X.
- **b.** α is a specifier that is not c-commanded by any other specifier in XP, and that precedes the edge

domain of X.

Müller claims a number of empirical advantages for his approach, and it has inspired other analyses of verb movement, including V2, in Germanic, notably in Wiklund and Bentzen (2007), Wiklund et al. (2007), Bentzen (2007, 2009).

(p. 209) In an independent development, a number of papers on the syntax of inversion and verb movement in French and other Romance varieties have suggested replacing earlier head movement analyses with alternatives based on remnant movement. Poletto and Pollock (2004), Pollock et al. (2003), and Pollock (2006) argue for a remnant movement analysis of verb movement into the C-system in French (and other Romance varieties). Their arguments are based on Kayne's (1994: 42–6) discussion of the landing site of clitics in Romance. Consider first a basic example with a direct-object clitic (here from Italian):

(22)

Voi	lo	vedete.
you(pl)	him/it	see
'You see him it.'		

Kayne adopts three postulates. First, that morphologically derived forms such as *vedete* are syntactically formed, possibly by syntactically combining the root *ved-* with the theme vowel -*e*- and the ending -*te*. Second, that the LCA applies to sub-word-level operations, and, third, that the LCA bans multiple head adjunction. Given these three postulates, the clitic would have to adjoin to the verb root *ved-*, followed by adjunction of *[lo ved-]* to (the functional head occupied by) -*e*- and then adjunction of *[[lo ved-] -e-]* to -*te*. Where the verb bears a prefix, as in *lo prevedete* ('you foresee it'), the clitic would have to attach to the prefix.

Kayne goes on to suggest that a more plausible option is to assume that clitics adjoin to empty functional heads. Kayne further observes enclisis to infinitives and imperatives of the type in (23):

(23)
a.

Fais-le.	(French)
Doit.	

b.

Parlargli	sarebbe	un errore.	(Italian)
To-speak.to-him	would-be	a mistake	

Since it is very likely that the verb moves to C in imperatives like (23a) (see e.g. Rivero 1994a, b), and that the infinitive is in a 'high' position in (23b) (Belletti 1990, Kayne 1991), Kayne concludes that in general verb movement to C does not 'carry along' clitics. It then follows that, in a French example like (24), involving 'subject-clitic inversion' with an object proclitic on the inverted auxiliary, the clitic+auxiliary combination has not moved to C:

(24) L'as-tu fait?
it.have-you done
'Have you done it?'

Kayne follows Sportiche's (1999) proposal that there may be V movement to C at LF, hence accounting for the root nature of the construction (he suggests that the clitic may delete at LF; see his n. 16). Finally, Kayne observes high-register examples, which show the order Clitic—Adverb—Infinitive (e.g. *le bien faire* 'it well to-do'), support the idea that the clitic and the verb do not have to combine.

(p. 210) Poletto and Pollock (2004), Pollock et al. (2003), and Pollock (2oo6)endorse Kayne's general conclusion that clitics and verbs cannot and do not combine in syntax, but propose that, instead of covert verb movement into the C-system in examples like (24), there is overt remnant movement. The derivation of (24) would proceed as follows:

(25)

a. Tu $[_{XP}$ le $[_{YP}$ as $[_{ZP}$ fait$]]]$ → (movement of ZP)
b. Tu $[_{ZP}$ fait$]$ $[_{XP}$ le $[_{YP}$ as (ZP)$]]$ → (remnant movement of XP)
c. $[_{XP}$ le $[_{YP}$ as (ZP)$]]$ tu $[_{ZP}$ fait$]$ (XP)

Remnant XP movement is triggered by the interrogative feature of the attracting head, which is part of an articulated C-system. It is unclear what the trigger (or the landing site) of ZP movement (probably vP movement) is, as well as the cliticization operation itself.[8] Kayne and Pollock (2001) propose a similar analysis of French Stylistic Fronting, and Pollock et al. (2003) extend the approach to interrogatives in various Romance varieties.

A third case where remnant movement has influentially replaced an earlier head movement analysis comes from verb-initial languages, in particular Macronesian languages showing an alternation between VSO and VOS orders (Massam and Smallwood 1997, Massam 2000, Rackowski and Travis 2000, and many of the papers in Carnie et al. 2005). In her study of VOS and VSO in Niuean, for example, Massam (2000) argues that there is an operation fronting a verbal constituent, and that this constituent is fronted to a position within TP. She then shows that there is a general operation which fronts non-verbal predicates which are clearly larger than heads, e.g. relative clauses. Third, Massam shows that what has been called noun incorporation in Niuean (e.g. by Baker 1988) cannot be movement of N into V (*pace* Baker), since there are clear cases where a constituent larger than N undergoes this operation. She proposes instead that putative noun incorporation is really the absence of object shift to a VP-external position. In that case, the fact that the apparently incorporated noun moves with the verb shows that what is moved is VP rather than V. VOS order is thus derived by VP fronting, and VSO by object shift to a VP-external position combined with remnant VP fronting, as shown in (26):[9]

(26)

a. $[_{TP}$ $[_{VP}$ V O$]$ T $[_{vP}$ S v.. (VP)$]]$ – VOS
b. $[_{TP}$ $[_{VP}$ V (O)$]$ T $[_{vP}$ S v $[_{AbsP}$ O (VP)$]]]$ – VSO

(p. 211) As (26) shows, the landing site of VP fronting is taken to be SpecTP; Massam argues that this is motivated by essentially the same property as that which causes the subject to raise to SpecTP in languages like English, French, and Mainland Scandinavian: the operations ‘can be seen as two reflections of a single EPP predication feature’ (Massam 2000: 111). This type of analysis, first put forward by Massam and Smallwood (1997), and developed by Rackowski and Travis (2ooo) as well as several of the papers in Carnie et al. (2005), has been applied to a number of languages which display both VOS and VSO orders (mainly but not exclusively Macronesian and Mayan languages; unlike rigidly VSO languages such as the Celtic languages, where it is at the very least much harder to motivate a remnant VP fronting analysis). Here too, though, the question of the trigger for some of the movements arises (see Chung 2005).

Let us now briefly evaluate remnant movement approaches against Chomsky's (2001) arguments, given in section 9.3. First, it is clear that remnant movement avoids the problems head movement causes for the Extension Condition and the definition of c-command. Since it is XP movement, it presumably extends the root. Similarly, as XP movement, the issues concerning the Head Movement Constraint and successive-cyclic vs. roll-up movement are directly solved by the postulation of XP movement (although it is not clear why remnant movement should, even apparently, obey a strong locality condition such as the Head Movement Constraint). On the other hand, remnant movement is expected to have LF effects; it is entirely unclear why this particular type of XP movement should be exempt from these.

Where the remnant movement approaches are problematic, as has frequently been pointed out, is in relation to the question of movement triggers. This may not always be a serious problem in the case of the movement of the remnant itself, but it is frequently difficult to see what drives the other movements (indicated as movement of Z and Y in (16)) exactly where the larger XP moves to a higher position. In the case of the derivation in (25), for example, this would apply to ZP movement, as mentioned above.

A further issue arises in connection with the apparent LF effects of head movement mentioned at the end of the previous section. Recall that we observed that the NPI *anybody* in subject position in (13b) is licensed by the auxiliary raised to C. This argument depends on the assumption that NPIs must be c-commanded by their licensers at LF. Movement of the auxiliary in examples like (13b) above affects LF by altering c-command relations involving the moved item. However, if we consider the remnant movement alternative, this conclusion would not follow. To see this, suppose that English subject auxiliary inversion does not have the form in (27a), the 'traditional' T-to-C movement analysis, but rather that in (27b), involving remnant TP-movement:

(27)
a. T+C .. [$_{TP}$ Subj (T) vP]
b. TP … C … Subj … vP … ([$_{TP}$ (Subj) T (vP)])

(p. 212) Obvious questions arise here concerning the nature and landing sites of subject movement and vP movement, but let us leave these aside here. Now, if remnant movement of the general type in (27b) were involved in subject—aux inversion, and if c-command is retained as the relation determining polarity licensing, then the definition of this relation would have to be complicated so as to allow the auxiliary to c-command out of TP into the complement domain of TP here. So negation inside TP, possibly attached to an auxiliary in T, does not c-command the subject, and cannot do so on any plausible, simple definition of c-command. That T rather than TP counts as the relevant licensing element can be seen from cases where *not* does not raise with the auxiliary and subject—auxiliary inversion does not license the NPI, such as **Which of them does anybody not like?* To the extent that there are LF effects associated with head movement (*pace* Chomsky), then, remnant movement approaches may have difficulties with c-command effects.

In conclusion, reanalyzing head movement as remnant movement avoids a number of the problems Chomsky pointed out for head movement, although if anything the 'trigger problem' may be exacerbated. Reconsidering head movement in this light has been productive in the cases of N-to-D movement, creation of verb clusters of the West Germanic/Hungarian type (see Koopman and Szabolsci 2000 and the papers in Kiss and van Riemsdijk 2004), and has led to new ideas in the case of V2 and certain cases of inversion in Romance. This approach is almost certainly, then, an alternative in some cases, but it remains unclear to what extent it represents a global alternative to head movement, and, given the 'trigger problem', it is not clear that it is conceptually simpler.

9.4.3 'Reprojective' movement

Still another strand which has been pursued as part of the general reconsideration of the nature of head movement is represented by a class of analyses which we can collectively label 'reprojective'. This approach has been developed primarily by Bury (2003, 2007), Donati (2006), Koeneman (2000), and Surányi (2006, 2007, 2008). The basic idea is to take head movement to be syntactic movement, but to treat it as arising from a different set of conditions from XP movement. Chomsky (1995b: 256–60) argues that where a new category γ is formed by movement of α to β, γ must always project the target of movement: hence DP movement attracted by T will create a new projection of T, *wh*-movement attracted by C will create a new projection of C, etc. Bury, Donati, and Surányi suggest that this may not always be the case, and that 'reprojective' movement may arise, where the moving category gives its label to the new category formed by movement. Bury develops this proposal in a very interesting way in connection with phenomena connected to both V-initial and V2 languages, treating V movement as 'reprojective' in this sense. He also applies the approach to free relatives. Here I will briefly summarize the (p. 213) main proposals made by Donati (2006) in her analysis of free relatives and related constructions.

Donati's main concern is the basic conceptual question of why we find phrasal movement at all, since head movement involves moving less material. She suggests that Chomsky's (2001) proposal to eliminate head movement is inadequate, since it cannot in principle rule out head movement to a specifier position (this point is also made by Matushansky 2006, Roberts 2005, Toyoshima 2000, and Vicente 2007). Again similarly to what we have seen here, she suggests that the Head Movement Constraint is irrelevant to the question of the existence of head and phrasal movement, in that locality constraints act on the search operation, not on movement itself, hence a single set of locality constraints should govern both types of movement. She further observes that there are empirical doubts about the HMC, citing 'long verb-movement' in Breton (see Borsley et al. 1996) and the similar cases in South Slavic and archaic Romance discussed by Ćavar and Wilder 1994, Lema and Rivero 1990, 1991, Rivero 1991, 1993a, b, 1994a, b, 1997, and Rivero and Terzi 1995). Instead, Donati adopts the chain uniformity condition of Chomsky (1995b: 253) in (28) and the minimality condition on Merge in (29):

(28) 'A chain is uniform with regard to phrase structure status' (where 'phrase structure status' means the '(relational) property of maximal, minimal or neither').
(29) Merge just enough material for convergence.

(29) applies to both Internal and External Merge. Finally, she assumes that a head, when merged either externally or internally, projects; XPs, on the other hand, do not. Thus, for Donati, head movement is always and only 'reprojective'.

Donati then goes on to show a minimal contrast involving movement of a [+wh]D. This gives the two possibilities in (30):

(30)
a. [$_{CP}$ DP C …
b. [$_{DP}$ DC …

In (30a), we have the derived structure of a *wh*-interrogative. Donati argues that the *wh*-feature cannot move as a head in this case, as it would turn the interrogative clause into a DP. Thus, requirements of LF convergence (the structure must be interpretable as an interrogative clause) cause the non-minimal DP movement option to be taken. However, 'in a context compatible with DP-selection and showing no phrasal pied-piping' (Donati 2006: 32), the option in (30b) should be available. This, Donati argues, is what we find in free relatives and comparatives.

For free relatives, Donati's evidence comes from paradigms like the following:

(31)
a. *I will visit [what town] you will visit.
b. I wonder [what town] you will visit.
c. I will visit [what] you will visit.

(p. 214) (31b) clearly contains an indirect question, i.e. a CP complement to *wonder* with *what town* in its specifier. On the other hand, *visit* does not take an indirect-question complement, or indeed any kind of CP, but only a DP. The complement in (31c) is thus a DP, a free relative. Pied-piping to the edge of a free relative is impossible, as (31a) shows (following Kayne 1994, Donati assumes that *whatever*- type relatives—as in *I will visit whatever town you will visit*—are not in fact free relatives). (31c) thus involves 'reprojective' movement of a [+wh]D, giving rise to a derived structure like (31b).

Donati goes on to argue the same for comparatives, known to involve *wh*-movement since Chomsky (1977b). The idea that comparatives are complex nominals is supported by the fact that they express a description of a degree, and by the fact that this expression can enter into scope ambiguities of the type first discussed in Russell (1905):

(32) I thought your yacht was bigger than it is.

If the comparative expression (*than it is*) is outside the scope of *think*, we have a non-contradictory reading for (32); if it is inside the scope of *think* we have contradictory reading (cf. 'I thought your yacht was bigger than itself'). Third, comparatives are strong islands for extraction, suggesting they are complex DPs (here *eat* is elided while *what* and *x-quickly* are *wh*-moved):
(33) *What do you eat the soup more quickly than Paul does (eat) (what) (x-quickly)?

Finally, Donati gives evidence that in Romanian and Bulgarian the same *wh*-element moves overtly as a head in comparatives but as a phrase in interrogatives. In fact, the same can be shown with non-standard varieties of English which allow *what* to appear in comparative sub-deletion and to act as an adnominal *wh*-determiner:[10]
(34)
a. Mary ate more cookies that what she ate [(what) candies].
b. *Mary ate more cookies that what candies she ate (what candies).
c. What candies did she eat (what candies)?

d. *What did she eat [(what) candies]?

Donati (2006: 39) concludes: 'there is no principled reason for wh-movement to be restricted to phrases.'

Once again, let us consider these proposals in relation to Chomsky's (2001) arguments. 'Reprojective' movement does not target heads, and so the Extension (p. 215) Condition and c-command problems Chomsky raises do not apply in this case. The triggering problem appears to be dealt with by Donati by LF: if the movement is not reprojective, one kind of structure and interpretation must result; if it is, then a different one results. The syntax itself allows either option. In a sense, then, Donati has LF act as a filter on the syntactic derivation. Regarding onward movement, presumably the possibilities here are determined by the reprojection option. Sticking to Donati's example with [+wh]D, we can note that free relatives are unbounded and subject to island constraints, and therefore must involve standard, non-reprojective *wh*-movement on earlier cycles prior to a last step of reprojective movement:

(35)

a. I will visit what Tom says Bill thinks Mary believes you will visit.

b. ?*I will visit what Tom believes the claim you will visit.

Conversely, the entire DP formed by reprojection can move, and indeed undergo A-movement:

(36)

a. What you will visit, (Tom says) I will visit.

b. What you will visit seems to have been visited by many tourists.

So it seems that each step of movement can in principle be either reprojective or not, but once reprojection takes place, it cannot be 'undone'. The latter constraint can arguably be seen as an instance of the general 'no-tampering' condition, in that once D has projected the label cannot be 'unprojected' but, conversely, until reprojection takes place, it is always possible in principle. The locality properties are directly tied to the nature of projected category: DPs, as in (36b) can undergo A-movement and the *wh*-phrases can undergo A′-movement, with the moved category obeying standard locality conditions in each cases. The LF properties of the structure resulting from movement category are crucial, as we have seen; on the other hand, PF appears to play no role in this approach.

Again, reprojection appears to be a valid alternative approach which avoids the general difficulties discussed by Chomsky. It leads to an interesting account of free relatives, and, in Bury (2003), of some cases of verb movement. How far it can be extended as a global alternative to head movement remains to be seen, however (see Koenemann 2000, Biberauer and Roberts 2010 for a reprojective account of V-to-T movement).

9.4.4 Conclusion

Here we have looked at the three main alternatives to standard head movement that have been discussed in the literature, in many cases directly responding to Chomsky's (2001) comments. No single version is entirely free of problems, and none appears to be a global alternative to 'traditional' head movement in the sense that it is clear that all former cases of head movement can and should be (p. 216) reanalyzed in the relevant terms. This may in fact be a good state of affairs: it is quite possible that the mechanisms of head movement were overextended in the earlier approach. At least in the DP and in the areas of verb clustering it really seems that XP movement analyses represent a valid alternative, while reprojection looks promising for some cases of verb movement, with perhaps PF movement valid for others. Here I leave these questions open: now I want to turn to more conceptual issues raised by some of the thinking behind the minimalist program.

9.5 Head Movement and the Minimalist Program

What the above sections have shown, I hope, is that there are at present a range of views on the question of head movement. One could claim that for certain core cases, say French 'V-to-T' movement or Germanic V2, upto four competing analyses are available: the traditional, GB-style head movement one, a remnant vP movement one, a PF one, and a reprojection one. The question is then, obviously, which of the available analyses is the most successful, both empirically and theoretically? To be unable to answer this question in any immediately straightforward way seems to me to be a healthy state of affairs: the phenomena are complex and the implications of and relations among the various types of analysis not easy to tease out. So there is no reason to expect an immediate or simple answer.

But the question I want to address here is a slightly different one: which, if any, of these approaches is likely to be the most successful one given the overall goals of the minimalist program, as Chomsky has articulated this in his recent work (Chomsky 1993, 1995b, 2000a, 2001, 2002, 2004a, b, 2005, 2007, 2008a)? To put it another way, do the particular goals of minimalist theory contribute anything to deciding which, if any, of the alternatives we have seen might be the best overall approach to the phenomena of head movement?

Obviously we cannot answer that question without reminding ourselves of the goals of the minimalist program. The simplest way to put this is to say that, having got an inkling of the nature of UG through the GB version of Principles and Parameters Theory, the minimalist program has as its goal to refine and axiomatize that conception by asking: why, among all conceivable UGs, do we have the one we have? To an extent, as Chomsky (2005b: 1) has pointed out, 'the issues can be recast as metaphysical rather than epistemological: Is that how the world works?' We move from asking questions about knowledge of language, questions whose answers have led to the postulation of UG, to asking questions about the kind of world which produces a mental object like UG with the properties we observe it to have.

(p. 217) This, in turn, has led to an emphasis on the 'third factor' determining the nature of the adult language faculty. To see what this means, observe that adult competence is the result of the interaction of three factors: (i) experience of the primary linguistic data (PLD), which we need to learn the vocabulary and set the parameters of our native language; (ii) universal grammar, the innate endowment which makes it all possible, construed as a set of principles with parameters initially open; (iii) principles not specific to the faculty of language.[11] These principles constitute the third factor in language design, and include:

> (a) principles of data analysis that might be used in language acquisition and other domains; (b) principles of structural architecture and developmental constraints that enter into canalization, organic form […] including principles of efficient computation […] It is the second of these subcategories that should be of particular significance in determining the nature of attainable languages. (Chomsky 2005a: 6)

The first and second questions may answer the epistemological question, but third-factor postulates seem to be implied in answering the metaphysical question. In a sense, we have to move beyond UG and, so, beyond explanatory adequacy (in the Chomsky (1964c) sense).

In pursuing the axiomatization of GB principles, we subject every postulate to a 'minimalist critique': do we really need it? Can it be reduced to something else? We want to get back to the first principles of syntax. We want to reduce the theoretical postulates to those which are (virtually) conceptually necessary. At the same time, we want our explanatory postulates to relate to the higher level of explanation constituted by the attempt to answer the metaphysical question by invoking third-factor considerations. The Strong Minimalist Thesis (SMT) expresses one hypothesis which can do this:

(37) Language is an optimal solution to legibility conditions.
(Chomsky 2000a: 97)

The notion of 'legibility conditions' here relates to interface properties. So the idea is that the core computational system of syntax provides the optimal way of relating an arbitrary set of lexical items to the interfaces (PF and LF, for simplicity) in such a way as to satisfy whatever conditions the intrinsic properties of the lexical items and the interfaces may impose.

Evaluating head movement, in any of its potential technical guises, against the SMT is difficult, since movement in general appears to be an unnecessary complication. Surely a system which lacked movement operations (of any kind, A′, A, head-or anything we might imagine) is simpler and more optimal than a system with such operations. Chomsky (2004a: 110) provided a compelling negative response to (p. 218) this conjecture: 'SMT entails that Merge of α, β, is unconstrained, therefore either *external* or *internal*. Under external Merge, α and β are separate objects; under internal Merge, one is part of the other Merge yields the property of 'displacement' '. To the extent that movement reduces to Internal Merge (IM), then, we expect to find it in natural language.

Concerning the status of narrow syntactic head movement, we might then reason that, all other things being equal, IM and EM are supposed to be exactly the same operation except that IM takes place 'within' a structure in the process of being while EM introduces the element to be merged from outside. Since EM quite uncontroversially applies to heads, i.e. single lexical items or feature bundles, we need a very good reason to treat IM in a different way (this point is made by both Donati 2006 and Roberts forthcoming). If head movement is absent altogether, or restricted to the PF interface, there must be an explanation for this in terms of what differentiates IM and EM.

Of course, Merge is restricted to a search space. EM can only look to the Numeration; IM is subject to syntactic locality constraints. So if we can find a reason in the theory of locality for the absence of syntactic head movement, we would have a principled reason to exclude it from narrow syntax. The A-over-A principle is a good candidate. Consider the formulation of this condition given in (38) (from Chomsky 2006: 45):

(38) If a transformation applies to a structure of the form
$[_S \ldots [_A \ldots] \ldots] \ldots$
for any category A, then it must be interpreted so as to apply to the *maximal* phrase of the type A.

(Here 'maximal' is not intended in the X′-theoretic sense, but simply as the largest phrase of type A, in the sense that A should not be dominated by further occurrences of A.) A non-maximal occurrence of A in (38) could be construed as the head of A. Then (38) would in general block head movement. Hence the principled exclusion of head movement from narrow syntax might depend on the extent to which a version of (38) can be integrated into the theory of locality. Rackowski and Richards (2005) make an interesting proposal in this direction, arguing that a version of the A-over-A Principle is a condition on Probe-Goal Agree, and may derive some of the effects of the Phase Impenetrability Condition (PIC). If this can be fully achieved, then we can eliminate many cases of head movement from core syntax on principled grounds, while maintaining that movement is IM. However, at least some of the cases discussed by Donati would remain. There is also evidence that some kinds of predicate-cleft constructions in some languages involve unbounded, island-sensitive, and hence A′-like, verb movement to a specifier position (Vicente 2007, Landau 2006). Assuming the verb is extracted from VP in these cases, the A-over-A Condition would be violated; hence it is not in fact clear that the A-over-A Condition applies in such a way as to ban head movement in general.

(p. 219) Where does this leave the different approaches we have looked at? If Move is reduced to IM, it arguably cannot apply in PF. Therefore 'PF head movement' must really be something else, such as concatenation of heads and affixes, or rebracketing along the lines of Marantz's (1984, 1988) conception of morphological merger (see also Embick and Noyer 2001, Roberts 2005, Matushansky 2006), and can not have LF effects (see note 5 above). As already suggested, the remnant movement approach is compatible with the total elimination of head movement from PF and narrow syntax: however, this approach too may have problems accounting for some kinds of LF effects, as we saw in section 9.4.2. On the other hand, 'reprojection' approaches may involve a complication of the theory of movement.

Very tentatively, then, we can perhaps conclude that the 'pure PF' and remnant movement alternatives are equally attractive in terms of the SMT, since they add nothing to what we appear to have to assume anyway regarding movement/IM, and this seems to be most compatible with the SMT. However, both approaches appear to have problems with some LF effects of head movement. This suggests a combined approach: we could extend the operation of head-to-specifier movement (independently needed for some types of predicate cleft) and combine it with a PF rebracketing operation along the lines of Marantz's (1984, 1988) notion of merger. This would allow us to retain the idea that IM can apply to heads, and allow an account of the observed LF effects, while at the same time acknowledging that these cases of head movement are partially morphological. This kind of approach is advocated in Roberts (2005) and Matushansky (2006); the difficulty with it is that the merger operation really has to be part of the head movement operation in order to avoid the difficulties with c-command, the Extension Condition, and successive cyclicity pointed out by Chomsky. This entails something of a departure from 'pure' IM, and so again creates a conceptual difficulty. It seems, then, that some alternative notion of incorporation may after all be needed.[12]

Notes:

(1) This section summarizes some of the main points in Roberts (2001).

(2) Chomsky excludes incorporation here because it has rather different properties from the other cases of head movement (in particular, according to Baker, it is implicated in the core cases of grammatical function-changing phenomena).

(3) This was noticed in Chomsky (2000a: 137), where it is concluded that the Extension Condition should be weakened in this case.

(4) The definition is as follows:

((i)) X c-commands Y iff *X and Y are categories* and X excludes Y and every category that dominates X also dominates Y (emphasis in original).

(5) On the other hand, Affix-Hopping can be handled as a purely morphological reflex of Agree among local heads in the verb auxiliary system of English, and Prosodic Inversion may well be similar to V2 in being a case where an inflection-bearing head is attracted to C with concomitant XP-attraction to SpecCP (see Starke 1993 on the similarities between clitic-second and V2). It may be that PF movement is not found. If, as suggested in Chomsky 2004a, movement is Internal Merge, this would make sense to the extent that Merge is not a PF operation. In Distributed Morphology, Merge of feature bundles in narrow syntax is distinguished from Vocabulary Insertion, which takes place post-syntactically. Vocabulary Insertion should not be seen as a case of Merge, because it is not combinatorial, it does not build structure, and it is not recursive. It is thus formally quite distinct from Merge. There is also evidence that LF is sensitive to Affix-Hopping; see Siegel (1984).

(6) Traces/copies of moved elements are in round brackets.

(7) Similarly, Boeckx and Stjepanović have no obvious way of ruling out (i):

((i)) *Debbie ate chocolate, and Kazuko milk drank.

(8) It is also worth pointing out that although Poletto and Pollock (2004) and Pollock (2006) share with Müller (2004a) the basic idea that V2-type verb movement (full in the latter case, residual in the former) should be reanalyzed as remnant movement, they do not suggest that remnant movement should be a global replacement for head movement, as they continue to assume both V-to-T movement of the type argued for in Pollock (1989) and a head movement analysis of (subject) cliticisation, hence *tu* in (25c) moves to the head whose specifier the fronted remnant XP occupies. In that case, XP movement can be seen as remnant TP movement.

(9) AbsP here stands for Absolutive Phrase, which Massam suggests may correspond to AgrOP in more familiar languages. My summary here glosses over the complication that Niuean is an ergative language and Massam's treatment of the assignment of ergative and absolutive case.

(10) The grammaticality of (34a) implies that even some varieties of English allow left-branch extraction, at least in comparative sub-deletion cases like this. The comments in Donati (2006: 37–8) could provide a basis for understanding why the left-branch extraction is possible in (34a) but not (34d). For discussion of whether comparative sub-deletion involves movement or unbounded deletion, see Bresnan (1976), Chomsky (1977b).

(11) This isn't an entirely new idea: cf. 'there is surely no reason today for taking seriously a position that attributes a complex human achievement entirely to months (or at most years) of experience, rather than to millions of years of evolution or *to principles of neural organization that may be even more deeply grounded in physical law*' (Chomsky 1965: 59, emphasis added).

(12) A further issue concerns Agree: why is this operation not subject to the A-over-A Condition and what, if any, is its connection to head movement? One might expect there to be some connection since Agree is a head—head relation. Perhaps we could envisage an approach which derives all the properties of head movement from Agree, adding essentially nothing to our conception of that operation. Such an approach would be conceptually very appealing, and is developed in Roberts (forthcoming). But this is not the place to evaluate that alternative.

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Abstract and Keywords

The theory of locality is a major topic of generative grammar, and the discovery of locality principles has enriched the scientific debate of precise details on linguistic computations, providing critical evidence on how the brain computes structures, and raising fundamental questions on the generality or task specificity of computational principles embodied in language. This article focuses on Intervention locality. Intervention locality is expressed in the different effects subsumed under Relativized Minimality or the Minimal Link Condition/Minimal Search and also, to a certain extent, the A-over-A tradition itself, as well as certain interpretive locality effects in multiple *wh*-constructions, in anaphor binding, in the licensing of polarity items, and, perhaps more straightforwardly, by such principles as the Minimal Distance Principle.

Keywords: intervention locality, syntactic process, generative grammar, Relativized Minimality, Minimal Link Condition, Minimal Search

10.1 Introduction

Syntactic processes cannot relate positions placed at an indefinite structural distance, or across any arbitrary syntactic configuration: these limitations fall under the rubric of 'locality'. The theory of locality is a major topic of generative grammar, and the discovery of locality principles has enriched the scientific debate of precise details on linguistic computations, providing critical evidence on how the brain computes structures, and raising fundamental questions on the generality or task-specificity of computational principles embodied in language.

Ever since Chomsky's A-over-A condition (Chomsky 1964c), a number of formal locality principles have been put forth. If we abstract away from many differences of detail, they are by and large amenable to two broad and rather different intuitive concepts:

(1) Intervention:
A local relation is disrupted by the intervention of an element with certain qualities which make it a potential participant in the local relation.
(2) Impenetrability:
Certain syntactic configurations are impervious to local rules, which cannot take place across their boundaries.

Intervention locality is expressed in the different effects subsumed under Relativized Minimality (Rizzi 1990b and much subsequent work) or the Minimal Link Condition/Minimal Search (Chomsky 1995b, 2000a) and also, to a certain extent, (p. 221) the A-over-A tradition itself (Chomsky 1964c, Kayne 1975), as well as certain interpretive locality effects in multiple *wh*-constructions (Beck 1996, Pesetsky 2000), in anaphor binding (by and large characterizable as 'intervention of a subject': Burzio 1991), in the licensing of polarity items (with intervention induced by a quantificational element: Linerbarger 1981 and much subsequent work), and, perhaps more straightforwardly, by such principles as the Minimal Distance Principle (control can't skip a potential controller: see C. Chomsky 1969 on aspects of acquisition). The Impenetrability concept has been implemented in very different forms in Ross's (1967a) Island Constraints, in Chomsky's (1973) Subjacency, in terms of Bounding Nodes or Barriers (Rizzi 1978b, Chomsky 1986a), in Huang's (1982) Condition on Extraction Domain, and more recently in the Phase Impenetrability Condition of Phase Theory (Chomsky 2001, Nissenbaum 2000).

This chapter will focus on Intervention locality. We will look at classical cases of intervention, touching upon impenetrability only to a marginal extent, and not raising the issue of the possible unification of the two concepts (for possible lines of unification see Rizzi 2009, and, for a comprehensive assessment of the issue of locality in minimalism, Boeckx 2008a).

10.2 Relativized Minimality

Certain *wh*-elements strongly resist extraction from indirect questions, while they are freely extractable from declaratives in many languages, a contrast clearly illustrated by adjuncts like *how*:

(3) How do you think [he behaved ___]?
(4) *How do you wonder [who behaved ___]

This contrast is naturally amenable to an intervention effect: what seems to be going wrong in (4) is that another element of the same kind as *how*, the *wh*-element *who*, intervenes in the path connecting *how* and its trace, while no intervention of a similar element is observed in (3). Relativized Minimality generalizes this observation to all local relations (Rizzi 1990b):
(5) In the configuration
… X … Z … Y …

a local relation cannot connect X and Y if Z intervenes and Z is of the same structural type as X.

(p. 222) Intervention is hierarchically defined: Z intervenes between X and Y when Z c-commands Y and Z does not c-command X. In the original system, the structural typology was very straightforward:

(6) Structural types:
A′ positions
A positions
Heads

Assuming binary branching and a general ban on phrasal adjunction, a complement will never intervene in terms of c-command, hence the two phrasal position types reduce to specifiers, A and A′.

This system generalizes the model of *wh*-islands in two ways. First, within the A′ system, interveners are not just *wh*-elements, but A′ specifiers in general. Hence one expects similar intervention effects induced by negation and at least certain adverbials, which arguably occupy A′ positions, thus explaining negative islands (Ross 1983) and certain adverbial intervention effects, best illustrated by *combien* (how much/how many) extraction in French (Obenauer 1983, 1994):

(7)
a. How did he (*not) solve the problem ___ ?
b. Combien a-t-il (*beaucoup) consulté [___ de livres] ?
‘How many did he a lot consult of books?’

These are the environments illustrating the so-called weak islands, environments selectively blocking certain kinds of movement and giving rise to asymmetries (on which see section 10.4 below).

Second, the system generalizes to dependencies other than A′: an A dependency will be blocked by an intervening A specifier, a subject, thus explaining the ban against long-distance Raising jumping across an intermediate clause:

(8)

a. *John seems [that it is likely [___ to win]]
b. John seems [___ to be likely [___ to win]]

And a head dependency will be typically blocked by an intervening head. Hence, in a language like Italian, in which both auxiliaries and participles can be independently attracted to C in certain non-finite clause (9a–b), the participle can never be attracted across the auxiliary, as in (9c):

(9)
a. Essendo lei ___ tornata a Milano,…
'Having her come back to Milan,…'
b. Tornata lei ___ a Milano,…
'Come back her to Milan,…
c. *Tornata lei essendo ___ a Milano,…
'Come back her having to Milan,…'

(p. 223) 10.3 Feature-based Relativized Minimality

The structural typology given in (6) is too coarse in many respects. Perhaps the most straightforward problem is that it is not the case that any intervening A′ position causes a minimality effect on any other kind of A′ movement. For instance, while quantificational adverbs like *beaucoup* (a lot) determine an effect in cases like (7b), other adverbs like manner adverbial *attentivement* (carefully) do not (Laenzlinger 1998):

(10) Combien a-t-il attentivement consulté [___ de livres] ?
'How many has he carefully consulted of books?'

Another case is that in Italian and many other languages a (Clitic Left Dislocated) Topic can freely escape from an embedded clause introduced by a contrastive focus:

(11) A Gianni, credo che QUESTO gli volessero dire (non qualcos' altro)
'To Gianni, I believe that THIS they wanted to say' (not something else)

Under standard assumptions on the nature of positions, both Topic and Focus are left peripheral A′ positions, and manner adverbs as well as quantificational aspectual adverbs are clause-internal A′ positions, so that the simple typology in (6) is unable to draw the required distinctions.

Chomsky (1995b) proposed a revision of Relativized Minimality in terms of the Minimal Link Condition, involving a much finer typology of positions:

(12) Minimal Link Condition (MLC): in the configuration
… X_{+F} … Z_{+F} … Y_{+F} …
X_{+F} cannot attract Y_{+F} if there is an element Z_{+F} specified with the same feature +F and closer to X than Y.[1]

(p. 224) MLC specifies the intervention effects in terms of featural identity: the intervener blocks the local relation if it is specified with the same feature as the attractor and the attractee in the movement configuration. Hence, attraction of *how* by the main clause complementizer in (13) is blocked by the intervention of *who*:

(13) C_{wh} you wonder [who_{wh} behaves how_{wh}]

In cases like (10)and (11) the intervener is featurally specified in a way different from the target of movement, i.e. the Focus feature is an A′ feature distinct from Topic, hence the lack of intervention effect in these cases is expected; and the manner adverbial is not specified as a +wh element.

The MLC achieves the desired result in these cases by defining the typology of the intervener in a more fine-grained, feature-based manner than (6). On the other hand, a system based on the featural identity of the intervener is too fine grained to capture some of the generalizations originally captured by RM. For instance, the negative island illustrated by (7a): clearly the intervening negation is not specified with the *wh*-feature, otherwise it would be attracted to an interrogative C and it is not; still it blocks attraction of a *wh*-element. Similarly, the quantificational adverbial in (7b) is not specified as a *wh*-element, i.e. it does not get attracted to the C system, and still it blocks *wh*-attraction. If we want to maintain an explanation of these generalizations in terms of a formal intervention principle, we need a definition of the typology of the intervener more refined than the original (6), but less fine-grained than a system based on identity of the attracting feature.

Such a system is proposed in Rizzi (2001a, 2004a), based on ideas and analytic results of the study of the cartography of syntactic structures (Cinque 1999, 2002, Belletti 2004b, Rizzi 1997, 2004c, Cinque and Rizzi 2010). The idea is that attracting features are classified into distinct classes, each one defined by some kind of superfeature, along the following lines:

(14)

a. Argumental: Person, Number, Gender, Case,…
b. Quantificational: Wh, Foc, Neg, Measure, Frequency
c. Modifier: Evaluative, Evidential, … Manner, Measure, Frequency, Neg,…
d. Topic

Argumental features are the typical defining features of A positions: Phi and Case features. Quantificational features characterize quantifier—variable dependencies: question operators, focalized constituents, and other scope-taking elements like negation, etc. Modifier features identify adverbial positions; so, by and large, we express here the Cinque hierarchy (Cinque 1999). Topics form a class of their own, as they are not argumental, not quantificational, and not adverbial. There is a certain amount of cross-classification, for instance, in that certain adverbials (measure, frequency, neg, …) are also quantificational, while other adverbials are not (manner).[2]

(p. 225) We then stipulate that the feature classes in (14) define the structural typology relevant for the functioning of Relativized Minimality: interveners belonging to the same featural class as the target block local relations, while no effect is observed across classes. So, for instance, an intervening quantificational adverb blocks a Quant(ificational) A′-chain in a case like (7b), repeated below, while the 'pure' Mod manner adverbial does not have a blocking effect (here I adopt the copy theory of traces, as in Chomsky 1995b, representing traces as full but unpronounced copies of the moved element, expressed within angled brackets):

a. Combien$_{Quant}$ a-t-il beaucoup$_{Quant, Mod}$ consulté [<combien$_{Quant}$> de livres]?

*

b. Combien$_{Quant}$ a-t-il attentivement$_{Mod}$ consulté [<combien$_{Quant}$> de livres]?

OK

Click to view larger

(15)

Similarly, *wh*-extraction from an indirect question involves an A′ Quant chain crossing over a Quant element, a violation of RM, while Topic extraction over a Focus in Italian involves a Top chain crossing over a position belonging to a distinct feature class, Quant, which yields a permissible configuration:

a. How$_{Quant}$ do you wonder [who$_{Quant}$ behaved <how$_{Quant}$>]

*

b. A Gianni$_{Top}$, credo [che QUESTO$_{Quant}$ gli volessero dire <a Gianni$_{Top}$>]

OK

Click to view larger

(16)

In conclusion, RM effects are triggered not only by interveners bearing identical features to the elements which should enter into the local relation, but also by interveners bearing 'similar enough' features, where similarity is precisely expressed by the membership of the different feature classes, identified by the 'superfeatures' in (14) (see also Starke 2001, Boeckx and Jeong 2004 on feature-based intervention locality).

(p. 226) **10.4 Asymmetries**

Not all cases of wh-extraction from indirect questions yield an ungrammatical result. Consider well-known pairs like the following:

(17)
a. Which problem do you wonder how to solve?
b. *How do you wonder which problem to solve?

Traditionally, this contrast has been described as an argument/adjunct asymmetry, a contrast concerning arguments vs. adverbials of various kinds; but other considerations suggest that the empirical generalization should be stated somewhat differently.

On the one hand, in cases like the French construction illustrated in (7b), if the *wh*-specifier is sub-extracted from the direct object DP, the intervening quantificational adverb gives rise to an intervention effect, but not if the whole object DP is pied-piped (Obenauer 1983, 1994):

(18)
a. *Combien a-t-il beaucoup consulté [___ de livres]?
'How many did he a lot consult of books ?'
b. [Combien de livres] a-t-il beaucoup consultés ___ ?
'How many of books did he a lot consult ?'

On the other hand, there are argument—predicate asymmetries like the following, involving *wh*-movable adjectival predicates of small clauses (Baltin 1992):

(19)
a. ?How many students do you wonder whether Bill considers___intelligent?
b. *How intelligent do you wonder whether Bill considers these students___?

These facts suggest that the asymmetry is between arguments and everything else: *wh*-arguments can be extracted from weak island-creating environments, while everything else (adjuncts, predicates, pieces of arguments…) cannot.

In fact, not all arguments are equally extractable. There is an additional interpretive property that must be met, as the following pair of Italian examples shows:

(20)
a. Quanti problemi non sai come risolvere?
'How many problems don't you know how to solve?'
b. *Quanta benzina non sai come procurarti?
'How much gas don't you know how to get?'

Why does the second kind of extraction sound definitely worse than the first? The difference appears to lay on the interpretive properties of the restriction of the *wh*-variable. In the case of (20a) it is easy to imagine a context of utterance in which there is a well-identified, presupposed set of problems, and the question concerns the numerosity, and indirectly also the membership, of a proper subset of (p. 227) the presupposed set of problems: a natural answer would be something like 'three problems, that is to say problems 1, 3, and 4, out of the set of six problems that you know about'. On the other hand, in the case of the mass expression 'how much gas' of (20b), the presupposition of a specific amount of gas is unlikely: we are just asking for a mere quantity, not for the identification of a particular subset of a known set. Even in this case, though, a presupposed reading can be forced, most clearly by using a partitive expression 'how much of … 'which makes the presupposed character of the range of the variable explicit; in this case, extraction substantially improves:

(21) ?Quanta della benzina che ti serve non sai come procurarti?
'How much of the gas you need don't you know how to get?'

This is the property often called Discourse-Linking (or D-linking), the property of *wh*-phrases in which the range of the variable is presupposed, salient in discourse, or anyhow assumed to be familiar (Pesetsky 1987, Cinque 1990).

The purely interpretive property is not sufficient, *per se*, to permit extraction of an argument. Consider the following contrast in Italian:

(22)

a. Quanti di questi problemi non sai come risolvere?
'How many of these problems don't you know how to solve?'
b. *Quanti non sai come risolverne (di questi problemi)?
'How many don't you know how to solve of-them (of these problems)?'

In the acceptable (22a), the whole *wh*-phrase including the D-linked, partitive lexical restriction is moved as a unit to the left periphery. In (22b), the lexical restriction is pronominalized by *ne* cliticization within the island, and the remnant is *wh*-moved. The interpretation is D-linked in both cases (that the restriction is presupposed is made clear by the right-dislocated tag in (22b)), but the two structures clearly differ in acceptability: D-linking is not sufficient—it is also required that the presupposed lexical restriction be pied-piped with the *wh*-phrase to the left periphery; if the two are separated, as in (22b), *wh*-extraction remains excluded.

Very different proposals have been put forth to capture these effects. Some of these postulate a non-local connecting device only available to D-linked *wh*-phrases (Cinque 1990, Rizzi 1990b, 2001b, among many others). Here I would like to present a view consistent with a single connecting device, the one involved in all kinds of A′ chains, and systematically constrained by Relativized Minimality. This requires a particular assumption on the functioning of RM, which has been put forth in Starke (2001). According to this view (here I phrase things in terms slightly different from Starke's original proposal, and closer to the analysis in Friedmann et al. 2008, which I will return to), there are three significant set-theoretic relations between the featural specification of the intervener Z, and of the two elements which should enter into a local relation: X, the target, and Y, the intervener (where (p. 228) +A, +B are morphosyntactic features defining positions and potentially triggering movement).[3] 'OK' or '*' associated with each line indicates whether or not that particular configuration is permitted by RM. Cases 1 and 3 are straightforward: when the intervener has an identical featural specification as the target (case 1), the local relation is blocked; when the intervener is featurally disjoined with respect to the target (case 3), the local relation is permissible. The interesting case is 2, in which the intervener has a featural specification which is a proper subset of the specification of the target. In this case, according to Starke's proposal, the local relation is permissible. In order to achieve this result, I will adopt here the following revised form of (5):

(24) In the configuration
… X … Z … Y …
a local relation cannot connect X and Y if Z intervenes and Z fully matches the specification of X and Y in terms of the relevant features.

where we understand 'relevant features' as the 'superfeatures' involved in the classification given in (14). So, the weak island can be voided when the target X is more richly specified than the intervener Z in terms of the relevant features; if the specification is equal (or, a fortiori, if the intervener is more richly specified than the target, the fourth logical case), the local relation is disrupted.

(23)	X	Z	Y
1. Identity: *	+A	+A	+A
2. Proper inclusion: OK	+A,+B	+A	+A,+B
3. Disjunction: OK	+A	+B	+A

To return to D-linking, many researchers have underscored the topic-like character of D-linked *wh*-phrases (Richards 1997, Rizzi 2001a,b, Bošković 2002b, Boeckx and Grohmann 2004). Topics are presupposed entities, salient in discourse or somehow familiar, and this is obviously reminiscent of D-linking (in the system of Rizzi 2006b, topics are characterized by the combination of aboutness and D-linking, and as such they minimally differ from subjects which involve pure aboutness). So it is tempting to attribute the greater freedom of D-linked *wh*-phrases to their topic-like character, and to the greater freedom of topics with respect to other A′ constructions. Still, D-linked *wh*-phrases cannot be fully assimilated to topics. On the one hand, D-linked *wh*-questions involve genuine operator-variable structures, (p. 229) much as non-D-linked *wh*-questions; they differ from pure topics in Italian (and other Romance languages) in that they are by and large incompatible with clitic resumption, while direct object topics obligatorily require clitic resumption:

(25)

a. Quale problema credi che (*lo) potremmo risolvere?
'Which problem do you believe that we could solve (*it)?'
b. Questo problema, credo che *(lo) potremmo risolvere.
'This problem, I believe that we could solve *(it).'

(see Cruschina 2008 for a recent discussion of this issue). In order to express the partial topicality of D-linked *wh*-phrases we may reason as follows. Suppose that *wh*- and topical heads can combine in the left periphery of the clause through head movement, giving rise to the composite head +Wh, +Top. Such a composite head may then attract a phrase which simultaneously has *wh*- and topical properties, a D-linked *wh*-phrase, which is specified as *wh*- by its interrogative specifier, and as topical by its presupposed lexical restriction. According to Cinque (1990), the obligatory presence of a clitic with pure topics like (25b) is due to the fact that topics are not (quantificational) operators, hence they are unable to bind a variable, so that a pronoun is required (in English a topic is presumably able to bind a gap through the intermediary of a null operator, much as in Chomsky 1977b). But then a mixed phrase +Wh +Top qualifies as an operator through the +Wh specification, and as such it can and must bind a variable, hence a gap, in (25b). The clitic is therefore excluded, at least in languages which have not grammaticalized the resumptive strategy.

Why can a +Wh, +Top phrase escape a weak island? We are back to the logic of Starke's (2001) system. A relevant representation is the following, where we express the relevant 'superfeatures' labeling the classes of (14) (we are using Top both as a superfeature and as a regular feature here; presumably, if it is necessary to distinguish different types of topics, as in Frascarelli and Hinterhoelzl 2007,Benincà and Poletto 2004, Cruschina 2008, the typology of topics should be expressed by a more refined feature system within the Top class):

(26)

Which problem	do you wonder	[how	to solve	<which problem>]
X		Z		Y
Quant, Top		Quant		Quant, Top

The target of the local relation is more richly specified than the intervener in terms of the 'superfeatures' of (14) because the target belongs to both the Quant and the Top class; we thus are in the proper inclusion configuration of (23), and the local relation is not disrupted by the pure Quant element in the embedded C-system, as it does not 'fully match' the specification of X and Y, in terms of the revised definition of RMin (24).

If a pure Quant *wh*-phrase is moved across a D-linked phrase, the result is ill-formed:

(p. 230) **(27)**

How	do you wonder	[which problem	to solve	<how>]
X		Z		Y
Quant		Quant, Top		Quant

This is expected: the intervener fully matches the specification of the target, Quant (in fact, it also has the additional specification Top, but this is irrelevant for the calculation of locality under (23), (24)), hence the local relation is blocked.

What happens when a D-linked phrase crosses over another D-linked phrase? Extraction sounds acceptable in this case too:

(28) A che impiegato non sai quale compito affidare?
'To which employee don't you know which task to entrust?'

This is unexpected, as the target and the intervener would seem to be equally specified as Quant, Top, so that the structure should be ruled out by RM (24). On the other hand, we should consider that the intervening *wh*-element, while being interpretively D-linked (as a consequence of the lexical choice of the *wh*-element *quale)*, does not have to be moved to the Spec of a composite head +Wh, +Top: the only requirement of the construction, an indirect question, is that it moves to the Spec of a head specified as +Wh, for the satisfaction of the *Wh*-Criterion. The movement of the D-linked phrase to a head specified +Top is optional, as is generally the case with D-linked phrases, which do not necessarily appear in a Top position. If the *wh*-phrase in the indirect question in (28) can target a 'pure' +Wh position (hence, a 'pure' Quant position), we expect that further extraction of a D-linked phrase should be well-formed, and the expectation appears to be correct, as (28) shows. Under the assumptions just spelled out, (28) could have the following well-formed representation:

(29)

A che impiegato non sai	[quale compito	affidare	<a che impiegato>]?
X	Z		Y
Quant, Top	Quant		Quant, Top

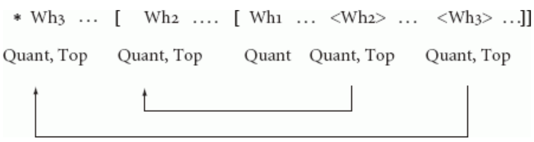
Still, the system makes a clear prediction, which it would be desirable to test: movement of a Quant, Top phrase across an intervener in a Quant, Top position should be ill-formed. If there is a way to force the intervener Z to be in a position of this kind in a structure analogous to (29), the prediction can be tested.

A case in point may be provided by double *wh*-island violations (see Rizzi 1978b, 1982: ch. 2, for the first discussion). Consider the following abstract situation: suppose that we extract a *wh*-phrase, say Wh2 in the following schema, from an indirect question (whose C position is filled by another *wh*-phrase, Wh1); under our assumptions, for the extraction to be successful, Wh2 must target a Quant, Top position:

(30)

… Wh2 …	[Wh1	… <Wh2> …]
Quant, Top	Quant	Quant, Top

(p. 231) In this way we have forced Wh2 to be in a Quant, Top position; if we now try to extract another *wh*-phrase Wh3 from this configuration, we predict that the extraction would fail: movement of Wh3 should be to a Quant, Top position to escape from Wh1, but it would inevitably cross Wh2, which also is specified as Quant, Top:



(31)

The prediction appears to be borne out. Consider a baseline structure in which we have a double indirect question, with Wh2 extracted from the lowest indirect question, as in (29):

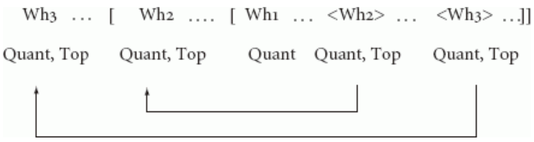
(32) Non mi ricordo [quale libro$_i$, abbiamo deciso [quando consegnare ___$_i$ al bibliotecario]]
'I don't remember which book we decided when to give back to the librarian'

Further extraction of a *wh*-phrase from the most deeply embedded *wh*-island (Wh3 in (31)) gives rise to an ill-formed structure:

(33) * A che bibliotecario$_k$ non ti ricordi [quale libro$_i$ abbiamo deciso [quando consegnare ___$_i$ ___$_k$]]?
'To which librarian don't you remember which book we have decided when to give back?'

In (32), extraction of the phrase *quale libro* from the most deeply embedded indirect question is possible, provided that it targets a position specified as Quant, Top in the higher indirect question. But then further extraction from this configuration is clearly degraded: even if the phrase to be extracted *a che bibliotecario* in (33) is D-linked and targets a Quant, Top position, it will inevitably move across another Quant, Top position, and RM will be violated. (33) thus reproduces the abstract scheme (31). The double *wh*-island context forces the intermediate *wh*-phrase to move to a Quant, Top position, and this precludes any further extraction:

(34) *A che bibl.… quale libro … quando … ___ … ___



(p. 232) Of course, some caution is needed, given the complexity of the relevant structure. Nevertheless, that the degraded flavor of (33) may not just be a matter of complexity is suggested by the fact that if the equivalent of Wh3 is A′ moved through some other mechanism, for instance as a topic in Clitic Left Dislocation, the structure sounds acceptable, in spite of the complexity of the dependencies involved:[4]

(35) Al bibliotecario, non mi ricordo quale libro abbiamo deciso quando consegnargli
'To the librarian, I don't remember which book we have decided when to give back to him'

10.5 Constraints on multiple movement

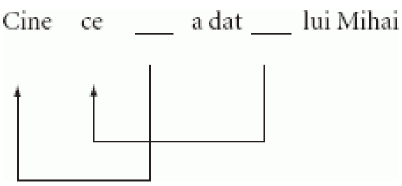
Certain languages permit (and require) movement of all the *wh*-elements in multiple questions. Some such languages display an interesting constraint on the cases of multiple movement, originally discussed in Rudin (1988): the *wh*-elements must reproduce the (hierarchic and linear) order of the respective extraction sites. This is shown, for example by the following pattern in Rumanian, in which the *wh*-subject obligatorily precedes the *wh*-object (Alboiu 2000):

(36)

a. Cine ce a dat lui Mihai?
'Who what gave to Mihai?'

b. *Ce cine a dat lui Mihai?
'What who gave to Mihai?'

Richards (1998) proposed the 'tucking in' approach, according to which the higher *wh*-element (the subject in (36)) moves first, in compliance with the Minimal Link Condition, and the lower *wh* (here the object) moves second, creating a new specifier position closer to the attracting C head, in violation of the strict cyclic principle (the extension condition in Chomsky 1995b; on the same phenomenon, see Bošković 2002b, Fox and Pesetsky 2004, among many other recent references). A distinct approach not requiring a weakening of the cyclic principle is proposed by Krapova and Cinque (2004) for the Bulgarian equivalent of (36). If one considers (p. 233) the two movement steps involved in the derivation of (36a), each of them violates RM, as stated in (34):



(37)

Nevertheless, in the crossing chain structure that is derived, the intervener always is only one member of the relevant chain: a complete chain never intervenes on any link of the other chain. So, the option that this state of affairs suggests is (1) that RM applies on the derived representations, rather than on each application of movement (possibly at the end of each phase: Chomsky 2001), and (2) that 'Z intervenes' in (24) is to be understood as 'all the occurrences of Z intervene': in (37), only one occurrence of *ce*, but not the whole chain, intervenes between *cine* and its trace, and only one occurrence of *cine*, but not the whole chain, intervenes between *ce* and its trace. The ungrammatical order is correctly ruled out by this interpretation:

*Ce cine ___ a dat ___ lui Mihai

(38)

Here, both occurrences of *cine* intervene between *ce* and its trace, hence the structure is ruled out by RM, under the current interpretation. So, intersecting chains are allowed, while nested chains are excluded.

The observed ordering constraint holds for chains of non-D-linked *wh*-elements. D-linked chains enjoy more freedom, permitting both orders (Alboiu 2000):

(39)

a. Cine cu care candidat a votat?
'Who for which candidate voted?'

b. Cu care candidat cine a votat?
'For which candidate who voted?'

The order to be explained here is (39b), the one which alters the order of the extraction sites. This possibility is in fact expected if D-linked *wh*-phrases have the option of targeting a position distinct from the one targeted by non-D-linked phrases. In fact, we have assumed that D-linked phrases can target the composite +Wh, +Top position, a position distinct from and higher than the pure +Wh position. Here *cu care candidat* can skip *cine* for familiar reasons, the composite nature of the attracting position, under Starke's interpretation of RM, as expressed e.g. in (24):

Cu care candidat cine ___ a votat ___?
For which candidate who voted?
Quant, Top Quant

(p. 234) **(40)**

Here the chain of *cu care candidat* crosses the whole chain of *cine*, but the configuration is allowed because the target position is specified as Quant, Top, hence it can bypass a 'pure' Quant element under (24).[5]

10.6 Appendix: intervention effects in acquisition and pathology.

Constructions involving movement across a constituent bearing a certain similarity with the target, even when grammatical, raise problems brought to light by the experimental study of adult performance, and which become particularly severe in special circumstances, such as in the course of language acquisition, or in agrammatism. A typical case in point is the contrast between subject and object relatives: object relatives are comparatively harder to understand than subject relatives by normal adult subjects (as reaction time experiments show: Frauenfelder et al. 1980 and much subsequent literature), and are clearly problematic for children between 3 and 5 or older, and for agrammatic patients. Grillo (2005, 2008) put forth the hypothesis that the problematic character of object relatives and a number of other A′ constructions involving objects for agrammatic patients may be looked at in (p. 235) terms of intervention ultimately to be traced back to the operation of Relativized Minimality. In his approach, linguistic representations in agrammatic speakers are featurally impoverished, to the extent that the featural distinctions diversifying A and A′ chains may be missing, or hard to keep in operative memory. This has the effect of causing much stricter RM effects in agrammatism, thus basically ruling out all chains in which a nominal crosses over another nominal.

Friedmann et al. (2008) have developed the core of this idea to capture the asymmetry between subject and object relative comprehension in children from 3.5 to 5. Experiments using picture-matching (or scenario-matching) tasks show that children in this age range properly understand who did what to whom in subject relatives such as (41a), but are in trouble with object relatives such as (41b), which elicit random answers (the experiments were conducted with learners of Modern Hebrew; percentages of correct answers are indicated after each example):

(41)

a. Show me the lion that ___ wets the elephant (90% correct)

b. Show me the chick that the cow kisses ___ (55% correct)

This is a well-known pattern, but Friedmann et al. show that the effect does not generalize to all object relatives: it depends on the nature of the head of the relative and of the intervening subject. If they both have the shape [D NP], as in (41b), the structure is problematic; but if the form of either the head or the subject is manipulated, comprehension improves. This is shown by free object relatives, in which the head of the relative is a bare *wh*-pronoun, or by headed object relatives in which the subject is impersonal *pro* in modern Hebrew, arguably a bare D; some examples of stimuli:[6]

(42)

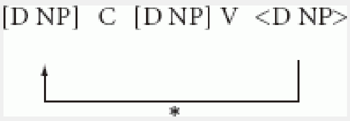
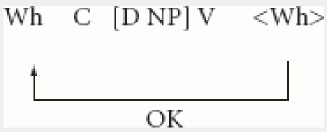
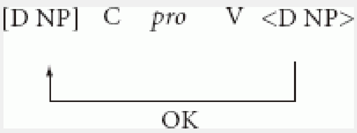
Tare	li	et	mi	she-ha-yeled menadned.	(79%correct)
Show	to-me	ACC	who	that-the-boy swings	
'Show me the one that the boy is wetting.'					

(43)

Tare	li	et	ha-sus	she *pro* mesarkim oto.	(83%correct)
Show	to-me	ACC	the-horse	that-brush-pl him	
'Show me the horse that someone is brushing.'					

Both these kinds of object relatives are properly understood by children. In short, while subject relatives (headed and free) are understood above chance, object relatives give rise to the following more complex pattern:

(p. 236) **(44)**

Configuration	Comprehension
a. Headed relative across a D NP subject	
	CHANCE
b. Free relative across a D NP subject	
	ABOVE CHANCE
c. Headed relative across a *pro* subject	
	ABOVE CHANCE

Movement of a relative head with a lexical restriction across an intervening lexically restricted subject, as in (44a), appears to be barred in the early grammar: a well-formed representation can't be computed, and the child tries to guess the proper assignment of thematic roles to the nominal expressions involved in the structure, which gives rise to a chance-level performance. If the target and the intervener are made sufficiently different, by dropping the lexical restriction on the relative head (44b), or on the intervening subject (44c), the antecedent—trace relation can be properly computed by the child, as the above chance success in the interpretation shows. The fact that the intervention effect is sensitive to the similarity of shape of the intervener and the target immediately recalls Relativized Minimality; in order to ascribe this contrast to RM, it must be assumed that the presence or absence of the lexical restriction (the NP constituent) plays a decisive role.[7]

This is made plausible by the fact that the property of possessing or not a lexical restriction plays a role as a possible attracting feature for movement (i.e. in the northern Italian dialects analyzed by Munaro 1999, lexically restricted and bare (p. 237) *wh*-elements appear to target clearly distinct positions in the left periphery of the clause; and in the Tromsoe dialect of Norwegian, bare *wh*-phrases do not trigger V-2, while lexically restricted *wh*-phrases do, a fact naturally amenable to a difference in the position targeted by the two types of *wh*-elements: Westergaard and Vangsnes 2005), so that, like any other feature capable of triggering movement, the feature expressing possession of a lexical restriction, which we will express as [+NP], may be responsible for a RM effect.

It remains to explain how early and adult grammars differ; why is it only in the former that a RM effect is triggered in object relatives like (41a)? Consider the relevant configuration (44a). Clearly, one attracting feature will be some kind of criterial feature marking the relative construction, call it +R; and, on the basis of the above considerations, the feature designating phrases with a lexical restriction [+NP] may be involved, too. Headed and free relatives may be differentiated in terms of presence vs. absence of this feature: +R, +NP for headed relatives, just +R for free relatives. So, the relevant configurations are:

(45) The lion that the elephant wets <the lion>

X	Z	Y
+R, +NP	+NP	+R, +NP

(46) who that the elephant wets <who>

X	Z	Y
+R	+NP	+R

(47) The lion that *pro* wet <the lion>

X	Z	Y
+R, +NP		+R, +NP

Case (45) corresponds to the proper inclusion case of table (23), while the free relative and *pro* subject constructions in (46)–(47), acceptable and interpretable for the child, may correspond to the disjunction case of table (23) (repeated for ease of reference as (48)):[8]

An attractive speculation to capture the difference between (45) and (46)–(47) is that the child grammar may adhere to a very strict version of RM permitting only the disjunction (case 3) of table (48), the one which avoids even the partial featural intervention which is permitted by the adult grammar with the proper inclusion (case 2): (p. 238) This stricter way of functioning may in turn be functionally motivated, as it allows the developing system to optimally avoid intervention, and the complexities connected to the computation of cases of partial featural overlaps between the target and the intervener. Of course, the price to be paid is a reduction in the set of structures which are ruled in by such a more restrictive locality principle, excluding object headed relatives and other A′ constructions moving a lexically restricted object across a lexically restricted subject (and also all the cases of selective violations of weak islands permissible by the adult grammars, always traceable to the proper inclusion (case 2) of (23)–(48), which are predicted by this approach to be problematic for the child in the relevant age range). The difficulties inherent in the computation of the proper inclusion case may also be the source of the problems experienced with object A′ constructions in language pathology, and may surface in normal adult performance in the slower comprehension of such constructions, as opposed to A′ constructions not involving the crossing of a (structurally similar) subject.

(48)	X	Z	Y	Adult grammar	Child grammar
1. Identity:	+A	+A	+A	*	*
2. Proper inclusion:	+A,+B	+A	+A,+B	OK	*
3.Disjunction:	+A	+B	+A	OK	OK

This approach could be criticized, in terms of the classic competence/performance divide, as trying to address a spurious empirical object: it seems to improperly put together strictly grammatical phenomena, such as the selective effects of weak island environments, and performance phenomena such as the slower comprehension time that adults manifest with object relatives, thus incurring into a category error. My impression is that it is useful and desirable to invert the perspective here: the fact of the matter is that there is a striking resemblance between certain grammatical principles and certain parsing strategies which have been proposed in the psycholinguistic literature, and this similarity should be captured.[9]

An approach adhering to a strict competence/performance, or grammar/parser, divide may be unable to capture, or make sense of, such similarities. The alternative approach pursued in the references cited in this appendix adopts a strongly integrated view of the grammar/parser interaction. It claims that the very same principle, RM, is operative in all sorts of intervention configurations, except that the exact featural constitution of the elements involved may give rise to distinct outcomes: full well-formedness or ill-formedness in the cases of disjunction or identity in the featural constitution of the intervener, and complexity effects in the case of partial identity (or subset—superset relation) in the specification of the intervener and the target. Such complexity effects may give rise to observable behavioral consequences in adult subjects, and may make the structure inaccessible to computation in special circumstances (acquisition, agrammatism). The outcomes are partially different, but a single underlying element, the grammatical principle, allows us to capture the 'family resemblance' between these phenomena.

Notes:

(1) MLC differs from RM in two other respects:

(1.) it is a condition on derivations, not on representations;
(2.) it applies on a specific process, movement, while RM tries to provide a general, rule-independent characterization of locality.

The derivational vs. representational nature of the constraints is a notoriously difficult question, which we will not try to address here. As for the generality of intervention effects, locality effects in processes different from movement (e.g. in phonological processes) seem to be amenable to a unified intervention principle, arguing for the plausibility of a general, process-independent characterization (see Rizzi 2004a, based on Halle 1995, and Nevins 2004 for a general discussion of intervention in phonological processes). Chomsky (2000a) further revises the MLC/RM as a locality condition on the relation Agree, sometimes referred to as Minimal Search. In that system the intervention principle does not restrict movement (Internal Merge) directly, but the identification of the candidate for movement through the prerequisite Agree relation.

(2) In a sense, topics are arguments, and share certain interpretive properties with referential subjects (in particular the 'aboutness' relation: Rizzi 2006b), but topic chains are fundamentally different from subject chains in that they obey much more relaxed locality constraints than subject chains.

(3) Some approaches to weak islands characterize the violation as purely interpretive (semantic or pragmatic): Szabolcsi and Zwarts 1997, Szabolcsi 1999, Kuno and Takami 1997). But a locality intervention principle like RM or MLC is needed for purely formal reasons as a general constraint on movement and other local processes, so it would be surprising to discover that some purely interpretive constraint mimics the effects of an independently needed, and more general, formal principle. The same kind of consideration holds for proposals to stipulate a 'scope intervention principle' which simply looks like the particular case of RM applying to scope-bearing elements.

(4) (35) raises the question why a topic chain is insensitive to the intervention of a position specified as Quant, Top which, according to (23), (24) should give rise to a minimality effect. In fact, the problem is more general: topic movement (at least in the Romance Clitic Left Dislocation construction) never seems to be sensitive to the intervention of another Top position, i.e. multiple topics are possible, and extraction of a topic from a clause introduced by another topic is also possible. See Rizzi (2004a) for discussion of this fact.

(5) The mechanism permitting crossing chains should be restricted to avoid overgeneration. Consider the following kind of example:

((i)) *How$_i$ do you wonder who$_k$ I said ___$_i$ we should meet ___$_k$

with the *wh*-adverb *how* construed with the intermediate verb *say*. This is clearly impossible, but the *how* chain and the *who* chain cross, if we consider them globally, so that not all the occurrences of *who* intervene in the *how* chain, and RM as expressed in (24) would not be violated. In order to rule (i) out, we can assume, following Chomsky (2001), that (24) is evaluated on the representation resulting at the end of each phase: so, at the end of the phase corresponding to the clausal complement of *wonder, how* must move to the phase edge in order to permit further extraction (let's say, it moves to the Spec of Force, if Force defines the clausal phase in a split-C system like the one in Rizzi 1997), but the resulting chain link crosses the only occurrence of *who* in this phase. *Who* intervenes in this link of the *how* chain, in the adopted technical sense, and therefore RM is violated.

(6) The structure with the impersonal *pro* subject is a resumptive relative, as the resumptive strategy sounds more natural with impersonal subjects; in Friedmann et al. (2008) it is shown that resumption *per se* is not a decisive improving factor: resumptive object relatives across a lexically restricted subject are not properly understood by the child. In this respect, resumptive and gap object relatives pattern alike.

(7) The chapter does not address the question of whether or not, on top of the formal property of the presence of a lexical restriction, the interpretive property of D-linking is also involved in the computation of intervention effects in children, as we have assumed in section 10.5 for the analysis of the asymmetries; but it should be noticed that, given the experimental setting, the phrases involved have been systematically introduced in the immediately preceding context, hence they are always D-linked ('here is a lion, here is an elephant, and then another lion and another elephant; now, show me the lion that the elephant wets'). The experimental evidence available so far does not tease apart the possibility that the relevant property may be 'having a lexical restriction' or 'having a D-linked lexical restriction'. In this section I will stick to the assumption that the weaker requirement 'having a lexical restriction' suffices, as is assumed in Friedmann et al. (2008), but the issue remains open.

(8) Whatever featural specification impersonal *pro* may carry (possibly limited to a set of Phi features), it is plausible that *pro* won't carry a lexical restriction, so it will not be specified +NP.

(9) For instance, the Minimal Chain Principle (De Vincenzi 1991) essentially guides the parser not to skip any potential position for a trace while scanning a structure, and this looks very much like a rough application of RM, one not paying attention, at the level of a first pass, to certain subtle featural distinctions. Similar considerations hold for the structure-sensitive interference effects on agreement highlighted in elicited production experiments, and amenable to familiar grammatical properties and relations (Franck et al. 2006, and many references cited there).

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[–] Abstract and Keywords

This article clarifies the concept of cycle, or phase, in minimalist parlance. Cyclicity is a derivational condition if there is one, a strong constraint if derivational timing is so relevant that chunks of structure abandoning the derivation become opaque to further computation. The challenge continues to be to understand the exact nature of this condition, which may be rather more widespread than it might at first seem.

Keywords: cycle, grammar, minimalism

11.1 Foundational Concerns

Generative grammar has long assumed that syntactic computation proceeds in cycles, although relevant such units have lived different incarnations ('domain', 'bounding node', 'barrier' and more, leading to the felicitous 'phase' of the minimalist program). A 'cycle' indicates a certain periodicity in the system, which correlates with empirical phenomena ranging from rhythmic patterns to interpretive regularity pockets. Such ideas originally pertained to phonological rule ordering: Chomsky et al. (1956) showed how multi-level accentual contours in English can be accounted for by rules applying in a transformational cycle (a proposal elaborated in Chomsky and Halle 1968). Extended to syntax more generally, the idea amounts to preventing a type of rule from recurring within a given derivation until a different type of rule has had a chance to apply, in the relevant domain deemed to be 'cyclic' (see Chomsky 1973: 243 for a statement, and Lasnik 2006 for perspective). We need not be more explicit at this point, since the basic intuition has had several lives that we will be exploring in what follows.

(p. 240) Summarizing earlier work, Lasnik (2000: ch. 2.7) emphasizes that a system containing obligatory and optional rules, or extrinsically ordered rules, is unlearnable.[1] Given difficulties of this sort, the theory moved in the direction of *intrinsic* orderings among rules, all of which were taken to be optional and limited by universal filtering devices. To the extent that no learning option is left to decide on the obligatoriness of any given rule, and both filters and ordering conditions are taken to apply universally, the amount of learning left is drastically reduced—to just arbitrary lexical choices. Chief among the intrinsic ordering conditions is the derivational cycle, as Freidin (1986) showed. This is all to say that, although cyclic conditions were not hypothesized to deal with these matters, they soon gained a critical role, over and above their descriptive value for first-order data.

The latter cannot be emphasized enough. Aside from their classically stressed value within phonology, it is hard to mention an area of syntax where cyclicity does not play some role. Conditions on theta-role assignment (theta-domains), case/agreement checking, successive bounding of long-distance displacement, binding, control, etc., all relate to cycles in some form. Indeed, rather faithful snapshots could be taken of the various theoretical models attempting to deal with syntactic nuances by asking what the relevant cyclic node was for that model, and whether it held without patching across modules. This state of affairs hasn't changed within minimalism, although it is harder to find a consensus as to what relevant cycles are and how they generalize across phenomena. This situation is not uncommon in the sciences, particularly if cyclic conditions are seen as conservations of some sort. That is, if phases (or bounding nodes, domains, or for that matter metrical feet, etc.) determine a structural zone where interesting relations for some reason happen to take place (successive cyclic movement, binding, stress assignment, etc.), then these zones reveal an otherwise hidden symmetry within our object of study. This should help us unearth its underlying architecture.

11.2 Recalling the Empirical Base

We will only be reviewing a handful of cyclic effects here, inasmuch as they remain central to minimalist theorizing (see Boeckx 2007 for a more detailed presentation of these matters). Let's start with successive cyclicity.

Successive-cyclic movements are well attested, as shown in (1a), where the italicized (unpronounced) copy of the boldfaced who indicates the displacement path. (p. 241) As Crain and Thornton (1998: 192) observe, many toddlers acquiring English actually go through a stage where they utter expressions like (1b) (*sic*):

(1)

a. [$_{CP}$ Who do you think [$_{CP}$ *who* Martin believes [$_{CP}$ *who* Susan likes who]]]

b. What do you think what Cookie Monster eats?

(1b) would appear to be a version of (1a), where relevant ‘copies’ of what, left in the edge of every clause that this element visits in its way to its scope site, are in some sense activated (contrary to what happens in adult English). This state of affairs obtains in many languages, as the German (2), from Kroch and Santorini (2007), illustrates:[2]

(2)

Wen	denkst	du,	wen	Martin meint,	wen	Susan magt?
who-acc	think	you	who-acc	Martin believes	who-acc	Susan likes
‘Who do you think Martin believes Susan likes?’						

Phenomena of this sort are common across the world's languages,[3] constituting prima facie evidence for successive cyclicity.

Recall also ‘reconstruction’ effects, as illustrated in (3):

(3)

a. Which picture of herself do you think Martin believes Susan likes?

b. Which picture of himself do you think Martin believes Susan likes?

c. Which picture of yourself do you think Martin believes Susan likes?

The analysis of reflexives as, roughly, clause-mates to their antecedents is rather straightforward if it proceeds successive-cyclically, as in (4):

(4)

a. … [$_{CP}$ Susan likes which picture of self] ?

b. … [$_{CP}$ Martin believes [$_{CP}$ [which picture of self] Susan likes wh-…]] ?

c. … [$_{CP}$ you think [$_{CP}$ [which picture of self] Martin believes [$_{CP}$ wh- …
Susan likes wh-…]]] ?

If the movement of the *wh*-phrase containing the anaphor proceeds (at least) across each CP edge, the displacement path will place the anaphor in each situation as a clause-mate of its antecedent (*Susan, Martin*, and *you*, respectively).

Those arguments involve successive steps in a long-distance relation, taken just so that cyclic conditions are met, and so cyclicity is thereby presupposed. Evidence for a derivational cycle can be even more indirect. One argument in that direction was implicit in the very clause-mateness just alluded to. The very fact that reflexives have an antecedent within a binding domain (or pronouns refuse it) is a cyclicity effect of sorts, where the cycle is the relevant binding domain. Consider another such argument, involving so-called uninterpretable features, and in particular case values.

(p. 242) Structural case values change from one verbal dependent to the next, typically involving accusative, nominative (or absolutive, ergative), and dative specifications. The list ends there, and it is not obvious why, when more straightforward alternatives exist: a system with *no* case values (the situation in Predicate Calculus, for instance) or just one case value (the situation within standard DPs, for nominal dependents); or alternatively a system marking each verbal dependent differently (as in arithmetic for different numbers). This puzzle compounds when we observe that case valuation obeys some sort of hierarchy, so that accusative (or absolutive) values are fixed before nominative (or ergative) ones, and both of these before dative ones. Let's illustrate that in detail.

San Martin and Uriagereka (2002) show that clausal embedding in Basque (e.g. in obligatory control) effectively renders a transitive verb unaccusative:

(5)

a.

Ni [PRO	pisua	galtzen]	saiatu	naiz.
I.ABS	weight-ABS	lose-nominalizer-LOC	try-part	I.be
‘I have tried to lose weight.’				

b.

Nik	hori	saiatu	dut.
I.ERG	that-ABS	try-part	I.have.III
‘I have tried that.’			

The transitive *saiatu* ‘try’ normally correlates with auxiliary *dut*, coding agreement with the ergative subject *nik* ‘I’ and the absolutive object *hori* ‘that’ (5b). But this case/agreement array is altered when the direct object is an embedded clause (5a): then the auxiliary accompanying *saiatu* ‘try’ is the unaccusative *naiz*, signaling agreement only with the, now absolutive, subject *ni* ‘I’. In other words, the clause is ‘skipped’ by the case/agreement system.[4] Consider also (6):

(6)

a.

Jonek	ni [PRO	ogia	egitera]	bidali
Jon-E	I-A	bread-DET-A	make-nominalizer-ALL	send-part
nau.				
I.be				
‘Jon has sent me to make bread.’				

b.

Jonek	niri	ogia	bidali	dit.
Jonek.ERG	I-D	bread	sent	III.have.I.III
'Jon has sent me bread.'				

For the ditransitive *bidali* 'send', case values depend on whether its dependents are regular DPs (in which instance (6b), the values are standard: absolutive for the direct object, ergative for the subject and dative for the indirect object) or instead an embedded clause is involved. In the latter instance, as (6a) shows, the (p. 243) system somehow bypasses the clause, assigning absolutive, instead, to 'the first DP it encounters' within the VP. So simply put: the grammar of Basque shows us how (structural) case is interested in DPs, not arguments. But how does the distribution of case values work for these DPs?

A mere configurational decision (sister to *v*′, sister to T′) cannot determine the case values, for configurationally all of these structures with different case values are identical. Rather, the grammar seems to be sensitive to the DPs it contains within a given cycle, assigning as few case values as possible, in some kind of ordering. The ordering is sensitive to a 'first case', which varies parametrically. For 'nominaccusative' languages, the first case value the grammar releases is within *v*P, the *accusative* value. In contrast, in 'ergabsolutive' languages there is no case valuation within *v*P: the first case to be active is external to this domain. The next case to become active also differs in each language. In nominaccusative languages it is the nominative value, i.e. a *v*P external case—the first case to become active in an ergabsolutive language, there called absolutive. Thus in the latter type, the next case value to become active is external to even the TP domain, which is commonly called ergative. As a consequence of this parameter, if there is a single case in a given sentence (its verb being unaccusative), this case will be different in each sort of language: the first to be active in an ergabsolutive language, but not in a nominaccusative one. However, from a different perspective it will also be the same: the case activated externally to *v*P, so that in effect nominative and absolutive case values are identical (i.e. nominative case is absolutive case).

It would not be accurate to blame the situation just described on what is external to, or internal to, *v*P. Clearly, dative case intervenes between the lower (accusative and nominative) and upper case values (absolutive and ergative). This 'in-between' status of datives is not reached configurationally, as (6) shows. Thus, suppose one were inclined to suggest that dative is merely associated to some v_{DAT} site, generated in between *v*P and TP, with some thematic import of the GOAL/SOURCE sort. The question for that approach would be why (6a) doesn't exhibit dative in the DP that gets this case in (6b), in a well-behaved fashion. Instead, this DP receives absolutive, just what a DP in direct object position would receive if there were one. The point is: the grammar seems sensitive to all these parallel considerations, somehow scanning a domain for the most efficient assignment of case values within that domain. That is the argument for cyclicity then: the domain where case valuation is evaluated is a cycle.[5]

While those arguments for cyclic domains are straightforward, the difficult task is to find ways in which they can be unified. Ideally, a cycle for case valuation should be the exact same domain where binding conditions are evaluated and, moreover, successive cyclicity is displayed. If, in contrast, this optimal result may (p. 244) not be achieved, it may mean that there are ultimately various reasons why cycles emerge in the faculty of language.

11.3 Theoretical takes on the Cycle

Several minimalist constraints have tried to express both cyclicity and successive cyclicity. The following is a list of the most obvious:

(7)

a. Extension Condition (Chomsky 1993: 22)
[Syntactic operations] extend K to K′, which includes K as a proper part.
b. Virus Theory (Chomsky 1995b: 233)
Suppose that the derivation D has formed Σ containing α with a strong (nowadays, 'uninterpretable') feature F. Then, D is cancelled if α is a category not headed by α.

(8)

a. Minimal Link Condition (MLC) (Chomsky 1995c: 311)

K attracts (nowadays, 'agrees' with) α only if there is no *β*, *β* closer to K than α, such that K attracts (agrees) with β

b. Phase Impenetrability Condition (PIC) (Chomsky 2000a: 106)

After a phase (cycle) is completed, its head cannot trigger any further operations.

(7a) demands that operations target the root of a phrase marker, while (7b) introduces the idea that extraneous features in a derivation must be eliminated as soon as detected. These constraints overlap, as Bošković and Lasnik (1999) show: if the Virus Theory in (7b) forces the system into the excision of the uninterpretable features (by way of agreeing elements already in the derivation), and moreover this process is immediate after the extraneous material is introduced, then it will target the place where the offending feature is placed: namely, the root of the phrase marker (7a). In turn, (8a) revamps Rizzi's (1990b) notion of 'Relativized Minimality', preventing operations if more local versions could take place, while (8b) defines cyclic domains and declares their opacity. Rizzi (2009) wonders to what extent 'impenetrability' (8b) can be derived from 'intervention' (8a). These matters are delicate, as none of the conditions in (7) or (8) seems any more natural than the others—and they are also subtly interdependent.

Consider in that respect locality effects involving displacement:

(9)

a. [What$_i$ does [Martin think [that [Susan [bought t_i]]]]]

b. ??[What$_i$ does [Martin wonder [why$_j$ [Susan [bought t_i]t_j]]]]

c. [Why$_j$ does [Martin think [that [Susan [bought books] t_j]]]]

d. *[Why$_j$ does [Martin wonder [what$_i$ [Susan [bought t_i]t_i]]]]

(p. 245) Long-distance movement is possible as in (9a), but it becomes degraded across 'islands' contexts (e.g. as in (9b), see Ross 1967a). The contrast is even sharper when the displaced element is an adjunct ((9c) vs. (9d)), as opposed to an argument (as in (9a) vs. (9b)). In fact, as Torrego (1984) showed, in Spanish the marginality of (9b) disappears, but the island effect remains for the adjunct:[6]

(10)

[Qué$_i$ piensa [Martín __ [que compró [Susana [__ t_i]]]]]

a.

what thinks Martin that bought Susan

'What does Martin think that Susan bought?'

[Qué$_i$ preguntó [Martín __ [por qué$_j$ compró [Susana [__ t_i] t_j]]]]

Click to view larger

b.

what asked Martin why bought Susan

'What did Martin ask why Susan bought?'

[Por qué$_j$ piensa [Martín __ [que compró [Susana [__ los libros] t_j]]]]

Click to view larger

c.

why thinks Martin that bought Susan the books

'Why does Martin think that Susan bought the books?'

*[Por qué$_j$ preguntó [Martín __ [qué$_i$ compró [Susana [__ t_i] t_j]]]]

d.

why asked Martin what bought Susan

'Why did Martin ask what Susan bought (for that reason for buying)?'

How do the different conditions in (7)and (8) bear on facts of this sort?

The approach to these facts since Chomsky (1977b) is that islands arise in relevant cycles if the normal, successive-cyclic, mode of extraction is prevented. The intuition is that the intermediate *wh*-element in the unacceptable examples ‘caps off’ its cycle, thereby preventing an otherwise valid displacement. Under these circumstances, either the Extension Condition in (7a) or the Virus Theory in (7b) are responsible for timing the various operations, preventing lethal ‘interleavings’ as in (11) to generate the ungrammatical (9d), a ‘countercyclic’ derivation:

(11)

[Why$_j$ does [Martin wonder [t$_j$ [Susan [bought what] t$_j$]]]]

a.

[Why$_j$ does [Martin wonder [t$_j$ [what$_i$ [Susan [bought t$_i$] t$_j$]]]]

b.

(11b) does not target the root of the phrase marker at that derivational stage, in violation of condition (7a); from the point of view of the Virus Theory, and (p. 246) redundantly, whatever uninterpretable feature may be responsible for the displacement of *what* has not been taken care of immediately—it had to wait for the derivation to go all the way upto (11a) first, in violation of condition (7b).

The redundancy may in fact be greater. Setting aside the ruled out-derivation in (11), representation (9d) also runs afoul of the MLC (8a), inasmuch as the top C is attempting to agree with *why*, prior to its movement, over and above the already displaced, and closer, *what*, which in principle also agrees (in relevant *wh*-features) with the very same C. There is more. Suppose that the lower CP is a cycle in the system, as has been the intuition ever since Chomsky (1973)—throughout Chomsky (1977b, 1981a, 1986a) and more. Then after this cycle is completed (presumably by ‘capping it off’ with *what*, in some sense) and abandoned, being deliberately naïve about the matter, it should not be able to involve in any further operations, such as the displacement of why. So we can rule out the ungrammaticality of the lower modification in *why does Martin wonder what Susan bought?* in four different, reasonable ways.

Moreover, our conditions require parametrization, to distinguish the English (9b) and the Spanish (10b).[7] At the same time, the effect remains robust with adjunct extraction across any islands, universally; indeed in such robust terms that it obtains even in languages where *wh*-movement is not overt, like Chinese (Huang 1982):

(10)

a.

Zhangsan	yiwei	Lisi	mai-le	shenme?
Zhangsan	thinks	Lisi	bought-asp	what
‘What does Zhangsan think Lisi bought?’				

b.

Ni	xiang-zhidao	[wo	weishenme	mai	shenme]
You	wonder	I	why	buy	what
‘What do you wonder why I buy?’					
[What is the x such that you wonder why I buy x?]					

c.

Zhangsan	yiwei	[Lisi	weishenme	mai-le	shu]
Zhangsan	thinks	Lisi	why	bought	book
‘Why does Zhangsan think Lisi bought books?’					

d.

Ni	xiang-zhidao	[wo	weishenme	mai shenme]
You	wonder	I	why	buy what
‘Why do you wonder what I buy?’				
[*What is the reason x such that you wonder what I buy for that reason x?]				

(12a) shows how the *wh*-element, *shenme* ‘what’, occupies the position it would if it had been the direct object *shu* ‘books’.[8] The same is true of (12b), involving a question verb, *xiang-zhidao* ‘wonder’. The fact that (12b) is possible in Chinese, (p. 247) involving a matrix question about the object of *mai* ‘buy’, indicates that in this language the relevant facts about argument extraction are possibly along the lines of the Spanish (10b), instead of the English (9b). But the interesting example is (12d), if compared with the grammatical (12c)—a long-distance question involving the adjunct *weishenme* ‘why’. In the surface (12d) is word-for-word identical to (12b), but the intended reading is different: here we want *shenme* ‘what’ to associate to the question verb, while we expect the adjunct to take widest scope, as a bona fide question. Huang showed that this is impossible, in line with what we saw both for the Spanish (10d) and the English (9d).

Researchers have speculated that parsing difficulties might underlie the explanation to these paradigms (see e.g. Culicover and Jackendoff 2005). In particular, the unacceptability of (9d) or (10d) is often taken to follow from the parser being confused about locating the variable that the *wh*-adjunct is to bind, which could be in either the matrix or the embedded clause. For argument *wh*-phrases, selectional restrictions in the verb locate the relevant site. A parsing approach to these matters is taken to be reinforced by the fact that examples like (9b) ameliorate when specific *wh*-elements (often referred to as ‘discourse-linked’) are invoked. For example:

(11) ? [Which book$_i$ does [Martin wonder [why$_j$ [Susan [bought t_i]t_j]]]]

While the point is well taken, this approach has difficulties explaining why the Spanish (10b) is perfect, even when no ‘discourse linking’ appears relevant. Matters get even trickier for examples like the Chinese (12d). Here the parser does not have to locate a gap, as there hasn't been any overt *wh*-displacement to start with. Still, the relevant interpretation is unavailable. This suggests that, contrary to what a purely parsing-theoretic view expects, the difficulty with the examples is grammatical in nature.

That said, the redundancy in the minimalist explanation remains troubling. Chomsky's most recent approach, reviewed in section 11.5, has been essentially to deny (7a) in favor of some version of (7b), and to reduce the role of (8a) in favor of (8b). The latter is in large part a consequence of the fact that, if cyclic conditions should be unified, it is hard to see how a ‘Relativized Minimality’ approach can say anything about conditions that are at right angles to displacement and similar relations. The MLC in (8a) is a condition telling us what contexts make dependencies licit, and the major cut in possible relations has to do with other putative relations ‘on the way’. But that is not what seems relevant in determining case values: for these only a notion like ‘clause-mateness’ (in some form) matters. Similar concerns could be raised about construal domains. It is when cyclicity is extended to these considerations that condition (8b), in whichever form turns out to be appropriate, would appear to have better chances of success. We return in section 11.7 to whether the reductionist desideratum has been met already, and if so to what extent an appropriate version of (8b) makes (8a) unnecessary.

(p. 248) 11.4 Cycles as Emergent Domains in Grammar

The ultimate question underlying this discussion is why the grammar has cycles. Since Chomsky (2000a: 99 and ff) at least, Chomsky suggests that cyclicities— in the form of what he then called the ‘lexical array’ (LA), the defining basis for what are nowadays called phases—are part of grammar as a design feature to avoid complexity.

> Suppose automobiles lacked fuel storage, so that each one had to carry along a petroleum-processing plant. That would add only bounded ‘complexity,’ but would be considered rather poor design. […] [W]e may take [the computational system] to be a mapping of Lex [the lexicon] to [relevant] LF representations […]
>
> Is it also possible to reduce access to Lex […]? The obvious proposal is that derivations make a one-time selection of a *lexical array* LA from Lex, then map LA to expressions, dispensing with further access to Lex. […] If the derivation accesses the lexicon at every point, it must carry along this huge beast, rather like cars that constantly have to replenish fuel supply. Derivations that map LA to expressions require lexical access only once, and thus reduce operative complexity in a way that might well matter for optimal design.

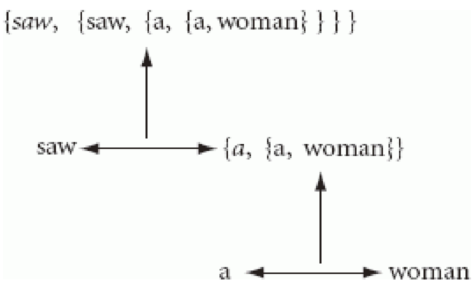
Chomsky may have been reacting to a controversy started by Lappin et al. (2000), and in particular the possibility that linguistic derivations comparing alternative computations running in parallel could lead to some computational blow-up. If derivations start in an LA, computations will not have to deal with an entire lexicon in the order of several tens of thousands of symbols. In addition, in (2000a: 106) Chomsky suggests that, ‘taking the derivation more seriously’, LA should be accessed cyclically, so that computational procedures reduce access to the lexical domain:

> Suppose we select LA [and] the computation need no longer access to the lexicon. Suppose further that at each stage of the derivation a subset LA_i is extracted, placed in active memory (the ‘workspace’), and submitted to the [derivational] procedure […] When LA_i is exhausted, the computation may proceed if possible; or it may return to LA and extract LA_j, proceeding as before. The process continues until it terminates. Operative complexity in some natural sense is reduced, with each stage of the derivation accessing only part of LA.

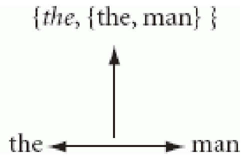
Although Chomsky made his move in order to avoid derivational difficulties with parallel computations,[9] what matters now is that, involving such a small amount of (p. 249) items, derivations will be very constrained—even if they do, in point of fact, run in parallel. A different issue, however, is whether it is *because* of the design feature of reducing operative complexity that cyclicities are present in grammar. The obvious alternative is that such conditions emerge in their own terms, whatever those may be, and they have the reduction in complexity as a desirable *consequence.*

It is worth bearing in mind that, in the sort of bottom-up, and very streamlined, bare phrase structure (BPS) system that Chomsky assumes from (1995a), the ‘workspace’ is articulated in a curious way.[10] The head-complement relation is what the system captures most naturally. In contrast, the head-specifier relation forces the system to go into a separate ‘derivational workspace’. Consider deriving *the man saw a woman:*

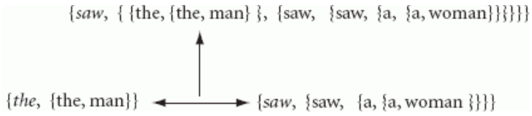
(14)



a.



b.



c.

We must assemble *the man* in its own workspace within the derivation (14b), place it on a ‘memory buffer’, and then assemble the results to the structure still active in (14a), as in (14c). Resorting to this buffer is a virtual definition of specifier. The question is whether the system treats material within the buffer, vis-à-vis the rest of the structure.

(p. 250) First Merge (i.e. the association of a selected head with some structure) yields the head-complement space that defines the ‘spine’ of any given derivation, while subsequent instances of Merge, particularly if they are to involve complex steps of their own, arise off this spine. Such an elsewhere condition seems to be what underlies specifiers, counterpoints of sorts to the more primordial head-complement relations. The distinction is just part of how the derivational dynamics work under Chomsky's assumptions, and the only issue left is whether these necessary derivational conditions (given the way the matter is framed) have interesting design consequences—even properties of the cyclic sort that interest now.

In that regard, one cannot help but notice that head-complement relations of the sort in(14a) are equivalent to what computer-scientists call ‘tail recursion’, whereas when complex specifiers are added to the picture, what obtains instead is called ‘full recursion’ instead. A well-known theorem equates tail recursion to flat structures, which are generable by very simple Finite State Automata (see e.g. Abelson and Sussman 1984: ch. 1.2). In other words, an object like (14a) can be (weakly) generated by even barer assumptions than those implied in the Merge procedure sketched in (14), yielding the string *saw*⁀ *a*⁀ woman.This string is quite different from the bracketed [saw[*a woman*]], which makes extra structural assumptions about constituency (e.g. establishing a parallelism with [saw [*her*]] that *saw*⁀ *a*⁀ woman would not sanction). Nevertheless, the point is that the string itself is *weakly* generable by an automaton with very limited resources, incapable of systematically signaling internal groupings. This equivalence does not hold for the full object in (14). The absolute simplest way to weakly generate an element exhibiting full recursion (in more familiar terms, with a complement *and* a complex specifier, at least) is in the BPS terms already outlined, or any formal equivalent. This could be left as a curiosity, or we can make use of the idea to deal with a problem that has worried syntacticians since Lucien Tesnière (1959) raised it in the 1930s: how to relate presumably intricate linguistic thoughts to their manifestation in speech.

The most recent systematic treatment of such matters goes under the name of ‘linearization’, and is due to Kayne (1994). The question is how to map complex phrases, expressing at least precedence and dominance relations among them, to a word sequence. Of course, if the linguistic system is computable in any standard sense,[11] there has to be some solution to this problem: any computable function can be written as a sequence of symbols. But the question that interests linguists, who are concerned with the strong generative capacity of their computational system, is which solution the language faculty deploys. This is relevant not just in itself, as a design feature of the system. If that actual solution is unique and **(p. 251)** universal, there will be a direct learnability gain, just as we saw in the first section for optional/obligatory or (un)ordered rules.[12]

That said, it seems interesting that, for portions of structure that are weakly reducible to Finite State dependencies, the implicit flatness of the structure already entails a direct organization in the required linear terms, for speech purposes. If the language faculty takes advantage of this formal fact, inherent to its ‘bare’ phrase structure derivations, it will be acting most efficiently—computationally so at least. Granted, that only linearizes chunks of structures (those weakly reducible to symbol strings), not the entire kind if sufficiently complex (fully recursive) elements present in syntax. But one can easily suppose that linearization is not produced in a wholesale fashion, and that a cyclic alternative in fact obtains: one chunk at a time. This is what interests us here, the point being:

cyclic Spell-Out for linearization purposes, if each Spell-Out chunk obeys the Finite State limit discussed above, would seem to be a natural way for the grammar to proceed in the task of externalizing its intricate structures.

Needless to say, cyclicity cannot be all there is to the complete linearization task: the chunks, useful as they may be for linearization purposes, must be pieced back into units that make the intended sense. This is what implies some kind of operational memory buffer, so that the chunks placed in a separate workspace as in (14) can be addressed to the correct portions of the structure, for full recursion to obtain. But what matters now is that the cyclic proposal has bite, much in the spirit of earlier ideas by Joan Bresnan (1971), who argued that the rules of the phonological cycle apply in the course of the syntactic cycle, the basis of present-day systems to track intonation patterns in PF. Similar considerations were raised by Jackendoff (1972) and Lasnik (1972), with regard to the computation of semantic structures. For them, the syntactic cycle plays a role in the mapping to the LF interface, in terms of such issues as the interpretation of anaphora, as we saw for the examples in (4). Many of these efforts were partially overlooked within early versions of the Principles and Parameters theory (Chomsky 1981a), until they were recast within that framework in Sloan (1991), a proposal that dovetailed naturally into Watanabe (1993), already a minimalist piece. Recasting these sorts of general ideas in explicitly minimalistic terms, Uriagereka (1999) argues for the convenience of simplifying the theory to involving, if necessary, multiple applications of the Spell-Out process. A model along the lines discussed above could then predict the asymmetries outlined in Huang (1982), between complements (part of the sentential spine) and the rest.

The basic contrast can be illustrated as in (15):

(15)

- **a.** Madonna is who everyone has seen pictures of.
- **b.** ?*Madonna is who pictures of have caused a public stir.

(p. 252) Intuitively, in (15a) the trace is within the spine (complement to a complement of the main verb), while in (15b) the trace is off the spine (complement to a non-complement of the main verb). In the structural terms discussed above, the relationship between *who* and its trace happens within the same workspace in (15a) and not in (15b). The Multiple Spell-Out (MSO) approach suggests that complex non-complements quite generally cannot be linearized at Spell-Out in terms of a correspondence between their possible Finite State analysis and a PF string, contrary to what happens with arbitrarily large chunks of the spine. So when hitting a non-complement, the system must go into a separate application of Spell-Out, however that is implemented technically. At that point, crucially for the contrasts in (15), the relation between material within this separately spelled-out phrase and whatever is outside enters a new sort of difficulty.

What that difficulty amounts to is hard to pin down. It could range from a radical problem (an impossible grammatical connection between a gap inside the spelled-out material and the outside) to merely more taxing conditions (having to involve post-syntactic relations to accommodate a failure in establishing a dependencyon a first derivational pass). Difficulties could even vary from language to language, for as we saw in the contrast between (9b) and (10b), languages differ with regards to how deviant their speakers judge island conditions. What remains factual is the structural asymmetry itself. Importantly for our purposes, a cycle of this sort is not proposed in order to reduce computational complexity; rather, under the conditions just described it simply arises.

In that regard, and going back to observations pertaining to the discussion of (9)/(11), it is worth bearing in mind that one condition never varies. Although the equivalent of (16b) can be improved in particular languages and specific contexts, this is never the case when the element that is displacing from inside a non-complement is an adjunct:

(16)

- **a.** Hatred is the reason why people believe that Dr King was assassinated.
- **b.** Hatred is the reason why it is believed that Dr King was assassinated.

(16a) is ambiguous, one pragmatic reading being more plausible than the other: the reason Dr. King was assassinated was hatred (so people believe), or alternatively the reason people believe (the particular belief about Dr King) is hatred. One hopes that the second reading is actually false in the real world. Now, when it comes to (16b), one reading is immediately lost—the plausible one. The description of this contrast falls directly into the pattern above. For modification of the *assassinated* event, the adjunct *why* must have started its derivational life within the lower clause; in (16a) this clause is part of the spine, the complement of *believe*, while in (16b) the clause is, instead, off the spine, associated to the pleonastic *it* subject of *is believed*. Then presumably only in (16b) does the lower clause spell-out separately. And the rest follows, as discussed for (16).

(p. 253) **11.5 The Still Puzzling Nature of Phases**

The best-known cyclic property of minimalist derivations goes by the name of 'phase'. This notion was first proposed in Chomsky (2000a: 106), and defined in terms of the subarray LA_i (in the sense above) that can be selected for active memory:

> LA_i should determine a natural syntactic object SO […] LA_i can then be selected straight-forwardly: LA_i contains an occurrence of C or of *v*, determining clause or verb phrase— exactly one occurrence if it is restricted as narrowly as possible […] Take a *phase* of a derivation to be a syntactic object SO derived in this way by choice of LA_i. A phase is CP or *v*P, but not TP or a verbal phrase headed by H lacking N-features and therefore not entering into Case/agreement checking: neither finite TP nor unaccusative/passive verbal phrase is a phase.

Once he characterized matters this way, Chomsky introduced the PIC, which he described as 'a still stronger cyclicity condition'. He informally stated it as in (8b) above (and see (20) below for an accurate presentation). We will discuss details in a moment, but for now let's proceed intuitively, showing Chomsky's proposed phrases as brackets:

(17) [$_{CP}$ John [$_{vP}$ *t* thinks [$_{CP}$ Tom will [$_{vP}$ *t* win the prize]]]]

Even if this is the correct array of facts, the question is why *v*P and CP should be the relevant phases. While claiming that these domains have a 'propositional' nature, in (2001: 11) Chomsky emphasized that: 'A subarray LA_i must be easily identifiable; optimally, it should contain exactly only lexical item that will label the resulting phase.' This is consistent with phases clustering around *v* and C, but it does not tell us why those should be the relevant categories. In (2004a: 124), Chomsky suggested that these domains 'are those that have an [Extended Projection Principle, EPP] position as an escape hatch for movement and are, therefore, the smallest constructions that qualify for Spell-Out'. This was under the assumptions that the PIC holds and, moreover, that EPP positions bypass it, for reasons that we return to. Now it may be true that only *v* and C happen to display EPP positions— but the question remains why. By (2005: 17), Chomsky was suggesting that phases should 'at least include the domains in which uninterpretable features are valued'. It is fair to say, judging from these works alone, that the matter of why *v* and C happen to determine phases is unsettled.

The view favored by Chomsky in recent years is the last one mentioned. He asserts in (2008a: 154), after reasonably reminding us how striking uninterpretable features are in a minimalist system, that phasal cyclicity has to do with the derivational need to deal with these features right away, a version of the Virus Theory in (7b):

> (p. 254) Since these features have no semantic interpretation, they must be deleted before they reach the semantic interface for the derivation to converge. They must therefore be deleted either before Transfer [of syntactic material to the interfaces] or as part of Transfer. But these features may have phonetic realization, so they cannot be deleted before transfer to the phonological component. They must therefore be valued at the stage in computation where they are transferred—by definition, at the phase-level.

The question still remains why the *v* and C projections happen to be the locus of uninterpretable features (where structural case and agreement for object and subject are determined, respectively).

The possibility that 'phases are exactly the domains in which uninterpretable features are valued, as seems plausible' (Chomsky 2008a: 155) echoes an earlier proposal examined, and originally rejected, in Chomsky (2000a: 107): to define phases in terms of convergence. A phase in this regard can be seen as the minimal domain where convergence holds, meaning that no uninterpretable feature is left unchecked. This prevents a local determination of what a phase is (i.e. a simple statement like 'the projection of a category with uninterpretable features'). Nevertheless, lacking a theory of where and why uninterpretable features show up, the gain may be a mirage: it depends on whether the presence of said features in relevant categories is principled, or a larger domain needs to be examined to determine that (see Gallego 2008).

In (2000a) Chomsky also worried about examples of the general format in (18):

(18)

a. Who did you see t?

b. [you [saw who]]

Chomsky assumed that the *wh*-phrase has an uninterpretable feature analogous to structural case for nouns, which requires it to move to its final landing site. That feature would be trapped inside the lower phase, and the reasoning is that if we were to define 'phasehood' in terms of convergence, then no domain containing a *wh*-phrase would ever be a phase—until the site where the proposed uninterpretable feature is checked. Then again, perhaps the grammar has an option, upon encountering the uninterpretable feature in *who* inside *v*P: to move the *wh*-phrase out of that domain, so that convergence is not compromised. This strategy would only work if there is a mechanism for displacement (here Object Shift), which may otherwise not have been invoked. Then one could still define the relevant phase (here *v*P) as a minimal domain of convergence.[13]

One more thought is worth bearing in mind with regard to examples of the form in (18), which presumably involve movement steps as in (19):

(p. 255) **(19)**

a. [$_{vP}$ you [v [saw who]]]

b. [$_{vP}$ who$_i$ [you [v [saw t$_i$]]]]

c. [$_{TP}$ you$_j$ [$_{vP}$ who$_i$ [t$_j$ [v [saw t$_i$]]]]]

d. [$_{CP}$ who$_i$ [$_{TP}$ you$_j$ [$_{vP}$ t$_j$ [t$_j$ [v [saw t$_i$]]]]]]

As is discussed in more detail in the next section, the domain of syntactic accessibility for the CP phase is everything higher than the first *v* projection—everything outside the circled material in (19). The question, then, is what the timing is for the movements in (19d). Does displacement of *you* (from a *v*P specifier to the TP specifier, for EPP reasons) precede displacement of *who* to CP, respecting the Extension Condition in (7a) (and see Hornstein 2001 for arguments in this direction)? Or is it the case, instead, that *who* moves first to its scope site, and later on the system adjusts the position of *you*, involving a process of so-called tucking-in—for operations targeting non-root domains (and see N. Richards 2001 for a defense of this approach)? In (2007) and (2008a), Chomsky is sympathetic to a view first discussed in Collins (1994) and defended in Uriagereka (1998: 365ff.): perhaps the question is ill-posed as above, and in fact all movements within a phase are parallel (see Gallego 2008). If this possibility does obtain, then at least within phases there is, in point of fact, parallel computation, and therefore enough look-ahead to be able to execute such computations within those cyclic confines. If so, the Extension Condition ought to be epiphenomenal: to the extent that it holds, it follows from the interaction of deeper systemic properties.

Now defining phases as minimal domains of convergence is not much deeper than achieving the task in terms of highlighting domains where uninterpretable features emerge. In both instances the issue is why the system is designed in such a way as to have C and *v* head those domains, as opposed to T or any other category. Unfortunately, up to now all theorists have been tracking the facts here, not really predicting their pattern. A theory could have been imagined to make T phases central to the system—if the empirical need had emerged—by calling that domain 'prepositional', the locus of EPP effects (as has always been assumed), the domain of uninterpretable features, or the minimal domain where a derivational portion converges. The chunk of the subarray LA_i that T would identify would be every bit as optimal, containing a single lexical item to determine it, as is the case with C and *v*, or any other category for that matter. So we do not know why the cyclic domains we call phases exhibit the periodicity they do.

Other theorists have weighed in on this matter, in various ways. For some (e.g. Svenonius 2001, Fox and Pesetsky 2004, den Dikken 2007a, b, Gallego 2007), what determines the relevant 'phasehood' is some interface condition that the system is (p. 256) attempting to meet in just those terms. (Chomsky has been skeptical about this sort of approach, although he himself invited this inference by relating phases to ‘propositionality’—in his original formulation, not so much as a consequence as a *raison d ’être.)* Others have, instead, attempted to correlate the particular choice of phases within the system (whether *v*, C, or others) to some sort of periodicity in the design construction of the cycles, which exhibits in this view a certain structural rhythm (see in particular Boeckx 2009c,e, M.D. Richards 2007, Gallego 2008, and Uriagereka forthcoming for this sort of approach). The hope in this theoretical move is that something is computationally natural about this particular periodicity, although what this is turns out to be far from obvious. A third group effectively denies the relevant cyclicity, either by embracing it to its limit (every phrase is a phase: see Manzini 1994, Takahashi 1994, Epstein et al. 1998, Fox 2000, Boeckx 2001, Bošković 2002a, N. Richards 2002, Epstein and Seely 2002b, Fox and Lasnik 2003, and Abels 2003 for perspective) or by questioning this notion of cyclicity altogether, at least in terms of phases (Collins 1997, Grohmann 2003a,b, Boeckx 2005, Boeckx and Grohman 2007, Jeong 2006, Chandra 2007).

11.6 Multiple Layers of Cyclicity?

We have sketched two different cyclic notions. The MSO approach carves out the spine of sentences from the rest, while phases, instead, chop that spine into further slices. These are obviously compatible (as are, in fact, other cuts we return to shortly). It is even likely that each of these systems needs the other. Without phases of some sort, the MSO approach only gives us derivational layers without any principled upper-limit; so it needs phases (of any periodicity) to be computationally feasible. At the same time, the MSO approach is designed to separate complements from non-complements, and more concretely specifiers. This is not a natural distinction in the phase system, even though grammatical conditions hold of specifiers. For example, Chomsky (2000a: 108) states his Phase Impenetrability Condition as in (20), fleshing out (8b) above:

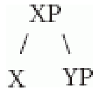
(20) Phase Impenetrability Condition
In phase α with head H, the domain of H is not accessible to operationsoutside α; only H and its edge are accessible to such operations.

Where β is the complement of a phase head H, Chomsky takes ‘β to be the *domain* of H, and α (a hierarchy of one or more SPECs) to be its *edge*’ SPEC[ifier]s at the edge determine the cyclic domain's ‘escape hatch’, the essence of successive-cyclicity effects.

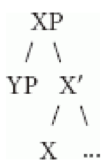
(p. 257) The obvious question, raised in skeptical studies too numerous to mention, is why specifiers are not transferred to interpretation when the complement of the phase head that hosts them does. Phase-wise, the edge does not go hand in hand with the rest of the computation. But this is somewhat expected if the derivational buffer alluded to in section 11.4 is real. First Merge yields a *domain* in the sense just described, which works in harmony with the Probing head that seeks a valuating goal within that space. In contrast, the phase *edge* is a boundary to the phase space that the domain establishes. Now the domain/edge counterpoint is what having both a MSO and a phase system directly ensures.

The issue here is what constitutes the basic design of the Agree operation and how it interacts with systemic Merge in its two manifestations: externally and internally. (21a) below arises for External Merge when it involves a basic head-complement relation, while (21b) does under similar circumstances still involving External Merge, but for spine-specifier relations (recall (14)). Conditions reduce when it comes to Internal Merge. While (21d) is straightforward (cf. (21b)), (21c) is not:

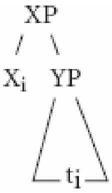
(21)



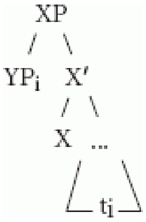
a.



b.



c.



d.

As a movement operation, (21c) is generally taken to be ungrammatical since Chomsky (1995c). Interestingly, the relation in (21c) does directly obtain, albeit in the form of Agree. This is contrary to what happens in (21d), which never involves an Agree relation.[14] In other words, the space of possibilities working for External Merge ((21a) and (21b)) is distributed in such a way that the equivalent of (21a) only holds for Agree (21c), while the equivalent of (21b) only holds for Internal Merge.

Stipulations are easy to concoct as to why that is. For example, the operation in (21c) arguably violates the Last Resort Condition (LRC), unlike (21d).[15] But this hardly constitutes a solution to the puzzle, particularly when there are situations where displacements as in (20d) obtain without any last resort licensing (e.g. in stylistic shifts). A more plausible approach capitalizes on a fact noted in section 11.4: subcase (21b), involving a complex specifier, is very different from subcase (21a). Concretely, only in (21b) does the grammar need to invoke two separate (p. 258) derivational workspaces. It maybe this condition, involving its own memory buffer, that licenses Internal Merge. This could relate to the complex derivational condition that emerges for 'copies', or more accurately occurrences of internally merged items, which we mentioned in section 11.2 when examining successive cyclicity. The most worked-out computational analysis of these 'context-sensitive' elements, Kobele (2006), invokes a derivational memory that seems to be of roughly the same sort—at least of the same computational complexity—as the one involved in externally merging complex-specifiers, if BPS conditions are assumed. It is even possible that situation (21b) never emerges, a view explored in Chomsky (2007) and (2008a) that leaves 'external arguments' as oddballs. This would imply that only (21a) obtains for the spine of head—complement relations, and only (21d) does for specifier relations off the spine. (In this view, only (21a) and (21d) are grammatical.) In either instance specifiers literally occupy their own 'derivational dimension'.

Implicit in the approach just sketched is again the idea that different cyclicity conditions obtain for objects like (21a)—generated by External Merge—and objects like (21d)—generated by Internal Merge. Cyclicity for simple objects as in (21a) may reduce to the monotonicity of the Merge operation when it obtains without any memory buffers, which gets to recur with each deviation from this condition that the grammar requires. In contrast, cyclicity for complex objects as in (21d) may relate to some form of the Virus Theory, a condition that gets activated when Agree operations are at stake.

Possibly still other cyclic conditions exist. One such situation involves compounding, inasmuch as relevant 'headedness' conditions (as in Di Sciullo and Williams 1987) are not easy to express within a compound in a BPS system. Without annotating 'bar-levels', 'left/right-highlights', or similar notational exercises, just how do we determine that *bird* is the head in *blackbird?* Uriagereka (forthcoming) defines the compound head by resorting to the cyclic transfer to interpretation of the member(s) of the compound that are not the head. A similar situation obtains for small clauses, analyzed in the symmetric terms discussed in Moro (2000), as in *I consider this bird* black. Again, here we want to be able to distinguish (at some point in the derivation) the subject from the predicate—but it is unclear how to do this in BPS terms. Uriagereka (forthcoming) suggests that the element that transfers to the interpretive components is the subject, again a cyclic solution.[16] None of these cyclic conditions has anything obvious to do with essential Merge conditions or the Virus Theory. For that matter, the same general point arises for phonological domains like metrical feet, for which standard syntax would appear to be at right angles. This is all to say that the grammatical conditions that generate cyclicity—setting aside (p. 259) now *successive* cyclicity, which seems like a hybrid of cyclicity and transformational syntax—may well be of a rather deep sort.

11.7 Conclusions

Cyclicity is a derivational condition if there is one, a strong constraint if derivational timing is so relevant that chunks of structure abandoning the derivation become opaque to further computation. The challenge continues to be to understand the exact nature of this condition, which may be rather more widespread than it might seem at first. It is important to keep asking how the situations discussed here (or others) relate to what constitutes a cycle in the system. Why is it C or *v* that head phases and why does a phase transfer to the interpretive components have to be as momentous as it appears to be for opacity effects to emerge (this being the ‘virus’ intuition)? In other words, when, how and why particular cycles arise in grammar continues to be a plentiful research well, which half a century of theorizing hasn't dried out. Chomsky (2000a: 99) is careful to emphasize that this computational architecture of language is a ‘curious and puzzling fact about the nature of the mind/brain’, adding that the matter raises ‘difficult and obscure questions about what this means for a cognitive system’. Rather than a mystery, however, we can safely say that cyclicity is now ‘just’ a fascinating problem, which our theories may well resolve within a generation. In the process, this may tell us much of what it ultimately means for a cognitive system to exhibit computational properties.

Notes:

Thanks to Cedric Boeckx, Ángel Gallego, Kleanthes Grohmann, Norbert Hornstein, Atakan Ince, Howard Lasnik, and Terje Lohndal for help with research for this chapter.

(1) Lasnik shows this to be so both quantitatively (for *n* the number of rules, 2^n obligatory/optional possibilities and *n!* possible orderings among them), and qualitatively: children would not be able to acquire such extrinsically ordered, possibly obligatory, rules without negative data.

(2) Although see Lohndal (2010) for some critical perspective, suggesting that this approach is too simplistic.

(3) Specific marks of the phenomenon range from agreement in sites akin to those of the copies in (2) to syntactic operations taking place there, as in (10)below.

(4) It is implausible that the *v* element responsible for case within the VP is missing in these instances. In the terms presented in Hale and Keyser (1993, 2002), this *v* element is responsible for the agent theta-role associated to *saiatu* ‘try’—and observe that the theta roles haven't changed from (5a) to (5b).

(5) Note that the argument for cyclicity remains even if one insists (disregarding the argument just presented) on a configurational treatment of case. After all, the relevant sequence of case values (say, <accusative, dative, nominative>) clearly recurs as embeddings proceed, with an obvious periodicity.

(6) Note that verbs in this language ‘gravitate’ towards the sites which the moved phrase visits in its outward displacement, which is signaled with arrows; this too is a successive cyclicity effect, as Torrego argued.

(7) The first parameter ever proposed explicitly, by Rizzi (1978b), was hypothesized in just this regard; see also Rudin (1988) for the possibility that the parameter is based on whether a language allows multiple specifiers.

(8) The question word has not been displaced, as in English (9a) or Spanish (10a).

(9) Given a shared LA (ia) (‘numeration’ if lexical tokens matter) and a common structure (ib), Chomsky compared derivations (iia) and (iib), expecting (iib) to outrank (iia):

((i))
(a.) {there, was, believed, …}
(b.) [to be a man here]

((ii))

(a.) *There was believed [a man to be t here].

(b.) There was believed [t to be a man here].

Basically the system prefers merging *there* (in the position of the trace in (iib)) to moving *a man* (as in (iia)). But then observe (iii):

((iii))

(a.) [A ballroom was [where there was a monk arrested]]

(b.) [There was a ball room [where a monk was t arrested]]

Starting in the same LA these sentences are predicted to compete, (iiia) outranking (iiib). But both are grammatical. To avoid this difficulty, Chomsky suggested accessing LA in a cyclic fashion. If *there* falls in each of two separate cycles, the comparison conditions will not be met. Note that right then the issue arises of what is a possible cyclic access to LA. If the embedded TP (as opposed to CP) could define a cycle, (iia) would not compete with (iib), and the entire reasoning would dissipate (and see section 11.5).

(10) The discussion that follows is adapted from Uriagereka (forthcoming), where relevant contextualization and references are presented in more detail.

(11) That is, if its ultimate symbols are manipulated by way of a genuine computational procedure of the classical Turing sort.

(12) A different issue, of course, is whether there are some open parameters for Kayne's linearization procedures (e.g. for the linearization of heads vs. complements), or for that matter whether his proposal is empirically correct.

(13) As Lasnik et al. (2005: ch. 6) point out, if in the next phase up there continues to be an unchecked *wh*-feature —which will be the case until the final resting site for the *wh*-element is reached, where the offending feature is eliminated—then the offending item will continue to be pushed out, thereby modeling this form of successive cyclicity. As Boeckx (2007) shows, there is no entirely worked out theory of what gears intermediate steps of long-distance movement (and see also Bošković 2007).

(14) Setting aside the special case of pleonastics (where the specifier is a neutralized head).

(15) The idea being that only in the latter instance is there a target that can gear the movement, for instance in terms of a prior Agree operation involving X there and some feature within YP prior to movement (a circumstance that would not obtain in (20c)).

(16) In this work the idea is extended to other predications involving adjuncts and possibly even external arguments, if something along the lines of the thematic theory in Uriagereka (2008a) is viable.

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Anti-Locality: Too-Close Relations in Grammar

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[–] Abstract and Keywords

This article discusses a new area of research within locality (and minimalism) – that of anti-locality – which deals with the minimal distance that dependencies must span to be licit. It first positions the idea of 'anti-locality' with respect to both the current understanding of the notion and its roots in generative theorizing. This eventually leads to an exposition of the Anti-Locality Hypothesis in the context of a structural tripartition through Prolific Domains, and, subsequently, to the notion of Copy Spell-Out as a 'repair strategy' to void an anti-locality violation that would arise otherwise. This is finally related to the conceptual underpinnings of Prolific Domains, and a specific explanation of the Anti-Locality Hypothesis stemming from interface conditions on the derivation and the computational system at large.

Keywords: dependencies, minimalism, Anti-Locality Hypothesis, structural tripartition, Prolific Domains, Copy Spell-Out

12.1 Overview

Anti-locality, in the sense used here, provides a perspective on locality conditions imposed on movement dependencies that pays close attention to a possible lower bound on legitimate distance.[1] In essence, it is encapsulated by the Anti-Locality (p. 261) Hypothesis stated in (1), which will be picked up in the next section from a quasi-historical vantage point, put in perspective throughout this chapter, and embellished with a presentation of more recent theoretical and empirical work pertaining to it.

> **(1)** Anti-Locality Hypothesis (Grohmann 2003b: 26)
> Movement must not be too local.

This take on (anti-)locality did not emerge from a vacuum, however. On the one hand, a dissertation written just a little later (Abels 2003) dealt with 'too-close' relations in syntax under the same name, but for a somewhat different purpose and within quite a different set of theoretical assumptions. In addition, there are several obvious precursors in work from the mid-1990s that was never, as far as I am aware of, pursued any further (Murasugi and Saito 1995, Bošković 1994, 1997, Ishii 1997, 1999, Saito and Murasugi 1999, all cited by Abels).

In this chapter, I will first position the idea of 'anti-locality' with respect to both the current understanding of the notion and its roots in generative theorizing. This will eventually lead to an exposition of the Anti-Locality Hypothesis (1) in the context of a structural tripartition through Prolific Domains and, subsequently, to the notion of Copy Spell-Out as a 'repair strategy' to void an anti-locality violation that would arise otherwise. This will finally be related to the conceptual underpinnings of Prolific Domains, and a specific explanation of the Anti-Locality Hypothesis, stemming from interface conditions on the derivation and the computational system at large.

Despite the emphasis on my own formulation of anti-locality, I would like to stress from the outset that, while presented in tandem, the tripartition of the clause (p. 262) into Prolific Domains and its relation to anti-locality (including Copy Spell-Out) are quite independent from each other. With the discussion of anti-locality in a broader context, I intend to show that domain-partitioning serves as one possible evaluation metric for anti-local relations (perhaps even the wrong one), with the latter related to, yet conceptually independent of, the former; I will also make the attempt to relate anti-locality to Phase Theory (Chomsky 2000a et seq.). The upshot of this chapter will be a comprehensive discussion and much-needed characterization of anti-locality—the Cinderella of contemporary theorizing on locality effects in natural language, which as such requires, and deserves, more thorough attention.

12.2 Too-short (movement) dependencies

The term 'anti-locality' was first used and couched within a larger theoretical framework in Grohmann (2000b) to capture the intuition not only that movement is delimited by an upper bound, but that the distance between two syntactic objects in a given (movement) dependency or chain is also subject to a lower bound in order to be licit. This intuition was then picked up by Abels (2003), who (correctly, yet only in passing) points to intellectual precursors in the literature engaged in attempts of ruling out too-close relations in phrase structure. These, and related issues, will be presented first. One of the central questions here regards the evaluation metric for anti-locality: If anti-locality, as understood here, is a ban on 'too-short movement steps' or 'too-close structural dependencies', over which derivational-structural part or domain is it to be computed, and when does this evaluation take place?

12.2.1 Ban on specifier-adjunction dependencies

Possibly the earliest discussion of a too-close relation between two syntactic objects in phrase structure in the relevant sense, or a too-short movement step in derivational terms—certainly the first in the minimalist literature—is the central concern of Saito and Murasugi (1999), originating in a 1993 manuscript. Given an economy condition on derivations such as (2), also known as Minimize Chain Links (MCL),

(2) Economy of Derivation (Chomsky and Lasnik 1993: 546)
Minimize chain links.

discussed in Chomsky and Lasnik (1993: 540ff.; cf. Chomsky 1995b: 90) and also aimed at capturing Relativized Minimality effects (Rizzi 1990b), the question arises not only how to compute well-formed chain links, but also how minimized they (p. 263) can be. This is particularly relevant in the context of Takahashi's (1994) influential framework, where, rather than checking a feature in intermediate positions, successive-cyclic movement follows from the requirement that individual movement steps be local (see Bošković 2002a and especially Boeckx 2003 for implementations and further refinements of Takahashi's proposal): Each chain link must be as short as possible, as per (2); see also Manzini (1994) for the suggestion that movement must pass through the domain of every head. Even though they do not explicitly say so, with this background, one can interpret one of Saito and Murasugi's goals as that of formulating a minimal satisfaction of minimizing chain links, i.e. a lower boundary of distance to be traversed in any movement step—or: an anti-local requirement on dependencies.[2]

One of the side-effects of Saito and Murasugi's work, as observed by Bošković (1994, 1997), was that 'vacuous topicalization' of subjects in English can be ruled out. Within the GB framework, where topicalization (3b), was—following Ross's (1967a) Chomsky-adjunction analysis (see also Iwakura 1978)—(free) adjunction to S (Baltin 1982) or IP (Lasnik and Saito 1992), thus derived from (3a), we can observe that subjects, unlike objects, cannot undergo this free adjunction step (3c).

(3)
- **a.** John likes Mary.
- **b.** Mary$_i$, John likes t_i.
- **c.** *John$_i$, t_i likes Mary.

The relevance of subject topicalization becomes more imminent in embedded contexts, where an IP-adjunction analysis was, at the time, even more undisputed (discussed in detail by Lasnik and Saito 1992, who also provide (4) below, taken in this presentation from Boeckx 2007: 102, citing Bošković 1994).[3]

(4)

a. I think that [$_{IP}$ Mary, [$_{IP}$ John likes <Mary>]].

b. *I think that [$_{IP}$ John, [$_{IP}$ <John> likes Mary]].

Frampton (1990) had already ruled the IP-adjoined position out as a possible landing site for subjects in English, but Saito and Murasugi's intention was to capture such a ban on independent grounds. The ban can be represented schematically (p. 264) as in (5), where 'specifier-adjunction' is meant to denote adjunction of a specifier within the same maximal projection XP—here, IP.[4]

(5) Ban on Specifier-Adjunction

*[$_{IP}$ XP$_i$ [$_{IP}$ t_i I^0 …]]

This ban might be interpreted more generally as a filter disallowing phrasal movement from [Spec, XP], a specifier position of some phrase, to [Adj, XP], an adjoined position of the same phrase. If so, it would be nice to deduce this filter as a property of the computational system, that is (if it holds, of course) on independent grounds and not as a language- or even construction-specific constraint.

Saito and Murasugi capture the Ban on Specifier-Adjunction more formally through the following two-clause constraint, which I choose to call the obvious Constraint on Chain Links (CCL) and frequently return to throughout the chapter:

(6) Constraint on Chain Links (classic) (after Saito and Murasugi 1999: 182)

a. A chain link must be at least of length 1.

b. A chain link from α to β is of length n iff there are n 'nodes' (X, X′ or XP, but not segments of these) that dominate α and exclude β.

They employ (6) to explain the ungrammaticality of a number of extractions within the clause IP and the nominal layer DP. One example is the contrast in (7):

(7)

a. Who$_i$ does John think [$_{CP}$ t'_i [$_{IP}$ t_i fixed the car]]?

b. *Who$_i$ does John wonder [$_{CP}$ how$_k$ [$_{IP}$ t_i fixed the car t_k]]?

Here, the argument goes, *who* cannot adjoin to IP as an escape hatch from the embedded clause and then *wh*-move to the matrix [Spec, CP] which, all things being equal (both being A′-movements), might otherwise be a conceivable derivation.

The same argument can be made to successfully rule out complementizer-less subject relatives in English, as Bošković (1997: 26) notes, picking up an idea from Law (1991) and arguing that, under economy considerations, an IP-adjunction analysis of the null operator Op in (8a) should be preferred over a CP analysis in (8b)—hence barring Op in subject position from adjoining to IP, as in (9).

(p. 265) **(8)**

a. the man [$_{IP}$ Op; [$_{IP}$ John likes t_i]]

b. the man [$_{CP}$ Op$_i$ [$_{C'}$ C[$_{IP}$ John likes t_i]]]

(9) *the man [$_{IP}$ Op$_i$ [$_{IP}$ t_i likes Mary]]

Apparently, some languages seem to allow a violation of the CCL in exactly these contexts. Bošković (1997: 27) provides an example from fifteenth-century Italian (Rizzi 1990b: 71, citing Wanner 1981), and mentions comparable short zero-subject relatives in the period of English shortly after (Bošković 1997: 185–6, n. 30), found in Shakespeare, where, similar to the null-subject language Italian, null subjects are also found (data and discussion communicated to him by Andrew Radford):

(10)

Ch'è faccenda [$_{IP}$ Op$_i$ [$_{IP}$ *pro* [$_{VP}$	tocca	a noi *t*$_i$]]] (Italian, 15th c.)
for is matter	concerns	to us
'For this is a matter (that) concerns us.'		

(11)

a.

There is a lord will hear you play tonight.	(*Taming of the Shrew, c.*1590)

b.

Youth's a stuff will not endure.	(*Twelfth Night, c.*1601)

Bošković suggests that such data do not run counter to the CCL after all, since in null-subject languages, the null operator Op can be moved from its base-generated position ([Spec, VP] in earlier stages of the theory, nowadays commonly assumed [Spec,*v*P], a point to be returned to presently), with the subject position [Spec, IP] or [Spec, TP] respectively filled by a null expletive (instead of Rizzi's *pro* in (10)—if this is needed for null-subject languages to begin with; cf. Alexiadou and Anagnos-topoulou 1998).

Note, however, as Volker Struckmeier (p.c.) reminds me, that several dialects of present-day Modern English also allow short zero-subject relativization, such as the so-called 'subject contact relatives' (Doherty 1993) in (12), termed 'contact-clauses' by Jespersen (1961: 132ff.), as den Dikken (2005: 694) points out, where these are taken from (cf. Jespersen 1961: 1444).[5]

(12)
a. There's one woman in our street went to Spain last year.
b. It's always me pays the gas bill.
c. I have one student can speak five languages.
d. He's the one stole the money.

From a conceptual perspective, if the CCL were a grammatical constraint, it would also not be desirable to allow its violation in some languages.

(p. 266) 12.2.2 Ban on complement-specifier dependencies

If there is anything to anti-locality in terms of a minimal distance requirement on movement steps—that is, if a constraint on syntactic operations, or the computational system more generally, should derive the Anti-Locality Hypothesis—it should involve more than just a ban on XP-adjunction from [Spec, XP].[6] This would be expected if the CCL is a good means of capturing the data presented in the previous subsection; in fact, under a multiple-specifier approach to phrase structure (Chomsky 1995b), not available to Murasugi and Saito at the time of writing either work, the constraint already rules out, as desired, movement from one specifier position to another within the same maximal projection. In this subsection, and especially the next, more empirical evidence will be collected for something like an anti-locality constraint on linguistic structures in grammar and possible formulations of it will be sketched based on the CCL, used from now on in a slight reformulation, as the Modified Constraint on Chain Links:

(13) Constraint on Chain Links [modified][7] (after Bošković 2005: 16)
Each chain link must be at least of length 1, where a chain link from α to β is of length n iff there are n XPs that dominate α but not β.

(p. 267) Bošković (1994) introduced this modification (see the Original Constraint on Chain Links in note 7) in order to rule out the illicit movement in (14c): the internal argument cannot move to the external argument position to yield (14a) with a plausible hypothetical interpretation of (14b).

(14)
a. *John likes.
b. John likes himself.
c. [$_{VP}$ John likes ~~John~~]

Since there is an additional complication in terms of phrase-structural assumptions, these cases will be discussed in more detail in the next section, which will lead to a new take on the Anti-Locality Hypothesis.

Note first that, if the CCL rules out movement from α [Spec, XP] to β [Adj, XP] because no maximal projection is crossed that dominates α but not β, thus resulting in a length of the chain link less than 1, movement from the complement position to the specifier within one and the same projection should likewise be ruled out. This is exactly what Abels (2003) seems concerned with for the most part.[8],[9] He (p. 268) presents data such as the following (as well as comparable examples from French and Icelandic; cf. Abels 2003: 116–17):

(15)
a. Nobody believes that anything will happen.
b. That anything will happen, nobody believes.
c. *Anything will happen, nobody believes that.

What we observe here is that CP is 'mobile', in Abels' terms, but TP, the complement of the phase head C (*that*), is not.[10] Thus there is a ban on moving TP out of CP, across and stranding its head, as in (15c), while extracting the entire CP, complement of a higher verb, is legitimate (15b).

On a par with the Ban on Specifier-Adjunction in (5) above, this ban might be interpreted more generally as a filter disallowing phrasal XP movement from a complement position of some phrase, [Compl, XP], to a specifier of the same phrase, [Spec, XP]. Structurally, we can then think of it as a Ban on Compl-to-Spec (cf. the Revised Ban on Specifier-Adjunction from (ii) of n. 4 above for the notation now used):

(16) Ban on Compl-to-Spec
*[$_{XP}$ YP_i [$_{X'}$ X^0 $\cancel{YP_i}$]]

If so, it would be nice to deduce this filter as a property of the computational system, that is—if it holds, of course—on independent grounds and not as a language- or even construction-specific constraint. Abels suggests he does (but see immediately below and n. 9 above). His approach ties in with Phase Theory (Chomsky 2000a et seq.), more specifically, with the need to move within a feature-checking/valuation framework. In the approaches stemming from (either (p. 269) version of) the CCL, it would arguably be by stipulation: within a specified phrase-structural configuration, movement minimally must cross a particular number/type of nodes. The understanding of anti-locality developed at length in Grohmann (2000b) and related work will give the 'independent grounds' a twist which will be presented and discussed in subsequent sections.

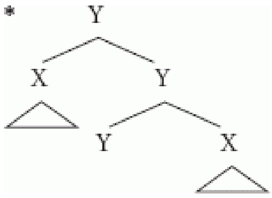
First, let us consider Abels' own 'historical precursor', the Head Constraint (HC) of van Riemsdijk (1978) in (17), which he resurrects and—rightly, I believe (Boeckx and Grohmann 2007)—relates to the Phase Impenetrability Condition (PIC) first formulated in Chomsky (2000) in (18):[11],[12]

(17) Head Constraint (adapted) (from Abels' version of van Riemsdijk 1978)
No rule may involve X_i (Xj) and Y in the structure … X_i … [$_\alpha$ … Y …] … Xj … if Y is c-commanded by the head of α, where α ranges over V, N, A, P.
(18) Phase Impenetrability Condition (strong) (Chomsky 2000a: 108)
In phase α with head H, the domain of H is not accessible to operations outside α, only H and its edge are accessible to such operations.

Abels (2003: 41) notes that '[t]he only difference between the two conditions is that van Riemsdijk includes a list of categories to which the [HC] applies, whereas Chomsky does not consider this list (i.e. the specification of which XPs are phases) part of the definition of the [PIC]'—but there is no need to, since phase heads are independently defined by Chomsky (2000a, 2001); see Boeckx and Grohmann (2007) for discussion and further references. HC and PIC are thus virtually identical, both banning an element outside (phase) XP relating to the complement of the (phase) head or any other element properly contained in it.

Essentially, given the two abstract phrase-markers in (19a, b), Abels' (2003: 104–5) explanation of why only (19b) is possible follows a two-step rationale, with the upshot that anti-locality is trivially driven by Last Resort considerations. (p. 270)

(19)



First, a constituent X merged as complement or specifier enters into the most local relation with the head Y (‘mutual total c-command’) meaning that, if Y and X bear a feature they should check against one another, the feature can be licensed right then and there (see also the revised notion of ‘natural relations’ in Grohmann 2001, 2003b: ch. 2). Second, the relevant movement step in (19a) is illicit because no feature can be licensed that could not be licensed prior to the movement step. (The same state of affairs, Abels continues—as already mentioned above—holds for the movement steps Spec-to-Spec, Spec-to-Adj, and Compl-to-Adj within one and the same phrase.) In other words, being merged into [Compl, YP] in (19a), X is already in the relevant structural configuration with Y to license any formal feature that needs checking—and as such should not move to [Spec, YP]. In essence, the Stranding Generalization (cf. note 9) then follows from ‘a more general constraint barring vacuous Merge (a violation of Last Resort)’, as Gallego (2007: 72) nicely puts it.

As far as I can see, all of Abels' cases (including those not discussed here[13],[14]) are covered by the CCL—whether the Original CCL from (i) in n. 7 above, the Classic CCL from (6), or the Modified CCL from (13). The question is whether the CCL is a ‘deep’ enough explanation (if any at all; see also n. 9 above). As just suggested, if correct, economy considerations would be enough to rule out vacuous Merge and derivational steps akin to (19a)—i.e. Spec-to-Adj and Compl-to-Spec (or even Compl-to-Adj, not discussed here) would be inadmissible movement steps from a checking perspective. However, such a conclusion rests on the legitimacy of assumed checking configurations and the stipulation that all movement is feature-driven; not entering into a discussion on Spec-Head checking versus Agree, see also n. 18 below for further complications. In sum, the status of the CCL does not appear crystal clear at this point. In the next subsection, I will thus present yet another take on anti-locality and initially suggest a reformulation of the CCL (p. 271) to cover the new sets of data. This will lead us to a fresh look at the relevant domain within which movement is banned. Since I will concentrate on a slightly expanded view of anti-locality, the kind of ‘deeper’ explanation on independent grounds I am looking for will be formulated in somewhat different terms, which requires additional discussion. This is what sections 12.3 and 12.4 will be concerned with.

12.2.3 Ban on domain-internal dependencies

The first attempt to systematically investigate and capture too short (or as used here, anti-local) movement steps—that is, too-close relations in phrase structure—was originally presented in Grohmann (2000b). There the term ‘anti-locality’ was also first offered to this class of linguistic structures, which I refined in follow-up work (much of it carried out in collaboration with, but also independently by, a number of other researchers). It centered around the ungrammaticality of ‘paradigmatic structures’ such as the a-examples in the following, and a very particular interpretation thereof.

(20)

a. *John likes.
b. John likes himself.

(21)

a. *Him (softly) kissed her.
b. He (softly) kissed her.

(22)

a. *What/Which vegetable, Mary detests?
b. What/Which vegetable does Mary detest?

To put the intended interpretation into context, one may wonder why—within the Copy Theory of movement (Chomsky 1993, Nunes 1995) and in a framework that reinterprets the Theta Criterion derivationally (Bošković 1994, Hornstein 2001, Nunes 2004)—the theta-marked object John in (20a) cannot move from its base-generated patient position [Compl, VP] to the position of external argument [Spec,*v*P] to get theta-marked again: (20a) cannot be derived as in (23) with the interpretation of (20b), or any other. In other words, objects may not move to subject-position within the thematically relevant part of the derivation, argument structure within *v*P.

(23) #[$_{vP}$ John V^0 [$_{VP}$ likes-V^0 ~~John~~]]

As noted for (14) above, the same hypothetical derivation was used by Bošković (1994) to support the ban on Compl-to-Spec movement; however, Bošković assumed a bare VP to host internal and external arguments alike, which, in the light of Larson's (1988) VP-shell or Hale and Keyser's (1993) light-verb approaches, is probably not the most accurate analysis for verbal argument structure. Once 'blown **(p. 272)** up' to include (at least) two separate projections, none of the versions of the CCL discussed so far would help in ruling out (14a)/(20a).

Likewise, employing a Pollock-inspired analysis of agreement projections in the split-Infl part of the part of clause structure below CP and above VP (Pollock 1989, Belletti 1990, Chomsky 1991), the question arises why the external argument may not be inserted into the derivation with an accusative case feature, move to the accusative-licensing position—such as [Spec, AgrOP] in Checking Theory (Chomsky 1993)—and then move to [Spec, TP] in order to satisfy the EPP; it could be argued that, in the absence of moving to a case-licensing position, the in situ object receives accusative case as some default option: (21a) cannot be derived as in (24) with the interpretation of (21b), or any other. In other words, (21a) might represent an example of the impossibility from moving a maximal projection within a split Infl, or expanded TP.

(24) #[$_{TP}$ him T^0 [$_{AgrOP}$ ~~him~~ AgrO^0 [$_{vP}$ softly [$_{vP}$ ~~him~~ v^0 [$_{VP}$ kissed-V^0 her]]]]]

It goes without saying that in the absence of a fully fledged framework, a 'default case' story for the in situ object seems unlikely. Not focusing on licensing case of *her* in (24a), however (perhaps via Agree or covert movement), the point is that within a single clause, an argument cannot move to two case-licensing positions and check two different cases—one possible motivation for the hypothetical movement of him in (24); other illicit instances to illustrate this point can be imagined.

Lastly, considering (22), we see that it is not possible to topicalize a *wh*-phrase within the same clause. Assuming a split Comp-layer (see Rizzi 1997 and much subsequent work[15])—where Foc(us)P would license interrogative *wh*-elements and Top(ic)P would license syntactically displaced topicalized expressions—it looks as though the movement indicated, from one specifier position to the other, is not allowed: (22a) cannot be derived as in (25) with the interpretation of (22b), or any other. Or in other words, phrasal movement within split Comp, an articulated CP, is not permitted.

(25) #[$_{TopP}$ who Top^0 [$_{FocP}$ ~~who~~ Foc^0 [$_{TP}$ Mary T^0 … ~~who~~]]]

(22) also includes a D(iscourse)-linked *wh*-phrase (Pesetsky 1987), *which vegetable*, which is as bad as the bare *wh*-item *what* (or any other). This is relevant as far as the syntax of D-linking goes, which has been likened to topicalization (see Grohmann 2006 for discussion and references). Neither should one worry about the fact that the a- and b-structures in (22) do not form a symmetrical minimal pair as they do in (20) and (21), where (22a) lacks T-to-C movement/*do*-insertion (Lanko Marušič, p.c.): (22b) with comma intonation after the intended *wh*-topic is as bad as (21a):

(26) *What/Which vegetable, does Mary detest?

(p. 273) What I argued in earlier work (Grohmann 2000b, 2003a, 2003b) was that data such as (20)–(22) are instantiations of a common pattern—hence 'paradigmatic structures', each exemplifying one type of anti-locality domain on the clausal level—allowing us to generalize over the relevant derivational steps highlighted in (23)–(25). Note that this need not be the case, of course: Apart from the specific analytical suggestions raised here, other reasons may come to mind to not even group hypothetical constructions such as those in (20)–(22) together in the way summarized above. But if these three types of linguistic structures can—or perhaps even should—be related, this pattern might be captured by a condition like (27), repeated from (1):

(27) Anti-Locality Hypothesis (Grohmann 2003b: 26)
Movement must not be too local.

This is to say that within a domain of clause structure yet to be specified, too-close movement is not allowed, very much in line with previous approaches to anti-locality discussed above. The Anti-Locality Hypothesis could then be presented schematically as in (28), where $|\alpha|$ is some structurally relevant domain within which movement is banned, or more theory-neutrally, within which dependencies cannot be formed.

(28) Ban on Domain-Internal Dependencies (Grohmann 2007: 183)

$$[_{AP|\alpha|}\ XP\ A^0{}_{|\alpha|} \ldots [_{ZP|\alpha|}\ \cancel{XP} \ldots]]$$

In line with the work presented in the previous subsections, α (or rather, $|\alpha|$ in this specific notation) in (28) could perhaps be identified as a maximal projection XP; it would then be captured by the CCL in (13), or its variants in (6) and note 7. However, it is quite obvious that by any standards, most, if not all, of the structures in (23)–(25) go beyond a single maximal projection (the ‘XP’ referred to in the CCL definitions). That is, if cases such as (20)–(22) cannot be analyzed as involving a single projection, which I take to be a reasonable view, the illicit movement steps in (23)–(25) do not fall under the structural bans the current versions of the CCL would capture, be it Spec-to-Adj (29a), Comp-to-Spec (29b), or Comp-to-Adj (29c):

(29)

a.

*[$_{XP}$ YP [$_{XP}$ ~~YP~~ [$_{X'}$ X …]]]	*Ban on Spec-to-Adj* (cf. (5), note 4)

b.

*[$_{XP}$ YP [$_{X'}$ X ~~YP~~]]	*Ban on Comp-to-Spec* (cf. (16))

c.

*[$_{XP}$ YP [$_{XP}$ Spec [$_{X'}$ X ~~YP~~]]]	*Ban on Comp-to-Adj* (not discussed)

In other words, a ban generalized from (23)–(25) would have to be expressed as something along the lines of (31)—and the question is how, since this clearly runs counter to virtually all standard assumptions about movement.

(30) *[$_{XP}$ YP [$_{X'}$ X (…) [$_{ZP}$ ~~YP~~[$_{Z'}$ Z …]]]]

(p. 274) A first approximation of an updated version of the CCL required to capture the cases of anti-locality presented in this subsection could look as follows, where ΠΔ stands for ‘Prolific Domain’—for now, a relevant chunk of phrase structure within which movement is ruled out that, unlike (30), integrates $|\alpha|$ from (28).

(31) Constraint on Chain Links (extended)
Each chain link must be at least of length 1, where a chain link from α to β is of length n iff there are nΠΔs that contain α but not β.

The next section is going to introduce the notion of Prolific Domains and ties in their relevance to anti-locality. As a consequence, a differentiated identification of anti-locality and ways of capturing it vis-à-vis the CCL will be presented first, followed by a discussion of some issues arising for a formulation of the Anti-Locality Hypothesis within Phase Theory (Chomsky 2000a et seq.).

12.3 Prolific domains

What (28) expresses is the idea that a structural configuration can be found that may include a number of projections (such as AP … ZP)—which all, however, share a common ‘contextual index’, indicated by $|\alpha|$—i.e.

something I baptized a Prolific Domain, understood as follows (originally proposed in Grohmann 2000b: 58):

(32) Prolific Domain (Grohmann 2007: 183, adapted from Grohmann 2003b: 75) A Prolific Domain is a contextually defined part of the computational system,
i. which provides the interfaces with the information relevant to the context, and
ii. which consists of internal structure interacting with derivational operations.

Now, if we can identify a group of projections that share some common contextual properties, we might have at our hands a deeper explanation for anti-locality, given that the following consequence can be demonstrated to be viable: each such Prolific Domain directly feeds the interpretive components of the grammar (i.e. LF and PF, viz. (32.i)), yet is structurally complex, allowing for movement operations applied within and beyond (viz. (32.ii)).

I suggest that there exists a natural class of such domains within the clause which make up a tripartition of the clause along the lines laid out here (originally proposed in Grohmann 2000b: 55):

(33) Clausal Tripartition (Grohmann 2003b: 74)
i. *Θ*-Domain: part of derivation where thematic relations are created
ii. *Φ*-Domain: part of derivation where agreement properties are licensed
iii. *Ω*-Domain: part of derivation where discourse information is established

(p. 275) Structures contained within each of these Prolific Domains share what I call a 'common context' (|*α*| from (28), where the specific context is thus |*Θ*|, |*Φ*|, or |*Ω*| for Theta-/*Θ*-Domain, Agreement-/*Φ*-Domain, or Discourse-/*Ω*-Domain, respectively):
(i) *Thematic relations* correspond to the kind of argument structure known to be created within *v*P and projections contained within it (at least VP, but possibly also other projections such as inner aspect or applicatives).
(ii) *Agreement properties* correspond to the kind of structural configurations known to be licensed within TP and projections contained within it (agreement, outer aspect, negation, and possibly other projections in a split Infl).
(iii) *Discourse information* corresponds to the kind of semantically/pragmatically prominent elements, including quantification/scope-relations, known to be established within CP and projections contained within it (such as TopP, FocP, and possibly other projections in a split Comp).

In other words, we yield a clausal tripartition into a finer articulated *v*P, a finer articulated TP, and a finer articulated CP.[16]

In addition, I suggest the Condition on Domain Exclusivity (CDE) to be exactly the kind of 'viable consequence' alluded to above (originally proposed in Grohmann 2000b: 61):

(34) Condition on Domain Exclusivity (after Grohmann 2003b: 78)
An object O in a phrase marker must have an exclusive Address Identification
AI per Prolific Domain ΠΔ unless duplicity yields a drastic effect on the output.
i. An AI of O in a given ΠΔ is an occurrence of O in that ΠΔ at LF.
ii. A drastic effect on the output is a different realization of O at PF.

The idea behind anti-locality thus understood is that Prolific Domains are structural chunks relevant for interface computations, that is, Spell-Out/Transfer (see especially Grohmann 2007), and that any given XP may only have one occurrence within such a Prolific Domain. The CDE, and sub-clause (34.ii) in particular, will be discussed presently, in the next section.

First, let us return to the introduction of this chapter, where I pointed out one of the central questions for a theory, or even a meaningful formulation, of anti-locality: the evaluation metric to be considered. So far we have considered two types (p. 276) of metric, one concerning the domain within which movement is ruled out and one concerning the length to be measured.

Concerning the former, the XP-motivated approach to anti-locality (sections 12.2.1 and 12.2.2), from Saito and Murasugi (1999 [1993]) to Abels (2003), bans all kinds of phrase-internal movement steps: Compl-to-Spec, Spec-to-Spec, Spec-to-Adj, and Compl-to-Adj presumably (not shown here, cf. (29c)). Note that within the 'classic'

Checking Theory of Chomsky (1993, 1995b), these movement steps would arguably be ruled out independently, since no new checking configuration would be achieved by the intended movement. Kayne (2005b) discusses some aspects bearing on this issue; see also n. 18 below. Under such reasoning, where the relevant barred movement steps are ruled out independently, it is on the one hand not clear whether anti-locality should actually enjoy a special status in syntactic theorizing. On the other hand, this would leave us without an account for the additional sets of data discussed in the previous subsection.

The second type of evaluation metric is the domain-motivated approach from that previous sub section (12.2.3), namely my own (Grohmann 2000b et seq.), where chunks of phrase structure larger than a single maximal projection are considered some kind of contextually defined domain (Prolific Domain). Here, these movement steps are not ruled out by virtue of applying within a single phrase, but by being ΠΔ-internal. This approach is in line with classic Checking Theory and thus would make a substantive contribution to identifying a new concept relevant for syntactic theorizing. Yet, as it is formulated, it actually seems to rely exclusively on a licensing mechanism captured by Checking Theory and, unlike extensions of the CCL approach (especially Abels 2003), does not contribute anything to Agree.[17]

The first question to ask, then, is whether we can already choose between either of these two suggestions on the size of the anti-locality domain. It may be worth noting, as Richards and Biberauer (2005) observe, that the Ban on Compl-to-Spec as understood in the CCL-driven approach, especially Abels' take on it, does not prohibit the complement of a (phase) head to raise into its specifier in every single case; it only does so if followed by feature-checking/valuation in the traditional sense, discussed at length by Abels. But what if the feature to be licensed by Compl-to-Spec movement is of a different nature, one that could not have been licensed prior to such movement? Richards and Biberauer suggest exactly this (see also (p. 277) Biberauer and Richards 2006: 63,11. 9) and propose, in the wake of Alexiadou and Anagnostopoulou's (1998) opening of the discussion on EPP-licensing cross-linguistically, that the EPP of T may in some language be satisfied by raising *v*P into its specifier position.[18] Arguably, whatever the ‘classic EPP’ is, this requirement seems to be met only by establishing a Spec-Head configuration and no other feature-checking or valuation. For the domain-motivated approach, this question never arises as (the specifier of) T is in a different Prolific Domain from *v*P to begin with (see the discussion around (33) above).

Concerning the latter type of metric, we also have a two-way split, namely the CCL-driven measurement in terms of chain links (based on the concept ‘(Form) Chain’) and the CDE-driven address identification in terms of occurrences (per Prolific Domain ΠΔ).[19] This is not the right place to engage in an extended critique of chains; Hornstein (2001) and Epstein and Seely (2006), among others, have already done this admirably, questioning the very concept of ‘chain’ as a formal device to be entertained in syntactic theorizing. Such concerns aside, one problem immediately arises for chains in the context of anti-locality and Spell-Out domains (such as phases, as assumed, even intended to be deduced, by Abels). As Terje Lohndal (p.c.) puts it, if one assumes the Merge/Move distinction—understood as External Merge and Internal Merge, respectively—and a cyclic mapping from narrow syntax to the interpretive interfaces, it seems that it becomes hard to use chains, at least chains that span such Spell-Out domains. After all, a chain would have to be formed containing one link in a spelled-out chunk of structure, inaccessible to (p. 278) further operations outside the phase (cf. the PIC in (18) and note 12 above), and one new link in the ‘window of opportunity’, the current derivational phase-chunk. Since Prolific Domains are taken to be Spell-Out units *en bloc*, as per (32.i)—to be expanded presently, in section 12.4.1—this is not a problem that the CDE-driven understanding of anti-locality faces: Once a ΠΔ is formed, that chunk of structure undergoes Spell-Out (or rather, Transfer, in the sense of Grohmann 2007); anti-local movement within will then be ‘detected’ when the domain contains both occurrences, the original copy and the anti-locally moved copy.

It thus might be suggested that a larger domain within which movement is to be banned is advantageous over the more restrictive formulation.[20] The next section is going to develop this idea and discuss core cases of repairing anti-local movement, a concept that could not even be formulated in the strict CCL-driven approaches. After developing my own framework of Prolific Domains in the computation, I will sketch the beginnings of possible synthesis between CCL- and CDE-driven approaches to anti-locality.

12.4 A theory of anti-locality

This section will inspect more carefully and look at more evidence for the existence of Prolific Domains, and by doing so provide additional arguments to be made for anti-locality, in natural language syntax across languages and linguistic structures. This presentation is largely based on my own work and collaborative research efforts, but also on contributions by other colleagues—some even critical, to be examined more closely towards the end, jointly, with an attempt to align the CDE-driven approach to anti-locality with phase-theoretic considerations.

12.4.1 Interface conditions and Spell-Out

As alluded to above, the Condition on Domain Exclusivity (CDE) from (34) is meant, or was at least proposed, as an explanatory way of capturing anti-locality effects in grammar—or rather, preventing them. (32.i) is stated such that Prolific Domains are by definition Spell-Out units that, once formed, send 'the information relevant to the context' (i.e. their content) to the LF- and PF-interface components. In Grohmann (2007), I made a distinction between the operation Transfer and the operation Spell-Out in the sense that one takes a sub-part of the derivation and ships it to PF *cyclically* (where operations like building prosodic domains apply; cf. (p. 279) Grohmann and Putnam 2007), and the other feeds the sensorimotor system once the PF-branch is complete—*uniquely*, i.e. once the computation has assembled all Prolific Domains.

(35) Transfer (Grohmann 2007: 190)
Transfer cyclically sends the structure of each Prolific Domain to PF.
(36) Spell-Out (Grohmann 2007: 190)
Spell-Out phonetically interprets the final PF output once.

The CDE, then, is an interface condition. In fact, the CDE, repeated here for convenience, can be expressed structurally in a simplified manner, as in (38):

(37) Condition on Domain Exclusivity (after Grohmann 2003b: 78)
An object O in a phrase marker must have an exclusive Address Identification AI per Prolific Domain ΠΔ unless duplicity yields a drastic effect on the output.
i. An AI of O in a given ΠΔ is an occurrence of O in that ΠΔ at LF.
ii. A drastic effect on the output is a different realization of O at PF.

(38) The CDE @ PF (Grohmann 2007: 184)
$^*[_{\Pi\Delta}$ XP … ~~XP~~], unless Copy Spell-Out applies to ~~XP~~.

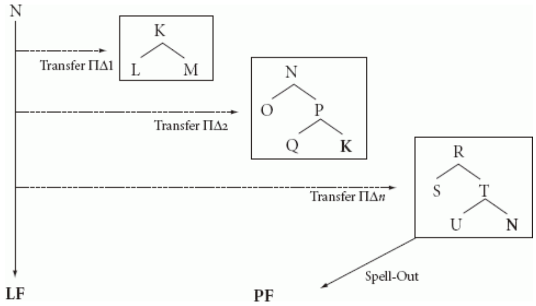
However, the shorthand illustration in (38) does not make explicit reference to the 'different realization' of Copy Spell-Out, which means that Copy Spell-Out should be defined separately. (39) is a first informal approximation:
(39) Copy Spell-Out
In a given Prolific Domain ΠΔ, spell out the lower of two copies of some object O through the insertion of a (minimal feature-matching) grammatical formative.

section 12.4.2 will address Copy Spell-Out in more detail and provide implementations of this concept in anti-local contexts for selected structures in selected languages, but the parenthetical specification will not be discussed here any (cf. Grohmann forthcoming a).

As summarized and embellished with a host of references elsewhere (see e.g. the overview provided in Grohmann 2009), according to Chomsky (1995b: 169ff., 219ff.), the architecture of the grammar comprises those entities which are either motivated by '(virtual) conceptual necessity' (VCN) or follow from what used to be known as 'bare output conditions', now referred to as Interface Conditions (ICs) in the terminology introduced by Chomsky (2004a: 2). One VCN component of the grammar is arguably the lexicon, the collection of lexical items and functional elements in the human mind/brain; the two components which follow from ICs are LF and PF, (linguistic) levels of representation which the relevant language-external systems read off—the conceptual-intentional system (C-I) and the sensorimotor (p. 280) system (SM), called the articulatory-perceptual system in Chomsky (1995b), are clearly VCN if there is anything to the characterization that language is the pairing of sound and meaning.

What the CDE-driven approach to anti-locality through the concepts introduced above assumes is depicted in (39): transfer to the phonological component applies to each Prolific Domain $\Pi\Delta_1$, $\Pi\Delta_2$ … $\Pi\Delta_n$, and the entire 'collection' undergoes Spell-Out (see Grohmann 2007, 2009, and relevant references cited for details).[21]

(40) Architecture of the grammar (adapted from Grohmann 2007: 189)



Understood this way, CDE-driven anti-locality is an IC-motivated constraint on the grammar that demands that each transferred ΠΔ contain at most one occurrence of any given syntactic object. Apart from the condition pertaining to the unique Address Identification in the sense of the CDE in (34)/(37), it does not require any additional stipulations.

12.4.2 Copy Spell-Out and anti-locality

section 12.2.3 presented the primary evidence for XP-extended, domain-motivated anti-locality. With the IC-motivated, CDE-driven approach to anti-locality—a 'theory of anti-locality' perhaps even—on the table, I would now like to showcase some of the data that, in my view, display the interplay of domain- and CDE-driven anti-locality best, namely, anti-local movement 'repaired' by Copy Spell-Out (Grohmann 2000a, b, 2003b).[22] What these data share is a dependency between a contentful, fully projected syntactic object XP and a functional (usually pronominal (p. 281) or otherwise copied/doubled) element ~~XP~~, where both occur within one and the same Prolific Domain ΠΔ at some point in the derivation. By the CDE (37), this should be ruled out: if there exists a real dependency in the sense used here ('movement'), it is anti-local and as such should be clearly ruled out on IC-motivated grounds.

There are two major approaches to such structures (as per note 22 above, for some perhaps more reasonable than for others, but that will not be of concern here):

(i) the 'Big-DP approach' which generates the two elements within a single, large projection and moves one out, stranding the other (see Boeckx 2003 for discussion and references[23]), and
(ii) the 'Spell-Out approach', where the pronominal element is interpreted as the phonetic realization of a copy of XP (going back to at least Pesetsky 1998[24]).

It should be obvious that the theory of anti-locality defended here adopts option (ii), indicated by strikethrough of the lower copy or occurrence of the object O referred to in the CDE: ~~XP~~. The fact that the copy is realized as a pronominal or other functional element from a restricted set derives from common assumptions about economy. It is the smallest linguistic unit that contains a minimal feature-specification of the copy, such as phi-features (dubbed 'grammatical formative' in Hornstein 2001 and adopted in the characterization of Copy Spell-Out in (39); see also Lidz and Idsardi 1997). Starting with Grohmann (2000a), and in several co-authored works, I continuously tried to make the case that the approach under (ii) is more prevalent in grammar than restricted to instances of 'standard' resumption, and call it Copy Spell-Out, where a lower, anti-local copy gets spelled out (for similar options for pronouncing non-top copies and discussing other issues around multiply pronounced copies/occurrences, see e.g. Nunes 2004, Kobele 2006, Boeckx et al. 2007, Kandybowicz 2008; for a larger perspective of such copy modification, see Grohmann forthcoming a).[25]

(p. 282) The paradigmatic case—because it is in a sense the least contended analysis to some extent—is so-called contrastive left dislocation, as found in Germanic, which is to be distinguished from the (at first glance) similar topicalization and hanging topic constructions (which show similar interpretive effects in English, not discussed here). The following is a brief, very rough guide to contrastive left dislocation in German, and relevant extensions across structures, along these lines (Grohmann 2000a et seq.), where the structure in (41b) entertained for the datum in (41a) goes back several decades in different guises:[26]

(41)

a.

[Seinen$_i$ Vater],	den	Mag	jeder$_i$	Junge.
his.ACC father	RP.ACC	Likes	every	boy
‘His father, every boy likes.’				

b. [$_{CP}$ seinen Vater C^0 [$_{TopP}$ den mag-Top^0 [$_{TP}$ jeder Junge T^0 …]]]

An alternative derivation takes the demonstrative *den*, glossed as resumptive pronoun (RP), to be a grammatical formative that, following (39), surfaces as the Copy Spell-Out of the anti-locally moved XP *seinen Vater*, indicated by the circled arrow ‘’; as per the conditions in (37) to (39), Copy Spell-Out applies to rescue the otherwise illicit movement within a single ΠΔ, here the *Ω*-Domain:

(42) [$_{CP}$ seinen Vater C^0 [$_{TopP}$ ~~seinen Vater~~ den mag-Top^0 [$_{TP}$ jeder Junge T^0 …]]]

This analysis allows us to capture all the interesting properties of this type of left dislocation as compared with other constructions with comparable interpretation, such as topicalization and hanging topic left dislocation, that have been discussed for many years (for extensive discussion and references, see Grohmann 2003b: ch. 4, but also e.g. Salzmann 2006 for significant critical remarks).

A similar line of analysis has been proposed for VP-topicalization in Hungarian (Lipták and Vicente 2009). For other left-peripheral phenomena, the CDE in the *Ω*-Domain has also been appealed to (with or without Copy Spell-Out). These include related constructions, such as emphatic topicalization in Bavarian German (Mayr and Reitbauer 2004, 2005), where a topic cannot undergo left dislocation in the embedded clause but has to move to the matrix in order to avoid anti-local movement, and embedded left dislocation structures in Southern Dutch (Tem-merman 2008), to which the Copy Spell-Out repair structure may reasonably be extended.

(p. 283) Grohmann and Nevins (2005) identified pejorative *shm*-reduplication in English as another relevant phenomenon, in which a (discourse) topic moves anti-locally in order to lend a pejorative interpretation to the linguistic expression:

(43) [$_{PejP}$ Binding Theory Pej^0 [$_{TopP}$ ~~Binding Theory~~ Shminding Theory Top^0
[$_{TP}$ we already have the theory of movement]]]

In addition, several studies on the left periphery in Bantu benefited from anti-locality considerations, even if with slight differences from the system presented here. For example, Cheng (2006) extends the applicability of Copy Spell-Out in interesting ways to Bantu relatives (see also Henderson 2007) and Schneider-Zioga (2007) finds anti-agreement in Bantu (Kinande) to relate to domain-motivated (anti-)locality.

These structures all concern anti-locality (possibly followed by Copy Spell-Out) in the *Ω*-domain. At the other end of the spectrum, we are also on a sound footing when we look closer at the *Θ*-domain. Taking my cue from Hornstein's (2001) revival of an old idea to transformationally relate antecedent and local anaphor,[27] something like a derivational history for (20a) sketched in (23) with the interpretation of (22b) can be extended after all—only, the lower copy spells out to satisfy the CDE:

(44) [$_{TP}$ John T^0 [$_{vP}$ John likes-*v*^{0} [$_{VP}$ t$_v$ ~~John~~ himself]]]

Other instances of *Θ*Δ-internal Copy Spell-Out are discussed in Grohmann (2003b).

What makes the tripartition of phrase structure into Prolific Domains doubly attractive, in my eyes (though not necessarily from a phase-theoretic perspective), is that evidence for the existence of the same Prolific Domains can be found within the nominal layer.[28] Starting with Grohmann and Haegeman's (2003) implementation of a CDE-driven account of prenominal possessive doubling across Germanic varieties, evidence accumulates that once again, the underlying assumptions are indeed more widespread.[29] CDE-driven anti-locality effects have also been successfully (p. 284) used to deal with demonstrative doubling in Modern Greek (Grohmann and Panagiotidis 2005). An interesting aspect of that analysis is that the notion of ‘syntactic object O’ from the CDE in (37) does not require phonetic content. As (45b) shows, the relevant O triggering Copy Spell-Out may also be a null operator. In addition, a finer inspection of the Greek DP allowed us to distinguish the two possible word orders in (45) on discourse-interpretive grounds. A fronted demonstrative has a strong deictic reading, understood as nominal focalization, whereas the Op-structure is discourse-anaphoric, taken to be an instance of nominal topicalization. Both points come out in the derivations (46a) and (46b) underlying (45a) and (45b), respectively:

(45)

a.

afta	ta	nea	~~afta~~	fenomena

b.

OP	ta	nea	afta	fenomena
	ART	new	these	phenomena
‘these new phenomena’				

(46)

a. [$_{\Omega\Delta}$ afta … ~~afta~~ ta [$_{\Phi\Delta}$ … ~~afta~~ …]] *deictic/focus-movement*

b. [$_{\Omega\Delta}$ *OP* … ~~*OP*~~ ta [$_{\Phi\Delta}$ … afta …]] *anaphoric/topic-movement*

This chapter is meant to provide an overview of anti-locality in grammar rather than full coverage of Copy Spell-Out analyses (for the latter, see the references cited). This also applies to the non-trivial question of which languages may make available which grammatical formative in which structures.

12.4.3 Extending the anti-locality hypothesis

Intuitively, it might be nice—though by no means necessary—if the lower bound of distance that may span a well-formed dependency (‘anti-locality’) and the upper bound (‘standard locality’) could be related. Standard locality is discussed elsewhere,[30] and it concerns much more than just ‘distance’ (islands, for one, which so far still have resisted a satisfactory minimalist formulation). In this sense, the following are just preliminary remarks, meant as food for thought in the same sense (p. 285) as the CDE-driven take on anti-locality extended CLL-driven formulations. In the same vein, the next subsection can be seen as an even more speculative follow-up on this one, all aimed at getting to the root of the status of anti-locality in grammar.

What we have seen with respect to anti-locality is that movement within a defined domain (whether a single XP or extended to Prolific Domains) is banned—but what is the minimal possible movement, and can this somehow be forced to be the optimal one as well? In Grohmann (2003a), I suggested it could be, and called it intra-clausal movement, to be forced to always target a position in the next higher Prolific Domain, expressed by the Intra-Clausal Movement Generalization (see also Grohmann 2003b: 307):[31]

(47) Intra-Clausal Movement Generalization (adapted from Grohmann 2003a: 305)

i. A-movement of arguments proceeds from *Θ*- to *Φ*-Domain.

ii. A′-movement of arguments proceeds from *Θ*- to *Φ*- to *Ω*-Domain.

iii. A′-movement of non-arguments proceeds from *Θ*-/*Φ*- to *Φ*-/*Ω*-Domain.

On a par, the Inter-Clausal Movement Generalization was meant to capture movement across clause boundaries, also in an attempt to come to grips with socalled successive-cyclic movement, which not only appears to be a well-established fact of displacement in natural language but also poses a serious challenge for economy-driven computations of Last Resort within virtually any version of Checking Theory:

(48) Inter-Clausal Movement Generalization (adapted from Grohmann 2003a: 305)

i. Successive-cyclic *Θ*-movement proceeds from *Θ*-Domain to *Θ*-Domain.

ii. Successive-cyclic A-movement proceeds from *Φ*-Domain to *Φ*-Domain.

iii. Successive-cyclic A′-movement proceeds from Ω-Domain to Ω-Domain.

The interplay of these two generalizations might give us a first clue as to how to go about finding a possible unification of locality conditions imposed by UG.[32]

(p. 286) Other than that, further extensions of a clausal tripartition into Prolific Domains, of the dynamic interplay between narrow syntax and interpretive interfaces, and of anti-locality derived through the CDE, followed or not by Copy Spell-Out—or any combination thereof—have been proposed in recent years. In an empirically driven study, Putnam (2007) adopts the general framework and applies it to Germanic clause structure in order to get a handle on scrambling data. The guiding intuition is here that, while scrambling through movement may result in 'free' word order, anti-locality constrains possible displacement operations in this empirical domain as well, and Prolific Domains contribute to the types of displacement observed. In joint work, we then took some of these details and suggested a ΠΔ-compliant, dynamic mechanism for prosodic stress assignment in Germanic (Grohmann and Putnam 2007). It is too early to judge the viability of either approach yet, but initial results look promising.

As does a fresh look at mixed projections (or mixed categories). If the data from English, Dutch, Greek, and Spanish discussed in Panagiotidis and Grohmann (2009) hold more widely, across more languages, an interesting twist on the licensing of such mixed projections (as in nominalizations) emerges. Verbal roots may 'become nominal' (or vice versa) specifically in those places where a Prolific Domain is created. In other words, if one would assume a 'switch'—some functional head that turns something verbal into something nominal (and the other way round)—as argued there, the position of this switch seems to coincide with boundaries of all three ΠΔs. Switch may take a *v*P, a TP, and a CP as its complement (as well as their nominal counterparts)—but not, say, a VP or an AspP, or a TopP.

Let me close this section by saying a couple of things concerning the conceptual underpinnings of Prolific Domains. Holding fast to the minimalist desiderata to only invoke constraints following from ICs, if they are not VCN in human language, anti-locality derived from the CDE seems to be well motivated—in other words, as I hope to have demonstrated, the CDE looks like a reasonable candidate to explain something 'deeper' about anti-locality.[33] This section has raised some interesting (p. 287) possibilities of extending a CDE-driven theory of anti-locality—but it has also highlighted some issues. Both will be elaborated next, where a return to the CLL-driven approach to anti-locality will be discussed within Phase Theory. The young age of anti-locality studies is responsible for the tentative nature of the present discussion, but it may also prompt much-needed further research activities in this area.

12.4.4 Alternative conceptions of anti-locality

Section 12.2 devoted considerable space to the origins of anti-locality considerations in recent syntactic theorizing. The distinction between what I call here XP-and domain-motivated anti-locality on the computational chunk to be evaluated, respectively, along with the technical implementations *qua* a CCL- or a CDE-driven approach, already makes room for maneuvering conceptions of anti-locality—with possibly diverging predictions as well as empirical and theoretical consequences. Some discussion on distinguishing different approaches to (either type of) locality effects can be found in Boeckx (2008a: 111ff.), but much more can, and indeed should, be said.

In addition to these two camps, and the empirical phenomena they cover, so-called 'syntactic OCP' effects may also be considered potential interest for anti-locality studies. Haplology effects in syntax have been investigated for quite some time (see e.g. Neeleman and van de Koot 2006 for an overview and Norvin Richards 2006 for additional references). More recently, Norvin Richards (2006) proposed a distinctness condition that rules out the linearization of <*α*, *α*>, two elements 'too close together'.

Virtually the same idea underlies identity avoidance in the form of a *XX-filter (van Riemsdijk 2008), reporting on a series of works spanning at least 20 years. These go beyond phenomena captured by independent principles proposed in the (pre-) GB-literature, such as the Doubly-Filled Comp Filter (Chomsky and Lasnik 1977) or the Avoid Pronoun Principle (Chomsky 1981a), and deal, among others, with resumption—or rather, reduction in the sense of *XX, where a sequence of XX reduces to a single X—in Swiss German relative clauses.

What makes Richards's cases interesting is that reference is not restricted to phonetically identical elements, but includes adjacency of elements of the ‘same (p. 288) type’ that are somehow ‘too close together’, such as the post-verbal two DPs in (49c) as opposed to the grammatical alternative (49b), taken from N. Richards (2006: 1):

(49)
a. ‘It's cold’, said John.
b. ‘It's cold’, said John to Mary.
c. *‘It's cold’, told John Mary.

From a theoretical perspective, one of the bigger questions concerns a possible formulation of anti-locality in Phase Theory, first proposed in Chomsky (2000a) and captured by the PIC in (18), reformulated in Chomsky (2001), as mentioned in note 12 above, and further explored in subsequent work (Chomsky 2004a, 2005, 2007, 2008a). The CDE-driven anti-locality theory has little to say about such an integration; for an explicit rejection of phases within this framework, see Grohmann (2003b: 296–303).

The CCL-driven version, however, could possibly be integrated. Abels (2003), of course, formulates his Stranding Generalization in phase-theoretically compatible terms (see note 9 above). Still, the details between the various versions of the CCL provided here and the Stranding Generalization differ. An adjusted, phase-theoretic formulation of the CCL might look as follows, where XP is now replaced by the phase PH:

(50) Constraint on Chain Links [adjusted]
Each chain link must be at least of length 1, where a chain link from α to β is of length n iff there are n PHs that dominate α but not β.

The Adjusted CCL is now very similar to the CDE except that it does not give prominence to (a split) T/Infl—which, as briefly mentioned in note 28 above, is arguably the weakest point empirically for the CDE-driven approach (at least on the clausal level). It builds on a domain-motivated evaluation metric for the size of computational chunk within which anti-locality applies. However, since phases are commonly restricted to *v* and C on the clausal level (in addition to D and P, possibly), one obvious difference is the bipartition within Phase Theory as opposed to the tripartition into Prolific Domains. Note, though, that, as Scheer (2009: 36) assesses the situation, for example, the phasehood of ‘TP is also under debate: [W]hile Chomsky (e.g. [2000a]: 106, [2004a]: 124) is explicit on the fact that TP does not qualify as a phase head (because it is not propositional), den Dikken [2007a] points out that according to Chomsky's own criteria, this conclusion is far from being obvious.’[34] And indeed, suggestions have been made to include a larger number of phase heads, such as VoiceP (Baltin 2007) or AspP (Hinterhölzl 2006). So perhaps the jury is still out on the possible success of the Adjusted CCL.

(p. 289) Another difference concerns the edge of phase heads, (possibly more than one) [Spec,*v*P] and [Spec, CP], respectively. The issue that now arises is that in the ΠΔ-based approach to the computation, multiple specifiers are ruled out (Grohmann 2000b, 2001, 2003b). The obvious problem for the Adjusted CCL is the frequently observed movement, in Phase Theory, from a direct object in [Compl, VP] to the phase edge (outer) [Spec,*v*P]—at least needed even in English for overt movement of object *wh*-phrases to abide by the PIC. Perhaps this could be fixed by stipulating (or, with additional work, possibly deriving) that the first-merged, inner specifier of a phase head is part of PH in the sense used in (50), but that later-merged ones are not. This would include the external argument in the relevant evaluation domain of ν for anti-locality, but exclude moved objects, for example. Note that the CDE-driven analysis of local reflexives could still be adopted, where Copy Spell-Out applies as illustrated in (44) above. But what may make a more thorough investigation of the Adjusted CCL, or some alternative, appealing is the important message. The length relevant for calculating the anti-locality domain is defined over ICs or IC-motivated chunks (with ΠΔs or PHs as the relevant units for Spell-Out and/or Transfer).

A very different route of banning anti-internal movement within Phase Theory, possibly more in the sense of Abels (2003), might be to extend it to phrase-internal movement. Starting with Epstein et al. (1998), an interesting line of research takes the computational ‘window of opportunity’ to be even more restricted—basically, to every instance of Merge or, in Muller's (2004b, 2007) approach, to every phrase (where the PIC is replaced by Phrase Balance); for discussion, see e.g. Boeckx (2003, 2008a), Epstein and Seely (2006), den Dikken (2007a), Preminger (2008), Epstein, Seely and Kitahara (Chapter 13 below), and Uriagereka (Chapter 11 above). These works are concerned with the identification of the computationally relevant structural domains.

In addition, there are other explorations of Phase Theory with potential consequences for formulating anti-locality in slightly different terms—whether, as pointed out in section 12.3 above, concerning the domain within which movement is ruled out or the length to be measured. Marušič (2005), for example, alters the effect of phasehood by suggesting non-simultaneous Spell-Out, where a given phase may send its information to one interpretive interface component but not the other. Gallego (2007) proposes the powerful tool of phase sliding, and den Dikken (2007a) argues for other ways of extending phases and the effects on the computation.

12.5 Outlook

Locality conditions *qua* upper bound on the distance (movement) dependencies may span in a given derivation/phrase marker are well established in syntactic theorizing. (p. 290) This chapter provided an overview of the more recent line of investigation pursuing the possibility that there is also a lower bound on distance—anti-locality. In this context, just posing the question should appear justified. What remains to be seen is whether 'standard locality' and 'anti-locality' follow from similar design specifications and, if not, whether they can still be related in any other meaningful way. This remains a task for the future.

What the preceding sections have done is offer an overview of existing work on anti-locality along two dimensions: (un)grammatical phenomena in language that may fall under the rubric 'anti-locality' in general, and ways of formulating something like an anti-locality constraint in grammar. Despite the differences among the approaches reported, they all share the same intuition: syntactic operations adhere to something like the Anti-Locality Hypothesis, that movement must not be too local.

The existing literature on anti-locality can be grouped into two camps, where anti-locality is either understood by some version of the Constraint on Chain Links (CCL) or results from the Condition on Domain Exclusivity (CDE). The former camp was further characterized by the Classic CCL (Saito and Murasugi 1999[1993]), the Modified CCL (Bošković 1994), plus basically either version phrased in a slightly different framework (Abels 2003), all of which capture anti-locality by banning movement within one and the same maximal projection XP. In addition, I offered a formulation in the form of the Extended CCL, subsequently revised further as the Adjusted CCL, to deal with a larger structure within which movement is barred, be it the derivational sub-unit identified here as a Prolific Domain ΠΔ (the Theta-/*Θ*-, Agreement-/*Φ*-, and Discourse-/*Ω*-Domains, i.e. *v*P, TP, and CP on the clausal level) or in terms of Phase Theory (with the phase heads *v* and C on the clausal level).

The latter camp follows the CDE more closely in both aspects concerning the structure over which anti-locality applies (Prolific Domains) and deriving the Anti-Locality Hypothesis on independent grounds (motivated by interface conditions)—my own version, described in most detail in Grohmann (2003b).

Despite some (I hope) appeal and success, a number of issues need to be addressed first and put on a sound footing, as remarked throughout. This said, various interesting research avenues can be envisioned that either follow up on anti-locality studies in grammar from one of the perspectives outlined here (XP-based and CCL-driven or domain-based and CDE-driven, or perhaps a unification) or take the matter even further by zooming in on the nature of the phase edge (with the innermost position possibly differentiated from the rest), the structure of the clause (a radically reduced repertoire of labeled nodes with a vastly refined understanding of standard and anti-local relations), and so on.

Notes:

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(1) Anti-locality will be used in this chapter exclusively for syntactic relations that are too close, as laid out presently. The term was first coined as such and described in detail by Grohmann (2000b), subsequently revised as Grohmann (2003b) and summarized in Grohmann (2003a). This chapter is based on these discussions, enriched by intellectual precursors, related work, and follow-up research presented or at least referenced here.

The notion of anti-locality does not denote, or bear any relation to, McCloskey's (1979) observation (called 'anti-locality'), according to which languages that rely heavily on resumption in relative clause constructions prevent the highest subject position from being lexicalized as a resumptive pronoun, even though Boeckx (2003: 85, in passing) compares the two.

Likewise, the term is not related to an 'impossibility of the existential dependency under clause-mate negation' (Giannakidou 2006: 371) pertaining to pronouns (e.g. Progovac 1994) nor to anti-locality effects observed in processing either (see Phillips 1996 for discussion and further references). Lastly, anti-locality is also not used for *what*-constructions in so-called 'partial movement' or 'scope-marking' constructions, as done by Muller (1997: 276), who says that '[p]artial *wh*-movement is 'anti-local', in the sense that the scope marker and the *wh*-phrase cannot be clause-mates', and Fanselow (2006: 453), who characterizes the structural configuration as one in which 'the [*what*-phrase] cannot appear in the clause in which the real *wh*-phrase originates'.

I am not aware of other technical uses of the term 'anti-locality', but simply state that, if they exist, they may also fall outside the scope of this chapter. This said, however, if future work can establish a connection between anti-locality as described here and one or more of the aforementioned phenomena—or potentially, if not obviously, related issues pertaining to distinctness ('syntactic OCP') put forth by Norvin Richards (2006) or the *XX-filter ('star double-x') of van Riemsdijk (2008) about which I have unfortunately nothing to say here—such work may help sharpen our understanding of too-close relations in grammar.

(2) I frequently use the term 'dependency' rather than 'movement' to at least leave the door open to a more theory-neutral capture of the relevant phenomenon. Suffice it to say that I do not discriminate among 'movement', 'dependencies', the compound 'movement dependencies', or any (parenthesized) version thereof.

(3) But as Volker Struckmeier (p.c.) notes, embedded topicalization in English does seem to be co-dependent on the presence of CP. Topicalizing within an ECM-structure, arguably of label IP/TP, is infelicitous:

((i))
(a.) I want [$_{TP}$ him to clean the car].
(b.) *I want [the car [$_{TP}$ him to clean *t*]].

For this reason, and many others, I will subsequently assume that topicalization targets a CP-related specifier position, such as Top(ic)P under Rizzi's (1997) split-CP approach or its non-cartographic counterpart if desired.

(4) Substituting IP with its current projection TP and employing the present-day notation for movement in terms of copies (represented throughout with strikethrough or, as in (4), angled brackets), rather than traces, yields a more 'modern' representation of (5), perhaps as in (i):

((i)) $^*[_{TP}\ XP_i\ [_{TP}\ \cancel{XP_i}\ T^0\ \ldots]]$

More generally, the ban can be schematically expressed as follows:
((ii)) Ban on Specifier-Adjunction [generalized]
$^*[_{XP}\ YP_i\ [_{XP}\ \cancel{YP_i}\ X^0\ \ldots]]$

(5) Among others, Henry (1995) discusses this construction in Belfast English and proposes an analysis—like Doherty, though with a different implementation (Agbayani 2006: 89, n. 2)—that does not assume *wh*-movement as in regular relative structures. Den Dikken (2005: 694) notes that Appalachian English also has subject contact relatives. None of these varieties are null-subject or *pro*-drop languages, of course.

(6) Observe, for example, that Bošković (1994) employs the CCL to rule out adjunction of a head to its own maximal projection as well as substitution of it into its specifier—those illicit phrase-structural configurations known as self-attachment (Chomsky 1995b: 321). It also remains to be seen how much of Murasugi and Saito's (1995) 'adjunction paradox', not discussed here, relates to the purported phenomenon of anti-locality in grammar and whether it

allows a sharper perspective on it.

Outside the 'Connecticut School' (the cited work by Abels, Bošković, Murasugi, and Saito), another line of research is also often cited in the context of anti-locality, provided by Ishii (1997, 1999) who discusses *that-t* effects. However, as Abels (2003: 132) points out, in essence, 'Ishii suggests that subjects are adjoined to TP (as in Kayne 1994) and that they are thereby in the minimal domain of the embedding complementizer'. In other words, rather than capitalizing on phrase-structural properties, this approach is built around the notion of Minimal Domain (Chomsky 1993: 11, 1995b: 299), within which movement is illicit due to Last Resort (Chomsky 1993, 1995b)—which is essentially Abels' conclusion as well, albeit on different grounds (see also the following discussion in the text and note 9 below). For Ishii (1999), the illicit movement step is from a position adjoined to [Compl, XP] to [Spec, XP].

I will not discuss such cases any further, since they touch on anti-locality only in the very periphery—as far as I can assess, an anti-locality account for one linguistic structure (*that-t* effects) not based on more general, architecturally or theoretically interesting, considerations (other than Minimal Domains and the general condition Last Resort in grammar). But note that Bošković (2005) adopts a similar structure for cases of illicit left-branch extraction involving adjectival modifiers, where the AP is adjoined to the NP-complement of the (phase head) D.

(7) This formulation is slightly different from Murasugi and Saito's (1995) published definition (see Bošković 1997: 27), apart from replacing the original A/B with α/β; Bošković (1994: 261, n. 19) credits Mamoru Saito in his 1993 University of Connecticut class lectures (see also Bošković 1997: 184, n. 27) for what I refer to here as the Original CCL, which was phrased as follows:

((i)) Condition on Chain Links (original) (Bošković 1994: 261)
(a.) Each chain link must be at least of length 1.
(b.) A chain link from α to β is of length n if there are n XPs that cover β but not α.

See Bošković (1994, 1997) for additional discussion on the exact definition; it will not play a role here, although I will return to it later and (at least try to) adapt it in order to capture the Anti-Locality Hypothesis in terms of more current syntactic theorizing.

(8) Pesetsky and Torrego (2001), Kayne (2003, 2005b), and Jo (2004), among others, address Compl-to-Spec movement as well. See also Boeckx (2007: ch. 6, 2008a: ch. 3) for more discussion, including issues concerning anti-locality and Abels' generalization, which can be restated as below (adapted from Boeckx 2007: 105 who reports on Abels' work), where (i.b) is the result of a combination of (i.a) and PIC, the Phase Impenetrability Condition (Chomsky 2000a, 2001; see also note 11 below and section 4.4 for more):

((i)) Given α, the head of a phase
(a.) Always: $^{*}[\alpha\ t]$
(b.) Always: $^{*}[_{\alpha P}\ \beta_i\ [\ \alpha\ t_i]]$

(i.a) bans stranding a phase head, while (i.b) rules out Compl-to-Spec movement within the projection of a phase head, something Abels generalizes to all heads.

Pesetsky and Torrego, developing an idea that goes back to their 1999 LSA Summer Institute course, suggest the Head Movement Generalization in (ii) to rule out movement of TP to [Spec, CP], among other things—i.e. the very same structural configuration captured by the ban on Compl-to-Spec. In other words, this is yet another different way of formulating anti-locality configurations in grammar through a ban on phrasal movement from complement to specifier position.

((ii)) Head Movement Generalization (Pesetsky and Torrego 2001: 363) Suppose a head H attracts a feature of XP as part of a movement operation.
(a.) If XP is the complement of H, copy the head of XP into the local domain of H.
(b.) Otherwise, copy XP into the local domain of H.

(9) In fact, he aims much higher: Abels' primary goal is to present and unravel what he calls the 'Stranding Generalization'—the idea that the complement of a phase head α may not move to the specifier position of α, whereas any material in the complement domain of α must move through the edge of α—thereby capturing

Chomsky's (2000a et seq.) Phase Impenetrability Condition but ‘enforced by general considerations of [l]ocality’ (Abels 2003: 13). Richards points out that this reduces PIC-effects to Relativized Minimality (Rizzi 1990b, Ch. 10 above), casting doubt on Abels' success in ‘deriv[ing] rather than stipulating] the anti-locality condition’ (Richards 2004: 58, fn. 3).

Abels' take on locality includes a particular perspective on anti-locality he expresses through the ‘Anti-Locality Constraint’ (see also (i) of note 8 above)—which again, as far as I can see, is nothing more than a generalized formulation of the Stranding Generalization, but this time it applies to all syntactic heads. I thus leave it open in how far Abels' self-assessment is met in his work: ‘The discussion of these two cases [presumably illicit cases of phrase-internal movement between complement, specifier, and adjunction positions] shows again that movement from complement to specifier within the same phrase is systematically ruled out by the Last Resort condition. Longer movements may be allowed. Similar anti-locality conditions are also assumed in Bošković (1994, 1997); Grohmann (2000b); Murasugi & Saito (1995); Ishii (1997, 1999) [and arguably Saito & Murasugi (1999[1993])]. *Unlike these authors I have tried to derive rather than stipulate the anti-locality condition*. The [a]nti-locality conditions assumed by these authors have various degrees of overlap with the condition derived here, but I will not pursue the matter.’ (Abels 2003: 106; references adapted and emphasis added—KKG)

(10) Abels (2003: 121ff.) shows that TPs are, in principle, mobile—crucially, however, only when they are not embedded under a phase head. Raising structures, for example, can be fronted, suggesting that these are TPs rather than CPs. See Boeckx and Grohmann (2007), but certainly also M. D. Richards (2007), for discussion in the context of Phase Theory (Chomsky 2000a et seq.), which is not the focus of the present chapter.

(11) The original formulation of the HC is given in (i):

((i)) Head Constraint (van Riemsdijk 1978: 160, his (58))
No rule may involve X_i/Xj and Y_i/Yj in the structure
$\ldots$ Xi $\ldots$ $[_{H^n} \ldots [_{H'} \ldots Y_i \ldots H \ldots Y_j \ldots]_{H'} \ldots]_{H^n} \ldots X_j \ldots$
(where H is the phonologically specified (i.e. non-null) head and H^n is the maximal projection of H).

(12) In later work, Chomsky actually proposes a weaker version of the PIC:

((i)) Phase Impenetrability Condition (Chomsky 2001: 14)
The domain of H [H = phase head] is not accessible to operations at ZP [next phase], only H and its edge are accessible to such operations.

I will address phases briefly in section 12.4.4, but for this chapter, neither version of the PIC plays a role; see also e.g. Abels (2003), Müller (2004b), Richards (2004), Gallego (2007).

(13) For example, instances of preposition stranding, originally the main concern of Abels, also going back to van Riemsdijk (1978); see also Koster (1978) and Emonds (1985) as well for pioneering contributions. The recent literature on P-stranding is rich, also with respect to anti-locality; see Bošković (2004) for additional references, and on the acquisition side, see Isobe and Sugisaki (2002) and Sugisaki (2008). (See also notes 6 and 8 above.)

(14) The other aspect of Abels' work, PIC effects, are not discussed here either (and arguably are not captured by the CCL), since it seems to relate more to ‘standard’ locality than too-close relations (cf. note 9 above).

(15) As a historical footnote, it should be pointed out that precursors of the split-Comp approach can already be founded in the context of CP-recursion (see Vikner 1995) and, in terms of a dedicated functional projection within CP, Müller and Sternefeld's (1993) T(opic)P.

(16) See also Platzack (2001) for virtually the same insight (clausal tripartition), but differing details (each feeding a separate interface to the system of thought). Ernst (2002), too, implicitly assumes such a tripartition in classifying adverbial modifiers to relate to either VP or TP or CP, as pointed out to me by Lanko Marušić (p.c.). In a sense, the idea behind such a tripartition of the clause goes, of course, back to at least Chomsky's (1986a) *Barriers*-framework, which originally implemented a structure for the entire clause conforming to X′-bar Theory, replacing the (R)EST S/S′-notation (cf. Chomsky 1965, 1973, Chomsky and Lasnik 1977, among many others; see e.g. Jackendoff 1977 and Stowell 1981).

(17) The anti-locality framework of Grohmann (2003b) was couched in a revised Checking Theory that introduced the notion of 'natural relations' (see also Grohmann 2001), yet it stuck to 'classic' Checking Theory in that licensing of grammatical properties was done in local structural configurations (Spec-Head, Head-Head, Head-Compl). This stands in stark contrast to the long-distance licensing operation Agree, under which a Probe in need of valuing some (uninterpretable) feature scans its c-command domain to find an appropriate goal with a matching (interpretable) feature. The Checking/Agree issue will not be discussed here any further, but see Adger and Svenonius (ch. 2 above) and Bošković (ch. 15 below) for valuable background discussion and references.

(18) Consider also Lee's (2005: 68) footnote 20 for the same observation: 'It is suggested in Abels (2003) that Last Resort categorically disallows movement of the complement of some head to the specifier of that very same head, since the Head-Complement relation is the closest relation which may have all features satisfied in the relation, thus no reason to move a phrase from the complement position to the specifier position of the same head. However, this claim is incorrect. The EPP-feature, which attracts an element to a specifier position, is in fact a Spec-requirement, not a feature of a head, i.e. EPP is not a feature in the technical sense of the word. Thus, as far as Last Resort is concerned, a complement should, in principle, be able to move into its own specifier position.' See also den Dikken (2007a, 2007b: 153), who makes the same point.

In his 'toxic syntax' approach, Preminger (2008) goes one step further. There may be many more instances of Compl-to-Spec movement, namely, every time Compl moving to Spec does not enter any Agree or checking relation with Head (where [Compl-to-Spec, XP] in these cases serves to 'purify' XP). See also note 20 below.

Alternatively, as Željko Bošković (p.c.) points out, Abels' analysis could actually be taken as an argument against Chomsky's non-feature-checking approach to the EPP. This would open up the question again how intermediate movement steps are triggered, one of the difficulties with the classic Checking Theory of Chomsky (1993, 1995b); see Boeckx (2007) for a comprehensive overview of '*what-, where-, when-*, and *why*-questions' of successive cyclicity and intermediate movement steps in particular.

(19) Here again, the first type corresponds to what was called 'XP-motivated' above and the second to 'domain-motivated'. Since an explanation of anti-locality seems to be the more interesting question, I will subsequently refer to the two approaches by their motivating name, i.e. 'CCL-driven' (referring to any version of the Constraint on Chain Links) versus 'CDE-driven' anti-locality (from the Condition on Domain Exclusivity in (34)).

(20) Note that the more restrictive version heavily depends on one's take on phases. Müller (2004b, 2007) provides a framework in which both phases and anti-locality domains are highly restrictive—every phrase is a phase. See also section 12.4.4 below.

(21) The boldfaced symbols are shorthand notations corresponding to the previous 'box' (K = [$_K$ L M] and N=[$_N$ O [$_P$ Q [$_K$ L M]]]). That is to say, what is spelled out at the end is the entire phrase-marker from R to M.

(22) Space does not permit detailed discussion of the data collected and phenomena reported in this subsection and the next, or a justification of the specific analyses applied; the reader is thus referred to the sources listed for a full set of references, discussion, and critical issues raised. Suffice it to state that by amassing all this potential evidence, the particular interpretation favored here receives at least some interesting support. The 'repair' aspect of Copy Spell-Out is picked up most prominently in Grohmann (forthcoming b).

(23) Boeckx's approach is part of a larger framework on chain formation, which follows his Principle of Unambiguous Chains in (i), where a Strong Occurrence is 'a position where a strong/EPP feature is checked' (Salzmann 2006: 285, which (i) is adapted from):

> ((i)) Principle of Unambiguous Chains (after Boeckx 2003: 13)
> A chain may contain at most one Strong Occurrence (the instruction for PF to pronounce α in that context), where an occurrence OCC of α is a sister of α.

But given the additional difficulties concerning chains raised above (see the end of section 12.3 above), this will not be pursued here any further; see also Hornstein (2009).

(24) See also note 27 below.

(25) Building on much research concerning the fate of syntactically created copies in the computation, I develop an explicit differential of several competing strategies, depending on the syntactic context, what I call (aspects of) ‘copy modification’ (Grohmann forthcoming a). In its narrowest understanding, Copy Spell-Out can be conceived as a ‘repair’ strategy (Grohmann forthcoming b).

(26) CP designates the highest clausal projection, which Rizzi (1997) calls ForceP; for the details of this analysis, see Grohmann (2000a, 2003b) and other references mentioned in the text. The same applies for the phenomena subsequently touched on which cannot be discussed in any detail here for reasons of space.

(27) See e.g Lidz and Idsardi (1997) and Kayne (2002) for proposals in the same spirit, and Lees and Klima (1963) for a historical precursor.

(28) The Prolific Domain in the middle is at the same time rather dubious with respect to testing for CDE-rescue effects in terms of Copy Spell-Out: the *Φ*-domain. For some—admittedly, somewhat problematic—attempts to reanalyze clitic left dislocation of the type found in Greek (but also Romance and Arabic), see Grohmann (2003b: ch. 5). However, in relation to nominal structures, there might be many more phenomena that lend themselves rather naturally to a CDE-driven account of anti-locality, including Copy Spell-Out (see the following text for some references).

Another issue that cannot be discussed here concerns Copy Spell-Out with head movement, which was excluded from being relevant for anti-locality (Grohmann 2003b: 80); see e.g. Quinn (2009), Beys (2006), and section 12.4.4 below, where some alternatives will be briefly presented (cf. Nunes 2004, Bošković and Nunes 2007, and Kandybowicz 2007a, b, 2008).

(29) In fact, in an impressively detailed and well argued dissertation, Ticio (2003) provides a lot of strong evidence for the existence of Prolific Domains—and subsequently, domain-motivated anti-locality—in the Spanish DP. The type of evidence comes from all kinds of DP-internal operations and relations, yet does not invoke any instances of ‘repairing’ illicit movement steps by Copy Spell-Out. This makes the study attractive for anyone intrigued by anti-locality phenomena in general but, for whatever reasons, skeptical concerning Copy Spell-Out and/or grammatical formatives in the computation of natural language.

(30) In terms of minimality, see e.g. Rizzi (ch. 10 above). As Željko Bošković (p.c.) points out to me, however, the conflict between standard and anti-locality is particularly obvious if standard locality is Takahashi's (1994) locality, briefly mentioned in section 12.2.1 above (see also Chomsky 2008a). Bošković (1994) already noticed a serious problem for this approach. Once some syntactic object is adjoined to XP, it should keep adjoining to XP, since this is the shortest step. Bošković appealed to anti-locality in order to prevent this from happening (so, in addition to Spec-to-Adj and Compl-to-Spec, this ban on Adj-to-Adj within the same projection could be added to the list of too-close syntactic structures).

(31) But see Bošković (2008c) for the ban that ‘operators in operator-variable chains cannot undergo further operator movement’ (his generalization (i) on p. 250)—something the CDE-driven approach to anti-locality captures for clause-internal movement (e.g. a version of (22a) with a *wh*-phrase touching down twice, to check [Foc]- and [*wh*]-features in separate positions within the same *Ω*-Domain). However, as Bošković (2008c: 269, n. 31) notes, the generalization seems to hold even of attempted movement into a higher clause—something that anti-locality under any understanding presented here cannot capture, as far as I can see. See also Rizzi (2006a) for related discussion.

(32) Following the Inter-Clausal Movement Generalization, Buesa García (forthcoming) makes a strong empirical case for the locality-unifying view in his treatment of subject-gap restrictions in questions in dialects of Castilian Spanish.

In Grohmann (2003a,2003b), I also addressed the issue of sideward movement (see Hornstein 2001, Nunes 2004, and references cited), but a discussion of this would lead us too far afield:

((i)) Sideward Movement Generalization (adapted from Grohmann 2003a: 305)
Weak thesis: Movement targeting *Θ*-Domain is merged next in workspace.
Strong thesis: Movement from *Ω*-to *Θ*-Domain is merged next in workspace.

And Haddad (2007) reformulates (i) in his dissertation on adjunct control as follows:

((ii)) Sideward Movement Generalization [revised] (adapted from Haddad 2007: 180)

If an element X targets a domain α when it undergoes first merge, X targets α when it undergoes sideward movement; a domain α can be a *Θ-, Φ-*, or *Ω*-Domain.

Further exploration of tying in sideward movement with standard and anti-locality issues looks like another exciting project for the future, as would a fresh look at the classic GB-distinction between A- and A′-properties (positions, movement, dependencies), also in tandem with the generalizations on Inter- versus Intra-Clausal Movement. On a related note, see Boeckx (2007) for detailed discussion of intermediate movement steps not triggered by feature-checking.

(33) In a pre-publication version of Medeiros (2008), the author suggests that his concept of C(omputational)-complexity might derive anti-locality—at least the type captured by the CCL banning phrase-internal movement—for free, trivially so. Working out this suggestion, again, has to be left for future research.

It should also be pointed out, however, that anti-locality as conceived within the CDE-driven approach—even when not accompanied by Copy Spell-Out (cf. n. 29 above)—faces some serious challenges as well. A lot of work on applicative constructions across languages seems to suggest that *v*P, even in a phase-based approach, may contain more structure. Lee (2004), Jeong (2006), and Boeckx (2007) in particular provide arguments for necessary movement operations within a *v*P that would crucially involve anti-local movement steps. This challenge remains to be addressed more seriously.

(34) Note that in the most recent versions of Phase Theory, TP *cannot* be a phase, as shown convincingly by M. D. Richards (2007) and adopted by Chomsky (2007). Richards in particular shows how in Phase Theory, TP *should* be able to become a phase—but doesn't (see also Richards 2004). Thanks to Terje Lohndal (p.c.) and Gereon Müller (p.c.) for bringing this point to my attention.

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Derivation(S)

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[–] Abstract and Keywords

This article examines the nature of derivations. Derivation plays a critical role in minimalist inquiry. But what is the nature of syntactic derivation, and specifically of operations? Just how is the form and application of derivational operations determined? And what criteria can be used in formulating the 'right' type of derivation? For the minimalist program, the strong minimalist thesis (SMT) plays a central role in formulating and evaluating derivation. Under SMT, we expect the recursive part of language faculty to be a system that not only satisfies minimal requirements imposed by the interface systems, but does so in accord with principles of efficient computation. Computational efficiency assumes computation, and the computation equipped with Merge goes some great distance in meeting this expectation, both identifying and satisfying the hypothesized third-factor principles (such as binary merger, no-tampering, inclusiveness, minimal search, and phase-based cyclicity).

Keywords: minimalist derivation, strong minimalist thesis, minimalist program, language faculty

13.1 Introduction

13.1.1 Human knowledge of language

The human capacity to acquire knowledge of language and to use it in speaking, understanding, and thinking distinguishes us from all other members of the animal kingdom. What is this uniquely human, virtually defining, capacity? It is a biologically determined, genetically endowed cognitive capacity present in each individual, barring pathology. 'Language study' in this biological sense was introduced by Chomsky in 1955, and has revolutionized the field of 'language'-study as a branch of human bio-cognitive psychology or 'Biolinguistics'.

Seemingly paradoxical is the fact that an individual's knowledge of language is infinite, yet is acquired on the basis of only finite exposure to 'the language' (see below) spoken in the community. That is, we are not like parrots or tape-recorders capable only of mimicking or recording and playing back the sound strings we have been exposed to; rather we can create and understand novel stimuli (p. 292) like this sentence, which you have probably never encountered before, and which was perhaps never before produced in the history of humankind. Thus, knowledge of language is knowledge, by a non-infinite organism, of an infinite domain, acquired on exposure to finite environmentally provided inputs. In addition to being infinite, the knowledge (by virtue of being knowledge) is entirely different from the environmental input to the organism. *Contra* standard presentations, the input to the child is not 'the language' or even 'parts of the language', it is not, for example, 'sentences' or 'noun phrases' or 'morphemes' or 'phonemes' or 'meanings', but rather consists of acoustic disturbances (of a particular kind, to be determined and characterized by a theory of human language acoustic phonetics), literally nothing more than perturbations of molecules in space hitting the eardrum, in contexts. Or, for signed languages, the input is visually perceived hand shapes in motion (again of a particular kind, not those of someone washing their hands). Thus the input itself, the acoustic disturbances or moving hand shapes the child is exposed to, must be somehow separated out by the child and analyzed as linguistic unlike other non-linguistic acoustic disturbances/visual stimuli. Entities defined by linguistic theory as in fact linguistic, thus sentences, noun phrases, morphemes, phonemes, meanings, are part of your unconscious knowledge of language and do not propagate through the air when I speak to you, or when we speak to our young children; e.g. no meaning floats from my lips to your ears, or from a signer's hands to a child's eyes. Thus, we have at least two fundamental bio-psychological mysteries (if not paradoxes) traditionally overlooked but revealed by Chomsky's revolutionary insights and reorientation: (i) How can any organism have knowledge over an infinite domain? and (ii) How can a being have knowledge not present in the input (e.g. knowledge of meanings, words, prefixes, suffixes, syntactic constituents, phonemes, etc.)?

Chomsky not only reveals these questions (not part of the traditional study of languages, e.g. Latin), but he answers them. As for the first question, a (finite) being or entity indeed can have knowledge over an infinite domain. There is no paradox here, but rather the empirical scientific motivation for the assumption that humans have finite, yet recursive symbolic rule systems. For example, a phrase structure rule such as 'X→X (optional)' is itself finite, but since it can re-apply to its own output, it has infinite generative capacity. To determine the nature of such rule systems has been and remains the goal of syntactic research throughout the modern era; rule systems provide the basis for the notion 'derivation', which it is the goal of this chapter to explore. As for the second question: how can an organism develop knowledge which is not in the input? Chomsky 'solves' this problem by identifying it as 'nothing other than' the central problem presented by all cases of biological growth: How is it that, for example, a tadpole develops into a frog, yet I am not inputting frog(ness) into the tadpole? How can an acorn develop into an oak tree, even though I am not giving the acorn an oak tree, but rather soil, water, light, etc.?

(p. 293) Knowledge, like any other property of an organism, develops as a result of an organism's genetically determined properties interacting with environmental inputs of a particular kind. Crucially, what can be an input (perhaps more accurately, 'intake') is in fact determined by the organism's genetic endowment (UV light is not input to my visual system, nor can I use bat-emitted return echoes to construct an exquisitely precise sonar-determined mental map of my surroundings). Thus, for Chomsky, there is a 'mental organ,' a faculty of language in humans. It grows with variation—arguably slight variation, under formal analysis—determined by varied input. Like biological growth in other domains, the input (French 'noises' vs. visual perception of hand shapes in motion) can in fact exert only very limited influence on the outcome. An adult frog's size might depend on food given to the tadpole, but a tadpole has the capacity to develop only into a frog; no variation of the input can induce the tadpole to become a horse. Similarly, humans develop only human language, the differences, though seemingly vast between them, must be in fact minor, a criterion for a successful theory (the quest for unifying generalizations, as sought in any science).

Part of the human capacity for developing language knowledge is, then, the innate capacity to formulate recursive (linguistic) symbolic rules, operating on a predetermined inventory of linguistic entities (phonemes, meanings, etc.) which the child brings to the grammar-growth task and 'superimposes on the input' patently lacking these properties. Thus a syntactic analysis is imposed upon the acoustic disturbances, as is a meaning, even though the input—an acoustic disturbance—contains neither.

13.1.2 What exactly is knowledge of language? A seemingly 'simple' case

No data or evidence in e.g. physics is 'simply explained'. Rather, explanation invariably requires a sufficiently explicit and articulated theory. If we postulate rules and entities upon which the rules operate, we need to specify the form of the rules and the class of entities. How do we go about doing this? Let's begin with a seemingly simple case. What does someone know exactly when someone knows a particular language? We proceed to construct a theory—the course of all rational scientific inquiry. We then hope to understand our theories, which, as a defining characteristic, elude common-sense intuition. Someone who knows 'English' (but not a monolingual Hindi knower) knows something about, for example, an acoustic disturbance of the form:

(1) CATS

But formal analysis leads us to apparent paradoxes. We can ask: What do you know about 'it'? How many entities does the sound CATS consist of? Contradictory (p. 294) answers all seem true: 'There is one thing, the word *cats*'. 'No, what is (or was) here is, as you said, an acoustic disturbance.' 'Yes, but I know that it is a noun not a preposition, hence word-class analysis is imposed.' 'No, there are two things here the word *Cat* and the plural suffix *s*.' 'No, there are four things, the word *cat*, its meaning, the plural suffix and its meaning.' 'Nope, there are five things; you forgot the combined meaning of *cat* and of *s*.' 'No, there are four things here, not five (the sounds C, A, T and an S in that order).' None of these percepts (determined by our knowledge) are in fact present in the continuous non-phonemic acoustic signal, yet all seem roughly true. If not present in the input, nor 'taught', then they must be imposed by the organism. Thus the child comes to the language learning task pre-equipped with an inventory of uniquely linguistic analytical entities; phonemes, words, meanings, and recursive symbolic rules for combining them, and these constructs are analytically imposed on the raw data input and thereby determine the linguistic percept. To overcome the apparent contradictions regarding *cats*, we assume everyone is in fact right—there is no contradiction, but rather human knowledge of language, i.e. my linguistic module, is itself modularized and imposes multiple levels and kinds of analysis.

Cat is indeed a single word: more specifically, I analyze an acoustic disturbance of this sort as a noun and it is also known by me to be a subpart of the noun *cats* (thus words can contain words, suggesting again recursive properties). The morpheme -*s* is known by me to be capable of functioning as the plural suffix, attached only to (a subclass of) nouns. The meaning of the word *cats* is determined by the meaning of its subparts (i.e. the meaning of *cat* and the meaning of -*s*) and the mode of combination that created them. (Although 'I saw a cat and some dogs' contains both *cat* and -*s*, there is no plural meaning *cats* imposed here, because *cat* and -*s* do not 'go together here'—they were not assembled together—no rule applied concatenating *cat* and *s*.) *Cats* means 'greater than one cat'. (We leave aside your knowledge of the confounding generic interpretation, as in 'Cats meow', which you know is not synonymous with 'greater than one cat meows.') *Cats* also consists of four (abstract) sound units, in a specific order.

Returning now to the infinitude problem, if I tell you there is a thing called a 'plog' and ask you for the plural, you do not say 'I don't know, I have never encountered *plog* before, so how can I know anything about its plural form?'[1] Rather you say 'plogs' (and interestingly, you seem to know the plural form even though I didn't tell you what *plog* means). How can you possibly have such knowledge regarding properties (its plural form) of a previously unencountered stimulus?[2] The hypothesis is that you know abstract symbolic algebraic rules that can apply to new data.

(2) If x = count NOUN, then plural = count NOUN + S

(p. 295) Attributing the rule to an individual, however much the attribution of abstract mental rules might offend common sense, as did the postulation of gravitational forces offend Newton himself,[3] explains the experimental result indicating that you do know the plural of *plog*—and hence could readily pluralize any of the infinite number of nonsense syllables I could give you. These plural forms cannot possibly be a memorized infinite list, since lists are by definition finite, but rather are by hypothesis, generated by finite rule, with an infinite domain of application (algebra).

In addition, you know the plural of *plog* is pronounced plogZ while the plural of *cat* is pronounced catS. We therefore confront another paradoxical question. Is the s at the end of *cats* the same as the *z* sound at the end of plogs? More paradoxes: Yes—they are each the plural suffix and are identical in this regard. No—one is an *s*-sound, the other *z*, and *s* ≠ *z*. The solution again leads us to levels, and rules mapping one to the other: a derivation. You know the *s* in *cats* and *z* in *plogs* are the same; each is the plural suffix, thus you have mental representations of the form:

(3)
- **a.** Cats = CAT + PLURAL
- **b.** Plogs = PLOG + PLURAL

Is the plural suffix pronounced *s* or *z*? Neither is correct for all cases, and as just noted we cannot store the plural form for each (one of an infinite number of) possible nouns, but a rule can capture this, explaining the ability to know the plural of an unencountered case. Suppose then that we say the stored finite form of the plural that you know is:

(4) PLURAL suffix = 1/2 s, 1/2 z

This suffix, of course, is abstract i.e. is not pronounceable—i.e. if we are on track, human phonological knowledge is not entirely knowledge of ‘pronounceable things’. Whether the plural suffix surfaces as *s* (cats) or *z* (dogz) is determined by rule, by the voicing feature of the noun-final Consonant (so-called Voicing Assimilation). Everything you seem to know about an acoustic disturbance like *cats* then can be captured in a derivation mapping one representation to another representation by algebraic rule application. All terms appearing in this analysis (noun, voicing, consonant, etc.) refer not to properties of the acoustic disturbance but to innate analytical apparatus imposed upon it (by humans). Nor are rules part of the input, hence their specific form (English rules) and general format (for any human linguistic rules) are also by hypothesis innately determined. Assuming there are such rules, then a new question emerges. Do the rules apply in any particular order, or are they unordered? If ordered, then there exists a specifiable assembly procedure: a derivation.

(p. 296) What could the different assembly procedures possibly be? Here are two, for example:

(A) (1st) Choose CAT from the lexicon, (2nd) interpret its meaning, (3rd) add the plural suffixs/z, (4th) interpret N+s/z (plural) meaning, (5th) apply Voicing Assimilation, e.g. s/z → s.

What is another possible ordering?
(B) (1st) Choose plural suffix s/z, (2nd) add CAT to it yielding CAT + s/z, (3rd) apply Voicing Assimilation, (4th) interpret the meaning of the subpart CAT, (5th) interpret the meaning of the subpart -s, (6th) interpret the meaning of CATS, given the meanings of each subpart.

As concerns (A), we might observe that there is no empirical motivation for its applying semantic interpretation before Voicing Assimilation. Voicing Assimilation does not need to know the meaning in order to apply correctly (as we saw above with plogs). Similarly, in (B), there is no reason to apply the opposite order in which Voicing Assimilation applies before semantics. The semantics (of plurality) seems to be insensitive to the s/z distinction. Assuming that both parsimony and empirical motivation matter to the central goal, in all science, of explanation, the sound and meaning systems are by hypothesis ‘divorced’—in the sense that there are separate phonological and semantic systems operating on disjoint sets of primitives. Another question: Is it odd in (B) to build CATS, and then go back and interpret just CAT, as if CATS had not yet been assembled—even though it had been? (B) exhibits two independent cycles: build X and Y, then interpret X alone and interpret Y alone, then interpret {X + Y}. Is this oddity worth eliminating—by integrating or intertwining structure-building and interpretation of structures thus far built?

Note, in addition, that some orderings are logically impossible, so should not be excluded by specifically Linguistic Laws/Universal Grammar (UG), but rather are rightly excluded by more general factors, in this case, logic; for example, all the orderings that begin with (1st) apply Voicing assimilation between the final consonant of the noun and s/z suffix (an impossibility, before ‘previously’ selecting a noun). It would be wrong to exclude such derivations by specifically linguistic principle. Rather, here, we would appeal to a more general non-linguistic principle. No rule R can operate on X and Y, unless X and Y are present to be operated on by R. Voicing Assimilation cannot (possibly) apply until I have ‘in hand’ at least two consonants that can undergo the rule. Similarly, any order in which CATS is interpreted before CATS is constructed/assembled, is excluded.[4]

Thus, as concerns rules and their possible order of application, we have illustrated a number of questions with our seemingly simple example:

(p. 297) **(5)**

a. Does the order of rule application matter empirically, i.e. do certain orders make the right predictions while others don't?
b. Assuming (with for example Newton and Einstein and countless others) that simplicity matters in depth of explanation, is rule ordering itself ‘simpler’ or more illuminating than unordered-rule approaches?
c. If it is, then among the logically possible orderings, which ones seem to provide the most insight or illumination (and suffice empirically)?
d. If ordering has empirical consequences, are some nonexistent orderings in fact excluded by, hence explained by logic or by overarching, domain-general bio-cognitive or physical principles not specific to the human language faculty (this latter class what Chomsky (2005) would call ‘the 3rd factor’)?

13.1.3 Syntactic derivation

Specific to syntax, Chomsky distinguishes derivational (rule-ordering) procedures from representational (unordered) ones as follows:

> By a ‘derivational’ approach, I mean one that takes the recursive procedure literally, assuming that one (abstract) property of the language faculty is that it forms expressions step-by-step by applying its operations to the pool of features: … it assembles features into lexical items, then forms syntactic objects from these items by recursive operations to yield an LF representation L, at some point applying SPELL-OUT and the operations of PHON to form the PF representation P. By a ‘representational’ approach I mean one that takes the recursive procedure to be nothing more than a convention for enumerating a set of expressions, replaceable without loss, by an explicit *definition* in a standard way. (Chomsky 1998: 126; emphasis added)

As concerns (5b), does ordering vs. non-ordering have any effect on explanation? That of course depends on what constitutes an explanation, always an extremely vexing question. As discussed by Epstein (1999) and Epstein and Seely (2002a), definitions are not explanatory and invariably prompt the question: Why do we find this definition, and not some other definable notion? Syntactic filters, the central government-binding (GB) theoretical construct in the ‘virtually rule-free’ (but not rule-free, argue Epstein and Seely 2002a) GB system, are axiomatic. Vergnaud's Case Filter (though undeniably a major breakthrough) is axiomatic. We invariably confront the question: Why this filter, and not some other equally definable one? One answer is to derive the filter's ‘truth’ by formulating an underlying generative procedure that yields the macro-configurations satisfying or described by the filter. The question ‘Why this filter?’ is then answered as follows. This descriptive generalization obtains precisely because the rules are formulated in such a way that we generate representations satisfying this generalization—this is generative explanation (see J. M. Epstein 1999, for its application to what he calls ‘Generative social science’).

(p. 298) The macroscopic syntactic regularities, macro tree structures or set representations are explained, by appeal to simple local rules the iterated application of which—the derivation—grow the macro phrase-structure tree, or assemble step by step the set (of sets). But, with such rule systems, we conversely confront the question: Why do we find these rules and not other definable rules? If the rules are axiomatic, we have no answer. The minimalist program recognizes and engages this apparent explanatory barrier, namely: if we have a filter-based GB type system, the filters are unexplained, but if we have instead a rule-based system, then we might explain the filters but the rules themselves are unexplained. If there are both rules (assembly) and representations (the objects assembled by the assembly routine)—as we think there must in fact be (i.e. no system can be ‘purely representational, nor purely derivational’, and it's not clear that anyone has ever proposed such a system)[5]—then how can we maximize explanation?

Suppose we have maximally simple rules and maximally simple (so-called ‘bare’) output conditions. The strong minimalist thesis (SMT) regarding the human language faculty can then be succinctly stated as ‘computationally efficient satisfaction of the bare output conditions’ by such maximally simple rules. This arguably maximizes explanation. Why these rules? Because they are maximally simple, neither construction-specific nor language-specific nor incorporating non-explanatory technicalia (as Chomsky notes, see below, the stipulation-free formulations of External and Internal Merge, in essence, come for free). Why these bare output conditions? Because they are maximally simple (e.g. a semantic representation contains only semantic entities, which follows from a more general, not specifically linguistic, law that e.g. a color representation specifies only color information, not e.g. shape). In addition, not only is the format of the rules themselves maximally simple, but the rule applications (i.e. the nature of derivation) is reduced to 3rd factor constraints (e.g. minimal search, no-tampering, binary merger) along with the overarching (inter-modular) hypothesis that rules apply so as to satisfy the bare output conditions. This, we believe, extends Chomsky's ‘language faculty as a mental organ’-hypothesis to embrace also inter-organ function; i.e. the narrow syntax (NS) interacts with other systems (conceptual-intentional and sensorimotor) and operates in order to produce outputs that are interpretable inputs to these NS-external systems. This approach includes not only formal specifications of single organ properties (e.g. the anatomy of the narrow syntax) but also inter-organ function or physiology.[6]

(p. 299) 13.2 Derivations and the strong minimalist thesis

Derivation plays a critical role in minimalist inquiry, as outlined above. But what is the nature of syntactic derivation, and specifically of operations? Just how is the form and application of derivational operations determined? And what criteria can be used in formulating the 'right' type of derivation? For the minimalist program, the strong minimalist thesis (SMT) plays a central role in formulating and evaluating derivation.

13.2.1 Basic statement of SMT: syntax looking outward

The SMT, a defining tenet of the minimalist program, can be characterized as follows:

> … to what extent is the human faculty of language FL an optimal solution to minimal design specifications, conditions that must be satisfied for language to be usable at all? We may think of these specifications as 'legibility conditions': for each language L (a state of FL), the expressions generated by L must be 'legible' to systems that access these objects at the interface between FL and external systems—external to FL, internal to the person.
>
> *The strongest minimalist thesis SMT would hold that language is an optimal solution to such conditions*. The SMT, or a weaker version, becomes an empirical thesis insofar as we are able to determine interface conditions and to clarify notions of 'good design.' (Chomsky 2001: 1, emphasis added)

The human faculty of language (FL) does not operate in a vacuum. Rather, it produces (syntactic) objects that are delivered to the phonological and semantic components, PHON and SEM respectively, and ultimately to performance systems, sensorimotor (SM) and conceptual-intentional (CI). Crucially then, FL interacts with these external systems. The minimalist program puts a premium on the relation between FL and the external systems, and what emerges is a central role for derivation. Think of FL as an input-output device, and then ask: What is the nature of the input? What is the nature of the output? What is the nature of the mechanisms of FL producing that output from the given input? Throughout, the minimalist approach asks one additional question: And why does FL take the form that it does?

13.2.2 SMT and the nature of derivation: the input

The minimum design requirement is that FL produces objects that the external systems can in fact use; thus, parts of at least some of those objects must be legible (p. 300) to the interfaces. But SMT requires much more than that some objects (or some parts of objects) are legible to external systems:

> Suppose that a super-engineer were given design specifications for language: 'Here are the conditions that FL must satisfy; your task is to design a device that satisfies these conditions in some *optimal* manner (the solution might not be unique).' The question is, how close does language come to such optimal design? (Chomsky 2000a: 92)

The hypothesis is that FL optimally meets conditions imposed on it from outside. An infinite number of mechanisms could produce legible objects. The minimalist hypothesis, however, is that FL is 'perfectly designed' to meet these conditions, and this obviously puts important design constraints on the form and function of the mechanisms of FL producing objects for the interfaces. And this, in turn, determines the nature of derivations.

The atomic units of FL are the arguably ineliminable and irreducible properties of sound and meaning, i.e. linguistic features. If the products of FL are to be usable to the external systems, these products must at least contain some interface-legible features; if there were no legible features at all, then it would be useless to, since 'unreadable' by, the interfaces. Minimalism takes an even stronger view, however: each and every element of the products of FL must be legible to one or the other interface; if *any* feature of the input to the interfaces is illegible to either SM or CI, then the input crashes. This follows from Full Interpretation (FI), a principle carried over from the GB predecessor of minimalism, proposed in Chomsky (1986b). FI requires that every element of a semantic representation and every element of a phonological representation receive an appropriate interpretation; elements cannot simply be disregarded. FI is the 'convergence condition' of more recent minimalist literature; Chomsky (1995b), for example, states that a phonological representation must be constituted entirely of elements that are legitimate to SM; similarly for a semantic representation at CI.

The initial point of the optimization of derivation required by SMT, then, is the postulation of a set of linguistic features each of which is 'usable' by one or the other interface, and a set of features that operations can access to generate expressions. The computational procedure doesn't operate directly on these features, however, since SMT demands ‘optimization’ of the derivation. Interestingly, and perhaps counterintuitively at first glance, relative to the linguistic features of sound and meaning, the reduction of formal complexity involves the establishment of a lexicon. A set of lexical items is created out of the full set of linguistic features. Moreover, the derivation proceeds by making a one-time selection of a lexical array from the lexicon, namely the lexical materials to be used in the derivation at hand. The atomic units of FL are linguistic features; the atomic units of the narrow syntax are presumed to be lexical items. SMT determines that the relation between the narrow syntax and linguistic features be mediated by the lexicon containing lexical items (i.e. assemblages of linguistic features) and that a derivation selects an ‘array’ (p. 301) of lexical items, from which the derivation builds an expression whose destination is the interfaces, where it serves as ‘a set of instructions’ to the language-external systems.

13.2.3 SMT and the nature of derivation: the output

What about the output of FL? The output is an infinite set of syntactic objects. The relation between a lexical array and the interfaces of sound and meaning is mediated by the narrow syntax. SMT hypothesizes that this mediation is carried out in an optimal way. Thus it is not the case that ‘anything goes’ in building syntactic objects. Although there is an infinite set of alternative mechanisms one could posit to ‘cover the data’, the minimalist approach, with SMT at its heart, seeks a far more difficult goal (one of explanation, as in other sciences):

> Tenable or not, the SMT sets an appropriate standard for true explanation: anything that falls short is to that extent descriptive, introducing mechanisms that would not be found in a ‘more perfect’ system satisfying only legibility conditions. (Chomsky 2001: 2)

The questions remain: (i) What does ‘optimal’ mean exactly? and (ii) How is derivation involved? A linguistic expression is not just the pairing of a phonological representation and a semantic representation, each formed by a convergent derivation; the derivation of this pairing must be optimal (Chomsky 1995b). Note first (and crucially for present purposes), if the derivation of an expression must be optimal for SMT to be met, then there must be a derivation (*contra* for example, the letter or at least spirit of the ‘rule-free’ GB theory). Meeting the condition of ‘good design’ determines, in large part, the nature of the mechanisms of the derivation.

Clearly, some mechanism is required to form lexical items into phrases, as in fact there is an infinite set of linguistic expressions (i.e. phrases):

> The simplest such system is based on an operation that takes *n* syntactic objects (SOs) already formed, and constructs from them a new SO. Call the operation Merge. Unbounded Merge or some equivalent (or more complex variant) is unavoidable in a system of hierarchic discrete infinity, so we can assume that it ‘comes free’ in the present context. (Chomsky 2008: 137)

The simplest form of Merge, consistent with SMT, is as an operation that puts two, and no more nor fewer than two, elements into a relation; fewer than two will not create an expression at all (beyond a lexical item itself), and more than two is beyond what is minimally required to create an expression (larger than a single lexical item). In large part, then, the form and the function of this ‘free’ operation, essential for a derivation, is determined by SMT.

(p. 302) 13.2.4 Minimal derivations

So far, a derivation is a partially ordered set of applications of the simple, binary operation Merge. Another property of I-language at the center of linguistic research since the inception of generative grammar is displacement: a single category, in one position for meaning but pronounced in a different position. The operation of movement has long been the primary mechanism by which displacement phenomena have been captured, and the seeming paradox that it raises (a single element in two places at once) resolved.

The initial merger of two elements is generally considered unconstrained. But a particularly intriguing domain of derivation involves the merger of categories X and Y, where X is an element properly contained within Y, referred to as Internal Merge (IM). IM is involved in such classic instances of movement as passive (e.g. *John was arrested*) and *wh*-movement (e.g. *Who did Sue arrest?*). Research questions regarding IM include (i) What are the constraints on IM? (ii) What features factor into IM, and why? (iii) What are the structural consequences of IM? This is a robust area of current research, and current literature provides various answers. One of the enduring hypotheses regarding IM is that it obeys the principles of 'least effort' seeking to eliminate anything unnecessary—superfluous elements in representations, and doing so without any superfluous steps in derivations—in short, simplify and reduce formal complexity as much as possible (see Chomsky 1991).

Derivation is a fundamental property of current minimalism. At the most general level, minimalism is nothing more than a willingness to seek scientific explanation, not just description. A research goal is to determine the exact nature of derivation, at a deep, explanatory level, and not to describe data using whatever stipulated technicalia seem to do the job, nor are convenient (non-explanatory) definitions over representations allowed even if they seem to do the trick. Rather we try to explain the properties of FL by deducing whatever defined stipulations promise empirical coverage.[7] For example, structure-building is the core of a derivation. Merge is thus 'required.' SMT determines that we posit as little beyond Merge as possible, while still maintaining 'empirical coverage'. Epstein et al. (1998) and Epstein (1999), for example, seek compliance with SMT in trying to maximize the explanatory impact of Merge and its iterative application, by deducing from it the relation of c-command. C-command is not stipulatively defined over representation, but instead falls out of the derivation, i.e. the independently necessary, and maximally simple iterative application of the structure-building operation, Merge. Research on the nature of derivation within the minimalist program then goes hand in hand with criteria for evaluation. This in turn entails taking SMT and various 3rd factor constraints (common to all sciences) very seriously, and it raises the bar for what counts as an explanatory analysis.

(p. 303) 13.3 The mechanisms of minimalist derivation

13.3.1 Introduction

As discussed in the preceding sections, generative grammar has long recognized that the human faculty of language FL is recursive, and the minimalist program has advanced a derivational approach under SMT by taking the recursive part of FL to be 'a step-by-step procedure for constructing Exp[ression]s, suggesting that this is how things work as a real property of the brain, not temporally but as part of its structural design' (Chomsky 2000a: 98). In this section, couched within the framework of Chomsky (2007, 2008a), we ask: (i) What is the minimum machinery specific to this recursive system (the genetic endowment for FL, the topic of UG)? and (ii) How does such machinery generate a stepwise derivation, in accord with the principles of efficient computation (the subcategory of what Chomsky (2005) calls 'the 3rd factor' that enters into the design of any biological system)? The answers we give constitute current (minimalist) hypotheses regarding the structure and operation of the human FL.

13.3.2 Minimum machinery

The recursive system allows FL to yield a discrete infinity of structured expressions, and 'the simplest such system is based on an operation that takes *n* syntactic objects (SOs) already formed, and constructs from them a new SO' (Chomsky 2008a: 137). This elementary operation is called Merge. SOs not constructed by Merge are lexical items LIs (= heads), provided by the lexicon. For an LI to be able to permit Merge, it must have some 'mergeability' property. This property is called the edge feature EF of the LI. In the simplest case, this EF either always deletes when used or never deletes. The empirical facts suggest the latter case, which allows an LI to have both a complement (a result of first Merge) and a specifier (a result of second Merge). As SO retains EF, Merge can iterate (in principle) without limit (= unbounded Merge). The minimum machinery, therefore, includes (at least) Merge and (mergeable) LIs each bearing an undeletable EF.

13.3.3 Computational efficiency

Under SMT, the minimum machinery equipped with Merge generates a derivation, in accord with the principles of efficient computation. Let us then ask what these principles are (or might be) and how they operate in this Merge-based system.

(p. 304) 13.3.3.1 Merge and its applications

Recall the operation Merge. In the simplest case, Merge takes two SOs already formed, and constructs from them a new SO. The limitation 'n = 2' yields Kayne's (1981b) unambiguous paths—the binary structures that Chomsky's (2000a) minimal search and Kayne's (1995) LCA-based linearization operate on.[8] Unless shown otherwise, we assume this limitation (= the simplest binary operation) to be on the right track. Another principle of efficient computation is the no-tampering condition NTC: 'Merge of X and Y leaves the two SOs unchanged' (Chomsky 2008a: 138). Intuitively, Merge of X and Y does not alter X or Y, but places the two SOs in a set. That is, Merge of X and Y results in 'syntactic extension' (forming a new SO = {X, Y}), not 'syntactic infixation' (embedding X within Y, for example). Thus, Merge invariably applies to the edge,[9] and the effects of Chomsky's (1993) extension condition (largely) follow from NTC.[10] We also assume the inclusiveness condition: 'no new objects are added in the course of computation apart from rearrangements of lexical properties' (Chomsky 1995b: 228). It is a natural principle of efficient computation, which eliminates bar levels, traces, indices, and any similar non-explanatory encoding technicalia introduced by NS. Under these three conditions, NS takes two SOs X, Y (keeping to the simplest binary operation) and merges X and Y to form a new SO, leaving them unchanged (satisfying NTC) and adding no new features to them (satisfying the inclusiveness condition).

Suppose X is merged to Y (introducing the asymmetry only for expository purposes). Then, either X originates external to Y, call it External Merge (EM), or X originates internal to Y, call it Internal Merge (IM). Under NTC, IM yields two copies of X: one external to Y and the other within Y (as in [X [$_Y$ … X …]]). There is no need to stipulate a rule of formation of copies (or remerge), and Chomsky's (1993) copy theory of movement follows from 'just IM applying in the optimal way, satisfying NTC' (Chomsky 2007: 10).

How does Merge get access to SOs? In the simplest case, only the label (i.e. head) of the full SO—either the 'root' SO thus far constructed or the 'atomic' SO (= LI) not yet merged—can be accessed to drive further operations. If X and Y are two separate full SOs, then their labels x and y can be accessed with minimal search. But if X is internal to Y (where Y is the 'root' SO thus far constructed), then the accessed label y of Y carries out the task of finding X; specifically, y probes into the complement of y (= the smallest searchable domain of y) and finds X as a goal of the probe.[11] We assume this probe-goal analysis to be part of minimal search.

(p. 305) Minimal search reduces operative complexity by infinitely restricting the searchable domain of the probe to just its complement domain. But complements can (in principle) be unbounded due to recursion. A further restriction (see below) rendering the search domain not only finite but quite small is then implemented to limit the smallest searchable domain to a more localized sub-domain of the complement domain of the probe—this is the general property of strict cyclicity.

13.3.3.2 Phase-based cyclicity

At the advent of the minimalist program, postulation of linguistic levels beyond conceptual necessity was taken to be a departure from SMT. There were two linguistic levels assumed to be indispensable: the levels accessed by the two distinct interface systems: sensorimotor (SM) and conceptual-intentional (CI). In versions of EST/Y-model, however, five linguistic levels had been postulated along with five separate cycles (taking LF to be the output of narrow-syntactic operations and the input of the mapping to CI, as originally defined in EST). In the past two decades, the multiplicity of levels has been subject to a minimalist critique,[12] and our conception of the architecture of FL has undergone a series of changes with some remarkable results. Chomsky writes:

> Optimally, there should be only a single cycle of operations. EST postulated five separate cycles: X-bar theory projecting D-structure, overt operations yielding S-structure, covert operations yielding LF, and compositional mappings to the SM and CI interfaces. With the elimination of D- and S-structure, what remains are three cycles: the narrow-syntactic operation Merge (now with overt and covert operations intermingled), and the mappings to the interfaces. As noted earlier, optimal computation requires some version of strict cyclicity. That will follow if at certain stages of generation by repeated Merge, the syntactic object constructed [or some subpart of it, SDE, HK, TDS] is sent to the two interfaces by an operation Transfer, and what has been transferred is no longer accessible to later mappings to the interfaces (the phase-impenetrability condition PIC). Call such stages *phases*. Optimally, they should be the same for both subcases of Transfer, so until shown otherwise, we assume so (the mapping to the SM interface is sometimes called 'Spell-Out'). LF is now eliminated, and there is only a single cycle of operations. The cyclic character of the mappings to the interfaces is largely captured, but not completely: there may be—and almost certainly are—phase-internal compositional operations within the mappings to the interfaces.

(Chomsky 2007: 16)

There is only a single cycle of operations in NS: one cycle per phase, where CP and v*P each count as a phase (the smallest possible working domain).[13] As NS completes each phase, Transfer reduces the phase-head-complement PHC to PHC* (p. 306) by deleting all CI-offending features (such as unvalued features and phonological features) from PHC. Then, NS sends the PHC* (lacking CI-offending features) to the semantic component SEM. The PHC itself (with those CI-offending features to be replaced by some phonetic features each receiving in principle some interpretation at SM) is sent to the phonological component PHON.[14] The subsequent mappings to the CI and SM interfaces by SEM and PHON proceed in parallel, and the phase-impenetrability condition PIC makes the 'transferred' PHC inaccessible to any syntactic operations in later phases. Intuitively, PIC explains 'syntactic inertness' and locality as deeply as possible by simply saying that the transferred PHC is gone (from the working domain of NS). Thus, no minimal search can probe into the complement of any earlier phase-head, predicting that there is no interphasal agreement.[15] Although there are still phase-internal compositional operations within the mappings to the CI and SM interfaces by SEM and PHON, phase-based cyclicity has contributed significantly to the reduction of computational load by restricting the working cycle to the size of the phase (minus the PHC of any lower phase).[16]

13.3.3.3 Phase-level operations

Every time NS reaches a phase level—where the values of uninterpretable features (such as structural case and redundant verbal agreement) can be determined by context—it executes a series of operations; one such operation is called Agree. Chomsky (2007: 18) argues: 'the simplest assumption is that they [uninterpretable features] are unvalued in the lexicon, thus properly distinguished from interpretable features, and assigned their values in syntactic configurations, hence necessarily by probe-goal relations.' Specifically, a phi-probing head H and a nominal goal N (the latter bearing inherently valued phi and unvalued Case) match in feature-type (namely phi), and they undergo Agree, which values phi on H and Case on N. To be visible for Agree, SOs must be active, bearing at least one unvalued feature (e.g. if structural Case of N has been valued, then N is 'frozen in place', unable to implement an operation).[17] Chomsky (2008a: 150) then extends this activity condition to syntactic operations generally (in effect, strengthening the principle of (p. 307) Last Resort).[18] Under this generalized activity condition, once N is assigned a case value, it becomes invisible for operations such as Agree and Merge; hence, it no longer participates in valuation or movement.[19]

Prior to Agree, such unvalued features are (by definition) offending features at the two interfaces; hence, they cannot be transferred before they are valued. Once valued, however, they may yield a phonetic interpretation (e.g. present singular 'be' = IS, present plural 'be' = ARE), but they will never yield a semantic interpretation (i.e. IS and ARE are otherwise synonymous). Thus, even if they are valued, they must be deleted when transferred to SEM. But such deletion cannot take place after they are valued; since once valued, they are (by definition) indistinguishable from those interpretable features with which they agreed. Thus, Chomsky (2007: 19) concludes that they must be valued at the phase level where they are transferred, and such derivationally valued features are deleted when transferred to SEM, but they remain intact when transferred to PHON.[20]

Still keeping to the phase level, T exhibits a phonetic reflex of redundant (syntactic) agreement (vacuously in some cases) if and only if T is selected by C. With this observation, Chomsky (2007: 20) proposes T's inheritance of such agreement features (and possibly some other inflectional features) from C, and he assigns this feature-transmitting property to phase heads generally: C and v*.[21] Furthermore, N. Richards (2007) deduces that feature transmission exists and precedes (p. 308) probe-goal agreement; otherwise, a directly agreeing phase-head (PH) retaining its phi features (and occupying the phase-edge) would come to bear derivationally valued features that would (by hypothesis) induce crash at the next phase level (given that PIC makes inaccessible any historical record of their 'previously unvalued' status). If probe-goal agreement is accompanied by IM, then a goal of the probe moves to the specifier of the probing head with which it agreed.[22] A-movement is then derivationally defined as IM contingent on probe by uninterpretable inflectional features, and A'-movement as IM driven solely by the EF of PH.[23,24]

13.3.4 Sample derivations

With this much as background, let us examine how exactly the Merge-based system generates a stepwise derivation for a transitive *wh*-interrogative like (6):

(6) Who saw John?

In the first phase cycle, repeated EM has constructed the v*P phase as follows:

(7)

a. EM merges V(see) to NP(John), forming [$_{VP}$ see John].

b. EM merges v* to [$_{VP}$ see John], forming [$_{v^*P}$ v* [$_{VP}$ see John]].

c. EM merges NP(who) to [$_{v^*P}$ v* [$_{VP}$ see John]], forming [$_{v^*P}$ who [$_{v^*}$ v* [$_{VP}$ see John]]].

In each application of EM, NS takes two full SOs (keeping to the simplest binary operation) and merges them to form a new SO, leaving them unchanged (satisfying NTC) and adding no new features to them (satisfying the inclusiveness condition). Also, NS operates in a 'bottom-up' fashion by the accessibility condition and merges to the edge by NTC. At the v*P phase level (constructed by EM(7c)), only the label v* of v*P can be accessed by the accessibility condition, and the label v* drives the following phase-level operations (where indices are introduced only for expository purposes):

(8)

a. Feature-transmission transmits unvalued phi from v* to V(see).

b. Agree values unvalued phi on V(see) and unvalued Case on NP(John).

c. IM raises NP(John) to Spec-VP, forming [$_{v^*P}$ who [$_{v^*}$ v* [$_{VP}$ John$_2$ [$_{V'}$ see John$_1$]]]].

d. Transfer reduces VP to VP* by deleting all CI-offending features and sends VP* to SEM and VP to PHON.

(p. 309) Under current assumptions, Feature-transmission(8a) exists and precedes Agree(8b) and IM(8c). By the generalized activity condition, Agree(8b) cannot precede IM(8c), but IM(8c) also cannot precede Agree(8b) because this ordering would place NP(John) with unvalued Case out of the searchable domain of the phi-probe V(see), inducing a failure of phi-matching (hence, neither phi-valuation on V(see) nor Case-valuation on NP(John) would take place).[25] Thus, Agree(8b) and IM(8c) apply simultaneously,[26,27] and Transfer(8d) completes the v*P phase cycle.[28]

In the second phase cycle, repeated EM has constructed the CP phase as follows (where '—' is the site of the transferred VP, no longer accessible to any syntactic operations, given PIC):

(9)

a. EM merges T to [$_{v^*P}$ who [$_{v^*}$ v*—]], forming [$_{TP}$ T [$_{v^*P}$ who [$_{v^*}$ v*—]]]

b. EM merges C to [$_{TP}$ T[$_{v^*P}$ who [$_{v^*}$ v*—]]], forming [$_{CP}$ C[$_{TP}$ T[$_{v^*P}$ who [$_{v^*}$ v*—]]]]

Again, in each application of EM, NS takes two full SOs (keeping to the simplest binary operation) and merges them to form a new SO, leaving them unchanged (satisfying NTC) and adding no new features to them (satisfying the inclusiveness condition). At the CP phase level (constructed by EM(9b)), only the label C of CP can be accessed by the accessibility condition, and the label C drives the following phase-level operations:

(10)

a. Feature transmission transmits unvalued phi from C to T.

b. Agree values unvalued phi on T and unvalued Case on NP(who).

c. IM raises NP(who) from Spec-v*P to Spec-TP, forming [$_{CP}$ C [$_{TP}$ who$_2$ [$_{T'}$ T [$_{v^*P}$ who$_1$ [$_{v^*}$ v*—]]]]].

d. IM raises NP(who) from Spec-v*P to Spec-CP, forming [$_{CP}$ who$_3$ [$_{C'}$ C[$_{TP}$ who2 [$_{T'}$ T[$_{v^*P}$ who$_1$ [$_{v^*}$ v*—]]]]]].

e. Transfer reduces CP to CP* by deleting all CI-offending features and sends CP* to SEM and CP to PHON.

Agree(10b) and IM(10c), just like Agree(8b) and IM(8c), carry out valuation and A-movement. At this CP phase level, however, there is an additional application (p. 310) of IM: IM(10d) raises the goal (= NP(who)) from Spec-v*P to the specifier of the EF-probing head (= C). IM(10d) is A′-movement, driven solely by the EF of C. Notice, Agree(10b) cannot precede or follow IM(10c), for the reason already discussed, and the same reasoning applies to the ordering between Agree(10b) and IM(10d). Thus, IM(10d) applies in parallel with Agree(10b) and IM(10c). As is generally assumed under the derivational approach, NS establishes syntactic relations in the course of a derivation, and (*contra* GB theory's unifying representationally defined concept 'government') no syntactic relation can be arbitrarily defined on output structures.[29] It follows, then, that there is a direct relation between who_2 and who_1 (established by IM(10c)), and between who_3 and who_1 (established by IM(10d)), but there is no relation between who_3 and who_2 since no application of IM involved these two positions (i.e. there is no feeding relation between IM(10c) and IM(10d)).[30]

Finally, Transfer(10e) applies to the entire phase CP. Although how Transfer applies at the very final phase (= the root) remains to be spelled out, it is natural to assume, under SMT, that Transfer sends the largest transferable domain to SEM and PHON, and the largest transferable domain may be the phase-head-complement when there is a continuation of the derivation, but the phase itself when there is no continuation of the derivation. With this assumption, Transfer(10e) completes the CP phase cycle, and the derivation of (6) converges.[31]

13.3.5 Summary

Under SMT, we expect the recursive part of FL to be a system that not only satisfies minimal requirements imposed by the interface systems but does so in accord with principles of efficient computation. Computational efficiency assumes computation, and we have seen how the computation equipped with Merge, as advanced by Chomsky (2007, 2008a), goes some great distance in meeting this expectation, both identifying and satisfying the hypothesized (and arguably explanatory) 3rd factor principles (such as binary merger, no-tampering, inclusiveness, minimal search, and phase-based cyclicity).

Notes:

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(1) See Halle (1981) and Berko (1958).

(2) This is Plato's problem; see Chomsky (1986b).

(3) See e.g. Chomsky (2005), and Epstein (2000: 3–5) for a brief explanation of Chomsky's mentalism and his rejection of ‘the mind-body problem’.

(4) More on this below: namely, the search for overarching non-linguistic constraints that apply to human syntactic systems, due not to UG but to more general laws. See Chomsky (1965: 59), cited by Chomsky (2007, 2008a), and see also Chomsky (2005).

(5) See Epstein and Seely (2002a) on derivations within the allegedly ‘rule-free’ GB theory.

(6) As is standard in the biological sciences, physiological function to some extent explains anatomical properties of the individual interacting organs (e.g. the pump-like properties of the heart are to some extent explained by considering the larger circulatory system within which this single organ performs a function). See Hempel (1965[1959]) regarding the form of such explanation, and Epstein (2007b) regarding its use within the SMT.

(7) See Epstein and Seely (2002a) and the citations.

(8) See also Barss and Lasnik (1986), Hale and Keyser (1993), and Larson (1988, 1990).

(9) For discussion of other interpretations of ‘Merge to the edge’ (including local Merge), see Chomsky (2000a) and Richards (1997).

(10) See Lasnik (2006) on the history of formulations of the cycle condition.

(11) See Epstein et al. (1998) for an explanation of the choice of complement (over e.g. Spec) as the sole search domain of the head.

(12) See e.g. Epstein et al. (1998), Epstein and Seely (2002a, b, 2006).

(13) It is an active area of research to determine what are (and are not) phases and (of course) to explain why

(see Epstein 2007a for recent discussion). Whether DP should count as a phase is an open question.

(14) For discussion of the operation Transfer and its application, see Chomsky (2004a). See also Chomsky (1998) for discussion of assignment of phonetic features to a bundle of formal features (including CI-offending features).

(15) It has been pointed out, however, that there seem to be cases of probe into a phase that has already been passed, as in Icelandic quirky case, with T agreeing in number with the embedded nominative nominal (e.g. me(dat) thought(pl) [t_{me} [they(pl, nom) be industrious]]). These properties remain to be explained. For relevant discussion, see Chomsky (2001) and Sigurðsson (1996).

(16) For discussion of the notion phase and its comparison with the notion barrier (Chomsky 1986a), see Boeckx and Grohmann (2007).

(17) The activity condition restricts the class of SOs eligible for Agree, thereby contributing to efficient computation in some natural sense, but it has been argued that it can be eliminated. See Nevins (2005) for critical discussion.

(18) Chomsky (2008a: 150) notes '[the edge-feature] EF of C cannot extract the PP complement from within SPEC-T: if it could, the subject-condition effects [exhibited by (i) and (ii)] would be obviated.'

((i))
(a.) it was the CAR (not the TRUCK) of which [they found the (driver, picture)]
(b.) of which car did [they find the (driver, picture)]?

((ii))
(a.) *it was the CAR (not the TRUCK) of which [the (driver, picture) caused a scandal]
(b.) *of which car did [the (driver, picture) cause a scandal]?

He continues '[i]t must be, then, that the SPEC-T position is impenetrable to EF, and a far more natural principle would be that it is simply invisible to EF.' The activity condition is then generalized to capture the observed subject-condition effects. For a different attempt to provide a derivational account of this invisibility, see Epstein et al. (2008).

(19) If the generalized activity condition does not apply to matching, such a nominal (with valued case) may still match in phi, blocking any further searching by a phi-probe (i.e. once the probe matches in phi, it cannot search any deeper). This analysis receives support from Icelandic expletive constructions, in which dative NP blocks further search by phi-probe of matrix T (e.g. *expletive seem(pl) [some man(dat) [the-horses(pl, nom) be slow]]). For relevant discussion, see Chomsky (2008a) and Holmberg and Hróarsdóttir (2003).

(20) Epstein and Seely (2002b) suggest that Chomsky's analysis, under which Transfer applies neither before nor after valuation converges, leads to the hypothesis that Transfer operates derivationally, more specifically, inside the application of a single rule application, with Transfer thereby 'seeing' the value change from minus (in the structural description) to plus (in the structural change). Another solution to this problem is to assume that all phase-internal operations apply simultaneously, hence it is as if there is just one rule application within a phase, and Transfer can see any unvalued feature change to valued (its vision spanning the single rule application) and thereby knows to spell out the just valued feature.

(21) In effect, feature transmission renders an embedded head H accessible to NS by transmitting a feature from the accessed label of the phase head to H.

(22) What forces IM to raise the goal to the specifier of the phi-probing head—the reside of EPP—is still an open question.

(23) For some potentially interesting consequences of these featural-derivational characterizations of movement types, as compared to previous representational definitions, see Obata and Epstein (2008).

(24) The operation Agree would be dispensable if there were no unvalued features. But the empirical fact suggests that they do exist. Under SMT, then, we seek some functional justification for the presence of such features (i.e. they are an inevitable part of efficient computation). For discussion of this issue, see Chomsky (2007).

(25) See Chomsky (2000a) for discussion of the 'invisible' status of A-movement traces to the probe-goal relation.

(26) Chomsky (2008a) argues that movement to Spec-VP (e.g. IM(8c)) is obligatory, as in ECM constructions (e.g. 'they believe him to be intelligent'). See also Lasnik and Saito (1991), Koizumi (1995).

(27) In addition to (8c), V(see) must adjoin to v* to restore the original VO order, thus head movement must take place at the v*P phase level, but the nature of head movement and its exact mechanism (esp. its relation to NTC) remain to be spelled out.

(28) See Epstein and Seely (2002b) for discussion of problems regarding simultaneous rule application within a purportedly derivational theory. See also Grewendorf and Kremers (2008). Chomsky's (2007, 2008a) analysis thus strikes a delicate balance between the derivational and representational approaches.

(29) See e.g. Epstein et al. (1998), Epstein (1999), Epstein and Seely (2002a, b, 2006). For recent discussion on this issue, see also Epstein et al. (2008).

(30) The relation between who_2 and who_1 corresponds to an A-chain(who_2, who_1), and the relation between who_3 and who_1 corresponds to an operator-variable chain (who_3, who_1), though the notion chain need not be introduced, apart from expository reasons (cf. Chomsky 2007, 2008a and Epstein and Seely 2006 and the references cited).

(31) If the derivation of simple sentences appears to be complex, that should not be taken as an argument against SMT; a principled system—being explicit/falsifiable—may well yield intricate-looking derivations i.e. simple (explanatory) laws can give rise to complex phenomena (= science).

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No Derivation Without Representation

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Abstract and Keywords

This article is divided into two main parts. The first goes over some fairly general considerations about derivations before descending into some more specific discussion of some derivationalists. It then focuses on a series of appealingly intricate and internecine comments on c-command. The article argues that c-command and representationalism fit together very well. A concomitant point is that a purported major success of derivationalism in this domain is rather more problematic than is often supposed. If c-command is judged useful and desirable, then this finding is discommodious for the alleged supernumerary status of representations.

Keywords: derivations, representations, c-command, representationalism

> Efficiency is increased as effort is decreased, as though the former approaches infinity as the latter approaches zero, and in the ideal case, which is obviously the impossible promise of Taoism, one should be able by doing nothing to achieve everything.
>
> (Arthur C. Danto, *Mysticism and Morality*)

A spectre is haunting minimalism—the spectre of representationalism. Many (most?) Minimalists seem to agree that derivationalism is the Way: that in building a syntactic object 'step-by-step', everything (syntactically?) useful and important simply falls out of this derivation, and the object being so perfectly and stepwisely built itself has no (syntactic) efficacy.[1] Yet the object, the representation, apparently persists, only now as a ghostly remnant, unable to have effects (in the syntax?). (p. 312) I hereby suggest that this incapacitation is theoretically premature.

The overall point—and the argument—is conceptual / theoretical. The argument is not that *within* the narrow syntax / computational component / derivation (whatever this may mean exactly) the objects/representations are necessarily potent, but rather that even on the derivationalist conception they *are*, and that if they *are*, it should be odd (and surprising) that they have no effects ever anywhere. Perhaps the narrow syntax / computational component / derivation can get away without talking (much?) about the objects. But whereof one does not speak, does this *thereby* not exist? This seems not merely unparallel in expression, but unlikely in fact.

The chapter itself has two main parts. The first goes over some fairly general considerations about derivations before descending into some more specific discussion of some derivationalists. Then the focus shifts to a series of appealingly intricate and internecine comments on c-command. A point on which I recurringly harp is that c-command and representationalism fit together *so very well*. A concomitant point is that a purported major success of derivationalism in this domain is rather more problematic than often supposed. What is not argued for is the usefulness of c-command. Rather, the argument is that if c-command is judged useful and desirable, then this finding is discommodious for the alleged supernumerary status of representations.

14.1 Don't Stop Till You Get Enough

Let's pretend. Let's pretend there's a numeration and let's pretend there's Merge.[2]

Now suppose we are deriving the sentence in (1).[3]

(1) The shirt on the floor looks very dirty.

Whatever else might be true, it seems inescapable that Merge has to put [shirt on the floor] together as a unit unto itself. That is, [shirt] can't be merged with [looks very dirty] and then [on the floor] merged with that, (nor, for that matter, can [on the floor] be merged with [looks very dirty] and [shirt] merged with that). At least, not if Merge is 'always at the root' and obeys Extension, and there's strict cyclicity and other good things like that.[4] The point here is that Merge is going to have to (p. 313) derive, and the grammar allow for, (complex) structures that are not immediately composed one with another.[5]

Put it this way: if the derivation has got to the point of [looks very dirty], no matter what comes out of the numeration next, [the] or [shirt] or [on] or [the] or [floor], this cannot be merged with [looks very dirty] if the derivation is going to proceed successfully. Instead, these bits will have to be merged with one another, then that result merged with [looks very dirty]. After that, for various Greedy-type reasons,[6] these structures will (have to) compose.

And?

Well, this does have *some* implications. One concerns binary branching. If there are going to be little structures that are not (immediately) composed, then it is not clear why all branching must be binary. Merge really cannot be the reason by itself, unless one stipulates that it be what Chametzky (2000) calls Noahistic (two at time); once there can be more than two syntactic objects available to Merge, it is an open question whether they must combine two-at-a-time. Perhaps there are independent reasons for Noahism; there had better be, for those who require that it be true.[7]

A different question is this. Why should all these objects compose into one object anyway? Greedy ('purposeful') reasons were assumed above. Good. But what about if there are no such reasons in a particular case? Is that possible? Why not? Suppose, for example, that the numeration had only [Kim] [leave] [Pat] [arrive] in it. Again, why not? Suppose the first two merge with each other, and the second two with each other—is there any reason this cannot happen? At this point, there would presumably be no (further) Greedy reasons to merge more. And what would be wrong? Is there any strictly syntactic reason why this would not be a successful derivation?

There is a role for our earlier discussion here: we know that it is independently necessary for there to be syntactic objects around (in a derivation or workspace or …) that are not immediately merged with one another. So it cannot be that this derivation is illicit just because there are objects that have not composed. Aha! The answer, then, must be that a derivation cannot end with such uncomposed objects around. But why not? Recall that objects composed on account of Greedy-type requirements of one or the other. But recall, also, that at this point there are (p. 314) no further such requirements, and our question is: what could be wrong with that here? Surely all derivations that succeed have the property of lacking further such requirements at their end. Why cannot this be the end of the derivation?

The suggestion is that there is nothing in the (narrow) syntax that requires such final composition. But it is not really possible to have [Kim leave] [Pat arrive] be a successful syntactic derivation, is it? Is it? Beats me. But it certainly seems that if you have only a derivation and composing driven by (local) syntactic Greed, then you will end up here, like it or not.

So, if you do not want to go there, something else is necessary.[8] Maybe an extra something in the syntax that says: 'Oh, and, by the way, all the stuff in Numeration, it has to compose into one Big Object.' Or maybe it is not strictly in the syntax, but is 'imposed' from the interface, as one says, but still with the same effect: 'If you want to cross this line, you have to do it as one Big Object.' Psychometricians and their confreres like to talk about 'face validity'; maybe minimalists have to talk about 'interface validity', and to have that, syntax has to present a single Big Object to the world.

It is not clear that anyone really disagrees with this, when put this way.[9] For example, Epstein et al. (1998) talk about this idea, calling it the First Law of Syntax, and they are derivationalists. But then they want to 'construe it derivationally'.[10]

But, still, there is something. Various derivationalists (Chomsky 2001, Epstein and associates, Uriagereka 1999) want, in various ways, to disable the final (big) object (as noted at the outset). There is no single 'interface' as such, but rather periodic/cyclic/phasic/whateveric smallish packages of syntax that are sent off and closed off throughout the derivation. This gets to be fairly complex stuff, and different in its various guises. Two underlying ideas seem to be fairly simply stateable, however. One might go this way for a negative reason, or one might do so for a positive reason (or both). The negative reason is a general anti-representationalism: having eliminated some levels (viz. DS, SS), now the right and true minimalist thing to do is to eliminate all levels (viz. LF, or its descendent).[11] The positive reason is derivational advocacy: that it would be better for a derivation as a derivation to have such a property, and having it results in there being no role for the Big Object (though there is no prior commitment to eliminating an LF-like level).[12] On the first (p. 315) motivation, the apparent continued requirement for a Big Object seems a matter of regret, while on the latter more one of indifference. On either, it might well be a matter for puzzlement, perhaps even embarrassment.

And so, to reiterate an idea from the preamble: it just seems passing strange that there must be this Big Object, and that it must not be allowed to do more than just be. A system that appears to specially and specifically require an epiphenomenon seems, for that reason, peculiar.

And just to be clear that I'm not imagining things, or still just pretending, it does seem that this is a view that is held. Thus, Epstein and Seely (2006: 178–9, n. 6) write: 'It is important to note in this regard that in an optimal derivational model, it shouldn't be merely non-explanatory to define relations on trees or representations. It should be formally impossible to do so.'

I, at least, need to pause a bit here.

What can 'It should be formally impossible to do so' mean? Can it be that the representations lack any or adequate information to formally define relations? I can't really imagine how that could be. So, presuming there is *something* that could be used to define relations, apparently the optimal derivational theory is, *qua* optimal, unable to access or use this something. I suppose this is coherent, and I guess it might even be true, and maybe not just by stipulation. But what I fail to see is why it is *desirable*. That is, if the 'optimal derivational theory' just does build representations and the representations are (among other things) 'information structures', and yet the optimal derivational theory is in principle debarred from accessing these structures, why should one *want* the—or only the—optimal derivational theory?

As Epstein and Seely themselves write 'Thus, in the rule-based Minimalist approach, iterative application of well-defined transformational rules is assumed … Thus, it would be odd indeed to pay no attention to the form of the rules, *intermediate representations*, and the mode of iterative rule application' (2006: 6, emphasis added). If attention must be paid, how so, given the optimal derivational theory? And why only to the intermediate representations? Epstein and Seely are explicit that the 'end-of-the-line LF representation' or 'the final LF representation' has 'no special status', and that their model 'is a satisfactory alternative … only if all points in the derivation are treated alike' (2006:180). But surely we are talking symmetric predicates here, ones that cut both ways. We have not merely due process (viz. derivationalism) but also equal protection: if it would be 'odd indeed to pay no attention to … intermediate representations' then it must also be odd indeed to pay no attention to … the final [LF] representation given that 'all points in the derivation are treated alike' and that 'the final LF representation' has 'no special status'. Or so it would seem.

At various points in their writings, Epstein and Seely assert that derivationalism is explanatory. I agree, at least sometimes (e.g. Chametzky 2000: 155). But they also (p. 316) assert that representationalism is (always?) non-explanatory. Here I do disagree, both generally and more particularly.

Generally: I don't know where or how or why the next explanatory whatever is. Nor do they. Nor does anyone else. So there.

Particularly: They write (Epstein and Seely 2006: 7, emphasis added): 'For us, if you define relations on (*or appeal* *in any other way directly to the macrostructure*) tree representations, you have failed to explain their properties.' While one might agree with them that 'definitions in general do not explain', the italicized parenthetical in the quote goes rather beyond this point. Other than a commitment to derivationalism, what could motivate such a blanket statement? And as for definitions, if they themselves are not explanatory, still they have a *role* in explanations—for example, might one not use a definition of, let us say, a 'derivation' in constructing an explanatory theory of, let us say, something syntactic (see (4) and (5) below)? But why then is it impossible for a definition that appeals in 'any other way directly to the macrostructure' to play an essential role in constructing an explanatory theory of something syntactic? How on earth could one know that this is *in principle* impossible?[13],[14]

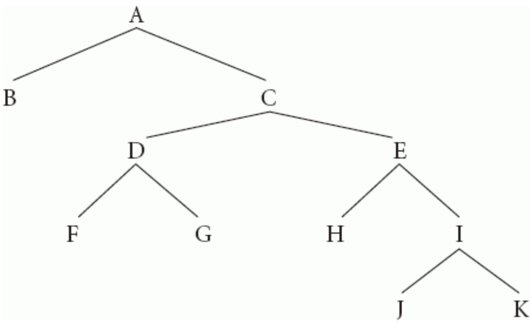
One final point before we turn to c-command. Brody (2002) has analyzed derivational and representational theories. He argues—demonstrates, it seems fair to say—that 'current (apparently pure) derivational theory is equivalent to a restricted multirepresentational theory (2002: 25).' That is, there is no pure derivational theory without representation(s).[15] It appears that Epstein and Seely's only response to this is to argue that 'representational theories with enriched derivation-encoding representational mechanisms, e.g., trace theory, are thus really "just" a kind of derivational theory … but, we would suggest, the wrong kind' (2006: 8; see 2002: 6–8) But while this seems to be "just" playing with words, it actually illustrates something about the utility of definitions. If you define 'derivation' in a usefully strict way, then the sorts of theories Epstein and Seely dislike are not derivational **(p. 317)** ones, not even of 'the wrong kind'. If the sorts of phenomena that exist and the kinds of information required to analyze them are most insightfully captured in derivational theories, then other sorts of theories that analyze these phenomena by encoding this information do not thereby become 'derivational'. Rather, they will be theories that, just because they are not derivational, are not the best (kind of) theory. I really do not see any point to Epstein and Seely's assertion.

Let's sum up before moving on. Derivations seem to require representations. Trying to say otherwise does not seem to make much sense, and actually saying that representations are there, but in principle impossible to see or use, seems to be even worse. Syntax seems to require a Big Object, but (some) syntacticians seem to have Big Objections. It might be nice to understand why a Big Object seems necessary. We turn, therefore, to c-command.

14.2 If You Could See C-Command Like I Can See C-Command

Some 25 years ago, John Richardson had the fundamental insight into c-command. Richardson and Chametzky (1985, R&C hereafter) then tried to start an explanatory inquiry into c-command.[16] In the event nothing much followed from this[17], so Chametzky (1996, 2000) gave it another shot. This has received a little more response, about which presently. First, though, we can refresh our memories about what R&C were trying to do.

The usual question with respect to c-command is 'Does node X c-command node Y'? R&C invert this, 'taking the point of view of the C-commandee', asking 'What nodes are the C-commanders of some node X?' This has an immediate, and salutary, consequence: the c-commanders of some node X are all and only the nodes which are sisters of all the nodes which dominate X (dominance reflexive). See (2), and (3), G as our 'target node'. **(p. 318)**



Click to view larger

(2)
(3) For any node X, the c-commanders of X are all the sisters of every node which dominates X (dominance reflexive).

For G, this returns the set {F, E, B}. This is evidently the correct set. C-command is a generalization of the sister relation, complementary to, and parasitic on, dominance.

In R&C it is assumed that phrase markers (PM) are totally ordered by the combination of dominance and precedence, and that the Exclusivity Condition holds, so that any pair of nodes is related by either dominance or precedence, but never both. This means that R&C understand c-command to condition linguistic relations between nodes in a precedence relation (because not in a dominance relation). These days, precedence as a syntactic relation and Exclusivity have fallen rather far out of favor.[18] However, on account of those assumptions, R&C call the set of c-commanders the 'minimal string' for a given node. Chametzky (1996) renamed it the 'minimal factorization', and that will be asked to do some heavy lifting below.

Anyway, in either case here is what is going on. There is at least one formal relation characterizing an object such as (2), viz. dominance. But there is nothing specifically or peculiarly linguistic about that relation or that object. There are also substantive linguistic relations among nodes in a dominance relation, viz. ones having to do with projection (X-bar theory, or its remnants/descendents perhaps). What then about nodes not in a dominance relation? What about substantive linguistic relations among these nodes? Well, this is what c-command does: it provides the set of nodes which are not in a dominance relation with some given node and with which that node can be in some substantive linguistic relation or other.[19]

I'm going to go on a bit about this. The minimal string/factorization is minimal in that there is no other set of nodes that both is smaller than (has fewer members, (p. 319) a lesser cardinality) than it and, when unioned with the set containing just the target node, provides a complete, non-redundant constituent analysis of the PM. C-command is the minimal string/factorization of a PM with respect to a target node. There are, of course, other sets of nodes that don't stand in a dominance relation with a target node. Why single out this one?

How about the set of all nodes not in a dominance relation with, say, our target G: {B,F,E,H,I,J,K}? But such a set will generally just ignore the fact that a PM is hierarchically structured, in that it contains nodes which are constituents of other member nodes. If dominance really is basic to syntax, to the structuring of syntactic objects, then we ought to be surprised, and chagrined, to find it utterly ignored in this way: if our inquiries suggest that such a set (relation), which is entirely indifferent to the dominance-induced structure, is the most useful one, then our original commitment to dominance is thereby undermined.

Going to the other extreme, how about the smallest set of nodes relatable to a target by dominance but not in a dominance relation to it, viz. the set of a target's sisters? This is a good set, but, in a sense, too good: it's just too constrained for our initial goal of finding a set of candidate nodes in the PM for further substantive relations with a given node. Grammar seems to have an 'incest-only taboo': it is just obviously false that a given node can only have substantive linguistic relations with its sister(s).[20] What we really want is a set that both is relatable to our target by means of dominance, though there is no dominance relation between the target and any set member, and which utilizes the full PM while respecting the hierarchical structure induced by dominance.

Looking at (2), there are really only three candidates with respect to G: {B,F,E}, {B,F,H,I}, and {B,G,H,J,K}. Only these sets provide complete, non-redundant analyses of the PM with respect to G. {B,F,H,I} is just an arbitrary analysis, but the other two are distinguished: {B,F,E} is minimal and {B,G,H,J,K} is maximal. But the maximal set has a now-familiar problem: it denies the relevance of the full hierarchical structure induced by dominance: it is just the set of (pre)terminals not in a dominance relation with the target G. In fact, it is arguably worse on this score than the set of all nodes not in a dominance relation with G. That set *ignores* the dominance-induced structure, but at least redundantly contains all the nodes; this set *denies* the structure, entirely leaving out the nodes indicating hierarchical structure. In other words, the minimal string/factorization set is the only non-arbitrary set which requires and respects the full branching hierarchical structure induced by dominance on a PM.

The chapter in Chametzky (1996) that said all this is called the 'The explanation of C-command'. It's still an appropriate title. If this ain't explanation, well, it'll have (p. 320) to do, until the real thing comes along.[21] One final bit of stage-setting, and then we can see whether the real thing *has* come along, as we compare R&C with the 'derivational explanation of C-command'.

Brody (2002: 27–33; 2003: 195–9) has analyzed—eviscerated might be better—the derivational explanation of c-command in some detail. I shall pick over the bones myself in a bit, but do not rehearse Brody's performance here. Instead, I want to draw attention to what he calls 'the core of the c-command problem' (2002: 32; 2003:198):

the arbitrary asymmetric conjunction in its definition: x c-commands y iff the following two conditions of somewhat different nature obtain: (a) there is a z that immediately dominates x and (b) z dominates y. It is crucial, but unexplained, that the two subclauses make use of different notions of domination.

Let's now recall our own (3):

(3) For any node X, the c-commanders of X are all the sisters of every node which dominates X (dominance reflexive).

Notice that we don't have two subclauses or two notions of domination. All we have is dominance, giving us the generalization of the sister relation. This is because R&C takes the point of view of the c-commandee. So, it appears that the core of the c-command problem just goes away. But maybe not. Maybe Brody would say that there are still two different relations there in (3): dominance and sisterhood. And that's still unprincipled and arbitrary, even if worded so as not to be exactly an asymmetric conjunction. But also notice this: if syntactic objects are hierarchically structured, then there's no way to avoid either the dominance relation or sisterhood. There is nothing any more basic than these when dealing with hierarchical structure. So, if there is going to be another relation, these are what you'd really, really want it to come directly from; and if you can't have even *that* relation with *that* provenance, then what can you have?[22]

There's something here worth talking about. It seems that if there is going to be a c-command relation, it will have to have more than one aspect to it: what could it mean to say that there is c-command, but it's *only* dominance? Or *only* sisterhood? To be a new relation, it has to have something about it that is different, after all. So what Brody appears to be objecting to, really, is just there being a new relation. (p. 321) And indeed, when he gets around to his own proposal, it turns out to be precisely that: there isn't any c-command. There is, instead, 'the accidental interplay between two (in principle unrelated) notions, one of which is domination'. The other is the specifier-head relation or the head-complement relation (Brody 2002: 32–3; 2003: 198, 226–7). Maybe Brody is right about this.I don't know. Butlet's be clear that his objection to c-command is really no objection once we understand c-command as in (3). C-command is a generalization of the sister relation, and that's as un-new as a new relation can get. Maybe it's unnecessary. But it's not illegitimate.

Now we can ask whether R&C's 'representational view of C-command' has been overtaken and surpassed by the 'derivational explanation of C-command' of Epstein et al. (1998). This is germane to our larger concerns because if it has not been, then there is, apparently, at least this much work for a representation still to do.

14.3 If You Build It, Will They C-Command?

EGKK = a derivational approach to syntactic relations (Epstein et al. 1998)

TPM = a theory of phrase markers and the extended base (Chametzky 1996)

EGKK have two criticisms of TPM (p. 174).[23] The first is that the maximal factorization of a PM does require the full branching hierarchical structure of the PM, *contra* TPM. The second is that the concept 'minimal factorization' is basically ad hoc.

For the first, they are wrong. At least, they are wrong given what is intended, though perhaps not clearly enough conveyed, in TPM. The argument there is that the maximal factorization is just the set of (pre)terminals not dominated by the target node (TPM: 31). The point is that this set does not require that there be any hierarchical structure in a sentence; it is compatible with a 'flat structure' in which all these (pre)terminals are daughters of the mother node of the representation. EGKK (p. 174) write: 'the factorization of any phrase-marker requires the existence of a phrase-marker to factorize, in which *all* nodes are in a Dominance relation with (at least) the mother node of the representation.' Fine. But the point about the maximal set is that it is consistent with their parenthetical '(at least)' being *at most*, and if the maximal set is what is called for analytically, this can be seen as evidence that sentences are not hierarchically structured, because, as noted, this set is consistent with the lack of such structure. EGKK seem to be already assuming that there is hierarchical branching structure; this is a perfectly good assumption, (p. 322) maybe a true one. But it is not one which the maximal set requires or necessarily leads one to embrace. This is the point of the claim in TPM. So, while there is a sense in which EGKK are right—the maximal factorization does require a PM—it is a sense that misses the point—PMs might, in general, have no hierarchical branching structure as far as the maximal factorization is concerned. Their second objection is that

> the notion of a factorization is needed to explain the naturalness of the representational definition of C-command, but to the best of our knowledge for nothing else. Though a factorization is easily defined, its definition serves only to facilitate a particular outlook on C-command; no notion of factorization is required independently. (EGKK: 174)

This is their big complaint. Let's grant that it *is* big. Let's also grant it for now, and look instead at EGKK's derivational approach.

They define—notice—three things: (4) a derivation and dominance (pp. 167–8, their (3)), and (5) c-command (p.170, their (5)) The important point is that the definitions for dominance and c-command are identical, except that 'input' appears in (5i) where 'output' appears in (4b.i). This exact parallelism is quite striking.

(4)

a. Definition of derivation
A *derivation* D is a pair <O, M>, where
i. O is set of operations $\{o_1, o_2, \ldots o_n\}$ (Merge and Select) on a set S of lexical items in a Numeration, and terms formed by those operations;
ii. M is a set of pairs of the form $< o_i, o_j >$, meaning o_i 'must follow' o_j.D is transitive, irreflexive, and antisymmetric (a quasi order).

b. Definition of dominance
Given a derivation D = <O, M>, let X, Y ∈ S.
Then X *dominates* Y iff
i. X is the output of some $o_i \in O$; (*we consider outputs*)
ii. X is not in a relation with Y in any proper subderivation D′ of D; (the *relation is 'new'*)
iii. Y is member of some $o_j \in O$ such that $< o_i, o_j > \in M$. (*the terms are in a relation only if the operations are*)

(5) Definition of c-command
Given a derivation D = <O, M>, let X, Y ∈ S.
Then X *c-commands* Y iff
i. X is the input of some $o_i \in O$; (*we consider inputs*)
ii. X is not in a relation with Y in any proper subderivation D′ of D; (*the relation is 'new'*)
iii. Y is member of some $o_j \in O$ such that $< o_i, o_j > \in M$. (*the terms are in a relation only if the operations are*)

(p. 323) However, EGKK note that, in actual fact, (4b.ii) is redundant (p. 168). This being so, we should rewrite it without the redundancy as in (6).

(6) Definition of dominance
Given a derivation D = <O, M>, let X, Y ∈ S.
Then X *dominates* Y iff
i. X is the output of some $o_i \in O$; (*we consider outputs*)
ii. Y is member of some $o_j \in O$ such that $< o_i, o_j > \in M$. (*the terms are in a relation only if the operations are*)

Now two points. First, of course, dominance and c-command are no longer exactly parallel in their definitions. Second,

> the notion of a 'new relation' is needed to explain the naturalness of the *derivational* definition of C-command, but to the best of our knowledge for nothing else. Though a 'new relation' is easily defined, its definition serves only to facilitate a particular outlook on C-command; no notion of 'new relation' is required independently

Now where have we heard something like *this* before?

So, once you untrick their definitions, they are guilty of exactly the worst sin they locate in the representational account of c-command in TPM. This is what the philosopher G. A. Cohen has called a 'look who's talking argument'. It isn't that the point being made is, as such, a bad one; it's that, for various reasons, the person/persons making it are especially badly situated to be bringing it forward.[24]

It seems, then, that at best there's a stand-off here. But maybe not. First, a smallish, nearly empirical point. EGKK notice that their c-command is reflexive (p. 179, n. 7). They point out that 'with respect to semantic interpretation, no category is ever "dependent on itself" for interpretation'. They say this is no problem. But isn't it? After all, if, for example, binding domains and relations are licensed by c-command, why shouldn't, say, an anaphor be its own binder given this reflexive c-command? Moreover, it was argued in R&C that c-command is anti-reflexive and non-symmetric, and this led them to derive (7) (their (38)),

> **(7)** All predicates which contain C-command as a necessary condition for their satisfaction will be antireflexive and non-symmetric.

Moving along, EGKK say that all one can do, after looking at outputs of operations, as in their dominance, is look at inputs, as in their c-command 'if we are to conceive of intercategorial relations as properties of operations (rule-applications) in a derivation' (p.169) But why should we do *that*? If there were the strict parallelism between dominance and c-command that they try to palm off, that would be a reason, no doubt. *But there isn't*. So, what's left—other than an a priori commitment to derivationalism?

(p. 324) Well, how about the conceptual underpinning for their dominance definition, viz. the 'First Law' of syntax: that everything has to get together as one syntactic object. But this does not require that 'inputs' be looked at; once you've got outputs/dominance, you do not need c-command for the First Law to take effect. And this just leads back to the conclusion that really dominance and c-command are not on a par, unlike what EGKK want us to believe.

A bit of stock-taking: if the representation is being built anyway, to not allow it some role is for that reason to make the theory conceptually worse than it ought to be, with, as we have now seen, no compensating theoretical advantage with respect to new, special purpose notions. A further point in the R&C approach's favor, not mentioned elsewhere,[25] is that by taking the viewpoint of the c-commandee, you align the relation with (some? most? all? of) its significant applications, e.g. it is anaphors or pronominals or traces or predicates that have a 'be c-commanded (or not)' requirement on them—it's not that there are some inherent binders that have a 'c-commanding' requirement on them.

But all of this is, surely, beside the point. What EGKK have done is nonsense. They have taken a name 'minimal factorization' and mistaken it for some kind of essence, or at least a (significant) concept. The important idea, the point, is that c-command is a generalization of the sister relation. The set in question has been given a couple of different names, in order to facilitate discussion, and, so it was naively hoped, understanding. But the name is really quite irrelevant. Maybe this isn't as clear as it should be in R&C, or in Chametzky (1996 or 2000). Maybe; but I think it is there.

Now, the deep puzzle about (almost) all derivational approaches to syntax is this. Why is there what EGKK call 'cyclic structure-building' if the resultant built structure is going to be syntactically impotent? As I've stressed, isn't it odd to build this object and yet not allow it any positive role? Shouldn't we expect the structure so built to do *something*? And the basic problem for (almost) all derivational approaches to c-command is this. *It is a representational relation*. EGKK concede as much when they say they 'are looking for relations between terms, such as C-command' (p.165). Their attempt to squeeze c-command out of their derivational approach is valiant, but it leaks. It illustrates the principle I like to call 'If all you have is a hammer, everything looks like a thumb.'[26]

These problems are related, of course. Once you accept that there is the Big Object, then c-command ceases to be a mystery. Not a necessity, surely, but, as I've harped on, if there are to be other than dominance-mediated substantive linguistic relations, then lack of c-command would be more in need of explanation. Indeed, if there's no peeking at the whole Big Object, then there's just no reason for c-command, EGKK to the contrary notwithstanding. And the idea of peeking is a **(p. 325)** suggestive one. If where the peeking is done is 'from the interface(s)', where presumably the whole Big Object could be available, and what gets peeked at are individual constituents that may or may not have needs to be met (i.e. are dependents, require some kind of licensing, whatever), then it does seem that this particular keyhole would naturally reveal the 'minimal factorization' from 'the point of view of the C-commandee'. We are ready now for our leavetaking.

14.4 But What Would Zeno Say?

As noted above,[27] there's a very widespread idea that grammars are essentially 'local', in the sense that what's really involved are basically just mother-daughter/sisterhood relations. Proponents of such views (and they come in various guises) then find themselves suggesting various ways to 'string together' their favored form of baby step to make a long march. But why? Why shouldn't grammar be satisfied with just what, on these views, are the basics/essentials? That is, are there any grammars of any languages that do not seem to manifest any non-local dependencies? *If locality is truly the be-all, why isn't it (ever) also an end-all*? Minimalists especially ought to wonder.

I am not aware that anyone has come up with a good answer to this question—but then I am not sure anyone has bothered to ask it, either—and I'm not betting that there will (can?) be one. What we see instead are various after-the-fact rationalizations for stringing together the strictly local bits in order to graft onto these treelet collections a result from a differently premised approach. If grammar isn't 100% local, a (the?) 'first step beyond' is the generalization of sisterhood advocated in R&C. It is at least arguable that just about everything interesting in grammar is tidied up fine with this first step beyond. Why fight it? What's the point? But, if you accept that c-command (= the generalization of sisterhood) is a real part of grammar, then you're likely stuck with the Big Object. They go together: if there is a Big Object, you'd expect c-command; if there is c-command, you need a Big Object.

The only other viable option, as far as I can see, is to deny that c-command is in fact relevant to grammar.There are two ways to do this. One is to deny that this is the right kind of grammar and to build a different kind. The other is to keep the kind of grammar but to deny c-command. Within broadly minimalist approaches, Brody (2002, 2003), Hornstein (2009) in different ways take the latter course, while Collins and Ura (2001) take the former.[28] Either of these is OK by me. Evaluating (p. 326) these positions is more a matter of the best analyses, I think, than one of theory, per se, so I have no ideas or recriminations to contribute.[29]

The Big Picture, then, is just this: it's kind of impossible to make much sense of a 'purely derivational' approach to syntax. And insofar as one tries to, as it were, asymptotically approach that as an ideal, one finds that progress slows, conceptual puzzles arise, and confusions mount—not usually considered hallmarks of a promising set of initial assumptions. And yet, despite Brody's (2002) scolding of those who advocate 'mixed theories' that are both representational and derivational,[30] I really have no objection to Hornstein and Uriagereka's (2002: 106) suggestion/conjecture 'that grammars are (at least in part) derivational systems'.[31] And just imagine how painful it must be to publicly end on so conciliatory a note.

Notes:

This chapter began as a presentation at the 2007 MayFest on 'Hierarchy' sponsored by the Linguistics Department at the University of Maryland/College Park. I thank the organizers for inviting me and the other participants for listening and, when appropriate (mostly), laughing. Thanks to John Collins for his comments on some earlier versions of some of this material. My special thanks go to John Richardson, who agreed to take some time away from his animal rescue work to do another joint presentation with me. John, however, must not be blamed for what I have done to the ideas he has shared with me, and no one else could be, so it's all on my head, as is appropriate.

(1) Boeckx (2008a) is a prominent exception, purveying a mixed, derivational-cum-representational minimalism.

(2) For my views on how much pretending would be necessary, see Chametzky (2000).

(3) Here and throughout, I ignore functional heads and categories; this is both convenient and, quite possibly, correct (Chametzky 2003).

(4) There is a point not being made here: that subjects can be arbitrarily complex syntactically—and, indeed, the fact that a subject might contain, e.g. a tough-movement construction was one of the central empirical facts motivating first-wave minimalism and the overthrow of DS—that is nonetheless worth noting in passing (or passing

in a note).

(5) De Vries (2009: 346) refers to an ‘auxiliary derivation’ when noting this fact.

(6) In a more current idiom: ‘all rule application is purposeful’ (Epstein and Seely 2006: 5).

(7) John Collins (p.c.) has addressed some of this: ‘This is difficult, but at a first stab, I'd say that this [non-binary branching/Merge] would make merge non-uniform, since we know that binary branching is fine for say verbs and their objects, etc. Non-uniformity or a lack of symmetry is kind of an imperfection. Also, given that branching is required, binary is the most economical, as only two things at most need to be in the workspace, as it were. So, perfection is the most economical and uniform meeting of interface conditions by the most general operations we find throughout nature. The interfaces rule out singletons and the empty set, and general uniformity/economy considerations go for binary, given that there must be branching for composition (lack of composition would be an imperfection as set formation would need to be restricted). Thus, something other than binary would thus be an imperfection, as far as we can tell.’

(8) We could perhaps call it the Wobbly or IWW principle: ‘One big union’.

(9) So, de Vries (2009: 357) notes in passing that Merge applies ‘until a final single-rooted structure is created’. On the other hand, linguistics seems to require that every possible position be occupied (as well as every impossible one).

(10) Good luck.

(11) I wonder about this. I don't see that there are the sorts of conceptual arguments contra an LF-ish thing that there were vis-à-vis DS and SS. So, the argument will have to be pretty much that one can do without such a thing, and with some advantage empirically. I simply do not know about this, though I am curious with respect to e.g. the treatment of inversely linked quantificational sentences.

(12) The idea, perhaps, is a kind of uniformity: if there are syntactic ‘steps’ in the ‘step-by-step’ derivation, then each step should define its own complete little universe, so to speak, not just a bit of merging. See Boeckx (2008a) for some discussion.

(13) Well, they do say ‘for us’; maybe Iowa isn't heaven because Michigan (apparently) is.

(14) Epstein and Seely invoke (2006: 7) Joshua Epstein's slogan for ‘generative social science’: ‘if you haven't grown it, you haven't explained it.’ This is actually a misquotation of Epstein (1999: 43), who writes ‘if you didn't grow it, you didn't explain its emergence.’ They do this also in their (2002c: 5). This, I suppose, is a *liberté* licensed by *fraternité*, so we could let it pass. On the other hand, I have a brother who is an art historian, and so I could probably find something positive about ‘representations’ in his work, if I looked hard enough. But so what? And further, Epstein (1999: 46) also characterizes this work as ‘connectionist social science’: ‘distributed, asynchronous, and decentralized and hav[ing] endogenous dynamic connection topologies’, a ‘social neural net’. And the actual phenomena studied, and the models used, have both a spatial and a temporal aspect that are each crucial. Perhaps the analogy still holds, or perhaps not. Additional irrelevant biographical detail: the Chametzky and Epstein brothers have known one another for close to 50 years.

(15) Brody also (2002: 27–33) dismantles some of the explanatory pretensions of the derivational approach to c-command, about which more directly.

(16) This maybe isn't quite right. We would have been happy enough if everyone had simply acknowledged we had explained c-command. And Kayne (1981b) did ask the right question; see Chametzky (1996) for why his answer is wrong.

(17) This maybe isn't quite right, either. John does honorable work in animal rescue.

(18) Though I understand why this has happened (I think), and have even made some of these sorts of arguments myself, I'm not totally convinced that precedence is not syntactic. But we need not worry that here and now. My own arguments, for those who care, are in Chametzky (1995, 1996). I do not actually argue that precedence is not syntactic.

(19) De Vries (2009: 367) appears to have discovered this: ‘essentially, it [c-command] identifies possible dependencies.’

(20) One can, however, try to make it true, and a sizable amount of syntax is devoted to trying to, the idea being that all non-local relations are actually composed of linked non-local ones. See section 14.4 below for some comment.

(21) What we've got is not, after all, nomic necessity or whatever, and, if, as per current phylotaxistic longings, someone can come up with a way to explain c-command using the Fibonacci series (Medeiros 2008, Soschen 2008), I'll bow out gracefully. I'm proud, but not stubborn. Also not losing lots of sleep.

(22) I guess this is what lies under Chomsky's (2000a) attempt to derive or explain c-command by means of ‘the elementary operation of composition of relations’ operating on sisterhood and immediately contain (actually, on contain, the transitive closure of immediately contain). I've denigrated this attempt elsewhere (2003: 200–1), and while I certainly don't mind repeating myself, I'll forbear this once. It's enough to remark that from our perspective this looks like an unwitting attempt to arrive at the fundamental insight of R&C.

(23) There's also something on Epstein et al. (1998: 106–7), but it's not important.

(24) Famously: ‘Dear Pot, You're black. Signed, Kettle.’

(25) It almost made it into R&C, but that was way over-long already.

(26) The parenthetical ‘almost's are due to Collins and Ura (2001). See n. 28, below.

(27) See n. 20, above.

(28) Collins and Ura give up structure-building and the phrase structure representation, and offer a ‘search algorithm’ analogue of c-command. The problem here is that it's not clear that anyone would ever come up with such a thing except as a reconstruction of already existing, essentially classical c-command.

(29) This zero is just with reference to minimalist approaches; for other approaches, I would have to begin with negative contributions.

(30) Brody seems to play the severely responsible Confucian to the standard Minimalist's blithely wandering Taoist.

(31) As pointed out in n. 1 above, Boeckx (2008a) tries to synthesize the derivational/representational thesis/antithesis.

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Last Resort with Move and Agree in Derivations and Representations

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Minimalism assumes that language consists of a lexicon and a computational system, with the latter embedded in two performance systems: articulatory-perceptual and conceptual-intentional. Two linguistic levels, phonetic form (PF) and logical form (LF), interface with the performance systems. A computation converges at the interface levels if it contains only legitimate PF and LF objects. However, we cannot define linguistic expressions simply as PF/LF pairs formed by a convergent derivation and satisfying interface conditions. The operations of the computational system which produce linguistic expressions must be optimal, in the sense that they must satisfy some general considerations of simplicity, often referred to as Economy Principles. One of them, the Last Resort Condition (LR), prohibits superfluous steps in a derivation. It requires that every operation apply for a reason. It has often been argued that a similar condition constrains representations, prohibiting superfluous symbols. These conditions require that derivations and representations in some sense be minimal. This article discusses the working of LR, as it applies to both derivations and representations. It starts with the former, examining how LR applies to both Move and Agree.

Keywords: minimalism, language, phonetic form, logical form, Last Resort Condition

15.1 Introduction

Minimalism assumes language consists of a lexicon and a computational system, with the latter embedded in two performance systems: articulatory-perceptual and conceptual-intentional. Two linguistic levels, PF and LF, interface with the performance systems. A computation converges at the interface levels if it contains only legitimate PF and LF objects. However, we cannot define linguistic expressions simply as PF/LF pairs formed by a convergent derivation and satisfying interface conditions. The operations of the computational system that produce linguistic expressions must be optimal, in the sense that they must satisfy some general considerations of simplicity, often referred to as Economy Principles. One of them, the Last Resort Condition (LR), prohibits superfluous steps in a derivation. It requires that every operation apply for a reason. It has often been argued that a similar condition constrains representations, prohibiting superfluous symbols. (p. 328) These conditions require that derivations and representations in some sense be minimal. The goal of this chapter is to discuss the working of LR, as it applies to both derivations and representations. I will start with the former, examining how LR applies to both Move and Agree.

Before doing that, let me point out that there are various ways of implementing LR formally—it can be stated as an independent condition or built into the definition of Move and Agree. However, regardless of how it is formalized, as soon as the issue of driving force for a syntactic operation is raised LR considerations become relevant, since the issue of driving force really makes sense only given LR. Much of what follows will in fact be more concerned with the driving force of syntactic operations than with the exact implementation of LR.

15.2 The ban on superfluous steps in derivations

15.2.1 Last Resort and Move

Minimalism has insisted on the last resort nature of movement from its inception: in line with the leading idea of economy, movement must happen for a reason, in particular, a formal reason. Case provides one such driving force. Consider (1).

(1) Mary is certain *t* to leave

Mary cannot be case-licensed in the position of *t*. Raising to matrix SpecIP rectifies its case inadequacy, since the raised position licenses nominative. Once *Mary* has been case-licensed, it is no longer available for A-movement, to a case or a non-case position. This follows from LR, if A-movement is driven by case considerations. Since *Mary* is case-licensed in the position of *t* in (2), LR blocks further movement of *Mary*.[1]

(2)

a. *Mary is certain *t* will leave

b. *The belief Mary to be likely *t* will leave

One fruitful line of research regarding LR concerns the issue where the formal inadequacy driving movement lies. The options are: (a) always in the target (pure Attract); (b) always in the moving element (Greed); (c) in the target or in the moving element (Lasnik's 1995a Enlightened Self-Interest). Greed was the earliest approach (Chomsky 1993), revived recently in Bošković (2007). Under this approach X can move only if X has a formal inadequacy, and if the movement will (p. 329) help rectify the inadequacy. Under pure Attract, the target head always triggers movement (Chomsky 1995c), which means the target must always have a formal inadequacy to be rectified by the movement. Under this approach, movement of *Mary* in (1) is driven by T/I: Tense has a property (e.g. the EPP or case feature) that must be checked against an NP which triggers the movement of *Mary (Mary's* case-checking is merely a beneficial side effect of the satisfaction of the attractor's requirement). If the trigger is T's case (i.e. Bošković's 1997 Inverse Case Filter, which requires traditional case assigners to check their case), (2a–b) can still be accounted for: the problem with (2a) is that *Mary* is case-checked in the embedded SpecIP so that the matrix T's case remains unchecked, and the problem with (2b) is that nothing triggers the movement. A question, then, arises why (3) is unacceptable (the question also arises under the Greed approach).

(3) *the belief to be likely Mary will fail the exam

In a framework that adopts the EPP, where the EPP drives movement, (3)is easy. However, (2) is problematic: the LR account cannot be maintained since there is reason for movement of *Mary*, namely the EPP. If the EPP/Inverse Case Filter accounts are combined, (3)and (2a) can be handled (as EPP/Inverse Case Filter violations respectively; note that A-movement in (1) is now redundantly driven by the EPP/Inverse Case Filter), but (2b) is still problematic. In other words, something additional needs to be said under the pure Attract account. Recall that under the Inverse Case Filter version of this account, (3) is at issue: for accounts of (3) that conform with this account, and which can also be extended to the Greed approach to LR, see e.g. Epstein et al. (2004), Bošković (2002a). As for the EPP account (or the combined EPP/Inverse Case Filter), the additional assumption Lasnik (1995a) makes to bring (2) in line is that once the case feature of an NP (like *Mary* in (2)) has been checked, the NP is no longer available for A-movement. Note that the assumption is also necessary under the option (c) from above. As long as we allow the target to drive movement, we have to deal with the question of why (2)is unacceptable. The conceptually unfortunate consequence of this account is that it basically brings back Greed into the system that was intended to eliminate it.

To capture the facts in question within a target-driven system Chomsky (2000a) posits the Activation Condition, which says X can move only if X has an uninterpretable feature, i.e. a formal inadequacy. The approach is still sneaking in Greed into a system where movement is supposed to be target-driven. In fact, under this approach something essentially has to be wrong with both the target and the moving element in order for movement to take place.

Let us now compare Chomsky (2000a) and Bošković (2007) in more detail, as representatives of target-driven and moving-element-driven approaches. Both of these works adopt the Agree account of traditional covert dependencies. Under Agree, two elements—a probe, which initiates an Agree operation, and its goal—establish a feature-checking operation at a distance without actual movement. This (p. 330) is all that happens in traditional covert dependencies. Chomsky assumes that Agree is a prerequisite for Move. Before Move takes place, X and Y establish an Agree relation, which is followed by movement if X is specified with an EPP property. This property of the target is what drives movement for Chomsky.

Bošković, on the other hand, places the trigger for movement on the moving element. In a phase-driven multiple Spell-Out system, where phases are heads whose complements are sent to Spell-Out (Chomsky 2001), element X undergoing movement moves from phase edge to phase edge until its final position, the underlying assumption being that if X is ever going to move, it cannot be contained in a unit that is shipped to Spell-Out. X then has to move to Spec YP, where YP is a phase, in order not to get caught in a Spell-Out unit. The analysis implies that there is some kind of marking on X indicating its need to move. So, how do we know that X will need to move? The question is not innocent, since in many cases what is assumed to trigger movement of X may not even be present in the structure at the point when X needs to start moving. To deal with such cases, Bošković (2007) argues the marking indicating the need for movement, which is standardly taken to be a property of the target (the EPP property of Chomsky 2000a, 2001), should be placed on the moving element, not on the target.

To illustrate, consider (4).[2]

(4) What$_i$ do you think [$_{CP}$ t$_i$ [$_{C'}$ that Mary bought t$_i$]]?

Chomsky's (2000a) account of (4) is based on the PIC, which says only the edge (Spec/head positions) of a phase is accessible for movement outside of the phase. Given the PIC, since CP is a phase, *what* can only move out of the CP if it first moves to SpecCP. This movement is implemented by giving *that* the EPP property (Chomsky assumes complementizer *that* may, but does not have to, have the EPP property), which is satisfied by filling its Spec position. The EPP then drives movement to SpecCP, after which *what* is accessible for movement outside the CP.

(5) raises a serious problem for this analysis, given the derivation on which we have chosen the EPP option for *that*, just as in (4).

(5) *Who thinks what that Mary bought?

To deal with this, Chomsky (2000a, 2001) makes the assignment of an EPP property to heads that do not always require a Spec conditioned on it being required to permit successive-cyclic movement. The embedded clause head in (4) can then be assigned the EPP property, since this is necessary to allow successive-cyclic movement. However, this is disallowed in (5) since the assignment is not necessary to permit successive-cyclic movement. The obvious problem for this analysis is look-ahead. Both (4) and (5)atone point have the structure in (6).

(6) [$_{CP}$ what$_i$ [$_{C'}$ that Mary bought t$_i$]]

(p. 331) To drive movement to SpecCP, complementizer *that* must be given the EPP property at the point when the embedded clause is built. But at that point we do not know whether the assignment of the EPP property will be needed to make successive-cyclic movement possible. We will know this only after further expansion of the structure. If the structure is expanded as in (5), it won't be needed, hence disallowed, and if it is expanded as in (4), it will be needed, hence allowed. So, at the point structure-building has reached in (6) we need to know what is going to happen in the matrix clause, an obvious look-ahead problem.

The problem is quite general. To appreciate this, consider (7), where X is a cyclic head (and XP a phase) and Y needs to undergo movement to W. In accordance with the Activation Condition Y has an uninterpretable feature (*u*K), which makes it visible for movement.[3] (8) represents the same scenario before W enters the structure.[4]

(7)

W[$_{XP}$ … X … Y]	
*u*F	*i*F
K	*u*K
EPP	

(8)

[$_{XP}$ …X …..Y]	
	*i*F
	*u*K

Since XP is a phase, given the PIC, if Y is to move outside of XP it first must move to SpecXP. In Chomsky's system this is implemented by giving X the EPP property to drive movement to SpecXP, with the further proviso that X can be given the EPP property only if this is needed to make successive-cyclic movement possible. We then need to know at point (8) that W will enter the structure later, as in (7). Let us see how the look-ahead problem can be resolved. The problem here is that the EPP diacritic indicating Y has to move to SpecWP is placed on W, given that we need to know that Y will be moving before W enters the structure.The problem is quite general under the EPP-driven movement approach. The gist of the look-ahead problem that arises under this approach is that the EPP diacritic indicating Y moves is placed on an element (W) other than the one that is undergoing the movement in question, but Y often needs to move (i.e. start moving) before W enters the structure. The conclusion to be drawn from this state of affairs is obvious: we have been wrong in placing the diacritic indicating the need for movement on the target (W)—the diacritic should be placed on the moving element (Y). Bošković (2007) implements this as follows. It is standardly assumed that a probe (p. 332) must c-command the goal, and that the probe must have a *u*K; otherwise, there would be no need for it to function as a probe. Following an insight of Epstein and Seely (1999), Bošković (2007) assumes the correlation between functioning as a probe and having a *u*K is a two-way correlation: just like a probe must have a *u*K, a *u*K must function as a probe.[5] In other words, checking of a *u*K on X requires X to function as a probe (i.e. c-command the checker). This means Y in (7)–(8) will need to undergo movement outside of XP to license *u*K. In fact, Agree would not suffice for that even if Y is located in SpecXP. Most importantly, we now know that Y will need to undergo movement outside of XP before W enters the structure: already at point (8) we know the structure will crash due to *u*K unless Y moves outside of XP. In other words, Y will have to move to a position c-commanding the *u*K licenser to check the feature. Since the *u*K licenser is not present within XP, this means that Y will have to move outside of XP, hence has to move to SpecXP. Notice also that Bošković (2007) argues for the following formulation of LR: X can undergo movement iff without the movement, the structure will crash. Movement to SpecXP then conforms with LR although it does not involve feature-checking between Y and X—a desirable result in light of arguments against feature-checking in intermediate positions discussed below. Eventually, Y will have to move to a position c-commanding W. Given the Shortest Move requirement, it will move to the closest position c-commanding W, which means SpecWP.

The analysis also deduces generalized EPP effects. We have already seen that there is no need to mark intermediate heads (X in (7)) with the EPP property to drive movement to their Specifiers. The movement takes place so that the element undergoing movement escapes being sent to Spell-Out, which would freeze it for the possibility of movement, leaving its *u*K unchecked. Now, the generalized EPP effect is deduced in its entirety. Thus, Y in (7) has to move to SpecWP even if W does not have the EPP property, which is then dispensable.[6]

Since the beginning of minimalism there have been various ways of implementing the generalized EPP effect: in early minimalism this was done via strength, and in Chomsky (2000a, 2001) via the EPP diacritic, which indicates that certain heads need Specifiers. In the above approach, generalized EPP effects follow from the *u*K of the moving element, which is independently needed even in Chomsky's EPP system. The interesting twist of the analysis is that the effect is stated as a property of the moving element, not the target.

It is also worth noting the restrictiveness of the above system. Thus, marking the K feature in (9) uninterpretable on Y will always lead to movement of Y to XP, i.e. (p. 333) it will result in Move, while marking it uninterpretable only on X will always lead to Agree.

(9)

X … Y	
K	K

This, however, brings us to a difference between the Chomsky/Bošković systems. Under both approaches a probe X, which initiates an Agree operation, must have a *uK*. Without a *uK*, there would be no reason for X to probe. Since there is no need for it, X cannot probe, given LR. In Chomsky's system, (10) then invariably crashes, since *u*K of Y cannot get checked.

(10)

X ….. Y	
*i*K	*u*K

This is not the case in Bošković's system, whereY would move to SpecXP and probe X from there, checking *u*K. (10) thus yields different results in Chomsky's and Bošković's systems. Below, I discuss one case of this type, which favors Bošković's system.

However, the most important difference between Chomsky (2000a) and Bošković (2007) for our purposes concerns the driving force of movement: while for Chomsky movement is target-driven, for Bošković it is moving-element-driven. We have already seen one argument for the latter: it concerns the case where X must start moving before its target enters the structure. The target cannot drive movement in such cases for a very simple reason: it is not there.

Multiple *wh*-fronting (MWF), an example of multiple movement to the same position, provides us with another relevant test.

Consider (11):

(11)

a.

*Koj	vižda	kogo?
who	watches	whom

b. Koj kogo vižda?
(Bulgarian)

Bošković (1999) discusses how MWF constructions can be handled within a target-driven (TD) and a moving-element-driven system (MD). In TD, we need to adopt the Attract 1-F/Attract all-F distinction, where Attract 1-F heads attract only one element bearing feature F, while Attract all-F heads attract all elements bearing feature F. Interrogative C in English is an Attract 1-F head, and in Bulgarian an Attract all-F head, attracting all *wh*-phrases. In MD, all *wh*-phrases in Bulgarian are obligatorily specified with the *u*K feature that drives *wh*-fronting.[7] MWF is then implemented as follows within the two systems.

(p. 334) **(12)**

a. TD: Attract all-F C head.

b. MD: each *wh*-phrase has *u*K.

Suppose we try to implement optional MWF. In TD, the C head would optionally have the relevant Attract all-F property, while in MD, *wh*-phrases would optionally have the relevant *u*K. This provides us with a very interesting tool to tease apart the two systems. Consider a sentence with four *wh*-phrases. In MD, we could give a *u*K to only two *wh*-phrases, which would result in two wh-phrases undergoing fronting and two remaining in situ. I will refer to this pattern as partial MWF. The pattern is impossible in TD: the C head either has the Attract all-F property, in which case all *wh*-phrases front, or it doesn't, in which case they all stay in situ. Crucially, partial MWF is impossible in this system.[8] The question is then whether there are languages that allow partial MWF. Surányi (2006) claims that Hungarian is exactly such a language.

(13)

(Mondd el)mikor	ki	tévesztett	össze	kit	kivel.
tell-imp prt when	who-nom	confused-3sg	prt	who-acc	who-with
‘(Tell me) who confused who with who when.’			(Balász Surányi, p.c.)		

Partial MWF thus provides an argument for MD.

Another argument is provided by quantifier raising (QR). QR is somewhat controversial, but if it does exist it provides a strong argument for MD. Suppose QR involves IP adjunction. QR must be driven by the moving element. Clearly, there is nothing about I that would require adjunction of a quantifier. On the other hand, under the QR analysis quantifiers are supposed to be uninterpretable in situ. It must then be that an inadequacy of the quantifier, i.e. the moving element, rather than the target, i.e. I, drives QR.

There is a bit of a complication in the technical implementation of QR, though. The standard assumption that quantifiers are uninterpretable in situ can be interpreted as indicating that they have a *u*K which makes them uninterpretable in situ, requiring movement. The movement can either eliminate the *u*K, in a way releasing quantifiers for interpretation,[9] or we could complicate the feature-checking system by assuming that as a result of movement, the uninterpretable feature becomes interpretable (i.e. it is interpretable only in certain positions).

Another argument for MD is provided by Fitzgibbons (2010), who discusses negative concord in Russian, where she argues all negative concord items (NCIs) must move to the negative head. What could be driving this movement? In TD, it would be a property of negation, and in MD a property of the moving elements, (p. 335) i.e. NCIs. Fitzgibbons argues that it cannot be the former, since negation does not require NCIs; after all negation can occur without NCIs. Therefore, it must be a property of the NCIs themselves; in fact, in contrast to negation, which can occur without NCIs, the NCIs cannot occur without negation. The NCI movement then must be driven by the moving elements. This provides us with another argument for MD.

15.2.2 Freezing effects and Last Resort

I now turn to a discussion of freezing effects, where LR considerations are crucially involved. Consider again the issue of what drives successive-cyclic movement. Since in Chomsky's (2000a) system movement is driven by the EPP property and Agree is a pre-requisite for movement, successive-cyclic movement always involves feature-checking. This means *what* must undergo feature-checking with *that* in (4). On the other hand, in Bošković's (2007) system the reason why *what* in (4)moves to the embedded SpecCP is to avoid being sent to Spell-Out when the embedded IP, a phase complement, is sent to Spell-Out. This would freeze *what* for further movement, as a result of which the *u*K driving *wh*-movement would remain unchecked, given that a *u*K can only be checked if it serves as a probe. Although under this analysis successive-cyclic movement is still in a sense feature-checking-driven, since without it the *u*K of the moving element would remain unchecked, there is no feature-checking in intermediate positions—*what* and *that* do not undergo feature-checking.[10]

Bošković (2002a, 2007)and Boeckx (2003) provide a number of arguments that there is indeed no feature-checking under successive-cyclic (A and A′) movement. Isummarize here one argument from Bošković (2002a).

Lobeck (1990) and Saito and Murasugi (1990) note that functional heads can license ellipsis of their complement only when they undergo Spec-Head agreement (p. 336) (SHA), i.e. feature-checking (see Bošković 2010 for a deduction of this generalization). (14) shows that tensed I, *'s*, and +*wh*-C, which undergo SHA, license ellipsis, while the non-agreeing heads *the* and *that* do not.

(14)

a. John left and [$_{IP}$ Peter$_i$ [$_{I'}$ did ~~t_i leave~~]] too.
b. John's talk was interesting but [$_{DP}$ Bill [$_{D'}$'s ~~talk~~]] was boring
c. *A single student came because [$_{DP}$ [$_{D'}$ the ~~student~~]] thought it was important.
d. John met someone but I don't know [$_{CP}$ who$_i$ [$_{C'}$ C ~~John met t_i~~]].
e. *John believes that Peter met someone but I don't think [$_{CP}$ [$_{C'}$ that ~~Peter met someone~~]].

Significantly, intermediate C cannot license ellipsis of its IP complement.

(15) *John met someone but I don't know who$_i$ Peter said [$_{CP}$ t_i [$_{C'}$ that ~~John met t_i~~]].

This can be easily accounted for if passing through an intermediate SpecCP does not imply feature-checking (SHA) with the C. (15) then provides evidence against the feature-checking view of successive-cyclic movement, where *that* would undergo SHA in (15), just as in (14d) and in contrast to (14e).[11]

What about languages with overt reflexes of agreement with intermediate heads under *wh*-movement? As noted in Boeckx (2004), it is not clear that there are languages with true intermediate *wh*-agreement. In many languages of this type, *wh*-agreement is only indirect: instead of a *wh*-phrase directly agreeing with an intermediate head, *wh*-movement induces special agreement between intermediate verbs and intermediate complementizers. I refer the reader to Bošković (2008a) for an analysis of this pattern that does not involve intermediate feature-checking,[12] and turn to a much better candidate for such successive-cyclic movement, namely Kinande. In Kinande the morphology of the C covaries with the morphology of the *wh*-phrase.

(16)

a.

IyondI	y0/	ABahI	Bo	Kambale	alangIra
who.1	that.1	who.2	that.2	Kambale	saw
'Who did Kambale see?'					

b.

EkIhI	ky0/	EBIhI	By0	Kambale	alangIra
what.7	that.7	what.8	that.8	Kambale	saw
'What did Kambale see?'				(Rizzi 1990b)	

The agreement occurs with displaced *wh*/focus phrases and can be found in every clause on the path of movement.[13]

(p. 337) **(17)**

[ekihi kyo	Kambale a.si	[nga.kyo	Yosefu
what wh-agr(eement)	Kambale agr.know	C.wh-agr	Joseph
a.kalengekanaya	[nga.kyo	Mary'	a.kahuka__]]]
agr.thinks	C.wh-agr	Mary	agr.cooks
'What did Kambale know that Joseph thinks that Mary is cooking?'	(Schneider-Zioga 2005)		

However, Boeckx (2004) suggests an iterative prolepsis account of Kinande long-distance *wh*-dependencies, analyzing traditional long-distance *wh*-movement from (18a) as in (18b), where the apparent argument of the lower V is generated as a matrix clause dependent that undergoes local *wh*-movement, binding a null element that also undergoes local *wh*-movement. Instead of a single, successive cyclic *wh*-movement, where the *wh*-phrase agrees with two Cs, we then have two local *wh*-movements, with different elements agreeing with the two Cs. In each case the C agrees with the head of a chain; (18b) does not involve true intermediate feature-checking.

(18)

a. $[_{CP}$ Op_i $[_{CP}$ t_i $[t_i$

b. $[_{CP}$ Op_i t_i $[_{CP}$ Op_i $[t_i$

Schneider-Zioga (2005) conclusively shows that Kinande does not have true long-distance A′-movement. Consider (19).

(19)

a.

ekitabu	$kiwe_{j/k}$	ky'	obuli $mukolo_j$	a.kasoma__kangikangi.	
book	his	wh-agr	each student	agr.reads	regularly
'(It is) His_j book that [every $student_{j/k}$] reads regularly.'					

b.

ekitabu $kiwe_{k/*j}$		kyo	ngalengekanaya $[_{CP}$	nga.kyo
book	his	wh-agr	I.think	C.wh-agr
[obuli $mukolo]_j$		akasoma _ kangikangi.		
every student		read	regularly	
'(It is) $His_{k/*j}$ book that I think [every $student]_j$ reads regularly.'				

c.

ekitabu	kiwe$_{k/*j}$	kyo	[obuli mukolo]$_j$	alengekanaya
book	his	wh-agr	everystudent	agr.think
[$_{CP}$ nga.kyo	nganasoma _ kangikangi]			
C.wh-agr	I.read	regularly		
‘(It is) His$_{k/*j}$ book that [every student]$_j$ thinks I read regularly.’				

(19a) shows local A′-extraction allows reconstructed interpretation. However, reconstruction is impossible with a long-distance dependency. Under the standard view of reconstruction that ties reconstruction to movement, we are led to conclude that the focused element undergoes movement from its θ-position to SpecCP in (19a), but not (19b-c). (19b-c) then indicate Kinande does not have true long-distance A′-movement.

Consider also Schneider-Zioga's (20)–(21).

(p. 338) **(20)**

*omukali	ndi	yo	wasiga [$_{island}$	embere __ wabuga]	
woman	who	wh-agr	you.left	before	spoke
‘Which woman did you leave before (she) spoke?’					

(21)

omukali	ndi	yo	wasiga [$_{island}$	embere	Kambale	anasi
woman	who	wh-agr	you.left	before	Kambale	knew
[$_{CP}$	ko.yo	__	wabuga]]			
	C.wh-agr		spoke			
‘Which woman did you leave before Kambale knew that (she) spoke?’						

(20) is unacceptable due to extraction from an adjunct. Significantly, (21), where the extraction site is embedded within an agreeing complementizer clause, is acceptable. This conclusively shows that the *wh*-phrase in (2i)does not undergo *wh*-movement to the matrix clause from the gap site.

I therefore conclude that Kinande agreeing long-distance A′-movement constructions do not involve a *wh*/focus phrase moving clause-to-clause, with a single *wh*/focus phrase undergoing agreement with more than one C.[14]

We have seen that a *wh*-phrase undergoing successive-cyclic movement does not undergo feature-checking with intermediate heads. Kinande shows that intermediate Cs actually can undergo agreement. What is, however, not possible is that after undergoing agreement with an intermediate C, a *wh*-phrase moves and establishes an agreement relation with another C. Those intermediate Cs in Kinande are really final Cs, since once a *wh*-phrase moves to SpecCP undergoing agreement with the C it is frozen in this position. The most straightforward way of interpreting this is that feature-checking for the *u*K involved in *wh*-movement is possible only once; once a *wh*-phrase undergoes agreement for this feature it is frozen. A natural step to take is to generalize this to every feature, which in turn provides strong evidence against Chomsky's (2000a) view of successive-cyclic movement.

Consider again Chomsky's system, where Y must have a *u*K to be visible for movement. X and Y in (22) undergo F feature-checking, and as a reflex of this, the *u*K of Y is checked, which happens after movement of Y to SpecXP.

(22)

[$_{XP}$Y$_i$(goal)	X(probe)	t$_i$
*i*F	~~*u*F~~	
~~*u*K~~	~~EPP~~	

Successive-cyclic movement has forced Chomsky to complicate this system by adopting the concept of defective heads, which are defective in that they are unable (p. 339) to check off the feature of the goal that has made the goal visible for movement to the head in question.

As an illustration, consider *wh*-movement: for Chomsky, the embedded C in both (23)and (24) undergoes feature-checking with *what*. The difference is that the embedded C in (23) is not, and the embedded C in (24) is, a defective head. Consequently, only the C in (23) checks off the *u*K of *what*, freezing it for further *wh*-movement. Since the embedded C in (24) is defective, it does not check the *u*K of *what*, which can then move to another SpecCP.[15]

(23)

I wonder	what$_i$	C	Mary	bought	t$_i$.
	*i*F	~~*u*F~~			
	~~*u*K~~	~~EPP~~			

(24)

a. What$_i$ do you think [$_{CP}$ t$_i$ that Mary bought t$_i$]

b.

You think [$_{CP}$	what$_i$	that	Mary	bought	t$_i$]
	*i*F	~~*u*F~~			
	*u*K	~~EPP~~			

As noted in Bošković (2008a), under non-feature-checking approaches to successive-cyclic movement, it is not necessary to stipulate the defectiveness of intermediate heads with respect to feature-checking since such heads are not involved in feature-checking in the first place. In other words, if there is no feature-checking with intermediate heads, we do not need to assume some heads are defective regarding how they participate in feature-checking. We can then make the process of feature-checking completely uniform in that all feature-checking inactivates the moving element, deleting the *u*K that has made it active for movement. This immediately captures the freezing effect of agreement with C. If a *wh*-phrase moves to SpecCP and undergoes agreement even with an intermediate C like *that*, the *u*K that makes it active for *wh*-movement will be erased, freezing it in SpecCP. There are, then, two options to get legitimate long-distance *wh*-structures: (a) reanalyzing long-distance *wh*-dependencies as a series of local *wh*-dependencies, where the declarative C undergoes agreement with X in its Spec, freezing it in place; (b) a wh-phrase moves to the Spec of *that* without undergoing agreement with *that*, the movement being driven by the considerations from section 15.2.1. Kinande takes option (a) and English (b).

The above discussion should be generalized. It is not only that *wh*-movement (i.e. feature-checking movement to SpecCP) cannot feed another *wh*-movement. As shown in Bošković (2008c) and references therein, no instance of

A′-movement can feed another instance of A′-movement. Thus, Lasnik and Uriagereka (1988) observe that although it is standardly assumed that QR is clause bounded, many speakers (p. 340) allow *every problem* to have wide scope in (25a). Significantly, even for them *every problem* cannot have wide scope in (25b).

(25)
- **a.** Someone thinks that Mary solved every problem.
- **b.** Someone thinks that every problem, Mary solved.

Assuming *every problem* scopes over *someone* in (25a) as a result of QR into the matrix clause, (25b) indicates that topicalization cannot feed QR.

Grohmann (2003b) notes that *wh*-movement cannot feed topicalization based on (26), where *who* undergoes topicalization after *wh*-movement to SpecCP, with the comma intonation indicating a pause associated with topicalization.[16]

(26) *Who, does Mary detest?

Bošković (2008c) shows that focus movement cannot feed *wh*-movement based on MWF. I simply refer the reader to Bošković (2008c) and references therein for additional arguments that A′-movements like *wh*-movement, focus movement, topicalization, QR, and NPI movement cannot feed each other. Why is this the case? Bošković (2008c) argues that there is a general, operator (Op)-type feature that is shared by elements undergoing A′-movements. It is the Op-feature that makes a phrase visible for an operator-style (A′-) movement (a topic/focus/*wh*-phrase then has *i*Top/*i*Foc/iWH and *u*Op). Given that there are no defective heads, once a phrase undergoes feature-checking A′-movement, its Op-feature is deleted, as a result of which the phrase cannot undergo another A′-movement.

Bošković (2008a) (see also Rizzi 2006a) argues that the freezing effect is even more general. Above, we have come close to saying that no instance of feature-checking movement can feed another instance of feature-checking movement. The only feeding relation still allowed involves feature-checking A-movement feeding feature-checking A′-movement. There is evidence that even this is disallowed, which gives us (27) within Bošković's (2007) system (more theory-neutral, X undergoes feature-checking movement only once).[17]

(27) X probes only once (i.e. X undergoes feature-checking as a probe only once). Consider Q-float under *wh*-movement in West Ulster English (WUE).

(28)
- **a.** Who$_i$ was arrested all t$_i$ in Duke Street?
- **b.** *They$_i$ were arrested all t$_i$ last night. (McCloskey 2000)

(p. 341) WUE allows (28a) but disallows (28b). McCloskey argues that *who* in (28a) must move to SpecCP without moving to SpecIP, the reasoning being that if *who* were to move to SpecIP in (28a), it would be impossible to account for the contrast in (28). Whatever rules out movement to SpecIP in (28b) should also rule it out in (28a). (28) shows that what is standardly assumed to happen in subject *wh*-questions—the subject moves to SpecIP and then to SpecCP—actually does not happen: the subject moves directly to SpecCP. Notice now that on the ‘standard’ derivation (28a) would involve feature-checking A-movement feeding feature-checking A′-movement, violating (27) (see also (30)).[18]

There are two questions to answer now. How is the requirement that SpecIP be filled in English satisfied in (28a), given that *who* never moves to SpecIP. Does (27) follow from anything?

Consider the first question within Bošković's (2007) system. Recall that Bošković (2007) dispenses with the EPP: EPP effects follow from case considerations. *John* in (29) has *u*Case.

(29) [$_{vP}$ John left]

Since *u*K must be a probe, *John* must move to a position c-commanding the case-checker (I). Given Shortest Move, *John* moves to the closest position c-commanding I, SpecIP.[19] In principle, *John* could move to SpecCP instead of SpecIP. Both movements would result in the checking of all relevant features. However, movement to SpecIP is preferred by Shortest Move, which favors the shortest movement possible. Consider now (28a). If *who* moves to SpecIP, its *u*Case will be checked. However, given (27), its *u*Op-feature will never get checked since *who* will be prevented from moving to a position *c*-commanding C. This derivation therefore crashes. The derivation is then irrelevant for economy comparison, including Shortest Move, which compares only possible derivations. Notice now that if, instead of SpecIP, *who* moves to SpecCP, from this position *who* can probe both C and I, checking both its Case and Op-feature. Movement to SpecCP is then the only possibility.

(p. 342) Consider now Kinande, where canonical subject/object agreement are impossible when the subject/object undergo *wh*-movement (Schneider-Zioga 1995). This can be straightforwardly captured in the above system if we make the natural assumption that canonical subject/object agreement are triggered in Kinande when the subject/object probe I/v from SpecIP/SpecvP respectively.

(30)

a. [$_{\mathrm{IP}}$ Subject I-agreement]
b. [$_{\mathrm{vP}}$ Object v-agreement]
c. [$_{\mathrm{CP}}$ Wh-Subject [$_{\mathrm{IP}}$ I-(*agreement)]]
d. [$_{\mathrm{CP}}$ Wh-Object [$_{\mathrm{IP}}$ [$_{\mathrm{vP}}$ v-(*agreement)]]

Since under the current analysis subject and object undergoing *wh*-movement probe both C and I/v from SpecCP, it follows that canonical agreement cannot co-occur with *wh*-movement.[20]

Inow turn to adeduction of (27). Consider the line of reasoning employed in the discussion of freezing effects above. Suppose X must have a *u*K to make it active for movement Y. Once X undergoes feature-checking movement to a Y, the *u*Kwill get checked off so that X cannot undergo another Y-movement. As discussed above, the freezing effect can be generalized to all A′ feature-checking by generalizing the *u*K that is involved in A′-movement-checking. If the same feature of the moving element is checked under all instances of A′-movement, once X undergoes feature-checking A′-movement, the relevant feature will get checked off, freezing X for further A′-movement. (27) then suggests a further generalization: it is the same feature of the moving element that is checked in all instances of movement, A or A′. As noted in Bošković (2008a), this means that once X undergoes any feature-checking movement it will no longer be able to undergo another feature-checking movement. This requires changing the way we have been treating movement. We can no longer consider the specific features like *u*Case or *u*Op to be the driving force of movement, since the driving force needs to be generalized. What we need is a general property X which can be given to any element when it enters the structure. This general property is tied to probing: it indicates a need to function as a probe and **(p. 343)** is satisfied under successful probing.[21] An element A marked with X (which cannot probe in situ) would move to the edge of a phase to attempt a probing operation: if A successfully undergoes probing, X is deleted, freezing A in place. If A fails to probe due to the lack of a goal (so it still has X), it moves to the higher-phase Spec to attempt probing again. The X property is then used to drive successive-cyclic movement *(instead of u*K, as in Bošković 2007). Another way of looking at this is as follows. Suppose X is PF uninterpretable (after all, the property 'I need to function as a probe' is not a PF-related property). This means sending an element with the X property to Spell-Out would cause a PF crash. Assuming that what is sent to Spell-Out is the complement of a phase head, A in (31) will have to move to the Spec of the phase head B to avoid being sent to Spell-Out, which would cause a crash.

(31)

W[$_{\mathrm{BP}}$	B	A
	K	*u*K
		X

Successive-cyclic movement works as before, without feature-checking with intermediate heads. When A moves to SpecWP it successfully probes W, checking *u*K and deleting X, which is tied to feature-checking under probing.

(32)

A	W	[BP
~~uK~~	K	
~~X~~		

The result of this system is that A can move to probe only once. Once A undergoes feature-checking movement, X is deleted, freezing A in place. (27) is then deduced.[22] Rodríguez-Mondoñedo (2007) observes a rather interesting prediction of this system. Consider (33).

(33)

X	Y	Z
K	F	*u*K
F		*u*F

Z in(33) has two uninterpretable features, which need to be checked, more precisely, receive a value.[23] The closest valuator for F is Y. In the absence of freezing effects, we would expect Y to value the F feature of Z, which in Bošković's (2007) system would happen after Z moves to SpecYP. However, given the freezing effect, probing for F from SpecYP would freeze Z, leaving its K feature unvalued. In the above system, we **(p. 344)** would therefore expect Z to move to SpecXP, probing for all its features from there. Locality then requires that X rather than Y values the F feature of Z. Rodríguez-Mondoñedo argues that this rather interesting prediction of the freezing system, where the closest valuator (Y) unexpectedly fails to do the job, is borne out on the basis of some previously unexplained instances of obligatory *a*-object marking with Spanish inanimates.

15.2.3 Last Resort and Agree

Inow turn to theeffects of LR for Agree, focusing on the claim that X can only be a probe if it has a *u*K.

Chomsky (2001) argues that in addition to the interpretable/uninterpretable distinction, we need a valued/unvalued distinction, where some features are fully valued lexically while others receive their value during the derivation. Consider Serbo-Croatian (SC) (34) (*kola* is a pluralia tantum).

(34)

a.

Zelena	kola	su	kupljena.
green.fem	car.fem	are	bought.fem
‘The green car was bought.’			

b.

Zeleno	auto	je	kupljeno
green.neut	car.neut	is	bought.neut

c.

Zeleni	automobil	je	kupljen.
green.masc	car.masc	is	bought.masc

The gender of the adjective and the participle depends on the gender of the noun. *Green* can be feminine, neuter, or masculine; its gender depends on the noun it modifies. As noted by Pesetsky and Torrego (2007), the dependence of the gender specification of adjectives and participles on the syntactic context in which they occur can be easily captured if they are lexically unvalued for gender: they receive their gender value after undergoing agreement with a noun that already has a valued gender specification. In contrast to the adjective/participle in (34), nouns like *kola, auto*, and *automobil* have a fixed gender specification: *kola* is always feminine, *auto* neuter, and *automobil* masculine. The most straightforward way of capturing this is to assume that nominal gender is lexically valued; in contrast to adjectives and participles, nouns do not receive their gender value during syntactic derivation, hence their gender value does not depend on their syntactic context.[24]

Since SC gender is quite clearly grammatical (it depends on the declension class a noun belongs to), we also have here evidence for the existence of valued (p. 345) uninterpretable features, a possibility that is disallowed in Chomsky's (2000a, 2001) system essentially by a stipulation. Allowing for the existence of valued uninterpretable features also allows us to simplify the feature-checking process. Since in Chomsky's system uninterpretable features are always unvalued, the system does not allow feature-checking between two uninterpretable features. Feature-checking is supposed to result in valuation of unvalued features. If both the probe's and the goal's feature is unvalued, their feature-checking cannot result in valuation. Disallowing the possibility of checking two uninterpretable features against one another forces Chomsky quite generally to tie checking of an uninterpretable feature F of a goal to checking of a different uninterpretable feature K of its probe (note that interpretable features, which are always valued for Chomsky, cannot serve as probes due to LR; since there is no need for them to initiate probing they are not allowed to do it), which makes feature-checking rather cumbersome and leads to a proliferation of features involved in checking. Thus, since (35a–b) cannot result in the checking of the K feature of Y ((35a) because, being unvalued, the *u*K of X cannot value the *u*K of Y, and (35b) because X cannot function as a probe due to the lack of uninterpretable features), Chomsky is forced to posit (35c), where the *u*K of Y is checked as a reflex of F feature-checking. This kind of reflex checking considerably complicates the feature-checking mechanism and leads to a proliferation of features involved in checking (we cannot simply have K-feature-checking in (35); rather, we need to assume an additional feature F is involved in feature-checking between X and Y).

(35)

a.

X	Y
*u*K	*u*K

b.

X	Y
i K	*u*K

c.

X	Y
*u*F	*i*F
	*u*K

Allowing valued uninterpretable features enables us to simplify the feature-checking relations from (35c). In particular, (35a) is now allowed, if one of the K features is valued.[25]

Given this much background, let us reconsider the question of what drives Agree. It is standardly assumed that semantics cannot deal with uninterpretable features, hence such features need to be eliminated before entering semantics. The elimination takes place through feature-checking. A question, however, arises why such features simply could not be deleted, in which case they would not need to be checked. It is argued in Bošković (2009a) that such features indeed can be deleted (p. 346) without checking, but only if they are valued (see also Chomsky 2001). In other words, valuation is a prerequisite for deletion of uninterpretable features. But if a valued *u*K can simply be deleted, there is no need for it to undergo feature-checking (see below for evidence to this effect). Then, given LR, a valued *u*K cannot function as a probe. On the other hand, an unvalued *u*K can function as a probe, since such elements do induce a crash, hence there is a need for them to undergo Agree. Pesetsky and Torrego argue that, just as uninterpretable features can be either valued or unvalued, as we have seen above, interpretable features can also be either valued or unvalued. As an example of unvalued interpretable features, they give the Tense feature of the Tense node; for them it is the locus of semantic tense interpretation, but its value depends on its syntactic context, i.e. the verb it co-occurs with. They also implement clausal typing in terms of an unvalued interpretable feature of C. It seems natural to assume that an unvalued *i*K would still be a problem for semantics; i.e. semantics would know what to do with an *i*K only if K has a value (see also Pesetsky and Torrego 2007). Unvalued *i*Ks can then also function as probes. From this perspective, what drives Agree is valuation: only unvalued features can function as probes. (36) then shows which contexts can yield a legitimate Agree relation, where X is a probe and Y its goal.[26]

(36)
- **a.** X[*unval/u*K]…Y[*val/u*K]
- **b.** X[*unval/i*K]…Y[*val/i*K]
- **c.** *X[*val/u*K]…Y[*val/u*K]
- **d.** *X[*val/i*K]…Y[*val/i*K]
- **e.** *X[*unval/u*K]…Y[*unval/u*K]
- **f.** *X[*unval/i*K]…Y[*unval/i*K]
- **g.** *X[*val/u*K]…Y[*unval/u*K]
- **h.** *X[*val/i*K]…Y[*unval/i*K]

Agree cannot take place between X and Y in (36c–d) due to LR(there is no reason for X to probe). The same holds for (36g–h) in Chomsky's system (2001), though the structures would be treated differently in Bošković's (2007) system, as discussed below. Finally, the problem with (36e–f) is that the unvalued features of X and Y cannot be valued. An innovation of this system is that it allows interpretable features to trigger feature-checking (see also Pesetsky and Torrego 2007), which was not possible in Chomsky (1995c), where uninterpretability was the trigger for feature-checking. [27] Also, in contrast to Chomsky (2000a), two uninterpretable features can undergo feature-checking, as long as the probe is unvalued and the goal valued. However, even uninterpretable features fail to trigger Agree if they are valued.

(p. 347) Another important property of the valuation-driven system is that valued uninterpretable features do not need to be checked, given that they can be deleted. This is a departure from Chomsky (1995c), where all uninterpretable features have to undergo checking. (On the other hand, while in Chomsky 1995c interpretable features do not need to undergo checking, in the above system interpretable features do need to undergo checking if they are unvalued.) There is strong evidence that valued unintepretable features indeed do not need to undergo checking. Consider first conjunct gender agreement in SC.

(37)

Uništena	su	sva	sela	i	sve	varošice.
destroyed.neut	are	all	villages.neut	and	all	towns.fem
‘All villages and all towns were destroyed.’						

The participle in (37) agrees in gender (i.e. undergoes feature-checking for gender) with the first conjunct, which means the second conjunct is not involved in gender feature-checking. Notice also that the conjunct does not have default gender, which is masculine in SC. Its non-default gender feature simply goes unchecked in (37). This is exactly what is expected given the above discussion: the gender feature of the noun is uninterpretable, but valued. As a result, it can be deleted (so that it does not enter semantics, where it would cause a Full Interpretation violation) without checking.

Another relevant case concerns case-checking. Case-checking is rather cumbersome in Chomsky's (2000a) system. Case is quite clearly uninterpretable on both the traditional case assigner (e.g. Tense) and the assignee (NP), i.e. on both the probe and the goal. Since, as discussed above, Chomsky disallows Agree between two uninterpretable features, he cannot have direct case feature-checking between T and NP. Rather, he has to complicate the system by appealing to the notion of reflex feature-checking, where case-checking is tied to the checking of another feature. Thus, for Chomsky, phi-features of T in (38) probe the NP, and somehow as a reflex of this phi-feature-checking the case feature of the NP gets checked. The ‘somehow’ here is rather mysterious, given that T does not even have a case feature for Chomsky.

(38)

T	NP
*u*Phi	*i*Phi
	*u*Case

The current system makes possible a much more natural approach to case-licensing, where both Tense and the NP have a case feature, in line with the attempt to eliminate the undesirable concept of reflex feature-checking. The case feature of both Tense and the NP is uninterpretable. Furthermore, since (finite) T always governs nominative, and the case of NPs depends on the syntactic context in which they occur, T's case is valued and NP's case unvalued.

(39)

T	NP
*val/u*Case	*unval/u*Case

(p. 348) Case-licensing in (39) can proceed without any problems and without reflex feature-checking, but crucially only in Bošković's (2007) system. In the target-driven system of Chomsky (2000a), even if the above assumptions regarding valuation are adopted so that valuation drives Agree, Agree would fail in (39)because T could not function as a probe due to LR. On the other hand, in Bošković's (2007) system, the NP would move to SpecTP and then probe T from this position. Since the NP has an unvalued case feature, it can function as a probe.

The above account makes another prediction. Since the case feature of traditional case-assigners is valued, which means it can be deleted even without checking, it does not have to undergo checking. This is in contrast to the case feature of NPs, which is unvalued, hence needs to be checked. This amounts to saying that the traditional Case Filter holds, but the Inverse Case Filter does not hold. There is strong empirical evidence that this is indeed correct. It is pretty clear that the Case Filter holds. As for the Inverse Case Filter, all attempts to enforce it (e.g. Bošković 2002a, Epstein and Seely 1999) have come up short against persistent empirical problems which pretty clearly indicate that traditional case-assigners do not have to check their case, which means that the Inverse Case filter does not hold. For example, the existence of verbs that assign case only optionally, as in (40), goes against the spirit of the Inverse Case Filter.

(40)

a. John laughed.
b. John laughed himself silly.
c. Mary is dressing (herself).
d. Peter is eating (apples).

Slavic genitive of quantification/negation also provides evidence against the Inverse Case Filter (see Franks 2002). In a number of Slavic languages, verbs that assign structural accusative fail to assign it when their object is a higher numeral NP. (*Kola* in SC (41b), which must bear genitive, receives its case from the numeral.) The same happens when a verb is negated, as illustrated by Polish (42b), where genitive of negation is obligatory. (There are similar arguments against obligatory assignment of nominative as well as some lexical cases; see Franks 2002.)

(41)

a.

On	kupuje	kola.
he	buys	car.acc

b.

On	Kupuje	pet	kola.	
He	buys	five	cars.gen	(SC)

(42)

a.

Janek	czytał	książkę.
Janek	read	books.acc

b.

Janek	nie	czytał	Ksjażki.	
Janek	neg	read	books.gen	(Polish)

I conclude, therefore, that the valuation-driven version of Bošković's (2007) system not only captures case-licensing without additional assumptions that were required (p. 349) in Chomsky's (2000a) system, but also accounts for the fact that the Case Filter, but not the Inverse Case Filter, holds (i.e. only the former is enforced).[28]

15.3 Lexical insertion/pure Merge and Last Resort

I now turn to the question of whether lexical insertion, or more generally, pure Merge, should be subject to LR. Chomsky (1995c) assumes that no aspect of lexical insertion, including pure Merge, is subject to LR, the underlying assumption being that if cost is assigned to lexical insertion, the cheapest thing to do would always be nothing, which means no lexical insertion would ever take place, resulting in silence. On the other hand, Chomsky (2000a) suggests that pure Merge is subject to LR, and is motivated by selectional requirements. The assumption leads to a considerable enrichment of the theory of selection, since all lexical insertion/pure Merge now has to be driven by selection. This is unfortunate, since selection was previously shown to be close to eliminable.[29] In Bošković (1997) I took the position that falls in between Chomsky's (1995c) and (2000a) positions: only pure Merge of functional elements is subject to LR.[30] As discussed below, the literature contains a number of appeals to economy-of-representation principles intended to ban unnecessary projections (see (44)). Interestingly, in actual practice they are all applied only to functional elements; they are used to ban only unnecessary functional structure. This 'accident' can be made more principled by taking the position that only pure Merge of functional elements is subject to LR. The functional/lexical category distinction makes sense given that lexical elements determine what we want or choose to say, and functional elements merely help us build legitimate grammatical structures. Bošković (1997) appeals to the natural assumption that the latter (building legitimate grammatical structures), but not the former (what we want or choose to say), is subject to economy principles to justify subjecting only pure Merge of functional elements to LR. Functional elements are then inserted into the structure only to the extent that they are necessary to build legitimate structures. Another way to approach this issue would be to assume that only functional categories are selected, a natural consequence of which would be to require only (p. 350) pure Merge of functional elements to be motivated by selectional requirements. Bošković (2004a, 2008b) shows the assumption that only pure Merge of functional projections is subject to LR enables us to deduce a rather interesting generalization concerning scrambling.[31]

(43) Only languages without articles may allow scrambling.

SC, Latin, Japanese, Korean, Turkish, Hindi, Chukchi, Chichewa, and Warlpiri all have scrambling and lack articles. Particularly interesting are Slavic and Romance. Bulgarian, for example, has noticeably less freedom of word order than SC. Also, all modern Romance languages have articles and lack scrambling, while Latin lacked articles and had scrambling. I argued in Bošković (2008b, 2010) that article-less languages do not project DP; the traditional noun phrase in such languages is an NP. I also adopted Bošković and Takahashi's (1998) approach to scrambling, on which scrambled elements are base-generated without feature-checking in their surface position, and then undergo LF lowering to the position where they receive case/θ role. The main goal of this approach was to make scrambling conform to LR. Scrambling is standardly treated as an optional overt movement operation that takes place for no reason at all, which should violate LR. Under Bošković and Takahashi's (1998) approach, the optional, LR-violating overt movement is replaced by obligatory LF movement that conforms with LR. Now, given that the traditional NP is DP, a functional category, in non-scrambling languages, and NP in scrambling languages, inserting it into the structure must have independent motivation (i.e. involve feature-checking/satisfaction of selectional properties) in non-scrambling languages, but not in scrambling languages. Since scrambling is pure Merge that does not involve feature-checking/ satisfaction of selectional requirements under Bošković and Takahashi's analysis, it is then possible only in NP languages.

15.4 The ban on superfluous structure: Economy of Representation and Last Resort

The above discussion of LR as it applies to pure Merge has bearing on Economy of Representation principles that ban superfluous structure, which can be restated in terms of LR if pure Merge is subject to LR.

(p. 351) A number of authors have proposed principles whose goal is to ban superfluous symbols from representations.[32]

(44)

a. The Minimal Structure Principle (MSP)
Provided that lexical requirements of relevant elements are satisfied, if two representations have the same lexical structure, and serve the same function, then the representation that has fewer projections is to be chosen as the syntactic representation serving that function (Law 1991, Bošković 1997).
b. At any point in a derivation, a structural description for a natural language string employs as few nodes as grammatical principles and lexical selection require (Safir 1993).
c. α enters the numeration only if it has an effect on output (Chomsky 1995c).

The basic idea behind (44) is that superfluous projections are disallowed. Thus, the MSP requires that every functional projection be motivated by the satisfaction of lexical requirements (such as selectional requirements and checking of features specified in lexical entries). Among other things, MSP has been argued to force the IP status on control infinitives and finite relatives as well as declarative complements not introduced by *that* (see Bošković 1997). Such clauses, which are potentially ambiguous in that they can be either CPs or IPs, are disambiguated by the MSP in favor of the IP option, the null operator being IP-adjoined in the relatives in question, essentially undergoing topicalization.[33]

(45)

a. John tried [$_{IP}$ PRO to leave]

b. the man [$_{IP}$ Op$_i$ [$_{IP}$ John left t$_i$]]

c. We [$_{VP}$ think [$_{IP}$ John left]]

Bošković (1997) gives a number of arguments for the IP analysis. Thus, the analysis accounts for the ungrammaticality of short zero-subject relatives, which under this analysis reduces to the impossibility of short-subject topicalization (see Bošković 1997 for a uniform account of both of these).

(46) *the man [$_{IP}$ Op$_i$ [$_{IP}$ t$_i$ likes Mary]]

(47) *I think that [$_{IP}$ John$_i$, [$_{IP}$ t$_i$ likes Mary]][34]

The IP analysis also captures the contrast in (48), given Saito's (1985) claim that resumptive pronouns are not allowed in adjunction structures.

(p. 352) **(48)**

a. *The book [$_{IP}$ Op [$_{IP}$ I was wondering whether I would get it in the mail]]

b. The book [$_{CP}$ Op [$_{C'}$ that I was wondering whether I would get it in the mail]] (Kayne 1984)

Turning to declarative complements, the IP analysis provides a straightforward account of the Comp-trace effect, which has been a recurring problem for the CP analysis. The reason why (49a) does not exhibit a Comp-trace effect under the IP analysis is trivial: there is no Comp.

(49)

a. Who do you believe left?

b. *Who do you believe that left?

The analysis also accounts for the obligatoriness of *that* with topicalization.

(50)

a. [$_{IP}$ Mary, [$_{IP}$ John likes]]

b. Peter believes that [$_{IP}$ Mary, [$_{IP}$ John likes]]

c. *Peter believes [$_{IP}$ Mary, [$_{IP}$ John likes]]

Given that the embedded clause in (50c) is an IP and that topicalization involves IP adjunction, (50c) is ruled out because it involves adjunction to an argument, which is disallowed (Chomsky 1986a, McCloskey 1992, Bošković 2004b). The problem does not arise in (50a–b).

A rather interesting question addressed in Bošković (1997) is how (44) interacts with Chomsky's (1995c) ‘numeration’, which is defined as an array of lexical items that is mapped by the computational system into a linguistic expression. Chomsky's (44c) determines the numeration itself. This is problematic due to its globality. To determine the effects of (44c) we need to know PF/LF outputs. But the numeration, which is determined by (44c), must be present in the initial stage of the derivation. The problem can be solved if elements affected by (44) are not present in the numeration. Under (44a) all we need to do is define the numeration on lexical elements only. Under this view, only lexical elements are present in numerations.[35] Repeated access to the lexicon is then allowed to ensure that we have all functional elements that are necessary to build legitimate structures. Instead of positing (44a), we can then simply require that lexicon be accessed only when needed, i.e. when a certain functional category becomes necessary in structure-building. This amounts to assigning cost to merger of elements that are not taken from the numeration. Under this view, merger of such elements is subject to the ban on superfluous operations, i.e. LR. Moreover, we do not need to exempt lexical insertion from the numeration from LR: if derivations that do not exhaust numerations do not converge, inserting an element from a numeration into the structure is a step toward a well-formed derivation (see Collins 1997), in accordance with LR. A tacit assumption here is that selection of lexical elements into numerations is costless. (p. 353) Assigning cost to numeration formation, or trying to determine why one numeration is formed rather than another, would mean bringing the question of what we want to say into the domain of inquiry covered by the study of the working of the computational mechanism of human language. As Chomsky (1995c) observes, requiring the computational mechanism of human language to deal with the issue of what we choose to say and why we choose it would be no different from requiring a theory of the mechanism of vision to explain what we choose to look at and why we do it.

Under the above approach, the MSP can be dispensed with. Its effects are derivable from the ban on superfluous steps in a derivation, i.e. LR. This is desirable, since while the MSP has an element of globality LR applies locally. The representations that the MSP rules out in favor of more economical representations cannot even be built under the derivational approach, since they violate LR.

Notes:

(1) (2) involves A-movement from a CP, which is often assumed to be disallowed. See, however, Bošković (2007) and references therein for evidence that such movement is in principle possible, even in English.

(2) In what follows, I ignore vP as a phase for ease of exposition.

(3) Bošković (2007) shows the Activation Condition holds for Move as a theorem. As for Agree, I argue it does not hold for it.

(4) K is either checked as a reflex of F-feature-checking between W and Y (see below for discussion of reflex feature-checking) or W has a K feature that can check the K feature of Y. For ease of exposition, I adopt the latter option, returning to the issue below.

(5) The assumption has many empirical consequences; see Epstein and Seely (2006) and Bošković (2007).

(6) See Epstein and Seely (2006) for discussion of the traditional EPP in this context, which is generalized in Bošković (2007), with an exploration of a number of additional consequences and an extension to successive-cyclic and *wh*-movement. (Bošković's analysis of the traditional EPP is actually quite different from Epstein and Seely's. The latter crucially appeals to the Inverse Case Filter, which is dispensable in the former.)

(7) See Bošković (2002b) regarding the nature of this feature, which is not important for current purposes.

(8) As discussed in Bošković (1999), independently of the Attract all-F property that is responsible for MWF, in some MWF languages (e.g. Bulgarian) the C also has an Attract 1-F property that requires one *wh*-phrase to move. If the C here could have this Attract 1-F property it would be possible to force one *wh*-phrase to move, but not two.

(9) This could be done through valuation if the *u*K feature is lexically unvalued, given the discussion in section 15.2.3, where it is argued that only valued *u*Ks can be deleted.

(10) In this respect, Bošković (2007) represents a return to early minimalism, where successive-cyclic movement was not a result of feature-checking. Rather, it was a consequence of the Minimal Link Condition (MLC) (Chomsky and Lasnik 1993 and Takahashi 1994;revived in Bošković 2002a, Boeckx, 2003, Chomsky 2008a). The MLC forces X undergoing movement of type Y to stop at every position of type Y on the way to its final landing site independently of feature-checking. *What* in (1) then must pass through the embedded SpecCP on its way to the matrix SpecCP.

This analysis crucially relies on the Form Chain operation, where all relevant syntactic conditions, including LR and the Cycle, are stated with respect to the formation of chains, not chain links. Under this analysis, *what* in (1) starts moving only after *wh*-C enters the structure. The MLC forces formation of intermediate chain links. LR is satisfied since the formation of the whole chain, whose head is located in matrix SpecCP, has a feature-checking motivation. Since the whole chain extends the tree, the Cycle is also satisfied.

Chomsky (1995c) eliminates Form Chain, which has led to the abandonment of this analysis of successive-cyclic movement because with the elimination of Form Chain, formation of each chain link must satisfy LR and the Cycle. This means that *what* in (1)must move to the Spec of *that* before higher structure is built, and the movement must have independent motivation. As discussed above, Bošković (2007) and Chomsky (2000a) do not differ regarding the former, but do differ regarding how the latter requirement is satisfied, due to fundamentally different treatments of the issue where the formal inadequacy which drives movement is located.

(11) The argument extends to Chomsky's (2000a) system, where the SHA requirement would be restated as an EPP requirement.

(12) As discussed in Bošković (2008a), the analysis may be extendable to Irish.

(13) *Nga* occurs in the embedded clause because monosyllabic Cs are second position clitics.

(14) A modification of Boeckx's analysis is necessary to account for (19c): the focused NP should not even undergo local *wh*-movement, or the reconstruction would be possible. The contrast in (19a,c) indicates that local A′-movement is possible only from the θ-position, i.e. we are not dealing here with a proleptic object undergoing A′-movement. (17) then has a structure like (i), where only the lowest null element undergoes movement.

((i)) $[_{CP}$ Op_i $[_{CP}$ Op_i $[_{CP}$ Op_i t_i

(15) For actual features involved in feature-checking under *wh*-movement, which are not important for our purposes, see Bošković (2008c) (C/*that* may also have the K feature).

(16) *To Peter, what should Mary give* indicates the landing site of topicalization precedes SpecCP in matrix clauses.

(17) A similar claim is made in Rizzi (2006a). However, his treatment of the claim is quite different from the one developed below. For another approach that bans A-A′ feeding in a system quite different from the one adopted here, see Chomsky (2008a), who argues for the existence of parallel movement. (For ways of teasing apart the analyses in question, see Bošković (2008a, 2009b). The Kinande case in (30) actually cannot be captured within Chomsky's system. Moreover, the system does not ban A′–A′ feeding.)

(18) In Bošković's (2007) system, *who* would be a probe in both cases, probing C and I from SpecCP/SpecIP respectively.

(19) One argument for this system concerns (i).

((i)) *I know what John conjectured.

Conjecture is not a case assigner (cf. **John conjectured it*). A question, however, arises why *know* cannot case-license *what*, given that *know* has the ability to do that. (i) presents a serious problem for Chomsky's (2000a) system. It appears that nothing prevents establishment of a probe-goal relation between the matrix v and *what*, which should case-license *what*. From Bošković's (2007) perspective, (i) is straightforward: the derivation in question is blocked because *what* with its *u*Case must function as a probe. The only way this can be accomplished is if *what* moves to the matrix SpecvP. However, this derivation is blocked because *who* is located outside of its scope (embedded CP), which is disallowed (see Saito 1992). (i) thus provides evidence that case cannot be licensed in situ without movement to the case licensor (see Bošković 2007 for discussion of cases where case movement was previously assumed not to occur).

(20) The object would pass through SpecvP in (30d) due to considerations from section 15.2.1, but it would not probe v from there because of (27). The above analysis has many consequences, discussed in Bošković (2007). To mention one here, it captures the behavior of verbs like *wager*, which ECM *wh*-traces, but not lexical NPs (Postal 1974).

((i))
(a.) *John wagered Mary to be smart
(b.) Who did John wager to be smart?

Assuming overt object shift in English, Bošković (1997) argues that due to the presence of an additional VP shell with a filled Spec that occurs with this class of verbs, *Mary* cannot undergo A-movement to the matrix SpecvP in (i.a) without a locality violation. (i.b) is then straightforward in the above system, where *who* probes *wager* from the matrix SpecCP. Since there is no A feature-checking movement to the matrix SpecvP in (i.b), the locality problem that arose in (i.a) does not arise in (i.b).

(21) Giving X to an element Y without uninterpretable features, which then would not function as a probe, would

lead to a crash. But there is always the derivation on which Y does not get X.

(22) Natasha Fitzgibbons (p.c.) suggests an alternative deduction of (27). She suggests maximizing feature-checking under probing to the effect that if X probes, X must check all its *u*Ks. The A–A′ feeding relations, where X would first move to probe for *u*Case and then move to SpecCP to probe for *u*Op, are also ruled out under this approach, which means (27) is deduced. The deduction is consistent with the derivations discussed above. Thus, *who* in (28) still moves directly to SpecCP, probing for both the *u*Case and the *u*Op feature from there.

(23) In the current system, checking is interpreted in terms of valuation of unvalued features; see section 15.2.3. The technical implementation of checking has not been important until now.

(24) Recall that *kola* in (34a) is a pluralia tantum, i.e. its number is plural although it is interpreted as singular. This kind of lexical quirk also calls for full lexical specification of ϕ-features of nouns. As pointed out by Pesetsky and Torrego (2007), there are no pluralia tantum verbs or adjectives, which is not surprising if the ϕ-features of these elements are lexically unvalued: such treatment does not leave room for lexical quirks like the one exhibited by the number of the noun in (34a).

(25) See below, and Bošković's (2009a) analysis of SC gender, where the gender feature of both the gender probing head, which is responsible for participial gender, and the noun is uninterpretable, but unvalued only on the former.

(26) *Val* indicates valued and *unval* unvalued features.

(27) I am putting aside here strength, which was used to drive overt movement.

(28) As for default case, which clearly does not need to be checked, the most appropriate way to handle it is to assume that default case involves valued case on the NP, which means it does not need to be checked. Since the value of default case is fixed for each language for all constructions (i.e. it does not depend on syntactic context), it is clear that it should be valued. Since valued uninterpretable features do not need to be checked, we then also capture the fact that default case does not need to be checked (nouns with default case occur in environments where there is no plausible case-assigner).

(29) More precisely, it was shown to follow from the semantic properties of lexical items, which should not be driving syntactic computation; see Pesetsky (1982) and Bošković (1997).

(30) I am actually generalizing here the position I took regarding lexical insertion to pure Merge in general.

(31) Scrambling here is taken to be the kind of movement referred to as scrambling in Japanese, not German, whose 'scrambling' is a very different operation with very different semantic effects from Japanese scrambling. One of the defining properties of scrambling for the purpose of (43) is the existence of long-distance scrambling from finite clauses, which German lacks.

(32) For additional principles along these lines, see Grimshaw (1997), Speas (1994), Radford (1994).

(33) I assumed that complementizer *that* is nominal in nature, hence unaffected by the MSP.

(34) See Baltin (1982), Lasnik and Saito (1992), Rochemont (1989), and Saito (1985) for the IP adjunction analysis of topicalization and Lasnik and Saito (1992) for evidence that short-subject topicalization is disallowed. Thus, they observe that if short-subject topicalization were allowed we would expect that *John* and *himself* can be co-indexed in (ib), just as in (ia).

((i))

(a.) John$_i$ thinks that himself$_i$ Mary likes.

(b.) *John$_i$ thinks that himself$_i$ likes Mary.

(35) This seems natural if the contents of numerations are determined by what we want or choose to say, given that, as discussed above, this is determined by lexical elements.

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[–] Abstract and Keywords

This article analyzes instances of optional movement and how optionality can be motivated in a minimalist context. The discussion involves the notion of interpretation and the systems external to narrow syntax that further manipulate linguistic expressions. It takes up quantifier raising and scrambling, which appear to be quintessential optional operations. The discussion adheres to the idea that they are optional movement, and shows that by the application of Fox's economy condition on optional interpretation, we can predict which optional operations are well formed and which are not.

Keywords: optional movement, minimalism, interpretation, narrow syntax, linguistic expression, quantifier raising, scrambling

16.1 Introduction

Movement has been a major topic of research at every stage in the development of generative grammar. In GB, movement operations are thought to be entirely optional, Move α being able to move anything anywhere, anytime, which leads unavoidably to massive overgeneration. Independent universal principles extract from this overly generated set of strings the subset that constitutes the grammatical strings of a particular language. These independent principles make it possible to meet explanatory adequacy in that they ‘give a general theory of linguistic structure of which each [grammar of a particular language] is an exemplification’ (Chomsky 1955/75a: 77). In GB, this ‘general theory of linguistic structure’, or UG, is the Principles and Parameters approach, and it informs us how language acquisition proceeds from the initial state to the mastery of a language. This is a particularly attractive formulation in that we have, in theory, a description of UG's initial state—the state before parameters are set—which is a principal goal of linguistic theory. However, there is one problem. These so-called universal principles are often—perhaps always—a description of the problem. This is the basis for the minimalist program (MP). In MP, effort is made to rid the theory of any element that does not have a natural and independent justification. An attempt to live up to this ideal—although by no means the only possible approach—is to view operations not as (p. 355) optional as in GB but as strictly last resort (e.g. Chomsky 1995b). This reorientation naturally leads to the hope that there ought not be any unnecessary generation of strings of the kind we find in GB. Typically, movements take place in the presence of a formal feature in the structure—this feature enters into agreement with an item located elsewhere in the structure, and the item moves to where the feature resides. If all movements are to be characterized in this way, optional operations should never occur.[1] Nevertheless, there is a class of operations, quantifier raising (QR) in languages such as English and a subclass of scrambling in languages such as Hindi and Japanese, which appear to be truly optional. We need to formulate a theory of optional operations that is consonant with the tenets of Last Resort. There are operations such as heavy NP shift that appear to be just as optional as the two I mentioned, QR and (a subclass of) scrambling, but for this chapter, I will focus on these two because they appear to be closely matched in their properties and thus are open to a unified account. In general I will adopt Fox's (1995, 2000) approach to optional movement, showing its advantages, but at the same time fleshing out the issues in this approach in order to sharpen the assumptions and expand the range of empirical phenomena that it can account for. In so doing, I will be particularly informed by Johnson (2000b), whose work extends the work on quantifier scope in MP by Kitahara (1996), Hornstein (1995), and Pica and Snyder (1995), which we will take up briefly in section 16.3.

16.2 Some preliminary observations

Let us begin with some familiar points from the literature on QR and on scrambling. We can see the effect of QR (Chomsky 1977a, May 1977) in environments where one quantifier takes scope over another, as in (1).

(1) Someone loves everyone.

The two quantifiers have ambiguous scope relative to each other, and this is expressed by QR, which raises the object quantifier above the subject quantifier, giving the inverse scope of 'everyone > someone'.

(2) everyone$_j$ [someone loves t$_j$]

Further application of QR, this time to the subject quantifier, induces the surface scope of 'someone > everyone'.

(3) someone$_i$ everyone$_j$ [t$_i$ loves t$_j$]

(p. 356) May (1977) proposes the following to account for these scope facts.

(4) Scope Principle (May 1977)

QP A takes scope over QP B iff QP A asymmetrically c-commands QP B.

A particularly strong support for characterizing QR as movement comes from Antecedent-Contained Deletion (ACD) (May 1985, 1991, Sag 1976, Williams 1977, Fox 2002; see Hornstein 1994 for an alternative to the QR analysis of ACD).

(5) John read every book that Tom did [$_{VP}$ *e*].

Under normal circumstance, the elided VP should correspond to the antecedent VP in the matrix clause, but that would lead to infinite regress due to the fact that the antecedent contains the elided VP.

(6) John [$_{VP}$ read every book that Tom did [$_{VP}$ read every book that Tom did [$_{VP}$ read every book that Tom did [$_{VP}$ read every book that Tom did …

(6) clearly fails to represent the actual interpretation associated with (5)—in fact it misrepresents (5) as uninterpretable. May argues that the correct interpretation becomes available if QR first moves the object universal and everything that accompanies it.

(7) [every book that Tom did [$_{VP}$ *e*]]$_i$ [John [$_{VP}$ read *e*$_i$]]

Now the matrix VP is [$_{VP}$ read *e*], and by replacing the original VP ellipsis site with it, we are able to associate the appropriate interpretation to the string.

(8) [every book that Tom did [$_{VP}$ read *e*]] [John [$_{VP}$ read *e*]]

Finally, May (1977) notes that the application of QR is limited to the local domain in which the quantifier occurs.

(9) Someone thinks that every student failed the test.

The inverse scope interpretation (everyone > someone) is difficult, if not impossible, to obtain, showing that QR cannot move a quantifier beyond the clause in which it occurs. One exception to this is the following in which a quantifier successfully moves out of an infinitival clause (Johnson 2000b).

(10) Someone wants to order every item in the catalogue.

This sentence readily admits the inverse scope interpretation, 'every item > someone'. I will return to these examples below.

Scrambling in Japanese shows essentially the same properties as what we saw for QR, and the fact that scrambling is overt movement gives further credence to viewing QR as movement.[2] While a subject—object quantifier combination does not easily (p. 357) allow inverse scope of 'object > subject' (11a), this scope relation becomes possible if the object is scrambled above the subject (11b) (Kuroda 1971; see also Hoji 1985).

(11)

a.

Dareka-ga	daremo-o	aisiteiru.
someone-NOM	everyone-ACC	loves
'Someone loves everyone.'		
someone > everyone, ⋆everyone > someone		

b.

Daremo-o$_i$	dareka-ga	t$_i$	aisiteiru.
everyone-ACC	someone-NOM		loves
'Someone loves everyone.'			
someone > everyone, everyone > someone			

The scrambled string in (11b), in which the object can scope over the subject, is identical in form to the string that results from covertly moving the object by QR in English for the purpose of inducing inverse scope (*everyone*$_i$ [*someone loves* t$_i$]). I will return later to why the other interpretation of 'someone > everyone' is also available in (11b).

The locality observed for QR also finds its counterpart in scrambling. As noted by Tada (1993; see also Oka 1989), while local scrambling induces a new scope relation as we saw above, long-distance scrambling fails to do so.

(12)

Daremo-o$_i$	dareka-ga	[Taroo-ga	t$_i$	aisiteiru	to]	omotteiru.
everyone-ACC$_i$	someone-NOM	Taro-NOM		love	C	think
'Someone thinks that Taro loves everyone'. Lit.: 'Everyone, Taro thinks everyone loves.'						
someone > everyone, ⋆everyone > someone						

In this example, the subordinate object quantifier has scrambled long-distance to the matrix clause. While the surface form itself is grammatical (a point we will come back to later), the expected new quantifier relation does not obtain. Instead, the only interpretation available is one that results from reconstruction of the scrambled phrase to the lower clause (Tada 1993; see also Saito 2004).[3] Although this failure of long-distance scrambling to induce a new scope relation may appear to be different from the locality of QR, I will argue that the two can in fact be viewed as exactly the same phenomenon. Finally, just as we saw that QR can move a quantifier out of an infinitival clause, scrambling a quantifier out of such an environment also leads to a new scope relation.[4]

(p. 358) **(13)**

a.

Dareka-ga	[Hanako-ni	dono-hon-mo	yomu yoo ni]	itta.
someone-NOM	Hanako-DAT	every-book	read	told
‘Someone told Hanako to read every book.’				
Someone > every book, *every book > someone				

b.

Dono-hon-mo$_i$	dareka-ga	[Hanako-ni	t_i	yomu yoo ni]	itta.
every-book	someone-NOM	Hanako-DAT		read	told
‘Someone told Hanako to read every book.’					
someone > every book, every book > someone					

To summarize, both QR and scrambling can create a new scope by moving a quantifier above another quantifier. But in neither case is a new scope relation allowed to obtain across a tensed domain, although an infinitival domain does not impose such a barrier to QR or scrambling. These observations lead us to suspect that QR and scrambling are one and the same operation, the only difference being that QR is covert while scrambling is overt, both being an optional movement in the relevant sense. Johnson (2000b) essentially comes to this conclusion, and we will pursue a similar line using a different view of scrambling. I will begin with a discussion of Kitahara (1996), Hornstein (1995), and Pica and Snyder (1995), who independently proposed an analysis of QR that does not depend on optional movement, and an extension of their approach by Johnson (2000b), who introduces the idea that QR is a form of covert scrambling.

16.3 To QR or not

Kitahara (1996), Hornstein (1995), and Pica and Snyder (1995) propose to do away with QR by noting that the scope facts (and also ACD in the case of Hornstein 1994) fall out from independent properties of the syntactic structure. They focus on the proposal in Chomsky (1991, 1993) that DPs (subject, object) must move to the specifier of agreement heads, AgrS and AgrO, for reasons of Case and agreement (I have updated the structure to include *v*P).

(14)

$$[_{AgrSP}\ \text{someone}_i\ [_{TP}\ [_{AgrOP}\ \text{everyone}_j\ [_{vP}\ t_i\ [_{VP}\ \text{loves}\ t_j\ ]]]]]$$

Hornstein (1995) argues that the inverse scope (everyone > someone) is induced by reconstruction of the subject quantifier *someone* to the original position underneath *everyone*. Kitahara takes a slightly different tack, although by and large empirically equivalent, by extending Aoun and Li's (1989) Scope Principle, which itself is an **(p. 359)** extension of May's (1977) original principle of the same name. This approach is similar in spirit to Pica and Snyder (1995).

(15) Scope Principle (Aoun and Li 1989, as revised by Kitahara 1996)
A quantifier X may take scope over a quantifier Y iff X c-commands a member of each chain associated with Y at LF.

For these linguists, the scope relations are a function of the basic structure of the sentence after movement meets case and agreement requirements. The idea is that the subject leaves a copy under the object, and this copy is visible to the interpretive mechanism so that the object can scope over the (copy of the) subject and induce inverse scope. In this way we can dispense with QR.

Johnson (2000b) provides further evidence that it is the lower copy of the subject that contributes to the inverse scope interpretation, but at the same time he argues against the overall 'Case' approach. To set the stage, note that in the following example from his work, it is possible for the object quantifier to scope over the existential subject quantifier.

(16) Some student or other has answered many of the questions on the exam.

Johnson notes that an existential quantifier of the type found in (16) cannot occur under negation, so that in the example below, *some student* must be interpreted outside of the scope of negation.

(17) I have not met some student. (some student > not)

If, as Kitahara, Hornstein, and Pica and Snyder argue, it is the lower copy of the subject chain that participates in inverse scope in some relevant sense, we predict that if negation prevents reconstruction of a subject existential, inverse scope should be blocked. This is what we see below.

(18) Some student or other hasn't answered many of the questions on the exam.

As Johnson notes, the lack of inverse scope here results from the fact that the existential subject quantifier must be interpreted above the negation, hence its lower copy is not visible for the purpose of scope. The correlation between the lack of inverse scope and the impossibility of reconstructing the subject provides independent evidence that the lower copy of the subject chain is what is active in inverse scope. Of course, we want to know why the lower copy must be active in inverse scope; it is something that we will answer directly below.

While the point above supports the Kitahara/Hornstein/Pica and Snyder approach to scope, Johnson (2000b) notes a problem with their analysis as well (he specifically takes issue with Hornstein's approach, but the argument also is relevant to the others' analyses). He shows that an adjunct can take scope over the subject.

(19) A different student stood near every visitor.

(p. 360) Hornstein (1995) and Pica and Snyder (1995) are also aware that phrases that do not require accusative case may scope over the subject, but they argue that these are phrases that are merged above the *v*P that contains the original position of the external argument. On this analysis, *near every visitor* is adjoined to the *v*P above the *v*P-internal external argument position. However, Johnson (2000b) provides data to show that even adjuncts that are construed lower than the external argument position can scope over the subject (see also Fox 2000, Kennedy 1997 for other arguments against the Case approach).

Johnson (2000b) proposes that inverse scope requires two operations: reconstruction of the subject quantifier to its original *v*P-internal position, as we saw above, and the movement of the object/adjunct to a position above the external argument position.[5] He calls the latter 'scrambling' of the sort found in languages such as Dutch and German. Scrambling in these languages typically moves an object or an adjunct to *v*P, which puts it above the subject copy in Spec,*v*P. Recall also that QR can move a phrase out of an infinitival clause to the next higher clause but not from a tensed clause.[6] We can see the same in scrambling in the following Dutch examples from Johnson (2000b); the first example shows extraction out of an infinitival clause, and the second out of a tensed clause.

(20)

a. … dat Jan *Marie*$_1$ heeft geprobeerd [t_1 te kussen].
… that John Mary has tried to kiss
'… that John has tried to kiss Mary.'

b. *… dat Jan *boken*$_1$ heeft besloten [dat er t_1 gelezen heeft]
… that John books has decided that he read has
'… that John has decided that he has read books.'

In the next section, I will expand on this view of scope as involving scrambling.

16.4 QR as scrambling

Let us begin with two questions about Johnson's 'scrambling' approach to inverse scope. First, what triggers this scrambling, and why does it move a phrase typically (p. 361) to *v*P? Second, in the existential-negation example in (18), in which we saw that inverse scope is blocked because the existential subject is blocked from reconstructing by negation, what prevents the object quantifier from scrambling to the top of the sentence above the subject quantifier as in below?

$[_{TP}$ *many of the questions...*$_j$ $[_{TP}$ *some student...*$_i$ NEG$[_{vP}$ t_j $[_{vP}$ t_i $[_{VP}$ V t_j ...$]]]]]$

Click to view larger

(21)

If the second movement were possible, we ought to be able to detect the inverse scope interpretation despite the unavailability of the lower copy of the subject. One answer is that this 'scrambling' is of the Dutch/German type—which Johnson assumes—that disallows this kind of movement; but once we expand our analysis to include scrambling in languages such as Japanese, which easily allows scrambling beyond the *v*P, this question becomes relevant. I will address these questions below by laying out the basic assumptions of optional movement.

16.4.1 Scope Economy and Edge Feature

Let us begin with the question: is the application of QR to the object quantifier, or some adjunct quantifier, always possible even in contexts such as the following?

(22) Mary admires every teacher.

Here, there is only one quantifier, so that even if the object quantifier moves by QR to a position that can take scope over the subject, we would not be able to detect the movement. So, does the object quantifier move at all in this situation? Fox (2000) provides an answer to this question, namely, that the object quantifier does not undergo movement by QR in these kinds of situations. The principle he proposes is the following.[7]

(23) Scope Economy (Fox 2000: 3)
Scope-shifting operations (SSOs) cannot be semantically vacuous.

Scope Economy predicts that the movement of the object quantifier would not take place for the purpose of taking scope in (22) above (*Mary admires every teacher*) because this movement would be semantically vacuous owning to the fact that it moves across an R-expression, *Mary*.

A particularly compelling argument for Scope Economy is found in ellipsis constructions. Sag (1976) and Williams (1977) point out that in an example such as the following, inverse scope is impossible.

(p. 362) **(24)** A boy admires every teacher. Mary does, too.
a boy > every teacher, *every teacher > a boy

One possibility for the lack of inverse scope is that such an interpretation is not allowed in ellipsis constructions, but that turns out not to be the case (Hirschbühler 1982).

(25) One guard is standing in front of every building, and one policeman, too.
one guard > every building, every building, one guard; one policeman > every building, every building > one policeman

Based on the well-known fact that the ellipsis site and its antecedent must be parallel in form (e.g. Lasnik 1972, Chomsky and Lasnik 1993, Tancredi 1992), Fox argues that the lack of inverse scope in (24) (*A boy admires every teacher. Mary does, too.*) is due to the fact that in the elided site, the object quantifier *every teacher* does not undergo QR because the subject is an R-expression, and in adherence to the parallelism requirement, the object quantifier in the first clause is prevented from undergoing QR. In contrast, we obtain inverse scope in (25) because the subject in both conjuncts is a quantifier.

Scope Economy is consistent with the 'last-resort' tenet of MP insofar as, if optional movement does not take place, such as QR/scrambling for scope-taking, a new meaning (inverse scope) would not be possible. Optional movement is therefore a 'last-resort' effort on the part of the grammar to induce the otherwise unavailable meaning. While it is consistent with last resort, it is important to note that this expanded notion of Last Resort potentially conflicts with MP's core notion that movement operations only occur if they need to. This is because the idea of optional movement regulated by Scope Economy leaves open the possibility of an item moving improperly, that is, moving without inducing new meaning. Improper movement by nature constitutes overgeneration, something that we would like to avoid in MP. Fox suggests a form of look-ahead to prevent improper optional movement (Fox 2000: 5), but I will suggest another approach that comes from the work on scrambling in Japanese, particularly Saito (1989) and its extension in Tada (1993).

Can optional movement move a phrase anywhere in the structure so long as Scope Economy sanctions it? Recall that in Johnson (2000b), QR as scrambling moves a quantifier to *v*P, presumably adjoining to *v*P above the original position of the external argument. This is also the position to which scrambling typically moves an item in languages such as Dutch and German. What triggers this movement? Maybe there is nothing to cause the movement, but there is a theory of movement that makes the right prediction that optional movement like QR/scrambling would end up at *v*P (among other positions). Chomsky (2008a: 144) suggests that 'only phase heads trigger operations' because phase heads come with what he calls an 'Edge Feature' that attracts items to the edge of a phase. Presumably an obligatory movement such as *wh*-movement results from a combination of the Edge Feature (p. 363) on C and a question feature on this C that enters into an agreement relation with the moved *wh*-phrase. Let us suppose that optional movement occurs when there is an Edge Feature but nothing else to link the phase head with an item within the phase. We thus have the following (Miyagawa 2006b: 33).

(26) Optional Movement
An element may freely move to any position with an Edge Feature.

This answers one of the questions we posed at the beginning of this section about Johnson's analysis: why does QR/scrambling move an item to *v*P? The answer is that *v*P is a phase, and *υ* carries an EF. It also accounts for why QR and scope-altering scrambling are possible out of an infinitival clause; such a clause is not a (strong) phase. This still leaves open the other question of why QR/scrambling doesn't move an item to a higher position in Johnson's example involving a subject existential and negation. We will return to this question below.

Scope Economy in combination with the Edge Feature approach to optional movement makes it possible to provide a precise analysis of perhaps the most compelling argument for Scope Economy. Recall that May (1977) noticed that QR is clause-bound.

(27)
a. Someone loves everyone.
some > every, every > some
b. Someone thinks that Mary loves everyone.
some > every, *every > some

However, there are exceptions to the clause boundedness of QR. The following is an observation by Moltmann and Szabolcsi (1994) discussed by Fox (2000).

(28)
a. One girl knows that every boy bought a present for Mary.
one > every, *every > one
b. One girl knows what every boy bought for Mary.
one > every, every > one

In (28a) the universal quantifier in the lower clause cannot take wide scope over the matrix indefinite, which is what we expect if QR is locally bounded. But in (28b), a subordinate universal quantifier unexpectedly takes such wide scope over the matrix subject indefinite. Fox notes that in (28a), the movement of *every boy* to the lower Spec of CP (or adjoining to this CP) does not lead to a new scope relation. Hence Scope Economy does not sanction this movement. In (28b), moving the universal *every boy* over *what* does lead to a new scope relation—it makes a pair-list interpretation possible under a quantifying-in approach to this interpretation (*every* > *what*) (e.g. Karttunen 1977; see Krifka 2001 for a review of this literature including problems with the quantifying-in analysis). This, then, sets (p. 364) up the movement of the universal quantifier to the matrix clause, where ultimately it may take scope over the existential quantifier in the matrix subject position. On the EF approach, in (28b) the subordinate universal quantifier moves to C, probably adjoining to CP, due to the Edge Feature on this C (the EF need not be erased after the *wh*-movement, but can stay active (Chomsky 2008a); it needs to be erased before transfer to semantic interpretation). Scope Economy sanctions this movement, and the universal quantifier is free to move to the higher clause to take scope over the matrix indefinite.

If QR and scrambling are one and the same movement, we ought to be able to find a similar phenomenon with overt scrambling, and we in fact do. Recall that long-distance scrambling does not induce a new scope relation (Tada 1993; see also Oka 1989).

(29)

Daremo-o$_i$	dareka-ga	[Taroo-ga	t$_i$	aisiteiru	to]	omotteiru.
everyone-ACC$_i$	someone-NOM	Taro-NOM		love	C	think
'Someone thinks that Taro loves everyone.' Lit.: 'Everyone, Taro thinks everyone loves.'						
someone > everyone, ⋆everyone > someone						

Note that in this example, the long-distance scrambled subordinate universal first moves to the edge of the lower CP, but this does not lead to a new scope relation because it crosses an R-expression. Scope Economy predicts this precisely in the same way that QR was blocked from occurring in the English example in (28a) above, where the local movement does not lead to a new scope relation. Now, if we replace this R-expression with a scope-bearing item, we predict that it is possible to induce a new scope relation, just as we saw for (28b) above in English; note the following (Abe 2005, Miyagawa 2005b, 2006a).

(30)

Daremo-o$_i$	dareka-ga	[itsuka	dareka-ga	t$_i$	kisu-sita	to]	omotteiru.
everyone-ACC$_i$	someone-NOM	sometime	someone-NOM		kissed	C	think.
'Someone thinks that at some point someone kissed everyone.' Lit.: 'Everyone, someone thinks that at some time someone kissed.'							
someone > everyone, everyone > someone							

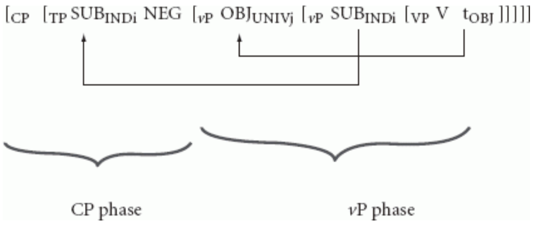
In this example the subordinate object universal first moves to the lower *v*P, where it takes scope over *dareka* 'someone' in Spec, *v*P, then to CP, where it again creates a new scope relation relative to *ituka* 'sometime'. This makes it possible for it to move to the matrix clause to take scope over the matrix subject indefinite, exactly in parallel to the English example in (28b) above. There are other points to discuss about (30) and I will return to the example below.

(p. 365) 16.4.2 Johnson's example again

Let us look again at the example from Johnson (2000b) that demonstrates that it is the lower copy of the subject chain that participates in inverse scope.

(31) Some student or other hasn't answered many of the questions on the exam.

The negation blocks reconstruction of the subject indefinite, which makes the inverse scope interpretation impossible. This example has two important implications for Scope Economy. First, what we see from this example is that Scope Economy must apply at the next phase, that is, in the phase subsequent to the phase that contains the optional movement. This is because in (31) we can tell that the lower copy of the subject in Spec,*v*P is unavailable only after the subject moves to Spec, TP across negation—hence, in the phase subsequent to that of the *v*P.



(32)

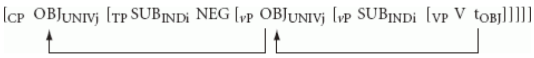
At the higher, CP phase, it becomes clear that the subject indefinite must be interpreted in Spec, TP, not Spec,*v*P. Under Scope Economy, this would invalidate the movement of the universal object to the higher Spec,*v*P in the *v*P phase since it would not lead to a new scope relation.[8] From this, we can deduce that Scope Economy applies as follows.

(33) Application Domain of Scope Economy

Scope Economy evaluates optional movement in one phase at the next higher phase. In the root phase the evaluation takes place simultaneous with the movement.

The movement of the universal object to the edge of *v*P across the indefinite subject in Spec,*v*P, is evaluated after the structure reaches the next phase, CP. By this time, the indefinite subject has moved to Spec, TP, across negation, and its (p. 366) copy in Spec,*v*P is no longer visible. Scope Economy applying at the CP phase would correctly deem the movement of the object quantifier inside the *v*P to be illegitimate.

Now we come to the second question posed at the beginning of this section: in Johnson's analysis, what prevents the universal object from moving to the higher phase, CP, to take scope over the indefinite subject in Spec, TP?



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(34)

While scrambling in Dutch/German typically does not move an item to such a higher position beyond the *v*P, scrambling in languages such as Japanese does. Particularly since we have drawn parallels between QR and Japanese scrambling, we should consider this movement to the higher position to be possible in principle. One possible account of why it is not available in this particular case is that the second movement of the object would be deemed illicit after Scope Economy deems its first movement illegitimate. However, a cleaner analysis would somehow prevent the second movement from taking place as a legitimate operation to begin with. As it turns out, the illegitimate nature of the second movement is something that Scope Economy predicts. The crucial point is that, although the second movement of the object crosses the indefinite subject, this fails to lead to a new scope relation because the scope relation of [OBJ_{UNIV} > SUB_{IND}] has already been established at the lower phase in which the object universal moves to *v*P above the subject indefinite in Spec,*v*P. The second movement of the object would constitute a semantically vacuous movement in violation of Scope Economy. So, to answer the question we posed, it is in principle possible for QR/scrambling to move an item to CP from *v*P, but it must be sanctioned by Scope Economy.[9]

The reasoning given above in fact provides an explanation for why it is that the lower copy of the subject chain must be active for inverse scope to be possible (Hornstein 1995, Kitahara 1996, Pica and Snyder 1995). The higher copy of the subject chain, in Spec, TP, cannot participate in inverse scope with the object quantifier (or some VP adjunct) because it would replicate the same scope relation already established by the two quantifiers at the *v*P phase.

This analysis also provides a straightforward account of the well-known pair below (May 1985).

(p. 367) **(35)**

a. What$_i$ did every student read t$_i$?

b. Which student$_i$ t$_i$ read every book?

(35a) allows a pair list (PL) interpretation while (35b) does not. Chierchia (1992) argues that in order for a PL interpretation to be possible, the universal quantifier must c-command the trace of the *wh*-phrase (see also Kitahara 1996).[10] This is true of (35a) but not of (35b). The Scope Economy approach to optional movement provides an explanation without having to make any additional assumptions such as that of Chierchia's that must invoke weak crossover. First, look at the *v*P phase of these two sentences.

(36)

a. $[_{vP}$ what$_j$ $[_{vP}$ every student $[_{VP}$ V $t_j]]]$

b. $[_{vP}$ every book$_j$ $[_{vP}$ which student $[_{VP}$ V $t_j]]]$

In both, the object phrase has moved to *v*P above the external argument, but the new scope that is induced is different. In (36a) the *wh*-phrase has undergone movement, and although this movement has nothing to do with scope-taking, but instead the *wh*-phrase must move so that it can ultimately end up in Spec, CP, it would be reasonable to view this as having established a *wh* > *every* scope relation (though not critical for our analysis). On the other hand, the movement of the object universal in (36b) establishes the scope relation, *every* > *wh*. At the CP phase, Scope Economy would evaluate the following two structures differently.

(37)

a. $[_{CP}$ every student$_i$ $[_{CP}$ what$_j$ $[_{TP}$ t_i … $[_{vP}$ t_j $[t_i$ …$]]]]]$

b. $[_{CP}$ every book$_j$ $[_{CP}$ which student$_i$ $[_{TP}$ t_i … $[_{vP}$ t_j $[t_i$ …$]]]]]$

In (37a), the movement of *every student* to CP is sanctioned because it induces the scope relation, *every student* > *what*, which is different—hence semantically not vacuous—from the earlier scope of *what* > *every student*. In contrast, the movement of the object universal in (37b) fails to lead to a new scope relation because the scope relation it creates, *every book* > *which student*, is identical to the one established already at the *v*P phase level. Thus, this movement of the object universal to CP cannot be sanctioned. Given that the *wh*-phrase c-commands the universal quantifier, a PL interpretation is correctly ruled out. This analysis upholds the ‘quantifying-in’ analysis of PL interpretation, which requires the universal to quantify into the *wh*-question.[11]

(p. 368) As the final note on the topic of PL interpretation, note the following from May (1985, 1988).

(38)

a. Which boy loves every girl? (no PL)

b. Which boy loves each girl? (PL)

(38a) is what we expect from the discussion above, but (38b) is unexpected. This example contains *each*, which is inherently focused (Culicover and Rochemont 1993). May (1985, 1988) argues that this inherent focus property causes the *each* phrase to move and adjoin to CP. This movement of the *each* phrase to CP is not an optional operation, but rather, it is an obligatory movement for the purpose of focus marking, which is not subject to the restrictions imposed by Scope Economy. The fact that this obligatory movement leads to the universal taking scope over the *wh*-phrase to allow the PL interpretation shows that Scope Economy applies only to optional movement. If Scope Economy were to apply even to obligatory movements, we would not expect the PL interpretation to emerge.

Let us return to the Japanese example that demonstrates that long-distance scrambling can induce a new scope relation if each movement is sanctioned.

(39)

Daremo-o$_i$	dareka-ga	[itsuka	dareka-ga	t_i	kisu-sita	to]	omotteiru.
everyone-ACC$_i$	someone-NOM	sometime	someone-NOM		kissed	C	think.
‘Someone thinks that at some point someone kissed everyone.’ Lit.: ‘Everyone, someone thinks that at some time someone kissed.’							
someone > everyone, everyone > someone							

In this example, there are two quantified expressions in the lower CP, ‘sometime’ and ‘someone'. This is important for Scope Economy to validate each movement. In the lower *v*P phase, the subordinate object universal 'everyone' scrambles across the subject indefinite in Spec,*v*P.

(40) [$_{vP}$ everyone$_j$ [$_{vP}$ someone [$_{VP}$ t$_j$ …]]]

At the subordinate CP phase level, this object universal would move across another quantifier, 'sometime', which creates a new scope relation, and 'everyone' is then free to move to the matrix clause to take scope over the matrix subject 'someone'. I will return to some issues that arise with this final movement later, but for now, note that if 'sometime' is removed, it is harder to interpret the long-distance scrambled object in the matrix position.

(41)

Daremo-o$_i$	dareka-ga	[dareka-ga	t$_i$	kisu-sita	to]	omotteiru.
everyone-ACC$_i$	someone-NOM	someone-NOM		kissed	C	think.
'Someone thinks that someone kissed everyone.' Lit.: 'Everyone, someone thinks that someone kissed.'						
someone > everyone, ??everyone > someone						

(p. 369) This is predicted because the movement in lower CP of the subordinate universal is not sanctioned by Scope Economy. Why is the 'everyone > someone' interpretation not completely out? As we will see below, long-distance scrambling may be sanctioned independently by focus (e.g. Miyagawa 1997, 2006a), so that the interpretation of the long-distance scrambled object at the matrix clause may be validated by focus, and scope can piggy-back on this just as we saw with the focus *each* in English above. For some reason, focusing does not lead to a clearly new scope relation, as we can see by '??' for the intended interpretation.[12]

As the final note in this subsection, we saw that the Edge Feature on a phase head triggers optional movement that results in QR and (a subclass of) scrambling. This is the reason why the two behave identically in the contexts we have observed. Although they behave identically as shown so far, there is one obvious difference—QR is covert while scrambling is overt. We will take up the consequence of this difference later, but for now, let us make clear our assumption about the nature of covert movement. Given that it is triggered by EF, and EF is a feature that occurs in narrow syntax, a reasonable assumption is that QR as covert movement and scrambling as overt movement both take place in an identical fashion, both triggered by the EF on a phase head. The difference arises with the decision to pronounce which of the copies that occur in the chain: in the case of scrambling, it is the higher copy that is pronounced while in QR it is the lower copy (see e.g. Bobaljik 1995a, Fox and Nissenbaum 1999, Groat and O'Neil 1996, and Pesetsky 1998 for this idea of overt/covert resulting from pronunciation). One consequence of this is that optional movement, either overt or covert, occurs in narrow syntax, not PF, so that there ought not to be any 'semantically vacuous' optional movement at PF, a point I will elaborate on later.

16.4.3 Does optional movement lead to overgeneration?

I began this chapter by noting the transition from GB, in which there is massive overgeneration, to MP, which, because of Last Resort, in principle is able to avoid overgeneration. However, by introducing optional movement into the grammar, we potentially set the stage for overgeneration to occur even in MP. An optional movement that violates Scope Economy would be tagged as an unacceptable derivation, and if we say that such derivation 'crashes', that takes us straight into the realm of (p. 370) overgeneration. One way to avoid this overgeneration is Fox's (2000) look-ahead, which prevents movements that violate Scope Economy from taking place to begin with by introducing a look-ahead mechanism. However, there is another approach available, from the literature on scrambling in Japanese, that avoids the difficulties associated with a look-ahead approach. This is the idea of radical reconstruction.

Saito (1989) argues that scrambling is semantically vacuous, and at LF, it is obligatorily put back into its original position, a process known as 'undoing' or 'radical reconstruction'.[13] I will use the latter term. To see this, let us again look at cases of long-distance scrambling that fail to induce a new scope relation, which Saito (2004) points to as a demonstration of radical reconstruction.

(42)

Daremo-o_i	dareka-ga	[Taroo-ga	t_i	aisiteiru	to]	omotteiru.
everyone-ACC_i	someone-NOM	Taro-NOM		love	C	think
‘Someone thinks that Taro loves everyone.’ Lit.: ‘Everyone, Taro thinks everyone loves.’						
someone > everyone, ⋆everyone > someone						

Tada (1993), who assumes the idea of radical reconstruction, gives an explanation that is similar to Fox's Scope Economy. He argues that the matrix landing site of the long-distance scrambling is not an operator position (he assumes that it is adjoined to the matrix TP following Saito 1985) so that the quantifier ‘everyone’ is unable to take scope in this position—in other words, it has no semantic role to play in this position. Consequently, it must be put back by radical reconstruction to its original position where scope is possible. One way to interpret Tada's analysis is that, by providing a repair to the illicit structure in the form of radical reconstruction, this string avoids crashing, thereby preventing overgeneration.

Radical reconstruction, as employed above, can avoid overgeneration in the case of QR as well. We assume that QR, a covert form of scrambling, is possible to the local phase head, its movement triggered by the Edge Feature on the phase head. If this movement meets Scope Economy, the movement is sanctioned (Fox 2000), but if not, it cannot be interpreted in that position (Tada 1993) and it must radically reconstruct to prevent overgeneration (based on a revision of Saito's original 1989 analysis).

One consequence of this way of viewing radical reconstruction is that no optional movement should occur in the PF component. In the literature, movement that has no semantic import is sometimes viewed as taking place in the PF component (p. 371) (see e.g. Sauerland and Elbourne 2002 for relevant discussion). At least for those cases of semantically vacuous movement that we have considered, this cannot be true because the movements are evaluated by Scope Economy, which is strictly a principle of the interface in narrow syntax.

16.5 Optional and obligatory scrambling

As noted earlier, scrambling leads to a new scope relation.

(43)

a.

Dareka-ga	daremo-o	aisteiru.
someone-NOM	everyone-ACC	loves
‘Someone loves everyone.’		
someone > everyone, ⋆everyone > someone		

b.

Daremo-o$_i$	dareka-ga	t_i	aisteiru.
everyone-ACC	someone-NOM		loves
'Someone loves everyone.'			
someone > everyone, everyone > someone			

Let us look closely at (43b) and see how the new scope relation becomes possible. Under the standard view of Japanese syntax (e.g. Saito 1985), the subject 'someone' resides in Spec, TP, and the scrambled object 'everyone' is adjoined to this TP. Note, however, that this structure violates Scope Economy. In the *v*P phase, the object universal moves to adjoin to *v*P, taking scope over the subject indefinite.

(44) $[_{vP}$ OBJ_{UNIVj} $[_{vP}$ SUB_{IND} $[_{VP}$ t_j $V]]]$

On the standard view, the subject then would move to Spec, TP in the next phase (e.g. Kishimoto 2001), and the object then moves above it. But notice that the movement of the object universal replicates the scope relation already established at the *v*P phase, hence Scope Economy would not sanction this movement for establishing a new scope. We would therefore expect it to undergo radical reconstruction; but quite to the contrary, the new scope relation is clearly available.

There is an alternative analysis that does not assume that the subject must always end up at Spec, TP. Using an idea originally proposed by Kuroda (1988), I (2001, 2003) proposed that the two word orders, SOV and OSV, are equivalent in the following way.

(45)

a. $[_{TP}$ S_i $[_{vP} t_i$ $[_{VP}$ O V]]]

b. $[_{TP}$ O_i $[_{vP}$ t_i $[_{vP}$ S $[_{VP}$ t_i V]]]]

The core idea is that Spec, TP must be filled due to the EPP feature on T, and this requirement can be met by moving the subject as in (45a) or the object as in (45b). (p. 372) In either case, 'the other phrase' remains inside the *v*P/VP. See Miyagawa (2001) for evidence that when the object raises, the subject can stay in Spec,*v*P. There are other items that can move into Spec, TP to satisfy the EPP, such as certain types of PP, but I will limit the discussion to subjects and objects. On this analysis the object-scrambled sentence in (43b) is associated with the following structure.

(46)

$[_{CP}$ $[_{TP}$	daremo-o$_j$ $[_{vP}$ t_j $[_{vP}$	dareka-ga $[_{VP}$ t_j	aisiteiru]]]]]
	everyone-ACC	someone-NOM	love

The scope of 'everyone > someone' is established at the *v*P phase level, and further movement of the object universal to Spec, TP is not an optional movement, but an obligatory one triggered by the EPP (see Miyagawa 2001 for evidence that the object is in Spec, TP in the OSV order).

Recall, too, that this surface form of object universal—subject indefinite not only allows the interpretation 'everyone > someone' but also the other scope of 'someone > everyone'. I will assume that the latter meaning reflects a different derivation in which the subject indefinite moves to Spec, TP to satisfy the EPP, then the object universal moves to CP by optional movement.

(47)

[$_{CP}$	daremo-o$_j$ [$_{TP}$	dareka-ga$_i$ [$_{vP}$t$_j$ [$_{vP}$ t$_i$ [$_{VP}$ t$_j$	aisiteiru]]]]]
	everyone-ACC	someone-NOM	love

The movement of the object universal does not lead to a new scope relation because it replicates the scope relation established already at the *v*P phase level; hence the object must be radically reconstructed to its lower position, which gives rise to the 'someone > everyone' scope interpretation because 'someone' in Spec, TP is the highest quantifier in the structure.

Finally, let us look again at the case in which long-distance scrambling successfully induces a new scope relation.

(48)

Daremo-o$_i$	dareka-ga	[itsuka	dareka-ga	t$_i$	kisu-sita	to]	omotteiru.
everyone-ACC$_i$	someone-NOM	sometime	someone-NOM		kissed	C	think.
'Someone thinks that at some point someone kissed everyone.' Lit.: 'Everyone, someone thinks that at some time someone kissed.'							
someone > everyone, everyone > someone							

We saw that Scope Economy sanctions the movement of the subordinate object universal 'everyone' to the edge of the lower CP thanks to the occurrence of 'sometime'. How does this subordinate object take scope over the matrix indefinite 'someone'? Based on what we saw above, a reasonable assumption is that the subordinate object universal adjoins to the matrix *v*P to take scope over the matrix subject indefinite.

(49)

[$_{CP}$ …	[$_{vP}$ daremo-o$_j$ [$_{vP}$	dareka-ga … [$_{CP}$	t$_j$ …]]]]
	everyone-ACC	someone-NOM	

(p. 373) From here, the universal moves to the matrix Spec, TP.

(50) [$_{CP}$ [$_{TP}$ daremo-o$_j$ [$_{vP}$ t$_j$ [$_{vP}$ dareka-ga … [$_{CP}$ t$_j$ …]]]]]

This last movement is not an optional one that needs to be validated by Scope Economy; rather, it is an obligatory movement needed to satisfy the EPP. This structure is what makes it possible for the long-distance scrambled subordinate object to take scope over the matrix subject indefinite.[14] For the other interpretation of 'someone > everyone', we can assume the same account as above—the subject moves to Spec, TP, and the object moves to C. The movement of the object is optional, but it fails to induce a new scope relation, so it must be radically reconstructed to the lower clause.

16.6 Why does QR apply only to quantifiers?

If QR and scrambling are the same operation, why is it that QR targets only a small subset of expressions that scrambling targets? QR only applies to quantifiers, but scrambling applies to virtually any kind of expression.

In order to answer this question, let us look again at Fox's Scope Economy. Fox (2000) actually generalizes his Scope Economy to what he calls Output Economy, by which the condition that licenses optional operation is one that has an 'effect on (p. 374) the output' (Fox 2000: 75). This notion has been adopted by others (e.g. Chomsky 2001, Miyagawa 2005b, 2006a). We can see that this substantially broadens the possibilities for licensing optional movement, although Fox himself is most concerned about operations that impact interpretation—what he calls 'interpretation-sensitive economy' (Fox 2000: 2). I will assume this 'interpretation-sensitive economy' as the principle that regulates optional movement, the idea being that an optional movement must lead to a new interpretation that would not be possible otherwise. I will call it Interpretation Economy for convenience.

Bearing this in mind, let us return to the question of why QR only targets quantifiers while overt scrambling can move all sorts of expressions. Interpretation Economy requires any optional movement to have an effect on interpretation. Covert movement such as QR can only have such an effect in one sense, that of altering scope relations. Consequently, the fact that QR, a covert operation, only applies to quantifiers follows straightforwardly from Interpretation Economy.

In contrast to QR, not only can overt scrambling affect scope, as we have seen, but it can also have an effect on another type of interpretation. As noted by Neeleman and Reinhart (1998), scrambling changes the focus potential of a sentence (cf. also e.g. Bailyn 2001, 2003, Ishihara 2001, Jung 2002, Miyagawa 1997, 2005b, Otsuka 2005, Yang 2004). Ishihara (2001) illustrates this for Japanese. Let us begin with a normal SOV word order.

(51)

Taroo-ga	[VP	**hon**-o	katta]
Taro-NOM	[VP	book-ACC	bought]
'Taro bought a book.'			

The focus here is on the object **hon** 'book', which is the phrase that bears the prominent stress. According to the Focus Rule of Neeleman and Reinhart (1998), which allows focus to project upward from the focused element, the focus domain of this sentence may be the object **hon**, the VP that contains it, or the entire TP. Thus, (51) can be used as an answer to the following three questions:

(52)

a. What happened? (focus on TP)
b. What did Taro do? (focus on VP)
c. What did Taro buy? (focus on object)

(53) below has a different focus domain set due to the scrambling of the object.

(53)

Hon-o$_i$	**Taroo**-ga	[$_{VP}$	t$_i$	katta]
book-ACC$_i$	Taro-NOM	[$_{VP}$	t$_i$	bought]

With neutral prosody, the focus domains are the subject NP **Taroo** and the TP, but the VP cannot be a focus domain because it does not contain the focus element *Taroo*. Therefore (53) cannot be used to answer 'What did Taro do?' Let us assume, (p. 375) quite plausibly, that altering the focus potential of a sentence counts as having an 'effect,' hence it can license optional movement.[15]

Now consider the following.

(54)

Hanako-ga	[$_{CP}$	Taroo-ga	hon-o	katta	to]	itta.
Hanako-NOM	[$_{CP}$	Taro-NOM	book-ACC	bought	C]	said
'Hanako said that Taro bought a book.'						

This sentence can be used to answer the following three questions, among others

(55)

a. What happened? (focus on matrix TP)

b. What did Hanako do? (focus on matrix VP)

c. What did Hanako say? (focus on complement CP)

Now consider the following LD-scrambling of the subordinate object, which is an ordinary nominal expression (*hon* ‘book’).

(56)

Hon-o_i	Hanako-ga	$[_{CP}$	t_i	Taroo-ga	t_i	katta	to]	itta.
Book-ACC_i	Hanako-NOM	$[_{CP}$	t_i	Taro-NOM	t_i	bought	C]	said
Lit. ‘Book, Hanako said that Taro bought (it).’								

A natural way to pronounce this sentence is to put focus stress on the LD- scrambled *hon-o* ‘book-ACC’ (Miyagawa 1997). This isolates the focus set to the highest node, and this sentence is used naturally to respond to the question, *What did Hanako say that Taro bought?*, with ‘what’ scrambled to the head of the sentence. It seems to me that (56) cannot be used as a natural response to any of the questions in (55) (*what happened?, what did Hanako do?, what did Hanako say?*), although it may be possible with a rich context. In any event, what is clear beyond doubt is that the LD-scrambling of the embedded object fundamentally alters the focus potential of a sentence, so that this LD-scrambling is licensed as an optional operation strictly on the basis of altering the focus potential, a form of altering the interpretation of the string.

16.7 Conclusion

The ‘last-resort’ tenet of MP requires the grammar to avoid overgeneration, a view that naturally leads to excluding optional movements. Optional movement conflicts (p. 376) with this tenet in two respects. First, being optional, it, in principle, need not occur, hence, when it does, it is not ‘last resort’; and optional movement can potentially lead to massive overgeneration of the type we find in GB. In this chapter I took up QR and scrambling, which appear to be quintessential optional operations. I adhered to the idea that they are optional movement, and showed that by the application of Fox's economy condition on optional interpretation, we can predict which optional operations are well-formed and which ones are not. The possible optional movements always lead to a new interpretation, which provides a kind of a ‘last resort’ view even of optional movement, albeit an extended and a somewhat weaker version. By fleshing out the assumptions behind the application of the economy condition, we extended the empirical coverage of this condition on optional movement beyond Fox's original dataset. I also suggested, contra Fox, that the economy condition does not prevent an illicit movement from taking place. Rather, such an illicit movement, if it occurs, is forced to undergo radical reconstruction because it cannot be interpreted in the moved position. The consequence of this is that, like in Fox's approach but without a look-ahead mechanism, we can avoid overgeneration even with optional movement.

Notes:

I am grateful to Danny Fox and Kyle Johnson for their input on earlier versions of this chapter. Earlier versions were presented at Kanda University of International Studies, MIT, and Nagoya University. I thank those in the audience for comments and suggestions.

(1) Chomksy (1995a) suggests that for operations such as object shift in Germanic, which is optional, a feature that triggers this operation is inserted just when the operation is to take place. On this view, the presence of a feature does not equate with last-resort movement, although at some deeper level one might be able to make such an argument.

(2) A number of linguists have noticed this correlation between QR and scrambling (e.g. Abe 2005, Beck 1996,

Diesing 1992, Johnson 2000b, Kitahara 1996, Miyagawa 2006a, Sohn 1995, and Tonoike 1997). The one exception to this correlation is ACD, which is not easily replicated in Japanese (but see Takahashi 1996).

(3) There are speakers who allow the new scope relation even with long-distance scrambling, a point I will return to later in the chapter.

(4) The element *yoo ni* in the following infinitival examples appears to be a C given that it occurs after the infinitival verb. That would make the lower clause a CP, which potentially would make it a phase, hence a potential barrier to A-movement. However, there is a reason to believe that this is not the right analysis. As shown by Nemoto (1993), it is possible for an element from within the infinitival *yoo ni* clause to undergo A-movement scrambling to the matrix clause, which clearly indicates that this environment is not a (strong) phase. See also Uchibori (2000) for relevant discussion.

(5) Johnson and Tomioka (1997) also assume a paired operation for inverse scope, subject reconstruction as we saw above, and the VP-internal item such as the object undergoing QR to *v*P. The latter is required to remedy type mismatch (Heim and Kratzer 1998).

(6) Hornstein (1995) deals with such cases by suggesting that the verb + infinitival undergoes restructuring, a process familiar from Romance. Kennedy (1997) points out, however, that the verb + infinitival combinations that allow wide scope go beyond the restructuring verb + infinitival combinations found in Romance.

(7) Reinhart (1995/2006) and Tada (1993) have noted similar ideas, although not as extensively developed as Fox (2000). Fox originally introduced the idea independently in Fox (1995).

(8) While Johnson (2000b) focuses on the unavailability of the lower copy in those constructions that does not allow inverse scope, cases like the following (Hornstein 1995) indicate that it is more accurate simply to say that if the subject must be interpreted in its higher copy position, inverse scope is not possible.

((i)) A boy danced with every girl before he left the party.

A boy here must be interpreted in Spec, TP in order to bind the pronoun in the adjunct clause, in turn depriving this sentence of inverse scope.

(9) One question about this analysis is how we deal with negation. In (32), while the movement of the object quantifier across the subject quantifier in Spec, TP violates Scope Economy as noted, this movement creates a new scope relation relative to negation, which arguably is in the higher phase. Certainly it is difficult, if not impossible, to get a reading in which the object quantifier scopes over negation (it does not scope over the subject either, of course): *Some student or other hasn't answered many of the questions on the exam*. It appears that there is a locality imposed on Scope Economy, in that if a quantifier A is moved across two scope-bearing items, Scope Economy is evaluated against the closest (higher) scope-bearing item. So, in (32), it is only the subject quantifier that comes into calculation of Scope Economy, so that negation cannot help to validate this movement.

(10) The idea is that the trace of the *wh*-phrase contains a pronoun-like element co-indexed with the universal quantifier that makes the PL interpretation possible, but this pronoun must be c-commanded by the universal quantifier in order to avoid a weak crossover violation.

(11) This analysis leaves a question about the following pair (Longobardi 1987; see also Cresti 1995).

((i)) What$_i$ do you wonder whether every boy bought t_i? (*PL)
((ii)) What$_i$ do you think every boy bought t_i? (PL)

As shown in (i), PL is not possible if a *wh*-phrase is extracted out of a *wh*-island that contains the universal. As shown in (ii), PL is possible if the *wh*-extraction is not out of an island. There are a number of possibilities, all with some problems, but I will not pursue this issue here.

(12) In Miyagawa (2005b, 2006a), I gave examples such as (41) as evidence for the relevance of Scope Economy to long-distance scrambling of quantifiers. As I noted, while many speakers found this construal possible, others did not. The addition of the second quantifier, ‘someone', as in (39), makes the interpretation more easily available.

(13) Saito's (1989) analyses all involve long-distance scrambling, which is solely A′-movement, as opposed to local scrambling, which may be either A- or A′-movement (Mahajan 1990, Saito 1992). The latter has been shown to be amenable to an analysis as obligatory, not optional, movement triggered by the EPP feature on T (see below for a brief discussion of this). See e.g. Kitahara (2002) and Miyagawa (2001, 2003, 2005a). See Miyagawa (2005b, 2006a) for a critical review of Saito's (1989) radical reconstruction.

(14) A problem with the derivation just given is that it forms what is standardly thought of as an improper chain—an A′-segment followed by an A-segment. I will leave this problem open. Related to this is the issue that long-distance scrambling is supposed to always be A′-movement, so that, for example, it does not create a new binder (e.g. Mahajan 1990, Saito 1992).

((i))

?*Futari-no	gakusei-o$_i$	otagai-no	sensei-ga	[Hanako-ga	t$_i$	sikaru	to]	omotteiru.
two-GEN	students-ACC	each other-GEN	teacher-NOM	Hanako-NOM		scold	C	thinks

‘Two students, each other's teachers think that Hanako will scold.’

Note that the LD-scrambled subordinate object has undergone an improper movement in the subordinate clause relative to scope. If one places a quantifier in the lower subject position, there appears to be an improvement.

((ii))

?Futari-no	gakusei-o$_i$	otagai-no	sensei-ga	[dareka-ga	t$_i$	sikaru	to]	omotteiru
two-GEN	students-ACC	each other-GEN	teacher-NOM	someone-NOM		scold	C	thinks

‘Two students, each other's teachers think that someone will scold.’

Although the judgment is not so clear, if this is correct, it gives us hope that even long-distance scrambling can have an ‘A’ version in the matrix clause and create a new scope/binding relation. See Uchibori (2000) for analysis that long-distance scrambling can form an A-chain.

(15) Ishihara (2001) makes two assumptions about (53). First, as argued by Miyagawa (2001), the object in an OSV order may move into the Spec of TP to satisfy the EPP of T. Second, there is verb movement to T (cf. Koizumi 1995, Otani and Whitman 1991), so that in (53), the lowest element is the subject in the Spec of *v*P. This is why the subject receives the nuclear stress, and it constitutes an argument that the verb raises in Japanese. In Dutch, in which there is no overt verb movement, scrambling of the object leads to the nuclear stress being assigned to the verb, which is the lowest element in the structure, unlike in Japanese. As a counterpoint, see e.g. Fukui and Takano (1998), Fukushima (2003), Fukui and Sakai (2003), and Takano (1996) for arguments that the verb does not raise in Japanese.

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Syntax and Interpretation Systems: How is Their Labour Divided?

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[–] Abstract and Keywords

This article looks at the division of labour between syntax and the interpretive systems by focusing on patterns of anaphoric dependencies. By eliminating the notion of an index, the minimalist programme enforces a strict distinction between how syntactic, semantic, and discourse factors contribute to anaphoric relations.

Keywords: syntax, interpretative systems, anaphoric dependencies, index, minimalist programme

17.1 Background

One of the driving forces behind the development of linguistic theory in the past five decades has been the modular perspective on the analysis of complex language phenomena. It involved factoring out the syntactic from the semantic or pragmatic as already in Chomsky's (1957) discussion of the contrast between (1) and (2).

(1) Colorless green ideas sleep furiously
(2) *Furiously sleep ideas green colourless.

It led to the endeavor to systematically factor out the language particular from the general in the representation of linguistic processes, and to a line of research exploring restrictions on the format of linguistic rules that enforce particular ways (p. 378) of 'cutting the pie', in such a way that rule formats correctly capture the properties of a module.

An enlightening discussion of the latter issue is given in Heny (1981), comparing a venerable version of the passive rule in English with the passive rule in Dutch. What these passives share is a passive auxiliary with a participial form of the main verb, movement of the object into the 'subject position' and the realization of the thematic subject as an optional by-phrase, as illustrated in (3).

(3)
a. (Leonard noted that) Jill saw Jack.
b. (Leonard noted that) Jack was seen by Jill.

However, what seems so straightforward now (and in fact was from the perspective of traditional descriptive grammar) was non-trivial to capture in the format of the transformational rules of the early 1970s. A Dutch version had to take into account that in Dutch the main verb comes after the object, and that (in subordinate clauses) the auxiliary comes after the main verb, as illustrated in (4):

(4)
a. (Leonard merkte op dat) Jill Jack zag.
b. (Leonard merkte op dat) Jack door Jill gezien werd.

If these contexts are represented in the rule, what one gets are in fact different rules. That they express the same process cannot be read off the rules themselves. Similar problems would show up if we were to formulate passive rules for other languages, or generalize over different construction types. So, the form of this type of rule forced us to treat what is language- and construction-specific in such processes on a par with what is general.[1]

What was needed, then, was to simplify the rule format, reducing its expressive power, and thus forcing the grammar to distinguish between various factors.

This is what indeed happened. Through the 1970s, there were systematic attempts to simplify the structural descriptions of transformations, thus reducing their expressive power. Chomsky (1973, 1977b) showed that structural descriptions (see note 1 above) can be reduced to three-term factorizations of the form *vbl—α—vbl* where only α has descriptive content (a categorial label, feature, etc.). Since the right-hand and left-hand terms are predictable, what resulted was a general operation Move α. The conditions on movement thus followed from the interaction between language-particular lexical properties reflected in source and target position and universal principles of computation. This, then, became the standard format of (p. 379) grammatical operations throughout the 1980s and the beginning of the 1990s, and revolutionalized the study of cross-linguistic variation. It formed the basis for our current understanding of the division of labor between grammatical and lexical properties. Thus, elimination of the context from the rule forced us to distinguish between a process such as dislocation itself, the factors *enforcing* it, and the factors *restricting* it.

The minimalist program as it developed since Chomsky (1995b) embodies a revolution in structure building. The lexicon-based Merge operation as the basic process of structure building effectively derived endocentricity and the properties of language hitherto captured by the X′ - schema. Dislocation as resulting from attraction/Internal Merge completed the line set out in the 1970s, making ‘transformations’ of the original more complex type impossible to state. This conception of grammar also had a fundamental impact on the view of the relation between the syntactic, computational system, and the interpretive components at the C-I side of the language system (as well as the realization components at the PF side). In this respect it turned out to be as revolutionary as the other changes discussed. I will show this on the basis of its impact on our conception of binding phenomena.

17.2 The canonical binding theory

All languages have anaphoric elements, i.e. elements that receive their interpretation in terms of some element that has been previously introduced. Any theory of language has to account for the fact that this is the case, and for the restrictions these dependencies are subject to. Ideally such facts should follow from the way the language system is designed; and our theoretical model should reflect this design.

The canonical binding theory (CBT, Chomsky 1981a) accounts for a particular set of such restrictions, illustrated by the pattern in (5):

(5)
 a. John expected [$_{\text{Clause}}$ Bill to admire him/himself]
 b. [$_{\text{Clause}}$John expected [$_{\text{Clause}}$ himself /him to be able to invite Bill]]

In (5a) *him* can have the value John, but not Bill, *himself* can and must have the value Bill but not John. In (5b) *him* can have neither the value John nor Bill, *himself* can and must have the value John, but not Bill. If we replace *John* by *John's mother* in (5a), nothing changes, but in (5b) *him* can now have the value John, but *himself* no longer can.

The CBT captures this pattern on the basis of two ingredients: a notion of binding, and conditions on binding. Binding, in turn, also consists of two ingredients: (co-)indexing and the structural condition of c-command. Technically, indices are (p. 380) numerals appended to a (nominal) expression (Chomsky 1980a, 1981a, Fiengo and May 1994).

C-command is defined as in (6):

(6) *a* c-commands *b* if and only if *a* does not contain *b* and the first branching node dominating *a* also dominates *b*.

Schematically: [a [$_\gamma$ …. b….]]

Binding is then defined as in (7):

(7) *a* binds *b* iff *a* and *b* are co-indexed and *a* c-commands *b*.

In the CBT the set of nominal expressions is partitioned into anaphors, pronominals, and R-expressions. Anaphors (*himself*, etc.) are referentially defective and must be bound, and in fact in a local domain. Pronominals (*he*, etc.) may have independent reference, but can be bound, though not in a local domain; R-expressions (*John, the man*, etc.) cannot be bound. The binding conditions making this explicit are given in (8):

(8)

(A) An anaphor is bound in its governing category.

(B) A pronominal is free in its governing category.

(C) An R-expression is free.

The definition of governing category/local domain is given in (9):

(9) γ is a governing category for α if and only if γ is the minimal category containing α, a governor of α, and a SUBJECT.[2]

Together, these conditions will derive the possible values of *him* and *himself* in (5). An example with possible and impossible indexings is given in (10).

(10)

a. John$_i$ expected [$_{Clause}$ Bill$_j$ to admire him$_{i/k/*j}$ /himself$_{*i/*k/j}$]

b. [$_{Clause}$ John$_i$ expected [$_{Clause}$ himself$_{i/*k/*j}$ /him$_{*i/*j/k}$ to be able to invite Bill$_j$]]

These binding conditions reflect a mix between syntactic factors and factors that belong to the interpretation system. The notion of a governing category is clearly syntactic. But Chomsky's original intuition behind the notion of an index is that it represents the referential value of an expression (Chomsky 1980a: 90, for instance proposes to reserve the integer 1 for arbitrary reference).

But even the interpretive component of the language system is not a unified system. As shown in Heim (1982) and Reinhart (1983), it is necessary to distinguish between binding and co-reference. Heim (1982) discussed the following pattern:

(11)

a. *This soldier* has a gun. Will *he* shoot?

b. *Every solder/No soldier* has a gun. *Will *he* shoot?

(p. 381) In (11a) *this soldier* and *he* can be assigned individuals from the conversational domain, and these individuals can be the same. However, in (11b) *every soldier* and *no soldier* are quantificational expressions and do not refer to individuals, hence do not introduce an individual into the conversational domain. Hence, *he* in (11b) cannot be anaphorically related to *every soldier/no soldier*. However, as is shown in (12), this does not preclude a dependency in another configuration:

(12)

a. *The soldier* was convinced that *he* would have to shoot.

b. *Every soldier/No soldier* was convinced that *he* would have to shoot.

The contrast between quantificational expressions and individual denoting expressions as manifested in (11) disappears in the context of (12). It reappears if the quantificational expression is further embedded as in (13):

(13)

a. The rumor that *the soldier* was a coward convinced *him* to shoot.

b. The rumor that *every soldier* was a coward convinced *him to shoot.

To cut things short, (12) exhibits binding, and binding requires the binder to c-command the bindee (not met in (11b) and (13b)) whereas (11a) and (13a) allow an interpretation based on co-reference.

Hence, within the interpretive component indices have a hybrid status. In the case of co-reference they really appear to reflect a value (as in Chomsky's original intuition), in the case of binding they just annotate a dependency.

But, in fact, the role of indices is not limited to the interpretive system. They also play a role in the syntactic system in that period, making their hybrid status even more pronounced. Chomsky (1980a: 53) considers an index to be part of the feature matrix of a lexical element (technically, the ‘complex symbol’ of a category) and also syntactic movement yields (co-)indexing between the moved element and its trace. On the other hand, in Chomsky (1980a: 90), not only indices but also an indexing procedure is part of the interpretive component, applying from top to bottom, assigning indices to elements that have not yet received one. Given all this, indices clearly fail to reflect a unified notion.

Fiengo and May (1994, henceforth F&M) start out departing from Chomsky's intuition. They explicitly state (p. 1) that the function of indices in syntax is to afford a definition of syntactic identity: elements are ‘the same’ only if they bear occurrences of the same index, ‘different’ if they bear occurrences of different indices. This syntactic notion of an index is in principle distinct from a semantic notion of identity (the system is explicitly taken to contain a mapping from syntactic to semantic identity). Yet, in F&M's system, indices effectively have a hybrid status as well. Although it is stated that indices express syntactic identity, the only justification for *John* and *him* and similar pairs to bear the same syntactic index is semantic identity. Syntactically, *John* and *him* are just different expressions.

One of the main issues arising in any syntactic conception of indices, and a major theme in F&M's important book, is the incomplete match between indices and (p. 382) values. That is, although co-indexing always represents an interpretive dependency between two expressions (although in the case of a quantificational antecedent, not precisely identity of value), the converse cannot hold. To take F&M's example, it must be possible for a speaker to remain uncommitted as to whether two expressions have the same value. For instance, if a speaker sees a person putting John's coat on, but cannot see his face, he may utter (14):

(14) He put John's coat on.

It may well be the case that the speaker in fact saw John putting on John's coat. The indexing cannot be (15a), lest a condition C violation ensues, but what then would the indexing in (15b) mean?

(15)

a. He_3 put $John_3$'s coat on

b. He_4 put $John_3$'s coat on

It cannot mean that he_4 and $John_3$ have distinct values; rather, this difference in indexing must leave open whether or not the values are distinct. This holds true in a wide range of cases, including all non-tautological identity statements. Although F&M's approach contains technical solutions for such problems, they do not eliminate what is the source of the problems to begin with, namely the assumption that syntax is the place to encode whether or not a pronominal has a particular value, or enters into a particular dependency.

Reinhart (1983) also discusses such issues, and focuses on the differences in semantic status of co-indexings brought up in Heim (1982), as discussed above. Consider for illustration the structures in (16)and (17):

(16)

a. $Jack_3$ hated the $woman_4$ that punished him_3

b. The $woman_4$ that $Jack_3$ hated punished him_3

(17)

a. Every boy_3 hated the $woman_4$ that punished him_3

b. *The $woman_4$ that every boy_3 hated punished him_3

In both (16a) and (17a) the co-indexing carries with it a particular instruction to the interpretive system, namely to create a variable binding interpretation. In (16b) the relation between *$Jack_3$* and *him_3* is one of co-reference, not binding, as is shown by the impossibility of the bound variable dependency in (17b). So, one and the same annotation is interpreted differently depending on the configuration in which it appears. From these and other facts, Reinhart concludes that in the simplest theory syntactic co-indexing is not interpreted at all, unless there is a c-command configuration where it is interpreted as binding (see Büring 2005). That is, there is no reason to assume that a co-referential interpretation of *Jack* and *him* in (16b) is encoded in the syntax. The upshot is that the hybrid notions of index and co-indexing that underlie the CBT are deeply problematic.

There is a further reason to be suspicious about an interpretation of co-indexing and the dependencies it annotates as syntactic. Canonical cases of syntactic (p. 383) dependencies are those involving movement/Internal Merge as in (18a). These dependencies are all characterized by their sensitivity to locality. Binding as such, however, is not, as is illustrated in (18b), where binding into an adverbial clause is impeccable:

(18)

a. *I wondered who$_4$ Jack got upset after Jill had kissed t$_4$

b. I wondered why every boy$_4$ got upset after Jill had kissed him$_4$

This indicates that resolving the hybrid status of indexing by pushing a syntactic reinterpretation has no promise. Rather, what is needed is a strict separation between the types of dependencies that are syntactically encoded and those dependencies that result from interpretive processes. This is precisely what the minimalist program brings about. It rules out indices as syntactic objects in a fundamental way.

17.3 A minimalist conception of syntactic binding

Within the minimalist program the computational system of human language (C_{HL} = syntax) reflects the combinatorial properties of a vocabulary of elements that only contain independently established phonological, morphosyntactic, and lexical features. Clearly, a lexical element as it is stored in the vocabulary cannot contain anything like an index as a feature, since that would prejudge its use. To put it differently, in the conception of an index in either Chomsky (1980a, 1981a) or F&M, virtually all nominal expressions are compatible with any index from an infinity of indices, clearly impossible to accommodate in the lexicon other than at the cost of unbounded multiplication of lexical entries.

Grammatical computations are taken to meet the Inclusiveness Condition: Any structure formed by the computation is constituted of elements already present in the lexical items selected. No new objects can be added in the course of the derivation. This excludes the possibility of adding indices to objects during the derivation as in Chomsky (1980a). Also, empirically there is no place for indices as morphosyntactic objects, since no natural language contains morphological objects/features remotely like indices, or marks co-reference morphosyntactically. Hence, indices, the core ingredient of the CBT, are not available within C_{HL}. This means that all earlier mechanisms based on indices must be reassessed.

In GB theory, indices played a crucial role in stating movement, as they marked the dependency between a moved element and its trace. The minimalist program pursues the idea that all movement can be reduced to Internal Merge: copying and merging, or re-merging an element that has been merged/put into the structure in an earlier stage of the derivation. The role of indices to mark this dependency (p. 384) has been taken over by the identity relation inherent in *being a copy of*, or even stronger, by the result of Internal Merge being that one element has two or more *occurrences* in the structure (where $x-y$ is the occurrence of a in xay).[3] So, the strictly syntactic use of indices as markers of identity has been taken over by identity as an inherent property of a in expressing that xay and uaw are occurrences of a. While this eliminates the problematic role of indices in movement, it leaves us with the task of reassessing the role of indices in binding, and seeing what can take over their role.

Having eliminated indices from the syntax, Chomsky (1995b) concludes that binding conditions must apply at the C-I interface, where a purely semantic indexing procedure could in principle still be available. Reuland (2001) shows that this cannot be correct. Conditions A and B are locality conditions. Locality is the hallmark of syntactic operations, and one would not wish to duplicate locality at the interface. Binding itself is not subject to locality, as the contrast between (18a) and (18b) shows. Hence it is proper to analyze binding itself as an operation that takes place at the interface, and the reasons for locality must be factored out.

We must, therefore, reassess the binding theory itself, and separate what is properly syntactic from what takes place in the interpretive component.

The minimalist architecture guides this endeavor, since it is entirely unequivocal in what can be syntactic. Syntax allows for three types of operations: i. Merge (external and internal); ii. Match; iii. Delete (up to recoverability). As discussed, Merge is the operation of building structure by combining more elementary objects. Match is a component of what is called checking in the earlier minimalist literature and subsequently Agree. It is a trivial property of any computational system with an identity predicate that it must be able to assess whether or not two occurrences of an object are occurrences of the same object. Such objects can be composite, but also elementary features. Delete is a component of feature-checking and Agree. The typical instance is one where an uninterpretable feature (such as case on a noun or person, number or gender features on verbal inflection) is deleted to ensure full interpretability.

Given the way deletion is conceived, and given the fact that it is subject to a recoverability requirement (deletion may not result in loss of information), deletion is always under identity with a matching object. For instance, a person feature on a subject may be used to delete an occurrence of the same person feature on the inflected verb.

It is important for our current concerns also that checking/Agree may convey syntactic identity. Logically, there are two ways in which the effect of deletion **(p. 385)** could be represented in the structure. One is to remove a particular occurrence of an element/feature from the structure, resulting in either an empty position/slot in a feature matrix or even the complete removal of the position. In either case, this original feature occurrence is entirely invisible for any subsequent step in the computation. Given the fact that in typical cases of checking, as in case-checking on nouns, or agreement-checking on finite inflection, the features remain visible for whatever determines morphological expression, this implementation can be discarded. The alternative is one in which the content of one occurrence of a feature is used to overwrite the content of another occurrence. In this case the slot/position remains, and is visible to realization processes as required. There are various ways to implement this (see Pesetsky and Torrego 2004 for a particularly well-thought-out implementation).

Importantly, all implementations based on overwriting (the content of) one occurrence with (the content of) another, copying a value, etc., induce identity between the objects these occurrences are occurrences of—just as the copying operation of Internal Merge yields different occurrences of the same object. Pesetsky and Torrego make this effect explicit in distinguishing between occurrences of a feature and instances. If one occurrence of a feature is overwritten with the content of another occurrence, or if a value of one occurrence is copied onto another occurrence, these occurrences now represent instances of identical objects. Consequently, checking/Agree also provides us with a representation of syntactic identity without indexing.

The architecture of the minimalist system thus enforces a demarcation between syntactic and semantic identity (or dependence). Therefore, the conceptual and empirical problems surrounding the hybrid notion of an index get resolved in a principled way.

What remains, then, is the task to implement the syntactic residue of binding, i.e. the processes involved in binding that are subject to locality, in a model allowing no more than (Internal) Merge and Check/Agree. This entails that any strictly minimalist approach is committed to the Feature Determinacy Thesis:

> **(19)** Feature Determinacy Thesis
> Syntactic binding of pronominal elements (including ‘anaphors’) in a particular environment is determined by their morphosyntactic features and the way these enter into the syntactic operations available in that environment.

Hence, being an anaphor or being a pronominal are not primitive properties, but derived ones. It also entails that one and the same element may behave differently in different syntactic environments, and that two elements with different feature content need not behave identically when they occur in the same environment, even if both are ‘defective’ in some respect. It also follows that two cognate elements in different languages, with a similar feature composition, may behave quite differently if there is just a minor difference in the grammar of these languages. **(p. 386)** A rather arbitrary selection of different binding patterns as in (20) may serve as an illustration:

(20) Some examples of variation in anaphoric systems:

- There is cross-linguistic and cross-anaphor variation in the binding domains:
  - Scandinavian *seg/sig* versus Dutch *zich* and German *sich* (Everaert 1986)
- Under certain structurally defined conditions certain anaphoric forms need not be bound:
  - free (‘logophoric’) use of *himself* in English (Reinhart and Reuland 1993)

- John was hoping that Mary would support *(no one but) himself;
  - free (‘logophoric’) use of *sig* in Icelandic.
- Certain languages allow locally bound 3rd person pronominals:
  - *him* in Frisian: ***Jan*** *waske* ***him*** ‘John washed’.
- Certain languages allow locally bound proper names:
  -

R-yu’làèa’z	Gye’eihlly	Gye’eihlly	(San Lucas Quiaviní Zapotec, Lee 2003)
HAB-like	Mike	Mike	
‘Mike likes himself.’			

  -

Pov	yeej	qhuas	Pov.	(Hmong, Mortensen 2003)
Pao	always	praise	Pao	
‘Pao always praises himself.’				

- Certain languages require a special form for local binding, but do not require that form to be locally bound:
  -

Malayalam (Jayaseelan 1997)			
raaman$_i$	tan-ne$_i$	*(tanne)	sneehikunnu
Raman	SE-acc	self	loves
‘Raman loves him*(self).’			

  - Peranakan Javanese (Cole, Hermon, Tjung 2008)

i.

[Gurue	Tono$_j$]$_i$	ketok	**dheen**$_{*i/j/k}$	nggon	kaca.
teacher-3	Tono	see	3sg	in	mirror
‘Tono's teacher saw him/her in the mirror.’					

ii.

Alij	ngomong	nek	aku	piker	[Tonoi	ketok	**awake**
Ali	N-say	COMP	1sg	think	Tono	see	body-3
dheen$_{i/j/k}$	nggon	kaca].					
3sg	in	mirror					
‘Ali said that I thought that Tono saw himself/him in the mirror.’							

The point is not that this type of variation cannot be described in pre-minimalist conceptions of grammar; many of these patterns were discovered and described quite early. The point is rather that the fact that this variation exists is totally mysterious in a pre-minimalist conception, and undermines its basic principles. Given the tools available in the CBT, with the notions of anaphor and pronoun (p. 387) being primitive, this variation could only be described by highly specific stipulations (including the stipulation of massive ambiguity between pronominals and anaphors, or even between proper nouns and anaphors). It is the type of ‘tools’ that a minimalist approach to syntactic binding is restricted to that facilitated developing the right perspective.

The Feature Determinacy Thesis shifts the focus in the investigation of binding from macro principles such as the CBT to the question of what types of feature clusters allow or enforce the formation of syntactic dependencies under which conditions.

It would lead us beyond the scope of this chapter to review the particular analyses that have been proposed for the variation at hand. Let me just briefly characterize the essential ideas behind some of the analyses.

Hornstein (2001) proposed as a general idea that a minimalist conception of construal should be based on movement. Boeckx et al. (2007) argue that the San Lucas Quiavini Zapotec (SLQZ) and Hmong facts present evidence for argument movement as a means to syntactically encode identity, following the line in Hornstein (2001).

Kayne (2002), also proposes that movement is a factor underlying syntactic binding, although his reduction is less rigorous than in Hornstein's conception. Zwart (2002) elaborates on Kayne's proposal.

In the system of Safir (2004), the notion of an index is abandoned in favor of a notion of dependency as the core notion underlying anaphoric relations.

Reuland (2001,[4] 2005) shows how Check/Agree encodes the anaphor antecedent dependency for SE-anaphors as a feature chain, and how a minor difference in the Case system explains contrasts of the type exemplified by Dutch versus Frisian. Reuland (2006, following the intuition of Everaert 1986) shows how the differences in binding possibilities for SE anaphors between Dutch and German (SOV) on the one hand and Scandinavian languages (SVO) on the other follow if the binding is established by chain formation which is disrupted by the syntactic movements leading to the ‘extraposition’ configuration of control clauses in Germanic SOV languages. Reuland (2001, 2008, forthcoming) shows how implementing binding of English SELF onto anaphors by covert movement of SELF onto the predicate explains the difference between exempt (‘logophoric’) and non-exempt positions (Reinhart and Reuland 1993) in terms of standard syntactic conditions on movement. The facts of Malayalam and Peranakan Javanese can be understood in terms of what triggers such movement. In all these cases, locality conditions on the binder-bindee dependency immediately follow from general locality conditions on the syntactic processes encoding it.

As discussed in detail in Reuland (2001, forthcoming), binding of 3rd person pronominals cannot be encoded by the formation of feature chains; as argued there, chain formation would violate the principle of recoverability of deletion, since (p. 388) different occurrences of a number feature need not be interpretively equivalent. Assuming that feature chain formation rather than movement is the mechanism of choice (*pace* Boeckx et al. 2007), this has two consequences. One is that pronominal binding cannot take place in the syntax, hence must take place in the interpretive system, which explains why it is not subject to locality. The other is that we must now find an independent explanation for why certain environments show complementarity between bound anaphors and bound pronominals. Invoking Conditions A and B no longer provides that, since it is these conditions themselves that must be explained. This issue will be taken up in section 17.5. First I will discuss some fundamental issues concerning the architecture of the system.

17.4 Dependencies beyond syntax

In section 17.2 we took a very broad definition of anaphoric elements as our starting point: elements that receive their interpretation in terms of some element that has been previously introduced. Even the notion of an element is very broad here. It covers both expressions such as *he* and individuals in the domain of discourse that serve as values for *he*. This is also what we want. In (21), nothing in the sentence determines whether *The old baron* and *The driver* are to be used to refer to the same individual or not.

(21) The old baron was crossing the bridge at dusk with a ramshackle carriage.
The driver was visibly tired. Suddenly, the carriage tipped over and the man fell into the swamp.

With *the man*, it's a bit different. It will be hard to use it for an individual other than the driver in the context given in (21). It is easy to conceive of an account for this based on pragmatic considerations—which I will not attempt to do here. But if (21) is only part of a story and in a prequel a third individual is very much highlighted, valuing it as this third individual is quite conceivable.

In principle nothing needs to change if one switches to pronominals, as in (22):

(22)
- **a.** The robber had entered the vault. John's accuser swore that he had taken the diamonds.
- **b.** He_1 had entered the vault. John's accuser swore he_2 had taken the diamonds.

In (22a) *the robber* can be John, John's accuser, someone else, *he* can be the robber, John's accuser, John, or someone else; in (22b) he_1 can be John, John's accuser, someone else, he_2 can be he_1, John's accuser, John, or again someone else. The subscripts on he_1 and he_2 serve no other purpose than marking them as different (p. 389) occurrences. In all these cases one expression may, but need not, have the same value as another expression, and none of this co-valuation has anything to do with syntax.

All this serves to illustrate the fact—well known, but not always sufficiently appreciated—that in principle valuing expressions is free within the limits set by how we can use the concepts they express. This freedom follows from the basic conception of what syntactic and interpretive systems are, which leads to an entirely natural division of labor. This freedom entails that two different expressions can have an identical value, or—introducing a slight asymmetry along a temporal dimension—that one expression can receive a value that already has been assigned to a previous expression, which makes the second expression *anaphoric* to the first one. This asymmetry motivated a manner of speaking in which the second expression is anaphorically dependent on the first expression. However, there is no need to assume that this type of dependency is in any way encoded in the linguistic system.

As we saw in the previous section, there are types of dependency that are syntactically encoded. From an architectural perspective—and in fact also from an evolutionary one—this is a non-trivial property of the language system. It means that natural language allows interpreting an expression in terms of another expression instead of assigning a value directly. The latter property characterizes a set of dependencies that is in fact broader than what is encoded in syntax proper, but constitutes binding in general.

One consequence of the elimination of indices is that a new definition of binding is needed to replace the index-based definition in (7). Reinhart (2000, 2006) argues that in fact a notion of binding that is based on the standard logical definition is all that is needed:

(23) Logical syntax binding: Binding is the procedure of closing a property
A-binding
a A-binds β iff α is the sister of a λ-predicate whose operator binds β.[5]

So, in (24b) John binds the Poss phrase in the first conjunct, and Bill does so in the second conjunct, yielding the sloppy reading. The strict reading is represented in (24c).

(24)

a. John loves his cat and Bill does too.
b. John (λx (x loves x's cat)) & Bill (λx (x loves x's cat))
c. John (λx (x loves a's cat)) & Bill (λx (x loves a's cat)) & a=John

(24b) represents semantic binding: the pronominal is translated as a variable, the λ-expression is formed by applying quantifier raising to the subject, and syncategorematically inserting the λ to bind the variable translating the trace. Binding, then results if *his* and the trace translate as the same variable. These representations bring out clearly the difference that is relevant from the architectural perspective (p. 390) discussed: In (24c) *his=a* and valued directly, in (24b) *his=x*, which is interpreted by equating it with another expression, namely the variable *x* which raising *John* gives rise to.

All this entails that the human language faculty allows three routes for establishing an interpretation. One involves establishing an anaphoric dependency in the discourse. The other routes deal with elements that are anaphoric in a stricter sense, namely that they may or must depend for their interpretation on properties of another *expression*—one based on identity of expression in logical syntax, the other on syntactic identity.

From one perspective, this seems to yield a language system that contains redundancies, and is therefore far from optimal. But, properly considered, this view is not warranted. Importantly, none of the components has anything specific built into it to accommodate anaphora. The systems each do what they can do, and would be hard-pressed to be prevented from expressing what they express.

Any interpretation system will have to be able to assign values to expressions. And it can do so independently of syntax. It is the system of use which specifies whether the expression *the ham sandwich in the corner* is valued as some greasy mass someone dropped in the corner on the floor, or gets Geoff Nunberg's famous interpretation of the customer in the diner ordering a ham sandwich, and nothing in syntax determines whether the ham sandwich in the corner can also be appropriately referred to as *the beefburger in the corner*—which it can if the person in the corner ordered a ham sandwich and a beef burger, perhaps from different waitresses.

Move and Agree create syntactic dependencies, and nothing in syntax can prevent chain-type dependencies from being interpreted in a particular way.

QR creates particular configurations, pronominals are interpreted as variables, and only ad hoc restrictions could prevent the option of translating the expression *his* in (24b) as a variable that ends up being bound by the QR-ed element.

Hence, intrinsically, no inefficiency arises. But the use of this modular system is in fact efficient, since—from syntax to discourse—the domain restrictions decrease, and each less restricted process is effectively used where some more restricted process is not available. But since there is overlap between domains, the question comes up how in cases of overlap labor is divided. This will be discussed in the last section.

17.5 Economy and division of labor

A widely discussed issue is the overlap in domains between variable binding and co-reference. As illustrated by the VP-ellipsis in (24), the possessive pronominal (p. 391) can end up being co-valued with the subject either by variable binding, or by co-reference. The distinction will show up in the interpretation of the second conjunct, but locally the two strategies yield the same interpretation.

In (25) this gives rise to a well-known problem. The reading where *him* is bound by *Oscar* is ruled out by condition of the CBT (we can ignore at this point how condition B is ultimately derived).

(25) **Oscar* admires *him*.

The question is, then, how the possibility of assigning the pronoun *him* Oscar as its referent can be blocked, since this would generally void the effect of condition B when the binder is referential, contrary to fact.[6] To this end, Reinhart (1983) formulates a 'traffic rule', Rule I, given here in the formulation of Grodzinksy and Reinhart (1993):

(26) Rule I: Intrasentential Coreference

NP A cannot co-refer with NP B if replacing A with C, C a variable A-bound by B, yields an indistinguishable interpretation.

In (25) the co-referential and bound variable interpretation would be indistinguishable, hence co-reference is blocked and condition B is free to apply. This rule correctly predicts that in cases where the interpretations are not indistinguishable, co-reference is possible, as in (27).

(27) I know what John and Mary have in common. Mary admires him and John admires him too.

Here *him* can be valued as John since the common property is that of *John-dmiration*, not that of *self-admiration*. However, as Reinhart (2000, 2006) discusses, Rule I as given in (26) faces an important problem, illustrated by cases of VP-ellipsis, as in (24), repeated here as (28):

(28)

a. John loves his cat and Bill does too.
b. John (λx (x loves x's cat)) & Bill (λx (x loves x's cat))
c. John (λx (x loves a's cat)) & Bill (λx (x loves a's cat)) & a=John

As we saw, such sentences allow both a strict reading, based on co-reference, and a sloppy one. The question is then why the strict reading is not blocked as it is in (25). One could say that the co-reference option is licensed by the difference it makes for the interpretation in the second conjunct—hence making the interpretations distinguishable as in canonical cases such as *Obviously, everybody hates Oscar. Even **Oscar** hates **him***. But, as Reinhart notes invoking this clause gives the wrong result for (29):

(p. 392) **(29)**

a. He likes Max's mother and Felix does too (he ≠ Max)
b. Max praised him and Lucie did too (him ≠ Max)

A distinct interpretation in the second conjunct cannot license a condition B or C violation in the first conjunct. Because of this, Reinhart modifies Rule I, and also reconceptualizes it. In Reinhart's new conception it is viewed as an effect of blocking: 'if a certain interpretation is blocked by the computational system, you would not sneak in precisely the same interpretation for the given derivation, by using machinery available for the systems of use' (Reinhart 2000, 2006). This still rules out (25), but it does not enforce a BV interpretation whenever it is available as in (28).

Reinhart's discussion is concerned with the contrast between binding (her computational system covers all cases of variable binding irrespective of how it is encoded) and co-reference. However, the same issue arises in the interplay between the presence or absence of syntactic encoding.

Consider the pair of Dutch sentences in (30), with dependencies encoded as in (31). The subscripts in (31a) represent the dependency formed by feature chain formation. This is just the mechanism needed for all instances of licit binding of simplex anaphors.

(30)

a. *Oscar* voelde [*zich* wegglijden]
b. **Oscar* voelde [*hem* wegglijden]
Oscar felt [him (self) slide away]

(31)

a. Oscar$_{\phi}$ voelde [zich$_{\phi}$ wegglijden]
b. Oscar λx (x voelde (x wegglijden))

Why is (30b), then, ruled out? No illicit process appears to be involved if (30b) is interpreted as (31b). Reuland (2001) resolves this by proposing that deriving (31b) from (30a) is more economical than deriving it from (30b). This is based on an economy hierarchy between components in the order: syntactic encoding < variable binding < co-reference which yields a preference for syntax where available.[7] In this form the proposal has a drawback, however, since it requires a comparison between two derivations from different numerations (one containing *zich*, the other *hem*) which the canonical approach to economy of derivation as in Chomsky (1995b) does not allow.

Exploring an economy-based account in a somewhat different case, Hornstein (2007) and Boeckx et al. (2007) resolve this problem by stipulating that functional material in general, including pronominals and anaphors, is not part of the numeration, but inserted whenever required. This solution's drawback is, then, that it violates the inclusiveness condition in spirit or form. The problem finds a natural solution, however, if we elaborate Reinhart's alternative approach.

(p. 393) The operative principle is that the language system is indeed based on a hierarchy in which syntax has priority over the components of the interpretation system, and variable binding in logical syntax priority over co-reference in the discourse system. However, the hierarchy does not require to use ‘syntax if you can’, etc.; rather, it disallows rescuing a derivation in which a syntactic principle is violated. Similarly, where syntax cannot be used—since the syntactic conditions for chain formation are not met for Poss phrases in languages like English and Dutch—as in (28), there is no need to use variable binding/logical syntax where possible. Again, Rule I only disallows using a discourse-based strategy to bypass a violation, as in (25).

In the case of an attempt to derive (31b) from (30b), the steps are: (i) represent the dependency syntactically by chain formation; (ii) establish that chain formation violates the principle of recoverability of deletions; (iii) cancel the derivation. As Chomsky (1995b) puts it, canceling a derivation means that no alternatives using the same numeration will be considered. No comparison with the derivation containing *zich* instead of *hem* is required. The derivation with *hem* is discarded entirely on its own (lack of) merits: ‘rejection is final.’

While the facts of (30) at least might seem to allow for a—less principled—account in terms of a more traditional version of the binding theory, the following paradigm from Brazilian Portuguese (BP) discussed in Menuzzi (1999) does not.

BP has two ways of expressing the 1st person plural pronominal: the expression *a gente* ‘the people’ and the canonical Romance pronoun *nós*. Its 1st person interpretation notwithstanding, *a gente* is formally 3rd person, as indicated by verbal agreement. This shows that *nós* and *a gente* differ in Φ-feature composition. Despite this fact, *nós* is a possible binder for *a gente* and vice versa. This indicates that for binding the semantic type prevails. The pattern is shown in (32):

(32)
a. Nós achamos que o Paolo já viu a gente na TV.
‘We think that Paolo has already seen us on TV.’
b. A gente acha que o Paolo já nós viu na TV.
‘We think that Paolo has already seen us on TV.’

This option also exists in a more local environment such as locative PPs.[8]

(33)
a. Nós tinhamos visto uma cobra atrás de nós.
‘We had seen a snake behind us.’
b. A gente tinha visto uma cobra atrás de nos.
‘We had seen a snake behind us.’
c. A genta viu uma cobra atrás da gente.
‘We saw a snake behind us.’

(p. 394) In the environment of (33), Dutch and English show no complementarity between pronominals and anaphors. But, in (34) a semantic match is not sufficient. Binding is ruled out unless antecedent and pronominal match in Φ-features.

(34)
a. Nós deviamos nos preparar para o pior.
‘We must prepare ourselves for the worst.’
b. *A gente devia nos preparar para o pior.
c. A gente devia se preparar para o pior.
d. *Nós deviamos se preparar para o pior.

A gente cannot bind *nos*, nor can *nos* bind the 3rd person clitic *se*, which would be the proper bindee for *a gente*. (34) represents a narrower domain than (33). (34) reflects the domain of chain formation. Since syntactic chains are based on Φ-feature sharing, non-matching features result in an ill-formed syntactic object, indeed a cancelled derivation that blocks alternatives (Chomsky 1995b).

Thus, the BP facts show three important things:

1. They provide independent evidence for a competition between narrow syntax and logical syntax.
2. They support ‘a rejection is final’ rationale for the role of economy in the division of labor.
3. They show how the syntactic micro structure down to the level of morphosyntactic features plays a crucial role in explaining conditions on binding.

17.6 Grammar and discourse

Many factors governing the interpretation process are lexical/structural, but as we saw, there is a domain where neither syntax nor the conditions on variable binding in logical syntax have anything to say. In such environments the role of discourse factors becomes visible. Interesting examples have been provided in Pollard and Sag (1992), presented below with some indexings added for expository purposes:

(35)
a. Bill$_j$ remembered that *Tom*$_i$ saw [a picture of himself$_{i/*j}$] in the post office.
b. Bill$_j$ remembered that the *Times*$_i$ had printed [a picture of himself$_{*i/?j}$] in the Sunday edition.
c. Bill$_j$ thought that *nothing*$_i$ could make [a picture of himself$_{*i/j}$ in the Times] acceptable to Sandy.

The anaphors in (35) are all in ‘exempt’ position. From the approach in Reuland (2001), the exempt property follows since the self-element cannot move onto the main predicate. As these examples indicate, *himself* must receive a sentential antecedent, but an intervening subect does or does not block a dependency across (p. 395) it depending on whether it qualifies as a potential antecedent. Note the striking contrast between the cases of (35), and cases where *himself* is a syntactic argument of a predicate, as in **Bill$_j$ remembered that the Times*$_i$ *had printed himself* $_{*i/*j}$ *in the Sunday edition*, where there is no escape from ‘impossible’ binding.

This pattern follows if grammar itself just says nothing about which antecedent an anaphor in exempt position must take, but the human processor follows an economy principle to the effect that early binding of an open position is preferred.

The role of discourse factors is further illustrated by the following contrast, also from Pollard and Sag (1992):

(36)
a. John$_i$ was going to get even with Mary. That picture of himself$_i$ in the paper would really annoy her, as would the other stunts he had planned.
b. *Mary was quite taken aback by the publicity John$_i$ was receiving. That picture of himself$_i$ in the paper had really annoyed her, and there was not much she could do about it.

There is a clear contrast between (36a) and (36b), although, structurally, the position of the anaphor *himself* is identical in both cases. Again, *self* cannot make a predicate reflexive for syntactic reasons, opening the door for discourse factors. In (36a) John's viewpoint is expressed, in (36b) Mary's. Hence, in (36b) *John* does not yield a proper discourse antecedent for *himself*. Such facts illustrate that where the computational system—broadly conceived—has nothing to say, discourse licensing comes in, with all the plausibility considerations and non-categorical distinctions this entails. Many more instances can be given, such as the contrast between logophoric and bound *sig* in Icelandic, but for present purposes these facts suffice.

17.7 Conclusion

By eliminating the notion of an index, the minimalist program enforces a strict distinction between how syntactic, semantic, and discourse factors contribute to anaphoric relations.

Notes:

(1) The English rule was stated as in (i), the Dutch ruled as in (ii):

((i))

X –	NP –	AUX –	V –	NP – Y	
	1	2	3	4	⇒
	4	2+BE+EN	3	by+1	

((ii))

X –	NP –	NP –	V-	AUX – Y	
	1	2	3	4	⇒
	2	door+1	3	PASS+WORDEN+4	

(2) The choice of SUBJECT is restricted by the accessibility condition (Chomsky 1981a), which I will not discuss here.

(3) Note that in versions of minimalist grammars that use numerations, there is a use of the term ‘index’ to mark the number of occurrences of a lexical item in the numeration, e.g. the numeration that will yield *the girl hit the ball* contains two occurrences of *the*. These have different indices, or alternatively, *the* will have the index 2 to represent that there are two *the*'s in the numeration. This use as a computational device has to be distinguished from the use of indices in the canonical theory.

(4) Preliminary version presented at GLOW 1995.

(5) With α, β, of the right—argumental—type.

(6) The fact that children appear to do so—the Delay of Condition B effect—has been widely discussed in the acquisition literature (e.g. Wexler and Chien 1985, Grodzinsky and Reinhart 1993).

(7) See Vasić (2006) and Koornneef (2008) for extensive discussion and experimental support for the existence of such preferences in sentence processing.

(8) Menuzzi does not give the *Nsós .. a gente* pattern in PPs, but a Google search instantiates this pattern as well.

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Minimalist Construal: Two Approaches to A and B

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[–] Abstract and Keywords

Until recently, mainstream minimalist theorizing has treated construal as a interface process rather than as a part of core grammar. Recently, a number of authors have resisted this categorization and have tried to reduce binding and control relations to those established by movement, agreement, or some combination of the two. This article compares and contrasts two theories that give the grammar a privileged position with respect to the establishment of binding relations. It discusses variants of Hornstein's movement-based analysis of construal and Reuland's Agree-based theory of reflexive binding.

Keywords: minimalism, construal, grammar, binding relations, Hornstein, Reuland, Agree-based theory, reflexive binding

Until recently, mainstream minimalist theorizing has treated construal as a (CI) interface process rather than as a part of core grammar.[1] Recently, a number of authors have resisted this categorization and tried to reduce binding and control relations to those established by movement, agreement, or some combination of the two.[2] In this chapter we'll compare and contrast two theories that give the grammar a privileged position with respect to the establishment of (at least some) binding relations. We'll discuss variants of Hornstein's (2001) movement-based analysis of (p. 397) construal and Reuland's (2001, 2005) Agree-based theory of reflexive binding. For ease of exposition, we'll refer to the former as Chain-Based Construal (CBC) and the latter as Agree-Based Construal (ABC).

18.1 Reasons to treat binding as a grammatical process

First, construal relations display the characteristic hallmarks of core grammatical processes. For example, both local reflexivization and local control are obligatory, respect a domain locality restriction, and (at least to a first approximation) adhere to a c-command requirement on antecedent—anaphor dependencies. Though it is logically possible that linguistic phenomena displaying these three signature properties of the grammar might fall outside the core, the fact that construal relations have them constitutes a prima facie reason for thinking that they all involve processes that lie within the core computational system.

This conclusion is especially persuasive with regard to locality. Given a minimalist mindset, conditions can arise in only two ways: as by-products of the computational system or as restrictions imposed by the interfaces (CI being the relevant one for construal). Thus, if locality restrictions on construal are not by-products of the computational system, they must arise from inherent interpretive properties of the CI interface. However, it is unclear what the source of such restrictions might be.

This is somewhat tendentious. Reinhart and Reuland (1993), following the original suggestion in Bach and Partee (1980), treat local reflexivization as an argument-changing process. It is thus defined over the argument structure of a predicate, a very local configuration. Conceptually, this is the right kind of move. However, there are two problems if this is extended to construal more generally. First, it cannot be extended to local control configurations, as control typically relates arguments of different predicates.[3] Second, as has long been noted, it cannot extend to cases of local binding like (1) (discussed further below), where the reflexive is clearly in a different argument domain from its antecedent.

(p. 398) **(1)**

a. John believes himself to be important.
b. John would greatly prefer for himself to win.
c. John heard himself leave the building.
d. The boys waited for each other to speak.

This is the only proposal we know of that conceptually reduces the locality of construal to a plausible interface property, viz. the locality that co-arguments of a single predicate enjoy. Note that other conceptions of argument structure (such as that used in Pollard and Sag 1992) have no plausible status as interface conditions, since the notion of 'co-argument' that they make available has no direct semantic significance.[4] Absent such a source, the only other option is to analyze locality in terms of features of the computational system, i.e. in terms of how construal relations are established rather than the interpretations they come to possess.

This conclusion is buttressed by two further observations. First, local construal relations interact with other parts of the grammar that are thought to be products of the computational system, such as agreement. This is particularly evident in control relations where controlled PRO functions with respect to ϕ-feature transmission very much like an A-trace due to movement.[5] Second, as has been known since Chomsky (1981a), there is a lot of overlap in the properties of movement and construal. For example, Chomsky (1981a) is in part based on the observation that A-traces that arise from movement distribute largely the way that local anaphors subject to principle A do. This is what enables GB to reduce movement effects to A-chain restrictions. Similarly, within GB, PROs and traces are understood to be identical at LF once indexing has occurred, thus allowing their substantial similarity to be theoretically accommodated. In sum, it has long been recognized that the outputs of the grammar (movement chains) and the subjects of construal (binding/control relations) are empirically very similar.

There was a second motivation behind the early resistance against banishing construal to the interface hinterlands. It stems from our present epistemological position with respect to our understanding of the properties of grammar versus those of the interfaces. We understand the first far better than we do the second. As such, treating construal as consequences of interface operations functions to weaken our theoretical obligations. All things being equal, grammatical proposals are easier to evaluate, develop, and understand than those based on interface principles and properties that we have barely begun to develop. Methodologically, then, treating phenomena in terms of the grammar—especially those that have grammatical fingerprints all over them—is the right way to go.

(p. 399) These two lines of reasoning have clearly resonated with the minimalist community, for it is now largely accepted that (at least some) construal relations reflect operations of the core computational system.[6] It is the basic outlines of these proposals that we will discuss in what follows.

One more point: we will focus exclusively on binding rather than control. The reason for this is strategic. There already exists considerable literature and debate about control theory and various ways of approaching it within a minimalist context.[7] There has been less debate about how binding is to be addressed within a minimalist framework. Given space limitations, we have chosen to travel the path less frequently explored.

18.2 The explanans: two grammatical approaches to construal

There are currently two minimalist approaches to construal. The first treats construal as parasitic on movement. The second ties construal to the output of agreement. To fix ideas, consider the abstract structure in (2):

(2) […Antecedent F^0. …. Anaphor. ….]

CBC treats the relation between antecedent and anaphor as that between links in a chain. Thus the theoretical representation of (2) would be (3).

(3) […Antecedent$_1$ F^0. …. Antecedent$_1$…]

What we see in (3) is a movement chain between two copies of the antecedent. An anaphor, in effect, is the morphological offspring of a copy of the antecedent. In cases of local binding, this can surface as a reflexive; in cases of control, as a phonetic gap. The object interpreted at the CI interface is, in this case, an A-chain bearing multiple θ-roles. Two points are worth keeping at the mind's forefront with respect to this proposal. First, the morphology is largely an afterthought. The form of the anaphor (e.g. the reflexive) plays no real role in the interpretation afforded. The relevant property is the multi-thematic chain at CI. This means that the agreement features are essentially bereft of semantic interpretation. Whatever one sees is essentially a reflex of low-level morphology.[8] Second, the antecedence relation, (p. 400) which is an inter-nominal relation semantically (the anaphor is dependent on the antecedent for its interpretation), is a relation between DP positions syntactically. There is thus a smooth path from the syntactic relation to its semantic interpretation. In the simplest case the syntax to semantics conversion rule is as follows: α antecedes β iff α and β are links in a common chain. We get binding/construal when both α and β links in a common chain, sit in θ-position.

The CBC has two distinctive 'minimalist' features. First, it relies on the possibility of movement into θ-positions. This is not an option in a theory that assumes D-structure (DS). As minimalism has proposed the elimination of DS, such movement is a conceptual option.[9] Second, it is reductive in that it aims to reduce reflexivization (a construal process) to another grammatical operation (movement). If successful, this constitutes a step towards the simplification of the grammar by eliminating a distinctive class of rules and thus advances the aim of simplifying UG that is characteristic of the minimalist program.

ABC analyzes construal differently. The relevant syntactic structure is something akin to (4) and the syntactic engine relating antecedent to anaphor is the operation of Agree.[10]

[... [... F^0 .... Antecedent ...Anaphor .....]

ϕ

ϕ

(4)

Note that in (4) the antecedent and anaphor relate to one another indirectly. The direct grammatical relations hold between the functional head F^0 and each of the antecedent and anaphor. The antecedent and anaphor only relate to one another in virtue of the ϕ-feature relations established with a common functional head. A DP is interpreted as an antecedent of an anaphor in virtue of this indirect ϕ-feature agreement. This contrasts with CBC, which sets up a direct relation between the nominal relata. Note three further contrasts. First, if there are chains at all in ABC they are feature chains, or so it appears at first sight (see below). Movement is not integral to the analysis, though it may also occur depending on the features of F^0, (e.g. an 'EPP' feature might force movement).[11] Second as construal is essentially parasitic on feature agreement, the specific features involved are likely to be of the utmost importance. To the degree that morphology reflects these features, then the morphology will not be at all adventitious, again in contrast with the CBC.[12] Third, (p. 401) reflexivization is syntactically an inter-chain relation, the antecedent and anaphor being independent chains related via Agree. Thus, the ABC is consistent with some version of the θ criterion, in contrast to the CBC.[13]

What makes the ABC minimalist? First, there is the reliance on Agree as the basic operation relating antecedent and anaphor. Second, ABC shares the reductive impulse we noted in the case of CBC. This approach too aims to analyze construal as just a special case of another kind of grammatical relation, taking it to be a byproduct of agreement via Agree.

In what follows we sketch versions of these theories by going over some core cases and seeing how they treat them. We will then return to highlight their differences and discuss their technology, raising problems for them. After a brief overview of post-GB developments in binding (section 18.2), we begin with a discussion of local reflexives (18.3), compare and contrast these with their counterparts in other languages (18.4), proceed to a brief discussion of some residual facts regarding reflexives (18.5), move on to a discussion of bound pronouns (18.6), and then end with a brief rhetorical flourish. Before proceeding we would like to make one point clear. Though we have our favorite strategy (as will become clear), we believe that both approaches are worthy of development, and that the respective technologies for implementing these proposals hide the large extent that the two converge concerning the proper way of treating binding in a minimalist framework. It might well be the case that the CBC and ABC are two implementations of the very same theory when one clears away the technological debris. In particular, both theories agree that a minimalist theory of binding will have the following characteristics (to be discussed in further detail):

i. In both theories, reflexivity is reduced to a chain relation and the anaphoric relation between reflexive and antecedent is formed by general grammatical operations used in other parts of the grammar. In this sense, both approaches are reductive in that they aim to eliminate a specific binding module. Where antecedents are taken to be heads of chains, and reflexives are interpreted as bound variables
ii. Something like a c-command restriction obtains between binder and bindee but the restriction follows from general principle of grammar rather than being a restriction specific to binding rules.
iii. Some ungrammatical sentences, such as *John expects herself to like Mary*, are ruled out by conditions on the well-formedness of chains, not the elementary operations that create those chains.

(p. 402) Given those similarities, it will be useful to state the points of divergence. The theories do not agree on the answers to the following questions:
i. Are chains primitive objects?
ii. How are chains formed?
iii. What is the utility of phi-features within derivations?
iv. Do anaphors all belong to a natural class? In particular, are simplex anaphors (like *zich* in Dutch) generated by the same operations and subject to the same principles as complex anaphors (like English *himself* or Dutch *zichself*)?

18.3 The explanandum

18.3.1 The core data

We take the target of explanation to be facts accounted for by the classical Binding Theory (Chomsky 1981a, 1986b), with a few modifications.[14] The modifications include the following:

• Following Reinhart (1983) we distinguish binding from co-reference and assume that the grammar concerns itself exclusively with the former. This leaves a residue of well-known problems concerning the distribution of pronouns, in particular the unacceptability of co-reference in cases like *John likes him*. We follow Reinhart in assuming that such cases are unacceptable because they are blocked by the availability of bound readings such as *John likes himself*.[15] This assumes that binding is subject to economy considerations and that bound readings of an inter-nominal relationship where available trump co-referential interpretations in the same context.

• We assume that a reflexive within a picture noun phrase that is bound from outside its containing noun phrase is not a 'true' reflexive subject to principle A. Rather, it is a pronominal with special logophoric requirements.[16]

With this much as background, let's look at the theories:

(p. 403)
18.3.2 CBC

Here are some core cases of local reflexivization.

(5)
a. John washes himself
b. John believes himself to be intelligent
c. *John believes himself is intelligent
d. *John believes Mary to be imitating himself
e. *John's book exposed himself to ridicule

The sentences in (5) exemplify the local binding requirements on reflexives. The CBC approach to local reflexives is based on the LGB analysis (Chomsky 1981a). Here Chomsky observes the parallel between the sentences in (5) and those in (6).
(6)
a. John washes *t*

b. John was believed *t* to be intelligent
c. *John was believed/believes *t* is intelligent
d. *John believes Mary to be imitating *t*
e. *John's book exposed *t* to ridicule

The sentences in (6) are A-movement analogues of those in (5). The unacceptability of the sentences in both (5) and (6) have the same source in an LGB account. They each violate Principle A of the Binding Theory on the assumption that both reflexives and A-traces are anaphors.

The CBC likewise collapses reflexives and A-traces, but does so by proposing that each is formed by movement. Just as movement is blocked in (6c, d, e), so too is reflexivization in (5c, d, e). There are various accounts of *what* blocks movement in the examples in (6). However, whatever it is, it can also do so in (5) if reflexives are residues of movement. For concreteness, let's assume that something like Case Freezing, Minimality, and Extension regulate A-movement. These three principles *seriatim* block (6c, d, e) as well as (5c, d, e). Note that CBC can remain agnostic as to the exact mechanisms forcing local A-movement. The crux of the proposal is that, whatever they may be, they extend to reflexives in virtue of the claim that they are also the outputs of such movement. Of course, any complete theory must provide an account of what drives the locality.

CBC requires one additional assumption. It presupposes that movement into θ-positions is possible. Sentences like (5a) have the interpretation in which John is both 'liker' and 'likee', i.e. John has two θ-roles. The underlying structure of (5a) is roughly like (7). Note that there is a copy of *John* as complement of *like* and the specifier of *v*. Thus the *John*-chain in (7) spans two θ-positions. This encapsulates the central claim of CBC: reflexive interpretations arise when a chain spans multiple θ-positions. Reflexivization is an *intra*-chain relation that arises when movement creates chains which contain multiple θ-roles.

(7) [$_{TP}$ John Pres [$_{vP}$ John v [$_{VP}$ like John]]]

(p. 404) Importantly, representing the syntax of reflexives in this way supports a trivial mapping to the logical form in (8). (7) represents the fact that *John* has two θ-roles. It is both the logical subject and the logical object of *like*.(8) says the same thing; reflexivization has the semantic effect of forming complex monadic predicates ('complex' as more than one variable is bound, 'monadic' as all the variables are satisfied by a single element) with the head of the reflexive chain serving as value for the roles abstracted over. The copy theory makes the translation of (7)into(8) a very simple process.[17]

(8) John λx[x likes x]

One last point: the reflexive itself plays no important semantic role in these structures. Its primary function is to check the case of the transitive verb. The interpretively active object is the A-chain spanning several θ-positions formed by *John*. The reflexive is not part of this object and so is semantically inert.[18]

18.3.3 ABC

ABC can avail itself of a similar explanatory strategy. Given standard assumptions, Agree is a sub-part of the complex operations Move. Move is the compilation of Agreement plus Internal Merge (the latter driven by EPP requirements of the probing head). In cases of local anaphoric binding, the probe is T (recall that probes are heads with uninterpretable features) and there are twin goals: the subject and the bound element. As will become clear shortly, Reuland does not treat English *himself* as a local anaphor in the normal sense, so this probing relation is best illustrated with Dutch zich.[19] According to Reuland's theory, (5a) has an initial structure like (9)prior to probing.

(9) [$_{TP}$ T[$_{vP}$ John v [$_{VP}$ washes zich]]]

T has unvalued ϕ-features and probes into its sister (*vP*) to find a DP with valued ϕ-features that can value them. *John* has valued ϕ-features and these can value those of *T*. For reflexivization to be an instance of Agree, we must further assume that *T* can *multiply* probe its sister, thus also coming to Agree with *zich*. The result of this multiple Agreement is (10).[20]

[$_{TP}$ T [$_{vP}$ John v [$_{VP}$ washes zich]]]

(p. 405) **(10)**

Whence the reflexive interpretation? It is assumed to arise from the fact that both *John* and *zich* Agree with a common probe, *T* in (10). There are some problems of detail that arise, however, if ABC is to be made compatible with much of the agree—probe—goal technology and if it is to undergird the antecedence interpretation witnessed in reflexives:

i. The anaphor *requires* probing by T.
ii. How does the binding relation supervene on the separate Agree relations established between T and the DPs?

Let's consider these.

First, as stated, the anaphor requires probing by *T*. After all, multiple Agree is a permitted grammatical option, not a universal requirement. Here, however, *zich* must be probed so as to (eventually) allow *John* to be interpreted as its antecedent. One way of forcing *T* to probe *zich* in (10) is to assume that *zich* has unvalued features. Multiple Agree would then be required in (10) to value the unvalued features of *zich*. Though this assumption would give us what is needed in (10), it is incompatible with standard versions of the probe—goal theory. For one thing, giving *zich* unvalued ϕ-features breaks the connection proposed in Chomsky (2001) between the valued/unvalued and interpretable/uninterpretable distinctions. However, following Pesetsky (2004), Reuland proposes that these are two orthogonal properties. Thus, *zich* is specified for unvalued—but interpretable—ϕ-features. The assumption that *zich* in (10) has unvalued features in turn creates problems for the standard cyclic assumption that unvalued features must be valued in the projection of the most local head.[21] This is not a problem if only heads have unvalued features, as these can be valued before the projection is 'closed off', but if anaphors have them as well, this is a problem because *zich* in (10) has clearly not had its features valued within VP or vP. One way of finessing this problem is to propose that while both unvalued and uninterpretable features license Agree, only checking the former is subject to cyclicity. On the whole, we feel that the additional technology required by Reuland's theory is costly within a minimalist setting.

The second problem we mentioned above had to do with the connection between Agree relations and binding. Even given the above technology, it is not clear how the antecedence relation between *John* and *zich* in (10) supervenes on the Agree relation that each has with *T*. The problem can be made more clear by comparing (5a) with (11)

(11) [$_{TP}$ T[$_{vP}$ John v [$_{VP}$ washes Bill]]]

(p. 406) After all probing and valuation is complete in (10) and (11), the feature values of *T* in each will be the same, as will the ϕ-feature values of the relevant DPs. *John, Bill, zich*, and *T* in both will have valued ϕ-features, presumably the ϕ-values {3rd person, singular, masculine}. However, whereas we want this three-way ϕ-feature identity to license the interpretation that *John* is the antecedent of *zich* in (10)/(5a) we don't want this interpretation to be licensed in (11). Given that the feature configurations are the same, how is this to be avoided? Clearly, what we want is some way for the ϕ-feature agreement in (10) to be understood as *the result of Agree applying* whereas this is not the case in (11). Thus, it is not merely the final state of the ϕ-feature configuration that counts but also its etiology; whether the features are the result of Agree or not. So, if it is assumed, as is the standard assumption (Chomsky 2001), that all features once valued are identical, then a problem arises of distinguishing those that count from those that don't.

There are two conceivable ways of solving this 'identity' problem. One is to assume that the features that get copied track the identity of their host. This would follow, if, for example, ϕ-features are indexed and thereby identify the expression that hosts them. In (11) this would mean that there is a difference between ϕ_{John} and ϕ_{Bill}. Given this assumption, the ϕ-valued structure of (10) would be (12) and (11) would be (13).

(12) [$_{TP}$Tϕ_{john} [$_{VP}$ John ϕ_{john} v[$_{VP}$ washes himself ϕ_{john}]]]
(13) [$_{TP}$Tϕ_{John} [$_{vP}$ John ϕ_{john} v[$_{VP}$ washes Bill ϕ_{Bill}]]]

These are clearly distinct and it would be easy to map a structure like (12) into the required semantic structure (8). Note what this indexing accomplishes. It allows the grammar/interface to retrieve the inter-nominal dependency between DPs that is only indirectly coded in their mutual agreement with T.[22] The inter-nominal dependency is recovered via the indexing of features.

Reuland (2001: 456–7) proposes that this is indeed what is taking place. Recall, the interpretable D-features of *zich* under Agree with T can be deleted as they are recoverable from those of the antecedent *John*.[23] Reuland states: 'by their very nature, formal features such as category and person are interpretive constants.' Hence, 'the contribution they make is not contextually determined (for person features, at (p. 407) least within one reprotive context). All occurrences of such features are therefore interchangeable.' This allows the members of one to delete those of another for the occurrences of such features are 'just copies of one another'. Antecedence occurs as a by-product of recovering the deleted features. Thus *Oscar* can bind *zich* in virtue of serving to recover its features; 'deletion of a feature F_α in DP_1 and recovery of F_α under identity with F_α in DP_2 is tantamount to treating F_α in DP_1 and F_α in DP_2 as copies, and in fact as occurrences of the same feature …' Thus, Reuland adopts a version of the indexed feature theory noted above, proposing that it follows from the recoverability of deletion.[24]

Despite its serviceability, this fix raises some important issues. First, it considerably expands the class of features, as well as expanding the kinds of features available.[25] In effect, there are as many ϕ-feature sets as there are possible DPs and each feature can be indexed as to its host (the latter being a feature of a feature like strength in previous accounts). There is little morphological evidence for this kind of feature multiplication. Furthermore, this proposal essentially reduplicates a movement account of binding, although here it is features that move rather than categories. As Chomsky (2000a) notes, this considerably complicates the grammar and so should be rejected *ceteris paribus*, for it duplicates the machinery of movement and complicates the definitions of chains. It will also complicate the mapping to PF and LF. Note that it is very unlikely that the morphophonology of agreement is affected by the indexed ϕ-value of a feature. Thus, there may be morphophonological effects for <{+1st person, +Male, −Plural}> versus <{+2nd person, −Male, +Plural}>, but it is dubious whether morphophonology ever distinguishes feature sets indexed to *Bill* from those indexed to *Sam*. To 'capture' this, it would be necessary to further complicate the grammar by having operations that strip the subscripts from feature sets when Spell-Out applies. In sum, though technically implementable, this solution is undesirable if avoidable. This said, it is worth noting that *if* adopted, it indicates a point of convergence between CBC and ABC; both exploit copies to mediate the semantic binding relation. The main difference is that in CBC, copies arise as a natural by-product of the Copy Theory of movement, whereas a novel and sophisticated theory of features and their values is required in the context of the ABC.

(p. 408) A second possible way of solving the 'identity' problem builds on a similar difficulty in Chomsky (2005). Chomsky here considers a puzzle put forward by Sam Epstein: how, if valued features are all identical, can one distinguish the interpretable ones from the uninterpretable ones at the CI interface? Chomsky suggests that this problem can be resolved by assuming that Agree is a sub-part of the Spell-Out operation. This means Spell-Out applies *as* valuation applies, i.e. simultaneously with Agree.[26] One could similarly take 'binding' to be part of Spell-Out. In effect, as the relevant features are valued they are spelled out and the antecedence relation is computed. This allows the grammar to track the fact that some features arise as a result of Agree and to tie interpretation to this fact. As Chomsky notes, this envisages a grammar in which many operations take place 'all at once'; series of Merges, Agreements, and Movements can all take place simultaneously.[27] We are not currently in a position to evaluate the workability of this proposal for we are not sure what it means exactly.

Let's assume that one of these approaches is workable and consider how ABC handles the data in (5c, d). If one assumes that Probing is subject to Minimality and something like Case Freezing, then the unacceptability of (5c, d) and (6c, d) can be made to follow. The relevant structures are provided in (14).

(14)

a. [T [$_{vP}$ John v [$_{VP}$ was believed/believes [$_{TP}$ himself is intelligent]]]]

b. [T [$_{vP}$ John v [$_{VP}$ believes [$_{TP}$ Mary to be imitating himself]]]]

The probe in (14) is *T*. Its ϕ-features are unvalued and it probes its sister to find a goal that can value these features. Multiple Agreement allows T to agree with multiple DPs in its domain. It can agree with *John* and its valued ϕ-features can value those of T. However, T cannot further probe *himself* as it is in a case frozen position. One way of understanding this is to suppose that *John's* ϕ-features are inaccessible once *John* has received case.[28] This is completely analogous to the assumption made in the CBC story above. Similarly, in (14b) T probes into its sister and finds *John*. What of *himself*? It is presumably 'too far away' and so is out of the reach of the matrix T. One option is to say that *Mary* or the embedded T restricts **(p. 409)** matrix T's reach due to Minimality. The desired effect is that T cannot see 'through' *Mary* or embedded T to probe *himself*.

This leaves one more case: (5e) represented in (15).

(15) [T [$_{vP}$ [$_{DP}$ John's book] v [$_{VP}$ exposed himself to ridicule]]]

To explain the unacceptability of (5e) we need to disallow T from probing *John* thereby valuing its features and then probing *himself* and thereby valuing its features. There are two ways of blocking this. The first is to assume that DP constitutes a phase and that the PIC prevents T from probing *John* in (15). This requires the further assumption that in the phrase *[$_{DP}$ John's book]*, *John* is not at the edge of the DP (*pace* the standard assumption, e.g. Abney 1987) for thus residing on a phase edge would make it visible. A second way of blocking Agree is to assume that it is subject to something like the A-over-A Principle. This is to suppose that the DP that contains *John* blocks *John* from valuing the features of T. This is plausibly just another instance of Minimality.[29]

18.3.4 Some remarks on *zich, zichself*, and *himself*

Reuland has the most fully worked version of an ABC theory of binding. Of particular relevance here is the distinction Reuland makes between *zich* and *him*. As we noted, *zich* is only specified for category and person features, whereas *him* has a category and a full set of ϕ-features (the same goes for *zichself* and *himself*). From this distinction, it follows that *zich* may be goal to higher probes, whereas the feature specifications of *him* forbid this. More accurately, the reduced feature set of *zich* can be deleted and recovered while the fuller ϕ-set of *him* cannot be. This is why, for example, sentences like (16a) will allow *John* to antecede *zich* while this 'binding' is not possible in (16b).

(16)

a. John$_1$ heard [zich$_1$ sing]

b. *John$_1$ heard [him$_1$ sing]

More specifically, because *zich* is only specified for person (we put category aside here) Agree in (17) allows the deletion of its person feature and its recoverability from the head of the chain formed with *John*. Recall that Reuland proposes that the person feature in *John* is (contextually) identical (i.e. an occurrence of the very same feature) as that in *zich* which was deleted under Agree with the person feature in the probe T.

[T [John heard[zich sing]]]

(17)

(p. 410) This contrasts with (18), with *him* in place of *zich*. *Him* has a full ϕ-feature set, and though the person feature may be contextually identical in *John* and *him*, the number and gender feature cannot be. Consequently, the full ϕ-set of features of *him* cannot be deleted as they cannot be recovered. Thus, binding cannot occur in this case.

(18) [T [John heard [him sing]]]

In sum, on the assumption that deletion of partial ϕ sets is illicit, the contrast above follows from the feature specifications of the two expressions.

Observe that Reuland adopts a few other assumptions. In particular, he assumes that different lexical items can never have the same (contextual) features, for otherwise it should be possible to optionally generate *him* with features recoverable from (i.e. contextually identical to) those of an antecedent. This must even hold for cases like *he heard him sing* (where *he/him* denote the same person) or cases of two names of the same person such as *Tully heard Cicero sing*. Different instances of the same lexical item (or different lexical items) can never have the same number and gender features.

Reuland notes that *zich* is often locally unbindable. Consider the contrast in (19).

(19)
a. John likes zichself
b. *John likes zich

Without *self* the *zich* cannot be co-valued with *John*. Reuland argues that the predicate that results would be ill-formed. But why? One possibility is a condition first proposed in Reinhart and Reuland (1993): predicates with co-valued arguments must be reflexive marked. The problem with (19b), then, is that the predicate is not reflexive marked. This would account for why predicates marked reflexive in the lexicon allow analogues of (19b), e.g. *John washed zich*. This, in effect, codes the antilocality condition on pronouns we find in Principle B. So, in addition to the Agree system and the feature system, Reuland requires a specification of how predicates become reflexive; either inherently (in the lexicon) or through some process like *self*-marking.

Interestingly, *self*-marking alone can support a reflexive interpretation. Recall that English does not have *zich*-reflexives. It employs a full pronoun form. Nonetheless, sentences like *John likes himself* and *John heard himself sing* carry reflexive interpretations. This cannot be because of an Agree relation relating *John* and *him* in these cases for *him* has a full ϕ-feature specification. Consequently, it must be that *self* suffices to provide the reflexive reading in these cases. This fact, however, must be handled gingerly, for it seems to obviate the need for agreement altogether in Reuland's system. After all, if reflexivizing the predicate yields co-valuation, then there will be co-valuation in (19a) even without the mediating effects of agreement.[30] This results in two entirely separate routes to reflexive interpretations: (p. 411) one via reflexively marked predicates and one via licensing of *zichs* via Agree. This is not an optimal state of affairs theoretically.

Space restrictions do not permit a full exploration of Reuland's subtle account. However, we hope to have demonstrated that the specific ϕ-feature specifications of the relevant anaphors contribute to how reflexive readings are derived within one ABC style account.

To conclude, we have outlined how the basic cases of reflexivization could be handled in terms of analyses based on Move or Agree. Though there are apparent differences between the two approaches, it is worth ending with a recap of their similarities. Both exploit the locality conditions on Move/Agree to restrict the reach of reflexivization. Both produce chains (in terms of which antecedence is interpreted) at the CI interface (see 18.2.2 for details). Moreover, both CBC and (some versions of) ABC define chains using copies. All in all, despite different technologies, the two approaches share key similarities.

18.4 Cross-linguistic variation

Perhaps the deepest and most interesting distinction between CBC and ABC is in their approach to cross-linguistic variation (though this may not be immediately obvious, given that the CBC has yet to be applied extensively outside English). Following LGB, the CBC assumes that anaphors are subsumed within a natural class. Rather than caching out this assumption in terms of a +/−anaphor feature, the CBC states that an anaphor is simply the overt Spell-Out of one of the copies in an A-chain which spans multiple theta positions. We must maintain that this notion of anaphor—suitably generalized—is one of universal significance, and not parochial to English or related languages. The property of ‘being an anaphor’, then, is essentially a relational one rather than a lexical one, in a manner reminiscent of the functional determination of empty categories (Chomsky 1982a).[31]

In contrast, if the ABC is on the right track, we expect (at least in principle) to find a wide variety of anaphoric/pronominal elements across languages. Since the properties of a dependent element are determined by its feature specification, we expect variation between languages simply as a consequence of lexical variation. The cross-linguistic implications of the ABC have already been explored in the literature, so we would like to say a few words on how various facts might be accommodated by the CBC. As should be clear, we face essentially the same problems as the LGB binding theory in accounting for cross-linguistic variation. However, (p. 412) the research of the past three decades has provided us with a box of tricks that was not available to the standard LGB theory.

To give one illustration, the economy-based nature of the theory allows us to account for a fact that was a genuine puzzle in the 1980s. In many languages, there are dedicated 3rd person reflexive forms, but ordinary pronouns double up as reflexives in the 1st and/or 2nd person. For example, Safir (2004: 61) points to the following Norwegian data:

(20)
a. Jon skammer seg/*ham
John shames self/*him
'John is ashamed'
b. Jeg skammer meg/*seg
I shame me/*self.
'am ashamed'
c. Jon fortalte Ola om meg.
Jon told Ola about me.

A locally bound pronoun is impossible in the third person—(20a)—but permitted in the first person—(20b)—even though *meg* can also function as an ordinary non-reflexive pronoun—(20c). In the CBC (and indeed the ABC), these facts can be made to follow on the assumption that the Norwegian lexicon simply does not contain a dedicated 1st person reflexive form. Thus, since no more economical derivation is available, a pronoun may be used instead as a last resort.

Continuing this extremely brief and incomplete cross-linguistic survey, let us consider Romance SE. Here, we can do little more than indicate the kinds of analysis that the CBA urges, since the associated phenomena are enormously subtle and complex. If we are to maintain that local anaphora is an essentially unitary phenomenon, we must reject the analysis of SE as a valence-reducing operator over argument structure (see e.g. Reinhart and Siloni 1999, Marantz 1984). Therefore, for genuine reflexive uses of SE (which by no means exhaust the range of its use), we expect that there will be a chain linking the external argument to the internal argument, and finally, to the matrix subject position:

(21) $[_{TP}$ DP… SE … $[_{vP}$ t_{DP} $[_{VP}$ … t_{DP} …]]]

This kind of analysis, argued for in Alboiu et al. (2004), is certainly not unreasonable as far as it goes. The real challenge is to integrate it with an overall account of how SE functions in its multitude of other roles. Alboiu et al. (2004) proposethat there are essentially two sources of SE: it may either be entered in the numeration as a DP specified only for person, or introduced at PF as the spell-out of a lower copy. These options correspond to the reflexive and indefinite/impersonal readings respectively. There are strong parallels here with the account of bound vs. referential pronouns we give in section 18.6. There are of course many other properties of SE which remain puzzling, and which we must hope will be explained as our (p. 413) understanding of verbs and VP/argument structure increases. For example, the distinction in meaning between Spanish *voy* ('I go') and *me voy* (approx. 'I go away/I leave'). Another problem is posed by 'impersonal' SE, which is not associated with any genuine reflexive meaning, and which forces 3rd person agreement on the verb.

It would be interesting to attempt an analysis of Dutch *zich* along the same lines as SE, though SE differs from *zich* in a number of problematic respects.[32] Much depends on the explanation ofthe factors governing the competition between *zich* and *zichself*. Though *zich* seems to be absolutely preferred to *zichself* in inherently reflexive environments, we suspect that in other environments, the competition may crucially involve pragmatic factors (see e.g. Geurts 2004). Thus, a narrow syntactic account of the distinction between the two may not be necessary for these cases.

18.5 Some additional facts

Let's now consider four additional kinds of data.

18.5.1 Reflexives are low, antecedents are high

First, how are we to block (22)?

(22) *Himself/Zich V John

A CBC approach blocks cases like these as follows: Since reflexives distribute like traces, a sentence like (22) must arise by lowering *John* from a higher position to a lower one. This kind of movement is banned in all current minimalist accounts by a principle analogous to the Extension Condition. Thus, *if* reflexives are residues of A-movement and A-movement like all movement obeys extension, it is impossible to generate sentences like (22) as they involve DP lowering in overt syntax, an illicit operation.[33]

How is (22) handled in an ABC account? The structure of (22)would be something like (23)with *T* probing both *zich* and *John*. What, if anything, is wrong with this?

[T [himself/zich v [V John]]]

(23)

(p. 414) At first blush, nothing. Note that the predicate is either reflexive marked or inherently reflexive so the co-valuation is licit. Note too that *T* establishes an indirect Agree relation between *John* and *zich*. If this sufficed to allow *John* to value the features of the reflexive and thereby bind it, we should be able to get the reflexive reading here, contrary to fact. To block this, we must add an additional assumption, made explicit in Reuland (2001). The Agreement indicated in (23) results in the formation of a chain like object at CI. Chains are defined as objects in which the expression whose features are recovered must be at the tail-end while the DP whose features drive the recoverability are in the head-position. Reuland executes this by defining a notion of an A-CHAIN which results from combining two A-chains. A-CHAINS are subject to the same 'chain' conditions that A-chains derived by movement are. The following definitions are employed.

(24) Chain: (α,β) form a Chain if (a) β's features have been (deleted by and) recovered from a and (b) (α,β) meets standard conditions on chains such as uniformity, c-command and locality.

(25) CHAIN: If (α_1,α_2) is a chain and (β_1,β_2) is a chain and $\alpha_2 = \beta_1$, then (α_1,α_2/β_1, β_2) is a CHAIN.

The definitions in (24) and (25) allow for chain composition so that the relevant CHAIN for evaluation is (α_1,β_2). What is important with respect to (22) is that the standard conditions on chains apply to the CHAIN (α_1,β_2) for this explains why the reflexive must be the second (lower) member of the chain.[34]

This proposal clearly descends from the Chain Condition in Reinhart and Reuland (1993). Both conditions function to extend the notion of 'chains' from structures derived via movement to structures derived using some other mechanism—agreement and feature recoverability in this instance. Note that the Chain Condition must be added to ABC to derive the facts in (22). It does not follow from how chains are constructed or how Agree functions in the context of construal. This contrasts with CBC accounts in which reflexivization is the product of movement ('reflexives' just being a species of trace/copy), and so we expect the output of movement to yield chains with the structure typical of chains, viz. with the head of the chain c-commanding the tail.[35] In other words, the Chain Condition follows from the CBC supposition that reflexives are residues of movement, while it must be added to ABC approaches to explain data like (23).

There remains one important question for the CBC with regard to (23), especially if one adopts the Copy Theory of movement (a mainstay of minimalist accounts). What CBC posits is that reflexives are (essentially) traces. However, minimalism (p. 415) does not have traces, it has copies, and, by assumption, all copies are created equal. How then does one go from (26) to (27)?

(26) [John T [John v [like John]]]

(27) John likes himself

One of the copies in (26) must be converted into a reflexive at the AP interface. Why is the bottom copy chosen for this honor and not the top one? There are several possible answers. One is that in (26) the top Copy of *John* has both case and theta marking and so is fully well-formed at the interface. It has the structure in (28):

(28) [John $_{\theta 2,+\mathrm{case}}$ T [John $_{\theta 2,-\mathrm{case}}$ v [like John $_{\theta 1,-\mathrm{case}}$]]]

This makes the top copy well-formed and hence non-deletable. Consequently, it is the bottom copy that is converted into the reflexive. In effect, reflexivization is an operation that saves an otherwise ill-formed PF structure. Note that the chain contains two θ-roles at LF.

A second option is to think of reflexives like doubling structures or a complex DP like *John's self*. The underlying form is (29):

(29) [John T [John v [like [John + self]]]]

John then moves from the lower theta position to the higher one getting case. The lower *John's* delete and *him* is inserted to support the bound morpheme *self*.[36] In either case, the higher copy cannot delete as there is nothing 'wrong' with it and the lower copy comes to have the form it does for morphological reasons. It is the target of this 'fix-up' operation as it is not well-formed.

In the end, both approaches account for the fact that a reflexive does not c-command its anaphor by adverting to the fact that reflexive structures instantiate chains. In the CBC, relevant chain properties are inherent in the basic account, in the ABC these properties must be added.

18.5.2 Reflexive binding without c-command

There appear to be cases in which the antecedent need not c-command the reflexive. Consider cases like (30):[37]

(30) After spotting the police, there quietly departed several men without PRO/themselves being detected.

(p. 416) The *without* adjunct presumably hangs at the VP level and *several men* is inside the VP, complement to *arrive*. Thus, it is plausible that *several men* does not c-command *themselves* or *PRO*.

Such cases are not particularly problematic for ABC accounts, as the relation between the antecedent and reflexive is mediated by agreement with T and so c-command between the two DPs is not required for the grammatical operation to be established. The relevant structure will be that in (31):

(31) …[there T [$_{VP}$ [$_{VP}$ arrive several men] [$_{adjunct}$ without themselves.…]]]

In (31) T can probe both *several men* and *themselves* and establish the relevant agreement relation. One more ingredient is required: to fully accommodate these data with the preceding facts, it will be necessary to redefine chains so that the c-command condition is relaxed. Recall that CHAINs were defined as species of chains, and that the latter incorporate a c-command condition on chain links. Here, *several men* does not, by hypothesis, c-command the reflexive and so an ill-formed chain (and CHAIN) is expected. We assume that the relevant technical changes can be made.

What of CBC accounts? Constructions like these are considered in Hornstein (2001) and are used to argue for the utility of sidewards (inter-arboreal) movement. Nunes (2001) proposed that it is possible to move between two sub-trees; or, more exactly, to copy from one sub-tree and merge to another. This is the source of (31). The derivation proceeds as follows:

(32)
i. Form the adjunct: [$_{adjunct}$ without several men. ….]
ii. Copy 'several men' and merge with 'arrived' forming two sub-trees: [$_{VP}$ arrived several men] [$_{adjunct}$ without several men. ….]
iii. Merge the two sub-trees: [$_{VP}$ [$_{VP}$ arrived several men] [$_{adjunct}$ without several men. ….]]
iv. Merge T and 'there': [There T [$_{VP}$ [$_{VP}$ arrived several men] [$_{adjunct}$ without several men. …]]]
v. reflexivize the lower A-trace copy: [There T [$_{VP}$ [$_{VP}$ arrived several men] [$_{adjunct}$ without themselves. ….]]]

(p. 417) Thus, to accommodate reflexive binding without c-command, CBC accounts rely on the possibility of sidewards movement, a possibility opened up with the minimalist analysis of movement as a complex process comprised of the simpler operations Copy and Merge. Note that if these cases actually involve movement, then the chain-like properties witnessed here are expected to hold.

There is another possible case of reflexive binding where c-command does not hold. This sort of binding travels under the name of 'sub-command' and occurs in many East Asian languages. Consider an illustration from Chinese.

An antecedent in Chinese can bind a local reflexive *taziji* even when contained within a DP (i.e. without c-commanding the reflexive).[38]

(33)

Zhangsan	de	guiji	hai-le	taziji/??ta
Zhangsan	de	trick	harm-perf	himself/him
Zhangsan's tricks harmed himself/him				

(34)

Zhangsan de shu zhi	jiaoyu-le	taziji/*ta
Zhangsan's book	educated-PERF	himself/him
Zhangsan's book educated himself/him		

Note that here the reflexive is in complementary distribution with a bound/co-referential pronoun, as it should be if it is truly a locally bound reflexive. This sort of binding is easily derived assuming sidewards movement. The derivation is in (35) (English glosses used):

(35)

a. merge: [John self]
b. merge: [educate [John self]]
c. copy *John* and merge (sidewards movement): [John book]
d. merge: [[John book] [educate [John self]]]
e. Finish derivation in usual way to check case etc.
f. Delete non-case marked residues and add pronoun to reflexive morpheme: [[John book] T [~~John book~~ [educate [~~John self~~]]]]

With this derivation *John* becomes the antecedent for the reflexive though it does not c-command it. It is another illustration of the possibility of binding without c-command which is expected if reflexives are formed by movement and if sidewards movement is a grammatical option.

What would an ABC analysis of sub-command look like? Presumably it would involve T probing and agreeing with both *Zhangsan* and *taziji*, as in(36):

[T [$_{vP}$ [$_{DP}$ Z's book] v [$_{VP}$ educate taziji]]]

(36)

(p. 418) This agreement pattern licenses a CHAIN headed by *Zhangsan* and footed by *taziji*.[39]

18.5.3 Binding within *picture* NPs

We noted at the outset that we would be assuming that reflexives within picture NPs are logophors. However, we need to be somewhat more precise here. Consider

(37) John likes (Mary's) pictures of himself/him

The reflexive related to *John* in (37) is a logophor. The relation between *John* and *himself* is logophoric. However, not all binding of a reflexive within a picture NP is logophoric. Consider (38):[40]

(38) John likes Mary's$_1$ picture of herself/*her$_1$

Here, *Mary* is antecedent of *herself* and note that it is in complementary distribution with *her*. This suggests that the relationship is one of binding. It is not logophoric. How do the two theories 'derive' these data?

CBC accounts simply treat this as another case of movement, this time within theDP. The relevant derivation is depicted in (39):

(39) John likes [Mary's$_1$ [picture of Mary]]

The lower copy assumes reflexive form in one of the ways discussed above.

Consider now an ABC account requires assuming that there is some probe within the DP that c-commands both *Mary* and *herself*. However, there is no obvious can-didate for this probe. Furthermore, it requires that *Mary* begin its derivational life in some position below D. The general assumption is that a possessive DP like *Mary's*, especially if interpreted as meaning something like *the one that Mary possesses/owns,* **(p. 419)** is directly generated in Spec D, where it receives the 'possessor' θ-role. If this is correct, it is not clear how an ABC account could establish the relevant binding relation. Note, in addition, that *John* is not a co-argument of *picture*, at least on the possessive/owner reading. The standard semantics for these sorts of genitives assume that the genitive introduces an additional relation so that the semantics is conjunctive with *John's picture of Mary* having roughly the logical form: 'John R x & picture (x, Mary)'. In other words, *John* is not the external argument of *picture*. Thus, whether or not *Mary* is an internal argument (a debatable assumption), as *John* is not, they cannot be co-arguments. Thus, the only option will be to assume that they are syntactically related via Agree through a common probe. The open question for an ABC account is what that probe is.

18.5.4 Adjuncts

Consider examples like (41) where there appears to be binding into an adjunct.[41] Note that with verbs like *tell* the reflexive is in complementary distribution with a pronoun, suggesting that the reflexive is not logophoric.[42]

(40)
- **a.** John told Mary about herself/*her
- **b.** John told Mary about himself/*him

For CBC accounts the problematic example is (40b), as it appears to involve a violation of minimality. (40a) is not particularly troublesome, as it simply involves movement from the PP to the object position. The derivation will either involve regular movement if the PP is a complement, or sidewards movement if the *about* PP is an adjunct. Two derivation sketches are provided in (41).

(41)
- **a.** [$_{TP}$ John T [$_{vP}$ Johnv[$_{VP}$ Mary [$_{V'}$ told about Mary-self]]]]
- **b.** [$_{TP}$ John T [$_{vP}$ John v [$_{VP}$ [$_{VP}$ told Mary] [about Mary-self]]]]

The problematic case is (41b), for here the derivation we need would appear to violate minimality if the *about* PP is a complement, or the CED if it is an adjunct. Of these two options, the adjunct possibility is the less troubling for it appears that such adjuncts are extremely weak islands given the acceptability of stranding *about*:

(42) Who did John tell Mary about?

(p. 420) This is not atypical of 'low' hanging adjuncts like commutative *with*, instrumental *with*, benefactive *for*, etc. Thus, CED concerns do not seem to apply to these adjuncts, for reasons that remain somewhat mysterious. Furthermore, if the *about* PP is a VP adjunct, then minimality does not arise as *Mary* in (41b) does not intervene between the PP and spec v as the direct object does not c-command the adjunct phrase.

Thus, these binding cases are problematic for CBC accounts only if the *about* phrase is a complement. The data in (43) suggests that it is not:

(43)
- **a.** John told Mary about Sue before I did about Jim
- **b.** John told Mary about Sue and tell Mary he should have about Sue

If this is correct, then CBC accounts can accommodate these binding into adjunct facts.

What of ABC approaches? If the *about* PP is an adjunct, then (40b) can be represented as (44), with *T* probing *John* and the reflexive. This then becomes converted into the relevant chain relation:

T [$_{vP}$ John v [$_{VP}$ [$_{VP}$ told Mary][about himself]]]

(44)

The cases in (40a) can also be easily accommodated if it is assumed that *v* can replace T as the probe. In this is acceptable, then the relevant agreement structure is (45):[43]

T [$_{vP}$ John v [$_{VP}$ [$_{VP}$ told Mary][about herself]]]

(45)

18.6 The complementary distribution of bound pronouns and reflexives

One of the key empirical findings concerning binding is the fact that reflexives are acceptable where bound pronouns are not and bound pronouns are acceptable where reflexives are not:

(46)

a. John$_1$ likes himself/*him$_1$
b. John$_1$ believes himself/*him$_1$ to be smart
(p. 421) **c.** John$_1$ believes *himself/he$_1$ is smart
d. John$_1$ expects Mary to kiss *himself/him$_1$

Historically, there have been two ways to track this fact within theories of binding in generative grammar. Within GB, the complementarity is coded in the basic structure of principles A and B. Reflexives and pronouns have identical domains within which they must meet conflicting requirements, viz. domains in which reflexives must be bound, pronouns cannot be. Within the earlier standard theory, the complementarity is the result of the following assumptions: (i) the rules of reflexivization and pronominalization are obligatory, (ii) the former precedes the latter, and (iii) the former bleeds the context for the application of the latter. What of ABC and CBC: how do they account for the complementarity? Interestingly, both assume that the acceptability of reflexive binding within a certain configuration blocks the availability of pronoun binding in that same configuration. In other words, both adopt a version of the second approach in which the application of reflexivization blocks pronominalization.[44] Let's consider some details.

Reuland's version of ABC assumes that pronoun binding is an extra-grammatical process. It takes place at the CI interface and is not grammatically coded. Reuland (2001) further assumes that relations that are established *within* a module are more economical than those that require the cooperation of more modules (see also Reinhart 1983). In particular, as reflexive binding is parasitic on the syntactic relation of Agree, which occurs within the grammar proper, it is more economical than pronoun binding, which is not the by-product of pre-interface syntactic relations having been established.[45] Reuland (2001) cashes out the proposed economy metric in terms of the relations of three modules: the syntax, the CI interface, and discourse structure. Syntactic chains are mapped onto variables at CI and then to discourse objects. The basic idea is that multiple links within a chain are mapped onto the same CI and discourse objects automatically. In contrast, links in different chains require a more cumbersome mapping. As reflexives form chains with their antecedents via Agree, while bound pronouns do not, the former create bound variable structures in a more efficient way than the latter. In short, expressions related to one another syntactically are cheaper to map to their semantic and discourse objects than are those related at CI. This is illustrated in (47). Note that just as intra-chain binding is preferred to inter-chain binding, so too binding is preferred to co-valuation without binding.

(p. 422) **(47)**

a.

Discourse storage (values)	a		a
	↑		↑
C-I objects (variables)	x_1		x_2
	↑		↑
Syntactic objects (CHAINs)	C_1		C_2
Basic expressions	*α*	…	*β*

b.

Discourse storage (values)	a		
	↑		
C-I objects (variables)	x_1	>	x_1
	↑		↑
Syntactic objects (CHAINs)	C_1		C_2
Basic expressions	*α*	…	*β*

c.

Discourse storage (values)	a		
	↑		
C-I objects (variables)	x_1		
	↑		
Syntactic objects (CHAINs)	C_1	>	C_1
Basic expressions	*α*	…	*B*

(Reuland 2001: 474; 71)

Hornstein (2001) proposes a similar analysis along CBC lines. The proposal is essentially an update of the Standard Theory analysis in Lees and Klima (1963). The proposal distinguishes two ways of establishing grammatical dependencies, one via movement, which yields the dependency between a reflexive and its antecedent, and the other via a rule akin to pronominalization, which establishes the relation between a bound pronoun and its antecedent. The latter is considerably more complex than the former, and is thus able to apply just in case it is not possible to set up a binding relation via movement. The relative simplicity of movement over pronominalization allow reflexives to pre-empt bound pronouns. As in Reuland's account, it is further assumed that binding trumps semantic co-valuation without grammatical intercession. Thus, co-reference is possible just in case one of the two forms of binding are not.

An illustration should help fix ideas. Consider (48). There is a movement relation that can be established between

John and *himself*. As it can be, it must be, and this blocks a binding relation between *John* and *him* (as well as a co-reference relation). This contrasts with (45), where there is no possible movement relation between *John* and *himself* and so the relation between *John* and *him* is permitted.

(48) John likes himself/*him
(49) John thinks *himself/he is smart

Note that the logic here is similar to the Merge over Move proposals made in Chomsky (2000a). Both hinge on taking simpler operations to be more economical (p. 423) than more complex ones and thus to pre-empt their application. In Hornstein (2001), reflexivization is simply the application of Copy and Merge, whereas pronominalization requires demerging of a chain and remerging the head in another θ-position substituting pronoun copies for the links of the demerged chain. This latter is a complex operation and so strongly dispreferred.

One curious consequence of this analysis is that it requires treating reflexives and bound pronouns as *non*-lexical elements.[46] In particular, they cannot form part of the numeration, for were they included in the numeration it would not be possible to compare derivations containing reflexives with those containing pronouns and thus it would not be possible to analyze reflexive structures as derivationally more economical than bound pronoun structures. Thus, there is a strong sense in which the morphology is secondary, in contrast to an ABC account like Reuland's. Interestingly, if one assumes that the Inclusiveness Condition is a property of UG, this implies that the ϕ-features expressed by bound pronouns and reflexives are not semantically interpretable. This appears to be correct. As Kratzer (2009) has noted, expressions interpreted as bound variables (reflexives and bound pronouns) have null feature values, as seen in (50). Why? Because on the bound reading, (50) can be contradicted by (51); *Mary* being feminine, *I* being 1st person, *you* being 2nd person, and *the 25 girls in grade 4* being plural. This should not be possible if the ϕ-features of *he* carried their obvious semantic interpretations of 3rd person, male, singular, for it would restrict values of the variable to those carrying these features. In sum, there appears to be some semantic evidence in favor of not taking the specific feature content of bound pronouns (and reflexives) as semantically viable, and this appears to follow from a version of CBC that explains the complementarity of pronouns and reflexives in terms of economy of derivation within a minimalist setting.

(50) Only John thinks that he is smart
(51) No, Mary thinks she is smart, I think that I am, you think that you are, and the 25 girls in grade 4 think that they are.

In sum, both ABC and CBC approaches return to the pre-GB intuition concerning pronouns and reflexives. Both approach the complementarity of these expressions as a reflection of the workings of economy considerations. Before leaving this topic, (p. 424) we would like to briefly outline another CBC approach to this fact that is not based on economy intuitions and is more akin to the GB idea that what underlies the complementarity are conflicting requirements in similar domains.

Grammars contain two kinds of XP movement, A versus A′.CBC proposes treating reflexivization as a species of A-movement. It is natural to then consider the possibility that bound pronouns are residues of A′-movement. This is natural in one further respect: A′-residues of e.g. *wh*-movement, are already interpreted as bound variables. Thus treating bound pronouns as products of A′-movement would serve to grammatically unify semantically identical items (see Aoun 1986). An analysis along these lines is given in Kayne (2002), though we will depart from his assumptions in a number of respects (most importantly, by restricting the analysis to bound pronouns).[47] Let's briefly consider what this kind of analysis might look like. To fix ideas, let's assume that bound pronoun structures arise by moving a DP via an A′-position to another A-position.[48] If something like this is the source of bound pronouns, what follows?

(52) John thinks [$_{CP}$ John [$_{TP}$] John T [$_{vP}$ John v [$_{VP}$ likes Mary]]]]

First, the c-command restriction on bound pronouns can be explained. Since Reinhart (1983) it has been widely accepted that for a pronoun to be interpreted as a variable it must be c-commanded by its antecedent. If extension is a condition on the construction of phrases, then the c-command restriction on bound pronouns is a by-product of the extension condition.[49]

Second, bound pronouns would be restricted by the availability of A′-positions. Thus, the unacceptability of (53) could be traced to the absence of an intervening complementizer position through which *John* could transit on his way to his final A-position. Similarly the acceptability of (50a, b) would be due to the availability of the embedded complementizer as an A′-transit point, as in (51).

(p. 425) **(53)** *John$_1$ likes him$_1$

(54)

a. John thinks that he is smart

b. John thinks that Mary likes him

(55)

a. John thinks [$_{CP}$ John [$_{TP}$ John(=he)…

b. John thinks [$_{CP}$ John [$_{TP}$ Mary likes John(=him)]]

Note that this account does not rely on economy calculations, unlike the two earlier proposals.

Third, ECM cases fall into line if in such cases the accusative case marked DP moves to a specifier in the higher clause for case-marking. Once this happens (56) reduces to (53).[50]

(56) *John$_1$ expects him$_1$ to win

This should provide a taste of what we have in mind. We find this option theoretically satisfying as it reduces the two kinds of bound variables to a syntactically unified class. It is also interesting as it allows for a non-economy account of the complementarity of pronouns and reflexives. This said, we raise it as an interesting curiosity, cognizant of the fact that it faces many empirical challenges. (Not least of these, to explain why pronominal binding does not obey island constraints.)

18.7 Conclusion

Both ABC and CBC approaches to reflexives assume that binding is mediated by a local syntactic relation: agreement with a higher functional head in the former and A-movement in the latter. This syntactic relation results in having the antecedent and the reflexive in a common chain-like object from which the antecedence relation is easily computed. The main difference between the two approaches is how the relation between antecedent and reflexive is established syntactically. In ABC accounts, the relation is indirect, mediated by agreement with a common functional head (usually T). The central technical relation is Multiple Agree; one head probing two different DPs, which typically are in a c-command relation (though they need not be). In CBC accounts, the relevant syntactic relation is movement. Thus there is (p. 426) a direct syntactic relation between antecedent and reflexive, the latter being the tail of the chain formed by movement of the former. Interestingly, at CI both analyses lead to identical outputs; chain-like objects. However, the derivations appear to proceed in very different ways syntactically. Here we have tried to outline how each would handle the central cases of reflexive binding while being responsive to minimalist concerns.

Notes:

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(1) See Chomsky (1995b: chs. 1, 3, 5).

(2) See Hornstein (2001), Landau (2003), Lidz and Idsardi (1998), Reuland (2001), Zwart (2002), Kayne (2002).

(3) The locality of obligatory control is often reduced to the selectional properties of a higher predicate. Thus, the locality witnessed in OC configurations is attributed to the locality of selection. This is an unsatisfactory account given minimalist predilections for several reasons. First, selectional accounts do little more than stipulate what should be explained. Second, we doubt that selection is the correct account for control construal as it fails to accommodate adjunct control, which also displays locality conditions. Third, the target of construal (PRO) is too far away to be selected by the higher predicate. Thus, even on a selection account the construal relation must be

very indirect if selection is involved (i.e. predicate controls C which controls T which controls PRO). The Rube Goldberg nature of this 'local' relation argues against so accommodating the locality of OC. For further discussion see Boeckx et al. (2010b).

(4) So too for Reinhart and Reuland's (1993) extension to syntactic predicates to handle the data in (1) above. We should note that argument structures of the Pollard and Sag type could in principle be related to semantic argument structure, given a sufficiently elaborate semantic theory (see e.g. Klein and Sag 1985).

(5) This sets aside complications arising in languages like Icelandic. For discussion see e.g. Landau (2003), Boeckx and Hornstein (2004), Bobaljik and Landau (2009), Boeckx et al. (2010a, 2010b).

(6) See the discussion of binding in Chomsky (2004a, 2005).

(7) See Hornstein (2001), Boeckx et al. (2010b), Boeckx and Hornstein (2003, 2004), for movement approaches to control and see Landau (2001), (2003) for a non-movement approach.

(8) This is essentially the position taken in analyses of construal in the standard theory; see Lees and Klima 1963 for details. Reflexives, bound pronouns and 'PRO' here are outputs of syntactic operations that, by assumption, have no impact on semantic interpretation given the Katz—Postal hypothesis (widely accepted at the time).

(9) Though current minimalism eschews DS, most do not allow movement into θ-positions. Such movement is banned in a variety of ways with the addition of further conditions on licit movement. This said, it is correct to say that the option of movement into θ-positions does not exist in a theory with DS.

(10) Construal here is binding. Control has a far more elaborate agreement structure.

(11) The scare quotes acknowledge that current wisdom has it that whatever drives the EPP is not a feature. Whatever it is, however, can combine with Agree relations to force movement.

(12) The centrality of the specific features is central to Reuland's (2001) proposal, for example. We return to borrow from his analysis and integrate it with the Agree-based system.

(13) This is not entirely correct. The system of binding outlined in Reuland (2001) assumes that Agree in these contexts results in the antecedent and reflexive forming chain-like objects. Thus, both ABC and CBC accounts result in multi-thematic chains at the CI interface. If this is correct, then there is a sense in which both accounts violate the θ-criterion, though ABC adheres to something like a conception of DS (preventing from movement into θ-positions) while CBC does not.

(14) We believe that these modifications enjoy wide acceptance and that most (if not all) current approaches to construal adopt them.

(15) A number of questions arise regarding the precise nature of the blocking relation, which we do not have space to address here. Reinhart herself proposed a significantly different account in later work (Reinhart 2006: ch. 4). See also Reinhart and Grodzinsky (1993).

(16) This follows a long tradition of analysis based on the observation that pronouns can felicitously replace these reflexives and can even be bound long distance over intervening subjects:

((i))
(a.) John likes stories about himself/him
(b.) John likes Mary's pictures of himself/him

See e.g. Ross (1970), Pollard and Sag (1992), Reinhart and Reuland (1993), Zwart (2002), Runner et al. (2003).

(17) Problems remain: e.g. why are (6a, c) out but (6a, c) in? The CBA would have to analyze (5a, c) as suffering from a kind of PF deficit. See Hornstein (2001) for some discussion in the context of inherent reflexives like 'John washed' and control complements like 'John expects to win'.

(18) This is not obviously a positive result, given the varying shades of meaning that reflexives can have when compared to 'control' structures that are formed similarly. See Lidz (1996) for discussion of the various kinds of

reflexive interpretation and Hornstein (2001) for a comparison with control clauses.

(19) See e.g. Koster (1985) for an overview of Dutch reflexives.

(20) Note that multiple Agree requires a reinterpretation of minimality so that *John* in (9) does not block access to *himself*. For discussion of multiple Agree see Hiraiwa (2001); for some criticism see Boeckx (2008a) and Hagemann and Lohndal (2010).

(21) So called ‘virus theory’. See e.g. Richards (2002).

(22) This is something that CBC codes directly by treating reflexivization as a relation between DP/θ-positions.

(23) The latest versions of Agree theory assume that what Agree does is ‘value’ unvalued feature sets. Interpretable expressions come with valued feature sets, non-interpretable expressions come with unvalued feature sets. This is intended to replace earlier discussion of deleted—interpretable features versus retained +interpretable ones. Given feature valuation, it is not clear what the status of *zich*'s feature is. Does T furnish a value for the unvalued person feature (i.e. treat *zich* as if it were uninterpretable), or does it overwrite the person feature of *zich*, this being permitted as the overwritten value is identical to the interpretable one it has? This is a technical issue that we put aside here.

(24) We are less convinced than he is that this is indeed the case, in particular that 2nd and 3rd person features are so indexed and that number (and gender?) features are not. However, what is relevant here is that assuming that the features are so individuated allows for antecedence to be recovered from an Agree relation that would otherwise underspecify them.

(25) Reuland might not agree with this point, given his views concerning recoverability above. However, we believe that a consequence of the indexing view is a multiplication of feature values for otherwise ‘3rd person’, for example, would be insufficiently refined to pick out the actual antecedent. Note incidentally that, strictly speaking, on the account provided above it is not the antecedent that ‘provides’ features to the reflexive, but the T that agrees with the antecedent. In short, it is the features of T received from the antecedent that are relevant. Presumably the distinctive indexing of these is carried from the antecedent to T under Agree.

(26) These assumptions stem from suggestions in Richards (2002).

(27) Note that grammars are no longer Markovian on this conception.

(28) Observe three points. First, we cannot allow the valued features of the matrix T to value those of the lower T and then the lower reflexive. Given that multiple Agree is permitted and given that T's valued features can probe and value those of *himself* in (10), it is not quite clear why this is not allowed. Perhaps some phase boundary intervenes, though this is unlikely in the passive version of (14a). At any rate, we must assume that this is not an option. Second, we do not have the option of dispensing with multiple Agree, as it is an absolutely necessary part of an ABC analysis of reflexives. Third, it is tempting to assume that, in the context of the proposal that Agree values unvalued features, the freezing witnessed in (14a) reflects the fact that the embedded T has had its features valued. However, this cannot be correct, for in the embedded clause the only DP available to value the features of the lower T are those of the reflexive and we are assuming that the feature values of a reflexive must themselves be valued. Hence, the embedded reflexive cannot value the features of the embedded T.

(29) See Hornstein (2009) for a way of reducing the A-over-A to minimality. See discussion below on cases where the illicit binding in (15) is fine.

(30) Reuland (2001: 483) argues that there are additional semantic effects that arise from Agree with *zich*. The relevant cases are also discussed in Lidz (2001).

(31) With the obvious difference that *himself* is not an empty category. *Contra* Chomsky (1982a), we are not convinced that empty categories are a natural syntactic class, and assume that relatively superficial principles determine whether and how a copy will be pronounced at the PF interface.

(32) To take but one example, *zich* cannot be used in the formation of middles, in contrast to Romance SE, which is typically obligatory in middle constructions. Intriguingly, there is a dialect of Dutch (Heerlen Dutch) that does make

use of *zich* in the formation of middles (Hulk and Cornips 2000). This dialect also associates certain aspectual properties with *zich* that are also found with Romance SE.

(33) There are various ways of coding this generalization. See Hornstein (2001), Lidz and Idsardi (1998) for two variants of this approach. Both key on the idea that reflexives are residues of movement.

(34) Reuland emphasizes that this approach to *zich*-type anaphors implies that they '*enter into a real dependency with their antecedents in* C_{HL}' (his emphasis). This again is a point of agreement between ABC and CBC.

(35) In fact, given a minimalist theory in which movement obeys extension, the c-command requirement on chains can be derived. For discussion see Hornstein (2009).

(36) The first option is roughly the one in Lidz and Idsardi (1998). The second is the proposal in Hornstein (2001).

(37) Reuland (2001) cites two cases from Norwegian and Icelandic ((iii) and (iv), respectively) that he says have a similar structure. They translate out roughly as (i) and (ii):

((i)) There was introduced a man to *zichself*
((ii)) There came a man with *zich* children
((iii)) Det *ble* introdusert *en mann*$_1$ for *seg*$_1$ selv /*ham$_1$ selv. [Norwegian]
((iv)) That *kom madur*$_1$ med börnin *sin*$_1$ / *hans$_1$ [Icelandic]

The case in the text does not involve *zich* but the English reflexive. However, the logic will be the same. We discuss (30) because it is a far clearer version of the example Reuland has in mind. This is because it is not clear that the PP containing the reflexive in (i) and (ii) is actually too high to be c-commanded by *a man* in each case. Moreover, Chomsky (1995b) cites examples like (30) as cases in which PRO is controlled by features in T. Following Cardinaletti (1997), he argues that these are indeed cases of control via T as similar structures in French prohibit such binding configurations. In French, the associate in existential constructions does not agree with T, as it does in English, Norwegian, and Icelandic. Note the example with an overt reflexive is less acceptable than the one with PRO. Nonetheless, the reflexive here is far more felicitous than is a pronoun similarly indexed.
((i)) *After spotting the police, there quietly departed several meni without themi being detected.

So, for discussion, we assume that the example in (30) with the reflexive is grammatical. One last point: we use *without* for it is able to have both an overt subject and PRO, unlike adjuncts such as *after* and *before*.

(38) The long form of the reflexive *taziji* is a local reflexive and contrasts with the short form *ziji*. We thank Ming Xiang for the Chinese data.

(39) The availability of sub-command in Chinese raises the question of what distinguishes English from Chinese. The key property that allows the Chinese facts and prevents analogous structures in English appears to be that Chinese reflexives require human antecedents, while English reflexives can be bound by non-human antecedents. This combines with the A-over-A Principle to yield the difference in behavior. This has the additional consequence of predicting that in Chinese sentences like (i) *Zhangsan* cannot antecede the reflexive, though it can antecede the pronoun. In other words, the effects in (35) and (i) are reversed.

((i)) Zhangsan$_1$ de Mama guiji hai-le *taziji$_1$ /ta$_1$
Zhangsan's mother harmed himself/him

The proposal that sentences like *John's mother loves himself* is out because of something like the A-over-A Principle is made in Kayne (1994: 25–6). There it is proposed that *John's mother* blocks *John* from being a possible antecedent as it is a more proximate potential antecedent. See Boeckx and Hornstein (2007) for implementation of this idea in a more general context.

(40) The Norwegian version of (38) has *sig selv* in place of *herself* and so should fall under the purview of Reuland's version of ACD.

(41) In languages with distinctive *zich* reflexives, examples like (40a) would not involve the *zich* form. These are limited to cases where the antecedent is a 'subject.' The scare quotes should be taken as warning that possessive DPs must count as subjects.

(42) This is of interest because with other cases that appear to be similar, it appears to be marginally possible to get well-formed co-valued readings with pronouns:

((i)) John talked to Mary about herself/?her

See Reinhart and Reuland (1993) and Büring (2005) for discussion of the status of *about* PPs in(i).

(43) Note all these analyses assume that adjuncts like these are not inaccessible to syntactic operations, be they movement or probing for agreement. In other words, they do not sit in 'another dimension' that is inaccessible to the syntax. See Chomsky (2004a) for such a suggestion.

(44) For a more detailed exposition of the intuitive kinship between standard theory approaches and the approaches outlined here, see Hornstein (2001).

(45) Recall that, strictly speaking, Reuland's account only applies to *zich(self)*. We ignore this detail here, for the complementarity clearly extends to English reflexives as well. We conclude that these too must be licensed within a single module, and so the economy reasoning outlined above carries over to these as well.

(46) Note that this proposal has nothing to say about referential pronouns. This raises the question of their status. One potential problem that an approach such as this must countenance is the formal similarity between lexical and non-lexical pronouns. On this approach it's not immediately obvious how we can account for this in a principled way. However, it's also possible that this might merely be a case where superficial similarity of pronominal types has led us to lump two distinct categories together. Evidence from Japanese *zibun* suggests that there may be languages that do formally distinguish between bound and referential pronouns. *Zibun* aside, this is really just the flipside of the problem faced by more traditional theories: why is it that the same lexical item can be interpreted either as a referential element or as a bound variable (in the latter case, with its phi-features being ignored for interpretative purposes)?

(47) We also offer a slightly different motivation for the intermediate movement (the requirement that variables be bound from an A′ position), and do not make use of doubling constituents. For Kayne, the requirement that there be movement to an intermediate position is to be derived from a general requirement on weak pronouns that they must move to the left (interpreted as a requirement that the entire doubling constituent containing the pronoun and its antecedent must move to the left before subextraction of the antecedent can take place).

(48) We know! This involves improper movement. For the nonce, assume that this is not a problem and that such movement is grammatically tolerated. After all, why shouldn't it be? We can move from A-positions to A-positions and from A-positions to A′-positions. Why then shouldn't we be able to move from A′ to A-positions? Currently, the impossibility of this movement is simply stipulated, often in terms of chain uniformity, though clearly A to A′ movement (which is permitted) violates chain uniformity too. For those truly exercised by this kind of movement, assume that somehow the pronoun left behind repairs the improper movement that we find. This might follow if improper movement is reduced to Condition C, as suggested in May (1985), and if the inserted pronoun obviates the effects of Condition C.

(49) See Hornstein (2009) for discussion of c-command and its relation to the Extension Condition. This abstracts from the question of whether there is sidewards, inter-arboreal movement. If there is, this claim must be tempered.

(50) A puzzle:

((i)) *John$_1$ wants/would prefer for him1 to leave

A possible avenue towards a solution: *John* moves to Spec P for case reasons and this prevents movement into CP for some reason.

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A Minimalist Approach to Argument Structure

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[–] Abstract and Keywords

This article reviews some of the many and varied arguments for the 'split-vP' syntactic architecture that has taken over most of the functions of theta theory in the old Government and Binding framework, and considers how it can account for the central facts of argument structure and argument structure-changing operations. It then reviews the framework-wide implications of the new approach, which are considerable.

Keywords: syntactic architecture, split-vP, Binding framework, argument structure

In the past fifteen years of minimalist investigation, the theory of argument structure and argument structure alternations has undergone some of the most radical changes of any sub-module of the overall theoretical framework, leading to an outpouring of new analyses of argument structure phenomena in an unprecedented range of languages. Most promisingly, several leading researchers considering often unrelated issues have converged on very similar solutions, lending considerable weight to the overall architecture. Details of implementation certainly vary, but the general framework has achieved almost uniform acceptance.

In this chapter, we will recap some of the many and varied arguments for the 'split-vP' syntactic architecture which has taken over most of the functions of theta theory in the old Government and Binding framework, and consider how it can account for the central facts of argument structure and argument structure-changing operations. We then review the framework-wide implications of the new approach, which are considerable.

(p. 428) 19.1 Pre-Minimalism *θ* - Theory

In the Government and Binding framework, a predicate was considered to have several types of information specified in its lexical entry. Besides the basic sound-meaning mapping, connecting some dictionary-style notion of meaning with a phonological exponent, information about the syntactic category and syntactic behavior of the predicate (a subcategorization frame) was specified, as well as, most crucially, information about the number and type of arguments required by that predicate—the predicate's *θ*-grid. This basic picture of a GB lexical entry for a transitive verb is outlined in (1) below.

(1) Pre-minimalist *θ* -theory: a lexical entry, ready for projecting
PHON: kiss
SYN: [____v NP_{ACC}]vp
SEM: [Agent, Patient] (or: [1, 2], or [kisser, kissee])
+ some notion of what 'kiss' means

Principles—universal constraints on well-formedness—such as the Theta criterion and the Projection Principle filtered out deviant syntactic representations, ensuring that the predicate *kiss* could not appear in a sentence with fewer arguments than required by the predicate's θ-grid, nor with more than required.

In cases where the verb can grammatically surface with fewer or more arguments than specified, the theory required a productive lexical operation to change the θ-grid. For example, a passivization rule might alter the representation in (1) to the derived representation in (2a) below, before the lexical entry projected any syntactic structure. Similarly, an agentive nominalization rule could apply to (l) to produce thelexicalentryin(2b):

(2)

a. The result of a lexical passivization operation applied to (1)
PHON: kissed
SYN: [___]v
SEM: [Patient] (or: [1], or [kissee])
+ some notion of what 'kissed' means
b. The result of an agentive nominalization operation applied to (1)
PHON: kisser
SYN: $[\_\_]_N$
SEM: indexed θ-role of the V—either $Agent_i$ or $Instrument_i$
+ some notion of what 'kisser' means[1]

(p. 429) Other argument-structure affecting lexical operations might include 'causative' or 'applicative', or 'dative shift'—any case in which the lexical item appears in a sentential context in which its core argument structure and subcategorization requirements appear not to be met. In GB, then, the theory included a generative lexicon, in which certain lexical entries are derived from or related to other more basic lexical entries by redundancy rules. These rules, besides their syntactic effects, could also have morphological and semantic effects.

One central locus of theoretical activity involved discovering what kinds of principles govern the relationship between the theta structure of the verb and the syntactic structure which projected from it. Principles like Baker's UTAH (Baker 1988), Tenny's Aspectual Mapping Hypothesis (Tenny 1992), or Levin and Rappaport's linking rules (1995) ensured that the appropriate participant in the event ended up in the appropriate place in the syntactic tree, accounting for theta-role/syntactic structure regularities. As noted above, the Theta Criterion ensured that no predicate could end up with the wrong number of arguments, and no argument could end up without an interpretation.

When the goals of the minimalist program were first articulated (Chomsky 1993 et seq.), however, it became immediately clear that the GB module devoted to argument structure failed to meet minimalist goals on a number of criteria. The division of labor between two generative components—syntactic and lexical, each with their own primitive operations—ran counter to the central notion of employing the minimally conceptually necessary set of tools for constructing complex constituents. Empirically, the theta-theoretic architecture of the VP led to problematic conclusions when combined with the bare phrase structure proposal of Chomsky (1995c). Within the developing conception of the syntax-semantics interface in which Fregean function-application is the semantic correlate of the syntactic Merge operation, as described in Heim and Kratzer (1998), the Theta Criterion was both redundant and imprecise, neither of which qualities are particularly minimalist. Finally, the problematic tension between morphology and syntax which is especially evident in the realm of argument-structure alternations, cross-linguistically, is highlighted even more in the context of minimalist sensitivities. In many languages the lexical redundancy rules necessary to account for argument-structure alternations introduce a lot of morphology, which behaves fairly com-positionally, i.e. syntactically, most of the time. Corresponding constructions in other languages can be purely syntactic, as highlighted especially, for example, by cross-linguistic variation in causative constructions, which are syntactic in English but morphological in Japanese. Having two parallel systems within the language faculty deriving identical effects on Logical Form via completely different means in different languages is antithetical to the minimalist program's theoretical goals.

Fortunately, most of the solutions to these problems had already come into focus in the early 1990s, from converging analyses proposed to deal with several divergent problems. Hale and Keyser's theory of l-syntax, aimed at explaining (p. 430) causative/inchoative alternations and denominal verb structure, Kratzer's work on agent asymmetries in idioms, Travis and Borer's work on event structure and syntax, Larson's proposals concerning the structure of ditransitive verbs, and Halle and Marantz's work on the morphology-syntax interface all conspired to provide the general answer to most of these issues almost as soon as they arose, which is that verbal predicates are made up of at least two projections—the ‘little v’ hypothesis.

19.2 A Minimal *θ* - Theory: None

It is in fact trivially simple to establish that the basic functions of GB's theta-theoretic module are subsumed within a modern understanding of the interpretation of LF representations. In the semantic architecture of the Fregean program, as described in Heim and Kratzer (1998), predicates are functions, which must compose with arguments in order to achieve interpretability at LF. Unsaturated predicates, or extra arguments which cannot compose with predicates, will result in type mismatch and interpretation failure (see Heim and Kratzer 1998: 49–53). Given that something like Fregean semantic composition is needed to understand the behavior of quantifiers and adverbial and adjectival modification in any case, it would be emphatically non-minimalist to propose a special interpretive mechanism and set of principles to capture the observation that predicates require arguments and vice versa. Within minimalism, and given a Fregean view of the LF interface, the single Full Interpretation requirement can do the work of the Theta Criterion and Projection Principle within minimalist theory.

What, then, of the argument-structure operations (and their morphological correlates) which formerly operated in the lexicon on *θ*-grids to create new lexical entries, with new argument structures, ready for syntactic Merge? How can the relationship between inchoative and causative forms of a verb, or between active and passive forms, be captured within a minimalist architecture? It would be possible to adopt the notion of optional application of specific pre-syntactic functions which would operate in much the same way that the lexical argument-structure operations did in the GB theory. However, given the converging evidence that the internal structure of even monomorphemic verbal predicates is syntactically complex, and that alterations to argument structure introduce additional syntactic complexity, minimalist theoreticians have come to the conclusion that such lexical generative mechanisms are unnecessary, and hence undesireable. Argument-structure alternations can be, and should be, treated entirely within the syntactic component, via the same Merge and Move operations which construct any syntactic constituent. One key idea that makes this proposal feasible is the notion that the external argument is ‘severed’ from the verb proper, i.e. is the argument of a separate predicate in the syntactic tree. In the next subsections, we review the converging proposals (p. 431) which lead to this conclusion, and consider the implications for argument structure generally.

19.2.1 Structural limitations on argument structure: Hale and Keyser (1993, 2002)

In the late 1980s and early 1990s, Ken Hale and Samuel Jay Keyser (H&K) formulated their first attempt at an explanation of a pressing lexical-semantic question about *θ*-roles. Why are there only 6 or 7 robust *θ*-roles? Why not as many as 50 or 60? Even 10 or 12 would be more consistent with the number of case-markers or prepositions or classificatory verb stems in various languages. Dowty (1991) argued strongly for just two basic roles, a ‘proto Patient’ and a ‘proto Agent’ role; in his approach, other apparent roles consisted of semantic feature combinations intermediate between the two. Further, many of the well-motivated extant 6 or 7 seem to come in roughly animate/inanimate pairs: Agent/Causer, Patient/Theme, Experiencer/Goal, plus perhaps Incremental Theme. As noted by Baker (1997), theories with three Dowty-like ‘macro-roles’ are adequate for most syntactic purposes. To the extent that finer-grained theta distinctions or elaborate Lexical Conceptual Structure are motivated (e.g. *CAUSE TO BECOME NOT ALIVE* = ‘kill’), they seem to be motivated on semantic, not syntactic, grounds. Three to six *θ*-roles were adequate to account for the syntactic data bearing on *θ*-theory within GB.

H&K linked this theoretical question to an apparently unrelated morphological one: In many languages, the class of unergative verbs—intransitive verbs whose single argument receives an Agent *θ*-role—show clear signs of being bimorphemic, derived by combining an event-denoting noun and an agentive ‘light’ verb which can be glossed essentially as ‘do’. Several examples of this phenomenon from Jemez and Basque are given by Hale and Keyser (1998: 115), repeated as (3) and (4) below. The difference between Basque and Jemez is simply that the nominal incorporates into the light verb in Jemez, while remaining independent in Basque.

(3) Jemez

a.

Z#x00E1;ae-’a	‘sing’
song-do	

b.

hiil-’a	‘laugh’
laugh-do	

c.

se-’a	‘speak’
speech-do	

d.

tu-’a	‘whistle’
whistle-do	

e.

shil-’a	‘cry’
cry-do	

f.

sae-’a	‘work’
work-do	

(p. 432)

(4) Basque

a.

lo	egin	‘sleep’
sleep	do	

b.

barre	egin	‘laugh’
laugh	do	

c.

lan	egin	‘work’
work	do	

d.

negar	egin	‘cry’
cry	do	

e.

eztul	egin	‘cough’
cough	do	

f.

jolas	egin	‘play’
play	do	

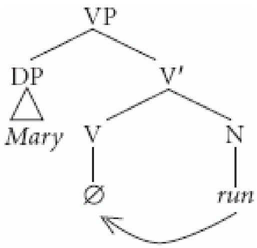
g.

zurrunga	egin	‘snore’
snore	do	

Even in English this relationship between unergative verbs and event nouns is quite transparent. Most if not all English unergative verbs have a zero-derived nominal counterpart:

(5) to laugh, a laugh; to walk, a walk; to run, a run; to work, work; to swim, a swim; to dance, a dance; to whistle, a whistle; to sneeze, a sneeze; to scream, a scream; to shiver, a shiver …

H&K considered the comparative data in English, Jemez, Basque, and other languages to indicate the presence of a uniform underlying structure, according to which there was a special verbal projection which introduced and assigned the Agent theta-role, translated roughly as ‘do’. They proposed that unergative verbs in English, as well as those in Jemez, are underlyingly transitive structures in which an agentive light verb selects for and optionally incorporates its bare N object. The Jemez and Basque light verb is morphologically visible while the English one is not. However, the presence of such a null verbal morpheme in English unergatives would explain the correlation between unergative verbs and bare nouns, and a single structure would account for English, Jemez, and other languages. Theyproposed the underlying structure below:



(6) Unergative verb derivation

(p. 433) Unergative denominal verbs of birthing, such as *calve, pup, whelp*, foal, and *spawn* would have the same structure as other unergatives, again accounting for the denominal character of such verbs.

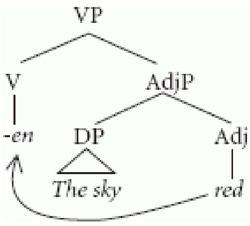
This proposal also provides an explanation for the lack of variation in the θ-roles assigned by unergative verbs to their single subject. If all unergative verbs are covertly composed of a null light verb 'do' and a nominal, then the real θ -role assigner—the element that truly selects the external argument—is the same in each case, the covert verb DO. There is only one θ -role assigned because there is only one θ -role assigner at work. The existence of several hundred different unergative verbs in English, for example, does not raise the spectre of several hundred different agent-selectors; there's only one, which occurs as a subconstituent of all of them. Hale and Keyser then went on to consider whether this idea could be fruitfully extended to other verbal predicates containing Agent arguments.

A similar situation arises with respect to causative/inchoative alternating verbs. In more languages than not, many inchoative verbs meaning something like 'become (more) ADJ' are morphologically related to or derived from the adjectival form. Some familiar English examples are below, as are some examples from Hiaki (Yaqui), a Uto-Aztecan language of Sonora, Mexico.

(7)

Verb	Adj	Verb	Adj
to redden	red	sikisi	siki
to fatten	fat	awia	awi
to soften	soft	bwalkote	bwalko
to sharpen	sharp	bwawite	bwawi
to warm	warm	sukawe	suka
…			

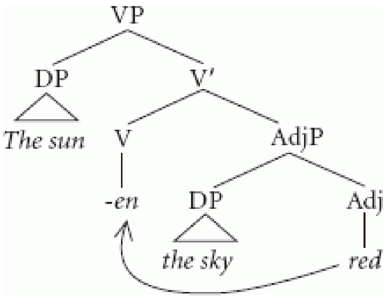
If inchoative verbs based on adjectives are derived by incorporating the underlying adjective into a verbalizing head, their morphological relatedness is predicted, as well as the additional morphology that shows up on the verbal form. Essentially, H&K proposed that deadjectival inchoative verbs are incorporated versions of unaccusative resultative constructions; a somewhat modified version of their structural proposal for an intransitive unaccusative verb is in (8) below:[2]



(8)

(p. 434) Here, the verbalizing element is semantically an inchoative raising verb; the construction is equivalent to *The sky turned red(er)* or *The sky got/became red(er).*No specifier of VP is present, and no agent θ-role is assigned.

These verbs, unlike unergative verbs, can alternate; that is, they may occur in a transitive structure in which an Agent theta-role does appear, as in *The sun reddened the sky*. In such a case, we could assume that, as in the case of the unergative verb, the verbalizer itself introduces the Agent, in its specifier position. The structure of a causative alternant of an inchoative verb, then, would be as in (9) below:



(9)

H&K's proposal thus suggested the beginnings of a way to get rid of θ-roles entirely. In (8) there is no specifier of VP, and there is no Agent in the structure—and it can be freely added, as in (9), to create a causative version. In the structure for unergative verbs in (6), on the other hand, there is necessarily already a specifier of VP, which receives an agentive interpretation; similarly in (9). Consequently no additional external argument can be added to such verbs, explaining the ungram-maticality of **John laughed the baby* and **John reddened the sun the sky*.

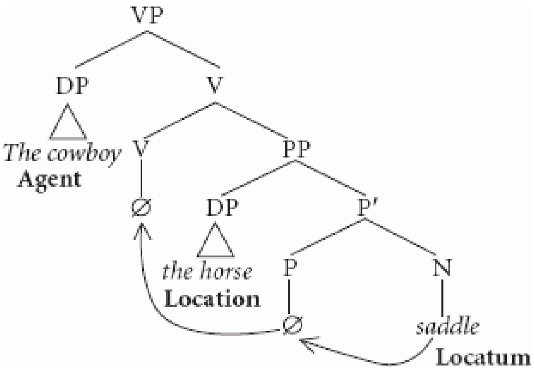
Hale and Keyser proposed that being an Agent simply means being in the specifier of VP, no more, no less. In the same way that identifying tense and agreement morphemes with functional syntactic heads renders the Mirror Principle unnecessary as a principle of the grammar (it becomes a descriptive statement that is explained by the syntactic process of head-to-head movement), identifying θ-roles bi-uniquely with syntactic positions renders linking principles such as UTAH unnecessary. UTAH also becomes a descriptive statement, explained by the semantic relationships between arguments, the positions they occupy in the syntax, and the functors that introduce them, rather than existing as a stipulated connection between an element in a θ-grid and a location in the syntactic tree.

H&K also proposed a natural extension of this system to a third class of verbs which pose a similar type of morphological problem as unergatives. In the struc-turesabove, we have seen what happens when an N is the complement of V with specifier (paraphrase: 'X DO N'), as well as what happens when an adjectival predication is the complement of V, both without a specifier (paraphrase: 'BECOME [X Adj]'), and with a specifier ('Y CAUSE [X Adj]'). H&K also argue that there are cases in which a PP is the complement of the external-argument selecting V (paraphrase: 'X CAUSE [Y on/at/with Z]'). When Z is incorprated into V, these are (p. 435) the location/locatum denominal verbs cataloged by Clark and Clark (1979). Some of these locatum verbs are listed in (10a) below; H&K's proposed structure is given in (10b):

(10)

a. bandage, bar, bell, blindfold, bread, butter, clothe, curtain, dress, fund, gas, grease, harness, hook, house, ink, oil, paint, pepper, powder, saddle, salt, seed, shoe, spice, water, word.

b. Structure: The cowboy saddled the horse = fit the horse with a saddle



The cowboy buttered the bread = smear the bread with butter

Again, the Agent argument occurs in the specifier of VP; the two inner arguments occur in the specifier and complement position, respectively, of the complement PP.[3]

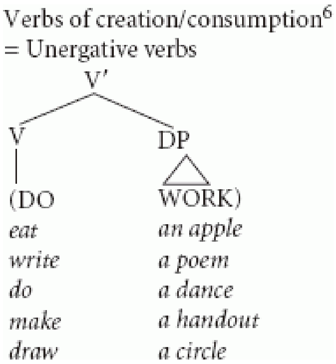
H&K thus were able to claim not only that spec-VP is reserved for Agent arguments, but also that what it means to be an Agent is simply to occur in specifier of a particular VP. The θ-role depends on the location in the tree, not the other way around.

H&K were also able to propose a specific invariant location for theme arguments. In all the structures containing *overt* direct objects above, namely (9) and (10b), the direct object undergoes a change of state or location, a canonical theme quality. In both cases, the direct object is the 'inner subject'—the subject of an embedded predication (a small clause, in the view presented here). H&K proposed that the locus for the canonical theme role is this inner subject position. The third robust θ-role—goal/location—is then associated with the position of 'inner objects': complements to P embedded under VP, as in *put the book on the table*.The assumption of an invariant spec-VP position for Agents, plus the exploitation of all X′-theoretical complement structures (N = X°, Adj = X° + Spec and P=X° + Spec + Comp) for the sister-to-V position, allows at most three arguments to appear with any (p. 436) given verb. This, H&K proposed, is the essential answer to the intial question of why there are so few theta-roles. It is because there are only three possible argument positions associated with any verbal predicate, namely (l) Spec-VP, (2) Spec of V's complement XP, and (3) Comp of V's complement XP, each of which receives a certain interpretation by virtue of its structural relationship to that predicate.

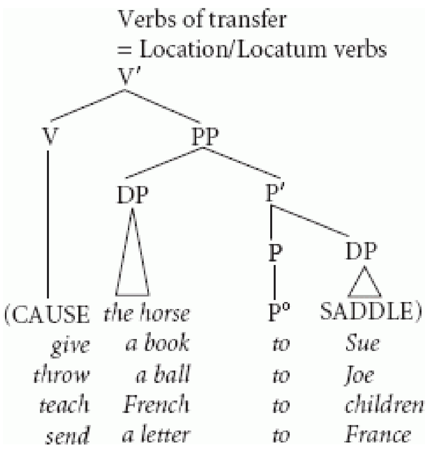
H&K had thus arrived at an inventory of structures for verbal predicates which maximally exploited X-bar theoretic structural possibilities. A 'light' V predicate selects a complement, either an N (non-branching), an Adj (binary branching), or a P (full X′ structure with specifier and complement).[4] The V predicate itself may or may not have a specifier position, which is the locus of the Agent argument, when present.

There are non-incorporated English counterparts of all of these structures, where the V position is filled overtly with a true verbal predicate, rather than acquiring its lexical content via incorporation of an adjective or noun. Each structure corresponds to a particular semantic class of verbs, whose interpretation is derived from the particular syntactic frame in obvious ways:[5]

(11) Verb classes with unincorporated instances of H&K's structures



a.

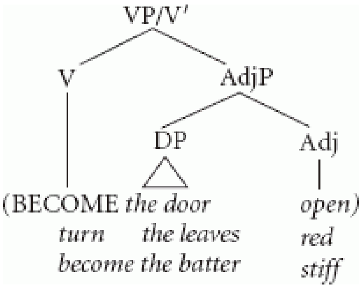


c.

(p. 437)

b. Verbs of change of state[7]

= Unaccusative verbs, with inchoative V)



= Causative verbs, with agentive V)

In the rest of this chapter, I will notate H&K's V category as *v*, and will usually notate complement AdjPs and PPs with their inner subjects as SCs (small clauses). Non-branching elements downstairs will continue to be labeled N for the moment. We can summarize the proposed structural correlations between *θ*-roles and syntactic position as follows:

(12)

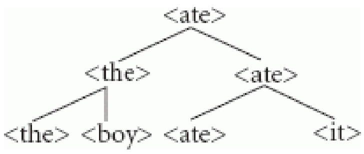
θ-role		Position of DP
Agent	≈	Spec-vP
Theme	≈	Spec-SC ('inner subject')
Goal	≈	Comp-SC
Incremental Theme	≈	Comp-vP

There are two logically possible verb classes which the combinatorial system proposed above makes possible, but whose existence was not addressed by H&K. We have thus far seen adjectival complements with both inchoative *v* and agentive *v*, but no cases of nominal complements or PP complements with inchoative *v*—that is, denominal predicates like *run* or *shelve* with no external argument. I would like to propose, however, that these two verb classes are instantiated, at least in a limited way, in English and other languages. The non-agentive counterparts to unergative verbs like *calve* or *dance* are the weather verbs: *it rained, it snowed*.These verbs simply represent incorporation of a bare N such as *rain* into the non-agentive v BECOME; an expletive must then be inserted to satisfy the EPP in Spec-TP in English. Similarly, unaccusative change-of-location verbs, as in *The plane landed* or *The boat docked* (=BECAME *theplane* P° LAND), are the non-agentive counterparts to the agentive location verbs.

One recurrent issue in this 'constructivist' view of thematic relations is the apparent lack of productivity of agent-deleting alternations (and, in the case of (p. 438) non-alternating unaccusatives like *arrive*, agent addition). All that is required is Merge of the embedded lexical structure with the specifier-less *v* category (BECOME), rather than with the agentive *v* which selects for a specifier (CAUSE), or vice versa. Why then are sentences like *#The city destroyed* and *#The captain arrived the ship* ill-formed? This question has been a fundamental issue in this framework and its relatives since its inception. In response, some, like Borer (2005), have proposed that in fact such mismatches are grammatical, and their uninter-pretability is of the same order as that of a phrase like *#colorless green ideas*— not a problem for the syntactic component at all. Others, like Harley and Noyer (2000) and Ramchand (2008), assume a system of featural licensing that determines which frames a verb root can appear in. Assuming that the problem of the productivity of alternation with specific verbs can be satisfactorily addressed, however, H&K's approach accounted straightforwardly for the morphosyntactic facts cross-linguistically, and addressed their theoretical question concerning the number of *θ*-roles available in natural language. As we will see next, it also provided a solution to two independent problems which would otherwise have impeded the development of modern minimalist theory.

19.2.2 Bare phrase structure and the vP proposal

The first problem concerned the development of a new framework for phrase-structure building. Chomsky (1995c), following a line first proposed by Speas (1986, 1990) and Fukui (1986), proposed to eliminate the X-bar component of the grammar, reducing all structure-building to the operation of Merge (see Fukui, Chapter 4 above). The set of two items constructed by Merge is labeled with a copy of the label of one of the items. The notions of 'head' and 'phrase' are then configurationally determined: a node is a head if it does not dominate a copy of itself, and it is a phrase if it is not dominated by a copy of itself (see Speas 1990: 44). Intermediate projections—bar-levels—have no status as objects of the theory. If an element meets both criteria, it can be both a phrase and a head simultaneously, as the object pronoun in (13) is. Clitics are the paradigm example of this: they behave like phrases in receiving a theta-role and checking case, and like heads in undergoing head movement. A tree notation of the sentence *The boy ate it* in this approach might look like this:

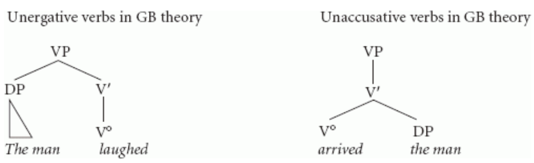


(13)'The boy ate it'

(p. 439) (Here, the words enclosed in angle brackets are intended to represent bundles of syntacticosemantic and phonological features, including category. Below, I will use the category labels as a shorthand to facilitate exposition, but they should be understood to represent the entire bundle.)

In bare phrase structure, therefore, non-branching nodes are a flat impossibility. Any X-bar theoretic analysis that requires a non-branching node requires reanaly-sis. For example, rather than propose a non-branching node, one could propose that a phonologically null element of the numeration has merged undetected.

This new conception of phrase structure created a significant problem for the standard treatment of the unaccusative/unergative distinction. Recall that unergative verbs are intransitive verbs with a single external argument; unaccusative verbs, in contrast, are intransitive verbs with a single *internal* argument, which becomes the subject by raising from its VP-internal position. This distinction could be naturally represented in X′-theory by exploiting a non-branching bar-level. In GB theory, the external argument would be base-generated in Spec-VP, sister to V′, while the internal argument would be base-generated in Comp-V, sister to V°, as illustrated in (14) below. The unaccusative/unergative distinction could thus be syntactically represented, as required by UTAH and allowing an account of the empirical distinctions between the two verb classes.



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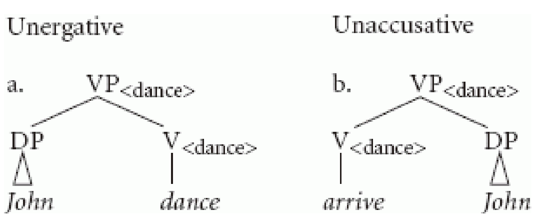
(14) Before the advent of the vP hypothesis

This is clearly problematic in the bare phrase structure approach, since the unergative/unaccusative structural distinction relies entirely on the presence of non-branching nodes. Within a BPS approach, the distinction presents a structural problem; eliminating non-branching nodes from (14) above produces (15) below:



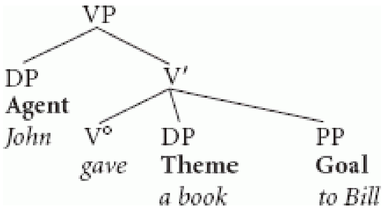
(15)

The unergative structure is supposed to contain a specifier (on the left) and the unaccusative one only a complement (on the right). But assuming that linear (p. 440) order does not matter in syntax, these two structures are indistinguishable, and the constellation of facts to do with the difference between the two classes of verbs has to be taken care of in some other, non-phrase-structural way (e.g. with reference to theta-roles or equivalents, as in LFG's f-structures).

Chomsky (1995c: 247–8) recognized this problem, and pointed out that the Hale and Keyser vP system provided a solution.[8] Since H&K proposed that unergatives actually are underlyingly transitive, with the structure in (6) above, while unac-cusatives are truly intransitive with the structure in (8), their system permitted the preservation of the unaccusative/unergative distinction without employing any non-branching nodes, thus allowing the elimination of X-bar theory.

19.2.3 Making room in the argument structure: Larson (1988) and VP-shells

At the same time that Hale and Keyser were developing their framework, Larson (1988) arrived at a bipartite structure for the VP based on the argument-structure requirements of ditransitive verbs. Given the VP-internal subject hypothesis of Koopman and Sportiche (1991), according to which external arguments are base-generated in Spec-VP rather than Spec-IP, a ditransitive verb like *give* requires a ternary-branching V′ constituent, to allow all arguments of the verb to receive a #-role under government by the verb, as illustrated in (16)below:



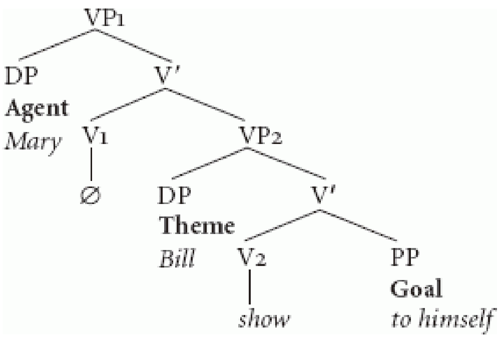
(16)

Following Kayne's (1984) suggestion thatX′-theory should be constrained to binary branching structures only, (16) is a theoretically inconsistent structure. Further, in (16), and its dative-shifted counterpart for *John gave Bill a book,* the theme and goal elements c-command each other, but Barss and Lasnik (1986) showed that binding-theoretic considerations suggest that the two internal arguments are in an asymmetrical c-command relation, such that the first argument in either the to-dative or ditransitive structure can bind into the second, but not vice versa, as illustrated in (17):

(17)

a. Mary showed Bill himself (in the mirror).
a′. *Mary showed himself Bill.
b. Mary showed Bill to himself (in the mirror).
b′. *Mary showed himself to Bill.

(p. 441) Larson's solution was to propose the ‘VP-shell’ structure in (18) as the base-generated syntactic structure for ditransitive verbs:



(18)

In this structure, the theme c-commands and precedes the goal, as required, and only binary-branching nodes occur. The innovation is to generate the external argument in a separate VP, or VP-shell, to which the lexical verb will head move to derive the final word order with the verb preceding the Theme. By now, the notion that the external argument appears in a separate VP projection from the rest of the argument structure should be familiar. Larson's work established that there were independent syntactic reasons to posit an extra VP for external arguments in the case of ditransitives, and the proposal connected straightforwardly with the vP framework developed by H&K and adopted for theory-internal reasons by Chomsky. The structure for ditransitives in (18) and the structure for location/locatum verbs proposed by H&K in (11c) are identical except for the node label on the lower shell.[9]

Having seen that the postulation of an independent verbal projection as the position of base-generation of Agent arguments can solve two thorny theory-internal problems, we now turn to consider some semantic repercussions of the bipartite VP proposal.

19.2.4 Semantic motivations for decomposing the VP: V-Obj idioms

Several independent arguments have also been made for a split-vP that build on facts about the semantics of eventive verbs. One primary class of such arguments derives from observations from the generative semantics literature concerning the scopal interpretations of VP modifiers; those are covered in section 19.2.5 below. A second argument builds on an independent observation due originally to Marantz (1984) (p. 442) and analyzed by Kratzer (1993, 1996) as involving the composition of verbal meanings through the conjunction of two separate predicates. Kratzer points out that if external, agentive arguments are in fact arguments of a separate v° functional projection, then Marantz's (1984) generalization about the restrictions on idiomatic composition can be explained.

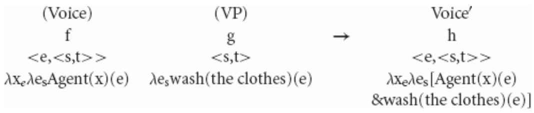
Marantz noted that while verb-object idioms/special interpretations are ubiquitous cross-linguistically, verb-agent idioms (that exclude the object) are close to nonexistent.[10]

(19)

kill a bug	=	cause the bug to croak
kill aconversation	=	causethe conversation to end
kill an evening	=	while away the time-span of the evening
kill a bottle	=	empty the bottle
kill an audience	=	entertain the audience to an extreme degree

Kratzer observes that if the subject and the object both compose directly with the verb *kill*, there is no principled semantic reason why there shouldn't be as many subject-verb idioms as there are verb-object ones. For example, *A bug killed the boy* could have one special interpretation (a non-‘kill’ meaning), while *The food killed the boy* could have another. However, these kinds of idioms, with free object positions and bound (idiomatic) agentive subjects, do not seem to occur.

If, however, Agents compose with a separate light verband then have their interpretation added to that of the lower predicate via a process Kratzer calls Event Identification, then the semantic independence of Agent arguments is expected. Event Identification combines the denotation of a Voice head (equivalent to v°, type <e,<s, t>>[11]) with the (argumentally saturated) denotation of the lower VP. This operation can be seen in (20) below (Kratzer 1993: ex. 19). In this example, it is asserted that there is an event (e) which is a wash-the-clothes event, and there is an event (e′) and an entity (x) and x is the Agent of e′. Event identification tells us that these two events are the same event, so x is the Agent of the wash-the-clothes event. The x argument never composes directly with the V predicate, only with the Agent predicate.



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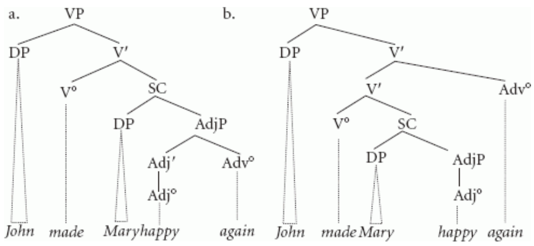
(20)

(p. 443) It is important to recognize that this treatment of Marantz's generalization only works if the object of the verb is truly an argument of the predicative V root, composing with it directly. A truly neo-Davidsonian analysis of the type proposed in a lot of later work (e.g. Borer 1994, 2005), in which there is a separate functor which introduces the object argument as well, won't work, or rather, will make the wrong predictions about idiomatic interpretations of the √: it will predict that verb-object idioms should be as rare as verb-subject idioms.[12]

19.2.5 Scope of modification: generative semantics redux

The vP hypothesis, particularly when enriched with an intuitive semantic content for the v° heads like H&K's DO, Kratzer's function *Agent(x,e),etc.*, draws exten-sively on insights first formulated within the generative semantics framework (e.g. McCawley 1976). The vP hypothesis is formulated within a somewhat more restrictive theory of phrase structure and the syntax-semantics interface, but it is adequate to capture many of the insights that the generative semantics decompositional structures were designed to explain.

Consider, for example, a biclausal sentence like *John made Mary happy again.The* adverbial *again* can be interpreted in two ways, as modifying *happy* or as modifying *make*. This ambiguity receives a straightforward structural analysis, since *again* can have two loci of adjunction: one on the embedded (small clause) predicate *happy* and one on the matrix predicate *make*, corresponding to the two interpretations. On the former, Mary was happy before (independently of John), had become sad, and then became happy again, thanks to John. On the latter, Mary had been made happy by John in the past, had become sad, and then been made happy by John again. The two structures are illustrated in (21)below:



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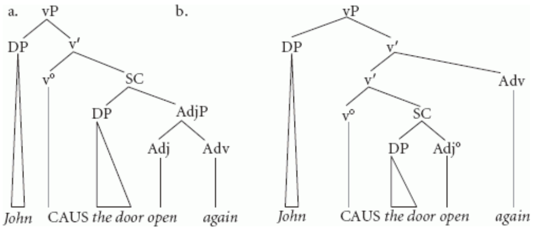
(21)

(p. 444) As shown in the generative semantics literature from the late 1960s and early 1970s, similar scopal ambiguities are present with simple ditransitive and other change-of-state verbs, as illustrated in (22)below:

(22)

- **a.** John opened the door again.
  - **i.** The door was open before, and now it's open again.
  - **ii.** John opened the door before, and he did it again.
- **b.** Mary gave the book to Sue again.
  - **i.** Sue had the book before and now she has it again.
  - **ii.** Mary had given the book to Sue before, and now she gave it to Sue again.

Von Stechow 1995 argued strongly for a generative semantics-type analysis of variable scope for adverbials like *again* in these sentences, within a vP-style syntax. By hypothesis, the causative verb *open* is made up of a predicate CAUSE (the null v°) syntactically taking a propositional complement headed by the intransitive predicate *open* (Adj°). The scope of *again* will then depend on whether it is adjoined to the embedded predicate or the matrix CAUSE predicate, just as it does in the clearly biclausal structure illustratedin(22)above.



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(23)

Beck and Johnson (2004) framed the same argument for ditransitive verbs, where *again* modifying the upper Larsonian VP-shell (vP) gives the reading of iterated causation of the event, and *again* attached to the lower VP shell (SC) gives an iterated result. In a ditransitive verb, the result denoted by the lower VP shell seems clearly to be stative location or possession. This can very clearly be seen in another dataset from McCawley (1968, 1979[1974]) and Ross (1976): the interpretation of temporal modifiers with ditransitive verbs:

(24) Temporal modifiers modifying the result of the event

a. Mary gave Bill the car until 3 o'clock (earlier this morning).

b. Mary lent her hat to Bill for 2 hours.

(p. 445) Here, it is not the action itself that lasts for two hours, but the state of Bill's having the hat, or Bill's having the car. A similar effect can be seen with *open* and related change-of-state verbs:

(25)

a. John opened the window for five minutes.

b. Mary turned the tap to 'cold' for five minutes.

If the resultant state is represented in the structure in a constituent independent of the constituent denoting the initiating action, in a VP-shell structure like those above, it is easy to treat the modification of that resultant state by a temporal adverbial; if it is not present, on the other hand, the syntax-semantics interface becomes remarkably complicated, as argued in detail by von Stechow (1995).

On this view of the contribution of the two different portions of the verbal predicate, the upper v° has its own semantic content, having to do with event initiation and causation. As argued above, the external argument, then, is semantically related to this upper v°, and is in fact not 'selected' by the root V° at all, though obviously the nature of the causation or event initiation in which the external argument engages will be affected by the content of the V° head, since different events require different types of initiation.

For at least certain verb classes, then, we have some semantic evidence that the verb is made up of a matrix external-argument-introducing projection, v°, involving causation or initiation, and a formally independent lexical projection, which seems to denote a result state and selects the internal arguments of the verb, and contributes the 'root' meaning of the predicate. The role of the vP hypothesis in accounting for event structure and event decomposition has been the focus of more than a decade of intensive study; see Ramchand (Chapter 20 below) for an extensive presentation. Some of the other key research in this area is represented in Travis (1991, forthcoming), Borer (1994, 2005), Alexiadou et al. (2004), and Ramchand (2008). See also Folli (2002), Pylkkanen (2002), Basilico (2003), Tomioka (2006), Baker and Collins (2006), Zubizarreta and Oh (2007), Merchant (2008) among many, many others, for related work.

A very well-known set of empirical objections to the decompositional project of the generative semantics framework were offered by Fodor (1970); space does not allow for a detailed rebuttal of these points in the context of this chapter, but for one explicit treatment of Fodor's arguments within the vP framework, see Harley (forthcoming). Although the vP hypothesis is at this point integral to the minimalist framework's treatment of argument structure, intra- and inter-framework debate continues. For contrary opinions and criticism from outside the minimalist program, see e.g. Kiparsky (1997), Wechsler (2005), Horvath and Siloni (2002).

(p. 446) **19.3 Alternatiyes Within Minimalism**

Alternatives within minimalism to the general proposal outlined above range from relatively minor amendments to wholesale rejections. Above, considerations of compositionality are taken to restrict unwanted configurations in the general spirit of the Theta Criterion: The notion that all θ-roles must be assigned, and that all DPs must bear a θ -role, follows immediately from the Full Interpretation requirement in combination with the semantic types of the constituents involved.

It is less clear that the uniqueness desideratum on θ-role assignment follows so directly. Does it follow that a single DP must bear only a single θ-role? Hornstein (2001) argues extensively for an approach according to which one DP may enter into thematic relations with more than one predicate, or indeed, may enter into thematic relations with the same predicate more than once. In his formulation, θ-roles are features of predicates, checked by DPs, and configurations in which a single DP checks more than one θ-role are the classic configurations of obligatory control and anaphor binding. A DP may merge with a predicate, checking its θ -feature, and subsequently undergo Move—Copy and re-Merge—to check the θ -feature of another predicate.

Interpreted in the terms of the present account, it seems clear that Copy and re-Merge could indeed result in a situation in which a single argument satisfied multiple predicates via function-application. Restricting the semantic possibilities opened up by the Copy and re-Merge treatment of Move would require additional stipulation. This aspect of Hornstein's proposal, then, is fully consistent with a Fregean approach to syntactic compositionality, assuming that other issues associated with the approach (distribution of overt vs. PRO realizations of traces, sideways movement, etc.) can be adequately worked out.

Hornstein's proposal that θ-roles are features, needing to be syntactically checked, however, is not consistent with the Fregean approach; syntactic features, like θ-roles themselves, would be additional mechanisms intended to replicate what the Full Interpretation constraint and a compositional semantics can already accomplish. Consequently, analyses like that of Siddiqi (2006) which critically rely on a featural conception of θ-roles are not consistent with the general picture presented here, and the phenomena accounted for thereby must be treated in some other way. Adger's (2003) approach, according to which semantic roles are necessarily associated with c-selectional features, may represent a middle road which could allow a reconciliation of the present approach and the syntactic feature-checking view of θ-roles.

A semantically decompositional yet syntactically more conventional approach to θ-roles is proposed in Reinhart (2002) and later work. In Reinhart's proposal, θ-roles are bundles of LF-interpretable features, analogous to the way that phonemes are bundles of PF-interpretable features like [±voice], [±velar], etc. **(p. 447)** Predicates in the lexicon bear clusters of these features, which are [±c] (for 'cause') and [±m] (for 'mental'); these features, in all combinations, define nine possible θ-roles. Reinhart's proposal is semantically decompositional, though not in the precise way proposed here, and can accurately characterize the class of verbs which participate in the causative/inchoative alternation (those with a [+c] external role— a Causer, rather than an Agent). A syntactic mapping procedure relates these clusters of features to particular syntactic positions, deriving a version of UTAH, and the syntax passes these features through to the LF representation, where they are mapped to neo-Davidsonian semantic interpretations, as illustrated in (26) below:

(26) ∃e (wash(e) & [+c+m](e) = Max & [-c-m](e) = the child) (= Reinhart's (4d))

Reinhart's system obtains its empirical results in a lexicalist paradigm, however, in which productive arity alterations apply presyntactically to the thematic feature bundles carried by verbs, altering the way in which they map to the syntax. In this sense, the propsal is really intended as a revision and improvement on the GB system, where separate, generative procedures changed lexical representations presyntactically. While Reinhart allows for the possibility that some morphological arity-affecting operations may apply in the syntax, she makes this a parameterizable option: there are lexicon languages, in which arity adjustments are presyntactic, and syntax languages, in which the same effect is obtained by a syntactic operation. In her system, for example, in Dutch, reflexivization reduction applies in the lexicon, while in German it applies in the syntax, accounting for the absence of lexical sensitivity in the latter case. In this regard, Reinhart's system is emphatically non-minimalist, espousing a separate, parametrically varying module of lexicon-internal operations, as well as syntactic equivalents of these operations. Reinhart's interesting empirical results notwithstanding, a single-engine approach like that outlined above seems to be more in tune with minimalist desiderata, and seems clearly also able to capture important empirical generalizations.

19.4 Conclusions

Although in this chapter I can only sketch the overall direction taken by a very large and empirically rich body of work spread over the past two decades, I hope at least to have motivated some of the core theoretical tools and concepts that are currently deployed in minimalist analyses of argument structure. In particular, it seems clear that it is possible and desirable to do away with the GB theta-theory; given that no theta-theory is more minimalist than some theta-theory, this is a desirable outcome. Further, I hope to have shown that semanticizing the original Hale and Keyser l-syntactic structures, in the appropriate way, gives robust and interesting results.

(p. 448) Many problems and questions remain, of course. Among other things, one open research question involves certain aspects of verb argument-structure flexibility that are not obviously accounted for by the three basic verb frames outlined above in (11). The parametric variation observed by Talmy (1985, 2000) in the availability of manner-of-motion constructions cross-linguistically has been a major topic of investigation, as has been the selected- vs. unselected-object distinction in re-sultative constructions (‘argument sharing’, see e.g. Levin and Rappaport Hovav 2001) but some of the core properties of these constructions remain mysterious—particularly how to account for argument-sharing effects in these structures. For relevant discussion, see e.g. Marantz (2007), Zubizarreta and Oh (2007), among others.

Finally, it is worth noting that the adoption of a neo-Davidsonian approach to argument structure interpretation, in combination with bare phrase structure, does not capture the core explanation that the H&K program was intended to discover, namely the reason for the apparent paucity of θ-roles. Recall that H&K wished to explain the observed restriction on the maximum number of arguments that a single verb can have—apparently around three. H&K's view of θ-roles was essentially purely configurational in nature, and consequently syntactic restrictions on possible configurations were the reason that there cannot be an arbitrary number of θ-roles. In the original formulation, X-bar theory provided a natural source of such a restriction—the most arguments that could be involved in the lower VP were two: a Spec and a Comp, and only one new argument could be introduced in the upper VP, in its Spec. Without X-bar theory, and with a neo-Davidsonian semantics and a bare phrase structure syntax, the limitation on available θ-roles must again be stipulated. Apparently, there is a functor Agent (e,x), but not other imaginable possible functors. It is possible that derivational constraints on syntactic structures (cyclic heads, phases, interface requirements) can yield the appropriate restrictions (see e.g. Boeckx 2008a for a proposal), but the original H&K explanandum still requires more work to understand.

Notes:

(1) Notice that there are two possibilities, both available in English: *kisser* can refer to a person who kisses, or to the mouth (the instrument of kissing). Examples like ‘transmission’ are similar, only with different possibilities for the indexed θ-role: Event_i, Theme_i, or Instrument_i

(2) The modification I have introduced here is just to turn H&K's complement clause from a mediated predication (with a lower V equivalent to something like Bowers 1993's PredP) to a small clause; this revision is intended as innocent here, to facilitate exposition. More substantive issues do depend on this modification, but unfortunately cannot be dealt with here. See Harley (2008a: 42–4, forthcoming) for discussion.

(3) At first, H&K proposed a structural account of the impossibility of certain location verbs (e.g. **church the money)*, but given the availability of syntactically and semantically equivalent verbs (e.g. *shelve the books, corral the horse)*, a different take on the productivity of this process seems appropriate (see Kiparsky 1997, Harley 2008b).

(4) See Mateu (2002) and subsequent work for extended development of this interpretation of H&K's proposals.

(5) In some languages, such as Persian (Farsi), such unincorporated ‘light’ verb plus non-verbal predicate constructions (‘complex predicate constructions’) are the primary exponent of verbal concepts, and, consistently with H&K's structures, can be sorted into the three primary classes shown here. For further discussion, see Folli et al. (2005).

(6) NB: The unincorporated ‘unergative’ structures above contain the only direct objects in this framework that are ‘inner subjects’. These are the arguments bearing Dowty (1991)'s ‘Incremental Theme’ theta-role. See Harley

(2005).

(7) The inner subject of these verbs, the theme argument, will raise to Spec-TP to check case features when the upper V is specifierless, as no higher argument is present in Spec-VP to intervene. Inchoative verbs are thus unaccusative, intransitive verbs with no external argument and with a derived subject created by movement from within VP.

(8) Speas (1990: 94–6) also adopts a version of the H&K proposal.

(9) Pesetsky (1995) and Harley (1995, 2002) propose prepositional lower shells for ditransitives; in the latter, a connection is drawn between the prepositional relation denoting 'have' identified by Kayne (1993) and Freeze (1992) and the identity of the lower shell. See discussion in section 19.2.5 below.

(10) Nunberg et al. (1994) argue that the asymmetric distribution of idioms is not indicative of any grammatical constraint but rather has an independent explanation in terms of a statistical conspiracy of the distributions of typical subject-predicate asymmetries involving animacy effects and topic-comment relations, and present some putative counterexamples; Horvath and Siloni (2002) also dispute the strength of the generalization. See Harley (in preparation) for a critique.

(11) e = individuals, s= events, t = truth values.

(12) An interesting ramification of Kratzer's proposal in conjunction with the framework described here concerns the denotation of PPs. Barry Schein (p.c.) notes that a Davidsonian treatment of modifiers entails that PPs are predicates of events, e.g. in sentences like *John buttered the bread in the kitchen*. However, this is incompatible with the notion that PPs can, at least in some cases, serve as arguments of the verb, as in *John put the food in the kitchen*, as such arguments are not predicates of events. If PPs can truly be arguments, as assumed here, and if Kratzer's approach is on the right track, it entails that a PP like *in the kitchen* is not univocal, but is sometimes of type <e> and sometimes of type <s, t>.

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Gillian Ramchand

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[-] Abstract and Keywords

This article presents a particular view of semantics which makes a strict separation between aspects of meaning that derive from the central computational system and those which belong in other, more general cognitive modules. While the details of the implementation are an empirical issue, it argues that the proposed model can handle many of the intricate issues involved in understanding verb meaning and argument structure. To the extent that the enterprise is successful, it is a model for the universal aspects of language structure and its close relation to abstract meaning relations. In this sense, the model is a proposal for a 'minimalist semantics'.

Keywords: computational system, argument structure, language structure, verb meaning, abstract meaning

20.1 Introduction

Recent years have seen a restaging of the generative semantics debate in the domain of verbal meaning. While a certain group of researchers maintain the importance of the lexicon as a repository of selectional and argument structure information (e.g. Gruber 1965, Levin and Rappaport Hovav 1995, Jackendoff 1990, Reinhart 2002), others have argued for a systematic decomposition of verbal meaning as reflected in the syntax, and have increasingly seen argument structure and event structure characteristics of verbal meaning as deeply 'constructional' (Hale and Keyser 1993, Ritter and Rosen 1998, Borer 2005). Rather than simply resuscitating the claims of generative semantics (Lakoff and Ross 1976), these new approaches have opened up fresh possibilities for understanding the systematicity of the compositional semantics associated with phrase structure. In enforcing a clear separation of encyclopedic, real-world information and contextual inferencing on the one hand from abstract semantic relations on the other, I argue that very clear and abstract patterns emerge in the syntax—semantics mapping that seem to be a pervasive property of the central combinatoric system. Thus, I see the proposals argued for in Ramchand (2008) as part of a minimalist agenda, which seeks to isolate the irreducible properties of the computational system that underlies our knowledge of language. The difference between my approach and more mainstream minimalist work in syntax is the belief that syntactic combinatorial primitives correlate with structural semantic **(p. 450)** combinatorial primitives, and that there is no way to make a principled modular difference between the core computation and structural semantic effects.

The issue of the syntax—semantics interface extends way beyond the issue of argument structure and event decomposition, of course. In this chapter I concentrate on the verbal domain because I have a concrete proposal to offer, but the architecture it advocates has implications for the grammar more generally. I take up these broader questions in the final section.

The central problem for the field as I see it is that syntacticians continue to assume a pre-minimalist view of argument structure classification and selection that is actually difficult to implement within current conceptions of grammar. The following are principles that many working syntacticians implicitly adhere to, as part of the background to whatever it is they happen to be working on.

(1) Syntactic Principle (*The Θ-Criterion*)
Each argument bears only one θ-role, and each θ-role is assigned to one and only one argument

Verbs and their arguments are thought of as being selected in a certain order. Consider the verb *put* in English.

(2) Calum put the two G&Ts on the dressing table.
(3) *put* (agent (location (theme))).
$\lambda x: x \in D_e\ [\lambda y: y \in D_e\ [\lambda z: z \in D_e$ [z put x (ON y)]]]

It is important to remember that nothing in the lambda formalism itself guarantees argument ordering on the basis of the denotation for a lexical entry. The order of combination must be stated on the lexical item specifically, or part of a more general lexical 'thematic hierarchy' (e.g. Larson 1988, Grimshaw 1990).

Further, we must assume some kind of classification of roles because unergative verbs and unaccusative verbs, for example, behave differently. The semantics of saturation does not give us this for free.

(4) Unaccusative verb: [[die]] = λx [x dies]
(5) Unergative verb: [[complain]] = λx [x complains]
(6) Transitive agentive verb: [[kiss]] = λx [λy[y kisses x]]
(7) Transitive experiencer verb: [[irritates]] = λx [λy[y irritates x]]

Given that there *are* generalizations to be captured in this domain, there are two clear strategies for implementing the generalizations we need:

(i) The lexical-thematic approach
This allows for the semantic classification of role types within the lexicon that is readable by a 'linking' theory that either (a) places these different roles in different places within the structure or (b) is independently readable by (p. 451) such modules as Binding Theory, Control Theory, and other semantically sensitive rule systems (Baker 1988, Larson 1988, Levin and Rappaport 1998, Jackendoff 1983).
(ii) The generative-constructional approach
This allows free base generation of arguments, but associates the interpretation of those arguments with particular structural positions (Hale and Keyser 1993, Borer 2005, Harley 1995).

Either (i.a) or (ii) are consistent with the strongest interpretation of the UTAH, given below in (8).

(8) *The Uniformity of Theta Assignment Hypothesis* (*UTAH*) (Baker 1988)
Identical thematic relationships between items are represented by identical structural relationships

Problems with taking the point of view in (i) surround the issue of selection. The lexical item must come coded with the information relevant to its argument structure, and this in turn must be enforced throughout the derivation. Within minimalism, feature-checking is the only mechanism clearly accepted in our arsenal. We are therefore forced to admit features such as [± Agent], or perhaps more parsimoniously [± external], into the derivation. We then have to allow such features to be optional on nominal projections, to ensure correct interpretation. They either have to be themselves interpretable (if we are to follow (i.a)) or they must force projection in a particular syntactic position (if we are to follow (i.b)). To my knowledge, such a system has not been actually implemented to date within the current minimalist framework, although it lies at the heart of many analyses of the GB era. Nor has there been explicit discussion recently of the options inherent in the choice between (i.a) and (i.b) (although it seems that (i.b) is more often implicitly assumed).

On the other hand, option (ii), the choice of the constructivists, has been popular in recent years. The reasons are not surprising. The problems of implementing thematic role selection through features (yet to be formalized) on lexical items seems technically difficult, and its success will no doubt require departures from minimal computational assumptions. The very existence of selectional features as a subgrouping with special properties (e.g. that they must be checked first, at the time of Merge (Chomsky 1995b)) is itself a major departure.

Consider too the thematic roles themselves, which have been argued over the years by various researchers to play a part in linguistic generalizations. The following list of thematic relations, informally defined, is drawn from the lists in Haegeman (1991) and Dowty (1989).[1]

(p. 452)

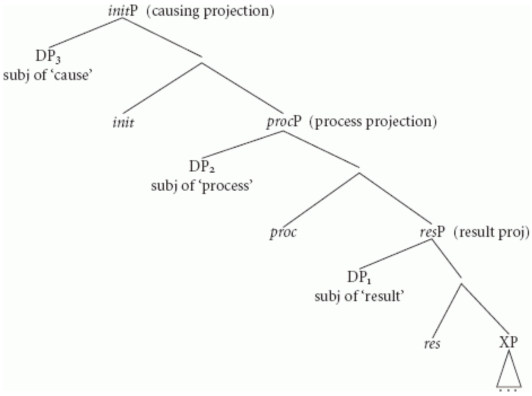
- THEME. A participant which is characterized as changing its position or condition, or as being in a state or position. Example: object of *give, hand*, subject of *walk, die.*
- AGENT (or ACTOR). A participant which the meaning of the verb specifies as doing or causing something, possibly intentionally. Examples: subjects of *kill, eat, smash, kick, watch.*
- EXPERIENCER. A participant characterized as aware of something. Examples: subject of *love*, object of *annoy.*
- BENEFACTIVE. Entity benefitting from some action.
  'John bought *Mary* a book.'
- PATIENT. A participant which the verb characterizes as having something happen to it, and as being affected by what happens to it. Examples: object of *kill, eat, smash,* but not those of *watch, hear*, and *love.*
- INSTRUMENT. Means by which something comes about.
  'John hit the nail *with a hammer*.'
- LOCATIVE. Place in which something is situated or takes place.
  'John saw the book *in the garden*.'
- GOAL. Object to which motion proceeds.
  'John gave the book *to Mary.*'/'John passed *Mary* the book.'
- SOURCE. Object from which motion proceeds.
  'John returned *from Paris.*'

As Dowty (1989) points out, the problems with this list of roles is that they are not clearly defined, and do not even represent replicable choices among researchers within the tradition of argument structure. One central problem is the mixing of abstract semantic constraints with purely encyclopedic information. In particular, it is difficult to decide whether a particular argument is for example a PATIENT because it is affected, or a THEME because it is affected by changing its location as a part of the event. In recent work, Baker (1997) claims that the notion of thematic role that is relevant for mapping to the syntax via UTAH is somewhat more abstract than the traditional list, and he offers the more abstract list of thematic categories as follows: Agent (specifier of the higher VP of a Larsonian structure), theme (specifier of the lower VP of a Larsonian structure, Goal/Path (complement of the lower VP). To separate grammatically relevant information from encyclopedic content, what we need is a classification that is based directly on the natural classes that emerge from the different structural positions available in the syntax. Baker (1997) is one move in that direction, and proponents of option (ii) more generally directly embrace that challenge as well.

(p. 453) In the spirit of option (ii), and with the goal of arriving at a more linguistically driven set of thematic relations, I will argue that VP (or vP) is actually somewhat more finely articulated, and is constructed from a recursive embedding of eventuality descriptors. Further, specifiers will be systematically interpreted locally as the 'thematic' element of each sub-description. A single natural and minimal relation between subevents ('leads to') will account for internal event complexity (the interpretations of 'cause' and 'telos') and also for the specific nature of the participanthood of those DPs in the complex events so formed. Complement positions will be systematically interpreted as not necessarily individuated with respect to the eventuality, but act to provide part of the eventuality description. This small but powerful subset of what we might call 'semantics' will be argued to be part and parcel of the central combinatoric system.

In Ramchand (2008) I argue for a small set of basic argument relations that are implicated in the linguistic construction of eventive predication, tied to a syntactic representation. The claim is that the generalizations at this level involve a kind of systematicity and recursion that is found in syntactic representations. The strongest hypothesis must be that the recursive system that underlies natural language computation resides in one particular module that need not be duplicated in other modules of grammar (i.e. not in the lexicon). At the same time, this means that the semantics that is compositionally built up by the syntax at this level can only include those aspects of meaning that are genuinely predictable and systematic—many aspects of meaning that are traditionally included in descriptions of lexical verbs (e.g. a certain subset of thematic information and semantic selection) must be excluded. The modularity that this involves has already been acknowledged within many theories of the lexicon as the difference between grammatically relevant lexical information and more general conceptual information, although the separation has mostly been argued to be internal to the lexicon itself (Hale and Keyser 1993, Jackendoff 1990, Grimshaw 1990, Kaufmann and Wunderlich 1998, Levin and Rappaport Hovav 1995). The approach here is a little different in that the grammatically relevant information actually comes from the interpretation of the syntactic structures that the verbs participate in. Any concrete proposal along these lines inevitably involves making a decision about which aspects of meaning should be represented in the syntactic system and which should be seen as coming from lexical encylopedic content. The proposal made here represents one set of choices, one that should be evaluated according to the usual standards of descriptive and explanatory adequacy.

The event structure syntax I propose contains three important subevental components: a causing subevent, a process denoting subevent, and a subevent corresponding to result state. Each of these subevents is represented as its own projection, ordered in the hierarchical embedding relation as shown below in (9).



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(p. 454) **(9)**

This 'decomposition' of V can be explained informally as follows. *Proc*P is the label for the projection at the heart of the dynamic predicate; it expresses a changing property, and it is present in every dynamic verb.[2] The *init*P exists when the verb expresses a causational or initiational state that leads to the process; it is present in (most) transitive and unergative verbs, but absent in unaccusative verbs. *Res*P expresses a result state, and only exists when there is a small clause predication expressing a property that comes into existence as a result of the dynamic change in the meaning of the verb. Although *res*P gives rise to telicity, it should not be exclusively identified with semantic/aspectual boundedness or telicity in a general sense, since telicity will also arise when the PATH argument of a *Proc* head is bounded.

In addition to representing subevental complexity, as motivated by work on verbal aktionsart (Vendler 1967, Parsons 1990, Pustejovsky 1991, Higginbotham 2001), this structure is also designed to capture a set of core argument roles, as defined by the predicational relations formed at each level. Each projection represented here forms its own core predicational structure, with the specifier position being filled by the 'subject' or 'theme' of a particular (sub)event and the complement position being filled by the phrase that provides the content of that event. The complement position itself of course may also be complex and contain another mini-predication, (p. 455) with its own specifier and complement. In this way, the participant relations are built up recursively from successively embedded event descriptions and 'subject' predications.

- *init*P introduces the causation event and licenses the external argument ('subject' of cause = INITIATOR).
- *proc*P specifies the nature of the change or process and licenses the entity undergoing change or process ('subject' of process = UNDERGOER).
- *res*P gives the 'telos' or 'result state' of the event and licenses the entity that comes to hold the result state ('subject' of result = RESULTEE).

This idea has antecedents, for example, in the work of Kaufmann and Wunderlich (1998), who argue for a level of semantic structure (SF) which is crucially binary and asymmetric and in which possible verbs are formed by constrained embedding.

POSSIBLE VERBS

In a decomposed SF representation of a verb, every more deeply embedded predicate must specify the higher predicate or sortal properties activated by the higher predicate. (Kaufmann and Wunderlich 1998: 5)

Kaufmann and Wunderlich see their SF level as being a subpart of the lexical semantics, and not represented directly in syntax, but the internal structure of their representations is very similar to what I am proposing here.

One way of looking at the primitives espoused here is in terms of the part—whole structure of events, which might serve to ground the intuition behind what is being proposed. If we think of a core dynamic event as representing the topological equivalent of a path, then the proposal here amounts to the claim that a verb must represent a single coherent path which can be assembled from a dynamic portion *proc* with or without endpoint *res* and beginning point *init*. The flanking state eventualities can be integrated with a process portion to form a coherent single event, by specifying its initial and final positions, but no *distinct* dynamic portion is possible without giving rise to a separate event path. As nice as this rationalization might be, however, it is important to stress that this choice of primitives is an empirical claim, not one that derives from any a priori notion of conceptual necessity.

20.2 The Semantic Interpretation of Structure

An important aspect of this proposal is the claim that there is a general combinatorial semantics that interprets this syntactic structure in a regular and predictable way. Thus the semantics of event structure and event participants is read directly off the structure, and not directly off information encoded by lexical items.

(p. 456) I see the proposals argued for in Ramchand (2008) as part of a minimalist agenda, which seeks to isolate the irreducible properties of the computational system that underlies our knowledge of language. Moreover, I have argued that the generalizations that emerge rely on a systematic event calculus, in an updated neo-Davidsonian theory. Once again, the view being pursued here is that semantics too is a generative system of combinatorial primitives that correlate in a simple, universal, and powerful way with syntactic combinatorics, as part of the central computational system.

In this way, the agenda pursued here is very similar to that pursued by Pietroski (2005a) (see also Chapter 21 below). Like Ramchand (2008), Pietroski is investigating what he sees as the core principles of combinatorial semantics; unlike Ramchand (2008), Pietroski (2005a) argues that in the light of the combinatoric complexity of syntax, the combinatorial properties of the structural semantics can be extremely austere, reducible completely (almost) to conjunction of monadic predicates. Essentially, Pietroski argues for a reining in of the full power of the lambda calculus in favor of a more restrictive set of recursive tools. Again like Ramchand (2008), the Pietroski proposals crucially rely on the event variable in the semantic representation to effect the simplifications he advocates.

The syntax—semantics mapping is constrained in theory by principles of compositionality. An example of how this can be stated can be found in the influential textbook by Heim and Kratzer (1998), which describes the scope of the interpretation function ‘[[]]’ as follows.

(10)

Terminal Nodes
If α is a terminal node, $[[\alpha]]$ is specified in the lexicon.

Functional Application
If α is a non-terminal node, $\{\beta, \gamma\}$ is the set of α's daughters, and $[[\beta]]$ is a function whose domain contains $[[\gamma]]$, then $[[\alpha]] = [[\beta]]([[\gamma]])$.

Semantic Principle
All nodes in a phrase structure tree must be in the domain of the interpretation function [[]].

However, given the full power of the lambda-calculus with no constraints on higher order functions, such a ‘Semantic Principle of Compositionality’ is not actually a constraint on human language, because it can always be satisfied (see also Higginbotham 2007). If we also admit type-shifting for individual 'lexical items', it is hard to see how it has any content at all.

Instead we need to ask, with Pietroski, whether distinctively human thought requires specific and special ways of combining concepts, special atomic concepts, or both, and investigate the minimal way in which this can be done. I make a very specific proposal here for the minimal semantic combinatorics. As we will see, it is somewhat less austere than the Pietroski toolbox, and I will compare it to the Pietroski proposal where relevant.

(p. 457) Thus, I lay out here how I have assumed that the general semantic combinatorial system works to interpret predicational structure.

20.2.1 Small clause complementation

Firstly, Ramchand (2008) assumes just one primitive rule of event composition, the 'leads to' relation.

(11) Event Composition Rule
e = e1 → e2: e consists of two subevents, e1, e2 such that e1 causally implicates e2
(cf. Hale and Keyser 1993)

There are two general primitive predicates over events corresponding to the basic subevent types as follows:

(12)

a. State(e): e is a state.

b. Process(e): e is an eventuality that contains internal change.

I have assumed that both the initiational eventuality and the result eventuality are states, and that their interpretation as causational or resultative respectively comes from their position in the hierarchic structure. In particular, in the *init* position, the state introduced by that head is interpreted as causally implicating the process; in the *res* position, the state introduced by that head is interpreted as being causally implicated by the process. We can therefore define two derived predicates over events based on the event composition rules.

(13) IF, $\exists$ e_1, e_2[State(e_1) & Process(e_2) & $e_1 \rightarrow e_2$], then by definition Initiation(e_1).
(14) IF $\exists$ e_1, e_2[State(e_1) & Process(e_2) & $e_2 \rightarrow e_1$] then by definition Result(e_1).

20.2.2 Specifiers

Further, the specifiers of each 'small clause' are interpreted according to the primitive role types defined by their position in the hierarchy, as given below:

(15)
a. Subject (x, e) and Initiation(e) entails that x is the Initiator of e.
b. Subject (x, e) and Process(e) entails that x is the Undergoer of the process.
c. Subject (x, e) and Result(e) entails that x is the Resultee.

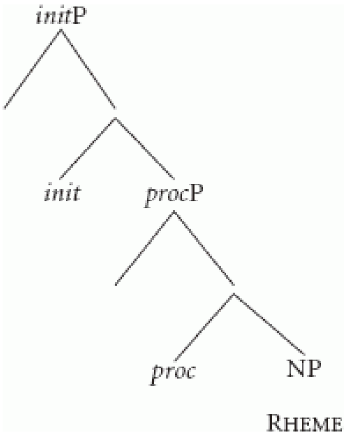
Using lambda notation for convenience, I spell out the denotations of the different pieces of structure, showing how they can be made to combine systematically to produce the required interpretations. The important point here is not (p. 458) the denotations in terms of lambda notation, but the idea that this dimension of skeletal semantics can be built up independently merely from the specification of the interpretation of pure labeled structure, *in the absence of lexical encyclopedic information.*[3]

(16) [[*res*]] = $\lambda P \lambda x \lambda e$[P(e) & State(e) & Subject (x,e)]
(17) [[*proc*]] = $\lambda P \lambda x \lambda e \exists e_1, e_2$[P($e_2$) & Process($e_1$) & e = ($e_1 \rightarrow e_2$) & Subject (x,$e_1$)]
(18) [[*init*]] = $\lambda P \lambda x \lambda e \exists e_1, e_2$ [P(e_2) & State(e_1) & e = $e_1 \rightarrow e_2$ & Subject (x,e_1)]

20.2.3 Complements

The proposal is the following. While the *proc* head can combine felicitously with a whole *res*P to create a result predication, it can also take a simple PP or DP in its complement position. In that case, the PP or DP does not determine its own independent subevent, but acts as a further modifier or descriptor of the *proc* event. In the terms of Higginbotham (2001), such NPs and PPs will combine by event 'identification', to further describe the properties of the relevant subevent.

The structures at issue here are those that have the form as in (19) below.



(19)

In fact, Ramchand (2008) proposes that event identification happens under a constraint of homomorphism, but a discussion of this is beyond the scope of this chapter.[4]

(p. 459) Taking the example of 'walking the trail' as representative of a rhematic complement DP filling the PATH role:

(20) [[walk$_{proc}$]] = $\lambda P \lambda x \lambda e$[P(e) & Process(e) & walking(e) & Subject(x, e)]
[[the trail]] = $\lambda e \exists x$['the trail'(x) & PATH(e, x))]

Although the specific relation invoked for the complement may differ (PATH under a homomorphic relation in this case), the Pietroski-style CONJOIN operation seems to be compatible with what is being proposed here.

20.2.4 How austere should we be?

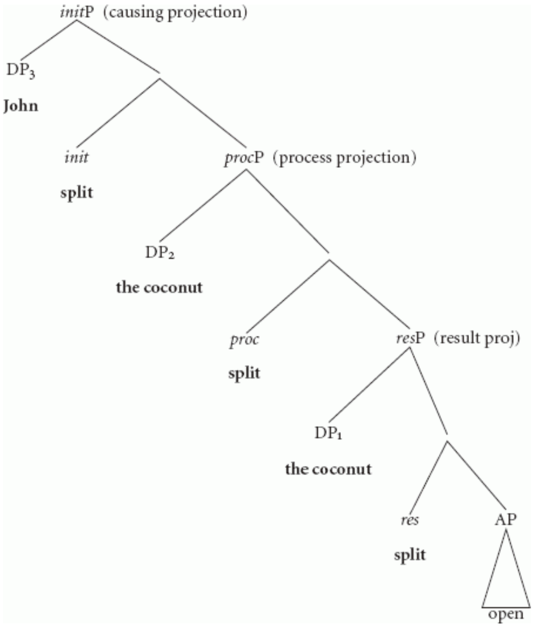
To summarize, the semantic combinatorial principles proposed here can be described as follows, each corresponding to a different syntactic configuration.

(21) Ramchand's 2008 (Recursive) Semantic Glue

i.	'Leads to/Cause' (→)	*Subevental embedding*
ii.	'Predication'	*Merge of DP specifier*
iii.	Event identification (conjunction)	*Merge of XP complement*

A natural question to ask here is to what extent all three combinatorial rules are necessary, and in particular, whether we can actually get rid of (i) and (ii) in favor of the conjunctivism that Pietroski proposes.

It turns out that (i) is fairly easy to get rid of, but at the cost of introducing the specific relations RESULT and CAUSE into the representation language (as opposed to the metalanguage). These concepts would have to be invoked explicitly as the interpretive concepts associated with *res*P and *proc*P respectively. Given the tree in (22) below, we could associate it with a semantics along the lines of (23) instead of the previous proposal in terms of 'leads to'.



Click to view larger

(p. 460) **(22)**

(23)

a. [[resP]] = $\lambda e \exists e_1$ [Result-Part(e, e_1) & open(e_1) & split(e_1) & State(e_1) & Subject(e_1,'the coconut')]

b. [[procP]] = $\lambda e \exists e_2$[Proc-Part(e, e_2) & splitting(e_2) & Dyn(e_2) & Subject(e_2,'the coconut')]

[[initP]] = $\lambda e \exists e_3$[Cause(e, e_3) & splitting(e_3) & & Subject(e_3, 'John')]

It seems to me that the cost of this is not prohibitive. Essentially, we give up the generality of the 'leads to' relation in favor of specific cognitive concepts called up by specific layers of the functional sequence. If the 'leads to' relation is confined only to these two embeddings within the clause, then it is not of great generality anyway. If, on the other hand, it proves to be a more general semantic relation in higher regions of the clause, or if it forms part of a natural class of semantic embedding relations for natural language, then the 'benefit' of reducing this to conjunction and an arbitrary set of relational concepts is somewhat less pressing.

In the case of Ramchand's (2008) relation (ii), I am less convinced that a move to a conjunctivist treatment is the right way to go. To do so, we would have to follow (p. 461) the Pietroski line, and invoke thematic relations during the course of 'recoding' the DP contribution. This recoding in Pietroski's system must be sensitive to the verbal lexical specification and/or structure. (It is never made entirely clear.) As discussed earlier in this chapter, I have not been convinced by traditional theories of thematic roles. Rather, I believe that the correct level of generalization emerges from the natural classes of items in specifiers of event predicational projections (and their combinations). In short, I suspect that eliminating (ii) in favor of CONJOIN misses an important generalization about the syntax—semantics mapping in this domain, i.e. that the specifier positions are uniformly interpreted as HOLDERS of the static or changing property denoted by their sister. Note also that Pietroski's conjunctivist system pairs the semantic rule of conjunction with *every* MERGE syntactic relation; if one believes in a primitive difference between specifiers and complements, then it is also plausible that a syntactic distinction of that kind might correspond to a systematically different mode of semantic combination. This is essentially what I am proposing here. The way (ii) is stated, it looks like an extremely general notion of predication, but I think it is important that we don't think of it in a technical sense, as defined by function—argument combination in an unconstrained lambda calculus. Instead, the claim here is that there is a primitive cognitive notion of PROPERTY ASCRIPTION that natural language symbolic structures hook up to systematically. Thus we need to place linguistic limits on what counts as 'predication' in this sense. To a first approximation, we can limit it to 'arguments' that denote simple entities in the model, not denotations of higher type.

20.3 Verbs and Argument Classification

I summarize the basic argument relations given by the primitives of this system including the composite roles that will be derived by Move, together with some illustrative examples.

INITIATORS are the individuated entities who possess the property denoted by the initiational subeventuality, which leads to the process coming into being.

(24)

a. *The key* opened the lock. **b.** *The rock* broke the window. **c.** *John* persuaded Mary. **d.** *Karena* drove the car.	PURE INITIATORS

The differences among the different initiators in the sentences above are due to the different lexical encyclopedic content of the verbs in question, and to the referential/animacy properties of the DP argument. By hypothesis, they are not related to structural position.

(p. 462) UNDERGOERS are individuated entities whose position/state or motion/change is homomorphically related to some PATH. UNDERGOERS are 'subject' of process, while PATHS are complements of process.

(25)

a. Karena drove *the car*. **b.** Michael dried *the coffee beans.* **c.** *The ball* rolled. **d.** *The apple* reddened.	PURE UNDERGOERS

(26)

a. Katherine walked *the trail*. **b.** Ariel ate *the mango.* **c.** Kayleigh drew *a circle.* **d.** Michael ran *to the store.*	PATHS

RESULTEES (Holders of result) are the individuated entities whose state is described with respect to the resultative property/Ground.

(27)

a. Katherine ran *her shoes* ragged. **b.** Alex handed *her homework* in. **c.** Michael threw *the dog* out.	PURE RESULTEES

GROUNDS of Result possess an inherent non-gradable property which describes the result state.

(28)

a. Karena entered *the room*. **b.** Kayleigh arrived *at the station.*	GROUND OF RESULT

UNDERGOER-INITIATOR is a composite role which arises when the same argument is the holder of initiational state *and* holder of a changing property homomorphic with the event trace of the *proc* event. (This can be modelled using the Copy Theory of movement.).

(29)

a. *Karena* ran to the tree. **b.** *The diamond* sparkled. **c.** *Ariel* ate the mango. **d.** *Kayleigh* danced.	UNDERGOER-INITIATORS

The example (29b) represents Levin and Rappaport Hovav's class of internally caused verbs, the (a) example is a motion verb which classically exhibits mixed behavior with respect to unaccusativity diagnostics. The (c) example deserves special mention because it is a case where the INITIATOR of the eating event is also somehow experientially affected by the process in a way that is only possible with animate/sentient causes. Because of this, we will see that the class of UNDERGOER-Initiators includes many cases of so called Actors or volitional Agents in the literature (see the next subsection for further discussion). RESULTEE-UNDERGOER is a composite role which arises when the same argument is the holder of a changing property homomorphic with the event trace of the *proc* event, *and* the holder of the result state.

(p. 463) **(30)**

a. Michael pushed *the cart* to the store. **b.** Katherine broke *the stick.* **c.** Ariel painted *the house* red.	RESULTEE-UNDERGOER

I have assumed that a composite role comprising a rhematic position and a role in specifier position is not attested. It has been proposed that there is a general prohibition against movement from the complement position of an XP to the specifier of that very same projection, which fits with my empirical findings here. I leave it open whether movements are possible from complement to specifier in larger domains within the first phase.

20.3.1 Mental participants

So far, I have been describing participant relations in terms of objectively observable causes, changes, and effects where intuitions seem more secure. However, initiation, process, and result are claimed to be the abstract structuring principles behind all eventive predications, and are intended to cover changes and effects in more subjective domains as well. Traditional thematic role systems often make a special case of Volitional Agents and Experiencers (Butt 1995, Belletti and Rizzi 1988), and the feature of mental state is one of the primitives used by Reinhart (2002) in her lexicalist theory of argument structure ([+m]). Cross-linguistically, animacy hierarchies play an important role in the syntactic realization of participant relations (see Ritter and Rosen 1998), and there is general cognitive evidence that humans interpret causational and affective relations differently when there are participants who possess sentience and will involved. I do not wish to deny the reality of these effects, but I propose to account for them without introducing additional heads or 'flavors' of initiational projections. Rather, I will argue that humans reason about sentient participants differently from the way they reason about inanimate objects and that this allows sentient creatures to participate in a wide variety of 'Subject' roles for subevents by virtue of their internal/psychological causes and effects, i.e. they don't have to be physical effects.

Often, the entailments of a particular participant differ systematically according to whether an animate or inanimate DP is in argument position, without any obvious change in the syntactic form of the verb phrase. In (31), the rock is a pure 'cause' or 'instrument', but John can be a volitional agent. In (32), the lever undergoes a physical change of orientation, while John is affected purely psychologically. In the latter case, the lexical encyclopedic content of the verb *depress* must be consistent both with physical and psychological motion 'downward' as a part of a pervasive analogy between physical and mental effects.

(31)
 a. The rock broke the window (*deliberately).
 b. John broke the window (deliberately).

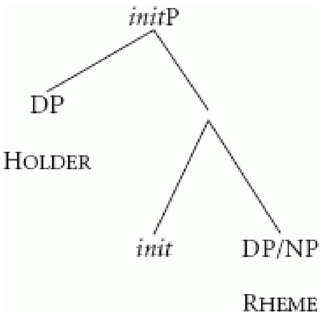
(p. 464) **(32)**
 a. Mary depressed the lever.
 b. The weather depressed John.

The point here is that animate/human referring DPs have the option of being interpreted as volitional causers, as willful controllers of a process, and as experiencers of static or changing mental states. For every sub-predication type and role type in specifier position that I have already proposed, I speculate that there is an analog in the psychological domain, but this will not be represented structurally here. For the stative subevents, it is clear what those interpretational inferences are: psych Initiators are 'intentional'; psych Resultees are experientially affected.

20.3.2 Stative predications

Finally, a word about stative verbs is in order here. The way the system has been built up so far, a stative verb cannot have any *proc* element in its first phase syntax, or any Undergoer argument, but only Rhematic or non-aspectual internal arguments. I will assume that stative verbs therefore consist simply of an *init* projection, with rhematic material projected as the complement of *init* instead of a full processual *proc*P. Since the *init* does not have *proc*P as its complement in this case, it is not interpreted as causational, but simply as a state. If there is an internal argument, it is in complement position and serves to further describe the state (without any path structure). The subject of *init*P is then straightforwardly interpreted as the holder of the state. Thus, a sentence such as the following (33) would correspond to the phrase structure shown in (34).

(33) Katherine fears nightmares.



(34)

Notating the first phase syntax of statives as *init* is not strictly necessary, since we could simply assume an independent verbal head corresponding to an autonomous state. However, I have unified the ontology because of the similarities in behavior between verbal statives and verbal dynamic verbs. Specifically, if we assume (as in one popular current view) that *init* (or rather, its analog, *υ*) is the locus for **(p. 465)** the assignment of accusative case as well as the licensing of an external argument (as per Burzio's Generalization), then statives are clearly verbal in this sense and have the equivalent of a little *υ* head in their first phase syntax.[5] Representing statives in this way also utilizes the ontology proposed here to the full—all possible combinations of heads and complements are attested and give rise to the different verb types we find in natural language. In particular, the phenomenon of *Rheme* nominal complements to *proc* heads (in complementary distribution to *res*Ps) exists side by side with *Rheme* nominal complements to *init* heads (in complementary distribution to *proc*Ps).

Given the existence of a functional sequence, then, whose order is forced by the semantic interpretation rules, we can assume that the syntactic structures are freely built up by Merge, but as we will see in the next section, they will have to be licensed by the presence of specific lexical items.

20.4 Lexicalization

Once we have admitted a more articulated functional sequence for the verb, a question arises with respect to lexical insertion, especially under a system where the syntactic features that are relevant for insertion each correspond to a distinct head in the structure. In line with minimalist thinking, and like the constructivist camp more generally, I retain the idea that the only truly generative system is the syntactic computation, and that no internally structured 'module' that one might call the lexicon is necessary. However, unlike extreme views of constructivism which embrace the notion of acategorial roots (Harley 1995, Marantz 1997), I do assume that lexical items possess syntactic features (restricted to features independently known to be present in the syntactic computation, and in fact confined here to just categorial features). For example, I assume that a verb like *run* possesses [*init, proc*] category features since it can lexically identify an initiated process, and a verb like *destroy* posesses [*init, proc, res*] category features since in addition it identifies a result. In addition, I will favor a non-projectionist view of these lexical items, seeing the association between structure and conceptual content as parallel map rather than a serial one. For convenience, this will be modeled as a system of late insertion, where a single lexical item can be associated not just to single terminal nodes but to systematic chunks of structure.

(p. 466) In Ramchand (2008), I argued that verbal lexical items come with a *set* of category features, and thus need to 'multi-attach' in the structures described above. This idea is consistent with the proposals recently formalized by Starke and Caha, rethinking the conditions of lexical insertion and extending it to nonterminal nodes. An alternative would be to use syntactic head movement or morphological merger to pre-bundle the necessary features under a particular terminal node, as in distributed morphology. Since these strategies simply mimic the effect of insertion in larger chunks of structure, I would argue that in a framework such as the verbal decomposition offered here, a rethinking of the terms of lexical insertion is both natural and necessary. I will follow Caha (2007) in advocating the more direct approach of lexical association to chunks of structure, while reformulating the notion of competitors. The proposal is that the choice of competitors is regulated by a 'Superset' principle, instead of the commonly assumed 'Subset' Principle of distributed morphology. It is important to realise that the Superset Principle can also be combined with a general 'Elsewhere condition' to give the generalized Superset Principle in (35). If the assumption of insertion under terminals is abandoned, then this principle gives equivalent results to the generalized Subset Principle in many cases (see Caha 2007 for discussion).

The Superset Principle is given below in full, and also decomposed into its minimized variety coupled with an Elsewhere condition, as articulated in Caha (2007).

> **(35)** The Superset Principle
> The phonological exponent of a vocabulary item is inserted into a node if the item matches all or a superset of the grammatical features specified in the node. Insertion does not take place if the vocabulary item does not contain all features present in the node. Where several vocabulary items meet the conditions for insertion, the item containing fewer features unspecified in the node must be chosen.
> EQUALS A1 PLUS B
> A1 Minimized Superset Principle
> A vocabulary item applies iff it specifies a superset of the features of a node
> B Elsewhere Condition
> Let R_1 and R_2 be competing rules that have D_1 and D_2 as their respective domains of application. If D_1 is a proper subset of D_2, R_1 blocks the application of R_2 in D_1. (taken from Neeleman and Szendroi 2006; see also Caha 2007)

Interestingly, the Superset and the Subset Principles make different assumptions about the architecture of the grammar and, in particular, of the relation between the syntax and the lexicon. As Caha (2007) puts it,

> **(p. 467)** The (Minimized) Subset Principle allows the spell-out procedure to ignore features of syntax, but not those of the lexicon. In other words, every feature specified in the lexical entry must have a matching feature in the syntactic structure, but not every feature of syntax must be 'spelled out' (i.e. targeted by a

vocabulary item that makes reference to it). The (Minimized) Superset Principle allows the interface to ignore features of the lexical entries (as not every feature specified in the entry for a given exponent has to have a matching feature in the syntax) but, on the other hand, all syntactic features must be spelled out.

I believe that the latter position is the correct one, and I will express it in (36) as follows.[6]

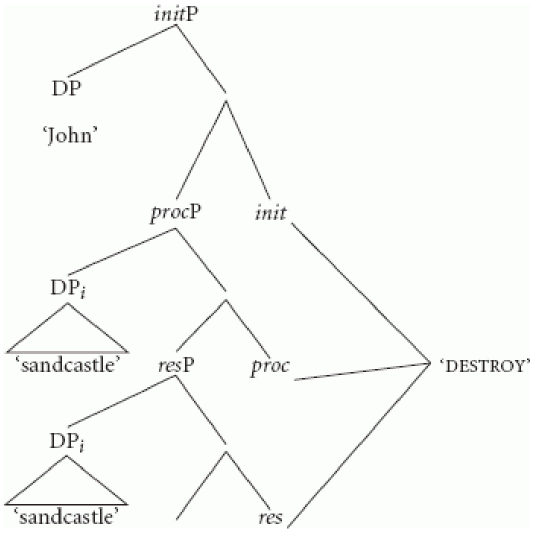
(36) Exhaustive Lexicalization
Every node in the syntactic representation must be identified by lexical content.

Given the modular independence of syntactic structures (with their structural semantic entailments) and encyclopedic content which is bundled in various sizes in the lexical inventory of different languages, we expect that the lexicalization of a particular structure could in principle look quite different from language to language. I believe that this is just as true for the verbalization of the first phase as it has been shown to be true for the typology of function-word expression across languages. Thus, I argue in Ramchand (2008), shown below, that for any particular first phase structure one can find analytic exponence as in the case of Bengali, agglutinative exponence as in the case of Russian, and synthetic exponence as in the case of the English verb *destroy.*[7]

For concreteness, I exemplify three different possibilities for the lexicalization of the full *init*P with embedded process and result below. These analyses are all more fully argued for in Ramchand (2008). The purpose of this section is to show the scope of the claims concerning lexicalization and the relation between encyclopedic content, lexical items, and semantico-syntactic structure.

The English verb *destroy,* having all three features *init, proc,* and *res,* identifies the full structure 'synthetically'.[8]

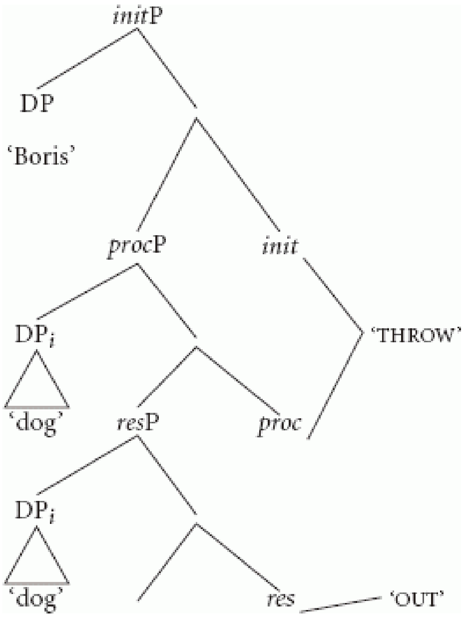
(p. 468) **(37)** John destroyed the sandcastle.



(38)

In Russian, I have argued that the lexical prefix *vy*- 'out' identifies the *res* of the predication and combines via head movement with the (imperfective) *init, proc* verb *brosil*—'throw' to create a telic resultative and perfective construction.[9] The mode of combination here is agglutinative and forms a single phonological word. This can be modeled by head movement, or alternatively thought of as linearized according to Brody's principles (Brody 2000a).

(39) Boris vy-brosil sobaku
Boris out-threw dog
'Boris threw out the dog'

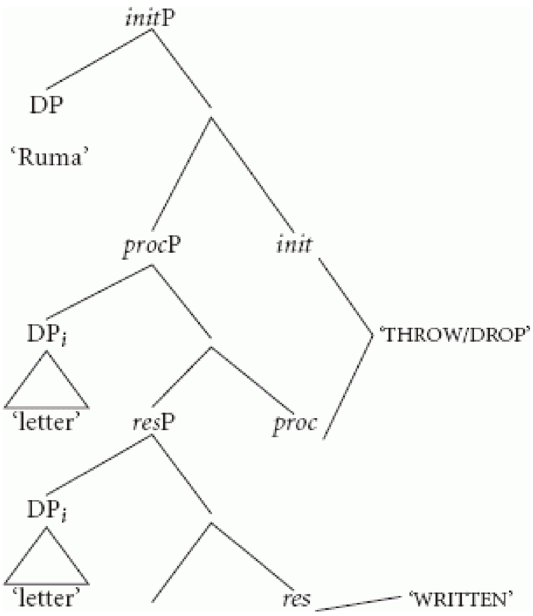


(p. 469) **(40)**

In Bengali, we find a completely analytic construction: the perfective participle *lekh-e-* 'written' identifies the *res* head, while the 'light' verb *phæla-* 'drop/throw' lexicalizes *init* and *proc.*

(41)

Ruma	cithi-ṭa	**lekh-e**	**phello**
Ruma	letter-DEF	write-PERFPART	drop/throw-3RDPAST
'Ruma wrote the letter completely.'			



(42)

(p. 470) Note that the Bengali complex predicate construction shown above and the Russian lexical prefix construction have otherwise posed paradoxes for lexicalist theories of argument structure. On the one hand, they are clearly morphemically compositional, and, in the case of Bengali it can be shown that the component parts are even independent syntactic units. On the other hand, the combination of lexemes changes the argument structure properties (something that lexicalists assume to be in the domain of the lexical module) and the constructions are mono-clausal by all diagnostics. The view proposed here accounts for the predicational unity of the complex predicates as well as their resultative semantics. The complex predicate construction of the resultative type, the verb—particle constructions, and the synthetic English verb *destroy* have essentially the same hierarchically organized components, but are just lexicalized/linearized differently.[10]

In all of the above examples, it is still possible to conceive of lexical insertion in a more traditional manner under terminal nodes, with head-to-head movement in the syntax, or in the morphology as the need arises. I present the multi-associational view here because I believe it requires fewer ancillary 'modules' (such as 'Fusion' in the morphology), and because it highlights the sharp difference between conceptual content and structural semantics. I will explain what I mean by that in the next subsection.

20.4.1 Structural semantics vs. encyclopedic content

Under this model, lexical encyclopedic information is seen as a parallel dimension of meaning to the structural semantics that is reflected in syntax. In theories like DM, the two types of meaning are treated 'in series' rather than in parallel, with lexical roots appearing at the bottom of the tree and functional elements built on top (Marantz 1997, Harley and Noyer 1999). In addition, because of the inexplicit mapping between syntax and semantics in those theories, the separation between the two dimensions of meaning is potentially compromised by leakage in the sense that functional items such as 'little v' are assumed to come in 'flavors' with different structural, but also semantic implications (Harley 1995, Folli and Harley 2004).

(p. 471) In the theory that I propose here, conceptual content and structural semantics are strictly located in different modules, conforming to a more strongly minimalist architecture where the properties of the computational system are kept distinct from other modules. The integration of conceptual content with the structural aspects of meaning is effected at a higher level of cognitive processing. I assume that the integration is analogous to the kind of integration that must be assumed for the visual system, for example, where different submodules control the construction of shape representations and color representations but where the object viewed is interpreted psychologically as being unified with respect to those dimensions. This is known as the 'binding problem' in the psychological literature, and I assume that it will apply here too.[11] The lexicon under this model is not a module in its own right, but consists of bundles of cross-modular associations between syntactic category (which links them to the central computational system), conceptual information (which links them to the general cognitive processing of information), and phonological information (about which I have nothing to say here).

20.5 Conclusion

I have presented a particular view of semantics which makes a strict separation between aspects of meaning that derive from the central computational system and those which belong in other more general cognitive modules. While the details of the implementation are an empirical issue, I have argued that the model proposed here can handle many of the intricate issues involved in understanding verb meaning and argument structure. To the extent that the enterprise is successful, it is a model for the universal aspects of language structure and its close relation to abstract meaning relations. In this sense, it is a proposal for a 'minimalist semantics'.

Notes:

(1) In fact, Dowty (1989) does not offer a list of thematic roles for our consumption, but uses it as the starting point for a discussion of the lack of secure definition for most in the list. In that paper, he argues strenuously for a new methodology based strictly on linguistic diagnostics and entailments.

(2) A *proc*P is present in this system regardless of whether we are dealing with a process that is extended (i.e. consisting of an indefinite number of transitions) or the limiting case of representing only single minimal transition such as that found with 'achievement' verbs. The difference between achievements and accomplishments in the traditional sense is captured by subsequent anchoring to the time variable, a topic that is beyond the scope of this chapter.

(3) The lexicalization process ends up associating encyclopedic information to this event structure via the lexical item—briefly, a lexical item with features for *init. proc. res* or some combination thereof, which contribute parallel conceptual content tied to the structural event positions they have features for. This information corresponds to the contribution of CONSTANTS in the lexical decompositional system of Levin and Rappaport Hovav (1995).

(4) Briefly, my proposal for complements of process builds on the formalism and intuitions regarding 'paths' and the material objects of creation/consumption verbs. The intuition is that a rhematic projection (in complement position) must unify with the topological properties of the event: if the event head is dynamic *proc,* the complement must also provide a topologically extended structure. In the case of directed paths that can be measured, the measure of the 'path' increases monotonically with the change expressed by the verb; in the case of (complement) Rhemes to stative projections, that Rheme must crucially *not* involve a path structure. DP/NP Rhemes in particular must provide structure in terms of the part—whole structure of their material constituency when combined with a dynamic event. Of course, rhematic elements are not just NPs; they can also be PPs and APs. In each case, however, the rhematic projection of process denotes an entity whose essential properties determine a scale which can be measured. PP Rhemes represent locational paths which are mapped onto the dynamic event (Zwarts 2003), and AP Rhemes provide a gradable property scale which functions as the mapping to the event-change path (see Wechsler 2001 for a claim along these lines). My claim is that the complement position of a process head is associated with the semantic relation of structural homomorphism, regardless of the category of that complement. The homomorphism has been noted before in different domains, and given formal representation. A unified semantics for homomorphism across categories is left unformalized here, but see Ramchand (2008) for a proposal.

(5) Here I leave open the issue of where one needs to distinguish 'unergative' from 'unaccusative' states, or whether that might correlate with the property in question being internally determined by the 'holder' (an individual level property) or simply an accidental or contingent property of that 'holder' (stage-level). It may well be that these differences also need to be structurally represented, but I have nothing more to say about these here.

(6) The name and formulation of the following principle emerged from collaborative conversations with Antonio Fábregas. See Fabregas (2007) for extensive discussion of its effects in the domain of Spanish directional complements.

(7) Just as in inflectional morphology, these categories are not parametric choices taken once and for all by a particular language—all languages represent mixed systems in some sense. I use the terms to describe a particular lexicalization pattern for a particular stretch of the functional sequence.

(8) In the three examples of tree structures that follow, I have uniformly drawn the phrase structures on the page as 'head-final'. I have done this (i) to emphasize visually the commonalities in the three cases, (ii) to visually separate the head contributions from the phrasal elements, and (iii) (most importantly), to emphasize the fact that these trees are intended to represent hierarchic relations with no implications of linear order. I assume that linearization for language is a language-specific and largely autonomous process that I put aside here.

(9) Ramchand and Svenonius (2002) also argue for this structure for the Germanic resultative verb—particle construction. The only difference is that the particle in English is analytic, and does not combine with the verb via head movement. Particle shift is accounted for under their analysis by either the particle or the small clause subject moving to identify *res*. The reader is referred to that paper for details.

(10) There is a further difference that I will not say very much about here, but which is discussed more extensively in Ramchand (2008). In brief, one can see that in some cases the higher subevents are lexicalized by an item with highly 'light' or abstract lexical content as in the Bengali light verb 'drop/throw' above, while the result subevent is lexicalized by a richly conceptually specified verb 'write'. In other cases the *init* and *proc* are lexicalized via elements that carry rich content, but the *res* is highly abstract and underspecified, as in the verb—particle constructions in English and Russian. Co-lexicalization of a single macro event is constrained so that lexical encyclopedic content must somehow be unifiable without conceptual contradiction. This means that at least one of the co-lexemes must be a fairly abstract and general item. So far I have found that this 'abstract' item can in principle be anywhere in the structure, but that the forms actually attested depend on the lexical inventories of the languages in question.

(11) My thanks to Michal Starke (p.c.) for alerting me to the possibility of thinking of the unification I propose in

these terms.

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[–] Abstract and Keywords

Chomsky's Minimalist Program (MP) invites a perspective on semantics that is distinctive and attractive. This article is organized as follows. Section 21.1 discusses a general idea that many theorists should find congenial: the spoken or signed languages which human children naturally acquire and use – henceforth, human languages – are biologically implemented procedures that generate expressions whose meanings are recursively combinable instructions to build concepts which reflect a minimal interface between the human faculty of language and other cognitive systems. Sections 21.2 and 21.3 develop this picture in the spirit of MP, in part by asking how much of the standard Frege–Tarski apparatus is needed in order to provide adequate and illuminating descriptions of the 'concept assembly instructions' that human languages can generate. It is suggested that we can make do with relatively little, by treating all phrasal meanings as instructions to assemble number-neutral concepts which are monadic and conjunctive. But the goal is not to legislate what counts as minimal in semantics. Rather, by pursuing one line of minimalist thought, it is shown how such thinking can be fruitful.

Keywords: minimalist program, semantics, human languages, Frege, Tarski

Chomsky's (1995b, 2000a) Minimalist Program (MP) invites a perspective on semantics that is distinctive and attractive. In section 21.1, I discuss a general idea that many theorists should find congenial: the spoken or signed languages that human children naturally acquire and use—henceforth, human languages—are biologically implemented procedures that generate expressions, whose meanings are recursively combinable *instructions to build concepts* that reflect a minimal interface between the human faculty of language (HFL) and other cognitive systems. In sections 21.2 and 21.3, I develop this picture in the spirit of MP, in part by asking how much of the standard Frege—Tarski apparatus is needed in order to provide adequate and illuminating descriptions of the 'concept assembly instructions' that human languages can generate. I'll suggest that we can make do with relatively little, by treating all phrasal meanings as instructions to assemble number-neutral concepts that are monadic and conjunctive. But the goal is not to legislate what counts as minimal in semantics. Rather, by pursuing one line of minimalist thought, I hope to show how such thinking can be fruitful.[1]

(p. 473) 21.1 Procedural Considerations

For better and worse, we can use 'language' and 'meaning' to talk about many things. As an initial guide to the topic here, let's tentatively adopt two traditional ideas: languages, via their expressions, connect signals of some kind with interpretations of some kind; and expressions of a human language have meanings—semantic properties that are recognized when the expressions are understood. Following Chomsky, I take each human language to be a state of HFL that generates expressions that pair phonological structures (the structures of PHON, hereafter PHONs) with semantic structures (the structures of SEM, hereafter SEMs), via which HFL interfaces with other cognitive systems that let humans perceive/articulate linguistic signals and assemble/express corresponding interpretations.[2] While the signals are plausibly gestures or sounds, in some suitably abstract sense, I assume that the interpretations are composable mental representations that may be individuated externalistically. On this view, SEMs can be characterized as instructions to assemble concepts, and meanings can be identified with such instructions in the following sense: to have a meaning is to be a certain kind of instruction, and thus to have a certain 'fulfillment' condition; and semantic theories for human languages are theories of the concept assembly instructions that HFL can generate. (Readers who find this banal may wish to skim ahead to section 21.2.) This mentalistic perspective permits versions of Truth Conditional Semantics. But the idea is that central questions about meaning concern the concepts and composition operations invoked via SEMs.

21.1.1 I-languages and interpretations

We need to distinguish generative procedures from generated products. So following Chomsky (1986b), let's say that I-languages are procedures that generate expressions, while E-languages are *sets of* expressions; cf. Frege (1980 [1892]) and Church (1941) on functions as intensions vs. extensions. As an analogy, note that (p. 474) '$|x - 1|$' and '$^{+}\sqrt{(x^2 - 2x + 1)}$' suggest different algorithms for determining a value given an argument, with 'x' ranging over whole numbers; yet each procedure determines the same set of argument-value pairs. We can use lambda-expressions to denote sets, and say that $\lambda x.|x - 1| = \lambda x.^{+}\sqrt{(x^2 - 2x + 1)}$. Or we can use such expressions to denote procedures, and say that $\lambda x.|x - 1| \neq \lambda x.^{+}\sqrt{(x^2 - 2x + 1)}$, adding that $\text{Extension}[\lambda x.|x - 1|] = \text{Extension}[\lambda x.^{+}\sqrt{(x^2 - 2x + 1)}]$. But whatever our conventions, different algorithms can have the same input—output profile. Likewise, distinct I-languages can in principle generate the same expressions. And in practice, speakers may implement distinct I-languages whose expressions associate signals with interpretations in ways that support workable communication.[3]

At least for purposes of studying the natural phenomena of human linguistic competence, including its acquisition and use, I-languages are importantly prior to E-languages. Each normal child acquires a language with unboundedly many expressions. So to even say which E-language a child allegedly acquires, one needs a generative procedure that specifies that set. And if a child acquires a set with unboundedly many elements, she presumably does so by acquiring (an implementation of) a procedure. Moreover, a biologically implemented procedure may not determine a set of expressions; but even if it does, there is no reason for taking this set to be an interesting object of study. Indeed, the acquired procedures may already lie at some remove from any stable target of scientific inquiry: the real generalizations may govern HFL, the faculty that lets humans acquire and use certain I-languages. But in any case, the theoretical task is not merely to specify the generable expressions that speakers can use. The task is to specify the expression-generating procedures that speakers implement.[4]

We begin, however, in ignorance. With regard to expressions of a human I-language (henceforth, 'I-expressions'), we don't know what the relevant interpretations are, or how they relate to reference and communication. But if spoken I-expressions connect sounds with mind-independent things, they presumably do so via mental representations. And for present purposes, I take it as given that human infants and many other animals have *concepts* in a classical sense: mental representations that can be *combined* in ways that can be described in terms of (p. 475) conceptual *adicities*; see e.g. Frege (1884, 1980 [1892]) and Fodor (1975, 1986, 2003). So if only for simplicity, let's suppose that spoken I-expressions connect (representations of) sounds with composable concepts, allowing for concepts that are distinctively human.

Matters are hard enough, even with this assumption, in part because a single I-expression maybe linked to more than concept, as suggested by the phenomenon of polysemy. But even setting aside examples like 'book'—which illustrates an abstract/concrete contrast that may distinguish kinds of concepts *and* kinds of things we can think about—it seems that a single lexical meaning can correspond to more than one concept. A speaker who knows that Venus is both the morning star and the evening star may have more than one concept of Venus, no one of which is linguistically privileged. Likewise, a speaker may have many ways of thinking about water. And as Chomsky (2000b) stresses, it is hardly obvious that some set is the extension of each 'water'-concept, given what competent speakers call 'water' when they are not doing science; cf. Putnam (1975). At a minimum, it would be rash to insist that each meaning privileges a single concept, or that concepts linked to a single meaning must share an extension. So let's say, tentatively, that each expression of a human I-language links a single PHON to a single SEM; where each SEM determines (and perhaps just is) a meaning that need not determine a single concept.[5]

21.1.2 Meanings as instructions

Chomsky describes PHONs and SEMs as instructions via which HFL interfaces with human articulatory/perceptual systems and conceptual/intentional systems. If we focus on comprehension, as opposed to production, words seem to invoke concepts that can be combined via operations invoked by phrasal syntax. So especially if a word can invoke different concepts on different occasions, one might describe each lexical SEM as an instruction to fetch a concept that meets a certain condition. Then a phrasal SEM can be characterized as an instruction to combine, in a certain way, concepts fetched or assembled by executing the constituent SEMs. The interest of this claim lies with the details: which concepts and combination operations are invoked by I-expressions? And eventually, the instruction metaphor must be replaced with something better, perhaps via analogies to programming languages and compilers. But the idea is that SEMs are Janus-faced: they are grammatical objects, whose composition (from a lexicon of atomic expressions) can be described in terms of formal operations like concatenation and labeling; yet **(p. 476)** they can direct construction of concepts, whose composition can be described in terms of semantic operations like saturation or conjunction. Or put another way: SEMs are generated, hence they exhibit a syntax; but these expressions are also apt for use in concept construction, allowing for an overtly mentalistic/computational version of the idea that meanings are 'directions for the use of expressions'; cf. Strawson (1950).

This leaves room for various conceptions of what these directions require. For example, one can hypothesize that 'brown cow' is (an I-expression whose SEM is) the following tripartite instruction: fetch a concept that applies to x iff x is a cow; fetch a concept that applies to x iff x is brown; and conjoin these concepts. This says nothing about where the concepts must come from. A speaker who links the words to suitable concepts, COW(x) and BROWN(x), might well conjoin those very concepts; but the instruction could be fulfilled by fetching any extensionally equivalent concepts. Or perhaps the instruction is to fetch a concept that applies to brown things, form a corresponding higher-order concept like & [BROWN(x), X(x)] and *saturate* it with a concept that applies to cows to obtain a concept like &[BROWN(x), COW(x)]. Fulfilling this instruction requires a certain process, culminating in the construction of a concept with a certain form. Or perhaps 'brown cow' calls for concepts from specific lexical addresses, but without imposing conditions on what the concepts apply to. Then twins might use the same I-expression to construct concepts that differ extensionally; although theorists can add that an I-language is an idiolect of English only if it meets certain externalistic conditions.[6]

Thus, many theorists should be able to adopt the idea that HFL generates concept assembly instructions, and that part of the task in semantics is to describe the 'I-concepts' that can be constructed by executing these instructions. Put another way, at least part of our job is to say which 'I-operations' are invoked by phrasal syntax and what kinds of concepts can be combined via these operations. We should not assume, a priori, that all human concepts are combinable via I-operations. The best overall theory may be one according to which few if any of our 'pre-lexical' concepts are combinable via the operations that I-expressions can invoke; see Pietroski (2010). But in any case, semanticists face a task that invites a minimalist question: what is the sparest inventory of operations and conceptual types that allows for rough descriptive adequacy with regard to characterizing the concept assembly instructions that HFL can generate? Here, we need to consider not just the syntactic operations employed in generating SEMs, but also the conceptual operations employed in executing SEMs.

(p. 477) My specific suggestion, developed and contrasted with others below, has two main aspects. First, an open-class lexical SEM is an instruction to fetch a *monadic* concept that need not be the concept lexicalized. Second, a phrasal SEM is an instruction to build a *conjunctive* monadic concept via I-operations that are limited to (i) highly restricted forms of conjunction and existential closure, and (ii) a few ways of converting one monadic concept into another. These conversion operations presuppose (a) some thematic concepts, associated with prepositions or certain grammatical relations, and (b) an analog of Tarski's (1933) treatment of 'closed' sentences as satisfied by everything or nothing, along with a number-neutral version of his appeal to sequences and variants. This is still a lot to posit, since concept construction has to be implemented biologically. Moreover, to even pose tractable implementation questions, we need theoretical notions of appropriate 'granularity' (Poeppel and Embick 2005); and while (i) *may* be in the right ballpark, at least some of (ii) seems worryingly grand. But I don't know how to make do with less—even ignoring lots of interesting details, in order to focus on highly idealized elementary constructions. And it is all too easy to posit far more: a richer typology of I-concepts, corresponding to abstracta like truth values and high-order functions; additional composition operations; type shifting, etc. But in the spirit of

MP, we can try to formulate the sparest proposals that have a prayer of descriptive adequacy, highlighting further assumptions that may be notationally convenient but replaceable with more economical alternatives.

21.2 Monadic Mentalese

In this section, I describe a possible mind with an I-language whose expressions can only be used to construct monadic concepts. Such a mind exhibits no semantic typology of the usual sort, though it deploys concepts of various types. Applying this model to human minds, given known facts, requires appeal to some additional operations for converting one monadic concept into another. But such appeal may be unavoidable and independently plausible. And in any case, it can be instructive to see which facts can be accommodated without assuming that human I-expressions/I-concepts exhibit a wide range of Fregean types.

21.2.1 Possible psychologies

For initial illustration, imagine a language whose syntax is exhausted by a unit-forming operation, UNIFY, which corresponds to a single operation of concept composition. In such a language, every complex expression is of the form [α, β], and the meaning of every expression can be specified as follows: SEM([α, β]) = (p. 478) **O**[SEM(α), SEM(β)]; where '**O**' stands for a 'macro' instruction to *execute* the two enclosed subinstructions, thereby obtaining two concepts, and then *compose* these concepts via the one invokable operation.

For example, suppose that *brown* and *cow* are atomic combinables whose meanings are instructions to fetch concepts from certain lexical addresses. Then SEM([*brown cow*]) = **O**[SEM(*brown*), SEM(*cow*)] = **O**[fetch@*brown*, fetch@*cow*]. Likewise, if *Bessie* and *cow* are atomic combinables, SEM([*Bessie cow*]) = **O**[fetch@*Bessie*, fetch@*cow*]. And if *Bessie* can be combined with [*brown cow*], SEM([*Bessie* [*browncow*]]) = **O**[SEM(*Bessie*), SEM([*browncow*])] = **O**[fetch@*Bessie*, **O**[fetch@*brown*, fetch@*cow*]]. If the operation invoked is monadic concept conjunction, then buildable concepts will all be of the following form: ⋅[Φ(x), Ψ(x)]; where '⋅' stands for a dyadic concept that can (only) connect two monadic concepts to yield a third, which applies to whatever both constituent concepts apply to. We can represent the meaning of [*brown cow*], for such a mind, as shown below.

SEM([*brown cow*]) = **CONJOIN**[fetch@*brown*, fetch@*cow*]

SEM([*brown cow*]) → ⋅[BROWN(x), COW(x)]

Here, '**CONJOIN**' indicates a kind of instruction, and '**→**' indicates the sort of concept that results from executing a given instruction. Ignoring polysemy for now, suppose that each lexical item is linked to exactly one fetchable concept, and hence that each lexical SEM is executable in exactly one way—viz. by fetching the corresponding concept.

Given this simple combinatorics, atomic SEMs have to be instructions to fetch monadic concepts. In this language, *Bessie* must be linked to a monadic concept, BESSIE(x). But this concept might apply to x iff x is a certain cow, who we call 'Bessie'; cf. Quine (1963), Burge (1973). And given a monadic concept for *Bessie* to fetch, executing SEM([*Bessie* [*brown cow*]]) is a way of constructing the concept ⋅[BESSIE(x), [BROWN(x), COW(x)]].

By contrast, suppose the sole operation of concept composition is saturation, which can combine a monadic concept like cow(x) with a singular concept like BESSIE to form COW(Bessie). Given this language, *Bessie* can fetch BESSIE, while *cow* can fetch cow(x). Executing SEM([*Bessie cow*]) is then a way of constructing cow(BESSIE), which is a 'propositional' concept. The singular constituent, BESSIE, is of a different type. Given these two types, <**t**> and <e>, one can say that COW(x) is of type <e, **t**> and the propositional concept is of the form <e, **t**> (<e>). More generally, the buildable concepts will exhibit the following abstract type: <α, β> (<α>), indicating that a concept of adicity n is formed by saturating a concept of adicity $n + 1$ with an argument of an appropriate type. (Concepts of type <**t**> and <e> have adicity zero.) If [*brown cow*] is also an expression of this language, this instruction cannot be executed by using *brown* to fetch BROWN(x), a concept of type <e, **t**>. But *brown* might fetch ⋅[BROWN(x), X(x)]; where this higher-order concept of type <<e, **t**>, <e, **t**>> was previously defined in terms of BROWN(x) and linked (p. 479) to *brown* as a second fetchable concept. Then executing SEM([*brown cow*])—i.e., **SATURATE** [fetch@*brown*, fetch@*cow*]—could be a way of constructing the concept ⋅[BROWN(x), COW(x)]; cf. Parsons (1970), Kamp (1975).

This second language, familiar in kind, permits lexical expressions that fetch dyadic concepts like CHASE(X, Y), which can be saturated by a singular concept to form a complex monadic concept like CHASE(X, BESSIE). Indeed, the operation of saturation itself imposes *no* constraints on which concepts can be fetched and combined with others: a concept of type $< \alpha, \beta >$ can be combined with *either* a concept of the 'lower' type $< \alpha >$, thereby forming a concept of type $< \beta >$, *or* any 'higher' type $<< \alpha, \beta >, \gamma >$ such that $< \gamma >$ is also a possible concept type.

If only for this reason, we should ask if we need to posit saturation as a composition operation in theories of I-languages. Even setting aside empirical arguments against such appeal (see Pietroski 2005a, 2010), one might prefer to explore hypotheses according to which there are severe restrictions on the concepts that can be fetched by atomic I-expressions. For even if the specific proposals explored are wrong, seeing why can provide insights about the actual typology. A theory that imposes few constraints on the fetchable concepts may be harder to disconfirm. But 'negative' facts, concerning nonexistent types and nonexistent meanings within a type, are relevant. And in any case, compatibility with facts is not the only theoretical virtue.

That said, incompatibility with facts is a vice. And monadic concept conjunction cannot be the *only* operation invoked by I-expressions for purposes of combining fetchable concepts. Expressions like 'chase Bessie'—'chase every cow', 'saw Aggie chase Bessie', 'did not chase Bessie', etc.—are not simply instructions to conjoin monadic concepts fetched with the lexical items. But given neo-Davidsonian proposals, one can plausibly say that 'Aggie chase Bessie' is used to build a multi-conjunct concept: a concept that applies to things that have Aggie as their Agent, are chases, and have Bessie as their Patient; see Parsons (1990), Schein (1993, 2002). In my view, this model of composition is basically correct and extendable to other cases. By way of exploring this idea, according to which I-languages differ in just a few ways from the first 'conjunctivist' language imagined above, let me describe a possible range of atomic I-concepts and I-operations that permit construction of complex monadic concepts. In section 21.3, I'll offer a proposal about how such concepts could be fetched and combined as suggested, given a syntax that adds a labeling operation to UNIFY; cf. Hornstein (2009). The resulting account may be compatible with the facts.

21.2.2 Lexicalization

Imagine an initial stage of lexical acquisition in which many concepts are paired with phonological forms, so that certain perceptions (of sounds/gestures) reliably invoke certain lexicalizable concepts. During a second stage, each <PHON, **(p. 480)** concept> pair is assigned a lexical address that is linked to a bin, which may eventually contain one or more concepts that can be fetched via that address. But if B is the bin that is linked to address A, then a concept C can be added to B only if C is a monadic concept that is the result of applying an available 'reformatting operation' to some concept already linked to A. Only a few reformatting operations are available. So there are constraints on which concepts can be fetched via any one lexical address.

Binned concepts must be monadic, because the computational system we are considering can only operate on concepts of this kind. The imagined mind has a language faculty that generates instructions to create complex concepts from simpler ones. But this modest faculty can only generate instructions of two kinds: those that call for conjunction of two monadic concepts, and those that call for conversion of one monadic concept into another. And while this limits the faculty's utility, the surrounding mind may be able to invent monadic analogs of nonmonadic concepts, thereby making the faculty more useful than it would otherwise be; cf. Horty's (2007) discussion of Frege on definition. For example, a dyadic concept like KICK(X, Y) might be used to introduce a monadic concept KICK(E), perhaps by introducing a triadic concept KICK(E, X, Y) such that KICK(X, Y) = ∃E[KICK(E, X, Y)] and KICK(E, X, Y) ≡ AGENT(e, x) & KICK(E) & patient(E, Y). Then given a proto-word of the form <PHON, bin-address, KICK(X, Y)>, the analytically related concept KICK(E) can be added to the bin, which will not contain the lexicalized dyadic concept.

More generally, this mind might create formally new monadic analogs of lexicalizable concepts as follows: use a concept C^n of adicity n to introduce a concept C^{n+1} of adicity $n + 1$; and use C^{n+1}, along with n 'thematic' concepts that are independently available, to introduce a monadic concept C^1. Suppose that given a singular concept like BESSIE, this mind can also create an analog monadic concept. For illustration, IDENTITY(X, BESSIE) will do. But given a proto-word of the form <PHON, bin-address, BESSIE >, one can imagine forming the corresponding monadic concept called(PHON, X), which applies to anything called with the PHON in question. And if CALLED(PHON, X) is added to the bin, it might later be fetched and conjoined with another concept—perhaps demonstrative—so that at least in the context of use, the resulting concept of the form •[CALLED(PHON, X), Φ(x)] applies to exactly one individual, like the one mentally denoted with BESSIE; see e.g. Burge (1973), Katz (1994), Longobardi (1994), Elbourne (2005b).

When a monadic concept is lexicalized, it may be added to its own bin. But this does not guarantee conjoinability with other concepts. Suppose the concept lexicalized with (the PHON of) 'brown' is a concept of surfaces, while the concept lexicalized with 'house' is not. Then the proto-word <PHON, bin-address, BROWN(s)> may, given suitable prompts, lead to introduction of a concept that applies to the *brown-surfaced:* BROUNS(x) ≡ ∃s[SURFACE(s, x) & BROWN(s)]. In which case, BROUNS(x) could be added to the bin, making it possible to coherently conjoin HOUSE(x) with a concept fetched via the address initially linked to BROWN(s); cf. (p. 481) Chomsky (2000b). As this example suggests, lexicalization might lead to a polysemous word that bins several monadic concepts. Whatever is initially lexicalized with 'book', the end result may be a lexical item that can be used to fetch any of several concepts—including at least one that applies only to certain spatio-temporal particulars created by publishers, and another that applies only to certain abstracta created by authors. Likewise, for mature speakers, the bin for 'country' may include at least two concepts: one that applies to the French *polis*, but not to the *terrain* inhabited by the citizens of France; and a distinct concept that applies to this *terrain*, but not the occupying *polis*. And perhaps only the *polis*-concept can be coherently conjoined with at least one concept fetched via the word 'republic', while only the *terrain*-concept can be coherently conjoined with at least one concept fetched via the word 'hexagonal'.[7]

21.2.3 Number neutrality

Especially given the possibility of reformatting, we need to be clear about kinds of *variables* that can appear in the concepts fetched/assembled via I-expressions. Other things being equal, one wants to posit as few kinds as possible. I see no way of avoiding appeal to various sortals, including sortals for 'eventish' things that can have other things as participants. Whatever one says about I-operations and adicities, we need distinctions among predicates; see e.g. Vendler (1959), Dowty (1979, 1991), Baker (1997), Svenonius (forthcoming). But one can try to minimize the number of logical types posited. And this quickly leads to questions about whether to accommodate plurality with one kind of variable or two.

One traditional approach treats all conceptual variables as singular, but sorted in a way that is usually interpreted in terms of a split-level domain: first-order variables range over whatever they do; and second-order variables range over 'plural entities'—sets, collections, or mereological sums—whose elements are things over which the first order variables range.[8] And we can certainly imagine a mind with I-concepts like COW(x_{-PL})and COW(x_{+PL}); where the fomer applies to cows and the latter applies to sets of cows. Or, in more explicitly Tarskian terms: COW(x_{-PL}) is satisfied by a sequence σ of domain entities iff the entity that σ assigns to the unplural variable is a (basic entity that is a) cow; and COW(x_{+PL}) is satisfied by σ iff the entity that σ assigns to the plural variable is a plural entity whose every element is a cow. From this perspective, a word like 'three' is used to fetch a nondistributive concept like THREE(x_{+PL}), which is satisfied by σ iff the entity that σ assigns to the plural variable is a plural entity with three elements. But at least if the focus is (p. 482) on I-concepts, which have whatever character they do, one need not think of each assignment of values to variables as a Tarskian sequence that assigns exactly one value to each variable.

Following Boolos (1998), theorists can allow for assignments that assign many values to a variable. Correlatively, we can imagine a mind with number-neutral concepts like COW($\mathbb{N}$), which applies to one or more things iff each of them is a cow; where 'things' exhibits grammatical agreement with 'one or more', with no suggestion of more than one. That is, an assignment **A** satisfies COW($\mathbb{N}$) iff the one or more things that **A** assigns to the number-neutral variable are such that each of them is a cow. From this perspective, 'three' is used to fetch a concept that applies to one or more things iff they are three (and hence more than one). Some things are three iff they correspond one-to-one with the points of a triangle, the words in the series 'one, two, three', etc. So given three cows, no one or two of them are three; but each is a cow, and any two of them are cows. So the concept ·[THREE($\mathbb{N}$), COW($\mathbb{N}$)] is as well-formed as ·[BROWN($\mathbb{N}$), COW($\mathbb{N}$)]. The former concept does not apply to any one or two cows; though likewise, the latter concept does not apply to any red or green cows. The concepts ONE($\mathbb{N}$)and ~ONE($\mathbb{N}$)—a.k.a. ~PLURAL($\mathbb{N}$)and PLURAL($\mathbb{N}$)—can combine with COW($\mathbb{N}$) to form the 'singular' concept ·[ONE($\mathbb{N}$), COW($\mathbb{N}$)] and the 'plural' concept ·[~ONE($\mathbb{N}$), COW($\mathbb{N}$)]; where the former applies to one or more things iff each is a cow and there is only one of them, while the latter applies to one or more things iff each is a cow and there are more than one of them.

For many purposes, we can adopt either view of I-concept variables: as essentially singular, always taking a single value relative to any assignment, but with variables of one sort ranging over sets of entities over which variables of the other sort range; or as number-neutral, ranging over whatever variables range over, but allowing that a concept can apply to some things without applying to any one of them. Following Schein (1993, 2002, forthcoming), I think the number-neutral approach is empirically superior; see Pietroski (2005a, 2006, 2010). But here, the more important point is that the sorted approach is not mandatory. And if we want to locate the sparest plausible assumptions about I-concepts, one might well start with the hypothesis that all I-concepts are number-neutral, allowing for specific concepts like ONE(x)/~ONE(x) that are not neutral. For if distinctively plural variables are required, there should be evidence of this.[9]

(p. 483) This point is especially important in the current setting, because Boolos ingeniously explored the resources available within *monadic* second-order logic. If these resources suffice for human I-language semantics, that is worth knowing. Assuming that the resources of first-order logic are inadequate, adopting a 'singularist' perspective on I-concept variables certainly invites—and it may require—the standard Fregean typology, given the limitations imposed by Tarskian sequences. And when asking *if* the typology is required, as opposed to convenient, we must not assume models of plural locutions that were designed to fit the typology. So in what follows, I will assume that appeal to number-neutral variables is legitimate, especially in a theory that already posits monadic reformatting as part of lexicalization.

21.2.4 Extending monadicity

It cannot be that all I-concepts are monadic. We can express relational thoughts. But this does not require a recursive combination operation, like saturation, that can take polyadic concepts as inputs. Conjunction can yield a simulacrum of polyadic thought given repeated—though not necessarily recursive—appeal to a severely restricted kind of dyadicity.

Earlier, I restricted '·' to combination of monadic concepts. But imagine a mind that allows for one exception: instances of **∃·** [θ(e, x), Φ(x)] are well-formed; where θ(E, X) is a dyadic concept whose second variable is *the* variable of the monadic concept Φ(x), and this variable is immediately *closed* to further conjunction, leaving a concept that applies to one or more things iff they bear the relation in question to one or more things to which Φ(x) applies. This permits concepts like **∃·** [AGENT (E, X), COW(X)], which applies to one or more events iff some cows were the agents of those events; cp. Carlson (1984), Schein (2002). Simplifying a little, we can say that some cows are the agents of some events iff each cow is the agent of an event, and each event has a cow as its agent; where each event has at most one agent. The more complex concept ·[KICK(E), **∃·** [PATIENT(e, x), ·[COW(x), PLURAL(x)]]] applies to one or more events iff: each is a kick; and one or more things are such that they are the patients of those events, and they are cows (i.e., each of them is a cow, and they are not one). Likewise, ·[**∃·** [AGENT(E, X), ·[RANCHER(X), FIVE(X)]], ·[BRAND(E), **∃·**·[PATIENT(e, x), ·[COW(X), FIFTY(X)]]] applies to one or more events iff in those events, five ranchers branded fifty cows.

Concepts of events are in no sense true or false. And *perhaps* concepts of type <e> and <**t**> will have to be introduced eventually, along with concepts of higher types. I discuss quantification in section 21.3.But Tarski (1933) provided a semantics for the first-order predicate calculus without appeal to truth values, and without treating closed sentences as instances of special type <**t**>, by effectively treating sentences as predicates: expressions satisfied by sequences of entities. So let's be clear that 'propositional concepts', which can be negated and conjoined, need not be concepts of truth/falsity.

(p. 484) Consider a pair of operators, ↑ and ↓, that create monadic concepts from monadic concepts; where for any one or more things, ↑Φ(x) applies to them iff Φ(x) applies to one or more things, and ↓Φ(x)applies to them iff Φ(x)applies to nothing.[10] More briefly, without fussing about number neutrality: for each entity, ↑Φ(x)applies to it iff Φ(x) applies to something; and ↓Φ(x)applies to it iff Φ(x) applies to nothing. One can think of ↑ and ↓ as polarizing operators that convert any monadic concept into a concept of everything or nothing, perhaps akin to EXIST(x)and ~EXIST(x). For example, given any entity, ↑COW(x) applies to it iff COW(x) applies to something; so ↑COW(X) applies to you, and likewise to me, iff there is a cow. By contrast, ↓COW(x) applies to you (and me) iff nothing is a cow. And for each thing, either ↑COW(x) or ↓COW(x) applies to it—since it is either such that there is a cow, or such that there is no cow. This mode of composition clearly differs from the always *restricting* operation of conjunction. But correlatively, nothing is such that both ↑COW(x) and ↓COW(x) apply to it. Hence, nothing is such that ·[↑COW(x), ↓COW(x)] applies to it.

Given a suitable metalanguage, we can say: $\uparrow\Phi(x) \equiv \exists_Y [\Phi(Y)]$, and $\downarrow\Phi(x) \equiv {\sim}\exists_Y [\Phi(Y)]$. But the idea is not that '$\uparrow$' and '$\downarrow$' are abbreviations for existential closure and its negation. For example, '$\uparrow$BETWEEN(X, Y, Z)' is gibberish, as is '$\downarrow$AGENT(E, X)'. The idea is rather that certain I-expressions, perhaps associated with tense and/or negation, invoke 'closure' operations that convert a monadic concept (say, of events) into a concept of all or none. So let's say that any concept of the form $\uparrow\Phi(x)$ or $\downarrow\Phi(x)$ is a *T-concept*, with 'T' connoting Tarski, totality, and truthy. Note that for any concept $\Phi(x)$ and any entity e, $\uparrow\uparrow\Phi(x)$ applies to e iff $\downarrow\downarrow\Phi(x)$ does, since each of these concepts applies to e iff $\uparrow\Phi(x)$ does—i.e., iff $\Phi(x)$ applies to something. Likewise, $\uparrow\downarrow\Phi(x)$ applies to e iff $\downarrow\uparrow\Phi(x)$ does, since each of these concepts applies to e iff $\downarrow\Phi(x)$ does—i.e., iff $\Phi(x)$ applies to nothing. And while $\uparrow\bullet[\Phi(x), \mathbf{\Psi}(x)]$ applies to e iff something falls under the conjunctive concept $\bullet[\Phi(x), \mathbf{\Psi}(x)]$, which applies to e iff e falls under both conjuncts, $[\uparrow\Phi(x), \uparrow\mathbf{\Psi}(x)]$ applies to e iff (e is such that) something falls under $\Phi(x)$ *and* something falls under $\mathbf{\Psi}(x)$. Thus, $\uparrow\bullet$[BROWN(x), COW(x)] is a more restrictive concept than $\bullet[\uparrow$BROWN(x), $\uparrow$COW(x)], much as the more familiar $\exists_Y$[BROWN(Y) & COW(Y)] implies $\exists_Y$[BROWN(Y)] & $\exists_Y$[COW(Y)] but not conversely. Correlatively, $\downarrow\bullet$[BROWN(x), COW(x)] applies to e iff nothing is both brown and a cow, while $\bullet[\downarrow$BROWN(x), $\downarrow$COW(x)] applies to e iff (e is such that) nothing is brown *and* nothing is a cow. So $\bullet[\downarrow$BROWN(x), $\downarrow$COW(x)] is a more restrictive concept than $\downarrow\bullet$[BROWN(x), COW(x)], and $\downarrow\bullet$COW(x) is more restrictive than $\downarrow\bullet$ [BROWN(x), COW(x)].

The basic idea is medieval: the default direction of inference is conjunction *reduction*—e.g. from $\bullet$[BROWN(x), COW(x)] to COW(x); but in the presence of a (p. 485) negation-like operator, this default is reversed.[11] And note that when the concepts conjoined are both T-concepts, which apply to all or none, 'closing up' has no effect. If **P** and **Q** are T-concepts, and so each is of the form $\uparrow\Phi(x)$ or $\downarrow\Phi(x)$, then $\uparrow\bullet[\mathbf{P}, \mathbf{Q}]$ is logically equivalent to $\bullet[\mathbf{P}, \mathbf{Q}]$: $\uparrow\bullet[\mathbf{P}, \mathbf{Q}]$ applies to e iff something/everything falls under both **P** and **Q**; $\bullet[\mathbf{P}, \mathbf{Q}]$ applies to e iff e/everything falls under both **P** and **Q**. By contrast, $\downarrow\bullet[\downarrow\mathbf{P}, \downarrow\mathbf{Q}]$ applies to e iff: nothing falls under both $\downarrow\mathbf{P}$ and $\downarrow\mathbf{Q}$; i.e., nothing is such that both **P** and **Q** are empty; i.e., something falls under **P** *or* something falls under **Q**. So propositional disjunction can be characterized, *à la* de Morgan, given T-concepts.

More generally, T-concepts provide resources for accommodating the meanings of sentential I-expressions without supposing that they exhibit a special semantic type <**t**>. So we should pause before assuming that HFL generates expressions of this type, as opposed to expressions that can be used to construct T-concepts, which can bear an intimate relation to existential *thoughts* of type <**t**>.While 'post-linguistic' cognition may traffic in complete thoughts, in which each monadic concept is saturated or quantificationally bound, HFL may interface with such cognition via formally monadic T-concepts. The notion of 'sentence' has always had an unstable place in grammatical theory. And especially within MP, one might want to preserve the old idea that each I-expression is (labeled as) an instance of some grammatical type exhibited by some atomic expression. One can stipulate that sentences are projections of some functional category. But no such stipulation seems especially good. So perhaps we should drop the idea that HFL generates expressions of type <**t**>, and adopt a more Tarskian type-free approach to human I-language semantics.[12]

21.2.5 Abstracting

At this point, our imagined mind can form many systematically related concepts. It can also convert an ordinary monadic concept—one that can apply to some but not all things—into a T-concept that must apply to all or none. But it cannot yet do the converse. And this is arguably the most interesting respect in which human thought is recursive. Given $\bullet[\mathbf{P}, \mathbf{Q}]$, the capacity to form $\bullet[\mathbf{P}, \bullet[\mathbf{P}, \mathbf{Q}]]$ is not that impressive. And while I cannot discuss the semantics of complementizer phrases (see Pietroski 2005a for a conjunctivist analysis), the kind of recursion exhibited by 'Aggie thinks that Bessie said that Aggie saw Bessie' is in many ways less interesting than the kind exhibited by 'who saw the cow that Aggie saw'. Embedding one sentence in another is a good trick. Using a sentence to create a concept that can apply to some but not all things is a great trick. Clearly, this requires more than mere conjunction. (p. 486) But as Tarski (1933) showed us, the requisite machinery is relatively simple, even if it initially seems complex. It involves a distinctive kind of composition, though not one that depends on an operation of saturation. So if appeal to this kind of composition is unavoidable, as I suspect, our question is whether we should appeal to it and saturation and conjunction.

Let's assume that our imagined mind can deploy *indices*, like '1' and '2', that can be used in two ways: deictically, as devices for temporarily tracking salient things perceived (cf. Pylylshyn 2007); or anaphorically, as devices for temporarily tracking things independently described. Some of these indices may be singular, but suppose that some are number-neutral. Let's also suppose that this mind also has some concepts like FIRST(X) and SECOND(X), which apply to whatever the corresponding indices are tracking. Such concepts are, in an obvious sense, context sensitive in a way that concepts like COW(X) are not. In an equally obvious sense, COW(X) applies to different things at different times, as cows come and go. But indices are, so to speak, designed as temporary tracking devices with no independent content of their own. So as an idealization, we can say that COW(X) simply applies to cows, without relativization to anything else; although CALF(X) is already more complicated. By contrast, FIRST(X) doesn't apply to anything *tout court*: FIRST(X) is satisfied by an assignment **A** iff the one or more things that **A** assigns to the conceptual variable are whatever **A** assigns to the first index; COW(X) is satisfied by **A** iff the one or more things that **A** assigns to the conceptual variable are cows.[13]

This allows for concepts like **∃·**[INTERNAL(E, X), FIRST(X)], which is satisfied by **A** iff: whatever things **A** assigns to the first index, those one or more things are the internal participants of whatever **A** assigns to the free conceptual variable. Likewise, the T-concept ↑·[**∃·** [EXTERNAL(E, X), SECOND(X)], ·[SAW(E), **∃·**[INTERNAL(E, X), FIRST(X)]]] is satisfied by **A** iff: whatever **A** assigns to the second index saw whatever **A** assigns to the first index; or more long-windedly, (all things are such that) there were one or more events of seeing whose external participants are whatever **A** assigns to the second index and whose internal participants are whatever **A** assigns to the first index. But let's suppress the eventish and conjunctive substructure, abbreviating this T-concept as follows: **2SAW1**. For regardless of whether T-concepts are formed by conjunction or saturation, T-concepts with constituents like FIRST(X) are concepts ripe for abstraction.

Given any index *i* and T-concept **P**, which can be evaluated relative to any assignment **A**, let **TARSKI**{*i*, **P**} be the semantic concept indicated below;

∃A*:A*$\approx_1$ **A**{ASSIGNS(**A***, X, 1) & SATISFIES(**A*, P**)}

(p. 487) where ASSIGNS(**A***, X, 1) applies to one or more things iff they are the things that **A*** assigns to the first index, and ‘**A*** $\approx_1$**A**’ means that **A*** differs from **A** at most with regard to what it assigns to the first index. To be sure, any natural concept of satisfaction is likely to differ from Tarski's. But the idealization is that a suitably equipped mind can use a T-concept with FIRST(X) as a constituent to form a concept that applies to one or more things (relative to a certain assignment of values to indices) iff making them the values of the first index (and holding everything else constant) satisfies the T-concept. Likewise, given a T-concept with SECOND(X) as a constituent, one can form a concept that applies to one or more things (relative to a certain assignment of values to indices) iff making them the values of the second index (and holding everything else constant) satisfies the T-concept. One can think of this as number-neutral lambda abstraction. But a Church-style construal of ‘λX.Φ(X)’ presupposes sequence variants and a Tarski-style construal of ‘Φ(X)’. And the goal here is to be explicit about theoretical commitments.

In the context of our example, relative to any assignment **A**:

TARSKI {1, **2saw1**} = ∃**A***:**A*** $\approx_1$**A**{ASSIGNS(**A***, X, 1) & SATISFIES(**A***, **2saw1**)}

and this concept applies to one or more things iff they were seen by whatever **A** assigns to ‘2’, since if **A*** $\approx_1$**A**, then both assignments assign the same one or more things to ‘2’; similarly,

TARSKI {2, **2SAW1**} = ∃**A***:**A*** $\approx_2$**A**{ASSIGNS(**A***, X, 2) & SATISFIES(**A***, **2SAW1**)}

and this concept applies to one or more things iff they saw whatever A assigns to ‘1’, since if **A*** $\approx_2$**A**, then both assignments assign the same one or more things to ‘1’. I readily grant that this kind of concept construction—from **2SAW1** to **TARSKI**{1, **2SAW1**}or **TARSKI**{2, **2SAW1**}—is more sophisticated than conjunction. Indeed, Tarskian composition violates some conceivable compositionality constraints respected by conjunction. Relative to **A**, the T-concept **2SAW1** may apply to nothing, while **TARSKI**{1, **2SAW1**} applies to many things; see Salmon (2006). Suppose that whatever **A** assigns to ‘2’, it/they saw many things, but *not* whatever **A** assigns to ‘1’. Then relative to a single assignment: **2SAW1** is false of each thing, and in that sense false, yet **TARSKI**{1, **2SAW1**} is true of many things; hence, ↑**TARSKI**{1, **2SAW1**} is true of each thing.

In this sense, **2SAW1** can be false while ↑**TARSKI**{1, **2SAW1**} is true. And since whatever **A** assigns to ‘2’ might have seen nothing, **2saw1** and ↑**TARSKI**{1, **2SAW1**} can both be false. Like it or not, this kind of ‘non-truth-functional’ composition is available to a mind that is equipped to perform the Tarski trick. And my claim is *not* that this kind of abstraction can be reduced to anything else. On the contrary, I see no way to avoid positing a capacity for such composition in the construction of I-concepts. But I also see no way to avoid appeals to a more mundane operation of conjunction, and at least a few thematic concepts; see Baker (1997). In my view, the question is whether we also need Fregean typology and an operation of saturation.

(p. 488) 21.3 Back to SEMs

Having described a possible mind with the capacities needed to construct a wide range of potential I-concepts, let me turn to the task of showing how I-expressions might be systematically described as instructions to build such concepts. In section 21.2.1, I imagined a language whose syntax is exhausted by a unit-forming operation, UNIFY. Let's now suppose that human I-languages make it possible to unify/concatenate expressions and *label* them so that a complex operation MERGE can be defined as follows (see Hornstein 2009): MERGE(α, β) = LABEL{UNIFY (α, β)} = LABEL{[α, β]}; where the new operation (deterministically) selects one of the two expressions just unified and appends a copy to the unified expression. The idea is that if α has the right properties to be the 'head' of [α β], then LABEL{[α, β]} = $[\alpha, \beta]_\alpha$. In which case, MERGE(α, β) = $[\alpha, \beta]_\alpha$, as desired. But what kind of instruction is $[\alpha, \beta]_\alpha$?

21.3.1 Conjunction and conversion

For any instructions **I** and **I***, let +[**I**, **I***] be a 'macro' instruction to execute the two subinstructions and conjoin the results, thereby creating a concept of the form ·[Φ(x), Ψ(x)]. Then examples like [brown_A cow_N]$_\text{N}$ and [cow_N [that Aggie saw]$_\text{C}$]$_\text{N}$, ignoring structure within the relative clause, conform to a very simple view: for any expressions α and β SEM($[\alpha, \beta]_\alpha$) = +[SEM(α), SEM(β)]. One might well endorse the medieval suspicion that modulo special expressions like negation, the general trend is for $[\alpha, \beta]_\alpha$ to be more restrictive than its constituents. This trend would be surprising if concatenation signifies an operation (like saturation) that is indifferent to whether or not a complex expression carries more information than its parts. And even many apparent counterexamples suggest complications of the trend, as opposed to wholesale departures. A big ant is an ant that meets a further condition; and even a fake diamond is a fake of a certain sort. Likewise, a chase of Bessie is a chase. But phrases like [chase_V Bessie_N]$_\text{V}$ suggest that the phrasal label—or more precisely, a mismatch between the phrasal label and the *other* constituent label—can play a significant role.[14]

At least for cases of combining constituents that correspond to concepts of different sorts, like a concept of events and a concept of an animal, a natural thought is that the phrasal label invokes an adapter that combines with one concept to form a concept of the same sort as the other. In terms of [chase_V Bessie_N]$_\text{V}$, perhaps the phrasal label V is an instruction to use the result of executing SEM(Bessie_N)in creating a concept that is sure to be conjoinable with the concept obtained by executing SEM(Bessie_N). There are various ways of encoding this idea. But consider the (p. 489) following principle of composition: SEM($[\alpha, \beta]_\alpha$) = +[SEM(α), ADAPT[SEM(β), α]]; where '**ADAPT'** stands for a macro instruction to execute the subinstruction and use the resulting monadic concept to form another, via some operation determined by the label α and the available conversion operations. Obviously, the work lies with specifying the specific instances of '**ADAPT**' in an empirically adequate and motivated way. But for [chasev Bessie_N]$_\text{V}$, in which BessieN is effectively classified as the internal argument of chasev, we already have what is needed. Suppose that classifiying an argument as internal is an instruction to use the argument to construct a concept Φ(x), and then a concept of things whose 'internal participants' fall under the concept. More explicitly, one can adopt the hypothesis below.

ADAPT [SEM(β), V] = **INTERNALIZE**:SEM(β)

For any expression β, let **INTERNALIZE**:SEM(β) be the macro instruction to execute SEM(β) and use the resulting concept Φ(x) to create a concept of the following form: ∃·[INTERNAL(e, x), Φ(x)]. The idea is that INTERNAL(E, X) is a 'thin' but formally thematic concept that groups together PATIENT(E, X), perhaps THEME(E, X), and any other 'thick' thematic concept—with independent conceptual content—that can be introduced by classifying an expression as the internal argument of a predicate. One way or another, the lexical item 'chase' can indicate that any internal participants of chases are *patients*, making it possible to replace ·[CHASE(E), ∃-[INTERNAL(E, X), Φ(X)]] with ·[CHASE(E), **∃·**[PATIENT(E, X), Φ(X)]].[15] In any case, [chase_V Bessie_N]$_\text{V}$ can direct construction of ·[CHASE(E), **∃·**[INTERNAL(E, X), BESSIE(X)]].

This kind of 'thematic conversion,' invoked to preserve a fundamentally conjunctivist conception of semantic composition, is formally similar to a more familiar kind. If the sole combination operation is saturation, then faced with examples like [brown_A cow_N]$_N$, one might adopt some version of the following view.

SEM([α, β]$_\alpha$) = **SATURATE**[SEM($_\alpha$), **ADAPT**[SEM(β), α]]

ADAPT[SEM([…]$_A$), $_N$] =**LIFT**:SEM([…]$_A$)

SEM([brown_A cow_N]$_N$) = **SATURATE**[fetch@cow_N, **LIFT**:fetch@brown_A] →·[BROWN(x), X(x)]{COW(x)} = ·[BROWN(x), ·COW(x)]

The idea here is that classifying brown_A as the 'inferior' constituent of [brown_A cow_N]$_N$ is an instruction to use the concept fetched (or constructed) via this consitutent into an analytically related concept of the higher type << e, **t**>, <e, **t**>>, which can be saturated by the 'head' concept fetched (or constructed) via the noun. But at least as a theory of I-languages, this assumes the availability of higher types, as well as an operation of conjunction.

(p. 490) From an E-language perspective, one can be less committal and say merely that words indicating two *functions* of type <e, **t**>—λx. *T iff x is a cow* and λx. *T iff x is brown*, where '*T*' stands for a certain truth value, and '*x*' is a singular variable—are combined to form a phrase that indicates a third function of the same type, λx. *T iff x is both a cow and brown*. From this extensional perspective, corresponding roughly to Marr's (1982) computational Level One, saturating-and-lifting is equivalent to conjoining. But from an I-language/procedural perspective, corresponding more closely to Marr's algorithmic Level Two, these are distinct operations: the former presupposes the latter as a sub-part; and while conjunction might be described as a very special case of saturation, restricted to concepts of one type, it might also be described as a basic operation. Moreover, since nouns can be modified with relative clauses, the requisite lifting operation would have to be available as a *recursive* operation. So especially if many adverbial modifiers have to be diagnosed in terms of monadic concepts of events, it seems that the requisite lifting operations will encode a conjunctivist principle of semantic composition *and more*. In which case, theorists may as well posit more than one basic composition operation; cp. Higginbotham (1985), Larson and Segal (1995), Heim and Kratzer (1998). By contrast, appeals to thematic relations seem unavoidable, if only to formulate the constraints on how they can project to grammatical relations in human I-languages. This invites the minimalist project of making do with conjunction by assigning thematic significance to certain cases of labeling.

Assigning such significance leaves room for the possibility that some I-expressions are unlabeled instances of the form [α, β]; Chametsky (1996) on adjunction. If [brown cow_N] or [cow_N [that I saw]] is an example, with the unlabeled constituent not being a candidate for the phrasal head, perhaps SEM([α, β]) = +[SEM(α), SEM(β)]; see Hornstein and Pietroski (2009) for discussion. One can add that SEM([α, β]$_\alpha$) = +[SEM(α), **ADAPT**[SEM(β), α]]. Alternatively, one can say that all phrases are labeled, and say that some phrasal labels call for a null' adapter. Perhaps **ADAPT**[SEM([…]$_C$), $_N$]= **RETURN**:SEM([…]$_C$). In which case, SEM([cow_N [that I saw]$_C$]$_N$) = +[SEM(cow_N), SEM([that I saw]$_C$)]. If [brown_A cow_N]$_N$ really means something like 'is a cow that is brown for a cow', there may be few if any cases of pure adjunction apart from relative clauses. But that would still leave endlessly many cases.

A related point is that words like cowN and chaseV may already be combinations of lexical roots with functional items that serve as labels. If cow_N =[✓ cow $_N$], then perhaps SEM(cow_N) = +[SEM(✓ cow), SEM($_N$)] = +[fetch@ ✓ cow, fetch@$_N$]; where $_N$ is a device for fetching a functional monadic concept like INDEXABLE, while $_V$ is a device for fetching a concept like TENSABLE, thus allowing for a distinction between SEM(chase_V) and SEM (chase_N). I cannot pursue these issues here, but simply raise them to note the kinds of resources still available *without* appeal to Fregean typology; see Hornstein and Pietroski (2009), drawing on Marantz (1984), Halle and Marantz (1993), Baker (2003), and Borer (2005).

(p. 491) Let's return to the idea that [chase_V Bessie_N]$_V$ is an instruction to build a concept like ·[CHASE(E), ∃-[INTERNAL(E, X), BESSIE(X)]], with the grammatical object of the verb used to fetch or construct a concept that can restrict the participant variable of a 'thin' thematic concept. There is an obvious analog for subjects, as in [Aggie_N [chase_V Bessie_N]$_V$]$_V$. Suppose we have a formally thematic concept, EXTERNAL (E, X), that groups together AGENT(E, X), EXPERIENCER(E, X), and any other 'thick' thematic concepts—with independent conceptual content—that can be introduced by classifying an expression as the external argument of a predicate. One way or another, 'chase' can indicate that any external participants of chases are agents, making it possible to replace ·[∃·[EXTERNAL(E, X), Φ(X)], ·[CHASE(E, X), …] with ·[∃·[AGENT(E, X), Φ(X)], ·[CHASE(E, X), …]].

The requisite conversion operation is easily defined. For any expression β, let **EXTERNALIZE**:SEM(β) be the macro instruction to execute SEM(β)and use the resulting concept Φ(x) to create a concept of the following form: **∃·**[EXTERNAL(E, X), Φ(X)]. But we can't say both of the following, at least not without qualification.

ADAPT [SEM (β), V] = INTERNALIZE: SEM (β)

ADAPT [SEM (β), V] = EXTERNALIZE: SEM (β)

Correspondingly, we can't say that SEM([Aggie$_N$ [chase$_V$ Bessie$_N$]$_V$]$_V$) = + [**ADAPT**[SEM(Aggie$_N$), V], + [SEM(chase$_V$), **ADAPT**[SEM(Bessie$_N$), V]. This doesn't make it clear *which* conversion operation, **EXTERNALIZE** or **INTERNALIZE**, goes with *which* grammatical argument. But there are three obvious possibilities to consider.

Perhaps labels should be viewed, not as atomic elements, but as standins for the entire 'head expression'; cf. Chomsky (1995b). If [Aggie$_N$ [chase$_V$ Bessie$_N$]$_V$]$_V$ = [Aggie$_N$ [chase$_V$ Bessie$_N$]CHASE] CHASE_BESSIE, which can be abbreviated as [Aggie$_N$ [chase$_V$ BESSIE$_N$]$_V$]$_{V(N)}$, then SEM([Aggie$_N$ [chase$_V$ Bessie$_N$]$_V$]$_V$) = + [**ADAPT**[SEM(Aggie$_N$), V(N)], +[SEM(chase$_V$), **ADAPT**[SEM(Bessie$_N$), V]. This effectively classifies external arguments as such, allowing for the obvious rules.

ADAPT [SEM (β), V] = **INTERNALIZE**:SEM(β)

ADAPT [SEM (β), V(N)] = **EXTERNALIZE**:SEM(β)

Or perhaps the syntax is nuanced in a different way—independently suggested by many authors, including Chomsky (1995b) and Kratzer (1996)—with external arguments as arguments of an independent verbal element, as in [Aggie$_N$ [*v* [chase$_V$ Bessie$_N$]$_V$]$_V$]$_V$. For these purposes, 'V(N)' can be replaced with '*V*'.

ADAPT[SEM(β), *V*]= **EXTERNALIZE**:SEM(β)

There are real empirical issues here. For example, is there a covert internal argument in 'Aggie counted'? But for better or worse, the conjunctivist framework does not force a particular stand on these issues.

(p. 492) A third option is that the conversion operation is not determined by head label alone. Perhaps cyclicity plays a role here: within a given 'phase' of instruction execution, the first occurrence of label 'V' triggers the first operation (**INTERNALIZE**), and the second occurrence of 'V' triggers the second operation (**EXTERNALIZE**). If there are at most two grammatical arguments per phase/cycle/whatever, one might imagine a binary 'switch' that gets 'reset' to its initial state at the start of each cycle; cf. Boeckx (2008a). If some such thought is correct, perhaps we can make do with formally thematic concepts that are super-thin: ON (E, X) and ~ON(E, X), instead of EXTERNAL(E, X) and INTERNAL(E, X).

In any case, an expression like [Aggie$_N$ [chase$_V$ Bessie$_N$]$_V$]$_V$ can be an instruction to build a concept like ·[**∃·**[EXTERNAL(E, X), AGGIE(X)], ·[CHASE(E), ∃·[INTERNAL (E, X), BESSIE(X)]]]. Adding adverbs and prepositional phrases is not without difficulties. But the leading idea, unsurprisingly, is that I-expressions like 'yesterday'—'on Tuesday', 'with a stick', etc.—are instructions to fetch/construct additional conjuncts. Prepositions, as functional elements, can be viewed as instructions to fetch adapters and convert concepts like STICK(X) into concepts like ∃·[INSTRUMENT (E, X), STICK(X)]. This provides a way of describing the massive polysemy of prepositions: there need not be a single 'thematizing' operation that 'with' invokes. And prepositional phrases may well have internal conjunctivist structure. I cannot pursue this rich topic here; but see Svenonius (forthcoming).

At this point, let me offer an explicit treatment of sentential expressions and relative clauses, before turning to quantificational constructions, which pose the most obvious challenge for a conjunctivist semantics. For simplicity, let's ignore tense. Eventish treatments are familiar; see Higginbotham (1985), Parsons (1990). It is also worth remembering that 'Aggie chase Bessie' can appear as an internal argument of 'see'. And the current proposal lets us treat both 'see trees' and 'see Aggie chase Bessie' as instructions to build concepts of seeings whose internal participants are one or more things that meet a certain condition: being trees, or being chases of Bessie by Aggie; cf. Higginbotham (1983). But at some point, a clause is treated as sentential. And if the concept built via [Aggie$_N$ [chase$_V$ Bessie$_N$]$_V$]$_V$ is prefixed with ↑, the result

↑·[∃·[EXTERNAL(E, X), AGGIE(X)], [CHASE(E), ∃·[INTERNAL(E, X), BESSIE(X)]]] is a T-concept that applies to all or none, depending on whether or not Aggie chased Bessie. Let's abbreviate this concept as **AchaseB**, recalling the discussion at the end of section 21.2.

There are many ways of encoding the idea that a tensed version of ‘Aggie chase Bessie’ can be an instruction to create a T-concept, depending on what one thinks about sentences and sentential negation. For instead of thinking about sentences as a special kind of grammatical category, headed by a special functional item, one might think of sentences as results *in thought* of ‘spelling out’ tensed instructions; cf. Uriagereka (1999). If a sentence corresponds to a cycle (or phase) of interpretation, (p. 493) the relevant I-expression may direct construction of a monadic concept that can be true of some things but not others. But this concept may be converted to a T-concept at the Conceptual-Intentional interface, given the demands of the judgment systems of external to HFL. And instead of thinking about overt negation in human I-languages as a modifier of sentences, as in familiar formal languages, one can hypothesize two ‘modes of closure’ at the interface: in the absence of an instruction to the contrary, use ↑; but given overt negation, use ↓.That said, one can also think of sentential classification as an instruction to adapt a subsentential instruction by invoking the positive T-operator.

SEM([…[Bessie$_N$ [chase$_V$ Aggie$_N$]$_V$]$_V$…]s) =

UP:SEM(…[Aggie$_N$ [chase$_V$ Bessie$_N$]$_V$]$_V$]…)

For present purposes, let's remain neutral about the details, and just say that executing the sentential instruction ‘Bessie did chase Aggie’ results in construction of the concept **AchaseB**. And one can still treat ‘Bessie did not chase Aggie’ as an instruction to construct ↓ **AchaseB**, which is a T-concept of the form ↓↑·[…]—or as an instruction to construct a logically equivalent concept of the form ↓·[…], which applies to none or all, depending on whether or not ·[…] applies to one or more things.

The more important point here is that if pronouns and traces of movement are instructions to fetch concepts like FIRST(x) and SECOND(x), as suggested at the end of section 21.2, then relative clauses are easily accommodated. Recall that relative to any assignment **A**, FIRST(x) applies to whatever A assigns to the first index; likewise for SECOND(x). And suppose that ‘which she chased’ is classified as the result of combining a displaced index-bearing expression with the very sentential expression from which it was displaced.

[which$_2$ [… [she$_{N1}$ [chase$_V$ ~~which~~$_2$]$_V$]$_V$…]s]$_2$

The embedded sentential expression can be treated as an instruction to construct the T-concept indicated below, which can be abbreviated as **1CHASE2**.

↑·[∃·[EXTERNAL(E, X), FIRST(X)], ·[CHASE(E), ∃·[INTERNAL(E, X), SECOND(X)]]]

Though instead of ignoring gender for simplicity, one can add that ‘she’ imposes a further constraint on external participants. The displaced *wh*-expression can also be treated as instruction to fetch a concept that applies a further restriction; ‘who’ plausibly adds a restriction to people. And crucially, one can treat the double-occurrence of the index as an instruction to invoke the Tarski trick, focusing on that index; cf. Heim and Kratzer (1998).

SEM([which$_2$ […]$_S$]$_2$) = + [SEM(which$_2$), **ADAPT**[SEM([…]$_S$), $_2$]]

= +[SEM(fetch@which), **Tarski**{2, SEM([…]$_S$)}]

Recall that for any index *i* and T-concept **P**, **Tarski** {*i*, **P**} is the semantic concept below.

(p. 494) ∃**A***:**A*** $\approx_1$**A**{ASSIGNS(**A***, X, 1)& SATISFIES(**A***, **P**)}

And the idea is that (the SEM of) ‘which she chased’ directs construction of a concept like ·[ENTITY(X), **Tarski**{2, **1CHASE2**}], which applies to one or more things iff they were chased by whatever A assigns to the first index. Similarly, ‘which chased her’

[which$_1$ [… [~~which~~$_1$ [chase$_V$ her$_{N2}$]$_V$]$_V$…]$_S$]$_1$

can be analyzed as an instruction whose execution leads to construction of a concept that applies to one or more things iff they chased whatever A assigns to the second index.[16]

21.3.2 Quantification

We're finally in a position to describe the meanings of quantificational constructions like 'Every cow arrived' and 'She chased every cow', by recasting the proposal in Pietroski (2005b, 2006) explicitly in terms of instructions to build concepts, without appeal to truth values.

(p. 495) The central idea is simple: a determiner like 'every' fetches a number-neutral concept of ordered pairs; where the ordered pair <X, y> can be identified with {X, {X, Y}}, with X as its 'external participant,' and y as its 'internal participant.' More specifically, let's say that EVERY (O) applies to some ordered pairs iff every one of their internal participants is one of their external participants; or put another way, (all of) their internals are among their externals. Likewise, MOST/THREE/SOME/NO(O) applies to some ordered pairs iff most/three/some/none of their internals are among their externals. And let's say that for any concept Φ(X), the concept MAX-Φ(X) applies to some things iff they are *the* things to which Φ(X) applies: MAX-Φ(X) ≡ ∀Y:Φ(Y)[AMONG(Y, X)].[17] Then ·[EVERY(O), ∃·[INTERNAL (O, X), MAX-COW(X)]] applies to some ordered pairs iff their internals are the cows, and each of their internals is one of their externals.

We can say that from a semantic perspective, being an argument of a determiner differs slightly from being an argument of a verb, in that the former imposes a maximization condition.

D-INTERNALIZE:Φ(X) =**INTERNALIZE**:MAX-Φ(X)

→ ∃·[INTERNAL(O, X), MAX- Φ(X)]

D-EXTERNALIZE: Φ(X) = **EXTERNALIZE**:MAX- Φ(X)

→ ∃·[EXTERNAL(O, X), MAX- Φ(X)]

And given a concept Ψ (X) that applies to one or more things iff they arrived, the concept ·[·[EVERY(O), ∃·[INTERNAL(O, X), MAX-COW(X)]], ∃·[EXTERNAL(O, X), MAX-Ψ(X)]] applies to one or more ordered pairs iff: their internals are the cows, their internals are among their externals, and their externals are the things that arrived. This concept applies to one or more things iff every cow arrived, assuming that ordered pairs exist if their participants/elements do. Likewise, ·[·[EVERY(O), ∃·[INTERNAL(O, X), MAX-COW(X)]], ∃·[EXTERNAL(O, X), MAX-·[COW(X), Ψ(X)]] applies to one or more things iff every cow is a cow that arrived.

I mention the possibility of restricting the externals to cows that arrived because this may be relevant to the *conservativity* of determiners—see Barwise and Cooper (1981), Higginbotham and May (1981)—and the ways in which external arguments of determiners *differ* from relative clauses; see Pietroski (2005a) for further discussion. It is easy to construct a concept of those that arrived, given a suitable T-concept and quantification over assignment variants.

(p. 496) MAX-∃$\mathbf{A}^*$:$\mathbf{A}^* \approx_1$ $\mathbf{A}${Assigns($\mathbf{A}^*$, X, 1) & Satisfies{$\mathbf{A}^*$, ↑ ·[ARRIVED(E), ∃·[INTERNAL(E, X), FIRST(X)]]}}

This concept just is MAX-**TARSKI**{1, ↑ ·[ARRIVED(E), ∃·[INTERNAL(E, X), FIRST(X)]]}. But given restricted quantifiers, we can severely restrict the appeal to assignment variants. Let's say that for any assignments **A** and $\mathbf{A}^*$, and any index *i*, $\mathbf{A}^* \subseteq_i \mathbf{A}$ iff: $\mathbf{A}^*$ differs from **A** at most in that $\mathbf{A}^*$ does not assign to *i* everything that **A** assigns to *i*; whatever **A** assigns to *i*, $\mathbf{A}^*$ assigns *one or more but perhaps not all* of those things to *i*. Given an assignment that assigns (all and only) the cows to the first index, the concept indicated below is a concept of those cows that arrived.

MAX-∃$\mathbf{A}^*$:$\mathbf{A}^* \subseteq_1$ $\mathbf{A}${Assigns($\mathbf{A}^*$, X, 1) & Satisfies{$\mathbf{A}^*$, ↑ ·[ARRIVED(E), ∃·[INTERNAL(E, X), FIRST(X)]]}}

And we can define **REDUCED TARSKI** {*i*, **P**} as follows.

∃$\mathbf{A}^*$:$\mathbf{A}^* \subseteq_i$ $\mathbf{A}${Assigns($\mathbf{A}^*$, X, *i*) & SATISFIES($\mathbf{A}^*$, **P**)}

Let me conclude by showing how the constituents of a quantificational expression can be instructions to build the requisite monadic concepts. As is standard within MP, I assume some version of the syntax shown below for 'She chased every cow'.

[[every$_{D2}$ cow$_N$]$_{D2}$ [...[she$_{D1}$ [chase$_V$ $\cancel{[\text{every}_{D2}\ \text{cow}_N]}_{D2}$]$_V$]$_V$ ...S]$_{D2}$]

For whatever reason—perhaps because 'every' needs an external argument—a copy of the indexed quantifier combines with the basic sentential expression, which then becomes the external argument of the quantifier. If the original/lower copy is interpreted as an instruction to fetch the concept SECOND(X), perhaps because that is the only coherent interpretation available, then the embedded sentential expression is an instruction to construct a T-concept like **1chase2**. But the whole I-expression, headed by every$_{D2}$, is the following instruction:

+ [SEM([every$_{D2}$ cow$_N$]$_{D2}$), **ADAPT**{SEM([… [she$_{D1}$ [chase$_V$ ~~[every$_{D2}$ cow$_N$]~~$_{D2}$]$_V$]$_V$ …S], D$_2$}.

This is an instruction to conjoin the concepts obtained by executing two sub-instructions: + [SEM(every$_{D2}$), **ADAPT**{SEM(COW$_N$), D$_2$}and **ADAPT**{**1CHASE2**, D$_2$}. The first subinstruction calls for conjunction of concepts obtained by (a) executing the indexed determiner instruction and (b) adapting a concept fetched with cow$_N$, in the way specified by classifying a noun as the internal argument of an indexed determiner. The second subinstruction calls for adapting **1CHASE2**, in the way specified by marking a sentential expression as the external argument of an indexed determiner. So one obvious hypothesis is given below.

SEM(every$_{D2}$) = fetch@every$_D$ → EVERY(O)

ADAPT{SEM(…$_N$), D$_2$} = **D-INTERNALIZE**:SEM(… $_N$)

ADAPT{SEM(…$_S$), D$_2$} = **D-EXTERNALIZE:TARSKI**{2,SEM(…s)}

(p. 497) This hypothesis has the desired consequences, assuming that every$_D$ fetches EVERY(O).

SEM([every$_{D2}$ cow$_N$]$_{D2}$) →·[EVERY(O), ∃·[INTERNAL(O, X), MAX-COW(X)]] **ADAPT**{SEM([… [she$_{D1}$ [chase$_V$ ~~[every$_{D2}$ cow$_N$]~~$_{D2}$]$_V$]$_V$ …S], D$_2$} → ∃·[EXTERNAL(O, X), MAX-**TARSKI**{2, **1CHASE2**}(X)]]

Conjoining the resulting concepts yields a concept of ordered pairs that meet three conditions: their internals are among their externals; their internals are the cows; and their externals are those things chased by whatever is assigned to the first index. And there are one or more such ordered pairs iff whatever is assigned to the first index chased every cow. So the corresponding T-concept can be the external argument of another determiner.

↑·[·[EVERY(O), ∃·[INTERNAL(O, X), MAX-COW(X)]],

∃·[INTERNAL(O, X), MAX-**TARSKI**{2, **1CHASE2**}(X)]]

Alternatively, one can hypothesize that [every$_{D2}$ cow$_N$]$_{D2}$ requires that (all and only) the cows be assigned to the second index. Then one could replace appeal to **TARSKI**—in the rule for external arguments of determiners—with appeal to **REDUCED TARSKI**.

ADAPT{SEM(…$_S$), D$_2$} = **D-EXTERNALIZE:REDUCED TARSKI**{2, SEM(…s)}

There are various ways to build in the restriction. But one possibility is that the determiner itself is understood as a reflection of a restricted quantifier.

SEM(every$_{D2}$) = +[fetch@every$_D$, **D-INTERNALIZE**:SEM(2)]

→ ·[EVERY(O), ∃-[INTERNAL(O, X), MAX-SECOND(X)]]

This effectively treats the *index* as the internal argument of the determiner. So one might well look for additional syntax; see Larson (forthcoming). Then one might say either that the noun cow$_N$ is *also* understood as specifying the internal participants, with the consequence that the cows must be the things being tracked by thefirstindex, orthatadaptinganoun to an *indexed* determiner just is a way of letting the index track the concept fetched with the noun.

ADAPT{SEM(COW$_N$), D$_2$}= **ASSIGN**{2, MAX-COW(X)}

Any such account highlights the analogy between external arguments of determiners and relative clauses; see Heim and Kratzer (1998). But it does not treat the arguments of determiners as expressions of type <e, **t**>. Hence, the proposal here does not predict that relative clauses can be understood as external arguments of determiners. And indeed, 'Every cow which Aggie chased' has no sentential reading according to which every cow is such that Aggie chased it. But in a relative clause, the index of the displaced relativizer invokes the Tarski trick. The index of a displaced determiner phrase may do the same; or it may invoke a more restricted trick that (p. 498) does not require consideration of any *new* values of the variable in question. But in any case, we need not suppose that the external arguments of determiners are sentential expressions that combine with covert relativizers, given the option of invoking **Tarski** or **Reduced Tarski** as part of the hypothesized significance of being an external argument of a determiner.

On this view, certain aspects of phrasal syntax are correlated with significant adjustments of the concepts fetched or assembled via the constituent expressions. One can call this a kind of type-shifting even if there are no types to shift. But if a conjunctivist semantics can handle quantificational constructions by appealing to simple operations like **INTERNALIZE** and **EXTERNALIZE**, given a maximalizing operator and **Reduced Tarski**, then it is hard to argue that such constructions support appeals to saturation—as opposed to conjunction, **INTERNALIZE** and **EXTERNALIZE**—in the semantics of subsentential constructions.

21.4 Conclusion

An unsurprising pattern emerges from this exercise. If one adheres to the idea that combining expressions is fundamentally an instruction to construct *conjunctive* concepts, along with the idea that open class lexical items are instructions to fetch concepts with independent content, one is led to say that certain aspects of syntax and various functional items are instructions to *convert* fetchable/constructable concepts into concepts that can be systematically conjoined with others. Perhaps this is the *raison d 'être* of syntax that goes beyond mere recursive concatenation: grammatical relations, like being the internal/external argument of a verb or determiner, can carry a kind of significance that is intriguingly like the kind of significance that prepositions have. These old ideas can be combined in a minimalist setting devoted to asking which conversion operations are required by a spare conception of the recursive composition operations that HFL can invoke in directing concept assembly. The list of operations surveyed here is surely both empirically inadequate, and yet already too rich. My aim has been to offer a specific proposal as one illustration of minimalist thinking in semantics, guided by two thoughts: this kind of inquiry has been fruitful in studying I-language syntax; and the study of I-language semantics has the same target of inquiry if I-expressions are instructions to build concepts.[18]

Notes:

(1) I understand MP broadly, not merely as an attempt to simplify extant conceptions of syntax; see note 2. But this chapter is not a review of the valuable literature that bears on attempts to simplify accounts of the 'syntax—semantics interface'; see Fox (1999), Borer (2005), Jackendoff (2002), Ramchand (2008). The focus here is on composition operations; cf. Hornstein and Pietroski (2009). Pietroski (2010) offers independent arguments for the view on offer, while exploring the implications for truth and the concepts that interface with HFL.

(2) For present purposes, I take it as given that humans have a faculty of language. But other things equal, one wants to posit as little as possible—especially in terms of distinctively human capacities—in order to describe and explain the linguistic metamorphosis that children undergo; cf. Hauser et al. (2002), Hurford (2007). This bolsters the general methodological motivation, already strong, to simplify descriptions of the states of linguistic competence that children acquire; cf. Hornstein (2009). If such competence includes knowing which meanings a given PHON can have (see note 3), then in evaluating attempts to simplify any other aspects of competence, we must consider implications for the semantic properties of expressions (cf. Hornstein and Pietroski 2009) and representations that interface with HFL in ways that let humans use this faculty as we do. Chomsky (1995b) argued, in particular, that the expressions generated by HFL just *are* PHON-SEM pairs. My proposal does not require this very spare conception of expressions. But if expressions have further ('purely syntactic') properties, that only amplifies the motivations for a spare conception of how SEMs are related to concepts.

(3) There are, however, many ways in which speakers *don't* compute interpretations. This is one moral of many 'poverty of stimulus' arguments, based on observations concerning (i) which sentences imply which, and (ii) logically possible interpretations that certain word-strings cannot support. See e.g. Higginbotham (1985), drawing on Chomsky (1965b). For reviews of some relevant psycholinguistic work, see Crain and Pietroski (2001).

(4) Cf. Marr (1982), Evans (1981), Peacocke (1986b), Davies (1987), Pietroski et al. (2009). Given that implementation matters, it seems obvious that explanations in this domain can and should be framed within a

‘biolinguistic’ framework; see Di Sciullo and Boeckx (forthcoming). Correlatively, we don't merely want theories that respect generic compositionality principles like the following: the meaning of expression α is determined by α's syntactic structure and the meanings of α's constituents. If the actual composition operations reflect innate aspects of human cognition, generic principles will be respected by languages that no child could acquire. In this sense, mere compositionality is multiply realizable (see Szabo 2000), raising the question of how it is realized in human I-languages; cf. Hurford (2007).

(5) Cf. Katz and Fodor (1963). One can still say that each concept has an extension in each context, and that in this sense, I-expressions link sounds to extensions. But if a lexical item L is polysemously linked to more than one concept, then an instruction to fetch a concept linked to L is fulfilled by fetching any concept linked to L—much as an instruction to fetch a rabbit from a room with rabbits is fulfilled by fetching any rabbit from the room. Though I have nothing to say about where polysemy ends and homophony begins.

(6) Perhaps some I-languages count as idiolects of English only if they are adequate tools for communication among certain people (including us). In which case, some I-languages may so count only if their lexical items are used to fetch concepts that are ‘extensionally similar’ in roughly the following sense: there is suitable overlap with regard to what the relevant concepts apply to; and for purposes of communication, disparities can be resolved or ignored.

(7) This would provide at least the start of an explanation for why ‘France is a hexagonal republic’ is defective in a way that ‘France is hexagonal, and France is a republic’ is not. See Pietroski (2005b), drawing heavily on Chomsky (1975b, 1977a).

(8) See e.g. Link (1983), Schwartzschild (1996). Letting ‘π’ range over plural entities and ‘$\in$’ have its usual meaning: $\forall\pi\forall\pi^*\{(\pi = \pi^*) \equiv \forall x[(x \in \pi) \equiv (x \in \pi^*)]\}$.

(9) I grant that adverting to lattices, with basic entities as terminal nodes, can be illuminating in various ways; see e.g. Link (1983), Schwartzschild (1996), and Chierchia (1998). But instead of interpreting each nonterminal node as a potential assignment of exactly one entity with elements to plural variable, one can interpret each such node as a potential assignment of more than one entity to a number-neutral variable; see Pietroski (2006). More speculatively, one might hope to accommodate mass nouns like ‘water’ and ‘wood’ in terms of a variable that is neutral as between one-or-more things or ‘some stuff’, with ‘chop (some) wood’ as an instruction to build ·[chop(x), **∃**·[PATIENT(E, X), WOOD(x)]]; where WOOD(x) applies, mass/count-neutrally, to (any sample of) wood. One could then distinguish PIZZA(x) from ·[PIZZA(x), countable(x)], or ·[PIZZA(x), COUNTABLE-AS-(X, PIZZA)], to distinguish ‘ate some pizza’ from ‘ate a pizza’; cf. Gillon (1987), Chierchia (1998).

(10) Or if you prefer, for any one or more things: the concept $[\uparrow\Phi(x)]$ (z) applies to them iff $\Phi(x)$ applies to one or more things; and $[\downarrow\Phi(x)]$ (z) applies to them iff $\Phi(x)$ applies to nothing. But omitting the extra brackets and variable position turns out to be at least as perspicuous.

(11) See Ludlow (2002) for discussion in the context of the ‘natural logic’ tradition as updated by modern conceptions of grammar, with particular attention to negative polarity facts.

(12) Partee (2006) raises the same kind of question, against a different background, though with less suggestion that a typology-free semantics might work.

(13) Think of an assignment as assigning one or more things to the free conceptual variable ‘x’ and one or more things to each index in the SEM. Other dependencies on assignments can be encoded in familiar ways, modulo number-neutrality. Larson and Segal's (1995) treatment is especially friendly to any Tarski-inspired theory.

(14) Nothing hangs on labeling names with ‘N’ and ignoring an internal structure, as opposed to $[ø_D\ \text{Bessie}_N]_D$, with a covert determiner and a lexical proper noun.

(15) For example, CHASE might be marked as an ‘action’ concept in a system that (for purposes of interfacing with SEMs) represents the agents/patients of actions as their external/internal participants; see Pietroski (2005a, 2008) for related discussion drawing on Baker (1997).

(16) As Heim and Kratzer's (1998) system nicely highlights, even if one appeals to saturation as a composition

operation, one still needs to posit Tarskian abstraction—often encoded as lambda abstraction (after Church 1941) —as a distinct operation. And this so, even given a Fregean typology. Suppose we treat indices and traces as constituents, as in [2 [she_1 chased_V $t_2]_S$], with the embedded sentence as an expression of type <**t**> and the larger expression as of type <e, **t**>. From an I-language perspective, one can say (modulo tense and gender) that relative to any assignment **A**: the concept formed by executing [she_1 chased $\mathbf{t}_2]_S$ denotes truth iff whatever **A** assigns to 1 chased whatever **A** assigns to 2; and correlatively, the concept formed be executing [2 [she_1 chasedv $\mathbf{t}_2]_S$]applies to xiff whatever **A** assigns to 1 chased x. But the idea *isn't and can't be* that the index denotes a fiinction-in-extension of type <**t**, <e, **t**>>, which maps the truth value of [she_1 chased t_2] onto a function of type <e, **t**>. Rather, '2' has to indicate a hypothesized (syncategorematic) instruction to convert a *representation* of one sort into a *representation* of another sort. Heim and Kratzer's third composition rule, in addition to rules for saturation and conjunction, makes this vivid. The attractive idea is that the higher copy of the lower index triggers quantification over assignment variants (taking assignments to be Tarskian sequences):

$$||[2\hat{}[\text{she}_1\ \text{chased}_V\ t_2]_S]||^A =$$

$$\lambda x.\ T \text{ iff } \exists \mathbf{A}':\mathbf{A}' \approx_2 \mathbf{A}[(x = \mathbf{A}'_2) \&\ ||\ [\text{she}_1\ \text{chased}_V\ t_2]_S]||\mathbf{A}' = T.$$

This has the desired result, taking the lambda-expression to be a theorist's representation of the hypothesized *concept* obtained in two stages: execute the sentential instruction, obtaining a concept that is doubly sequence-sensitive, and modify the resulting concept as directed by '2'. One can remain agnostic about the detailed forms of the concepts constructed. And from an E-language perspective, one can take the lambda-expression to be (only) a theorist's representation of the hypothesized *satisfaction condition*. But from an I-language perspective, the goal is to say how competent speakers represent the alleged satisfaction condition. And while theorists can abbreviate —as in $||[2\hat{}[\text{she}_1\ \text{chased}_V\ t_2]_S]||^{\mathbf{A}} = \lambda x.\ \textbf{CHASE(1, x)}$—we should remember that the corresponding psychological hypothesis presupposes some version of the Tarski trick. I just want to make such appeal explicit, so that we can ask what *other* mental machinery we need to posit in accounts of how I-concepts are constructed. For many purposes, it is fine to use a notation that effectively mixes appeals to saturation and abstraction. But this makes it harder, though by no means impossible, to see which operation does what work where.

(17) So given some but not all of the things a concept applies to, the 'maximized' concept does not apply to them. The number-neutral 'MAX-Φ(x)' can be cashed out with a first-order variable: $\forall x[X(x) \equiv \Phi(x)]$. But this says that for each domain entity, it is one of the one or more Xs iff it meets a certain condition. It *doesn't* say that there is a set s such that for each domain entity, it is an element of s iff it meets the condition: $\forall x[(x \in s) \equiv \Phi(x)]$. Suppose the domain entities are all and only the Zermelo-Frankl (ZF) sets. Then there *are* one or more entities (viz the ZF sets) such that each entity is one of them iff it is non-self-elemental; but there is no set whose elements are these entities. And a concept of 'being among' (or inclusion) could be used to introduce a concept of ordered pairs: EVERY(O) ≡ ιY:INTERNAL(O, Y){ιY:EXTERNAL(O, X)[AMONG(Y, X)]}; or in first-order/singular terms, ∀o:∃x[Oo & Internal(O, X)] {∃p[Op & External(P, X)].

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Wolfram Hinzen

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Abstract and Keywords

This article explores the relation between language and thought. The term ‘thought’ is an abstraction. It has its uses: for many philosophical purposes one may simply want to abstract from the linguistic forms that structure propositions, and concentrate on their content alone. But that should not confuse us into believing in an ontology of such entities as ‘thoughts’ – quite apart from the fact that, if we posit such entities, our account of them will not be generative and be unconstrained empirically. Where the content of forms of thought that have a systematic semantics corresponds to a so-called grammatical meaning – meaning derived from the apparatus of Merge, phasing, and categorization – minimalist inquiry is a way of investigating thought, with syntax–semantic alignment as a relevant heuristic idea. Having the computational system of language in this sense equates with having a ‘language of thought’, with externalization being a derivative affair, as independent arguments suggest. Thus, a somewhat radical ‘Whorfian’ perspective on the relation of language and thought is developed, but, it is a Whorfianism without the linguistic-relativity bit.

Keywords: thought, language, meaning, Whorfian perspective, semantics, minimalism

22.1 Introduction

Chomsky famously called language a ‘window into the mind’, echoing an early modern, rationalist tradition which underpinned the theory of grammar in the eighteenth century and had roots in Descartes. Recall that for Descartes, one's introspective awareness of one's own mind is the most certain of things—more certain, in particular, than the objects of sense perception and our knowledge of the external world. Language *expresses* this internal world. Thought thus takes priority over language, its medium.

It is not clear, though, in what sense we have an introspective awareness of this internal space of thought at all. Conscious awareness of a thought comes when we formulate or express it. As for thinking in others, it has given rise to the famous problem of ‘other minds’. I'll never know what you think, no matter what you say. Yet, thinking happens. In a way, it's the most certain of things. Furthermore, hypotheses about what and how we think are needed to explain how we act. Neither is there anything absurd in the rationalist view that language is the medium wherein thought takes on a phonetic or visual (in sign languages) shape.

Still, even on this view, where language is strictly distinguished from thought, language will be our main and perhaps only point of access for the study of structured thought. Radicalizing this conclusion, Gilbert Ryle and the later Wittgenstein argued that the ‘inner space’ was a myth to start with. Nothing is hidden, and where (p. 500) there is no shared language, there is no thought going on either (see Davidson 2001 for a recent formulation of this view).

Traditionally, then, the topic of ‘thought’ is defined through its contrast with ‘language’. One specific traditional issue is that of relative priority, and as we move into the twenty-first century, nothing much appears to have changed in the way the problem is set up. Thus, Jerry Fodor (2001: 2) presents us with:

> the chicken and egg problem as it arises in the philosophy of mind: namely: ‘Which comes first, thought or language?’ Or, by way of narrowing the field still further, ‘Which comes first *in order of explanation*, the content of thought or the content of language?’

This problem, I will here argue, is ill-formed. Language is what it is because of the mind—because of how we, in virtue of having a faculty of language, *think* about linguistic data. On the other hand, thought is also very much what it is because of language—clearly, the emergence of language in evolution and development has changed the way we think. Moreover, current linguistic theory effectively studies language as an internal computational system which is rationalized through its inherent use *as* a language of thought. If minimalism is on the right lines, there shouldn't be such a thing as ‘pure language’, viewed independently of how it interfaces with thought, at all.

A language of thought that is meant to be entirely independent of the computational system of language has been posited, though, as in Fodor's still prominent language of thought (LOT) hypothesis (Fodor 1975, Pinker 2007). According to this hypothesis, any natural language is irrelevant to how we think, for in order to learn any such language, we need to think already—in the LOT. Any human language such as English is therefore no more than an arbitrary system of sounds to which we map the ‘concepts’ that we already think.

If we think of language in universal terms, though, and assume single generativity—there is only one computational system underlying language, with phonology and semantics being ‘interpretive’ rather than independently generative—it is not clear what distinction the introduction of a separate LOT makes. The use of this computational system *is* a language of thought. Rather than dismissing this system as arbitrary and hypothesizing another, unknown, generative system, we might as well take our clues regarding the nature of thought from what we have learned about this system. I will argue that, methodologically, there seems to be no empirically constrained approach to thought (or ‘concepts’) that would *not* principally depart from the specific forms that we see them to have in the combinatorial system of language, where they are structuralized in particular ways and function. The naturalization of thought requires rethinking its connection to language, in this fashion.

The workings of this computational system are to be distinguished from aspects of language deriving from its externalization (the mapping of syntactic structures to phonetic forms that give instructions to sensorimotor systems). Minimalism (p. 501) has yielded the plausible conclusion that externalization is a process post-dating the evolution of language viewed as an internal computational system used for purposes of thought rather than communication (see Chomsky 2007). Independently of this suggestion, which is specific to minimalism, it has never been clear how much light can be shed on the architecture of language by looking at it as a communication system (Bickerton 1996, Hauser 1996). These are two other reasons for looking at (I-)language *as* a language of thought, rather than as an arbitrary externalization device.

Minimalism has less to say about another aspect of thought that the philosophical tradition has marveled at, and which is much harder to tackle, or even formulate. Parmenides was fascinated by the fact that truth could, ultimately, only be thought (rather than seen or heard). Moreover, thought has a connection to truth and reality—something *objective*—which perception critically lacks. The idea has been attacked as often as it has been formulated, but its persistence over the millennia suggests there is something to it. Plato's theory of Ideas is the first systematic way of tackling it. The relation of thought and reality has been a prime issue ever since, leading to such doctrines as realism and anti-realism: on a realist view, thought captures or mirrors reality as it really is; on an anti-realist view, commonly ascribed to Kant, it falls short of reality, necessarily reflecting our apprehensions of reality as well.

In the last 100 years, on the other hand, the ‘Parmenidean’ intuition of the objectivity of thought has obtained the status of a foundational axiom. It is felt that it is of the essence of a thought (say, the one that the Pythagorean Theorem expresses) that it, without any change in its content, can be shared between thinkers, at any point in time. Moreover, if it is true, it remains so, and it is not true because it happens to be thought, or happens to be thought true. This, Frege would contend, makes thoughts essentially abstract, language-independent and even mind-independent (‘non-psychological’) entities, and the philosophical tradition has largely adopted this axiom. Thus the very point of a ‘Representational Theory of Mind’ (Fodor 1975), for example, continues to be that the contents of thought are not *in* the mind but rather external entities which are merely *represented* by the mind (or by symbols within it, so-called ‘mental representations’). Propositions, in short, are nothing psychological or internal, but objects to which minds merely ‘relate’. They are radically external, be it that they are located in a ‘third realm’ (Frege), modeled in terms of sets of possible worlds (Stalnaker), or viewed as analogous to external facts or state of affairs (Russell, early Wittgenstein). In neither case do propositions have any particular connection with language, and Russell in particular argued vehemently against conflating the notion of a proposition as required in philosophical logic with linguistic entities such as sentences.

Such theories standardly miss out on the generativity of thought, which it crucially shares with language. On some versions of these views, propositions aren't even structured (Stalnaker 1987). But thought is discretely infinite and recursive (p. 502) much as language is. Generative mechanisms in the mind need to be posited that power this generativity. Once we lay down plausible mechanisms for this, it becomes obscure how these could differ from those posited within the computational system of language. Minimalism offers a highly constrained explanatory account of the latter, and that account has no use for mind-external entities such as propositions, over and above the structures that the computational system of language computes and that appear at the interfaces between the language system and the non-linguistic systems that use it (Hinzen 2006). This is simply to extend a claim that Chomsky (2000b) has made in regard to a putative relation of ‘semantic reference’, to putative abstract entities such as propositions and concepts: both disappear from a naturalistic approach to the mind.

If so, the ‘cathartic’ effects that the minimalist program has arguably had on linguistics may well reach further than that, into psychology and philosophy as well. In fact, minimalism has by now led us to rethink the very foundations of semantics.

22.2 Some matters of intellectual heritage

As Chomsky's *Cartesian linguistics* (1966) made clear, the intellectual allegiance of the generative enterprise is early modern rationalism. That language mirrors, expresses, or ‘externalizes’ human thought in a material medium is an essential aspect of the axiomatics underlying this tradition. Language ‘translates’ human reason, which is the ‘content’ of language, whereas language itself is the material or ‘sound’ side of human thought (‘form’). As systems of signs, languages stand in an arbitrary relation to the thoughts expressed. The latter live an independent life governed by the rules of logic and rationality rather than grammar. The ‘ideal language’, therefore, will be one whose grammar equates with logic, a declared aim of the ‘general’ or ‘philosophical’ grammars that were published in great numbers since the seventeenth century. The grammar of Port-Royal (*Grammaire générale et raisonnée*, first published 1660), in particular, sets out as follows:

> La Grammaire est l‘Art de parler./ Parler, est expliquer ses pensées par des signes, que les hommes ont inventez à ce dessein./ On a trouvé que les plus commodes de ces signes, estoient les sons and les voix. (Arnauld and Lancelot 1975[1660]: 5).

In short, language is an arbitrary system, resting on convenient conventions, whose essential function is the material rendering of (immaterial) thoughts. The unity of (p. 503) grammar across human languages is preserved by viewing this unity as an inherent expression of the unity of human reason. What is identical in all national languages is the logical construction of these thoughts.

The influence that this axiomatic basis has exerted on the philosophy of language as well as on linguistics and cognitive science is vast. That the organization of grammar merely expresses our thoughts and is not an aspect of their aitiology or inherent structure is an assumption unquestioned and even unaddressed in the kind of tradition tackled by any major philosophy of language introduction currently used in standard Anglo-Saxon curricula, including Morris (2008), Lycan (2000), and Devitt and Sterelny (1999). This is what we would expect given what this tradition owes to the logicist project of Frege and Russell, with its consistent and inherent demotion of language to an inaccurate and deeply ‘deficient’ expressive tool with regards to the logical form of our thoughts, for which an artificial language was to be designed to express them correctly. Ever since then, introductory and advanced logic courses in philosophy adopt the methodology of ‘finding the logical form’ of some grammatical construction, where the logical form is essentially unconstrained by what the forms of language are—a point I will illustrate in section 22.5.

Turning to the philosophy of mind and cognition, we have already discussed the LOT hypothesis, which is based on a rationalist view of language as an arbitrary means of externalization. Fodor, as late as 2001, confesses to believe 'that the function of sentences is primarily to express thoughts', and 'that the content of a sentence is, plus or minus a bit, the thought that it is used to express' (Fodor 2001: 11). And as we would also expect on a rationalist model, sentences are said to lack that content precisely to the very extent that they are not fully 'explicit' about the way the thought is put together (and that extent is said to be large).

A related major problem in the philosophy of mind, over all these years, has been that of developing 'theories of content' (e.g. Peacocke 1986a, Fodor 1990). Their main objective is to explain how we get from something perceived as arbitrary and meaningless (languages viewed as formal systems of signs) to something that carries content. On standard views, this problem of 'content-infusion' is solved by convention (Lewis, 2002), rules of rationality (Grice 1989, Brandom 1994), usage in a speech community (Wittgenstein 1956), or external causal relations (reference) (Fodor 1990).

One might even construe the history of philosophical thinking about language from antiquity to our days as a gradual departure from Plato's *Kratylos*, where the thesis of the *non-arbitrariness* (or necessity) of the linguistic sign is first systematically explored (Leiss 2009). Rorty's thesis that language describes or represents nothing is the culmination of this gradual loss of a realist interpretation of language. For Rorty, sentences are no more than 'strings of marks and noises used by human beings in the development and pursuit of social practices—practices which enabled people to achieve their ends' (Rorty 1992[1967]: 373). This thinking has its beginning in Occam's nominalist demotion of language in the fourteenth century, where **(p. 504)** language ceased to be a reflection or mirror of the structure of the world and became the arbitrary convention it is widely assumed to be today.

Within modern linguistics, too, it is part and parcel of cognitive and functionalist approaches to grammar that language is rationalized as a means to express independently given 'thoughts' as well: language structure is in the service of what's to be 'expressed'. In current work on language evolution, as well, it is frequently hypothesized that 'language evolved for the expression of complex propositions' (e.g. Pinker and Jackendoff 2005), while no attention is given to a blatant entailment of this view, that propositional thought is not a function of the organization of language and the operations inherent in its computational system. Fodor's (1983) modularity hypothesis also demotes language to a 'peripheral' system, whereas 'thought' is an independent and 'central' one (and see Fodor 2000). In Fodor's more recent work, inherently logical characteristics such as compositionality are reserved for 'thought', while 'language' is said to be deprived of them (Fodor 2001).

As for generative grammar, the thesis of the 'autonomy of syntax' usually ascribed to it implies that linguistic form can be (perhaps has to be) looked at from a purely formal point of view, as an arbitrary structure unmotivated in semantic terms. On one way of interpreting this, there is nothing inherently meaningful about the way language organizes itself structurally—a nominalist assumption. The emphasis of the Principles and Parameters framework of the 1980s on the Logical Form (LF) of a sentence as the essential aspect of what is universal across human languages may also be seen as a late reflection of an early modern rationalist assumption. The hypothesis of the existence of such a level of representation can be understood as the view that language design (to some extent) reflects the logical organization of thought, viewed as independently given (hence language structure is not quite as unmotivated as one thought). The 'inverted-Y' (EST) model of grammar explicitly conceptualizes narrow syntax or the computational system of grammar as a device that *mediates* between thoughts on the one hand and sounds on the other. In one current version of minimalist grammar, the same axiomatics is written into the very architecture of the language faculty, in that the computational system of language is rationalized by appeal to the expressive conditions 'imposed' by extra-linguistic systems of 'thought' ('Conceptual-Intentional' or C-I systems), with which it is said to 'interface'. I will critique this view in the next section.

Overall, then, we see how deeply assumptions about the relation of language and thought are enshrined into the development of linguistics and cognitive science, even where they are not obvious at first sight. It is important to see that there is nothing conceptually necessary in the assumption that language expresses thought. In fact, it is often forgotten that 400 years earlier than the early modern concept of universal grammar, there was another tradition targeting the same concept, which made quite different axiomatic assumptions: modistic grammar.

Modistic grammars were named 'philosophical', in that language was looked at as the way in which we come to have *knowledge* of the world, a vision completely **(p. 505)** foreign to the rationalist approach just reviewed (Leiss 2009). Language, for the modists, provides the *format* of that knowledge. The grammars were also 'speculative' in that they aimed at a science in the Aristotelian sense, i.e. to achieve explanatory adequacy of theory rather than to get mere observational adequacy or data recording; and in that they regarded language as a ‘speculum’, a mirror of reality filtered through the ‘modes of understanding’ (*modi intelligendi*) and the ‘modes of signifying’ (*modi significandi*), assumed to be the same in all languages. On this picture, linguistic form is intrinsically *motivated*. It is the way in which the world becomes known to us. Grammatical features in particular are called *modi significandi*, because each feature is an aspect of the way meaning is encoded (Covington 1984: 12). The sign is thus non-arbitrary with regards to the content, or reason. Thoughts are not simply ‘ready’, with language ‘printing them out’. They are what they are because of the way language formats them. The way that single lexical items are linked together grammatically in a sentence is a way of depicting how they are linked up in the world, if the *perspective* under which we see things in the world is taken into account. Modistic grammar is a realist theory of language in this sense, contrasting with generative views of language (Chomsky 2000b), which tend to see language as a representation of nothing.

Grammar, then, as seen by the modists, provides the means to build up systematic and structured perspectives on nature. The difference between the noun *cursus* (run) and the verb *currere* is a difference in the perspective we adopt towards the same *modus essendi* (lexical content; cf. Rosier-Catach 2000: 544). Parts of speech are therefore not definable by what they signify, but by the way they signify it (Covington 1984: 128). They emerge as first stages in the grammaticalization of the mind. Parts of speech allow for further categorial modifications, as through case, number, and gender in the nominal domain, and through tense, aspect, and diathesis in the verbal domain. The syntactic process in which such modifications are added is the process of building up *grammatical meaning*, in which grammatical features are added to lexical semantic features. The result allows for taking ‘perspectives’ on reality, which are not part of reality (ontology), but reflect different apprehensions of it. These do not depend on what happens to be physically real: thanks to grammatical means of combination, human beings are able to construct virtual realities (Leiss 2009).

No doubt this vision has some nostalgic flair to it. Yet, is there anything in it that we know today is wrong? Is language *not* that to which we owe systematic forms of knowledge of the world? Can we ascertain that linguistic form *is* arbitrary with respect to the thought expressed? Yet, only few people appear to take such views. According to Davidson, for example, non-human animals cannot think, since thinking depends on a propositional format; only this format has the power that we associate with human thought (Davidson 2001). Bickerton (1996) defends similar conclusions, as does, in other ways, Hinzen (2006, 2007), a view I will now develop.

(p. 506) 22.3 Problems of explanatory priority

Hinzen (2006), as Chomsky (2007) describes it, defends ‘a more radical conception of the FL-CI interface relation’—i.e. the relation between the faculty of language and the extra-linguistic C-I systems that predated the arrival of language. To see this radicalization, consider the Strong Minimalist Thesis (SMT), as currently pursued. According to the SMT, the design of the language faculty is a result of conditions imposed on this faculty by outside, non-linguistic cognitive systems. Put differently, what the syntax *qua* computational system of language produces has to ‘match’ what's independently there. The semantics ‘imposes’, ‘demands’, or ‘requires’, and the syntax ‘satisfies’. To whatever extent it does so optimally and with maximal efficiency, the SMT is vindicated and language finds what is called a ‘principled explanation’.

This model of generative grammar assumes the axiomatics of its functionalist opponents. It is what I have called a ‘matching complexity’ model of grammar (Hinzen 2009a). There is a task set, to which this particular cognitive system provides an engineering solution. The solution is rationalized by the task. A tight coupling between input and output is thereby assumed. Structures are specified for a particular function, contrary to what we expect in evolution, where such tight couplings will typically be too expensive to engineer and a premium is paid instead for adaptability—the reusability of generic forms when new purposes for them arise. Just as we do not want to assume purpose-engineered specialized rule-systems for every grammatical construction in every language, but overarching and underspecified principles (Boeckx 2008), we perhaps shouldn't assume either that syntax evolved to match a specific and pre-set semantic task.

Note further that the SMT model involves a certain methodological dualism with regards to syntax and semantics: it is as if semantics is essentially self-justifying, while syntax is not. ‘Thought’ is simply there, as it always has been in the rationalist tradition. The putative interface conditions therefore come for free—it is only extra conditions in syntax itself that have to be especially justified. This is effectively a nominalist conception of language, which the Port-Royal grammarians, but crucially not the modists, adopted.

The matching complexity model will also invite the assumption that there is a rich innate endowment to match the complexity in question. In Chomsky (2007), on the other hand, the innateness of language (the UG component) is programmatically minimized, while the explanatory scope of domain-general principles is maximized. We might think of this program as beginning from minimal relational codes (structures generated by Merge, say) that are as such unspecified for the output achieved by them, and yet accomplish it. This program is not consistent with the idea of setting an output that is to be matched. Indeed, as I will here contend, there is no independent task or output to be matched; the task accomplished by grammar arises with grammar itself.

(p. 507) Note that the existence of the C-I interface is no conceptual necessity. Conceptually necessary, as Hinzen (2009b) points out, is only that language is used. Any more involved assumption about the structure of the C-I systems will therefore have to be justified empirically and/or conceptually. One empirical hypothesis is that 'C-I incorporates a dual semantics, with generalized argument structure as one component, the other being discourse-related and scopal properties' (Chomsky 2008a). But as far as I know, the evidence that there are cognitive systems in nonlinguistic animals incorporating this very distinction—effectively, a fully human type of propositional semantics, with truth, intentional reference, quantification, and all—is non-existent. If so, the explanation that 'language seeks to satisfy the duality in the optimal way, EM [external Merge] serving one function and IM [internal Merge] the other, avoiding additional means to express these properties' (ibid.), is circular. It departs from a functional need to satisfy a semantic demand that may not be more than a shadow of a syntactic distinction, reified into an ontology of 'thought'.[1]

That ontology, quite possibly, could be a kind of cognitive illusion. Why do we feel that beyond the hierarchical and categorial organization of language—linguistic form—there is an independent system governing 'content'? That the structure of language is not what we should study when studying the systematic organization of thought, but instead an entirely independent structure, to which linguistic forms merely 'relate', or which they 'represent'? How do we even evaluate the empirical suggestion that, apart from the abstract organization of language as apprehended by minds, there are mind- and language-independent objects, propositions, out there, to which our minds mysteriously 'refer'?

Perhaps the most basic question to be asked here is why non-linguistic thought should obey a propositional format in the first place, which is not a logical necessity. Where it is argued that it is part of the job of a syntactic 'phase' (Chomsky 2008a) to compute a 'propositional' unit, it is assumed that the C-I systems are propositionally structured. But persistent problems in observing sentential organization in chimpanzee communication, either in the wild or under conditions of heavy training and tutoring (Terrace 2005), point against this conclusion. As for the presence of at least the right kind of *systematicity* (Fodor and Pylyshyn 1988) in primate *thought* (rather than communication), it may well be possible to 'bring on' some of it under extensive conditions of training, especially when it is scaffolded by a public medium of shared and stable signs provided to and (p. 508) manipulable by the monkey (McGonigle and Chalmers 2007). But this form of systematicity and compositionality does not in any way involve the more specific kinds of propositionality and referentiality that we find in human language, which depend on what was called 'grammatical meaning' above: the systematization of concepts into lexemes with categorial features and specific modes of composition opening up new perspectives on nature.

Older answers defending the existence of propositions appeal to their objectivity, which any 'psychological' or mind-dependent objects are not supposed to warrant. But what, *except* the systematic organization of thought through the faculty of language, could explain why we can reflect on experience in systematic, sharable, and objective ways? What if not *internal* relational codes and categories would allow us to reason propositionally about the world in the first place? Pursuing Frege's anti-psychologism in the philosophy of thought further, Russell offered a number of other putative arguments to the effect that propositions can't be sentences or internal to the mind. All of these arguments are based on an anti-realist conception of language and its study. On this conception, a sentence is a mere form, a meaningless entity or symbol (acoustic sound or graphic shape) which only receives meaning by being connected to external objects through a 'reference-relation'. This conception of the sentence is inconsistent with linguistics as it developed after Russell. I have discussed these and other arguments for externalism and propositions further in Hinzen (2006: 233, 2007).

Again, the picture of the human sentence as a meaningless shape or noise is a paradigmatic expression of a nominalism as inaugurated by Occam. Where language is meaningless or pure form, representing nothing in either the world or the mind, meaning has to come from somewhere else. It has to come from a theory of reference, for example, and I suspect it is here where the idea of the reference-relation as the determinant of meaning has its origin: the existence of semantics is what compensates for an anti-realism in our conception of linguistic form.

Yet, the very notion of 'form' is little more than an abstraction. One can look at any natural object, whether it is a plant, a coast line, a crystal, or a sentence, and analyze it according to its formal properties. That option is free, but, for the same reason, the claim that language is a 'formal system' is as empty as irrefutable. That meaning properties are any less intrinsic to a linguistic expression than its formal properties is not an empirical finding. It is a philosophical axiom, inspired by a largely implicit and perhaps unconscious nominalist epistemology of language.

Notice again that 'thought', if it exists as an independent module or domain, is generative as much as language is. There is no finite number of thoughts that are constrained to you. Hence, there needs to be a computational engine that powers this generativity. That requires thoughts to be internally structured, which in turn raises the question of whether these internal structures of thought could potentially be radically different from those that we see in linguistic expressions. Note, however, that the relations that structure thought are of a logical nature: predication and (p. 509) modification, for example, are logical relations. This is important, since, while it is entirely conceivable that physical relations can differ from possible world to possible world (e.g. causation could be a different relation in a world with a different physics), it is harder to see how logical relations can possibly differ in this fashion. Hence if these relations structure thought, and we know anyhow that they structure language (clearly, they are present in syntax and have been subject to syntactic study for many years), then the two systems can't fundamentally differ in the structures they exhibit. This will be *why* thought can be empirically studied in grammatical terms, and be naturalized in this fashion.

Discrete infinity, as an aspect of language, is achieved through the part—whole constituency structure of the underlying structures of sentences. Could there be a radically different solution to the same problem in the case of 'thought'? Again this is hard to see, especially if, in line with the explanatory aims of minimalism, we have minimalized the computational system of language to such an extent that it only exhibits those features that *any* system exhibiting the basic features of language has to have. How could thought, as a generative system, lack those features? We could see and easily model thought in abstraction from the forms of human language (Stalnaker, 1987). Whether this would illuminate any empirical aspect of the specific format of human thought or the origin of its systematicity seems doubtful. In the next section, I will flesh out these arguments further by emphasizing syntax—semantics alignment: the hypothesis that where semantics is systematic at all, semantics and syntax cannot independently vary.

Here I conclude that an idea quite foreign to the axiomatic basis of minimalism, namely that language formats thought as opposed to expressing it, deserves a hearing. The earlier generative tradition has, given its emphasis on modularity (Fodor 1983), thoroughly emphasized the independence of thought and language, so as to provide evidence for the status of the latter as an autonomous module with a strong 'innate' basis (e.g. Yamada 1990). Yet, what has proved particularly fruitful is recent minimalism's drive towards maximizing the use of domain-general principles in explaining the language faculty (Chomsky 2007). Old ideas of modularity do not fit naturally with how the field has evolved.

22.4 Syntax in thought's service

22.4.1 Creating 'concepts'

The more we make semantics modular and separate its generativity from that of syntax, as in the doubly generative system of Jackendoff (2002), the less we can (p. 510) use the form of language to explain the contingent features that its semantics exhibits (grammatical rather than lexical semantics). The more syntax is part of the aitiology of human thought, on the other hand, and in fact provides its format, the more the opposite is true. If semantics is in fact aligned with syntax—syntax carves out the ways semantics must follow (Uriagereka 2002)—we see again that a highly constrained theory of linguistic form might also provide an explanatory and naturalistic approach to the origin of human thought.

Why, if linguistic form is arbitrary with regard to the thoughts expressed, would we see syntax and semantics aligned to the large extent that we do? We see, in particular, apart from novel forms of grammatical meaning arising in systematic ways as lexemes get categorized and combined, that the structural complexity of meanings co-varies perfectly with the complexity of the linguistic forms involved. This is surprising on the view that there is double generativity, with the one not speaking to the other (Jackendoff 2002). Clearly, for example, to reach an expressivity of a certain type, we need a syntactic object structurally rich enough in exactly the right way. No full, truth-bearing proposition can be expressed by anything short of the full hierarchy of the human clause. Thus, none of the following can grammatically be said to be either true or false:

(1)

a.	[Fire!]	Exclamatives
b.	[Tom]	Names
c.	[Mary's smile]	Possessive Noun Phrase
d.	[Caesar's destruction of Syracuse]	Nominalization with full argument structure
e.	[You idiot!]	Expressive Small Clause
f.	[me, serial killer]	Appellative Small Clauses
g.	[to be an idiot]	Non-finite Tense Phrase
h.	[For John to love his mother]	Complementizer Phrase with non-finite Tense
i.	[if he is handsome]	Complementizer Phrase acting as modifier
j.	[John pale]	Small Clause Predication (cf.: ‘I saw John pale’)
k.	[who has no money]	Relative Clause

A person shouting ‘Fire!’, as in (1a), is not usefully described as saying ‘There is fire here’, i.e. as making a propositional claim (for example, alarm calls cannot occur embedded, unlike propositions). The same is true of interjections like ‘Ouch’, which are inherently affective rather than propositional, and again cannot be syntactically combined. As for (1b), single words that function as proper names each denote a person in some irreducible fashion: naming isn't making a claim of truth. Complex noun phrases like (1c, d) are paradigmatically referential rather than true or false—no claim *about* Mary's smile, etc. is made. Expressives of the type (1e) also hardly **(p. 511)** translate into the propositional claim ‘You are an idiot’ (Potts and Roeper 2007). (1f) does not change this basic assessment, though it is interestingly more complex than (1e): the determiner ‘an’, in particular, is excluded in constructions of the (1e) type. (1g) to (1k) are clausal constructions, all involving verbs, and yet, none qualifies.

Truth, in short, arises at a late stage in the build-up of hierarchical, structural complexity, which by itself indicates that it, as a semantic core primitive, is *syntactically conditioned*. At the other end of the complexity spectrum, we see that what is syntactically maximally simple is semantically maximally simple as well. This is the lexical root that is not yet categorized, assuming Marantz's view that categorization is a matter of syntactic configurations in which lexical roots appear in the context of special functional heads (*v*, *n*, and *a*) that categorize them (Marantz, 2001). These create the ‘lexical’ categories of verb, noun, and adjective, as in the case of the root √top, which can give rise to a verb, a noun, or an adjective (as in *top manager*), all with the same phonological signature, depending on the syntactic context in which they appear. This process differentiates the meaning of the root involved. By contrast, there is nothing to say about the meaning of √top, except that it means ‘top’: this is the view of the ‘disquotational’ (or ‘empty’) lexicon (Borer 2005, Fodor and Lepore 1998), on which the meaning of a non-syntactically complex lexical item is simply its reference, and cannot be illuminated beyond being stated disquotationally (and, *pace* the Fodorian tradition, it also cannot be illuminated by explaining reference further, as a relation to some independently identifiable object).

As grammar kicks in, then, new perspectives on the world become possible, which, as I will argue now, crucially have nothing to do with lexical or conceptual meaning—or, more importantly, with ontology. Ontology is the theory of what kind of objects the world contains. It is widely entertained (e.g. Davidson 1967, Higginbotham 1985) that it is a factual question (about the semantics of natural language, or, in a minimalist context, the structure of the C-I systems), whether language makes reference to events as entities. In short, it is assumed that we can answer ontological questions by studying language. However, it has nothing to do with the world that, as it happens, you can make use of a *v*-head to create the verb *destruct* and make the claim that *Caesar destructed Syracuse*, whereas, if you availed yourself of an *n*-head you could create a noun, *destruct-ion*, and speak of *Caesar's destruction of Syracuse*. The *perspective* you adopt changes: in the former case you refer to a time at which an event took place. In the latter, you refer to the event itself, treating it more as an object. But the event as such—the lexical content that the event involves—is the same in both cases. The change in perspective has nothing to do with the world but concerns how we signify what is the same lexical content: a mode of signifying.

Let us get into some details to drive the point home. Recall (2a), which allows the nominalization (2b) but not the intransitive form (2c). In contrast, (3a) does not (p. 512) have the nominalization alternation (3b), even though it allows for the nominalization in (3c), and allows for the intransitive form (3d) as well:

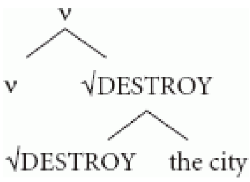
(2)

a. Caesar's *destroyed* Syracuse.
b. Caesar's *destruction* of Syracuse.
c. *Syracuse destroyed.

(3)

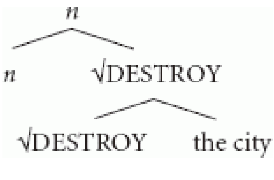
a. Caesar *grew* tomatoes.
b. *Caesar's *growth* of tomatoes
c. Caesar's *growing* of tomatoes
d. Tomatoes *grew*.

Now, in the idiom of 'concepts', every one of the italicized forms expresses one of these, I suppose. We now want to study these concepts, tell how they differ, and explain why they behave the way they do. As implied above, one explanation is that, before grammar enters, there are only concepts in the mind unspecified for category: the roots √DESTROY and √GROW. From the viewpoint of the computational system of language, these are atoms. As it happens, the encyclopedic content of the former is such that it entails an external agent, while the latter does not. Then (2a) involves the structure (4), where *υ* is a verbalizing and Agent-projecting head:



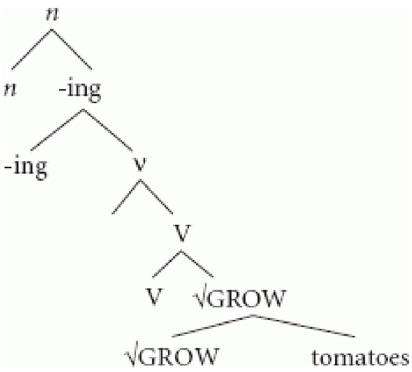
(4)

(2b), by contrast, involves the same structure, though with an *n*-head:



(5)

(3a) involves the same structure as (4) with √DESTROY replaced by √GROW. However, as the root √ GROW refers to an event that is internally caused, *υ* is merely a verbalizing, not an Agent-introducing, head. The nominalization of this root analogous to (5) thus does not project an Agent and yields an impossible structure if the Agent is illegitimately added (see 3b). By contrast, if we verbalized the root, the relevant verbalizer could assign an Agent, and we can then nominalize in a second step, obtaining (6), corresponding to the concept in (3c):



(p. 513) **(6)**

Arguing for these structures for the verbs and nouns in (2)–(3), Marantz (1997) points out that we can systematically explain, through an interaction of the lexical content of the roots involved and the syntactic configuration in which they appear, which forms are possible and which are not. Now, I am inclined to replace the ‘forms’ in the previous sentence by the word ‘concepts’, and think of (4)–(6) as structures of concepts. What would be wrong with that? Why is syntax not a (systematic and generative) study of concepts? Why would the same structures only be ‘representations’ and the concepts themselves something external to them, with unknown structures? What in that case would explain the patterns in which such concepts appear, as in (1) and (2)?

One could say, as Hinzen (2009a) has, that grammar creates the ontology of language: the distinctive set of basic categories in terms of which we think, such as object, event, state of affairs, or proposition, each of which correspond to lexical roots standing in particular syntactic configurations. For all of these correspond to distinctive configurations that the syntax avails us of. Crucially, the intuitive, conceptual, or paradigmatic complexity of the ontological entities involved increases with their syntactic complexity, supporting syntax-semantics alignment. But, we need to be careful: strictly, again, the derivational dynamics that creates such ‘entities’ has nothing to do with ontology. These are figments of the mind, and the world is unaffected by them. There is no linguistic road into the world of ontology. As Chomsky (p.c., 2006) has pointed out, Quine was wrong about the notion of ‘ontological commitment’: the use of language does not commit us to any ontology. We can choose the ontology as we wish: it is a matter of (linguistically fabricated) perspectives. The world does not tell us whether √DESTROY is an object, an event, or a proposition. Nothing but syntax can tell us that.

At this point, nothing prevents us from extending this approach to lexical categories to a principled theory of possible meanings. The project is to systematically ‘deflate’ terms such as ‘event’ or ‘proposition’ into notions such as that of a syntactic structure headed by *v* or C (Complementizer). I do not see what else would lend empirical content to these intuitive notions. We can't, say, use the methods of (p. 514) physics to answer the question whether the world contains events in addition to objects. This is a formal or categorial question relating to our experience of the world, not its ontology as revealed by physics.

22.4.2 Deflating propositions

How then *can* we make sense of propositional thinking in humans? If we look at the architecture of the human clause under current conceptions (see Fig. 22.1), we see it falling into two basic domains, which in contemporary terms correspond to *phases* of a derivation.

Here we assume that linguistic expressions have a basic part—whole structure, usually modeled in terms of binary sets containing other binary sets, the consequence of the operations of the basic recursive operation Merge. The sets generated by Merge are labeled, though this is perhaps a derivable notion (Collins 2002a, Chomsky 2008a). The label enters into semantic interpretation. Below we isolate two such labels as playing a special role in interpretation, *υ* and C, where *υ* is a verbal projection containing the verb's external argument. This makes intuitive semantic sense, since on current conceptions *υ* configures an event with its intrinsic participants (argument-structure), irrespective of quantificational and discourse properties of the sort that propositions have: the ‘duality of semantics' mentioned above. 'C' by contrast is shorthand for just these quantificational and discourse properties.

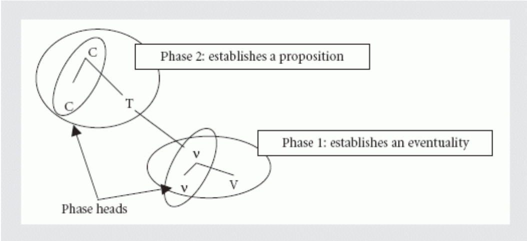


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Fig. 22.1.

In principle, there could be an independently given event/proposition distinction that motivates this syntactic duality, but this would be a misleading way of putting it. There is conceptual and empirical-syntactic evidence in favor of the existence of special or dedicated boundaries such as *υ* or C, which have little to do with their semantic function. That is, phase edges do make semantic sense—but we **(p. 515)** know independently that they make syntactic sense, namely as landing and reconstruction sites for movement, and phonological sense, as prosodic units (Uriagereka 1999; Franks and Bošković 2001, McGinnis 2004). In effect, their semantic identity seems inseparable from their syntactic and perhaps even phonological identity (Carstairs-McCarthy 1999). Our semantic intuitions, short of independent evidence for them, are more likely shadows of those independent syntactic facts.

The empirical evidence in question takes its departure from old observations to the effect that syntactic derivations are *cyclic*. A sequence of operations associated to one particular syntactic head has to be completed first, before a new head is inserted into the derivation. If an argument moves long-distance, moreover, it moves successive-cyclically, i.e. respecting the cycles that exist. The conceptual evidence is evidence to the effect that the computational system of language should be, and in fact is, efficiently designed. Thus, when a derivation continues beyond a completed cycle, that cycle cannot be erased or changed. For example, where NP is completed and merges with V, V would not merge inside NP, but leave it untouched, and merge at its edge. A cyclic system, moreover, only makes sense computationally where the information contained in a cycle is not retained upon its completion. In contemporary frameworks this is implemented by stipulating an operation of Transfer at each cycle boundary, where the construction so far is spelled-out (at PF) and interpreted semantically as well. After Transfer, the structure is thus gone from the derivational workspace (it is 'forgotten'). This is Chomsky's famous Phase Impenetrability Condition (PIC).

If we think of a cycle as a phase of a derivation followed by Transfer, then, clearly, the simplest (most minimal) system is one that only consists of a single cycle (Chomsky 2007). It begins operating with a head, which then projects up to a boundary after a (small) finite number of iterations of Merge. After the 'first phase' of the derivation, in which the lexical category of a root is settled ('lexical syntax'), the head of the first post-lexical phase, say a noun, will function as a predicate denoting simply a kind, like 'man'. 'Man' as such is virtually useless, however, since it doesn't refer to anything: not this or that man, not every man, not some man, etc. Hence the phase will be complete only once we know that: once reference is established. Similarly for an eventive head, like 'destruct'. Again, we don't know which destruction, by whom, and of what we are talking about, if *v*-destruct is the only thing we've got. Before this is settled, the derivation can't proceed. Once we know which event it is, say Caesar's destruction of Syracuse, we can proceed, and embed the event in time (add finiteness) and attach a truth value to it (proceed to the second clausal phase). Plausibly, not every head projects a phase (is a phase head); and there are conceptual and empirical arguments that the conceptually most plausible option, where phase-heads and non-phase-heads precisely *alternate*, is also the option which is realized in human language (M. D. Richards forthcoming): C-T-*υ*-V, the basic architecture of the human clause, where T and V are phase complements and C and *v* phase heads.

(p. 516) Nothing, now, stands in the way of taking this principled account of linguistic architecture and regarding it, looking at the inherent semanticity of the basic layers in the human clause, as the very skeleton (or template) of a human thought: the way a thought *has* to be structured. That skeleton expresses the formal constraints under which one systematic form of semantics has become possible in evolution (there might be others). To conclude this is to carry an idea about the architecture of the language faculty further that began right after the downfall of generative semantics in the early 1970s. The latter effectively trivialized the theory of grammar by reducing it to a mapping between an independently constituted system of 'thought' and a linguistic 'surface form': language is merely an 'expression' of 'deep thought'. In other words, there was no grammatical 'deep structure' that provides constraints on semantic information as expressed in language. A very abstract 'semantic level of representation' was thus posited. As Hinzen (2006: 156–8) points out, ever since then it has been a consistent development within generative grammar that semantics, rather than the *input* to grammar, is to be read off from what the grammar *outputs*. Its systematicity depends on the forms the grammar provides.

Huang (1995) tellingly observes that there was no *semantic* motivation for the introduction of the LF level in the early 1980s at all, and that there couldn't have been: conceptually speaking, it would have been equally simple or simpler to map surface structure to the logical forms of thoughts as depicted in semantic theory directly. Instead, the motivation for this level was, as he puts it, that 'quantificational sentences exhibit properties that are best captured by principles and constraints that have been independently motivated in overt syntax' (Huang 1995: 131). As he concludes, 'the notion that meaning is determined by form is amply demonstrated by the fact that many properties of quantificational sentences which are generally thought of as matters of interpretation are to a large extent seen to pattern on a par with matters of form and are explainable as such at little or no cost to the grammar' (p. 155).

Still, in these times of Government & Binding, these 'explanations' of meaning in terms of form were not yet principled in the way Minimalism would then force them to be. Curious as the dependence of meaning on grammatical constraints proved to be, what these followed from was unclear. If the organization of grammar could be seen as a natural or conceptual necessity, the research program changes more into the one hinted at above: to offer a principled account of meaning by offering a principled account of linguistic form, with which meaning aligns.

This is where generative grammar as a theory of thought comes into its own. Where grammar virtually *gives* us the structures that we need for the logical analysis of our thoughts, it becomes very hard to see how we can maintain the view that 'thought' has some independent origin, and logical structure is unrelated to the advent of language on the evolutionary scene, viewed as merely a means of expression. The entire conceptual distinction between language and thought becomes blurred.

(p. 517) Once that is understood, we might fantasize about various possible further results. For example, we might make it our vision to show that syntax is 'crash-free', hence that whatever representations it generates, they cannot fail to be interpretable (Frampton and Gutmann 2002; and see Hinzen 2006: 241, 249). A syntactic object that wouldn't mean anything couldn't be derived in the first place (it would be an impossible object). Even better would be the result that syntax doesn't contain truly uninterpretable features to start with (Sigurdsson 2007). In short, there are rich prospects for a grammatical theory of meaning, viewed as an internalist theory that is constrained enough to be explanatory, on the horizon.[2]

The bad news is that this conception of a theory of semantics tells us next to nothing about what has fascinated philosophers most about the mind over a period of 2,500 years: the externalist aspects of truth and reference, and the objective validity of the structures our mind can generate. Somehow, ideas generated in our head transgress the boundaries of our organism and actually 'hit' the world. On the picture above, selecting a phase head is like setting up a referential anchor, but not throwing it yet. For this we may need to build the phase further, until it is rich enough in descriptive terms in order to hit a target within a given discourse context in which the anchor is thrown. The discovery and study of phase boundaries, which on the present view correlate with crucial semantic cuts in the build-up of a grammatical meaning, is part of a study of reference in this sense. Yet, it says nothing at all about intentionality in the sense of a theory of what happens *between* the moment where the arrow (or anchor) has been pointed and thrown, and the moment where it lands. Anything I say in this chapter leaves this mystery where it hasalwaysbeen.[3]

22.5 The limits: reference and truth

Reference and truth are thus the outer limits of internalist inquiry in the sense of Chomsky (2000b), and unsurprisingly they are strongholds of externalist theorizing in semantics and the philosophy of thought (Hinzen 2007). Yet, we wish to explain how it is that different syntactic forms are put to different semantic uses. A phenomenon that has been at the focus of attention throughout the philosophical tradition and particularly in the analytic philosophy of language is the uses of proper (p. 518) names. These are the paradigm of semantic reference in the standard externalist philosophical sense: some kind of direct, basically causal, relation between a word and a mind-external thing (‘Fido’ the name and Fido the dog). The thing is the word's meaning. The relation between the word and the thing fully accounts for the word's meaning (see e.g. Lycan 2008 for a lucid exposition).

Having argued for the syntactic deflation of the notion of proposition, let us see what can be done about proper name reference, in naturalistic terms. On the above picture, the paradigmatic case of linguistic meaning is the use of a word as a mere ‘label’, where no conceptual information can intervene between the word and its reference, i.e. the word refers ‘directly’, unmediated by any description or any contents of our mind. So much is this a foundational axiom that cases where there is a name but no object (‘Hamlet’, ‘Pegasus’) have been rated as cases where there can be no meaning either. With all the rigor of logical consequence, the conclusion was drawn that where we *seem* to be using a name to refer to something—to Hamlet, say—we are not referring to anything (contrary to fact, for we are clearly not referring to God or Arnold Schwarzenegger, say: no one other than Hamlet is what we are referring to). The reason, on the standard view, for the fact that what we are doing is *not* meaningless, is that we are *describing* in these instances, and that describing is not referring. This is what Russell's theory of descriptions contends, which became the paradigm of philosophical logic and formal semantics, given its foundational assumption that ‘logical form’ can depart from linguistic form. In Russell's particular theory, language, however, is not merely misleading here and there: it is ‘*systematically* misleading’, in that every single name is, really, or deep down, or ‘logically’, a description. A better instance of syntax—semantics misalignment is hard to imagine.

Some constraints on the extent to which this can happen would therefore seem needed. In particular, it seems clear that when we use language to describe—say when we describe Mary as *the girl with the golden hair*—the linguistic form we are using is of a radically different kind from that of the word *Mary*, which we use when we merely want to name Mary. This makes a putative equivalence of a name to a description, as stipulated by the ‘description theory of names’, a rather astounding fact. If such an equivalence were maintained, it is clear that it could only be maintained ‘at a semantic level of representation’. If there was no such level, we would have to invent it. Why we should, is unclear. That we are not describing, when we refer to Mary merely as ‘Mary’, and that we are not naming when we refer to her as ‘the girl with the golden hair’, seems clear enough. Syntactic differences in the structures used are aligned with a semantic difference, and in fact help to explain it (Hinzen 2007).

Nor does this difference go away when we refer to Hamlet as ‘Hamlet’ rather than descriptively, as ‘Shakespeare's neurotic Prince’. Language doesn't notice the difference that existence makes. No grammatical operation, as far as I know, runs or not depending on whether the object referred to exists or not. Grammar is as (p. 519) blind to existence as arithmetic is. That by itself is one of its hallmarks: reference, as a phenomenon of language use, is not physically constrained. The fact that grammatical formatives catapult us out of the world of lexical meaning, opening up an entirely new and infinite universe of thought, is what its gift to human nature consists in (Roeper 2007). If the logical analysis of our thoughts is centrally at odds with our routine ‘reference to non-existents’ (as Lycan's 2008 review of the philosophy of language manifests), it's on the wrong path.

Russell's theory of descriptions was quickly combated (Kripke 1980), but his idea that direct reference in the sense above somehow is the paradigm of meaning (or what *constitutes* semantic ‘content’), has not so far given way. It has been more natural, for example, for philosophers to agree on the ontology and existence of ‘impossible objects’, in order for terms such as ‘round square’ to be meaningful, than to agree on the fact that meaning is determined internally to a mind which houses the relevant lexical items and combinatorial principles, and aligns linguistic meaning with linguistic form. In this particular case, the meaning would be an automatic consequence of the semantic lexical features of the lexical items involved, the fact that the structure is compositionally interpreted, and the fact that the two predicates are structuralized adjunctively, giving rise to a description that applies to an object if that object is both a square and round. With this explanation, no metaphysical issues arise, no Meinongian ontology of ‘impossibilia’ is needed. Issues of external reference don't even enter. Nor would there be a problem if we decided to call one such impossible object ‘Fred’ and argued (truly) that ‘Fred was impossible’. ‘Fred’ will be no more than an idiomatic or shorthand way of referring to Fred, discarding the description with which we picked him (it) out on this occasion.

This in itself again suggests, as Kripke correctly maintains, that the reference of names need not be descriptive:

the specific uses to which we put proper names indicate that where context is sufficiently rich for us to know who Bill is, the name 'Bill' will do the job of picking him out, and a complex description like 'the man with the prosthetic leg' isn't needed. In short, a name shortens a phase, given the conception above where a phase consists in the completion of an act of reference (Arsenijevic 2007). More technically, a proper name is a phase of which only the head (noun) is left: the noun is transferred immediately into the discourse, instead of projecting a more complex description first. Since whatever descriptive information went into the creation of the idiom in question is gone at this point ('forgotten'), it is then possible to discover that the referent identified as 'Bill' isn't, in fact, a man, say, without 'Bill' thereby starting to refer to anyone other than the person it referred to before. Thus reference is oblivious to the descriptions by which it is accompanied, a fact that equates with the 'rigidity' that Kripke ascribes to names—making rigidity a trivial consequence of how the linguistic system works, and independent of any of the metaphysical considerations that Kripke invokes (in particular, metaphysical essentialism). The explanation sketched is internalist throughout.

(p. 520) On the other hand, 'Bill' is only contextually and temporarily marked as a name and can, in a new derivation, act as a standard descriptive noun whose reference is restricted first, before any act of reference takes place. For example, in 'the Bill I knew', reference is to a stage of Bill, viewed as a whole of many such stages or parts.[4] Reference is descriptive, in short. Yet, once the reference is determined, again nothing but the head of the construction, Bill, will be left. Deprived of the restriction, it will act as a rigid anchor for the stage picked out, to which we can now refer, independently of the description that helped to pick it out in the first instance, as in the comment that 'the Bill you knew was in fact called Fred'.

Note that descriptive information always comes before any reference takes place, even though it is (partially) discarded once the reference has taken place. Thus, even if a noun has idiomatized into a name that everyone has memorized as such, like 'Stalin', the description that was there prior to idiomatization normally will not have entirely disappeared. We could discover, for example, that Stalin wasn't a mass murderer but only a harmless cobbler, like his father, contrary to how we have described him so far. The name could then easily retain its reference: it would refer to the same person, though a radically different one. But it would *not* retain its reference (to a person), if, say, we discovered that he was only a hallucination. While people remain persons if they do not murder, they don't if they turn out to be figments of the imagination. Reference is thus *always* based on the residues of a descriptive predicate, even if it is rigidified, once the referential anchor is thrown.

The point of this discussion is this. Even though reference, and particularly the reference of names, has been the stronghold of externalist theories of content, the explanation of specific acts of reference is almost entirely a matter of analyzing the relevant elements of linguistic form and the dynamics of the computational system involved. The environment and its physical features hardly enter into the explanation—and indeed we cannot possibly suggest that it is the physical world out there which explains how we refer to it, as this would make it a mystery why no other creature in the same environment intentionally refers to it as we do, or why the way we refer to persons, say, is essentially independent of their physical features. Our modes of signifying are directly correlated with the syntactic forms that we use. Thus when we refer to a part of an individual, like 'the young Bill', this depends on partitioning the 'Bill' space and quantifying over its various presentations at different points in time: we are thus creating an ontology of parts of individuals that is solely the result of having grammar and quantificational syntax, not of what physical features the environment contains.

(p. 521) Hinzen (2003) extends aspects of this analysis to the study of truth as the basic primitive of human semantics as well. An explanatory approach to human thought will have to tackle its semantic primitives, and apart from reference, there is truth to deal with (the reference, specifically, of sentences). Hornstein et al. (2002) observe that the syntax of predication changes depending on whether you quantify over the integral parts of an object, as in (7a), or else over features only accidentally related to it, as in (7b):

(7)

a. My Saab has a Ford T engine.
b. A Ford T engine is (located) in my Saab.

Whereas in (7a) we refer to the engine as an integral part of the car in question, in (7b) the engine might as well be merely stored in its trunk. (8) is ambiguous between these readings:

(8) There is a Ford T engine in my Saab.

Hinzen (2003), using Hornstein et al.'s syntactic analysis of the integral reading of this example, analyzes (9) in the same way:

(9)

a. There is some truth in/to what you say.

b. What you say has some truth to it.

If this analysis is correct, it means that, as language configures our thoughts in these instances, the way we quantify over the integral parts of an object in (7a) is also the way we quantify over (parts of) the truth in examples such as (9). The predication is the same in the two cases.

An interesting aspect of this, from a philosophical point of view, is that perhaps nothing, in nature, is intrinsically a part of a whole: this, again, is a matter of what referential perspective we take. So, if the referential perspective we take in the case of a predication of truth is the part—whole or integral one, nothing out there in the world will 'correspond' to such judgments. The judgments in question are not a reflection of outer reality, but a perspective we take up on it, which wouldn't exist without the specific kind of structures that the syntax of natural language allows us to configure. Looking at these structures tells us that truth-values do not attach to abstract propositions in some 'external' way: as speakers configure judgments of truth, truth values attach to the propositions in question as *integral* parts or aspects of them. Put differently again, we don't map propositions to external truth values when we judge the truth, as formal semantic theories suggest. Truth values are not external objects of reference, as in Frege, but integral 'parts' of the objects to which they attach. In (9a), we partition a proposition into parts, and refer to those parts, linking them to the space conceived as a whole in an integral way.

This does not solve or even address the metaphysical problem of thought that I mentioned in the introduction. Thought *has* that objectivity and abstractness which philosophers have so marvelled at. Somehow, our thoughts resonate to the **(p. 522)** world about which we think. Yet, if I am right, the structures in terms of which we do so are not what describe this relation, or what we stand in relation to, out there. Language does not reflect reality, but our perspectives on it.

22.6 Conclusions

The term 'thought' is an abstraction. It has its uses: for many philosophical purposes one may simply want to abstract from the linguistic forms that structure propositions, and concentrate on their content alone. But that shouldn't confuse us into believing in an ontology of such entities as 'thoughts'—quite apart from the fact that, if we posit such entities, our account of them will not be generative and be unconstrained empirically. Where the content of forms of thought that have a systematic semantics corresponds to what I have called grammatical meaning above—meaning derived from the apparatus of Merge, phasing, and categorization—minimalist inquiry is a way of investigating thought, with syntax—semantic alignment as a relevant heuristic idea. Having the computational system of language in this sense equates with having a 'language of thought', with externalization being a derivative affair, as independent arguments suggest (Chomsky 2007). I have thus developed a somewhat radical 'Whorfian' perspective on the relation of language and thought, but, as in the case of Spelke (2003), who closes with a similar remark, it is a Whorfianism without the linguistic relativity bit. The structures that thought co-opts are universal.

Notes:

(1) The same kind of remarks could be made with respect to other putative hypotheses about the C-I systems, such as that the 'C-I interface requires the A/A′ distinction', 'if expressive potential is to be adequately utilized'. The explanation offered is that syntax provides just that—via the device of inheritance. For a final example, Chomsky (2004a) argues that the workings of the C-I involve an operation of 'predicate composition', and the syntax therefore has to provide it, through the process of adjunction to a category, which means that adjunction has found a principled explanation. These putative explanations simply don't capture the generative principles that give rise to the phenomena in question, no matter whether we locate them on the linguistic or non-linguistic side of the semantic interface. Clearly, they may not arise from external conditions imposed on language at all.

(2) Note that the notion of grammatical meaning targeted here is radically different from the notion of a 'conceptual semantics' in the sense of Jackendoff (2002), even though the latter is internalist as well. It is, in particular, nothing

to do with lexical semantics, or encyclopedic knowledge.

(3) On my view, the behaviorist research program tackling this problem is the only one that ever existed. The framework of 'Cartesian linguistics' is not a replacement for it, and it has a different explanatory target.

(4) Again, independently of any ontological concerns: there are no stages of persons as independent objects lying around in the external world, as it were.

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Parameters

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[–] Abstract and Keywords

This article describes how the minimalist program approaches parametric variation. To do that, however, it is important to step back a little bit and offer some perspective, by considering the different ways in which linguistic variation was dealt with during the Government and Binding era. The discussion is divided as follows. Section 23.2 elaborates on the properties of the initial Principles & Parameters model that allowed investigation of linguistic variation. Section 23.3 discusses the most relevant consequences and proposals that resulted from parametric investigations. Section 23.4 pays special attention to the minimalist approach to universal grammar and its consequences for the study of language variation. Section 23.5 summarizes the main conclusions.

Keywords: minimalist program, parametric variation, linguistic variation, Government and Binding, Principles & Parameters, universal grammar

23.1 Introduction

Since its early stages, linguistics has been concerned—in one form or another—about comparative work, studying contrastive evidence to classify languages and establish paradigms. Particularly relevant was the endeavor carried out by nineteenth century grammarians and their 'comparative method'. Such a method, which devoted much work to reconstruct previous linguistic systems and engineer genetic networks of Indo-European languages, aimed at determining a common origin by looking at morphophonological properties of the type structuralists and neogrammarians focused on. An example of this line of research is offered in (1), which illustrates Grimm's Law on phonological correspondences between Germanic and non-Germanic languages.

(1) Phonological change (Proto-Indo-European to Proto-Germanic)[1] (Campbell 2001: 91)

	French	English
a. *p 〉 f	*pied*	*foot*
b. *t 〉 θ:	*trois*	*three*
c. *k 〉 h:	*cceur*	*heart*
d. *d 〉 t:	*dent*	*tooth*
e. *g 〉 k:	*grain*	*corn*
f. *bh 〉 b:	*frère* (from *bhrater)	*brother*

(p. 524)

A leading idea at that time was that different languages were related if they had a similar 'inner structure'—where 'inner' was interpreted in a mere morphophonological fashion. This notion was later reinterpreted by structuralists in order to develop and sharpen units such as 'phoneme', 'morpheme', or 'distinctive feature', all of which boosted a descriptive fieldwork leading to typologist studies of the Greenbergian sort (see Greenberg 1963).

Although such comparative concerns have also played a prominent role within the research agenda of generative grammar, this latter approach to language is characterized by two fundamental departures from the nineteenth-century tradition. On the methodological side, generative grammar capitalized on the study of syntax so that the creative and regular nature of language—its 'infinite use of finite means', to use von Humboldt's famous words—could be captured. This was already noted and underscored by Noam Chomsky in his *Aspects of the theory of syntax* (1965), where it was pointed out that 'valuable as they obviously are, traditional grammars are deficient in that they leave unexpressed many of the basic regularities of the language with which they are concerned. This fact is particularly clear on the level of syntax, where no traditional or structuralist grammar goes beyond classification of particular examples' (Chomsky 1965: 5).[2] Much effort was thus devoted to obtain a fine-grained characterization of particular grammars by constructing systems of rules (e.g. VP → V NP, NP → DET N, and so on and so forth) capable of describing the knowledge of a given language by a native speaker (i.e. his/her competence). To the extent that this aim was achieved, a grammar was said to be 'descriptively adequate'.

The proliferation of rule systems provided a very powerful tool to revamp and continue comparativist work, but at the same time gave rise to a conflict with 'a much deeper and hence much more rarely attainable level of adequacy, *explanatory adequacy,* which seeked for a grammar that could be feasibly acquired on the basis of primary linguistic data (PLD)' (Chomsky 1965: 24–7). Put differently, the rule-based approach to language-specific constructions was hard to square with fairly standard learnability conditions, posing 'the problem of explaining how we can know so much given that we have such limited evidence' (Chomsky 1986b: xxv): Plato's Problem (see Boeckx 2006, Chomsky 1986b, 1988).

> **(p. 525)** The two tasks just mentioned are in conflict. To achieve descriptive adequacy, it often seems necessary to enrich the system of available devices, whereas to solve our case of Plato's problem we must restrict the system of available devices so that only a few languages, or just one, are determined by the given data. (Chomsky 1986b: 52)

This tension led to the so-called Principles and Parameters (P&P) approach, which I will review in section 23.2. The description/explanation clash underlies one further distinction, which brings us to the main conceptual shift introduced by generative grammar: the distinction between I-language (the individual manifestation of an abstract linguistic capacity, the faculty of language; FL henceforth) and E-language (the actual outcomes of this capacity). The theory of the I-language is called universal grammar (UG), an initial state S_0 of such abstract capacity common to the species that yields relatively steady states S_S, particular grammars, if the child is exposed to the relevant experience.[3]

The P&P theory changed the way of thinking and asking questions about uniformity and variation of languages, stimulating far-reaching and very productive lines of inquiry that were in the spirit of traditional studies. In the early 1990s, this framework was given a reductionist twist whose goal was to investigate whether the properties of UG can be shown to follow from general properties of organic systems and the interface conditions imposed by the external systems within which FL is embedded. This twist is known as the minimalist program (MP), and it is the main goal of this chapter to discuss how it approaches parametric variation. To do that, however, it is important to step back a little bit and offer some perspective by considering the different ways in which linguistic variation was dealt with during the Government and Binding (GB) era. Discussion is divided as follows. section 23.2 elaborates on the properties of the initial P&P model that allowed investigation of linguistic variation. section 23.3 discusses the most relevant consequences and proposals that resulted from parametric investigations. In section 23.4 special attention is paid to the minimalist approach to UG and its consequences for the study of language variation. section 23.5 summarizes the main conclusions.

23.2 The Principles and Parameters approach

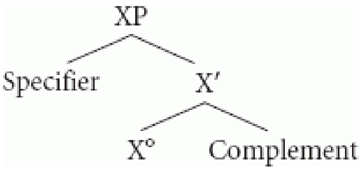
This section is devoted to exploring the most influential developments that the P&P approach introduced through the 1980s and early 1990s with respect to the study of linguistic variation. As will be seen, P&P endowed linguistic theory with a (p. 526) robust and precise apparatus that reconciled descriptivist concerns with explanatory demands by depicting an architecture that was theoretically elegant and capable of accounting for—and harmonizing—'points of invariance' (i.e. principles) and 'points of variation' (i.e. parameters).

23.2.1 From rules to (flexible) principles

As has been noted in the literature (see Boeckx 2006, Chomsky 1986b, 1995c, 2005, 2010, Chomsky and Lasnik 1993, and references therein), the P&P model allowed for the first time the conflict between descriptive and explanatory demands to be solved. To do that, the rule systems of the Extended Standard Theory were replaced by invariant (universal) principles that allowed for linguistic variation—parametrized principles. At the time, it was thought that UG had a rich and modular structure, with different subsystems (Binding, Case, Control, Government, etc.) whose principles would be underspecified, awaiting the relevant setting by linguistic experience. The gist behind this approach can be grasped through Chomsky's famous metaphor (attributed to James Higginbotham) of parameters as switches:

> We no longer consider UG as providing a format for rule systems and an evaluation metric. Rather, UG consists of various subsystems of principles […] Many of these principles are associated with parameters that must be fixed by experience […] Borrowing an image suggested by James Higginbotham, we may think of UG as an intricately structured system, but one that is only partially 'wired up.' The system is associated with a finite sets of switches, each of which has a finite number of positions (perhaps two). Experience is required to set the switches. (Chomsky 1986b: 146)

The modular flavor of the first P&P models, with different UG-internal components scattered—but at the same time interacting—was a highly attractive hypothesis about the nature of cognitive systems (see Boeckx 2009b, Fodor 1983), but it was also methodologically advantageous, as it allowed linguists to dissect phenomena and analyze their parts in a very careful way. X-bar Theory (see Chomsky 1970, Jackendoff 1977), whose schema imposed a head-complement dependency, could account for the Head Parameter if the switch positions were responsible for determining whether the head precedes (English *with us*) or follows (Latin *nobiscum*) its complement:



(2) X-bar schema (linear order irrelevant)
(3) Head Parameter (see Travis 1984)
A head X precedes/follows its complement.

(p. 527) A similar reasoning could be extended to Move a, the GB transformational rule, if the switch forced displacement after or before Spell-Out. Relevant applications of this parameter are the V-to-I Parameter or the *Wh*-Movement Parameter (see Chomsky 1991, 1993, Holmberg and Platzack 1995, Huang 1982, Vikner 1995).

Regardless of details and specific instantiations, notice that the kind of variation that was advocated for by Chomsky (1981a, 1986b) was already encoded in the universal principles. In other words, the very possibility for linguistic variation was already hardwired in UG, by means of parametrizable principles.

23.2.2 Parameter-setting effects

For the reason just mentioned, changes in a single parameter could have massive, cascade-like, effects, with 'proliferating consequences in various parts of the grammar' (Chomsky 1981a: 6). Such consequences, known as 'clustering effects' (see Boeckx forthcoming, Biberauer 2008, Biberauer et al. 2010, and references therein), were seen as a consequence of setting a principle in a specific manner. Take, for instance, Rizzi's (1982, 1986) influential Null Subject (or pro-drop) Parameter (NSP), which was originally related to the availability of rich morphology and pro licensing, but was nonetheless shown to have a cascade of syntactic consequences, as the list in (4) shows:

(4) Clustering effects (NSP) (Chomsky 1981a: 240)
- **(i)** missing subject
- **(ii)** free inversion in simple sentences
- **(iii)** 'long *wh*-movement' of subject
- **(iv)** empty resumptive pronouns in embedded clause
- **(v)** apparent violation of the *[*that-t*] filter

Similar cascade effects were attributed to the positive setting of Snyder's (1995) Compounding Parameter (CP), as can be seen in (5).

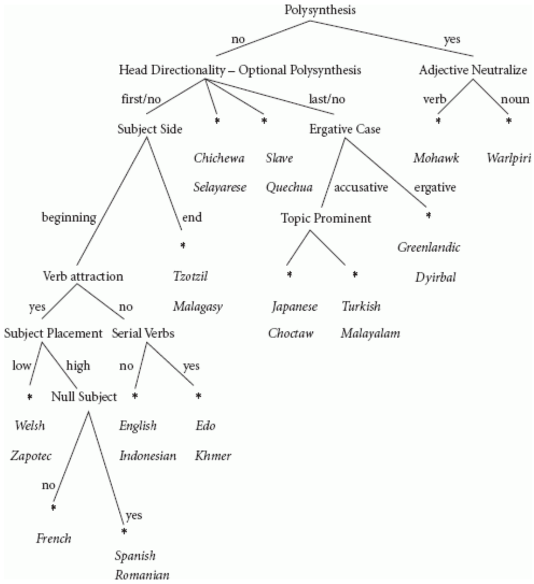
(5) Clustering effects (CP)
- **(i)** productive N–N compounding
- **(ii)** verb—particle constructions
- **(iii)** double object constructions
- **(iv)** manner incorporation (satellite-framed languages)
- **(v)** preposition stranding
- **(vi)** non-adverbial/true resultatives

However, perhaps the clearest example of cascade effects is to be found in Mark Baker's well-known Polysynthesis Parameter (see Baker 1996, 2001), whose formulation and consequences can be seen in (6):

(p. 528) **(6)**

a. Polysynthesis Parameter (Baker 1996: 14)
Every argument of a head element must be related to a morpheme in the word containing that head (an agreement morpheme, or an incorporated root).
Yes: Mohawk, Nahuatl, Southern Tiwa, Mayali, Chukchee, (Mapudungun)…
No: English, Spanish, Chichewa, Japanese, Quechua, Turkish, (Kinande)…

b. Parameter Hierarchy (Baker 2001: 183)



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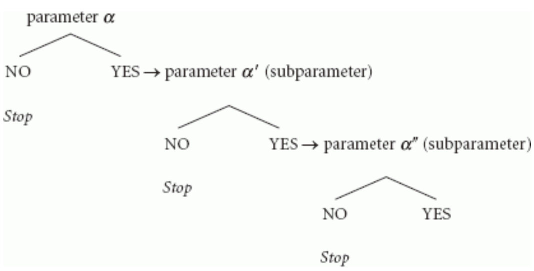
Viewed this way, parameters are nothing but constrained (relativized) principles. Therefore, anyone working on parameters could in fact be said to be working on principles—as Richard Kayne notes when observing that 'the study of syntactic parameters and the study of syntactic unversals go hand in hand' (Kayne 2000: vii). At first glance, one could tentatively try to pin down the characteristics of a (syntactic/structural) parameter as follows:[4]

(7) Properties of (syntactic) parameters
(i) they instantitate the degree of variation (i.e. flexibility) of universal principles;
(ii) they require experience (the relevant 'cues'; see Lightfoot 1991, 1999);
(iii) they can have syntactic consequences (clustering effects).

In Chomsky (1986b) it was further proposed that the degree of variation of parameters (i.e. the number of positions for the switches) was limited to the minimum: two, as said above (see Chomsky 1986b: 146). It must be noted that this is not forced by anything deep, and in fact empirical evidence could be interpreted to argue for a three-way splitting of parameters. This could be the case of the NSP, where radical pro-drop, non-pro-drop, and partial pro-drop languages have been identified (see Biberauer et al. 2010, Kayne 2000). Be that as it may, the binary option was a reasonable methodological assumption, apparently supported by more general factors, like Berwick's (1985) Subset Principle, stating that one of the choices (the unmarked one) generates a subset of the expressions generated by the other.

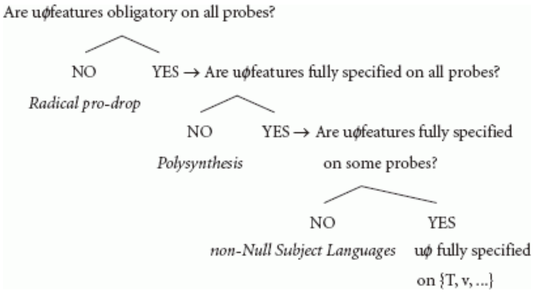
Such bifurcated nature of parametrized principles has been exploited in formulating different versions of 'parameter schema' (see (6)), which, as can be seen below, yields a species of embedding continuum in which more marked parameters (or sub-parameters, for which more positive data are needed) spawn (see Baker 1996, 2001, Boeckx forthcoming, Gianollo et al. 2008, Biberauer et al. 2010, Roberts and Roussou 2003, Uriagereka 1995).

(8) Parameter schema



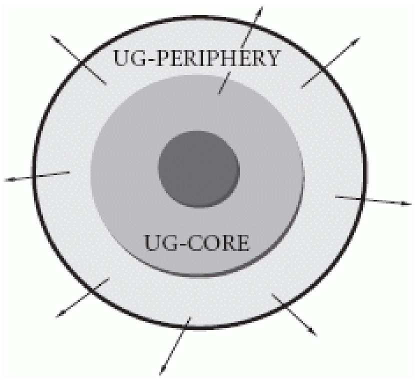
In the particular case of the NSP, Biberauer et al. (2010) propose the following application of (8):

(p. 530) **(9)** Null Subject Hierarchy



23.2.3 Core grammar and marked periphery

One final distinction of the P&P approach, also relevant for parametric investigations, has to do with Chomsky's (1981a) cut between core grammar and marked periphery (see (10)). The core grammar was taken to be the form adopted by UG after parameter setting, an idealized version of UG more or less homogeneous (regular) within a speech community. The fuzzy periphery was in turn seen as a repository of idiosyncratic properties, borrowings, irregular forms, and marked constructions, plausibly of the kind studied by Culicover and Jackendoff (2005) and many related construction-based and typological approaches (see Croft 2001, Goldberg 1995).[5]



(10)

(p. 531) The periphery shares, at the relevant level, many traits with a Bloomfieldian-Saussurian lexicon (see section 23.4), a list of exceptions and idiosyncratic phenomena that must be consciously learned, one by one, by the speaker.

Overall, the P&P approach proved stimulating, productive, and influential on different grounds. It not only offered an articulated and robust theory to approach the tension between diversity and uniformity without succumbing to the idea that ‘languages can differ from each other without limit and in unpredictable ways’ (as Martin Joos, a representative figure of American structuralism, claimed; see Chomsky 2010), but also—and more importantly—made it possible to overcome the tension created by the necessity to meet both descriptive and explanatory demands. In the next section, I would like to review the most prominent studies that stemmed from the P&P approach, as well as some problems that were quickly noted in the literature.

23.3 Parametric syntax

The P&P approach outlined in Chomsky (1981a) started a new wave of studies on comparative syntax within the generative enterprise whose goal was to provide detailed and comprehensive surveys of language-specific phenomena (clitics, *wh*-movement, subject positions, linear order, etc.) in order to attain a better understanding of UG (see Baker 2001, Belletti and Rizzi 1996, Cinque and Kayne 2005, Kayne 2000, and references therein for relevant discussion). Broadly speaking, most progress within the post-LGB era concerned two different domains, which I would like to discuss here: (i) the macro-micro-parameter distinction, and (ii) the growing interest on functional categories and syntactic representations.

23.3.1 The rising of microparameters

It was Borer (1984) who first asked the question of where (i.e. 'where in UG') parameters were. Recall that, as it was first outlined, the P&P approach endorsed the hypothesis that parameters and principles were not different entities: the former were nothing but adjusted versions of the latter. Note that, unless qualified, Chomsky's initial perspective would be consistent with different subsystems of UG (Case Theory, X-bar Theory, etc.) being the locus parametric setting. This was a somewhat unappealing idea from a perspective where syntax is universal (uniform or immune to variation, by definition). Borer (1984)—and also Fukui (1986), Fukui and Speas (1986), Ouhalla (1988), and Webelhuth (1992)—gave a slight but ultimately crucial twist to Chomsky's conception by proposing that parameters were to be located in the lexicon, an idea that was later embraced by Chomsky himself (see Chomsky 1995c, 2000a, 2001): 'there are no language-particular choices with respect to the realization of universal processes and principles. Rather, interlanguage variation would be restricted to the idiosyncratic properties of lexical items' (Borer 1984: 2). Following Baker (2008), let us refer to this thesis as the Borer–Chomsky Conjecture (BCC):

(11) The Borer–Chomsky Conjecture (Baker 2008: 353)
All parameters of variation are attributable to differences in the features of particular items (e.g. the functional heads) in the lexicon.

The BCC signaled a turning point with respect to previous attempts to approach variation by drawing a dramatic line between macroparameters and microparameters. Rizzi's (1982) NSP is the typical example of a macroparameter (but see Baker 2008 for qualifications): A parameter hard-wired in the syntax itself (actually, a principle after its setting) with large-scale consequences, typically visible in non-related languages. Microparameters in turn were parameters of the BCC type, with limited consequences that result from features being associated to specific lexical items, typically visible in closely related languages. As Baker (2008: 355–6) puts it:

> The standard microparametric view is that the primitive, scientifically significant differences among languages are always relatively small-scale differences, typically tied to (at most) a few closely related constructions […] Large differences between languages always reduce to lots of these small differences […] In contrast, the macroparametric view is that there are at least a few simple (not composite) parameters that define typologically distinct sorts of languages. For example, there might be a single parameter in the statement of Merge that induces the core difference between head-initial and head-final languages (Stowell 1981). There might be a single parameter that lies down the core structure of a nonconfigurational polysynthetic language as opposed to more configurational, isolating languages (Baker 1996). And so on.

Building on Baker (2008), the macro/micro cut can be narrowed down to the following three properties.

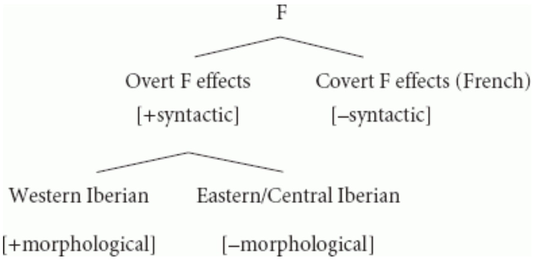
(12) Microparameters/macroparameters

	MACROPARAMETERS	MICROPARAMETERS
The locus of variation	Syntax (P&P principles)	Lexical (functional) items
The extent of variation	Strong (and manifold) effects	Small (easily detectable) effects
The methodology of comparison	Comparison of historically unrelated languages	Comparison of related languages (or dialects)

(p. 533) As has been noted in the literature (see Piattelli-Palmarini et al. 2009 and references therein), there were empirical reasons to abandon the macroparametric perspective, and regard macroparameters as nothing but a side effect of clustering effects (see Biberauer et al. 2010, Roberts 2006). To begin with, it was soon recognized that some principles (say, Condition C of Binding Theory) were largely not parametrizable. It was also noted that macroparameters were often adscribed to particular lexical elements (the (non-)pronominal nature of T, the morphologically defective status of clitics, etc.), thus virtually blurring the macro/micro distinction. And there were methodological reasons too. Pursuing the microparametric path was closely related to studying the acquisition task of the child (variation was restricted to the lexicon, where the assembling of simple and limited features holds the key to acquiring a language), which also made the study of UG more feasible (since the relevant linguistic tests are more revealing and easy to run when they target closely related languages); the latter point is better phrased in Kayne's observation that '[I]t is not that microcomparative syntax is eas[ier] to do, nor that one is guaranteed of success […] It is rather, I think, that the probability of correctly figuring out what syntactic property is parametrically linked to what other one […] is higher' (Kayne 2005c: 282).

But this said, the 'macro' and 'micro' approaches to parameters are not necessarily exclusive. One good example of this is Juan Uriagereka's work on FP (the F Parameter), a syntax-pragmatics encoding projection that aimed at capturing the so-called 'hot' behavior of Romance languages. The implementation of the FP undertaken by Uriagereka involved both a syntactic and a morphological parameter, both of them equipped with a binary setting: (i) Does a given language L have FP? (yes or no), (ii) if yes, is F morphologically rich? (yes or no). To be more specific:

(13) Uriagereka's (1995) F Parameter



As can be seen, the first binary choice was, in Uriagereka's formulation, syntactic (a macroparameter), determining whether F was or not projected in the CP layer of a given language, whereas the second one was morphological (a microparameter). (p. 534) Proposals along these lines show that the macro and micro perspectives, though apparently incompatible and contrary, can benefit from each other in trying to ascertain the properties of related languages.

23.3.2 The Cartographic Project

The BCC-based logic to corner variation into the lexicon became very popular and influenced different research projects that can be traced until present day (see Belletti and Rizzi 1996, Biberauer 2008, Biberauer et al. 2010, Cinque 2005, Holmberg and Platzack 1995, Kayne 1981a, 1989b, 1991, 1995, 2000, 2005c, Longobardi 1994, Thráinsson 1996, among many others). Furthermore, it was enhanced by a hotly debated topic in the literature of the late 1980s and early 1990s: the nature of functional categories. The pioneering work by Abney (1987), Chomsky (1986a), Pollock (1989), Fukui (1986), Fukui and Speas (1986), and many others opened the door to a full-blown inventory of both inflectional and semantic formatives that ultimately gave rise to what has been known as the Cartographic Project (see Belletti 2004a, Cinque 1999, Rizzi 1997, 2004a, and much related literature). This particular approach to the nature of syntactic structures, whose main goal is to 'draw maps of syntactic configurations as precise and detailed as possible' (Rizzi 2004b: 223), came in handy to work out highly articulated, very flexible, and language-specific typological descriptions that reveal ordering constraints within dedicated syntactic regions (the so-called Left Periphery).

The influential proposal by Rizzi (1997) that C should be split into different functional categories (as many as needed in order to make room for the distinct left-periphery-mates) was quickly pushed to other clausal and non-clausal domains, where functional elements (most remarkably, V, D, and P) underwent a thorough process of decomposition (similar in spirit to the lexicon project of Hale and Keyser 1993; see section 23.4) that tried to embed semantic properties—*qua* functional heads—in the syntax.

Perhaps the most exhaustive and systematic cartographic study corresponds to Cinque's (1999) adverb hierarchy, reproduced in (14):

(14) Cinque's (1999: 106) Universal Hierarchy [frankly $Mood_{speech\ act}$ [fortunately $Mood_{evaluative}$ [allegedly $Mood_{evidential}$ [probably $Mod_{epistemic}$ [once T(Past) [then T(Future) [perhaps $Mood_{irrealis}$ [necessarily $Mod_{necessity}$ [possibly $Mod_{possibility}$ [usually $Asp_{habitual}$ [again $Asp_{repetitive(I)}$ [often $Asp_{frequentative(I)}$ [intentionally $Mod_{volitional}$ [quickly $Asp_{celerative\ (I)}$ [already T(Anterior) [no longer $Asp_{terminative}$ [still $Asp_{continuative}$ [always $Asp_{perfect(?)}$ [just $Asp_{retrospective}$ [soon $Asp_{proximative}$ [briefly $Asp_{durative}$ [characteristically(?) $Asp_{generic/progressive}$ [almost $Asp_{prospective}$ [completely $Asp_{SgCompletive(I)}$ [tutto $Asp_{PlCompletive}$ [well Voice [fast/early $Asp_{celerative(II)}$ [again $Asp_{repetitive(II)}$ [often $Asp_{frequentative(II)}$ [completely $Asp_{SgCompletive(II)}$ …

(p. 535) Building on a careful empirical study, Cinque (1999) claims that the hierarchy in (14) is always present, and uniformly ordered, in all languages. Cinque's contention, and the Cartographic Project as a whole, has been criticized on empirical grounds (see Bobaljik 1999 and Nilsen 2003 for arguments against Cinque 1999; see Cinque 2004 for a reply). Conceptually, there are different questions that can be asked about cartographies: (i) What is the motivation for these functional heads? (ii) What is the nature of the hierarchical ranking? (iii) What is the nature of the linear order constraints?—and there are of course more. Not many attempts to provide principled answers to (ii) and (iii) have been made in the literature (see Boeckx 2008a, and Fortuny 2008), but (i) has indeed been addressed, to argue that the hierarchy is either part of UG (see Cinque 1999, 2004, Starke 2004, 2009) or part of the Conceptual-Intentional systems (see Ernst 2002, Fortuny 2008).

Synthesizing, the Cartographic Project can be seen as a fine-grained extension of GB-type analyses that allows for the sort of detailed descriptions that nicely fit with the prominent role of constructions in descriptive work.

23.3.3 Problems for microparameters

Though perhaps not at first glance, it is easy to see that the idea that parameters should be placed in the lexicon, coupled with the explosion of functional categories advocated for by the Cartographic Project, yields a scenario not too distinct from the one that motivated the investigations that led to the P&P model in the first place. As Roberts (2006: 4) emphasizes, a blind application of the BCC thesis results in a rather unrealistic upper bound of the number of grammars where ‘languages should appear to vary unpredictably and without assignable limits, even if we have a UG containing just 80–1000 parameters’ (see Kayne 2000 for related matters). In other words, whereas P&P was designed devoid of language-specific rules (passive transformation, *do*-insertion, *wh*-movement, etc.) it ended up putting forward a yet to be determined list of functional categories with a binary setting that yields a quite intricate system. There is not only the issue of how many functional categories there are (and what their settings can be), but also in what order they are fixed, or when they are spelled-out. Notice that this is not to deny the conceptual and empirical advantages of a microparameteric approaches (especially if considered vis-à-vis macroparametric ones), but, at the relevant level of abstraction, microparameter theorizing seems to pose a second (descriptive-explanatory) tension, analogous to the one P&P was meant to solve.

One other critique to microparameters arises from the fact that many of the expectations that the parametric approach helped create (the ‘clustering effects’ that they were supposed to trigger; see section 23.2) were never completely met. Let us go back and consider, for concreteness, the NSP, essentially as formulated (p. 536) by Chomsky (1981a), Rizzi (1982, 1986), and Taraldsen (1980), which established an implicational relationship between morphological richness of the verbal paradigm (still a controversial notion; see Bobaljik 2001, and references therein) and a cluster of morphosyntactic phenomena, like circumvention of *that*-trace effects, availability of free subject inversion, and of course pro-drop. However empirically well supported, a cursory look at cross-linguistic data suffices to see that these correlations are not too strong. Firstly, languages such as Chinese are not in any intuitive sense morphologically rich, but they license pro, whereas Russian, a morphologically rich language, fails to do so. Secondly, Catalan, Italian, and Spanish allow for subject inversion (VS and VOS sentences), but only the latter rules in VSO (see Belletti 2004a, Picallo 1998, Zubizarreta 1998). Finally, there are some languages, like Brazilian Portuguese, which manifest weak NSL effects, qualifying as a partial NSL (much like Finnish, as argued for by Holmberg 2005).

These mismatches have been seen by some as a failure of the entire parametric approach (see Newmeyer 2004, 2005, who takes clusters to be mere ‘tendencies’ that are better captured by functionalist explanations that deal with performance effects, such as parsing strategies). An alternative that has recently been put forward is Boeckx's (forthcoming: 11) claim that clustering effects are the result of (third-factor) strategies that highlight formal similarities in order to minimize the learning task:

> [T]he guiding intuition I would like to pursue is that clustering results from the child seeking to maximize the similarity across parameter settings, harmonize their values, and thereby economize what must be memorized (via the formation of a generalization across similar parameters). My suggestion goes in the direction of a recent trend expressing a renewed interest in the nature of learning and the interaction between nature and nurture (see Yang 2002, 2004, 2005; Pearl 2007). As Yang and Pearl have made clear (in a linguistic context; see Gallistel 1990, 2006 for more general remarks on learning modules), a proper characterization of the learning task requires paying attention to three important components: (i) a defined hypothesis space (for language, UG), (ii) data intake, and (iii) a data-update algorithm.

Boeckx (forthcoming) proposes a Superset Bias in order to implement this idea, underscoring its role in grouping (read 'clustering') similar formal patterns. If the child has evidence that V precedes its complement, and D and T do too, then he/she will hypothesize that all heads precede their complements (the Head Parameter), which will only be rejected if enough counterevidence is found.

(15) Superset Bias (Boeckx forthcoming: 12)
Strive for parametric value consistency among similar parameters

Whatever the interpretation of the facts, what should be out of the question is that the original outline of parameter clusters cannot be maintained. Newmeyer (2004, 2005) is then plausibly right in claiming that the alleged clusters do not (p. 537) belong to the UG core—the very point that microparameter theory tries to make—but this does not necessarily mean that the 'tendencies' can only be accounted for in terms of communicative or parsing strategies (see Hawkins 2004).

23.4 Minimalism

In the early 1990s, Chomsky sketched what a minimalist program for linguistic theory should be like (see Boeckx 2006 and Uriagereka 1998 for much discussion and a detailed introduction). The roots of the MP can be traced back to the P&P impulse to reduce the formal complexity of previous models through mechanisms and economy conditions that had a 'least effort' flavor to them (see Chomsky 1991, 1993, 2000a). Although this was already one specific aim of the P&P approach, it was not until the advent of MP that this simplication was really feasible. This was particularly clear in Chomsky (1986a), where the computational metrics postulated to unify the list of islands identified by Ross with a series of independent locality conditions were in fact quite complex. Such a methodological endeavor, which has always been present in generative models in one form or another, is not the key trait of minimalism, though. It is what Martin and Uriagereka (2000) dub 'ontological minimalism' that really defines the MP as something brand new: from the perspective of minimalism we are no longer asking how elegant our theory of the faculty of language can be (without government, without X-bar Theory, without construction-specific filters, without internal levels of representation, etc.), but rather how elegant the faculty of language itself is. Ontological minimalism can thus be regarded as a strong thesis about language design, embodied by the idea that language communicates with external systems of human biology in an optimal way (see Chomsky 2000a).

Notice, though, that entertaining Chomsky's Strong Minimalist Thesis (SMT) might tacitly entail abandoning—or, more precisely, sidestepping—the '& Parameters' component of the 'Principles and Parameters' model. Put another way, since linguistic inquiry now seeks to recast principles from computational efficiency considerations and external (i.e. interface) conditions, it is hard to see how parameters would fit in the minimalist picture—and, even if they did, how they could be approached. In brief, variation is not a priori expected in an optimal natural object (as it is 'a *prima facie* imperfection'; see Chomsky 2001), so the common practice has been to put parameters aside, and focus on UG principles.

In this section I would like to discuss how a core aspect of generative theorizing during the last thirty years (the parameters) can be accommodated under minimalist desiderata. As will be seen, the only route that can be pursued in fact forces us to persevere and develop the BCC, assuming that variation is restricted to the lexicon (p. 538) and related to the interaction between narrow syntax and the morphophonological component (what Chomsky 2007, 2008a, 2010 refers to as 'externalization').

23.4.1 Factors of language design

A good starting point to approach minimalism is Chomsky's (1965: 59) hypothesis that the faculty of language is regulated by three factors (see Chomsky 2004a, 2005, 2007).

(16) Factors of language design (Chomsky 2005: 6)
a. Genetic endowment
b. Experience
c. Principles not specific to FL

As we have seen, in P&P attention was mainly paid to overcoming the tension between Factor I and Factor II—between descriptive and explanatory adequacy. To the extent that the logical problem of language acquisition was appropriately addressed by P&P, minimalism focuses on Factor III. This takes us to the realm of *why*-questions that define Galilean inquiry (see Boeckx 2006). Minimalism is thus not only asking 'what the properties of language [Factor I] are but also why they are that way' (Chomsky 2004a: 105), investigating the role played by 'language independent principles of data processing, structural architecture, and computational efficiency' (Chomsky 2005: 9).

The SMT above compels minimalists to recast substantitve principles from considerations about computational efficiency and properties of the systems with which the faculty of language must interact, i.e. the Sensorimotor and Conceptual-Intentional systems. More generally, minimalism seeks to 'show that the basic principles of language [can be] formulated in terms of notions drawn from the domain of (virtual) conceptual necessity' (Chomsky 1993: 171). This leitmotiv had a clear impact on the architecture attributed to UG, which dispenses with the three (internal) levels inherited from the so-called 'Y-model' of Chomsky and Lasnik (1977), together with mechanisms and formatives that appear to have no counterpart in the biological world (indices, government, etc.).

When it comes to variation, minimalism appears to stick to some variant of the BCC (see Chomsky 2001: 2, 2007: 6, 2008a: 135), but such a variant has not been pursued in the minimalist literature in a comprehensive way (with few exceptions, like Gallego 2007, Mendívil 2009, Biberauer et al. 2010, and references therein). The reason is obvious. The interest is in Factors I and III, where variation does not have a natural place, as it does not seem to contribute, in any conceivably coherent way, to set specifications of optimal design. The question that could be asked is thus as follows: in what sense is parametric variation an indication of optimal design? Or, even more radically (and relevantly): 'why are there so many languages?' (Chomsky 2010).

(p. 539) Let us now go back to the three factors, and try to ask the question of where (among them) variation belongs. Factor I is the expression of our genetic endowment (the faculty of language), which should ideally be uniform across the species. Chomsky (2001: 2) captures this assumption by postulating a Uniformity Principle (variation is 'restricted to easily detectable properties of utterances') strengthened by Boeckx (forthcoming) as a thesis.

(17) Strong Uniformity Thesis (SUT) (Boeckx forthcoming: 5)
Principles of narrow syntax are not subject to parametrization; nor are they affected by lexical parameters.

The SUT amounts to the idea that whatever is part of UG is not subject to change.[6] One could reinterpret Boeckx's (forthcoming) SUT as the requirement that, for example, Merge cannot vary in the way it applies—it cannot be the case that Merge takes two syntactic objects in a language L_1, but three in a language L_2.Different aspects may conspire to determine the binary nature of Merge (see Chomsky 2008a), but once it is ascertained, it should not manifest cross-linguistic variation.

Factor II is the external data (E-language), which provides the experience necessary for the initial state of UG to stabilize, yielding a specific (I-)language and variation.

Factor III is at the heart of the MP, what truly distinguishes it from its predecessors. The SMT of minimalism tries to find out to what extent Factor III effects are operative in setting Factor I (UG). What are these effects? Chomsky (2005) distinguishes at least two types of Factor III effects:

(18) Factor III
a. Principles of data processing
b. Architectural/computational-developmental constraints

Let us focus on the second category, for it is the one that most clearly corresponds with principles of efficient computation and interface conditions. In the minimalist literature, they fall within five general groups:

(19) Architectural/computational-developmental constraints
 a. Inclusiveness Condition (see Chomsky 1995c)
 b. Minimal Link Condition/Relativized Minimality (see Chomsky 1995c, Rizzi 1990)
 c. Extension Condition/No Tampering Condition (see Chomsky 1993, 2000a)
 d. Phase Impenetrability Condition (see Chomsky 2000a, 2001)
 e. Full Interpretation Principle (see Chomsky 1986b)

(p. 540) Much like Factor I, Factor III is unlikely to be the source of variation, as it comprises 'principles of neural organization that may be more deeply grounded in physical law' (Chomsky 1965: 59), 'principles of a more general character that may hold in other domains and for other organisms, and may have deeper explanations' (Chomsky 2008a: 135). It has sometimes been noted that certain locality conditions (minimality or bounding nodes) can be weakened or even ignored by some languages (see Baker and Collins 1996, Rizzi 1982), but that seems conceptually problematic, even if empirically attested. It is almost impossible not to concur with Boeckx (forthcoming) that a picture where locality conditions were obeyed in Italian, but not in English, is highly implausible —it is of course possible, but if so, computational efficiency would not be efficient at all, and the goal of characterizing UG and its interaction with the external systems would become extremely hard, perhaps impossible.[7]

All in all, if variation cannot be due to Factor I (because of its uniform nature), nor Factor III (because of its deeply invariant character), then where should we look for it? Chomsky's answer to this question, to be elaborated in the next section, is that the lexicon is the locus of parametrization. Variations thus emerge through the interaction of Factor I (which provides lexical items, or the features necessary to assemble them; Chomsky 2000a: 100, 2001: 10, 2005: 4, 2007: 6) and Factor II, the experience (see M. D. Richards 2008 for explorations of this idea). Identifying the locus of variation is equivalent to answering how variation works, but we should not forget that minimalism also seeks to ask and answer *why*-type questions. I return to this in section 23.4.3, where I consider Chomsky's contention that the existence of so many languages may be due to the fact that the faculty of language is not optimal for externalization (i.e. the mapping to the morphophonological component), although arguably it is for interactions with the semantic component that connect with the Conceptual-Intentional systems.

23.4.2 UG: a look from below

In the preceding section I referred to the minimalist thesis that UG (Factor I) is supposed to be small and simple, not only for evolutionary reasons (see Hauser, et al. 2002, Piattelli-Palmarini and Uriagereka 2005), but also to comply with the SMT, which seeks to provide principled ('beyond' explanatory adequacy) accounts of language principles, where 'principled' means that the principles 'can be reduced to the third factor and to conditions that language must meet to be usable at all' (Chomsky 2008a: 134). In brief, the less we can attribute to UG, the more principled (i.e. less stipulative) our account of principles will be.

(p. 541) Pursuing this hypothesis, Chomsky has recently argued for what one could call bare version of UG, crucially different from the 'highly structured theory' that was postulated in the GB era (Chomsky 1981a: 3). UG should now be approached—Chomsky reasons—from below, and not from above:

> Through the modern history of generative grammar, the problem of determining the character of FL has been approached 'from top down': How much must be attributed to UG to account for language acquisition? The MP seeks to approach the problem 'from bottom up': How little can be attributed to UG while still accounting for the variety of I-languages attained, relying on third factor principles? The two approaches should, of course, converge, and should interact in the course of pursuing a common goal. (Chomsky 2007: 4)

Just to get an idea of what UG would look like from these different angles, consider the properties that are adscribed to it below (this is only a tentative sketch, and different alternatives easily come to mind). I will refer to these variants of UG as UG_{GB} (UG from above, rich, highly structured) and UG_{MP} (UG from below, small, uniform).

(20) UG from above (see Chomsky 1981a: 5)

System of rules	Subsystems of principles
(i) Lexicon **(ii)** Syntax: **(a)** categorial component **(b)** transformational component **(iii)** PF-Component **(iv)** LF-Component	**(i)** X-bar Theory **(ii)** Bounding Theory **(iii)** Government Theory **(iv)** Theta Theory **(v)** Binding Theory **(vi)** Case Theory **(vii)** Control Theory

(21) UG from below (see Chomsky 2007)

Factor I (UG)	Efficiency and interface conditions (not UG, but Factor III)
(i) Set of features F **(ii)** Merge	**(i)** Inclusiveness **(ii)** Minimality **(iii)** No Tampering Condition **(iv)** Phase Impenetrability Condition **(v)** Full Interpretation

(p. 542) From the comparison between (20) and (21) it should be clear that the structure of UG_{MP} is substantially smaller than that of UG_{GB}. The subsystems of principles of UG_{GB} are now regarded as part of Factor III, which already entails an important reduction. (21) furthermore proposes that UG_{MP} only consists of features (which, once assembled, yield a lexicon)[8,9] and the recursive operation Merge—nothing else.[10] Following M. D. Richards (2008), I will assume that Agree can be reduced to the existence of uninterpretable features, which require valuation.[11] In sum, the slim UG (i.e. UG_{MP}) depicted in (21) is nothing but the profile of a minimalist view where 'everything is Merge' (see Boeckx 2009a, Chomsky 2004a, 2008a).

As we noted above, minimalism can only seriously consider the kind of parametric variation that arises from the interaction of Factor I (more precisely, the set of features F) and Factor II, the latter helping select and assemble members of F into a language-particular lexicon, the true locus of variation. Consequently, before going ahead, we should try to be a bit explicit as to what properties we adscribe to the lexicon.

Mainstream minimalism endorses an idiosyncratic conception of the lexicon (a 'repository of irregularities', roughly in the sense of Bloomfield and Saussure; see Chomsky 1995c: 235), which fits nicely with the unprincipled nature of variation.[12] Recent investigations in the syntax—semantics and syntax—morphophonology interfaces (most notably, the exo-skeletal approach of Borer 2005, the Lexical Syntax project of Hale and Keyser 1993, and the distributed morphology framework founded by Halle and Marantz 1993) have made important contributions that invite us to reconsider the information that has traditionally been labeled as lexical. **(p. 543)** Hale and Keyser's (1993) thrust is that the lexicon is nothing but a syntax.[13] More precisely, the lexical entry of an item 'consists in the syntactic structure that expresses the full system of lexical grammatical relations inherent in the item […] here, as elsewhere, the syntactic structure itself is determined by general syntactic principles' (p. 96). A similar hypothesis is put forward by Halle and Marantz (1993) and their associates (see Embick 2007, 2008a, Harley and Noyer 1999, and references therein), who in turn claim that traditional lexical entries are not atomized but scattered in different lists: a list of morphosyntactic formal features (list A), a list of phonological features (list B), and a list of semantic-encyclopedic features (list C). Note that these authors are ultimately arguing against lexicalism, suggesting that the lexicon is to be approached from below too. Both Hale and Keyser (1993) and Halle and Marantz (1993) propose, in brief, that we 'put things where they belong'. If we do that, variation does not have to be seen as belonging to some ill-understood and messy pre-syntactic inventory, and we can truly aspire to give variation a more precise location—one (or more) of the DM lists.

Let us now consider the types of features that are typically associated with lexical items in more detail. Restricting the set F to the typology entertained by Chomsky (2001), we can distinguish: phonological ([±sonorant], [±voice], etc.), semantic ([±count], [±specific], etc.), and formal ([Case: x], [number: x], etc.) features.[14] The question that arises is: what features are involved in variation? Under the assumption that only the syntax—semantics mapping is optimal (stable, uniform, etc.), it would be odd for semantic features to be a source of variation, which leaves us with formal and morphophonological features as more likely suspects. Building on ideas by Chris Collins, Kayne (2005c, 2008) suggests that parameters are related to the former:

(22) Parametric variation is a property of unvalued features only.[15]

Note that (22) is reminiscent of the BCC, particularly if unvalued features are inflectional. Perhaps some subset of formal features (or all of them, if Kayne 2005c is right) could be responsible for language variation, but that is also unlikely. I take it, in line with Vergnaud's ideas on abstract case, that formal features are present in all languages, variation being determined by their phonological realization (see next section). If correct, then variation would boil down to the actual manifestation of formal features, which takes us to the last type of features: phonological features. In this respect, Chomsky (2001: 18) interestingly observes:

> **(p. 544)** Language design is such that [semantic features] and [formal features] intersect and are disjoint from [phonological features] […] It also seems that FL may retain something like [semantic-formal] and the narrow syntax into which they enter while the phonological component is replaced by other means of sensorimotor access to narrow-syntactic derivations.

Considered together, the observations by Chomsky (2001) and Kayne (2005c, 2008) appear to place variation in the morphophonological manifestation of closed classes (i.e. functional categories, which contain unvalued features).[16] At this point it is worth going back to Chomsky's remark about phonological features: unlike formal and semantic ones, they are added after the computation (after Transfer applies). In a specialized architecture like the one advocated for by Halle and Marantz (1993), Chomsky's words would entail that variation is restricted to the 'list B' (the vocabulary items). Notice, incidentally, that taking the list B as the unique locus of variation can make sense out of the idea that 'variety [is] restricted to easily detectable properties of utterances' (Chomsky 2001: 2). If so, then the conclusion is that parametric variation can indeed be located in the lexicon, but only in one of its lists: the one that contains phonological features (the vocabulary items that undergo late insertion, as Boeckx 2009a claims).

We can leave the discussion at this point. The goal of this section was twofold. Firstly, I introduced Chomsky's (2007) bottom-up perspective on UG, which clearly contrasts with the top-down version entertained in GB. UG is now reduced to a bare minimum, with a set of features [F] that gives rise to a lexicon, and the unavoidable structure building operation Merge. All the rest is dumped into the external systems—recast as interface conditions or efficiency principles. Under the quite reasonable assumption that Merge is not suspect of variation, the only suspect left is the lexicon. The second goal was precisely to investigate different minimalist lexicon projects: Borer (2005), Hale and Keyser (1993), and Halle and Marantz (1993). After briefly discussing the main claims made by these approaches, we have concluded that variation should be concentrated in the morphophonological part of the lexicon, which, in distributed models, is outside the narrow syntax, where optimal design traits (i.e. inclusiveness, full interpretation) are neither found nor expected (see Chomsky 2000a: 118). In what follows I would like to reinterpret this in the light of Berwick and Chomsky's (forthcoming) claim that language variation may be due to how externalization works.

23.4.3 Variation: addressing the *why*-question

In previous sections we saw that parametric variation can be handled through binary option schemas (to which I return). That concerns the *how* question. However, **(p. 545)** so far we have not asked the beyond explanatory adequacy question of *why* variation exists. Notice that this question was not considered, let alone formulated, in the GB period, since variation was regarded as an integral component of UG. Since UG was assumed, and variation was encoded in UG, there was really no *why* question to pose. Asking why variation exists was, from that perspective, not different from asking why UG exists—and no real difference obtained with the lexicalist approach to parameters stemming from the BCC.

Minimalism demands a different way to consider parameters, as UG (Factor I) cannot be the source of variation. In particular, minimalist logic makes us consider Factor III conditions as being responsible for whatever affects UG (Factor I), but in the case of variation (Factor I–II interaction, we are assuming),[17] this task becomes almost unrealistic: what efficiency or interface conditions would require that Factors I and II interact in different and (at first glance) unpredictible ways? In recent writings Chomsky has considered the idea, first introduced in his class lectures, and recently discussed by Massimo Piattelli-Palmarini (see Piattelli-Palmarini et al. 2009: 101ff.), that parameters emerge to solve a 'minimax' conflict. To see what is meant by this consider two potential scenarios, the opposite extremes in (23) and (24):

(23) Scenario A: All parameters are innately specified.
(24) Scenario B: Variation is unconstrained.

Scenario (A) implies a huge task being undertaken by UG, a costly option that puts a considerable burden on genetic encoding, reducing, or even eliminating, the acquisition task. Scenario B, on the other hand, puts an overwhelming burden on learnability. Since these two options are in clear opposition, parameters would emerge to solve the conflict, providing a system where variation is not already fixed in UG (there is underspecification), and it is not totally unconstrained either (it is located in the morphophonological component of the lexicon, as we argued above). We could restate this hypothesis as follows:

(25) Minimax Thesis
Parameters are a minimax solution to the (conflicting) scenarios A and B.

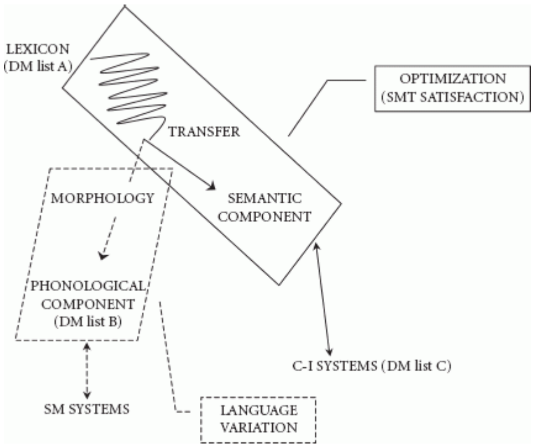
In Piattelli-Palmarini's words (Piattelli-Palmarini et al. 2009: 106–7):

> The suggestion is that maybe what we have is a minimax solution, where you minimize the amount of genetic information and at the same time you optimize the amount of learning that there has to be in an acquisition somehow. Mark Baker has this hypothesis that the reason we don't all speak the same language is because we want to be understood by people in the next tribe; which is a cute idea, but it really doesn't explain much, because you can only do that if you already have an organ that is predisposed to have a large but finite set of **(p. 546)** possible languages [...] The guiding (and interesting) idea [...] is that you have a minimax, you have something close to the perfect compromise between loading the biology, loading the genetics, and having a reasonably complex acquisition process.

Along with the minimax thesis, Chomsky (2010) has argued that variation belongs to the morphophonological component, to the process of externalizing internal computation.[18] In the remainder of this chapter I would like to briefly explore this idea. Let me begin by quoting the gist of the proposal.

> Still to be addressed is the second of the salient puzzling questions: Why are there so many languages? That seems curious, and a violation of the spirit of SMT. There have been suggestions over the years, ranging from sociological/cultural to possible minimax optimization. None seem compelling. Asymmetry of interface relations suggests an alternative: perhaps it is not a matter of evolution of language at all. Externalization is not a simple task. It has to relate two quite distinct systems: one is an SM system that appears to have been basically intact for hundreds of thousands of years; the second is a newly emerged computational system for thought, which approaches perfect design insofar as SMT is correct. We would expect, then, that morphology and phonology—the linguistic processes that convert internal syntactic objects to the entities accessible to the SM system—might turn out to be intricate, varied, and subject to accidental historical and cultural events: the Norman conquest, teen-age jargon, and so on. Parametrization and diversity too would be mostly—maybe entirely—restricted to externalization. (Chomsky 2010: 60)

Note that this would be consistent with what was discussed in the previous section, but also with Chomsky's claim that the faculty of language is only optimal to meet the requirements of the Conceptual-Intentional systems, doing 'its best' when it comes to the satisfaction of those imposed by the Sensorimotor systems (among which one might include functionalist pressures). Externalization (i.e. Transfer to the morphophonological component, or Spell-Out) would thus be an ancillary process, imposed in a secondary evolutionary stage. Chomsky has in fact suggested (see Chomsky 2007: 14, 2008a: 136) that once the faculty of language emerged (due to a sudden mutation), its use was primarily mind-internal, for inner speech—a 'language of thought', whose users 'would have the ability to think, plan, interpret, and so on in new ways, yielding selectional advantages transmitted to offspring' (Chomsky 2007: 14)—with externalization taking place before, in a contrived way, by recruiting the Sensorimotor systems. If we put all these pieces together, there emerges a system where variation is a mere Spell-Out issue, targeting the visible stretch after Transfer, indicated through punctuated lines.



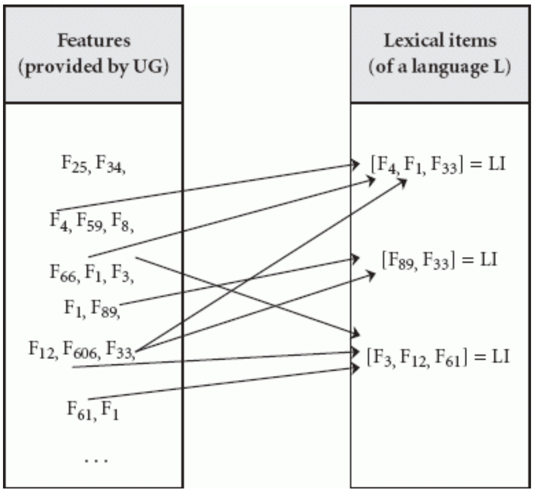
Click to view larger

(p. 547) **(26)**

There is more than one way to interpret Chomsky's idea that variation is related to externalization. There is a *lectio facilior*, whereby all variation (*qua* Spell-Out) is something trivial and random, not too different from the observation that Fritz Lang had a German accent when speaking English—just a matter of pronunciation, understood as 'easily detectable properties of utterances' (Chomsky 2001: 2). This perspective would be not only too simplistic, but also undesirable, as there would be no clear way of capturing the correlations that have been established in the literature (the 'clustering effects'). To be sure, it may well be the case that the parametric generalizations fail to pass systematic tests, but this does not mean that the phenomena observed with respect to parameters—say, pro-drop and free subject inversion—must be disregarded or regarded as accidental coincidences. The *lectio facilior*, however, favors this possibility.

There is also a *lectio difficilior*, compatible with some version of the (Borer–Chomsky) conjecture that variation is a matter of lexicon setting. Under this interpretation, we all have the same faculty of language, variation being a matter of lexicon setting. To be precise about it, one could envisage that the setting of the lexicon proceeds as Chomsky suggests, by means of a feature-selection-and-assembling process, carried out by UG in order to 'provide atomic elements, lexical items (LI), each a structured array of properties (*features*) to which Merge and other operations apply to form expressions' (Chomsky 2007: 6). If lexical items are nothing but feature bundles or complexes, then '[a] particular language is identified at least by valuation of parameters and selection from the store of features made (p. 548) available by UG, and a listing of combinations of these features in LIs (the lexicon)' (Chomsky 2007: 6–7).[19] This process could be as in (27).[20]

(27) Lexicon assembling

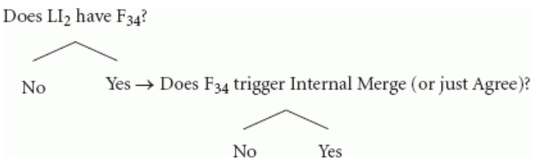


Assembling a lexicon in (27) involves one crucial step of feature-bundling that yields units for computation. importantly, the process above, to the extent that it is tenable, is not incompatible with the idea that variation is to be found within the morphophonology alone. What it tries to show is that variation may have two sources: (i) the way in which the UG features are assembled (variation *before* the syntax), and (ii) the way in which the outputs of syntactic computation are provided with morphophonological features (variation *after* the syntax). Only (ii) properly qualifies as strict externalization in Chomsky's (2010) sense, as it is the **(p. 549)** type of variation that deals with how syntactic objects are converted into speech units. The other type of variation is still lexical business because it concerns how features are associated to create units to be used by Merge. Let us state it as follows:

(28) Types of lexical variation
- **(i)** Combination of features provided by UG.
- **(ii)** Association of lexical items with vocabulary items.

It is crucial to note that only the first type of variation can capture clustering effects. For concreteness, these effects could follow from the fact that a lexical item LI in a language L is formed by combining (or not) a given feature F. Consider this idea in (29):

(29) Clustering effects (see Boeckx forthcoming)



(29) provided, one could assume that there are two versions of LI_2, one endowed with F_{34} (say, in English) and one without it (say, in French), giving rise to different clustering effects. This implementation would be consistent with, e.g. Kayne's (2005c) analysis of variation in terms of pronunciation (externalization) asymmetries. It could be the case that F_{34} is [±mass], thus teasing apart English *grape* from French *raisin.* Likewise, F_{34} could be [±affix], indicating that a given element must be incorporated into another one at the phononological component, distinguishing Spanish *dormir* (Eng. sleep) from Basque *lo egin* (Eng. sleep do). Many other asymmetries (i.e. parametric choices) could be accounted for in this manner.

In this section I have explored the possibility that language variation may be an effect of how externalization operates (see Chomsky 2010). In particular, I have suggested that a straight interpretation of this idea is, though plausible, unable to capture the regularities that parameters have been shown to manifest. I have thus sketched a second way to understand Chomsky's claim, taking variation to follow from (i) the way the features provided by UG are combined, and (ii) the way syntactic objects are spelled-out. The sources are undoubtedly different, but they both concern the lexicon, in accord with the BCC. The main advantage of this dual possibility, in any event, is that it can accommodate clustering effects while sticking to a view in which variation does not have a syntactic nature.

(p. 550) 23.5 Conclusions

Minimalism is probably not the best framework to investigate parameters, or at least not from a traditional—and even P&P-oriented—perspective. For two reasons. First, the theoretical apparatus has been reduced to the minimum (UG, looked at from below, has nothing beyond Merge and features), so it could hardly be useful for undertaking careful data-centric investigations, like those in the 1980s and early 1990s. Second, and most importantly, minimalism is not commited to the dynamics of language variation, but to 'what [one] might call the 'Galilean style': the dedication to finding understanding, not just coverage. Coverage of phenomena itself is insignificant' (Chomsky et al. 2002: 102).

In the preceding pages we have seen that the early P&P thesis that variation is encoded in the syntax (in UG principles that feature a binary 'yes/no' setting) is at odds with the SMT, and must be abandoned—there can be, strictly speaking, no parametric syntax (see Boeckx forthcoming). This is the very idea behind UG, where the 'U' should be interpreted literally: uniform, unique, and unparametrized. Variation (and other prima facie imperfections or design flaws) must therefore be placed where they belong, in the morphophonological component (see Berwick and Chomsky forthcoming: 15, Chomsky 2010: 60, Chomsky 2000a: 117ff., 2002: 109ff.), which, on the one hand, appears to ignore Full Interpretation, Inclusiveness, and other Factor III conditions and, on the other, nicely captures the aribitrary, language-specific, and non-deducible nature of the lexicon. Assuming this much, as Otto Jespersen noted, '[w]e are in a position to return to the problem of the possibility of a Universal Grammar, [since n]o one ever dreamed of a universal morphology[…] It is only with regard to syntax that people have been inclined to think that there must be something common to all human speech' (Jespersen 1992 [1924]: 52).

It is a well-known (but perhaps tacit) idea that minimalism has sidestepped the parametric enterprise that was the hallmark of the P&P framework in its attempt to ask deeper questions about the faculty of language. Since the goal is to understand how an optimal system interacts with the external components of the mind—body, the interest in variation (a non-optimal trait) decreased, and logically so. If what has been discussed in the previous lines is on the right track, it is no longer the case that minimalism is not provided with what is needed to undertake the study of variation. On the contrary, variation should actually be part of the minimalist agenda, if it is understood as an interface phenomenon.

Notes:

Some of the issues that are discussed in this chapter were presented at the workshop on *Linguistic Variation in the Minimalist Framework* (Barcelona, January 14–15 2010). I am grateful to the audience for questions and comments. For suggestions and discussion, I would also like to thank Noam Chomsky and Juan Uriagereka. Special thanks go to Cedric Boeckx, for his interest in this work and his unvaluable help. Needless to say, all errors are my own. This research has been partially supported by grants from the Ministerio de Educación y Ciencia-FEDER (HUM2006–13295-C02–02), and from the Generalitat de Catalunya (2009SGR-1079).

(1) The * in these examples indicates that the form has been 'reconstructed', that is, obtained through comparative strategies.

(2) Berwick and Chomsky (forthcoming) make a similar point when they observe that 'structuralist inquiries focused almost entirely on phonology and morphology, the areas in which languages do appear to differ widely and in complex ways' (p. 2).

(3) What is called 'language' in usual practice (say, Catalan) is some E-language-like entity, subject to sociocultural, historical, and political factors. The same holds for notions such as 'dialect', 'idiolect', or their multiple variants (Barcelona Catalan, standard Catalan, normative Catalan, and so on).

(4) I am putting aside the important issue of what the format of principles can be, which has never—to the best of my knowledge—been addressed in a systematic fashion (see Chomsky and Lasnik 1993: 27).

(5) The periphery was said to be subject to a theory of markedness (see Chomsky 1981a: 8), for which there has not been any explicit formulation (but see Speas 1997, and references therein). According to Chomsky, the structure of the periphery should relate 'to the theory of core grammar by such devices as relaxing certain [of its] conditions, processes of analogy in some sense to be made precise, and so on' (1981a: 8). Notions such as

‘analogy’, ‘tendency’, or ‘preference’ are often invoked when trying to account for linguistic change, especially by behavioral sciences (see Boeckx 2009b and Chomsky 1986b). As in the case of markedness, however, there is no consensus about how to formalize those intuitively correct but poorly understood notions. See Uriagereka (2007) for an attempt to formalize some of these psycho/sociological processes within minimalism.

(6) Another way of interpreting Boeckx's (forthcoming) SUT is to take it as part of Factor III, where principles analogous to the Superset Bias would reduce variation in order to minimize the genetic endowment.

(7) It is in fact unlikely for UG operations or principles to be parametrized. Such a view has nonetheless been held by Mark Baker, who has argued that ‘there are some parameters within the statements of the general principles that shape natural language syntax’ (Baker 2008: 354).

(8) Chomsky (2000a: 100, 2001: 10) argues that the assembling process is carried out by L (the Language, perhaps Merge). Things may be different in a distributed approach, like Halle and Marantz's (1993).

(9) Chomsky (2001: 10) distinguishes three types of features (see also Chomsky 1995c: 230ff.): (i) phonological, (ii) semantic, and (iii) formal. Types (i) and (ii) may be part of UG, but they only provide instructions to the semantic and phonological components, not the narrow syntax.

(10) M. D. Richards (2008) makes a more radical proposal, whereby UG consists only of features: edge features and formal features. The former activate Merge, whereas the latter do Agree (and Transfer too). I will follow Richards (2008) as far as Agree is concerned. The possibility that Chomsky's (2008a) edge features are responsible for Merge is also sensible, but it depends on how the notion feature must be modeled. For one thing, edge features (unlike e.g. ϕ-features) have no values, fail to trigger Match, and cannot delete. These three properties of edge features make it difficult for them to qualify as bona fide features (i.e. valued attributes), but not as mere properties (see Boeckx 2008a, 2009a).

(11) The same could be claimed for a mapping device (Chomsky's 2004a Transfer), but this is more likely to belong to Factor III (see Boeckx 2009, Chomsky 2007, 2008a).

(12) As pointed out in Boeckx (2009a), the idiosyncratic nature of lexical items does not immediately fit with the claim that computation is optimal—it makes computation also idiosyncratic, assuming that Merge operates with lexical items and their idiosyncrasies. The alternative proposed by Boeckx (2009a) in order to solve this tension is that Merge operates with atomic featureless units. Under this perspective, lexical items are subject to a process of generalized late insertion of semantic, formal, and morphophonological features after the syntax, which is where all variation takes place in Boeckx's (2009a) system.

(13) Hale and Keyser's (1993) approach raises one potential problem to the idea that variation is in the lexicon. If the lexicon involves some syntax, then variation should be out of it. One could invoke the (ill-understood) l-syntax/s-syntax cut to solve this tension, only the former being idiosyncratic, but this is just restating the problem. See next section for a more reasonable alternative.

(14) Formal features can be treated as attributes with a value. It is not clear that the binary (+ vs.−) option is applicable to this type of feature as Boeckx (2008a) convincingly argues.

(15) This would be consistent with Chomsky's (2001) idea that ‘interpretability of features is determined in the lexicon’ (p. 5).

(16) Chomsky (2001: 11) in fact argues that distribution is favored for closed classes.

(17) See M. D. Richards (2008) for arguments that there is another type of variation (‘points of optionality’, to use Richards's words) which emerges when ‘UG lacks specified instructions for [its] resolution such as the manner in which a given feature is satisfied or lexicalized’ (p. 140).

(18) Although this idea may appear to be entirely new, it is not (see already Chomsky 1993: 169). See also Chomsky (2010), and Piattelli et al. (2009: 386).

(19) It is important to point out that Chomsky (2001: 10–11) does not entirely subscribe a distributed perspective: ‘In the simplest case, Lex is a single collection […] Lex is distributed when departure from the simplest account is

warranted in favor of late insertion, typically for inflectional elements and suppletion’ (p. 11).

(20) As Cedric Boeckx (p.c.) observes, this presupposes an additional source of generative power, apart from syntax itself. There is, as Boeckx further notes, no comprehensive theory of how feature assembling processes take place—the only one is called ‘syntax’. This view is not too different from Starke's (2009) idea that feature bundles are actually full-fledged syntactic structures—in other words, (i) should be analyzed as in (ii):

((i)) $D_{[X,Y,Z]}$
((ii)) $[_{XP}$ X $[_{YP}$ Y $[_{ZP}$ Z]]] (→ spelled out as ‘D’)

In the Nanosyntax framework sketched in Starke (2009), lexical items can vary from language to language because Spell-Out does—e.g. perhaps D results from the Spell-Out of the features X, Y, and Z in English, while only from the Spell-Out of X and Y in Sinhala.

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Charles Yang and Thomas Roeper

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[–] Abstract and Keywords

Perhaps more clearly than any other field, the study of child language acquisition highlights the continuity from the Principles & Parameters framework to the Minimalist Program. This article is organized as follows. The first half lays bare the mechanisms whereby, assuming universal grammar (UG), the child is able to analyze and compare the input data. The fact that we are able to build a fairly intimate model of how UG extends to models that accept the raw primary linguistic data is a support both for the abstractions of minimalism and for the data-comparison systems which utilize them. The second half explores the promise of minimalism in the microscopic terrain of spontaneous acquisition. It provides an overview of where minimalist principles are at work: Merge and Label, Merge over Move, the Strong Minimalist Thesis, and its impact upon recursion.

Keywords: child language acquisition, Principles & Parameters, minimalist program, universal grammar

PERHAPS more clearly than any other field, the study of child language acquisition highlights the continuity from the Principles and Parameters framework (Chomsky 1981) to the Minimalist Program (Chomsky 1995b). As is the case for all meaningful theoretical developments, under minimalism new challenges emerge, traditional puzzles are cast under different lights, while important insights from previous work can still be retained; this chapter provides an overview of these issues. The first part builds on the continuity from P&P to minimalism, with focus on the role of parameters in the theory of language acquisition and the mechanisms of learning. The second part turns to the minimalist innovations, specifically how the new formulations of the syntactic system bring new tools to the explanation of child language.

(p. 552) 24.1 Formal issues in minimalism and language acquisition

The P&P framework, for the first time, gives a plausible solution to the logical problem of language acquisition: how does the child acquire a language so rapidly and accurately under limited linguistic experience? The principles, which are considered universal, are not learned, and can be expected to be operative in (early) child language; this opens up a wealth of topics for empirical research which continues in the minimalist era. The parameter values, which vary cross-linguistically, must be learned on the basis of specific linguistic evidence. Thus, the commonalities and differences in children's acquisition of specific languages receive a unified treatment. Moreover, if the number of parameters is finite, then there is only a finite—albeit large, so it appears—number of grammars that form the child's learning space, which at least formally sidesteps the well-known problem of inductive indeterminacy in an infinite hypothesis space associated with phrase structure rules (Gold 1967, Chomsky 1981).

Has minimalism altered the fundamental problem of language acquisition? We feel that the answer is both No and Yes. No, because minimalism has not supplemented the basic architecture of P&P for the task for language acquisition (section 24.1.1), and yes, in the sense that the minimalist approach to the language faculty in a broad context of cognition and evolution has led to new conceptions of learning, which may provide a more complete explanation of child language acquisition (section 24.1.2). These issues are closely related; for instance, the empirical evidence for or against parameters cannot be separated from the mechanisms by which parameters values are determined. For expository purposes, however, we shall discuss them in turn.

24.1.1 Parameters and child language

There are two senses in which the term 'parameter' can be understood, and it might be useful to draw their distinctions more clearly. In the conceptual sense, parameters simply denote the finite range of biologically possible linguistic forms, a claim about natural language upheld by most theories of grammar even though the term 'parameter' is typically associated with the GB/minimalism framework. Parameters, then, can be viewed as a type of anchor points for dividing up the linguistic space: the complex interactions among them would provide coverage for a vast array of linguistic data—more 'facts' captured than the number of parameters, so to speak—such that the determination of the parameter values would amount to a simplification of the learning task. This conceptual notion of parameters goes well with the perspective of machine learning and statistical inference, where (p. 553) plausible learnability can only be achieved by constraining the hypothesis space within some finite dimensions (Valiant 1984, Vapnik 1995; see Nowak et al. 2002 for review). These mathematical results hold under the usual assumptions of language acquisition (e.g. the learner only receives positive data though in some cases even negative data does not make learning more tractable) but are not dependent on the nature of the specific learning algorithm or other cognitive capacities that avail to the learner. So formally, an approach to language variation and acquisition by the use of parameters remains the best, and only, game in town.

Once instantiated as specific theories about human language, parameters can be understood in the sense of empirical statements, which of course can be verified, confirmed, or rejected. The failure of certain proposals of parameters does not mean that the whole theory of parameters ought to be rejected out of hand: we might not have figured out the correct ways of dividing up the linguistic space. In fact, many specific formulations of parameters in the theoretical literature have received support from language acquisition research, to which we turn presently.

The evidence for parameters comes in two lines. The first has been running throughout the history of P&P framework. Ever since Hyams's pioneering work (1986), parameters have been used as a tool to explain non-target grammatical patterns (see Crain and Pietroski 2002, Rizzi 2004d, Roeper 2000 for recent efforts). Take, for instance, the well-known case of null subjects in child English. For the first three years of life, English-learning children do not use subjects consistently (Valian 1991), and objects are occasionally omitted as well. Earlier research (e.g. Hyams 1986, 1991, Hyams and Wexler 1993) has attributed these omitted arguments to an Italian type pro-drop grammar (Rizzi 1986) or a Chinese type topic-drop parameter (Huang 1986), yet the usage frequencies of subjects and objects from the studies of (both child and adult) Italian and Chinese are significantly different from those in child English (Valian 1991, Wang et al. 1992). More recently, the null subject phenomenon has been interpreted as the presence of the topic-drop option gradually being eliminated (Yang 2002). One of the key observations here is the striking similarity between child English and adult Chinese. For instance, the availability of subject drop in Chinese is subject to an asymmetry in topicalization.

(1)
a. *Mingtian*, [__ guji [*t* hui xiayu]].
(__=John) Tomorrow, [__believe [t will rain]]
'It is tomorrow that John believes will rain.'
b. *Bill, [__ renwei [*t* shi jiandie]]
(__=John) Bill, [__believe [*t* is spy]]
'It is Bill that John believes is a spy'.

The main observation is that the null subject, which is identified by linking to the discourse topic, is not possible when the new topic is an argument (1b) but possible when it is an adjunct (1a); see Friedmann, Belletti, and Rizzi (2008) for related theoretical considerations. Such distributional patterns are virtually perfectly (p. 554) replicated in child English. For instance, during Adam's null subject stage (Brown 1973), 95% (114/120) of *wh*-questions with missing subjects are adjunct questions ('*Where$_t$ __ going t*?'), while very few (2.8%=6/215) of object/argument questions drop subjects ('* *Who$_t$ __ hit t*?'). Moreover, if the Chinese-type topic-drop option is available, and probabilistically accessed (see section 24.1.2 below for the use of probabilistic learning), then a certain level of null object, which is grammatical in Chinese when the topic happens to be the object, can be expected in child English as well. And we can make the stronger prediction that the relative ratio of null objects and null subjects to be identical across English children and Chinese adults, which is confirmed with data from Wang et al. (1992). The ratio of null objects over null subjects is 0.29 (11.6%/40.6%) for Chinese adults, and 0.32 (8.3%/25.9%) for English children during the subject-drop stage.

The second strand of evidence comes from the statistical correlates of parameters in child language acquisition. It builds on the observation that parameters are correctly set at different points of language development. Since parameter-setting requires language specific information, one can estimate the amount of necessary data for a parameter value in child directed input and relate it to the time course of parameter-setting. Several parameters and their development are summarized in Table 24.1 (see Yang 2002, 2009, Legate and Yang 2007 for additional discussion).

Table 24.1. Statistical correlates of parameters in the input and output of language acquisition. Very early acquisition refers to cases where children rarely, if ever, deviate from target form, which can typically be observed as soon as they enter into multiple word stage of production (e.g. finite verb-raising in French: Pierce 1992). Later acquisition is manifested through children's systematic use of non-target but parametrically possible options. References cited: a. Brown (1973). b. Yang (2002). c. Wang et al. (1992). d. Valian (1991). e. Pierce (1992). f. Hyams (1986). g. Lightfoot (1999), Yang (2002). h. Clahsen (1986). i. Thornton and Crain (1994).

Parameter	**Target**	**Requisite evidence**	**Input**	**Time of acquisition**
Wh-fronting	English	*wh*-questions	25%	very early[a]
Topic drop	Chinese	null objects[b]	12%	very early[c]
Pro-drop	Italian	null subjects in *wh*-questions[b]	10%	very early[d]
Verb raising	French	verb *adverb/pas*	7%	very early (1;8)[e]
Obligatory subject	English	expletive subjects[b,f]	1.2%	3;0[c,d]
Object verb second	German/Dutch	OVS sentences[b,g]	1.2%	3;0–3;2[b,h]
Scope marking	English	long-distance *wh*-questions	0.2%	>4;0[i]

(p. 555) Parameters for which the target value is expressed more frequently are learned faster by children than those which are expressed less frequently.[1] These findings provide support for the reality of parameters, adding to the traditional motivation from cross-linguistic generalizations. For an illustration that unites comparative syntax and language acquisition in a single stroke, see Snyder (2001).

Despite its considerable success, the parameter seems to have fallen out of favor in current minimalist theorizing and other theoretical frameworks (Newmeyer 2004, Culicover and Jackendoff 2005). It is certainly logically possible to recast the fact of language variation without appealing to syntactic parameters; we can point to variation in the lexicon, variation in the functional projections, features, feature strengths, feature bundles, etc. to ‘externalize’ the parametric system to interface conditions, presumably out of the syntactic system proper. For instance, consider the appeal to parameters in the acquisition of subjects above. The topic-drop option of the Chinese grammar (and early child English) may be construed as a discourse principle, the property of pro-drop can be attributed to the morphological system which may be partially connected to discrete categorization and other cognitive abilities, and the English-type obligatory use of subject is a reflex at some generalized EPP feature that is realized at the PF interface. But it is also important to realize that such a move does not fundamentally change the nature of acquisition problems: the learner still has to locate her target grammar in the space of finite choices with a naturalistic sample of data within a reasonable amount of time. And to the extent that syntactic acquisition can be viewed as a search among a constrained set of grammatical possibilities, the minimalist—and indeed, non-minimalist—alternatives to parameters ought to provide similar empirical coverage. In section 24.1.2 we provide some learning theoretic considerations for a plausible theory of parameters, with the possible implication that the mechanisms of acquisition may shift some explanatory burden out of the innate UG device. Nevertheless, the empirical evidence for parameters remains, and conceptual arguments for the elimination of parameters run the risk of losing important insights and discoveries through decades of fruitful research.

24.1.2 Minimalism and learning

One of the most revolutionary aspects of minimalism is the consideration of the language faculty in a broad cognitive and perceptual system, which marks a (p. 556) significant shift from the earlier inclination to attribute the totality of linguistic properties to universal grammar. Viewing a theory of language as a statement of human biology, one needs to be mindful of the limited structural modification that would have been plausible under the extremely brief history of *Homo sapiens* evolution. The minimalist program of language evolution (Hauser et al. 2002) seeks to isolate aspects of the linguistic system and identify their homologies in other domains and species. Likewise, one can raise the question how much of the Language Acquisition Device is specific to language—or acquisition.

Our review here focuses on the algorithmic mechanisms of language acquisition. First, consider the problem of parameter setting (or whatever formulation that parameters receive in the minimalist setting). Much of the discussion in generative literature has centered around domain-specific learning algorithms, the most prominent of which is the triggering model of Gibson and Wexler (1994), schematically illustrated below.

(2) At anytime, the learner is identified with a grammar *G*, i.e. a set of parameter values:
- **a.** Upon receiving an input sentence *s*, analyze (e.g. parse) *s* with *G*
- **b.** If success then do nothing; return to a.
- **c.** If failure then:
  - **i.** Randomly select a parameter and flip its value, obtaining a new grammar *G′*.
  - **ii.** Analyze *s* with *G′*.
  - **iii.** If success, then keep *G′*; return to a.
  - **iv.** If failure, revert back to *G*; return to a.

A model like triggering is designed to make full use of the structural properties of the linguistic systems. The so-called Single Value Constraint in (2c.i), for instance, reflects the view of parameters as an intrinsically interactive system such that the modification of the grammatical hypothesis ought to be minimal, as the learner changes the value of only one parameter. It is probably fair to say that domain specific learning is still the dominant approach in the generative study of language acquisition; to wit, virtually all learning models in the Optimality Theory use some version of Constraint Demotion which takes advantage of the structures of ranked constraints—and is indeed considered as a virtue of both the theory and the learning model (Tesar and Smolensky 1998).

But the domain-specific razor cuts both ways. A learning model that goes hand in hand with a linguistic theory must be modified, or completely abandoned, if the linguistic theorizing takes a different direction. Furthermore, if the defects of the learning model are revealed (see Berwick and Niyogi 1996 on the triggering model), the grammatical theory may be impeached as well (see Tauberer 2008 on OT learning). For quite independent reasons, general learning (p. 557) mechanisms have been applied to language acquisition in recent years (Labov 1994, Yang 2002). A prominent feature of this line of work is the introduction of probabilistic distributions over grammatical hypotheses, which may be, and are indeed believed to be, domain-specific. Consider the *variational* learning model (Yang 2002), which is borrowed from one of the earliest mathematical models of learning (Bush and Mosteller 1951)—from the behaviorist tradition, indeed—that has been observed across domains and species (Herrnstein and Loveland 1975):

(3) At any time, the learner is identified with a population of grammars with associated probabilities:
- **a.** Upon receiving an input sentence *s*, select grammar *Gi* with probability *Pi.*
- **b.** If success then increase *Pi*; return to a.
- **c.** If failure then decrease *Pi*; return to a.

In contrast to the triggering model, the variational learner doesn't actively participate in learning: the hypotheses themselves do not change, and the only thing that changes is their distribution. The conception of learning as a gradual and probabilistic process opens up the possibility of explaining child language as quantity-sensitive growth in response to the volume of necessary linguistic evidence in the environment; the developmental correlates of parameters in Table 24.1 are uncovered under such considerations.

The reader is directed to Yang (2002) for a more formal treatment of variational learning applied to a parametric space, and the most general convergence result can be found in Straus (2008). In some cases, the variational model is provably superior to alternatives such as triggering, with additional benefits of bringing the formal grammar model closer to the facts of child language. But there is no general result that probabilistic learning is inherently superior (see Yang 2008 for discussion). In fact, the plausibility of a learning model depends more on the structure of the grammar space and less on the algorithmic aspects of learning. One can easily imagine the worst-case scenario where parameters interact in arbitrarily complex ways such as all learning models become intractable. Here we consider the issue of plausible learning in several directions, some of them traditional while others stemming from minimalist considerations.

First, as Chomsky noted long ago (1965: 61), an explanatorily adequate theory of grammar is one in which the hypotheses can be 'scattered', i.e. distinguished by a reasonable amount of linguistic data in a computational tractable way. (Modern theory of statistical inference speaks of a similar notion called 'shattering', which refers to the requisite amount of data capable of locating the target hypothesis in statistical classification: see Vapnik 1995). In the generative literature, there have been several convergent lines of research that point to the advantage of a structured parametric space that favors the learner. One of the earliest efforts (p. 558) is the cue-based learning model of Dresher and Kaye (1990); see Dresher (1997) and Lightfoot (1999) for applications to syntax. Dresher and Kaye observed that ambiguity, which refers to the fact that an input token may be compatible with multiple grammars, could easily mislead the learner, who is presumed to make learning decisions locally. Their solution lies in a set of parameters whose values can be determined unambiguously but only following a predefined and thus presumably innate sequence. The work of Fodor (1998) and Sakas and Fodor (2001) is another response to the ambiguity problem. Here the learner hedges its bets more intelligently than the randomly guessing triggering learner. It avoids learning from input that is compatible with multiple hypotheses and only modifies the grammar on unambiguous data. The detection of data-grammar ambiguity is achieved by trying out multiple grammars with each input token. Finally, the idea of parameter hierarchy (Baker 2002), largely motivated from a comparative/typological point of view, would set the learner on course of a sequence of decision that starts from major divisions of languages—e.g. whether a language is ergative or not—to minor ones such as the placement of adjuncts on the left periphery (Cinque 1999). The hierarchy, like cues, is conjectured to be innate and thus solves the ambiguity problem from within. The natural question, of course, is to what extent the parameters required to describe the word's languages follow the ideal expressed in these works. In some cases, the probabilistic learning model of Yang (2002) can replicate the effect of parameter sequences and cues without assuming innate specification, but it does not remove the necessity of structured parameter space to achieve plausible ordering, especially if the parameter space is structured to simplifying learning.[2]

Second, it would be a mistake to suggest that child language can be entirely explained in terms of searching for a solution in a constrained parameter-like space. The most obvious case can be found in morphophonology: innate principles of UG notwithstanding, even the most enthusiastic nativist would hesitate to suggest that the English specific rule for past tense formation ('*-d*') is one of the options, along with, say, the '*-é*' suffix as in the case of French, waiting to be selected by the child learner. In the domain of syntax, we also find patterns of variation that may be governed by universal constraints but are realized in particular languages in highly specific and widely ranging ways such that the learner cannot but make use of (constrained) inductive learning mechanisms. An example of this type can be seen in the distribution of dative constructions such as double object and prepositional (p. 559) dative across languages. There is broad agreement that certain universal syntactic and semantic properties are the necessary conditions for a verb to participate in dative constructions in the first place (Pesetsky 1995, Hale and Keyser 2002, Rappaport Hovav and Levin 2008). Nevertheless, these constructions are productive in English but are limited to a lexically closed class of verbs in languages such as Korean (Jung and Miyagawa 2004) and Yaqui (Jelinek and Carnie 2003) that must be learned individually. Tellingly, English-learning children show an initial stage of conservatism, that they do not generalize these constructions to novel items (and thus do not make ungrammatical errors) (Gropen et al. 1989, Snyder 2006). The detection of productivity apparently takes place at around 3;0 (Conwell and Demuth 2007; cf. Snyder and Stromswold 1997). The course of acquisition strongly resembles the phenomenon of over-regularization in morphological acquisition, suggesting that a learning mechanism capable of detecting the productivity of linguistic productivity is at play. The questions here are again traditional; the acquisition of the universal and language-particular and construction-specific aspects of language was once at the forefront of language acquisition research (Fodor and Crain 1987, Pinker 1989), with special focus on the core periphery distinction drawn at the outset of the P&P framework (Chomsky 1981). A recent approach (Yang 2005, 2009a) draws inspiration from the principle of efficient computation under Minimalism and develops a decision procedure by which the processing of productive and exceptional items are jointly optimized. While still preliminary, this work gives an example of how optimal design principles of language (Chomsky 2005) may be applied to language acquisition.

Finally, it is equally important to recognize the limits of general computational mechanisms in language acquisition (Gomez and Gerken 2000), which appear to have been gaining popularity ever since the demonstration of statistical learning of artificial languages by infants (Saffran et al. 1996, cf. Chomsky 1955/1975a). Much of this work, however, remains confined to a laboratory setting at the moment. Still less effort has been made to test whether these mechanisms scale up in a realistic learning environment, and there have been negative results (Yang 2004).

Looking more broadly, a current theme in cognitive science has advocated a data-intensive and memory-centric approach to language learning (Tomasello 2000, Bybee 2006), which leads to claims about child language as ‘item-based’ and limited in syntactic creativity. Even though these positions have not been embraced by the minimalist community, one does find similar stances toward the division of labor between the grammar and the lexicon. Following Borer (1984), it is assumed that language variation, and thus acquisition, can be attributed to the properties of lexical items. (Surely words have to be individually learned.) Without taking ourselves too far afield, it is useful to discuss these issues briefly as minimalism has forced us to reconsider the relation between the language faculty and general cognitive systems.

(p. 560) At the very beginning stage of acquisition, the child's grammar is necessarily restricted to specific lexical items; after all, hearing ‘Hi baby’ once is not going to give the child the complete grammar of English. The leading questions, then, can be phrased as follows:

(4)

a. From a learning perspective, how does the child go from specific instances of the data to general grammatical properties?
b. From a developmental perspective, are any major aspects of grammar (e.g. verbal syntax, noun phrase structures, as held in the item-based learning approach) actually item-based even for the youngest children that can be assessed? (The child's grammar could be off target but productively so, as in the case of Null Subjects in English acquisition.)
c. From an methodological perspective, could the item-based approach in principle offer an adequate solution for the problem of language acquisition?

No complete answers will be given here, for they have not been fully explored at the present time. The generalization problem (4a) is a traditional one, as the discussion of productivity learning above indicates. To address (4b) and (4c), useful insights can be garnered from the statistical properties of the linguistic data from both adults' and children's linguistic production, which constitute the input to and the output of language acquisition. A strong and consistent pattern, one that is familiar in the field of corpus and computational linguistics, is the so-called Zipf's law (1949): most linguistic items, be they morphemes, words, morphological rules, or phrases, are used very rarely even when the amount of linguistic data is very large, and that the items that the learner receives reliable evidence in the input are relatively few, and these patterns hold for both child and adult languages (Chan 2008). Thus, regarding (4b), one cannot simply take the relatively low degree of usage diversity in child language (Tomasello 2000) to be an indication of the child's grammar as organized around specific lexical items and constructions (Yang 2009b). As for (4c), the study of the linguistic data reveals a major challenge, the so-called ‘sparse data problem’ (Jelinek 1998), that all acquisition models must face. As the linguistic model gets more complex, the amount of data required for the instantiation of the model, i.e. acquisition, increases rapidly such that even very large samples will not be sufficient. And there is suggestive evidence from computational linguistics that piecemeal learning using lexicalized grammar models pays little dividend compared to more general and overarching rules (Bikel 2005). So we are back to the heart of generative grammar: how should a theory of grammar simplify the learner's task in order to achieve successful acquisition with a relatively small quantity of data? We hope that child language acquisition can provide stringent but revealing conditions on further developments of minimalism, much the same way it has carried out its duty for the Principles & Parameters framework.

(p. 561) 24.2 Empirical evidence of minimalist principles

24.2.1 How acquisition evidence preceded or illuminates linguistic theory

We can now ask a sharper question: does minimalism itself offer some methods to capture the primary linguistic data? Can minimalist principles of sentence construction explain micro-steps on the acquisition path, especially those not seen in the target adult grammar? Note that one can ask a prior question: should acquisition data be expected to map naturally onto any linguistic theory? In the 1960s, it was doubted that acquisition research was possible because performance factors so profoundly clouded what happened in a child's mind that no relevant evidence could be expected. It remains easy to attribute deviations from adult grammar to external performance factors—it can be found in every acquisition journal and is often treated as a naturally superior explanation. Nonetheless, in almost every instance the seemingly deviant acquisition data has instead proved to be a reflection of grammatical constructions found in other languages.[3]

24.2.1.1 Small clauses

An instructive example was small clauses: it was often informally suggested in the early 1970s that children said ‘it big’ because they lacked the memory space to say ‘is’ with case-assignment as a default accusative in English (as in ‘him big’). If factors like sentence-length (Bloom 1990) were paramount and forced deletions, then the deletions might happen virtually randomly. If analyzed in grammatical terms—if grammatically describable at all—they would not tell us about the acquisition path.

Radford (1990) and Lebeaux (2000) argued more interestingly that expressions like ‘it big’ revealed that small clauses had an important status in grammar. Had the acquisition data been taken seriously right away, it might have led at a much earlier point to the recognition that small clauses are a significant form of complementation (see Moro 2000). Other examples follow.

24.2.1.2 Scope and partial movement

At first, sentences like ‘Only I like milk’ (meaning: ‘I like only milk’) were taken to be a performance failure to represent scope by children. However, with the advent (p. 562) of Logical Form (LF) they could be analyzed as an early reflection of movement operations to a pre-sentential LF position for quantificational elements (Lebeaux 2000). Similarly the unusual behavior of quantifiers in acquisition fit event-based semantic theory descriptions (Philip 1995), and has engendered a large literature of alternative syntactic and semantic analyses,[4] although at first (Crain and Thornton 1998) performance explanations were given as a way to deny the relevance of the phenomena.

Partial movement structures in acquisition (‘*what* did she say *what* she wanted’) (deVilliers et al. 1990, 2007 for comprehension and Thornton 1990, Oiry 2008, Strik 2009 for production) were analyzed at first as non-core phenomena that occurred only in dialects (McDaniel 1989) but have progressively been re-analyzed, such that they now form support for the Strong Minimalist Thesis (as we discuss below). Had the ‘performance’ or ‘marginal’ explanation not seemed to have priority, spontaneous acquisition evidence (not consistent with target grammars) can be, and should have been, seen as providing UG hypotheses directly (see Boeckx 2008a, 2008b for a compatible argument).

24.2.1.3 Principle B

Another example is the well-known delay of Principle B effect (Chien and Wexler 1990) (‘John washes him’ where ‘him’ is interpreted as ‘John’), which has been analyzed in pragmatic terms and Optimality Theory or cognitive egocentricity perspectives (Hendricks and Spenader 2006). Recently, Elbourne (2005), Verbuk and Roeper (to appear) and Hamann (to appear) have shown that Frisian/Old English continue to allow pronouns in a single clause, suggesting that there must be a parametric account. Hamaan (to appear) and Verbuk and Roeper (to appear) have argued that the presence of sentences like ‘John took a wallet with him’ in English, in contrast to their absence in German, indicates that even subtler parameters are at work. In German ‘with him’ is disallowed and *mit sich* (with self) occurs, so in German Principle B applies to both arguments and adjuncts. This leads to the correct prediction that children abandon Principle B errors earlier in German than in English, where a domain narrower than the clause is critical. The fact that Principle B is realized late in English is then, once again, not a reflection of a grammar-extraneous factor, but of the paucity and subtlety of data needed to set a refined parameter in English, as we have discussed above in section 24.1. This supports Kayne's (2005) claim that there are many subtle parameters (see also Lightfoot 1999, Westergaard 2009, and Bentzen et al. 2009), which means in effect that we need to have a more microscopic view of the acquisition process (see Roeper 2009 for an overview) along many dimensions.[5]

(p. 563) 24.2.1.4 C-command

Acquisition evidence can play a role in overcoming the obscuring impact of social and historical factors. Boeckx (2008b) argues that it is the child data, where real support for parametric analysis belongs, not adult data, which is too confounded by social and historical factors. An effort to state c-command perfectly for adult grammars (with alternatives like m-command) were proposed, but none could overcome the diversity of relevant data, so that it remains unresolved. Nevertheless, evidence for knowledge of c-command was shown by Lawrence Solan (1983), who showed that for sentences like:

(5)

a. The horse hit him_i after the sheep_i ran around.
b. *The horse_i told him that the sheep_i would run around.

children were twice as likely to allow backwards coreference when there was no c-commanding pronoun (‘him’) (5a).[6] Therefore, in the spirit of Boeckx's argument, the fact that the basic notion of c-command appears clearly in the work of Solan and subsequent work shows that the principle is essentially correct, despite the fact that it has been obscure in the adult grammar (with variants like m-command proposed).

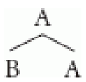
24.2.2 Acquisition and specific minimalist principles

The success of acquisition research in the terms of linguistic history leads, nonetheless, to a greater challenge: does it provide specific support for principles utilized in minimalism? What follows shows that a minimalist acquisition perspective provides direct evidence for several abstract principles: Asymmetric Merge, feature-checking, the labeling algorithm, the Strong Minimalist Thesis (phase-based interpretation), recursion, and the role of interfaces.

At first it might seem very unlikely that minimalism simplifies rather than complicates the acquisition problem. If a child seeks to analyze input substantively in terms of noun or verb projecting noun phrase or verb phrase, it would seem to be a step ahead of a child who begins only with the notion of Merge, which might seem to fit anything. A closer look reveals otherwise. In fact, minimalism allows grammatical principles, like asymmetric merge, to participate more directly in the analysis of Primary Linguistic Data than, for instance, phrase-structure rules.

24.2.2.1 Asymmetric Merge

Merge deviates in an important way from what might be called a general cognitive capacity for the act of combination or concatenation (Hornstein 2009) which (p. 564) applies to almost anything in life experience. Merge requires asymmetry: a label is chosen, usually seen as a projection of one lexical item, which allows one part of a binary Merge to dominate the other, following a labeling algorithm (Chomsky 2006).



(6)

In concrete terms, though hard to establish, it predicts that a child will perceive the difference between ‘ocean blue', which is adjectival, and 'blue ocean' which is nominal, (see Roeper to appear).

Children assemble a number of single words which involve nouns, verbs, prepositions, pronouns (although their category label may be obscure). When a child says 'up' is it a preposition or really a verb? While, as with one-word utterances, many theories could explain two- and three-word expressions, the fact that we get unique combinations in early language is captured directly by the notion of Asymmetric Merge and Label[7] here with no AGR node (following Roeper and Rohrbacher 1994):

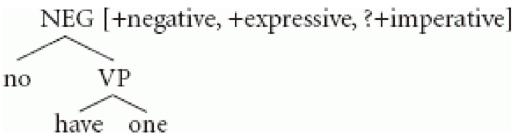
(7)

Adam 03: no play toy
Adam 01: no write on there/no kick box no do suitcase/no hurt head/no have one
Sarah 26: no it upsidedown
German: nein auto kaput (no car broken) nein dick baby (no fat baby)

This first representation seems to be a negative feature that subcategorizes the lexical categories of N or V or VP: Neg [V or N]. Such examples were initially—quite counter-intuitively—analyzed as reduced forms of Negative+Sentence, where huge amounts were deleted because of performance demands or the absence of lexical items. While Asymmetric Merge might allow any combination, the absence in English and German of *not* (*nicht*) (see Pierce and Deprez 1992) **Not run* suggests that other linguistic features drawn from UG limit the range of possible two-word utterances prior to the full expression of functional categories. Nonetheless, under minimalism it could be viewed as a grammatical expression—while earlier theories, which have been pursued over a much longer course of research, demanded either deletion or non-grammatical representations. Therefore, as a possible grammar within a theory of multiple grammars and competition, outlined above, it may not immediately disappear when more elaborated aspects of grammar emerge. The major question, of course, is to determine which way of interpreting children's language is better supported by empirical evidence.

(p. 565) An expressive pragmatic feature seems to be present as well, and therefore *no* can be seen as an Expressive word (Potts and Roeper 2005) that can be paraphrased as: *No way yoghurt* (Drozd 2001). If it is an expressive, it may have no lexical label at first (like *wow*, or *gee*) and still be subject to asymmetric Merge. It will grow (presumably by adding features) into a NegP that is embedded within a VP.

What then are the set of possible labels that asymmetric Merge can generate? It might, for instance, generate a Neg-feature, but possibly without all of the functional information that languages allow, and therefore it is an incomplete functional category:



(8)

We can guess that it may contain an Expressive feature, as we suggested above, linked to forms like *wow, gee, well* (Potts and Roeper 2005) and possibly a Force feature like imperative. The question is interestingly abstract, since something like 'no yoghurt' is often associated with an imperative impulse. However, wordlessly pointing a finger in a deli at a sandwich also conveys imperative import, but perhaps via a form of communication that does not invoke grammar. Therefore we do not know from this example whether we should add a Force feature to the node (or add a higher node) or leave the imperative property initially to inferential pragmatics of the sort that interpret gestures.

The important point here is that Asymmetric Merge allows an immediate representation of a child's first utterances and, more importantly, an abstract analytic instrument that enables a child to attack in a simple way what is a very complex set of inputs, before projecting the full array of functional categories (which is not to say that the capacity is absent). The significance of this point should be underlined: a virtue of the abstraction of minimalism is that it reduces the Primary Linguistic Data problem by giving the child representational tools[8] that allow first-stage efforts (p. 566) to represent linguistic forms whose full feature system has not yet been identified. In that sense, minimalism predicts that Stages can exist.[9,10]

24.2.2.2 Pair Merge and Set Merge: adjunction in child grammar

Merge further divides into two kinds: Pair Merge (adjunction) and Set Merge (argument subcategorization). One can ask whether both kinds of Merge appear in acquisition. Lebeaux (2000) and Roeper (2003) argue that at first all attachment may be Pair Merge or adjunction. Only upon a recursive 2nd Merge must the child decide exactly the higher label. Both Set Merge and Pair Merge are visibly present as soon as three-word utterances arise (Brennan 1991), where, interestingly, prepositions are absent only with adjoined elements:

(9)

I cried stairs (=on)	Shirley get meat dinner (=for)
I cut it a knife (=with)	Richard bring snack Shirley (=for)
feed baby fork (=with)	Shirley cut fork (=with)
I sleep big bed (=in)	Save some later (=for)

(10)
a. I cried [Set merge ⇒ verb requires subject]
cry stairs [Pair Merge: *stairs* is adjoined to *cry*]

Arguments in contrast have prepositions:
(10)
b. I played *with John/* Jim was *at Cooperstown/* putting Daddy *in wagon*

Brennan (1991) reports that there were 46 prepositions for arguments and only 3 for adjuncts, although adults and children both have more adjunct than argument PPs. 'For 3 of the 4 children studied, it was true that adjuncts never surfaced with PP's, while the distribution of PP's in argument position was haphazard.' The child is apparently able to adjoin *stairs* without any subcategorized feature linked to cry. It deviates from the adult grammar which requires a PP projection to introduce this information in order to assign case to the adjunct 'stairs'. If the case module, however, is not yet fully defined for prepositions (where transitives, intransitives, and particles must be differentiated), causing overuse of accusative Default case, (p. 567) then the theory predicts that such examples should occur. This in turn reflects modularity: the emergence of language-particular features within the case module may have its own acquisition path.[11]

Asymmetric Merge also allows a direction of complementation and therefore predicts word order invariance. Early suggestions by Bowerman (1973) that children's grammar was semantic and lacking syntactic ordering (because examples like 'Adam watch' and 'watch Adam' with the same meaning exist) was disproven by Bloom (1990), who showed that children at the two-word stage never made order errors with pronouns and predication. That is they said 'that big' or 'it big' but never *'big that' or *'big it'.

24.2.2.3 Pied-piping and economy of representation

Do we have evidence for feature-checking as a motive for syntactic movement?

Feature-checking and economy of representation receive another kind of specific support from spontaneous aspects of acquisition. If feature-checking motivates movement, e.g. if a *wh*-word carries a feature which matches a CP feature and moves to check it off, then it is only the critical feature that needs to move, not everything moved under pied-piping. Everything extra is a 'free rider' under a minimalist form of feature-checking. The sharpest spontaneous evidence comes from Gavruseva and Thornton (1999), who provide an extensive experimental evidence from at least ten children that they will break up 'whose' and move only 'who' in production:

(11)

Q: John saw someone's book. Ask him which book?
A: Who did you see t's book

This is precisely what ought to occur, but children never hear direct evidence for it in English since the choice of lexical items from the numeration offers only the contracted 'whose' which drags the object along: 'whose book did you see?'

Do-insertion, originally claimed to be a Last Resort phenomenon, fulfills economy of representation under feature-checking in early acquisition in precisely the same way. Hollebrandse and Roeper (1997) find that children prefer to insert 'do' rather than pied-pipe a V+Tense (as in 'painted') from the lower V-node to a higher tense node. This occurs for brief periods in various children who spontaneously produce non-target grammar do-insertion:

(12) 'do it be colored'
'I did paint this and I did paint this'

Do-insertion achieves immediate feature satisfaction without requiring percolation of the lexical feature to a higher node. (See Fitzpatrick 2005, Heck 2009, Roeper 2003, (p. 568) Cable 2007 for discussion of economy and pied-piping.) From this perspective, the child resorts to *do*-insertion as a First Resort, preferred over pied-piping a verb, and it converts *do*-insertion into an operation that preserves economy of representation for feature-checking rather than being a response to an imperfection in grammar. This can be seen as evidence that Merge is more economical than Move (Internal Merge).

24.2.2.4 Barrier theory and the Strong Minimalist Thesis

Barrier theory formed the crux of linguistic work for a quarter century, and its central tenet was clearly supported as a universal constraint in the evidence that extraction from NPs and strong islands were prohibited (e.g. Otsu et al. 1990, Baauw et al. 2006, Friedmanetal. 2009 among others; see also de Villiers and Roeper (in preparation).[12]

Children were given a choice of 'with' as part of the NP or VP in the following context (Otsu 1981):

(13) The boy fixed the dog with a broken leg with a bandage.
What did the boy fix the dog with?

They choose VP-with 'a bandage' 90% of the time, the form consistent with an NP-barrier, where [NP_1 a broken leg [PP with [NP_2 a bandage]] prohibits extraction of one NP from inside another.

They likewise do not allow long-distance adjuncts in cases like (de Villiers et al. 1990, 2007):

(14) The boy said in the morning he balanced the ball on his nose at midnight. When did the boy say how he balanced the ball?
Answer: In the morning. (*midnight)

Surprisingly, over 30% of children answer *how* ('on his nose'), treating *why* as a scope-marker, inviting a Partial Movement analysis. What feature mechanism exactly blocks movement? Possibly an outgrowth of Relativized Minimality is the right path—see Grillo (2008), Friedmann et al. (2009), and Schulz (2005), who posits an intermediate focus element that is distinct from a question element to satisfy and delete a *wh*-feature. The strong fact remains: children easily take a medial *wh*-to be either a barrier or a copy of a scope-marker (Partial Movement). Such findings have been reported in several other languages (Oiry 2008, Strik 2009). It remains to be seen how these locality effects in syntax are captured under minimalism.

(p. 569) Another angle on this problem emerges below when we consider the Strong Minimalist Thesis. Chomsky (2005) has proposed that Phase theory should include interpretation, i.e. that syntax, semantics, and phonology may all be bounded within the Phase:

Strong Minimalist Thesis (SMT)

> 'Transfer.....hands Syntactic Object to the semantic component, which maps it to the Conceptual-Intentional interface. Call these SOs *phases*. Thus the Strong Minimalist Thesis entails that computation of expressions must be restricted to a single cyclic/compositional process with phases. Chomsky (2005)

This principle offers a fresh perspective on Partial Movement in production and comprehension by children. We find productions such as:

(15) *What* did she say *what* she wanted?
What do you think *which animal* says woof woof?
What do you think *which Smurf* really has roller skates? (Thornton 1990: 246)

Here we find that the child applies the SMT to phonology and pronounces the intermediate trace. The spontaneous reflection of the SMT is underscored by the fact that Asian and Spanish second-language learners do the same (Schulz 2003, Gutierrez 2005).

In addition, interpretation of the lower clause is called for when the IP Phase Edge is met. We find the SMT interpretation for sentences like (de Villiers et al. 1990, 2007):

(16) When did the clown say how he caught the ball?

where children answer the 'how' question (e.g. 'on his nose'), providing a plausible within-first-Phase answer. Adults delay an interpretation until the next phase, where the CP properties inherited from the higher verb make the lower clause into an indirect question that is not answered.

Children in fact take an extra step, interpreting not only overt *wh-* words, but *traces* as well:

(17) [she bought a cake but said she bought paper towels]
'what did she say t she bought t?'

as if they are answering the lower clause without the impact of the indirect question feature on the CP inherited from *say* (see de Villiers et al. 2007) while adults correctly say 'paper towels'. These spontaneous deviations from adult grammar are precisely what the SMT promotes—immediate interpretation within each phase-and therefore they demonstrate the core role of *locality* in human grammar.

How does the child eliminate an overt medial *wh*-word or the phasal interpretation of a *trace*? The answer is not yet clear but it should follow from the formal representation of *opacity* at LF. Roeper (2009) argues that the child must learn (p. 570) to alter a trace—modify the unmarked interpretation required by the SMT—to prevent it from being interpreted in its original phase. It changes from a full trace to a converted trace when evidence arises that full reconstruction delivers the wrong interpretation.[13] This is forced when the child, in the example above, recognizes that what the mother bought and what she said she bought are in conflict. Once again, it is precisely where complex constructions elicit subtle spontaneous deviations that reflect basic principles that the stunning contribution of acquisition is evident.

24.2.3 Recursion

Chomsky et al. (2002) have argued that a core feature of minimalist representations is recursion. The operation of Merge creates recursive hierarchies in every language. Some categorical recursion is virtually invisible. Thus the fact that one article occurs inside of a structure containing another may not be detectable by the human computational system, for instance, the articles in: *the* man in *the* house.

However, there are language-specific forms of recursion which children do not acquire instantly and whose complexity is intuitively evident. Possessives, adjectives, and clauses require recursive generation and are systematically delayed in the grammars of children, and not uniformly present in the languages of the world. For instance children and L2 speakers find it very difficult to handle forms like:

(18) Cookie Monster's sister's picture

Three-year-old children regularly prefer a conjoined reading *Cookie Monster and sister*'s *picture* when faced with alternatives (see Roeper 2007).

Here is an illustrative dialogue, among many, where the parent does not perceive the difficulty (Childes, Brown Corpus, Sarah 039):

MOTHER: What's Daddy's Daddy's name?

SARAH: uh.

MOTHER: What's Daddy's Daddy's name?

SARAH: uh.

MOTHER: What is it? What'd I tell you? Arthur!

SARAH: Arthur! Dat my cousin.

MOTHER: Oh no, not your cousin Arthur. Grampy's name is Arthur. Daddy's Daddy's name is Arthur.

SARAH: (very deliberately) No, dat my cousin.

MOTHER: oh. What's your cousin's Mumma's name? What's Arthur's Mumma's name? And what's Bam+Bam's daddy's name?

SARAH: Uh, Bam+Bam!

MOTHER: No, what's Bam+Bam's daddy's name?

SARAH: Fred!

MOTHER: No, Barney.

Where exactly then, does the difficulty lie? The answer connects to the SMT. The child must not simply grasp the fact that a category is embedded inside an identical category, but also generate an interpretation at each Phase Edge. Thus the child interprets a possessive as possessive and the next point of interpretation calls for embedding that possessive meaning inside another. The alternative is the non-embedded conjoined reading noted above (choosing Cookie Monster's and sister's picture for (19)). This shift from conjoined to embedded occurs for every form of recursion where the Phase-Edge must be interpreted: PP, AP, CP, and DP (see Roeper 2007, 2009).[14]

Ultimately this means that, just as syntax must be connected to Interface Theory, it is precisely at the point of the interface between recursive syntax and the interpretive connection to the SMT that the child experiences a challenge. These results point at the idea that interfaces must be articulated to understand UG, yet it leaves a promising challenge for future research: how exactly do recursive structures engage computational complexity?

24.2.4 Interface connections

From an architectural perspective, there is evidence that the notion of an interface, still theoretically hazy, unlike the original theory of autonomous syntax, provides a scaffolding on which to place a crucial role for semantics and pragmatics in the emergence of grammar.

First some clarification of the concept. We take interfaces to be mechanical and biologically articulated, like the connection between the heart and the lungs. In that respect, we need to sharply differentiate an innately specified *interface* from a system-wide *interaction* in energy use that connects every part of an organism.

Acquisition theory has always tacitly assumed a rich interface whose mechanics were subtle and mysterious. There is no doubt that inferences about context must feed into both lexical and syntactic growth, but how? This is an old and common intuition: the challenge is to build it into a mechanism.[15] It is likely that (p. 572) the acquisition mechanism begins with an over-reliance on context—an interface choice—that is ultimately altered when grammatical comprehension becomes autonomous. For instance, children will misunderstand a sentence like (19a) (Bever 1972, Roeper 1981):

(19)
- **a.** The mouse was eaten by the cheese.
- **b.** The cheese ate the mouse.

guessing that the mouse ate the cheese for (19a), since world knowledge makes it plausible. At the same time, part of their grammar is autonomous: they already know that the active sentence (19b) is nonsense. Why? If the active is in the grammar, then context does not guide interpretation, but if passive is not controlled, then the child will make a pragmatic choice—a rather commonsense idea. When the child has a passive transformation as an hypothesis, then *context* can serve as *confirmation* of syntactic and semantic hypotheses:

(20)

Scene: baby drinks milk
Sentence: ‘The milk was drunk by the baby.’

can justify the conclusion that a passive sentence is involved if one assumes these dimensions coincide:

Pragmatics: baby drinks milk

Semantics: Agent verb Theme

Syntax: object ⇒ undergoes preposing to subject

where a real event is the backdrop to comprehension. The child may need to hear quite a number of sentences before he has sufficient evidence that these interfaces all match, supporting the view that frequency of exposure will correlate with point of acquisition. When passive is entered in the grammar, then it can be immune to context, allowing the child to understand nonsense, jokes, etc. Thus context can play a role in acquisition, where it is minimized in the final grammar. In sum, the pragmatics of the context is connected by innate stipulations to syntax and its semantics in a narrow range of options that the child searches through.

Outside of a role in acquisition, there are limited open parameters where the role of context participates in parameterization. Huang (1982) showed that there are language-particular choices: ‘hot’ and ‘cool’ languages allowing different amounts of contextual deletion. The acquisition path may allow the child to begin with a ‘hot’ language—where there is an over-reliance on context—and shift to a ‘cool’ language just in case she is speaking an English-like language. Thus important and precise openings to ‘context’ are themselves part of grammar and may be linked to subtle variation in, for instance, where object-drop is possible. We are just at the outset of discovering the acquisition path for context and pragmatic implicatures which may play a role in the recognition of syntax. These connections call for an enrichment, not a minimization, of the innate component and they feed other forms of efficient computation.

(p. 573) 24.3 Some conclusions

The first half of this chapter has laid bare the mechanisms whereby, assuming UG, the child is able to analyze and compare the input data. The fact that we are able to build a fairly intimate model of how UG extends to models that accept the raw primary linguistic data is a support both for the abstractions of minimalism and for the data comparison systems that utilize them.

The second half explored the promise of minimalism in the microscopic terrain of spontaneous acquisition. We have provided an overview of where minimalist principles are at work: Merge and Label, Merge over Move, the Strong Minimalist Thesis, and its impact upon recursion. Constraints, like barriers, are always obeyed, but we found children not only follow the barrier-like constraints of the Strong Minimalist Thesis, but as well, show spontaneous evidence of phase-based effects. We introduced questions about interfaces and argued that they are central to the acquisition process, allowing confirmation of syntactic analysis. It constitutes an important alternative to viewing children's language as the result of interfering performance factors. Drawn together, this evidence validates the core prediction of minimalism: if the theory is correct, then its mechanisms should be transparent in the acquisition process.

Notes:

Thanks to all of our students and colleagues. The authors' names are ordered as to reflect the organization of the chapter.

(1) By implication, these findings suggest that the claims about children's adult-like linguistic competence (e.g. Pinker 1984, Crain and Thornton 1999) and very early parameter setting (Wexler 1998) must be refined; see Yang (2002, 2009) for details with specific reference to null subjects and verb second (cf. Poeppel and Wexler 1993). In any case, it is important to note that these claims are generally made without an adequate theory of *how* the

learner manages to arrive at the target grammar, which does vary from language to language; see section 24.1.2.

(2) Note that the size of the search space may not matter as much as it appears. A system with 1,000 parameters seems harder to learn than one with only 50 parameters. But if the former consists only of parameters whose values can be determined independently (e.g. in the sense of Yang 2002), and the latter has massive ambiguity problems resulting from parameter interactions, then the larger space can be more plausibly learned than the latter. Recent work of Sakas and Fodor (2009) finds via computer simulation that in a linguistically realistic and complex domain of parameters, the majority of them may indeed be independent.

(3) Our discussion focuses on those data where we see principles most directly, primarily English, and where the arguments can be presented efficiently. Similar claims can be made about other realms and with other languages which we have not chosen to focus upon, including root infinitives, VP-ellipsis, passive, relative clauses, articles, and quantification among others. See the articles in de Villiers and Roeper (to appear).

(4) See Philip (2004) and references in (Drozd 2001).

(5) Conroy et al. (2009) continue to offer performance-based accounts of the evidence, but their discussion fails to include PP-pronoun violations ('he has it with him') discussed above, which changes the nature of the problem. Nevertheless they observe that the performance-based accounts still do not explain even the facts they consider, which suggests that the performance variables should be seen as allowing a grammatical option to surface.

(6) See as well Goodluck (1978) and more recent work by Crain and Thornton (1998) and references therein; see Lidz and Musolino (2002) for recent evidence of c-command effects with quantification.

(7) See Roeper (1996) and Roeper and Rohrbacher (2000) for extensive further evidence.

(8) Other abstract operations presumably delivered by UG are present in earlier theories but remain significant sources of spontaneous overgeneralization, e.g. Operators. Operators capture a wide variety of discontinuous connections in grammar. Are they among the primitives a child uses in early comprehension and production? Does a child who recognizes variation seek co-variation at a distance? Spontaneous acquisition suggests that Operators are among children's first analytic devices.

Two-year-olds say 'this is to eat' which can be analyzed as [This1 is [OP1 t1 to eat t1]] with an Operator linking the *trace* after *eat* to the lower CP to the upper subject.

The presence of Concord like 'I don't want none' without an adult model suggests that children seek and project Operator-variable relations. It is arguably present in many forms of 'overgeneralization' in children's grammar, from tense to plural to quantification: 'feetses', 'had came', and 'both rabbits are on both sides of the fence' (Partee, p.c.).

(9) Lebeaux (2000) had already argued that acquisition data supported structure-building notions like Adjoin-alpha, which were more abstract than either phrase structure rules or subcategorization would allow.

Lebeaux (2009 and references therein) argues that the early stages where no evidence of functional categories are present support a notion of subgrammars within a sequence of modules involving case assignment and movement. He points to evidence from L2 acquisition (Vaininikka and Young-Scholten 1994, in prep) and Tag-grammars (Frank 2002), which also are compatible with these early stages of acquisition. The acquisition data provides critical support for this nested conception of grammar (which, in turn, underlies his analysis of very complex binding structures in the adult grammar).

(10) In principle, Merge might allow any combination, but in fact we neither find any combination like ***want to* based on the frequent experience of ellipsis ('do you want to?') or *'yes want to', or 'said ate' = 'I said I ate'. Therefore more must be said about the features inside these first Merges.

(11) See Lebeaux (2009) for this argument. It may be a notion of Abstract Agree; see Roeper et al. (2003) for related evidence from language disorders.

(12) See also Baauw (2000) for cases where strong islands seem to be violated, pointing toward pragmatic factors that cause phenomena like subjacency to be called weak constraints. The fact that children can realize violable

constraints increases the necessity for a strong innate, input oriented bias because they must in effect overlook exceptions. Work in phonology on Optimality Theory is relevant to this line of reasoning.

(13) This approach converges, from an acquisition perspective, with work on trace conversion (Sauerland 2003) and multi-dominance (Johnson 2009) which is at the forefront of current syntactic research.

(14) See Boeckx (2008) for discussion of an Alternating Phase Constraint which entails the interpretive impact of the SMT. See Hollebrandse and Roeper (2009) for discussion and extension to acquisition.

(15) See Roeper (1981); Chomsky (1980) observes that acquisition follows ‘triggering experience’. An example of this interface is the view by Wexler and Culicover (1980) that deep structure is recognized from context. Lebeaux (2000) argued that children's capacity to map a deep structure meaning onto surface structure serves as a means to confirm the transformational mechanism that the acquisition mechanism proposed.

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A Minimalist Program for Phonology

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Abstract and Keywords

This article applies minimalist concerns to the domain of morphophonology. It advocates a bottom-up approach to phonology, made possible by treating the phonological module as a system of abstract symbolic computation, divorced from phonetic content. This approach has come to be called 'substance-free phonology'. The article argues that phonologists must stop the practice of 'substance abuse', or misguidedly mixing the study of phonological form with the properties of phonetic content.

Keywords: minimalism, morphophonology, abstract symbolic computation, substance-free phonology

25.1 Introduction

Perhaps the single most important facet of minimalism is that it turns the traditional, top-down approach to investigating universal grammar (UG) on its head. As Chomsky (2007: 3) describes it, the minimalist program 'seeks to approach the problem "from bottom up": How little can be attributed to UG while still accounting for the variety of I-languages attained[…]?' This shift in perspective is particularly apparent in more recent minimalist works (e.g. Chomsky 2004a, 2005, 2007, Boeckx 2006), but it is implicit in the Strong Minimalist Thesis, which dates back to the early 1990s. The Strong Minimalist Thesis (SMT) is, as Boeckx (2010) puts it, 'a challenge to the linguistic community: Can it be shown that the computational system at the core of [the faculty of language] is optimally or perfectly designed to meet the demands on the systems of the mind/brain it interacts with?' We hope to push the SMT as far as it can go because it encourages us to *make sense* of the language faculty's properties, not in isolation, but rather within the larger picture of cognition. What's more, pursuing this line of inquiry is bound to yield new understanding of the Conceptual-Intentional (C-I) and Sensorimotor (SM) **(p. 575)** systems, because it forces us to think about the legibility conditions imposed on the language faculty by those other modules.

Despite the way minimalism has radically reshaped our view of the interfaces, it seems odd that, as van Oostendorp and van de Weijer (2005: 3) remark, the minimalist program 'has not been applied to phonology'; similarly, Pinker and Jackendoff (2005: 220) state: 'The Minimalist Program, in Chomsky's original conception, chooses to ignore …all the phenomena of phonology' But there is no reason why this should be. The following quote summarizes one of the primary motivations behind the present work:

> For decades, generative linguists have viewed the internal grammar in terms of the interplay of two types of factors: genetic endowment, generally referred to as Universal Grammar (UG), and experience—that is, exposure to e-language. In recent years this picture has been augmented by a third type of factor: general principles of biological/physical design. This new focus tends to worry those who had been hoping for a rich and articulate UG (see Pinker and Jackendoff [2005]), but on the other hand it is fully in line with minimalist thinking. A particularly welcome effect produced by this shift of focus is that we may now reassess the issue of formal similarities and dissimilarities between syntax and phonology. For many years,

> the dominant view has been that syntax and phonology are fundamentally different. […] But general principles of design may very well be active in syntax and phonology in similar ways. (van Riemsdijk 2008: 227)

Investigating phonology from this perspective lays the groundwork for testing and refining the arguments made by Bromberger and Halle (1989) in support of the view that phonology is fundamentally different from syntax (*contra* van der Hulst 2005 and Anderson 2006). Such work allows us to focus not on the question of *whether* phonology is different, but rather *how* it is different and *why* this is the case. If we are correct to emphasize the role of Third Factor principles in the architecture of grammar (Chomsky 2005), then this should be a fruitful endeavor.

I argue further that phonologists should take seriously the idea advanced in many recent minimalist writings that phonology is an ‘ancillary’ module, and that phonological systems are ‘doing the best they can to satisfy the problem they face: to map to the SM interface syntactic objects generated by computations that are “well-designed” to satisfy C-I conditions’ but unsuited to communicative purposes (Chomsky 2008b: 136). Phonology is on this view an afterthought, an externaliza-tion system applied to an already fully functional internal language system. While some have taken this to suggest that phonology might be messy, and that we shouldn't expect to find evidence of ‘good design’ in it, there is another perspective which suggests instead that the opposite conclusion is warranted: phonology might be much *simpler* (less domain-specific) than has previously been thought, making use of only abilities that already found applications in other cognitive domains at the time externalized language emerged (see section 25.5; see also Mobbs 2008).

(p. 576) 25.2 The Substance-Free Approach

I advocate here (and in recent work, most notably Samuels 2009a) for a bottom-up approach to phonology, made possible by treating the phonological module as a system of abstract symbolic computation, divorced from phonetic content. This approach has come to be called ‘substance-free phonology,’ as first described by Hale and Reiss (2000a, 2000b). We argue that phonologists must stop the practice of ‘substance abuse’, or misguidedly mixing the study of phonological form with the properties of phonetic content.[1] As summarized by Reiss (2008: 258–9, emphasis original):

> [Hale and Reiss (2000a, 2000b)] conclude that the best way to gain an understanding of the computational system of phonology is to assume that the phonetic substance (say, the spectral properties of sound waves, or the physiology of articulation) that leads to the construction of phonological entities (say, feature matrices) *never* directly determines how the phonological entities are treated by the computational system. The computational system treats features as arbitrary symbols. What this means is that many of the so-called *phonological universals* (often discussed under the rubric of markedness) are in fact epiphenomena deriving from the interaction of extragrammatical factors like acoustic salience and the nature of language change. Phonology is not and should not be grounded in phonetics since the facts which phonetic grounding is meant to explain can be derived without reference to *phonology*.

The goal of substance-free phonology is to determine the nature of the universal core of formal properties that underlie all human phonological systems, regardless of the phonetic substance or indeed of the modality by which they are expressed. The substance-free approach is, like minimalism, a research program rather than a theory; multiple different theories are being explored. All these theories share the following set of assumptions:

(1) The common basis of substance-free phonology (Blaho 2008: 2)

- Phonology refers to the *symbolic computational system* governing the signifiant, i.e. the non-meaningful level of linguistic competence. Phonology is taken to be *universal*—common to all (natural human) languages and all modalities—and *innate*. Phonological knowledge is part of UG, but phonetics is not.
- Phonological primes are substance-free, in that their phonetic interpretation is invisible to phonology, and thus does not play a role in phonological computation.
- (p. 577) Markedness and typological tendencies (in the sense of Greenberg 1957, 1978) are not part of phonological competence, but rather an epiphenomenon of how extra-phonological systems such as perception and articulation work.

One of the most salient arguments in favor of maintaining a substance-free phonology concerns the nature of what a theory of UG, and of phonological UG in particular, should seek to explain. Hale and Reiss (2008: 3) set up the following hierarchy:

(2) Attested ⊂ Attestable ⊂ Humanly computable ⊂ Statable

a. Attested: Cree-type grammars, English-type grammars, French-type grammars

b. Attestable: ‘Japanese’ in 200 years, Joe's ‘English’

c. Humanly computable: p → s/_r

d. Statable: V → V: in prime numbered syllables: paka 2nu 3tipa 5forse 7 → paka:nu:tipa:fose:

Clearly, the set of attested grammars is inappropriately small: it is, I hope, un-controversial that the list of attested languages does not exhaust the possibilities provided by UG. Conversely, the set of statable languages is far too large: it seems like a pretty safe bet that no grammars refer to the set of prime numbers, or the sign of the Zodiac at the time of the utterance, or whether the interlocutor owns any blue-collared shirts, etc.

The more pressing question is whether it is correct for a theory of UG to zero in on the set of attestable languages, or the humanly computable ones. Advocates of substance-free phonology hold that the idea ‘that every possible [grammar] should be instantiated by some attested language …is naïve, just as it is deeply naïve to expect that all logically possible permutations of genetic material in the human genome are actually attested in individual humans’ (Vaux 2008: 24). In Newmeyer's (2005) terms, we maintain that synchronic phonological theory should characterize only the set of possible languages, not probable ones. Instead, the biases typically attributed to formal markedness should be explained by reference to properties of our perception and production systems, and to sheer accidents of history; this shifts much of the burden to the theory of sound change.

If synchronic phonological theory's sole task is to describe what is a possible synchronic phonological pattern/process—in other words, if markedness is not part of phonological competence—then what accounts for the fact that some patterns and processes are more common than others? Blevins (2004: 8–9; emphasis original) states the working hypothesis very clearly:

> *[R]ecurrent synchronic sound patterns have their origins in recurrent phonetically motivated sound change*. As a result, there is no need to directly encode the frequent occurrence of these patterns in synchronic grammars themselves. Common instances of sound change give (p. 578) rise to commonly occurring sound patterns. Certain sound patterns are rare or unattested, because there is no common pathway of change which will result in their evolution.

The locus of explanation for phonological typology largely shifts to diachrony, because ‘many of the so-called *phonological universals* (often discussed under the rubric of markedness)’ are not exceptionless, and ‘are in fact epiphenomena deriving from the interaction of extragrammatical factors such as acoustic salience and the nature of language change’ (Hale and Reiss 2000a: 167; emphasis original).[2] I will set the issue of diachrony aside here (but see Samuels 2009a: ch. 2), and concentrate on the consequences of this view for the phonological portion of UG in the rest of the present work.

25.3 A Minimalist Program for Phonology

The work undertaken here opens the door for future research into questions which are independently raised and particularly timely given another consequence of the minimalist program (see Boeckx 2009e): genuine variation within narrow syntax has been eliminated, being relegated instead to the lexicon and to morphology. As a result, there can be no more study of comparative *narrow* syntax, but careful investigation of phonological representations and processes can provide complementary data that is bound to inform our knowledge of syntax, both narrowly and broadly construed. Given what we now understand about syntactic architecture in light of minimalism, such a study lays the groundwork for testing and refining the arguments made by Bromberger and Halle (1989) in support of the view that phonology is fundamentally different from syntax, a view which has been opposed by van der Hulst (2005) and Anderson (2006); we can begin to focus not on the question of whether phonology is different, but rather how it is different and why this is the case.

A major theme which I explore in recent work (especially Samuels 2008, 2009a) is that, while phonology and syntax may look similar on the surface—and this is not likely to be a coincidence—upon digging deeper, crucial differences between (p. 579) the two modules begin to emerge. One area where surface similarities hide striking differences is in the comparison between phonological syllables and syntactic phrases. Syllables and phrases have been equated by Levin (1985) and many others, with some going so far as to claim that phrase structure was exapted from syllable structure (Carstairs-McCarthy 1999). I argue these analogies are false, and that many of the properties commonly attributed to syllabic structure can be explained as well or better without positing innate structure supporting discrete syllables in the grammar. And in keeping with purely syntactic accounts of prosodic domains such as those proposed by Kahnemuyipour (2004), Ishihara (2007), and others, I move to eliminate the prosodic hierarchy as well, instead arguing that phonological phrasing is directly mapped from the phase structure of syntax, as we will discuss in the next section. This means phonological representations are free to contain much less structure than has traditionally been assumed, and in fact that they are fundamentally 'flat' or 'linearly hierarchical'.

I posit only three basic computational operations for phonology (see Samuels 2009a):

- Search provides a means by which two elements in a phonological string may establish a probe—goal relation. The search algorithm, formulated by Mailhot and Reiss (2007), formalizes the system of simultaneous rule application proposed in Chomsky and Halle (1968: 344): 'to apply a rule, the entire string is first scanned for segments that satisfy the environmental constraints of the rule. After all such segments have been identified in the string, the changes required by the rule are applied simultaneously.'
- Copy takes a single feature value or bundle of feature values from the goal of a search application and copies these feature values (onto the probe of the search).
- Delete removes an element from the derivation.

The relationship of these phonological operations to their apparent counterparts in (narrow) syntax remain to be investigated in more depth. However, one crucial difference between syntactic and phonological structure-building operations—the fact that phonology lacks Merge (combine α and β symmetrically)—can already be noted; Samuels and Boeckx (2009) argue that this has multiple important consequences which are responsible for some of the ultimately distinct characters of these two systems. The counterpart of Merge at PF is what we call Concatenate (combine α and β asymmetrically), which is not a primitive operation, but rather is decomposable into Search and Copy creating new precedence relationships between segments (see Raimy 2000 on precedence and Samuels 2010 on this application of Search and Copy). Whereas iterative applications of Concatenate yield a flat structure, iterative applications of Merge yield a nested hierarchical structure: syntactic structures must be flattened, whereas linear order is a primitive in phonology (Raimy 2000). Also, since phonology lacks Merge, it also follows that (p. 580) it lacks movement, since movement is a subspecies of Merge (Internal Merge or Re-Merge; Chomsky 2004a). Without the possibility of re-merging the same element, the notion of identity is extrinsic in phonology, unlike in syntax (see Raimy 2003).

Further investigation of all these issues is, in my view, central to extending minimalism into the phonological domain. However, for reasons of space I must focus here on only one topic, namely how it is possible to construe the syntax—phonology interface in such a way that phonology need not build its own domains, but can merely operate over the strings it receives from the syntax directly. It is also worth noting here that the picture sketched below is quite far from the whole story concerning the syntax—phonology interface. The exact mechanisms underlying such PF operations as linearization, vocabulary insertion, and copy deletion remain to be uncovered, as does the ordering of these operations; the present work focuses on only one particular, seemingly late, stage of the mapping from syntax to phonology.

25.4 Phonological Derivation By Phase

Throughout the generative era, several cyclic models of phonology have been proposed. The first of these was introduced by Chomsky et al. (1956). The phonological cycle became a crucial component of Chomsky and Halle (1968) and was adopted in syntax by Chomsky (1957). In phonology, this concept was later implemented as the 'strict cycle' of Kean (1974) and Mascaró (1976). The tradition of lexical phonology (and morphology) begun by Kiparsky (1982) and Mohanan (1982) developed the idea of cyclicity further, building on Pesetsky (1979).[3]

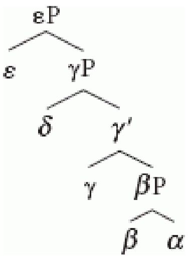
Recently, a new movement in phonological theory has emerged, attempting to combine the insights of lexical phonology with distributed morphology (Halle and Marantz 1993) and the phase-based theory of narrow syntax and the syntactic interfaces developed in Chomsky (2001, 2008b; derivation by phase or DbP). The theory presented here, which I call phonological derivation by phase (PDbP), falls under this umbrella. It takes as a starting point the conceptual argument laid out in the foundational work by Marvin (2002: 74): 'If we think of levels in the lexicon as levels of syntactic attachment of affixes, we can actually say that Lexical Phonology suggests that phonological rules are limited by syntactic domains, possibly phases.'

From a minimalist standpoint, a model of grammar with synchronous cycles across the various modules is highly desirable. Indeed, this is the 'best-case scenario' for computational efficiency according to Chomsky (2004a: 107). There is also a growing body of literature which argues that phases are required to regulate syntax's interfaces with the semantic and phonological components. See, for (p. 581) instance, Boeckx (2009e) on how phases facilitate 'wild-type' or 'free' Merge and a conjunctivist semantics of the type proposed by Pietroski (2005a et seq.).

Moreover, PDbP also allows us to recognize the important contributions of cyclic models of phonology. For instance, all attempts to account for phonological opacity effects in a monostratal theory suffer from serious empirical or technical problems (see Vaux 2008 and references therein for discussion). Since the model proposed here relies on a cycle that is not proprietary to phonology, it is insulated from one family of recurring criticisms of lexical phonology, namely that its levels were poorly motivated and allowed to proliferate in an unconstrained manner (see e.g. Itô and Mester 2003). In PDbP, by contrast, we expect evidence for the cycle to come from syntax and semantics in addition to (morpho)phonology. And there can be no *ad hoc* stipulation of cycles/levels if a phonological analysis must be responsible to, and grounded in, such external evidence; conversely, phonological phenomena should be able to provide evidence which is helpful for syntactic analysis.

25.4.1 Phases and spell-out

Before going any further, we should clarify how the basic phase architecture works. Consider a syntactic tree like the one below.



(3)

At certain points during the construction of this structure, the derivation is punctuated by the introduction of a phase head. What is crucial for present purposes is that phase heads initiate Transfer or Spell-Out, sending a chunk of the completed derivation to the semantic and phonological systems. Specifically, the complement of a phase head is the chunk that gets transferred, at the point when another phase head enters the derivation. Upon transfer, the 'spell-out domain' (transferred chunk) is rendered opaque to further syntactic operations. This is formalized in the Phase Impenetrability Condition:

(4) Phase Impenetrability Condition (PIC) (Chomsky 2001 version)
For [ZP Z …[HP α [H YP]]]: The domain of H is not accessible to operations at ZP, but only H and its edge.

Typically (or perhaps even necessarily; see M. D. Richards forthcoming), phase heads and non-phase heads alternate with one another, so the chunks being (p. 582) transferred are larger than a single terminal. For (3) below, let us assume that only γ and ε are phase heads. The derivation will proceed as follows:

(5)

a. Merge (β, α): α accessible to β.
b. Merge (γ, βP): β, α accessible to γ.
c. Merge (δ, γ'): γ accessible to δ.

d. Merge (ε, γP): δ, γ accessible to ε.βP transferred.

In the discussion to follow, I assume that Uriagereka's (1999) conception of Multiple Spell-Out (i.e. complex specifiers and adjuncts are spelled out alone) and Chomsky's phase framework can be simultaneously entertained. One means for accomplishing this is suggested by recent proposals such as Narita (2009b), Narita and Samuels (2009), and Boeckx (2009e), which argue that only simplex syntactic objects can undergo Merge: complex objects introduced on a left branch must therefore be reduced to simplex objects before they can be integrated with the main derivational spine. This is achieved by the transfer of all but the head of the mergee. That is to say, complex specifiers and adjuncts must be headed by phase heads.[4]

One important clarification is necessary in order to enable us to make broader use of the PIC. In narrow syntax, 'accessible to operations' essentially means eligible for movement (i.e. Internal Merge or Re-Merge), and able to participate in Agree. For phonological purposes, I will move forward under the assumption that an accessible string of phonology is visible to SEARCH and modifiable by COPY and DELETE. Now let us assume, then, that phase impenetrability holds in phonology, so each phonological string becomes inaccessible subsequent to the transfer of another string to the phonological component. By preventing reaching back too far into the derivation, the PIC derives the effects previously attributed to the erasure of morpheme boundaries ('brackets') at the end of every cycle (Siegel 1974, Mohanan 1982), opacifying the results of earlier cycles. In other words, a rule can only affect something on its own cycle and/or the previous one, nothing more. The solution adopted here is similar in spirit to the lexical phonology tradition: word-building operations and phonological rules interleave, and the PIC prevents modifying previous cycles after they are built.

Another idea which is crucial to PDbP is that phasal domains are identifiable not just at the clausal level (i.e. *v*, C, etc.) but also within words. Parallel to *v*, Marantz (2001) establishes {*n, a*} as phase heads. In distributed morphology terms, following (p. 583) Marantz (1997), these elements are the categorial heads to which a-categorial roots must merge, and derivational affixes also belong to these classes. Marvin (2002) and Di Sciullo (2004, 2005) argue on multiple independent grounds that the PIC holds for these 'morphological phases.' I argue in §5.2 of Samuels (2009a) that lexical rules are responsible to the PIC on this smaller scale—a lexical rule has as its domain two adjacent morpheme-level Spell-Out domains—while post-lexical rules are responsible to the PIC at the clausal level.

The strongest claim made by the PDbP approach is that Spell-Out domains are the only domains that phonology needs. In other words, both the levels of lexical phonology and the constituents of the prosodic hierarchy come for free when we assume distributed morphology and a phasal syntax: phonological domains are directly imposed by morphosyntactic structure, and phonology need not erect any boundaries. It has been recognized for at least forty years (i.e. at least back to Chomsky and Halle 1968) that phonological domains correspond—in some fashion—to morphosyntactic ones. If the correspondence is not one of exact congruence, then phonology must construct (or adjust) boundaries. But if the correspondence is exact, then phonology can simply 'read' the structures it is given. Theories that assume exact correspondence subscribe to the 'direct reference' conception of the syntax—phonology interface; see Kaisse (1985), Odden (1990), and Cinque (1993). In recent literature, it is common to read that direct reference can not be correct because there are apparent mismatches between syntactic and phonological domains. This is the position held by proponents of 'indirect reference' theories such as Selkirk (1984), Nespor and Vogel (1986), Truckenbrodt (1995), Seidl (2001), and many others. If PDbP is correct, there is no need to abandon direct reference for an indirect theory. In fact, the situation is even better: phonology doesn't have to read syntactic boundaries, it just applies to each chunk as it is received. PDbP can thus lead us to an understanding of phrase-level phonology that involves no boundary construction and eliminates the prosodic hierarchy.

25.4.2 Prosodic hierarchy theory

Since Selkirk (1978), and in classic works on prosodic hierarchy theory such as Selkirk (1984) and Nespor and Vogel (1986), a hierarchy of phonological constituents has been identified.[5] The most standard of these are (from smallest to largest, or weakest to strongest) the phonological word (ω), the phonological phrase (ϕ), the intonational phrase (I-phrase), and the utterance (U).

It is commonly (though not exceptionlessly) thought that this hierarchy of constituents obeys the conditions in (6)–

(7) (Selkirk 1984, Nespor and Vogel 1986):

(p. 584) **(6)** STRICT LAYERING HYPOTHESIS
A given nonterminal unit of the prosodic hierarchy, X^P, is composed of one or more units of the immediately lower category, X^{P-1}.
(7) PROPER CONTAINMENT
A boundary at a particular level of the prosodic hierarchy implies all weaker boundaries.

That is, the prosodic hierarchy is nonrecursive (though see Dobashi 2003 and Truckenbrodt 1995 et seq. for arguments to the contrary), and no levels can be skipped.[6]

The fundamental hypothesis of prosodic hierarchy theory is that the constituents that are suggested by these converging lines of evidence are correlated with, but not isomorphic to, syntactic constituents. For this reason, it is (proponents of the prosodic hierarchy claim) necessary to erect and adjust boundaries in the phonology, on the basis of syntactic information. Two general schools of thought have emerged on how this construction is undertaken: the relation-based mapping approach represented by Nespor and Vogel (1986), and the edge- or end-based mapping approach represented by Selkirk (1986) and, in Optimality-Theoretic terms, Truckenbrodt (1995, 1999). I briefly summarize below how ø is constructed in each of these theories.

(8) Relation-based ϕ-construction (Nespor and Vogel 1986: 168ff.)
a. *ϕ domain*
The domain of ϕ consists of a *C* [clitic group] which contains a lexical head (X) and all *C*s on its nonrecursive side up to the *C* that contains another head outside of the maximal projection of X.
b. *ϕ construction*
Join into an n-ary branching ϕ all *C*s included in a string delimited by the definition of the domain of ϕ.
c. *ϕ restructuring* (*optional*)
A nonbranching ϕ which is the first complement of X on its recursive side is joined into the ϕ that contains X.

(9) End-based ϕ-construction (Truckenbrodt 1995: 223)
A language ranks the two following universal constraints:
a. ALIGN-XP, R: ALIGN (XP, R; ϕ, R)
For each XP there is a ϕ such that the right edge of XP coincides with the right edge of ϕ.
(p. 585) **b.** ALIGN-XP, L: ALIGN (XP, L; ϕ, L)
For each XP there is a ϕ such that the left edge of XP coincides with the left edge of ϕ.

Dobashi (2003) shows how the theories in (8)and (9) make different predictions with regards to the syntactic structure in (10):

(10) $[_{IP}$ NP $_{subj}$ Infl $[_{vp}$ V NP $_{Obj}]]$

The relation-based model in (8)willconstruct (11a), and if the optional restructuring rule applies, (11b). The end-based model in (9), if ALIGN-XP, R outranks ALIGN-XP, L, will construct only (11b).[7]

(11) ϕ boundaries for (10)
a. (NP $_{subj}$)ϕ (Infl V)ϕ (NP $_{obj}$)ϕ
b. (NP $_{subj}$)ϕ (Infl V NP $_{obj}$)ϕ

The two prosodic hierarchy models therefore agree on one thing, namely that the subject must always be phrased separately, which a great deal of literature on prosodic phrasing in SVO languages has shown is generally true (see Dobashi 2003: ch. 2). However, they differ as to whether it is possible to phrase the object alone as well. The fact that prosodic hierarchy theory (in whatever its guise) predicts such a restricted set of prosodic constituents ('domain clustering') is often cited as an advantage. Inkelas and Zec (1995: 548) write:

> in making these predictions, the Prosodic Hierarchy Theory distinguishes itself dramatically from so-called direct access theories … in which each individual phonological rule may specify its own unique syntactic conditions. There is no expectation in such theories of any convergence or mutual constraining effect among rule domains.

In the remainder of this section, I attempt to show that, while Inkelas and Zec's criticism may be valid for the particular direct reference theories formulated prior to 1995, within a phase-based model of grammar, direct reference is actually more constrained and more accurate in its predictions than prosodic hierarchy theory, and far more parsimonious in its assumptions.

Another important difference between direct and indirect reference theories has to do with modularity. Paraphrasing Seidl (2000), both sides acknowledge that there are phonologically relevant domains at the phrasal level; direct reference theories state these domains in terms of syntactic primes, while indirect theories state them in terms of phonological primes. This is not a matter of mere preference; adopting indirect reference violates the modular architecture of grammar. For indirect reference theories, prosodic constituents are constructed from a syntactic representation, (p. 586) as should be obvious from (8)–(9). And yet, for Optimality-Theoretic approaches which use constraints like the ones in (9),

> prosodic structure is created by ALIGN and WRAP constraints *in the phonology*, i.e. the constraints at hand being interspersed with purely phonological constraints in the same constraint hierarchy. Mapping between morpho-syntax and phonology, which is what ALIGN and WRAP do, is a process that needs to be able to interpret morpho-syntactic structure—something that is impossible on modular grounds when sitting in phonology. (Scheer 2009: n. 14)

In short, if we want to maintain that phonological representations do not include syntactic information, then the indirect mapping approach is not viable (see also Scheer 2008: esp.§7.4).

Many other arguments against the prosodic hierarchy exist, particularly in light of bare phrase structure (Chomsky 1995a), in which it is impossible to refer to syntactic projections (i.e. XP), as both relation- and edge-based approaches must (see Dobashi 2003:10ff.). I will not attempt to recap these arguments here. The analyses presented by Seidl (2000, 2001) and Scheer's (2008) conceptual arguments are to my mind particularly devastating for prosodic hierarchy theory, and I encourage the reader to consult these works. I will limit myself to one very simple argument here: as I have already mentioned, the reason why indirect reference theories exist in the first place is that there are allegedly mismatches between syntactic structure and phonological domains. One famous mismatch, already noted by Chomsky and Halle (1968), is shown in (12). Brackets represent clause boundaries and parentheses represent I-phrase boundaries.

(12)

a. Syntax: This is [the cat that caught [the rat that stole [the cheese]]]
b. Phonology: (This is the cat) (that caught the rat) (that stole the cheese)

However, a phase-based approach to syntax fares much better when it comes to approximating both syntactic and phonological phenomena. In fact, I believe it fares so well that there are no longer mismatches, and the argument for indirect reference therefore disappears. For example, one diagnostic for the phrasing in (12b) comes from stresses on *cat, rat*, and *cheese*. But these stresses can be generated by a rule which accents to the highest element in each clause-level Spell-Out domain (Kahnemuyipour 2004); the purported mismatch is therefore illusory.

25.4.3 Direct reference and phase domains

Ultimately, direct reference can be maintained only if a Spell-Out domain corresponds *always* and *exactly* to ϕ, and only if ϕ is the unique level of phonological domain necessary above the word level can we have a truly direct reference theory of the interface. The purpose of the present section is to give proof of concept for a theory that, in pursuing this goal, eliminates the recourse to projections/labels that (p. 587) plague prosodic hierarchy theory. This pared-down syntax—phonology interface should be the null hypothesis, as has been stated most explicitly by Scheer (2008), given that Spell-Out is precisely the operation that connects syntax and phonology.

It would be a massive undertaking to show that phase domains suffice for every rule with a claimed domain of ϕ, but I attempt to give proof of concept in Samuels (2009a: §5.6) by using vowel assimilation in Lekeito Basque, obstruent voicing in Korean, and second-position clitic placement in Sebro-Croatian as case studies. I demonstrate that the PIC predicts the application and blocking contexts of both processes, both above and below the word level, in exactly the way I have just described. Further empirical studies in phase-based phonology include Seidl (2001) primarily on Bantu and Korean; Marvin (2002) on English and Slovenian; Kahnemuyipour (2004) on Persian, English, and German; Piggott and Newell (2006) and Newell (2008) primarily on Ojibwa; Sato (2006) on Taiwanese, French, Gilyak, Kinyambo, and Welsh; Ishihara (2007) on Japanese; Bachrach and Wagner (2007) on Portuguese; Michaels (2007) on Malayalam; Kamali and Samuels (2008a, 2008b) on Turkish; and more programmatically, Embick (2008b) and Scheer (2008, 2010). At present, however, I will restrict myself to describing how we can achieve the attested typology of phonological phrasing in SVO languages using a phase-based model.

Let us consider what prediction a phase-based system makes about ϕ-construction, compared to prosodic hierarchy theory. Dobashi (2003) argues that there are essentially four types of attested SVO languages, none of which exhibit $(SV)_\phi$ $(O)_\phi$ phrasing:

(13) Typology of ϕ-domains in SVO languages (Dobashi 2003: 38)
a. $(S)_\phi$ $(V)_\phi$ $(O)_\phi$
b. $(S)_\phi$ $(V)_\phi$ $(O)_\phi$ or
$(S)_\phi$ $(VO)_\phi$ if O is non-branching
c. $(S)_\phi$ $(VO)_\phi$
d. $(S)_\phi$ $(VO)_\phi$ or
$(SV)_\phi$ if S is non-branching

The type of language in (13a) is exemplified by French, as shown in (14), and the Aŋlɔ dialect of Ewe. Italian falls under (13b); see (15). Kimatuumbi (16) represents (13c), and Kinyambo (17) is of the type in (13d).[8] All the examples below are taken from Dobashi (2003: §2.2); I indicate the phonological phrasing with brackets.

(p. 588)

(14)

[L′	Immigré]	[envoyait]	[un	paquet]	[à	sa	famille]	
the	immigrant	sent	a	package	to	his	family	
								(french)
‘The immigrant sent a package to his family’								

(15)
a.

[Venderá]	[questo	leopardo]	[in	Dicembre]	(Italian)
sell.FUT.3SG	this	leopard	in	December	
‘He will sell this leopard in December’					

b.

[prenderá]	[tordi]	*or*	[prendera tordi]
catch.FUT.3SG	thrushes		
‘He will catch thrushes’			

(16)
a.

[Mamboondó]	[aawííle]	(Kimatuumbi)
Mamboondo	die.PST.3SG	
‘Mamboondo died’		

b.

[naamwéeni	nchéngowe	Maliíya]
see.PST.1SG	husband	Mary's
‘I saw Mary's husband’		

(17)
a.

[abakozi	bákajúna]	(Kinyambo)
workers	help.PST.3PL	
‘The workers helped’		

b.

[abakozi	bakúru]	[bákajúna]
workers	mature	help.PST.3PL
‘The mature workers helped’		

c.

[okubon’	ómuntu]
see	person
‘To see the person’	

However, this typology omits crucial data concerning the phrasing of clitics in French, as Narita and Samuels(2009) discuss. A subject and/or object clitic is actually phrased together with the verb:

(18)

[Donnez	en]	[à	Marcel]
give	some	to	Marcel
‘Give some to Marcel’			

(19)

[Nous	allons]
we	go.1PL
'We go'	

(p. 589) **(20)**

[Marie	le	voit]
Marie	him	Sees
'Marie sees him'		

The same is (optionally) true of simplex objects in Aŋlɔ Ewe, as shown below:

(21)

[mē]	[kpe˝	fle˝-gé]
I	stone	buy-prt
'I'm going to buy a stone'		

Thus, Dobashi's typology should be amended, as it turns out there are only two distinct types of phrasing, not four; furthermore, French and Aŋlɔ Ewe do not in fact pattern identically, while French and Kinyambo do.[9]

(22) Revised typology of ϕ-domains
a. $(S)_\phi$ $(V)_\phi$ $(O)_\phi$
$(S)_\phi$ $(V\ O)_\phi/(O\ V)_\phi$ if O is non branching (Aŋlɔ Ewe, Italian, Kimatuumbi)
b. $(S\ V)_\phi$ $(O)_\phi$ if S is non-branching
$(S)_\phi$ $(V\ O)_\phi/(O\ V)_\phi$ if C is non-branching
$(S\ O\ V)_\phi$ if S and O are non-branching (French, Kinyambo)

Now we will see how to generate this typology in PDbP. We start with the basic assumption, as introduced earlier, that a phase head's complement domain (minus what has already been spelled out) is transfered to the phonology as a unit, corresponding to ϕ in prosodic hierarchy theory. We begin with the fairly standard assumption that the inventory of phase heads includes C (or at least one head in an expanded Left Periphery), *v*, and D. Say the verb raises to v or to T. It so happens that in all of the languages mentioned above, there is evidence for V-to-T raising, but even if the verb only raises to *v*, an in situ object will be spelled out by *v* separately from the verb.

We also have to account somehow for the typological fact that branching arguments behave differently from non-branching ones. In (13) and (22) we see this asymmetry in object phrasing, which Dobashi (2003) accounts for by saying that object raising is optional and can only be undergone by simplex objects. Narita and Samuels (2009) provide an alternative account under which movement of a branching object will still result in its being phrased separately, if movement of a branching phrase requires prior spell-out of the phrase head's complement. In either of these accounts, a simplex object which raises out of VP may be phrased with the verb, for instance if it instantiates D (e.g. a pronoun) or undergoes N-to-D raising (e.g. a proper name; see Longobardi 1994); there is also the possibility that some languages which do not exhibit overt determiners may not even have D, as (p. 590) argued by Bošković (2005); we predict that in such a language, a raised NP object could phrase with a verb in T or *v* regardless of branchingness.

Comparing Italian with Ewe and Kimatuumbi, we note that they are alike in having the subject undergo A′-movement to TopicP (on the connection with pro-drop in Italian, see Alexiadou and Anagnostopoulou 1998 and Frascarelli 2007). For example, a subject can be followed by a complementizer in Kimatuumbi, which shows that subjects can be quite high in this language. The fact that subjects and preposed elements all behave alike with regard to phonological rules diagnostic of the ϕ domain in this language also suggests that this may be correct. It has been independently argued for Bantu that subjects typically appear in Topic (see Demuth and Mmusi 1997 and references therein on Sesotho and Bantu more generally; also Pak 2008 on Luganda). Simplex objects (at least; recall the previous paragraph) may raise in all three languages.

The difference between French and Kinyambo on the one hand and Italian, Ewe, and Kimatuumbi on the other is the height of the subject: in French, the subject has only moved as far as Spec, TP, so a non-branching subject can be phrased with a verb in T. In sum, the phonological phrasing exhibited in a particular language depends on the interaction of three things, holding constant the position of the verb in T/*v*: whether the subject is in an A′ position, whether the object has raised out of VP, and whether the arguments branch. There is no need for any readjustment of domains on the way from syntax to phonology, or for any purely phonological conditions tying prosodic phrasing to branchingness.

If this purely phase-based account of phonological phrases is correct, it makes explaining domain span rules very simple: they are simply post-lexical rules that clause-level phase heads trigger on their complement domains. By tying spell-out domains directly to prosodic phrasing, we also derive the Maximal ϕ Condition of M. D. Richards (2004, 2006):

(23) Maximal ϕ Condition
A prosodic phrase ϕ (…*ω*, etc.) can be no larger than a phase.

The same works for domain limit rules, or phenomena that take place at the edges of *ϕ*-domains. As we have just established, the domains in question are the complement domains of clausal phase heads. The edge of this domain can be marked by a post-lexical rule carried by the phase head since, as I mentioned earlier (and see Samuels 2009a: §5.2), a post-lexical rule sees the entire clause-level spell-out domain as a single string without any internal boundaries.

The ultimate message which I hope to convey is that, if we want to understand cross-linguistic variation in phonology, we need to understand cross-linguistic variation in morphosyntax better. This calls for collaboration between phonologists, morphologists, and syntacticians, all working together towards the common goal of describing the range of linguistic structures that are available. This could shed light on several outstanding issues, such as the intriguing phonological differences **(p. 591)** between polysynthetic and less agglutinative languages. In the languages we have seen in this chapter, a clause-level phase defines a phonological phrase which may consist of several words (recall the Maximal *ϕ* Condition). This provides an interesting contrast with the conclusions of Compton and Pittman (2010), who argue that in Inuktitut, the phase defines a single prosodic word; Piggott and Newell (2006) argue the same for Ojibwa. This suggests that at the opposite end of the spectrum are isolating languages like Chinese: for them, it is almost as if every terminal defines a prosodic word. This could perhaps be thought of as the prosodic word being defined as a morpheme-level phase rather than a clause-level one.

Many details of PDbP remain to be negotiated, given that the syntax upon which a theory of the syntax—phonology interface must necessarily depend remains in flux. Nevertheless, already there is a quickly growing list of empirical successes which have been achieved by tying phonological rule application directly to Spell-Out domains, and the phonological literature is rife with obvious candidates for phase-based analyses.

25.5 Phonology, Minimalism, and Evolution

To conclude, I would like to take a step back and consider the implications of this type of phonological theory for the minimalist program and for biolinguistic concerns such as how language may have evolved. This is an important, though understudied, area in light of criticisms of the minimalist program which hinge on arguments made concerning phonology. Most notably, in their rebuttal of Hauser et al. (2002), Pinker and Jackendoff (2005) take Chomsky to task for ignoring ‘all the phenomena in phonology’ (p. 220) in the original minimalist papers. But if the perspective for which I have advocated here is correct, then phonology need not be seen as a thorn in the side of the minimalist program.

In the previous sections we discussed some consequences of the minimalist perspective for phonological theory and the syntax—phonology interface, but in what remains we will turn the tables and ask not what minimalism can do for phonology, but what phonology can do for minimalism. I believe it makes sense to frame this discussion in evolutionary terms: in short, the idea that the language faculty is simple and the idea that language evolved quickly and with a minimum of genetic changes go hand in hand. As Hornstein and Boeckx (2009: 82) explain,

> [I]n light of the extremely recent emergence of the language faculty, the most plausible approach is one that minimizes the role of the environment (read: the need for adaptation), (p. 592) by minimizing the structures that need to evolve, and by predefining the paths of adaptation, that is, by providing preadapted structures, ready to be recruited, or modified, or third factor design properties that emerge instantaneously, by the sheer force of physical laws.

Along these lines, in Samuels (2009b) and Samuels (2009a: ch. 6), I demonstrate on the basis of behavioral and physiological studies on animal cognition that all the cognitive abilities necessary for the phonological representations and operations argued for in the previous chapters are present in creatures other than *Homo sapiens* (even if not to the same degree) and in domains other than phonology or, indeed, language proper. This implies that nothing required by phonology is part of the faculty of language in the narrow sense (FLN, as opposed to the faculty of language in the broad sense, FLB), in the terms of Hauser et al. (2002) and Fitch et al. (2005). In particular, the conclusion I draw from this investigation is that phonology may be entirely explainable through Third Factor principles pertaining to general cognition and the SM system (Chomsky 2005 et seq.).

This view accords with the evolutionary scenario developed by Hauser et al. (2002) and Fitch et al. (2005), who view language not as something that evolved gradually as an adaptation for communication (cf. Pinker and Jackendoff 2005, Jackendoff and Pinker 2005), but rather as something that emerged suddenly as a result of minimal genetic changes with far-reaching consequences. Particularly relevant is the distinction they make between the 'Faculty of Language—Broad Sense' (FLB), including all the systems that are recruited for language but need not be unique to language, or to humans, and the 'Faculty of Language—Narrow Sense' (FLN), which is the subset of FLB that is unique to our species and to language. At present, the leading hypothesis among proponents of this view is that FLN is very small, perhaps consisting only of some type of recursion (i.e. Merge) and the mappings from narrow syntax to the interfaces.

I therefore reject the claim made by Pinker and Jackendoff (2005: 212) that 'major characteristics of phonology are specific to language (or to language and music), [and] uniquely human,' and their statement that 'phonology represents a major counterexample' to the hypothesis proposed by Hauser et al. (2002), namely that FLN consists of only recursion and the mapping from narrow syntax to the interfaces. What I suggest, in effect, is that the operations and representations which underlie phonology were exapted or recruited from other cognitive domains for the purpose of externalizing language.[10]

Few authors have discussed phonology as it pertains to the FLN/FLB distinction. For example, Hauser et al. (2002: 1573) list a number of approaches to investigating a list of the SM system's properties (shown below in (24)), and these are all taken (p. 593) to fall outside FLN. However, none of these pertains directly to phonological computation.

(24)

a. Vocal imitation and invention
Tutoring studies of songbirds, analyses of vocal dialects in whales, spontaneous imitation of artificially created sounds in dolphins
b. Neurophysiology of action-perception systems
Studies assessing whether mirror neurons, which provide a core substrate for the action-perception system, may subserve gestural and (possibly) vocal imitation
c. Discriminating the sound patterns of language
Operant conditioning studies of the prototype magnet effect in macaques and starlings
d. Constraints imposed by vocal tract anatomy
Studies of vocal tract length and formant dispersion in birds and primates
e. Biomechanics of sound production
Studies of primate vocal production, including the role of mandibular oscillations

f. Modalities of language production and perception
Cross-modal perception and sign language in humans versus unimodal communication in animals

While all of these issues undoubtedly deserve attention, they address two areas—how auditory categories are learned, and how speech is produced—which are peripheral to the core of phonological computation. The most interesting two issues from my perspective are (c) and (f). These are of course very relevant to the debate over whether phonological features may be emergent and how phonological categories are learned; I discuss both issues in Samuels (2009a: ch. 3). And the instinct to imitate, addressed in (a) and (b), is clearly necessary to language acquisition. However, investigating neither these nor any of the other items in (24)has the potential to address how phonological objects are represented or manipulated, particularly in light of the substance-free approach to phonology, which renders questions about the articulators (e.g. (d, e)) moot since their properties are totally incidental and invisible to the phonological system.

Two papers by Yip (2006a, 2006b) outline a more directly relevant set of research aims. She suggests that, if we are to understand whether 'animal phonology' is possible, we should investigate whether other species are capable of the following:[11]

(25)
a. Grouping by natural classes
b. Grouping sounds into syllables, feet, words, phrases
c. Calculating statistical distributions from transitional probabilities
(p. 594) **d.** Learning arbitrary patterns of distribution
e. Learning/producing rule-governed alternations
f. Computing identity (total, partial, adjacent, non-adjacent)

This list can be divided roughly into three parts (with some overlap between them): (25a, b) are concerned with how representations are organized, (25c, d) are concerned with how we arrive at generalizations about the representations, and (25e, f) are concerned with the operations that are used to manipulate the representations. I would add three more areas to investigate in non-linguistic domains and non-human animals:

(26)
g. Exhibiting preferences for contrast/rhythmicity
h. Performing numerical calculations (parallel individuation and ratio comparison)
i. Using computational operations: search, copy, concatenate, delete

In Samuels (2009a, 2009b), I present evidence that a wide range of animal species are capable of the tasks in (a–i), though it may be the case that there is no single species (except ours) in which all these abilities cluster in exactly this configuration. I show (*contra* Yip) that there is already a substantial amount of literature demonstrating this, and that it is reasonable to conclude on this basis that no part of phonology, as conceived in the present work, is part of FLN.

It is beyond the scope of this chapter to review studies of animal cognition and behavior in any depth. However, let me briefly note that the evidence suggests that Pinker and Jackendoff's criticism of Hauser et al. (2002) concerning phonology is unfounded, at least if the theory of phonological representations and operations proposed in Samuels (2009a) is close to correct. A wide range of animal species have been shown to group objects, extract patterns from sensory input, perform sequential objects, perform searches, engage in copying behaviors, and manipulate sets through concatenation. As conceived of here, phonology thus provides no challenge to the idea that FLN is very small, perhaps consisting of just recursion and the mappings from syntax to the C-I and SM interfaces. Moreover, such a conclusion lends credence to the idea that the theory sketched in section 25.3 is a biologically plausible model of phonological competence. The human phonological system is, in short, a domain-general solution to a domain-specific problem, namely the ex-ternalization of language. Far from posing a problem for the evolutionary scenario posed by Hauser et al., or for minimalism, the evidence from phonology in fact supports the guiding hypotheses of both of these programs.

Notes:

(1) It is interesting to note that other cognitive scientists, such as Kaplan (1995 [1987]) and Pylyshyn (2003), also

caution against ‘the seduction of substance’ in their fields (computational linguistics and vision, respectively).

(2) Though one typical argument for dividing the labor between grammatical and extragrammatical explanation in this way is the fact that so-called phonological ‘universals’ typically have exceptions, I want to make clear that the presence of such exceptions is merely a *clue* that we should be looking to extragrammatical factors for an explanation of such tendencies; even exceptionless generalizations may not warrant grammatical explanations. As Hornstein and Boeckx (2009: 81) write, when we turn our attention to true ‘I(nternalist)-Universals’, or the laws of the faculty of language, as opposed to Greenbergian ‘E(xternalist)-Universals’, ‘the mere fact that every language displayed some property P does not imply that P is a universal in the I-sense. Put more paradoxically, the fact that P holds universally does not imply that P is a universal.’

(3) For a proper introduction to lexical phonology, see the works cited above, the papers in Kaisse and Hargus (1993), and McMahon (2000).

(4) Naturally, if we are going to pursue this type of theory, we must identify what is a phase head, and therefore what is a spell-out domain. Chomsky (2001 et seq.) takes C and transitive *v* to be phase heads; Legate (2003), Marvin (2002), Marantz (2007), and others argue that *v* must be a phase head in unaccusative and passive constructions as well. Crucially, T is not a phase head. Svenonius (2004), Bošković (2005), and Ott (2008), among others, argue for D as a phase head, and I will follow them here; I also follow Bošković (2005) in claiming that D need not be present in all languages or for all arguments. Other questions remain open, such as whether P is also a phase head (see Abels 2003). It is my hope that PDbP will open the door for phonological effects to shed some light on these unresolved matters.

(5) I will not seek to give a comprehensive primer in prosodic phonology/prosodic hierarchy theory here; I direct the reader to Inkelas and Zec (1995), on which I base the brief introductory discussion below, for an overview.

(6) The evidence for prosodic constituents falls into three major classes: (a) phonological rules for which they serve as domains of application, (b) phonological processes which occur at their edges (primarily suprasegmental, e.g. boundary tones), and (c) restrictions on syntactic elements relative to their edges (e.g. second position clitics).

(7) An OT implementation of end-based ϕ construction, the Strict Layer Hypothesis, and Proper Containment requires many more constraints than just the ALIGN family, such as WRAP-XP, NONRE-CURSIVITY, EXHAUSTIVITY, LAYEREDNESS, and HEADEDNESS. See Truckenbrodt (2007) for an overview.

(8) Dobashi takes only *v* and C to be phase heads. He notes that if the verb raises to *v*, as is commonly assumed, this will result in the subject and verb phrased together, and the object in its own ϕ domain. This prediction differs from that of prosodic hierarchy theory (recall (10)), and is undesirable from a typological perspective. Dobashi's answer to the mismatch between the *prima facie* predictions of the phase-based model and the typology in (13) is to modify the Spell-Out procedure. He argues that the leftmost element in a Spell-Out domain is actually not spelled out with the rest of the phase, but instead hangs back to establish the ordering of its phase with respect to the next phase, and only then is transferred. This has the effect of delaying Spell-Out of V (the leftmost element in *v*'s complement domain) and the subject (the leftmost element in C's complement domain), resulting in the desired separate ϕ domains for the subject, verb, and object. This captures (13a) and part of (13b). Languages like (13c, d) are claimed to have V-to-T movement and object raising to Spec, *v*P such that nothing remains in the lowest Spell-Out domain; for this reason, the verb and object are phrased together. To account for the second options in (13b, d), Dobashi proposes a rule of restructuring which combines two ϕ -domains if one of them fails to meet the requirement that each ϕ minimally contains two prosodic words. Languages that exhibit the alternations in (13b,d) allow restructuring, while (13a, c) do not.

(9) I ignore here the differences between languages with respect to allowing CV order with non-branching objects, since it is orthogonal to the issue at hand.

(10) On the possibility that language more generally is an exaptation, see e.g. Piattelli-Palmarini (1989), Uriagereka (1998), Boeckx and Piattelli-Palmarini (2005), Hauser et al. (2002), and Fitch et al. (2005).

(11) Yip mentions two additional items which also appear on Hauser et al.'s list: categorical perception/perceptual magnet effects and accurate production of sounds (mimicry).

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Minimizing Language Evolution: The Minimalist Program and The Evolutionary Shaping of Language

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[–] Abstract and Keywords

This article sketches the sorts of issues that arise within evo-minimalism. Lewontin reminds us that the evolutionary study of cognition, and more particularly the language faculty, is no easy game, given the distance between the proteins genes express and the cognitive functions in whose development or deployment such proteins may be implicated. The article attempts to sketch the sorts of difficulties that arise in relating all the necessary steps leading from one to the other extreme of such a causal chain, which casts some doubt on the viability of constructing true evolutionary explanations in the realm of cognitive novelties such as the 'Faculty of Language–Narrow Sense'. In addition, several difficulties exist in deciphering whether putative relations of hypothesized capacities in various species are genuine homologies, or instead, weaker analogies.

Keywords: evo-minimalism, language evolution, language faculty, cognition

> The less attributed to genetic information for determining the development of an organism, the more feasible the study of its evolution.
>
> (Chomsky 2007)

(p. 596) 26.1 Introduction: The Narrow Faculty of Language and The Factors of Language Design

Hauser et al. (2002) usefully separates the language faculty in terms of (i) interface (external) conditions that can be witnessed in other species—the faculty of language in a broad sense, or FLB—and (ii) internal conditions that seem unique to the human species—the faculty of language in a narrow sense, or FLN. Aside from reasonably suggesting that comparative studies with other species should be pursued to explore FLB, the piece also speculates on what might fall within FLN. Both aspects of the puzzle relate to the minimalist program (MP), which (since this handbook gives a perspective on it) we will be presupposing.

The external systems of FL present a relatively expected and observable continuity with regards to other forms of animal cognition and behavior. However, the internal conditions of the computational system—the basis for FLN, centered at least around the property of recursion—appear to be somewhat unique. This matter is perhaps best evaluated from the perspective of the automata that accept computations of various levels of complexity. A Finite State Automaton (FSA) only recognizes lists. In contrast, a Push Down Automaton (PDA) recognizes groupings, for instance of the phrasal sort in language. It does so with a crucial emendation vis-à-vis the FSA: it presents a so-called stack where it can store designated symbols, halting their computation for arbitrary periods of time, to resume work with them later on in the computation. Only a PDA recognizes truly recursive conditions, and only a stipulation prevents a PDA from recognizing these types of structures.

Seen that way, an important empirical question is whether language is the only genuinely recursive behavioral mechanism in nature. This is difficult to rigorously evaluate. The defining characteristic of *bona fide* recursion is that its structures can arbitrarily grow on different structural areas (not just the beginning or end of the structure), so that the output of a recursive system is in principle unbounded. When it comes to humans, although we obviously cannot directly demonstrate their truly unbounded behavior, we bank on the idea that at any given moment our interlocutor will understand us when we say '… and so on', resorting to their own linguistic imagination to hypothesize the relevant unbounded structures. However, we cannot do this with another animal's behavior, precisely because we do not function by their mental mechanisms. A recent controversy can illustrate this problem.

Gentner et al. (2006) documented the capacity of European starlings to discriminate, after extensive training, relevant signals with the pattern $(AB)^n$ vs. A^nB^n. If *n* in these patterns is arbitrarily large, the automaton necessary for recognizing $(AB)^n$ is an FSA, whereas the one necessary for recognizing A^nB^n is a PDA. Unfortunately, if *n* is finite—as in the case for the starlings, which didn't go beyond four repetitions (p. 597) in the experiments conducted thus far—an animal successfully processing A^nB^n could still be using a non-recursive FSA procedure, coupled with some pattern-recognition device for small numbers (see Perruchet and Rey 2005 for relevant discussion). So such a limitation in the study entails that this method of observation doesn't prove the point, one way or the other.

FSAs involve a fixed number of states, which predictably transition from one to the next. One can change such automata by rewiring the connections among their states, or less dramatically by modifying the relative weights that given transitional probabilities from one state to the next may carry. The latter is a *training*, which after some finite (usually very large) number of trials results in, effectively, a different physical 'brain', capable of different behavior. This is conceptually very different from what goes on inside a more flexible 'brain' with PDA capabilities, which is intrinsically capable of creative and plastic (recursive) behavior. One could mimic a finite number of recursive behaviors with ample usage of finite state devices, given unlimited resources (e.g. time) to reprogram them; however, this would not be realistically achievable without such a display of resources (or external manipulation). So then one can construct a different sort of argument to help us decide whether a given animal is deploying complex computational resources in displaying a physiological condition.

Suppose we witness some set of behaviors S in a given animal A. Suppose, moreover, that behaviors in S are not fully specified in innate terms, so that A actually has to acquire them in some period, or seasonally perhaps. Suppose, in particular, that some behavior *b* is describable in terms of an FSA *f*, another behavior *b′* in terms of a different (or differently weighted) FSA *f′*, and so on. Then the questions are: Could the animal be realistically trained to go from *f* to *f′* in terms of whatever the scientist can observe? Is the change achieved (very) fast or (very) slow? Is the change mediated by any observable training, by other animals or circumstances? Can the change be precipitated by conditions other than training? Etc. This is the line of reasoning explored in Gallistel (2009) and elsewhere.

We won't decide on these issues in this context, but we want to clarify their logic. At stake is whether the computational mind (with recursion as one of its characteristics) is a recent development related to language (Hauser et al. 2002:1573) or, instead, the phenomenon is ancient (Jackendoff and Pinker 2005: 214–15). Now granted, if the latter is the case, first of all we should witness recursive behaviors outside of language, whether in humans or in other species—difficult though this may be to establish. Second, a putative ancient emergence of the computational mind, aside from pushing back the burden of explanation (and its scope), still forces us to ponder the (very) recent emergence of directly observable linguistic behaviors, in just the FLN of the human species. In short, if the computational mind happens to be ancient, why did it take so long to manifest itself so clearly, in the FLN guise? This takes us back to our point of departure from a different route: the uniqueness of FLN, either as a readily observable form of recursive behavior or as its own, novel (p. 598) system, doesn't naturally square with the traditional adaptationist perspective in the theory of evolution.

From that point of view organic design is to be explained in terms of modifications to pre-existing genotypic structures in response to environmental phenotypic pressures. Structural adjustments of this sort are possible, to start with, thanks to the sort of diversity that arises from the natural tendency of genes to randomly mutate on their own. Ultimately, however, it is the environment that determines the adaptive value of given individuals, resulting from specific genetic combinations, in terms of their survival up to a reproductive age that allows those particular genes to be transmitted forward (see e.g. Dawkins 1986, 1996, Dennett 1995). Within this approach, evolution is possible only after the gradual, and very slow, accumulation of genetic modifications that address any given environmental issue. It is precisely because of this logic that this theoretical stance faces difficulties in explaining the sudden emergence of FLN, no matter what interpretation is given to this notion.

That need not cast a shadow on an evolutionary explanation of FLN (*pace* traditional skepticism, from Wallace 1864 to Lewontin 1998). It does mean, though, that a broader perspective must be entertained, a matter that relates to Gould's (2002) ‘adaptive triangle’, whose resemblance to Chomsky's (2005) ‘three factors of language design’ is clear, as is noted by Boeckx (2009c) (Figure 26.1).

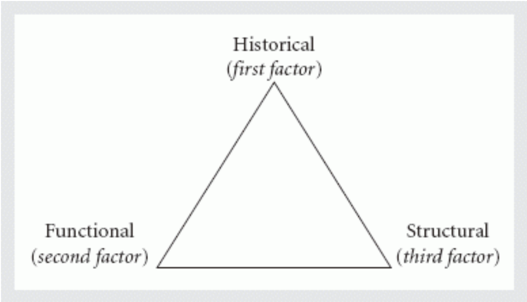

Fig. 26.1. Three factors in the evolutionary shaping of language.

In this entangled system, only the functional angle (or second factor) makes reference to environmental pressure—which again bears on the gradual shaping of structures, as just discussed. In turn, the historical angle (or first factor) points towards the capacity to evolve through re-using resources governing development. Evo-Devo approaches have shown, over and over, that relevant mechanisms in this regard are extremely conserved, thereby demonstrating how much weight sheer history has on the evolutionary system (see Hall 1999, Hall and Olson 2003, Robert 2004, Carroll 2005; Chomsky 2008b, 2010 also mentions interesting parallelisms **(p. 599)** between the Evo-Devo and minimalist programs). Finally, the structural angle (or third factor) appeals to very broad physical laws, which of course rule over the organic universe as much as they do over the inorganic one. Among these ought to be considerations about resource optimality, in energetic and spatio-temporal conditions, or even in less understood terms pertaining to computational considerations.

All of that said, Evo-minimalism is a liberating theoretical move: it allows us to bypass the customary dogma of having to shape evolutionary accounts of language in terms of the functional angle alone (Pinker and Bloom 1990, Pinker 1994, Jackendoff 2002, Pinker and Jackendoff 2005, Jackendoff and Pinker 2005). To be sure, MP itself can be pushed to the limit of the Strong Minimalist Thesis (Chomsky 2000a), thereby inviting us to concentrate more on third-factor considerations than anything else in the triangle above. But this is just a (reasonable) methodological move, inviting us to explore deeper explanations first, to introduce modifications to this desideratum as empirical need emerges, as we see below in more detail. This is all to say that Evo-minimalism, as we see it, is a broad perspective, where evolutionary ideas from various angles are entirely welcome if they contribute to explanation.

26.2 The Scope of Third Factors in Evolution

The most basic combinatorial procedure in human syntax, which often goes by the name of Merge, can be used to illustrate the scope of third factor considerations, within the Gould—Chomsky evolutionary triangle, in general evolutionary studies—including the putative limitations of such factors. Among its properties, Merge includes: (i) binarity, (ii) (asymmetric) labeling, (iii) structural preservation (in satisfaction of ‘extension’ and ‘no-tampering’ conditions on phrase-markers), and of course (iv) unboundedness, which underlies the property of recursion that we discussed in the previous section. Importantly, there is more.

We have spoken above of structures of the form $(AB)^n$ vs. A^nB^n; but language exhibits also structures of the form $A^n \ldots B^n \ldots C^n$ (where the dots represent arbitrary intervening material). A sentence like *these (simple) phrases (surely) are in and of themselves (rather) complex structures* directly illustrates discontinuous dependencies that no simple (even PDA) automaton can recognize without some computational enrichment. These kinds of dependencies are central in expressing the semantics of quantification, for example allowing a quantifier like *most* in *we like most theories* to relate the denotation of *theories* (its restriction) to the denotation of *we like x* (its scope). All of this leads to a condition (v) on flexibility (p. 600) for our characterization of Merge. Each new application of the operation can take stock from the lexicon (External Merge) or the computation (Internal Merge), which results in the effects of the operation being generalized from context-free to context-sensitive specifications. Now: What does Merge owe such universal specifications to?

The very universality of Merge conditions indicates either a genetic basis or an even more elementary origin, again in the sense of the Gould—Chomsky evolutionary triangle. But pondering formal conditions as just introduced pushes us in one of two directions already mentioned. If Merge is an ancient trait of animal thought, then its specifications are at least compatible with a genetic basis. In contrast, if Merge is a very recent trait, co-extensive with the emergence of FLN, then it is unlikely that properties of this sort should follow from genetic specifications. While in the second scenario there is no direct role for genetics, the wiggle room that exists in the previous scenario has to be handled with care. For suppose one were to claim that asymmetric labeling (to pick one of the properties above at random) aids communication, or some such function, for some reason R. To make such a claim contentful, one would also have to demonstrate how, say, *symmetric* labeling, or total absence thereof, not only does not invoke R, but moreover is so maladaptive for linguistic communication that hominids who may have randomly evolved this alternative structure would have been weeded out of existence, outcompeted by others which manage to transmit their genes.

We welcome such explanations, but this is not what one normally finds in the literature (for perspective on this issue when applied to the basic notion of c-command, see Uriagereka forthcoming: ch. 3). If the basis for a given condition is claimed to be genetic, arising in standard evolutionary terms, that line of explanation owes us, at least, a set of alternatives (which the winning one is supposed to have defeated across time), and a specific correlation between the winning strategy and a functional fitness that guarantees its having persevered. Not having either such a metric or, worse still, testable alternatives, is tantamount to not having enough time for the evolutionary scenario to deploy itself, or worse, not even having an account of the condition that meets the simple demands of the evolutionary logic (see Carstairs-McCarthy 1999 for illuminating discussion on this).

Now a genuine Evo-Devo account of, again for concreteness, asymmetrical labeling in third-factor terms also owes something more than just pointing out how things *ought* to follow from general design specifications. Otherwise the risk is to be equally vacuous. The issue in this instance is not so much having different hominids compete for a successful ecological niche; instead, putative alternatives should not even emerge in evolutionary history, for third-factor reasons (see Thompson 1942 [1917] for the *locus classicus* of this idea). For the explanation not to be circular, the design specification ought to have some independent motivation, ultimately rooted on the physical sciences or some related field. After all, if this ideal isn't met, proponents of the theory criticized in the previous paragraph, if persuaded (p. 601) by the conceptual point now being raised, could simply propose that the alleged semantics of human language, say, happen to be the relevant *absolute*—e.g. rooted in a metaphysical logic—and such a condition would, in effect, reduce to a third factor for the purposes of the explanation.

In all likelihood, matters will be complex or at least nuanced, given what can be surmised from comparison with other species. Certainly no analog of animal behavior, no matter how cautious or generous we are in the interpretation of the data, comes close to presenting the sort of formal complexity implicit in Internal Merge, and the evidence that can be surmised from the fossil record leaves the matter unresolved even for Neanderthals (Balari et al. 2008). But does this mean that only humans evolved Merge, or is it, rather, the case that only they evolved *Internal* Merge for some reason? Either way, though, in our view a serious interpretation of the facts gives credence to the discontinuity conjecture.

26.3 A Question of Time Forcing an Alternative

Faced with a situation as just described, it is naturally tempting for minimalists to try and blame the emergence of FLN on a single evolutionary event, some sort of major brain reorganization that is responsible for all the structural nuances associated to syntax (see Berwick and Chomsky forthcoming for a recent version of this idea). While this is consistent with at least putative discontinuities in the fossil record—inasmuch as these can really be associated to language—it is also somewhat doubtful that things could be so drastic. The main reason for skepticism has to do with the enormity of the task, if taken seriously. Aside from the fact that human language is *more* than just recursive (as mentioned, discontinuous dependencies are not recognized by the archetypical automaton that recognizes recursion), the other major property of this faculty has to do with the size of its vocabulary (in the order of tens of thousands, as compared to a few dozen signals even in elaborate communication rituals in other animals; Marler and Tenaza 1977, Goodall 1986).

If one admits that vocabulary explosion, context-sensitive syntax, *and* recursion (at least) are all different, in principle, then there doesn't seem to be enough time for all of that to happen within the last couple of hundred thousand years. The likelihood of a single event reorganizing a brain for some strange reason is admittedly minute, though possible; the likelihood of two independent, and combined, reorganization events approaches the infinitesimal. If three such events are supposed to be **(p. 602)** involved (recursion, vocabulary explosion, and context dependency), one arguably enters the realm of miracles.

It is faced with this dilemma that it seems to us profitable to pursue a different route. Our approach is based, first, on the idea that the divide between FLB and FLN need not be simplistic, in the sense that some components of language, as understood in present-day theories, neatly belong to each category. Take again Merge, with the observable characteristics just signaled (binarity, labeling, structural preservation, unboundedness, flexibility, etc.). In principle, each of those properties may follow from FLB or FLN demands—it is surely an empirical matter. Indeed, it seems to us likely that what is currently experienced as linguistic Merge is, in the end, the result of an intricate evolutionary pathway, involving both FLB and FLN considerations (Boeckx 2009c). We can think of this as a *distributive* property of the faculty of language, which actually corresponds rather well with the idea that various levels of computational complexity in its syntax associate to various levels of semantic complexity (Uriagereka 2008a).

Second, once that precision is made, there is no need to treat recursion as a recent emergence, simply because other aspects of Merge (or FLN more generally) may in fact have arisen recently. If recursion is strictly not a *bona fide* characteristic of FLN, we immediately do away with one leg of the worrying triad mentioned above (vocabulary explosion, recursion, and context sensitivity). Now we hasten to add that we are thinking of the manifestations of recursion (and more generally PD computational characteristics) that Gallistel and his associates have been exploring in various animal brains, which have nothing to do with language, communication, or other forms of externalized behavior. But this doesn't affect our argument—quite the opposite. What matters to us is that recursion in nature could have been a phenomenon dating back millions of years (Pinker and Jackendoff 2005).

The issue here is a PD-style automaton, within which only a stipulation (e.g. preventing the computational manipulation of symbols of the same sort) would limit its recursive capabilities. It is, no doubt, a great evolutionary problem to understand the emergence of such a system—but it is not our problem right now. Given that amount of time, in fact, it may have been that such an emergence is of the first, second or third factor sort, or even a combination thereof. Our true problem, within this perspective, is to understand how an ancient form of *thought* recursion got recruited for the faculty of *language*, into FLN.

Luckily, one of the major components of the faculty in point is logical form, which seems, at least in part, nothing but 'linguistically articulated thought'. If this (classical) picture is correct, we need not make a connection between thought and language. We should, instead, presuppose that language *is* thought of a certain form, which somehow presents the surprising characteristic of being 'externalizable'. If so, the recursive characteristic—by hypothesis present already in thought—is not something language would need to evolve a connection to. Rather, language could be seen as an emergent property of thought being externalized, in which case we **(p. 603)** need to understand the externalization conditions (see Hinzen 2006, 2009b on these and related matters). Although, to anticipate a possible objection, this reasoning need not entail that all of human thought was present in ancient form, and simply got externalized through language; we argue below that, in fact, the externalization itself may have yielded more complex forms of thought, in a sort of feed-back-loop into the overall system.

Externalization conditions leading to language seem to come in at least two guises: the ability to materialize 'bundles of thoughts' into identifiable external units, and the ability to put those together into a coherent whole that matches the originally constructed structure, even after it is translated into a set of motor instructions of a very different sort. The logic of how each of these relates to the other arguably tells us which of the corresponding evolutionary events ought to be prior. By definition, the externalization requirement arises at another interface, this time between linguistically articulated thought and the motor system, at least—it is immaterial for our purposes whether the modality is oral, gestural or of any other motor sort. In contrast, 'bundling thoughts' (into identifiable external units) seems to require at least one bit that is internal to the system: the bundling part.

We can then reasonably conjecture that this was the evolutionary event leading to FLN: the ability to somehow compile complex thoughts, articulated internal to a mental computation, into fixed atoms to manipulate further in the computation (Boeckx 2008a, Ott 2009). Although it is unclear how this happens at a neurological level, the compilation ought to be correlated with the vocabulary explosion we can witness in our toddlers, which halts by puberty—suggesting that some relevant component has endocrinal regulation.

Locke (1997) observes a correlation, around twenty months on average, between a child's analytical/computational abilities and the explosive stages of lexical acquisition. Indeed, a reduced inventory (under a couple of dozen words, about an eighth of what is normal) is a hallmark of a syndrome, particularly when sensorimotor or environmental factors are not deficient. One has to of course be cautious with extrapolating developmental correlations to the evolutionary considerations we are dealing with, but they are nonetheless worthy of note, and in any case they emphasize the subtle interaction of factors at play: The proposal presented so far 'simply' organizes bundles of articulated thought, as a consequence of a sudden third factor consideration; but in addition to that, we need to find a way to connect that brain reorganization to externalization conditions, involving the motor system.

The connection in point is not necessary. By hypothesis we are assuming an internal compiler of a PD sort, so one may wonder whether that isn't all one needs for a parser. However, an individual may present PD characteristics, and still not have the capacity to process the PD output from another individual which has been externalized into a one-dimensional speech channel. In particular, a PD grammar can produce paratactic (conjunctive) and hypotactic (subordinating) structures—but these come out identical once they are squeezed out of their internal conditions, (p. 604) one word at a time. In some sense, a parser must be able to reconfigure a structure that it is perceiving in a flattened version, in the spirit of the Analysis-by-Synthesis proposal (as in Townsend and Bever 2001, although these ideas date back to Halle and Stevens 1962 and Chomsky and Miller 1963). In turn, for that task to succeed, some significant attention span is required, so that a sufficient amount of cues are parsed; only then will the parser be able to decide on the internal structure of what is being processed. We show below the relevance of these matters, but to talk about them in a more informed fashion, let's make an excursus into bird brains, which begin to be reasonably understood when it comes to, at least, their song circuitry.

26.4 In Search of Bird Timing

While for some songbirds the particular song they woo females with is innate, many acquire it during a critical period, soon after hatching. The parallelism with human language had been pointed out long ago, observing how these birds (e.g. zebra finches) have two separate brain circuits involved in the singing process: an acquisition circuit (which, if severed, entails the inability of the bird to acquire his song) and a production circuit (which, if severed, entails the inability of the bird to produce an already acquired song). These circuits are linked by the so-called Area X, which is the human homolog of a sub-cortical brain area within the basal ganglia, the caudate nucleus. The putative homology got significantly strengthened when the bird version of the human gene *FOXP2*, isolated in relation to a language/speech pathology (Lai et al. 2001), was found to be expressed in Area X of the bird brain both during song acquisition (Scharff and Haesler 2005)and performance (Teramitsu and White 2006), just as it is expressed in human fetuses in the caudate nucleus (as well as the cerebellum and the frontal cortex; Ferland et al. 2003, Lai et al. 2003, Liégeois et al. 2003, Takahashi et al. 2003)—and it appears mutated in at least one well-studied family with a kind of severe dysphasia.

FOXP2 (and its bird analog *FoXP2*) is a transcription factor—that is, a regulating gene that turns other genes on and off (in human fetal development alone, 285 genes have been identified within the regulating chain of *FOXP2*: Spiteri et al. 2007). This means that whatever its role is in whatever it is that this gene controls—which goes well beyond the brain—is extremely complex. That said, the fact remains that the protein this gene transcribes is remarkably conserved, with only eight point variations between birds and humans. It is thus plausible for a deep homology to exist between brain circuits in the two species, even if separated by well over 300 million years of evolution. More specifically, a reasonable proposal, in light of facts as presently understood, is that somehow the relevant brain circuit, in both (p. 605) birds and humans, is related to the sort of working memory that the respective computational systems need to deploy. This needs some discussion.

The first reason to suspect that memory resources might be implicated stems from the proposal in Ullman and Pierpont (2005) concerning the syndrome observed in individuals of a family whose version of *FOXP2* presents a crucial mutation. Importantly for the reasoning, these individuals present difficulties not just in a variety of tasks associated to their linguistic ability (to the point that they were originally diagnosed with Specific Language Impairment, or SLI), but also in many activities whose underlying property could be described as 'rhythmical'. For example, these individuals cannot dance or tap their fingers in succession. Provided that the caudate nucleus has been known to be implicated in procedural memory regulation, of the sort required to keep track of serial behaviors of various sorts, Pierpont and Ullman proposed their Procedural Deficit Hypothesis (PDH). Without attempting to go into specifics here, we will show next that this approach can be revamped in terms that are sensitive to strict computational considerations.

We spoke above about two types of automata, the FSA and the PDA, and alluded to the fact that linguistic structures exist that go beyond the capacities of either one of those automata—requiring the computational power of what could be thought of, simply, as a PDA+ (Uriagereka 2008b), within the so-called Chomsky Hierarchy of automata (FSA > PDA > PDA+ > … > Universal Turing Machine). These are all specifications of automata within the computational architecture originally conceived by Alan Turing, based on a logical processor writing the results of mechanical operations on an unbounded tape, until it halts. One way to characterize the differences between the three sorts of systems discussed here is in terms of their memory, or perhaps more accurately *working* memory and 'attention' specifications, for each sort of device—each being progressively richer in this particular regard.

For FSA devices, there is as much memory (at most) as the computational machine itself (given *n* computational steps, the machine can only remember, at most, *n* steps). What one may think of as the machine's 'current state' or, as it were, attention, in turn, can be limited even further (in arbitrary ways). This means that, at any given step in the computation, the machine may be given the capacity to only look into the next two (three, four, etc.) steps.

As already mentioned, in contrast, PDA devices have access to a stack, a part of the Turing tape that is used just for the purposes of storing a chunk of the computation, which is halted while the computation results continue to be written elsewhere in the tape—to then return to manipulate the halted material. It is this mechanism that allows the system to deal with recursion. In turn, the stack itself comes with a memory regime. What can be placed on the stack is organized in terms of how the computation proceeded: material computed first is buried inside the stack, simply because it was dealt with first. The top of the stack effectively reflects the current state of the computation. Once again, one can limit access to (p. 606) the machine's current state or attention, by regulating how far down into the stack the system is allowed to scan. What this means is that, although further elements are buried deeper into the system's memory, the system must first deal with the *n* steps it has attention over.

Similar considerations apply to 'context-sensitive' conditions, which have all the nuances of 'context-free' ones recognized by a PDA, but in addition allow manipulations of material within the stack. The reason they are called 'context-sensitive' is because what is kept in memory in these instances is not just the actual computation as it was assembled, but rather that plus its *context*, meaning whatever else the system had put on the stack. If at any given point in the computation the system knows not just that it has operated with X (it has placed X on the stack) but furthermore the *history* of what went on in the computation to carry the system to X—i.e. the element Y that the system manipulated in order to get to X—then one can deduce everything that took the computation to that particular juncture. Obviously, this entails a more powerful form of memory, which for our purposes can be thought of as derivational—even if, again, attention limitations in these systems are systematically imposed, thereby drastically reducing their power.

Piattelli-Palmarini and Uriagereka (forthcoming) suggest that these computational notions, involving systemic memory and attention, could be profitably related to Ullman and Pierpont's conjecture about the role of *FOXP2* in procedural memory, which they propose should be considered relevant in the avian brain as well.

Todt and Hultsch (1998) proposed that the bird signal combinations they studied could be generated by an FSA, although as we mentioned in section 26.1, this may not be an ultimately fair assessment. While Gentner et al.'s (2006) conclusions about starling brains having PDA characteristics may be subject to questioning, extending Gallistel's general approach to the computational roots of animal behaviors to observations regarding the plasticity and rapid reorganization of bird songs (particularly in seasonal singers; see Eens et al. 1992 on starlings, Nottebohm and Nottebohm 1978, Nottebohm et al. 1986 on canaries) suggests that bird brain capacities go beyond the simplest FSA level. Be that as it may, to make our case now it is sufficient to speculate with the possibility that what *FOXP2* regulates is not so much the total memory in any given device (what carries the system from FSA to PDA capabilities, and so on) but rather the device's *attention* at whichever computational level (within the Chomsky hierarchy) it occupies. The question then is how a gene network putatively regulating an 'attention span parameter' (ASP) could be of any relevance to our evolutionary reasoning.

The analysis of the situation is easier in the case of birds, for which a variety of experiments are possible. All that is known so far is that knock-downs (birds for which *FOXP2* has been damaged) fail to acquire the song, if they are affected prior to the critical period, and fail to perform the acquired song, if they are affected after the critical period (Haesler et al. 2007). This situation has led a number of **(p. 607)** researchers to suspect that the underlying role that the gene network is playing has to do not so much with memory/attention regulation, but with motor issues (see Jarvis 2006 for perspective). However, Teramitsu and White (2oo6) showhow up-or down-regulation of the gene in Area X of the bird's brain, as he sings the song, depends on whether the song is being directed to the female (in which case it is very fixed and ritualized) or it is being 'played with' (including variations) with the bird singing alone. While the motor system is deployed in both instances, apparently the rhythmic control differs in each. One technical way to rationalize those facts is in terms of the gene network specifically regulating the hypothesized ASP in the computational system (even if it turns out to be an FSA). The larger the attention span, the more fixed a given series of chirpings will be; with a smaller attention span, the opposite effect will ensue.

A system of this sort would make sense if we think of the fact that bird song is actually a communication system that has to be 'squeezed out' into the airwaves. Not only is a bird producing a song; con-specifics must process it, to somehow gather from that enough information about territory, mating, or whatever else is involved. Minimally, one version of a birdsong must be distinguished from another, for which the animal must pay close attention to the succession of chirpings. In sum: rudimentary as it may be, a bird needs a parser of some sort, to reconstruct whatever song was perceived, perhaps then comparing it to mentally stored songs (or however else the ultimate successful communication ensues). It makes good sense for a parser to control the signal it perceives as time goes by, which in turn presupposes something like the ASP. One rationalization of the role of the *FOXP2* variant in birds is that this gene network is involved in the precise timing of such an attention span.

26.5 The Attention Span of an Ape

Supposing something along those lines is correct for birds, the evolutionary question boils down to why and how an ape brain having achieved the atomic compilation alluded to in section 26.3—which correlates with vocabulary explosion—can recruit the sort of rhythmic network associated to *FOXP2*. This would thus be a first factor effect.

As we understand it, whatever is responsible for the atomization of thoughts into words drastically augments expressive power. This is in part related to the computational analysis in Nowak et al. (2001), which shows how the size of a lexicon has to pass a certain (small) threshold for non-trivial combinatorial (context-free) syntactic communication to be advantageous in a population. This methodological point is important: the language faculty is, in part at least, a social phenomenon, and **(p. 608)** therefore it matters not only what sort of computational machinery can support it, but also how the output of these internal mechanisms are actually externalized *and reconstructed back* into hypothetical structures—just as we saw for the bird instance, at a (putatively) less complex computational level.

To be specific, we may suppose that there was once a relatively stable proto-lexicon in the order of a few dozen signals for given hominids, or perhaps clans thereof—which is not that different from what chimpanzees can be trained to control (Gardner et al. 1989). This is not to say that corresponding thoughts in individuals capable of externalizing these signals only involved whatever concepts these particular elements denoted. Internal thoughts may have been drastically more complex, as they probably are in animals that are capable of complex reasoning, and yet do not externalize much (including most higher apes). But if the externalization, at this hypothesized evolutionary stage, was not very different from what can arguably be witnessed for trained chimps, effectively these populations would live in a shared 'FS world', even if their internal thoughts would be, again by hypothesis, of a 'PD' sort.

It may seem paradoxical to say that an individual capable of PD thoughts would live in an effective FS world.

However, bear in mind a difficulty first pointed out by Lucien Tesnière in the 1930s (see Tesnière 1959): no matter how complex (even multi-dimensional) an individual's thought processes may be, if these internal computations have to be externalized through a one-dimensional channel of the sort motor systems allow, then much information will have to be compressed, and perhaps some may even be lost. Tesnière realized this conundrum is at the heart of the language faculty and how it may have evolved.

As they come out of our motor systems, into the airwaves, and back into them, linguistic representations have to be of an FS sort, even if briefly. The task of human parsing, in a nutshell, consists in regaining that amount of information that turns an FS representation back into an appropriate PD mode—hopefully the one a given speaker intended (turning his or her pronounced words into his or her putative phrases). One simply cannot presuppose a reconfiguration of this sort to summarily exist simply because the thought process which generates it is already of a PD sort. If that were the case, given the logic of the Gallistel approach to animal thought that we have been assuming, literally all thinking animals (all animals?) would be capable of parsing their respective thoughts—would be linguistic creatures. All we are suggesting now is that there could have been an effective FS state in hominid linguistic interactions (lacking a parser), perhaps corresponding to what Bickerton (1990) called 'proto-language', although with different presuppositions about internal thought.

Then, presumably for sudden third-factor reasons that we have not been able to specify, the brain reorganization leading to atomization of thoughts happened, again *internally to the thought processes*. Individuals with this particular brain reorganization would have faced both new opportunities and new challenges. The **(p. 609)** opportunity should be obvious: the ability to conceive of new thoughts simply by bundling and storing old ones, instead of having to construct them anew every time they need to be deployed, provides an extraordinary repository for an individual, which augments their expressive capacities orders of magnitude compared to its absence. But new challenges arise too, particularly for individuals who may feel the urge to externalize these new thoughts. Consider this in detail.

A solipsistic individual endowed with atomization capabilities, not interested or capable of communicating said thoughts to their con-specifics, would not encounter any particular challenges in that regard. But sociable individuals would, given Nowak et al.'s (2001) analysis: if the number of new thoughts these individuals attempt to associate with external signals is high, then it would be advantageous for these signals to organize themselves beyond FS conditions. In a manner of speaking, this would be the moment when it would be advantageous to take a gamble on the thoughts of a con-specific: instead of processing them solely as what they seem—dull FS representations—to assume they are the results of thought processes as complex as one's own. For this to be effective, one needs more than what philosophers call a 'theory of mind', or the mere assumption that others, too, are internally as complex as we are. The bottom line is that one needs a parser, or the task of reconstructing someone else's expressed thought would be hopelessly under-specified. This is, then, how we interpret Nowak et al.'s findings: an augmentation in vocabulary size, if shared by a population, would put an evolutionary pressure on that population to evolve a parser.

So that already shows how intricately related evolutionary factors must be. Some third-factor condition (resulting in vocabulary expansion), when channeled through the interactions of a population (a second-factor consideration), results in an evolutionary pressure, a situation whereby the emergence of a first-factor condition would be advantageous to the population. Needless to say, one has to show the details of the latter, but the point is simply to emphasize that these various evolutionary components do not act in isolation of one another, and they in fact plausibly feed, instead, into one another in intricate ways, essentially following the Chomsky—Gould evolutionary triangle: 3rd factor > 2nd factor > 1st factor.

As to what the particular first factor may have been that allowed hominids access to a parser, the *FOXP2* network discussed in the previous section seems like the obvious candidate: what works for the birds may work for the apes. In fact, there are indications that this particular network regulates other 'rhythmical' behaviors in at least mice (Shu et al. 2005, Fujita et al. 2008), bats (Li et al. 2007) and possibly all animals that might fall under the broad rubric of 'vocal learners'. Obviously for this speculation to be turned into a theory it would have to be substantiated in various ways, but the point of principle should be clear, and it is certainly a testable hypothesis.

If our discussion concerning bird brains in the previous section is on track, what would have changed in the hominid lineage after the evolutionary scenario **(p. 610)** sketched above has to do with what we called the ASP parameter within the bird brain. In other words, higher apes and presumably other hominids, by hypothesis, would not have had the right attention span to carry a parsing of the sort necessary to reconstruct the PD capabilities of a hominid thought at this stage. So the evolutionary pressure would be, precisely, in terms of regulating attention (to the computation), once it becomes advantageous to do so after vocabulary explosion. Note that other hominids without vocabulary explosion, or for that matter other higher apes, presumably would not have reached this evolutionary pressure to begin with, lacking the vocabulary size that makes this biological move evolutionarily effective.

Now even if the course of evolutionary events proceeded roughly as indicated, in rigor that has not yet given us all components of FLN. All that has been modeled so far is the externalization of an ancient PD-style thought, which grew explosively complex after a brain reorganization gave us the ability to atomize thoughts into words, by way of the recruitment (in conditions of selective pressure) of an equally ancient—yet unrelated—mechanism that regulates rhythm. This is all fine, and the vocabulary explosion may have been part of FLN (remember: we are assuming a distributive language faculty, so there is in principle no reason not to expect components of FLN arising at different times). But this has nothing to say, yet, about the other salient characteristic of language: its context-sensitivity.

Methodologically, note that we have basically introduced a 'computational morphospace', to use the concept in Balari and Lorenzo (2009), whose different observable phenotypes correspond to the various levels in the Chomsky hierarchy of automata. Moreover, we have shown how 'going up' in this space is not a simple task: it requires an intricate interaction of third-, second-, and first-factor considerations, which is what makes the language faculty distributive—and which presumably speaks to the rarity of its emergence. But we want to be honest here: if we are leaving a central aspect of FLN for a later emergence, aren't we again running out of evolutionary time?

26.6 Reasonable Scenarios for Context Sensitivity?

The question boils down to why, how and when context sensitivity (long-distance dependencies, crossed paths, parallelisms, and so on) emerged in language. The logic of the situation, given our computational morphospace approach to the matter, forces us to say that this aspect of FLN is either posterior or at the very least co-temporary to the emergence of everything discussed so far. Of course in rigor it (p. 611) could have been that context sensitivity is as much part of the ancient animal brain as context freeness, which fits well with the idea that all this amounts to is Merge—of an internal sort. That said, the mechanisms for externalizing Internal Merge pose computational difficulties that go beyond the 'squeezing' problem discussed in the previous section. Consider this seriously.

The problem for External Merge boils down to the receiver of a linguistic signal not being able to determine whether the next incoming element is to be associated to the structure hypothesized up to any given point as a mere concatenative dependent (i.e. paratactically, as part of a list) or as a subordinate constituent (i.e. hypotactically). Language allows both options, and if we carried along a blackboard to draw the tree diagrams for each, or had telepathy, distinguishing them would not be complicated—but instead we must deal with 'flat' speech. What the '$FQXP_2$-solution' seems to have done (in ways that are certainly not understood) is to provide us with some kind of 'scanning window' within which we make hypothetical decisions corresponding to our own internal cues. Let's say that much is true, in some fashion. It unfortunately won't suffice for context-sensitive matters.

Context-sensitive dependencies *distribute* a process through various chunks of structure. If there was any simplistic hope of hypothesizing phrasal structure from well-behaved words that keep their place in line, all of that is destroyed by the presence, precisely, of words out of their natural interpretive place, features that exhibit their presence across simultaneous words but with a single interpretation (so-called copies), or words that miss their phonetic support altogether (gaps, ellipsis processes). It is, of course, the syntax of all this silence (to use Merchant's 2001 apt expression) that carries human language to its unique expressive power, not just in terms of eloquence and beauty, but more simply for the everyday expression of quantification or focus. Take away context sensitivity and what is left has little to do with human language. Put context sensitivity into the picture and the parsing difficulty alluded to above, in terms of hypotaxis vs. parataxis, turns into a nightmare.

That is all to say that a hypothetically latent context-sensitive procedure in an ancient brain may have been much further away from externalization than a corre-sponding context-free procedure in a similar brain. In other words, the recruitment of the *FOXP2* network, allowing a useful regulation of the ASP parameter for apes that is usually reserved to vocal learners, may simply not have been enough to result in the externalization of context-sensitive dependencies. Then again, it may have. The problem is that nobody has a reliable picture, in strict computational terms, of how it is that whichever context-sensitive dependencies happen to manifest themselves in human language are to be understood, and therefore it is hard to make an informed speculation like the one involving the ASP parameter.

We can, however, lay out the boundary conditions of the problem, in the hope that a future understanding of these matters will help us sort them out.

(p. 612) Possibility number one is that, in fact, the parser that brought *sapiens* the ability to systematically externalize their ancient PD-style thoughts sufficed, in some form, to parse ancient PD+−style thoughts as well. We don't know how to specify this idea any further.

Possibility number two is that, in contrast, only PD-style thoughts were ancient, and only these got externalized as discussed in the previous section. In turn, PD+−style thoughts would have been merely latent, but not realized even in hominids prior to these evolutionary events. This, by the way, is quite consistent with the idea that, although one can have a reasonable discussion about whether other animals are capable of inferential thoughts that presuppose a complex recursive syntax of the PDA sort, it is harder to make a similarly sound case for non-human animals and all the hallmarks of PDA+ behaviors: quantification, 'knotted' behaviors, inferential ellipses, and so on (though see Balari and Lorenzo 2009).

Needless to say, the question for the second possibility is precisely why, how and when the merely latent PD+−style thoughts suddenly became available to *sapiens*. A tantalizing possibility, consistent with recent findings regarding the *FOXP2* network, is that this particular trait is specific to modern *sapiens*, and not even Neanderthals had it.

The matter is extremely topical, and relates to the issue of precisely when it is that the current human *FOXP2* network mutated to its present form. Although originally Enard et al. (2002) argued that this was in all likelihood within the last 200,000 years, more recently Krause et al. (2007) have persuasively shown the exact version of this gene in Neanderthals, which must mean that the mutation in point could be roughly twice as old. Now, as Benítez-Burraco et al. (2008)and Coop et al. (2008) discuss, that fact doesn't, in itself, entail that Neanderthals had FLN. The problem is that, as Balari et al. (2008) argue, the evidence that this sapiens subspecies was linguistic in the sense we give to this term for anatomically modern humans is very dubious. So the question is: could it be that recruiting the *FOXP2* network in fact does not immediately entail 'liberating' context sensitivity?

It would be pretentious for us to even speculate with an answer, but one should pay close attention to the resolution of these various issues and controversies to plot a reliable future path. Note, in this regard, that if the *FOXP2* network had been recruited by Neanderthals too, logically this need not entail that it was for the same reasons our subspecies did (in our hypothesis, after selective pressure stemming from the brain reorganization leading to vocabulary explosion). After all, *FOXP2* seems to be regulating very different sorts of rhythmical behaviors in various vocal learners, from ultrasound to sonar.

It is noteworthy, however, that if the results in Krause et al. (2007) hold (unlike previous results in this regard), it is very likely not just that *FOXP2* was recruited by our common ancestor with Neanderthals, but that the event in fact took place recently, possibly within the last 350,000 years. This suggests that, if our speculations about the origin of this event for anatomically modern humans have to **(p. 613)** do with selective pressures resolved relatively rapidly, basically the same sorts of selective pressures ought to have been present for Neanderthals. This would have meant, then, that this subspecies would have been capable of at least the sort of protolanguage we speculated with in the foregoing sections, including, importantly, the vocabulary explosion that precipitated the selective pressure in point. The good news, again, is that the matter is testable, and as we come to unearth the Neanderthal genome, comparative studies might help us in this regard—although in this instance we are very much in the dark as to what that putative (third-factor-driven) brain reorganization was.

That speculation leaves context sensitivity as the truly unique human (modern *sapiens*) evolutionary step. Logically speaking, it could have emerged from more than one source. One possibility is that the selective pressure encountered both by Neanderthals and modern *sapiens*, which by hypothesis resulted in a useful parser, did not get resolved with equal success in each subspecies. After all, we have merely suggested that the *FOXP2* network is implicated in the ASP parameter, but we do not know how. It is possible that the detailed regulation of this parameter, or related matters, could have been different in each subspecies. In this regard it is very encouraging that Vernes et al. (2008) should have isolated a *FOXP2* target, $CNTNAP_2$, which is apparently associated to SLI in unexpected ways.

$CNTNAP_2$, whose protein plays a role in neuronal communication, is expressed in the orbital gyrus and the superior frontal anlage, cortical areas related to language. Apparently, children with a point substitution in this gene (albeit in a region that *FOXP2* does not bind!) perform poorly in tasks related to the diagnose of SLI. When it comes to regulating genes and their targets, matters are extremely subtle and complex, and we are only hearing the first news of what will certainly turn out to be a complex saga. That said, it would of course be interesting to examine $CNTNAP_2$ in Neanderthals, as would be eventually important to know specific differences and similarities in all of the genes that might be implicated in the language faculty (Benítez Burraco 2009, after an exhaustive analysis of the literature, speaks of well over 150, distributed over five chromosomes).

Another logical possibility is that context-sensitive conditions correspond to an entirely different gene network, stemming from a separate mutation. The problem with this approach has already been mentioned: even if the recruitment of the FOXP2 network took place within the last 350,000 years (thus leaving the possibility open that a putative brain reorganization resulting in vocabulary explosion was prior to that), the likelihood of yet another genetic change, which arguably must be in place within the last 150,000–100,000 years, is not that great. The latter dates relate to two factors: (i) uncontroversial symbolic behavior within the fossil record, relating to the last human diaspora including navigation and varieties of jewelry that could be dated as early as 110,000 years before the present (Oppenheimer 2003, Henshilwood et al. 2002, d’Errico et al. 2005), (ii) the presence within the most distant community of anatomically modern humans, separated from the rest of (p. 614) humankind between 90,000 and 150,000 years ago, of context-sensitive procedures in the languages they speak.

We’re speaking of Khoisan languages. Genetic studies suggest that these communities separated from the rest prior to the last human diaspora. Moreover, they apparently remained isolated until about 40,000 years ago (Behar et al. 2008). That is consistent with the peculiar externalization one witnesses in these sorts of languages: the famous clicks, which have nothing to do with standard consonants and are practically non-existent elsewhere in the world. This fact argues both for the isolation of this language family, and moreover for the ‘stubbornness’ of its speakers in keeping their own linguistic system, effectively disregarding external influences or, at any rate, having had little contact with the rest of the world. However, judging from reports on the syntax of these languages as found in Collins (2001, 2002b, 2003), their context-sensitive processes are as clear as elsewhere in the languages of the world. The only reasonable conclusion one can reach about this state of affairs is that the logical forms of Khoisan languages evolved to their present status prior to their separation from the rest of the world, so between 90,000 and 150,000 years ago, or before.

That leaves some 200,000 years to get these systems after the supposed recruitment of the *FOXP2* network. It is certainly not a lot of time, and it is unclear what the selective pressure (if any) would have been in this instance. It is, of course, worth attempting to build one such scenario, on analogy to the one we sketched based on Nowak et al.'s (2001) results, but we leave that to other researchers, since we have nothing useful to add to this possibility.

Still a third logical possibility is that context-sensitive specifications are actually not genetically specified, but eventually emerged from yet another instance of third-factor considerations. Again, we won't even attempt to speculate on how this may have been, particularly because little is known, as already mentioned, about the sort of automaton that would realistically instantiate PD+ specifications. But once our ‘computational morphospace’ is roughly in place, it is legitimate to attempt an explanatory approach in these terms, perhaps worrying about the combined likelihood of such scenarios (recall: in our approach, the vocabulary explosion is also blamed on unknown third-factor considerations).

Just to defend the putative plausibility of the last scenario, bear in mind that nothing in the logic of what we have said forces the third-factor event correlating with the vocabulary explosion to have taken place *at the same time* as the putative recruitment of the *FOXP2* network. It could be contemporary or it could be prior— much prior. It's anybody's guess.

Still another, and in the end rather different, possibility is that our logic in this chapter has been slightly backwards, and actually the *FOXP2* recruitment pre-dated the vocabulary explosion. The good news of that sort of scenario would be that we could then attempt to correlate our two third-factor scenarios (vocabulary explosion and PD+ conditions) into a single one–not that we have an understanding (p. 615) of what, specifically, would be involved in either sort of situation. The bad news, however, would then be that we could not blame the recruitment of *FOXP2* on selective pressures of the sort sketched above. If we also want to blame that on third-factor considerations, then we face the same sort of problem we have just sketched. If, on the other hand, we want to continue to blame that evolutionary event on more standard aspects of the problem, the task would be to explicitly say what those are.

26.7 Coda: Evo-Minimalism as an Antidote against Skepticism—If Handled With Care

Our intention has not been to provide *the* evolutionary tale of FLN. We think we know enough about both language and evolution to realize that this is going to be a very complex exercise, which may actually never be fully realized. Our intention is more modest: we wanted to sketch the sorts of issues that arise within what we have called Evo-minimalism.

Lewontin (1998) reminds us that the evolutionary study of cognition, and more particularly the language faculty, is no easy game, given the distance between the proteins genes express and the cognitive functions in whose development or deployment such proteins may be implicated. We have attempted to sketch the sorts of difficulties that arise in relating all the necessary steps leading from one to the other extreme of such causal chain, which casts some doubt on the viability of constructing true evolutionary explanations in the realm of cognitive novelties such as FLN. In addition, several difficulties exist in deciphering whether putative relations of hypothesized capacities in various species are genuine ho-mologies (such as the one we expect for the *FOXP2* networks), or instead, weaker analogies.

More generally, we are not even sure whether the language faculty is subject to the sort of variation that is implicit in the logic of adaptationist scenarios, as we emphasized first negatively (observing how putative stories often do not consider alternatives such as ‘symmetric labeling’, to oppose as a competitive trait to what we observe), and next positively (discussing how a vocabulary explosion may have created a selective pressure for sociable individuals, vis-à-vis a solipsistic alternative). To make things worse, it is patent that the language faculty as we experience it manifests itself in thousands of guises (the various languages of the world), albeit in a way that is plainly not part of the evolutionary equation. We haven't even attempted to touch on this property of the system, hoping that it is a side-effect of (p. 616) one or more of the evolutionary scenarios that we have considered—but that itself again emphasizes the enormity of the task and of our ignorance in addressing it.

Still, these considerations all pertain to the historical and functionalist vortices of the evolutionary explanation, within the Gould—Chomsky triangle. To the extent that the minimalist proposals on the nature of language and linguistic computations are empirically correct, Evo-minimalism may turn out to be a fruitful program, inasmuch as it attempts to solidify the validity of evolutionary accounts from deeper, third-factor considerations. This is the positive side of our story, an invitation to continue theoretical work in this regard to keep specifying the details of the computational morphospace that is involved in the language faculty. We may not know, yet, how a PDA+ works, but we have unearthed much within simpler systems presupposed by it, in ways that allow evolutionary questions as posed here.

We want to finish with a note of caution to minimalists, the potentially troubling side of our story: we cannot let third-factor explanations become circular: universal property P in the system follows from third-factor considerations because … no simpler explanation is available. That, alas, is no explanation. We will be the last to invoke the reductionist bullying that takes a line of reasoning to be truly worthy only if it reduces to some known physics. But at the same time we exhort minimalists to remember that momentous third-factor events, though surely possible, are probably rare, and their signature should be obvious. Our own suggestion that vocabulary explosion ought to arise from third-factor conditions should be scrutinized in this very light, and we hasten to add that our account has left all the necessary details unspecified—although our claim is certainly testable. At any rate, if theorists start stringing multiple hypothetical such events within a few thousand human generations, the resulting evolutionary tale will have the explanatory force of: ‘Then a miracle happens!’

Notes:

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[–] Abstract and Keywords

While research in the 'principles and parameters' tradition can be regarded as attributing as much as possible to universal grammar (UG) in order to understand how language acquisition is possible, Chomsky characterizes the 'minimalist program' as an effort to attribute as little as possible to UG while still accounting for the apparent diversity of human languages. These two research strategies aim to be compatible, and ultimately should converge. Several of Chomsky's own early contributions to the minimalist program have been fundamental and simple enough to allow easy mathematical and computational study. Among these are (i) the characterization of 'bare phrase structure'; and (ii) the definition of a structure building operation Merge which applies freely to lexical material, with constraints that 'filter' the results only at the phonetic form and logical form interfaces. The first studies inspired by (i) and (ii) are 'stripped down' to such a degree that they may seem unrelated to minimalist proposals, but this article shows how some easy steps begin to bridge the gap. It briefly surveys some proposals about (iii) syntactic features that license structure building; (iv) 'locality', the domain over which structure building functions operate,; (v) 'linearization', determining the order of pronounced forms; and (vi) the proposal that Merge involves copying.

Keywords: universal grammar, Chomsky, minimalist program, structure building, locality, linearization, Merge, copying

WHILE research in 'principles and parameters' tradition (Chomsky 1981a) can be regarded as attributing as much as possible to universal grammar (UG) in order to understand how language acquisition is possible, Chomsky characterizes the 'minimalist program' as an effort to attribute as little as possible to UG while still accounting for the apparent diversity of human languages (Chomsky 2007: 4). Of course, these two research strategies aim to be compatible, and ultimately should converge. Several of Chomsky's own early contributions to the minimalist program have been fundamental and simple enough to allow easy mathematical and computational study. Among these contributions are (i) the characterization of 'bare phrase structure,' and (ii) the definition of a structure building operation Merge which applies freely to lexical material, with constraints that 'filter' the results only at the PF and LF interfaces. The first studies inspired by (i) and (ii) are 'stripped down' to such a degree that they may seem unrelated to minimalist proposals, but in this chapter, we show how some easy steps begin to bridge the gap.

This paper briefly surveys some proposals about (iii) syntactic features that license structure building, (iv) 'locality,' the domain over which structure building functions operate, (v) 'linearization', determining (in part) the order of pronounced forms, and (vi) the proposal that Merge involves copying. Two very surprising, (p. 618) overarching results emerge. First, seemingly diverse proposals are revealed to be remarkably similar, often defining identical languages, with recursive mechanisms that are similar (Theorems 2–5, below). As noted by Joshi (1985) and others, this remarkable convergence extends across linguistic traditions and even to mathematical work that started with very different assumptions and goals (Theorem 1, below). Second, all the mechanisms reviewed here define sets of structures with nice computational properties. In particular, they all define 'abstract families of languages' (AFLs) that are efficiently recognizable. This raises an old puzzle: why would human languages have properties that guarantee the existence of parsing methods that correctly and efficiently identify all and only the well-formed sentences, when humans apparently do not use methods of that kind? A speculation—perhaps supported by the whole cluster of AFL properties—is that this might facilitate learning, by facilitating the calculation of how lexical properties should be adjusted in light of input available to the learner.

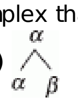
One methodological point should be noted immediately. One way to study generative mechanisms is by considering what they generate, what structures and what 'sentences'. Often this is the first, easiest thing to assess. This does *not* indicate that the linguists' task is to provide grammars generating all and only some set of pronounced sequences that are judged 'grammatical'. For one thing, syntactic principles (at least to a good first approximation) do not depend on phonetic properties, and so in our formal studies, the alphabet of pronounced complexes gets scant attention—in a certain sense, we do not care what the sequences are, but only what kinds of patterns they exhibit. And morphophonology also intervenes to obscure the sequences of syntactic heads in ways that are not well understood. But most importantly, in studying syntax, we abstract away from many 'non-syntactic influences on language, factors which are not known in advance. Spoken phonetic sequences, intuitive judgments about 'sentences', and corpus studies can provide our evidence, in part, but we do not know a priori, in advance of the science, what should count as a 'sentence', or which properties of language can be explained in which ways. This methodological stance is familiar from Chomsky (1965: ch. 1, sect. 1) and many other places.

27.1 Bare phrase structure

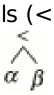
Following Muysken (1982), Chomsky (1995b: 242–3) suggests that the important insight of X-bar syntax is a relational one: a head X determines certain relevant properties of the phrase XP it is the head of. This idea, sometimes called 'endocentricity,' gets its content with a specification of what comprises a phrase XP and which properties are relevant. In generative grammar, the key questions are (p. 619) which properties of the complex XP are visible later in the derivation, and which are determined by the head? These properties are sometimes called the 'label' of the complex: Chomsky (2007: 17) says 'all operations are driven by labels.' If the labels encode what is visible to syntactic operations, they must encode at least those properties of the head that have an influence on later derivational steps, and any properties of other elements that can enter into other syntactic relations ('valuation', 'licensing') elsewhere in the derivation. Chomsky (1995b: 245) suggests that merging constituents α and β yields a set $\{\alpha, \beta\}$ together with label α,

(1) $\{\alpha, \{\alpha, \beta\}$

a complex that could be also be regarded as an unordered tree:[1]

(2) 

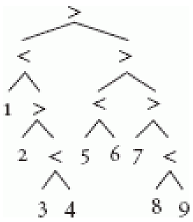
The set notation and the tree both represent α twice, but that is not necessary.[2] A slight variant of this notation from (Stabler 1997)represents α just once, with a tree notation in which internal nodes are labeled with order symbols (< or >) 'pointing' towards the head:

(3) 

We could label the leaves of these trees with lexical items, but we can further reduce clutter in the representation by assuming that all and only the syntactic features of heads are visible, where these features are 'erased' once they cease to play a role. In that case, the leaves will either be lexical items or simpler structures.

Some research in the minimalist program also assumes that when elements are merged, their linear order (and possibly even the question of whether they will be pronounced) may not be determined until later in the derivation. We will postpone (p. 620) consideration of this abstraction, and tentatively assume that linear order can be determined locally by the nature of the elements merged, forming complexes in which linear order can already be determined. So we will use the notation of (3), departing from (2) not only with the indication of headedness but also with the assumption that the tree is linearly ordered. Some adjustments to this tentative assumption, and other ideas about 'linearization', are briefly considered in section 27.5 below.

With these assumptions, consider a tree like this:

(4) 

The head of this whole tree is the node labeled 8. Every tree is regarded as a subtree of itself, and the leaves are subtrees too. Let's say that a subtree is a ‘maximal phrase’, or a ‘maximal projection’, or simply ‘maximal’, if it is not properly included in any larger subtree that has the same head. The ‘minimal’ elements in each tree are the leaves. Then the subtree of (4) with leaves 2, 3, 4 is maximal, while the subtree containing only 3, 4 is neither maximal nor minimal, and the subtree containing only the leaf 4 is both maximal and minimal. Furthermore, with the assumption that the tree is linearly ordered, the heads are pronounced in the order 123456789.

Considering the lexical items more carefully, let's assume that they have semantic and phonetic features, which are distinct from the formal syntactic features. We will put the non-syntactic features first (usually using the conventional spelling of a word to indicate what is intended), followed by a double colon ∷, followed by a sequence of syntactic features.

Phon ∷ feature_1 feature_2 … feature_n.

We use the double colon ∷ in lexical items, but for reasons mentioned in Appendix A.1 below, in derived structures a colon: will separate phonetic from syntactic features—in the body of this chapter this distinction will be respected but will not matter. And here we assume that syntactic features are ordered sequentially, but an alternative is considered in section 27.3 below.

Tentatively, let's distinguish four kinds of syntactic features: in addition to the usual ‘categorial’ features N, V, A, P, C, T, D, …, let's assume that a head which wants to select a phrase of category X has a feature =X. So then we have the ‘selector’ features =N, =V, =A, =P, =C, =T, =D,…. Ultimately, of course, we would like to understand the nature of these features: why some verbs select DP arguments, for (p. 621) example. But for the moment we simply give such a verb a feature =D.[3] Some other properties may require or allow a licensing relationship of a different kind, so we will have ‘licensee’ or ‘goal’ features −*wh*, −focus, −case,…. We assume, initially, that these features are all distinct; for example, no licensee feature is also a category. The heads that can license phrases with such requirements will be called ‘licensors’ or ‘probes’ and will have features +*wh*, +focus, +case,…. So a simplistic lexicon might have 4 items like this:

Marie ∷ D

who ∷ D −*wh*

praises ∷ =D =D V

ε ∷ =T +*wh* C.

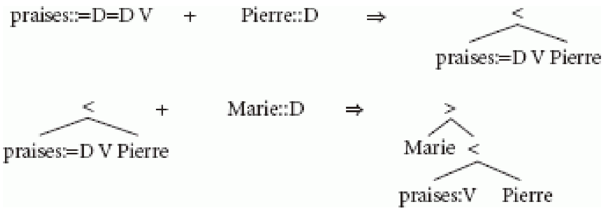
A minimalist grammar (MG) is simply a finite lexicon like this. That is, a minimalist grammar G is a lexicon,

(5) G ⊂ Phon × Features*, a finite set,

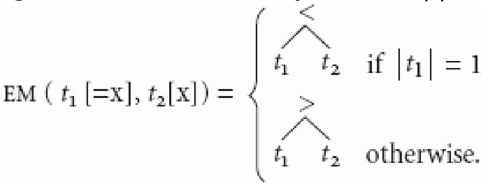
where Features* is the set of finite sequences of syntactic features, and where the elements of the lexicon are combined by the merge operation which is defined in section 27.2, just below. Later sections consider some variations in both the feature system and in the merge function.

27.2 Merge: first version

We will regard the lexicon as providing the labels for 1-node trees, so that Merge can be regarded as a function that applies either to pairs of trees (External Merge, EM) or to single trees (Internal Merge, IM). Since we are assuming that the derived structures specify linear order (an option we reassess in section 27.5 below), EM is specified with two cases as well. When a lexical selector combines with a first element, that element is attached on the right and is called the complement. When a derived expression selects another element, that element is attached on the left and is called a specifier. (Some linguists have proposed that each category can have at most one specifier, but these first definitions will not impose that bound, allowing any number of specifiers.) Furthermore, we assume that the selector features =X and X must be the first features of the heads of the arguments, and that they are (p. 622) both erased by merge. For example, applying Merge to the pairs of structures shown on the left, we obtain the derived structures on the right:



Let's write $t\,[\alpha]$ when the head of the tree has a sequence of syntactic features whose first element is α. Given a structure $t[\alpha]$, let t denote the result of (i) erasing feature α and (ii) if the head of $t[f]$ has a double colon ::, changing it to a colon. And for any tree t, let $|f|$ be the number of nodes in t. Then the function EM is given by

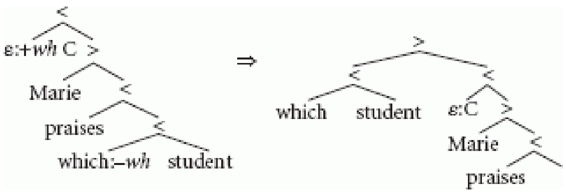
(6) 

To reduce notational clutter, we write a leaf node label *word*: ε simply as *word*, as in the *Marie praises Pierre* examples above. And leaf nodes with no features at all, ε:ε, are usually written just ε, or as nodes with no label. In particular, the tree consisting of one node and no (phonetic, semantic or syntactic) features is sometimes called ε.

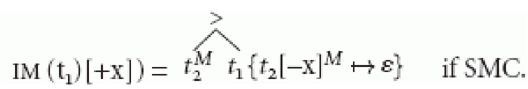
Internal Merge applies to a single structure t[+x] only if it satisfies this strong version of the 'Shortest Move Constraint':

(7) SMC: Exactly one head in the tree has −x as its first feature.

In that case, IM moves the maximal projection of the −x head to specifier position, leaving an empty subtree ε behind. (Section 27.6.5 considers the idea that IM involves a copy, leaving the original −x phrase in its original position. Covert movements that leave the phonetic material behind are briefly discussed in section 27.6.2 below.) So, for example,



To define this operation, given any tree t, let $t\{t_1 \mapsto t_2\}$ be the result of replacing subtree t_1 by t_2 in t, and given any subtree (possibly a leaf) t, let t^{M} be the maximal projection of the head of t. Then IM applies to a tree t_1[+x] containing subtree (p. 623) t_2[−x], by deleting the +x feature to obtain t_1, removing the maximal projection of the −x head to obtain $t_1\{t_2[-x]^M \mapsto \varepsilon\}$, and finally adding $t_2{}^M$ as a specifier. In sum,

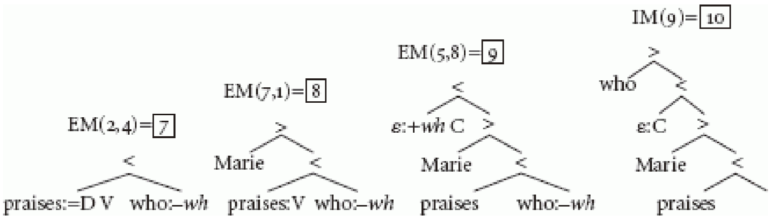
(8) 

Since EM and IM apply in different cases, their union is also a function, which we will call Merge.

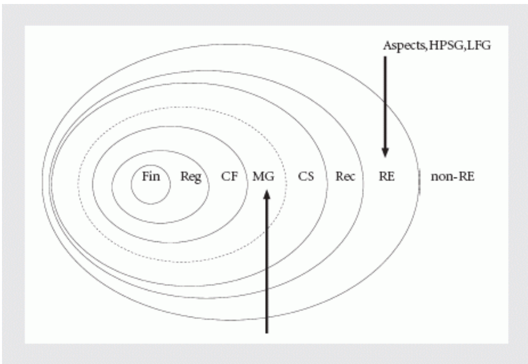
Example G1. Consider the grammar G1 given by these six lexical items, numbered for easy reference:

0	Pierre::D	who::D − *wh*	4
1	Marie::D	ε::=V + *wh* C	5
2	praises::=D =D V	knows::C =D V	6
3	ε::=V C		

With this grammar G, we can apply Merge (EM and IM) as follows, for example:



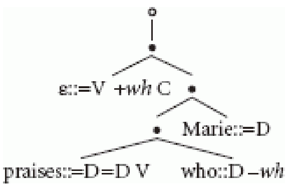
For any MG G, let the set structures(G) include all and only the structures that can be derived from the lexical items of G by applying Merge in all possible ways. Let a completed structure be one in which (i) there is exactly one syntactic feature, (ii) that feature is the ‘root’ or ‘start’ category, and (iii) that feature is at the head of the tree. And let the language L(G) be the set of phonetic sequences at leaves of completed structures of G.



Click to view larger

Fig. 27.1 . Chomsky hierarchy

The derivation tree of the four-step derivation of our example can be drawn like this:



Here we use • to represent EM and o to represent IM. Since these functions apply unambiguously, the derived structure obtained at each internal node of this derivation tree is completely determined. So if C is the ‘start’ category of our example grammar G, then this derivation shows that the sentence *who Marie praises* (p. 624) $\in L$ (G1). Notice that the derivation tree is not isomorphic to the the derived tree, numbered 10 above.

Minimalist grammars (MGs), as defined here by (5), (6), and (8), have been studied rather carefully. It has been demonstrated that the class of languages definable by minimalist grammars is exactly the class definable by multiple context free grammars (MCFGs), linear-context-free rewrite systems (LCFRSs), and other formalisms (Michaelis 1998, 2001b, 2004, Harkema 2001a). MGs contrast in this respect with some other much more powerful grammatical formalisms (notably, the ‘Aspects’ grammar of Peters and Ritchie 1973, and HPSG and LFG: Berwick 1981, Johnson 1988, Torenvliet and Trautwein 1995), as shown in Figure 27.1.The MG definable languages include all the finite (Fin), regular (Reg), and context-free languages (CF), and are properly included in the context-sensitive (CS), recursive (Rec), and recursively enumerable languages (RE). Languages definable by tree adjoining grammar (TAG) and by a certain categorial combinatory grammar (CCG) were shown by Vijay-Shanker and Weir (1994)tobesandwichedinsidethe MG class.[4] So we have:

Theorem 1: $CF \subset \boxed{TAG \equiv CCG} \subset \boxed{MCFG \equiv LCFRS \equiv MG} \subset CS.$.

When two grammar formalisms are shown to be equivalent (≡) in the sense that they define exactly the same languages, the equivalence is often said to be 'weak' and possibly of little interest to linguists, since we are interested in the structures humans recognize, not in arbitrary ways of defining identical sets of strings. But the weak equivalence results of Theorem 1 *are* interesting. For one (p. 625) thing, the equivalences are established by providing recipes for translating one kind of grammar into another, and those recipes provide insightful comparisons of the recursive mechanisms of the respective grammars. Furthermore, when a grammar formalism is shown equivalent to another one that is already well studied, many new facts about the new formalism may come to light; this in fact happened in the case of MGs.

One key insight behind Theorem 1 can be expressed as follows (see Michaelis 2001a, 2004, 1998, for full details).For any MGG, let's say that a derived tree in structures(G) is useful or relevant if and only if it is used in a derivation of a completed structure. That is, completed structures are useful, and so are all the structures involved in their derivations. Let useful(G) be the set of useful elements of structures(G). Then it is easy to see that every minimalist grammar G has the following property:

(9) (Finite partition) $\exists n \geq 0$, $\forall t \in$ useful(G), the number of heads in *t* with syntactic features is less than n. So every useful structure can be classified according to which features these heads have, providing a finite partition of useful (G).

MGs have this property because, first, there can only be a finite number of lexical items, and so there can only be a finite number of licensee features. Second, each head in a useful tree will have some suffix of the syntactic feature sequences in the lexicon, since syntactic features are checked and erased in order, from the front. And third, by the SMC, no useful tree can have two heads beginning with the same licensee. So there can only be finitely many heads with non-empty sequences of syntactic features. Classifying each useful tree according to these feature sequences, in a relevant sense, completely determines syntactic properties of that tree.

Michaelis (2001a: §5.2) shows that MG languages have a large set of nice closure properties that make the class a 'substitution-closed full abstract family of languages' (AFL) in the standard sense introduced in Ginsburg and Greibach (1969) and discussed by Mateescu and Salomaa (1997: §3).[5] This means, for example, that it is easy to represent the results of certain kinds of substitutions and other operations. Many standard parsing methods and probabilistic models depend implicitly on AFL properties (Hale 2003, Nederhof and Satta 2003, Goodman 1999).

It is also known that the MG definable languages are efficiently parsable (Seki et al. 1991), and that standard parsing methods can be adapted for them (Harkema 2001b).[6] In CKY and Earley parsing models, the operations of the grammar (p. 626) (EM and IM) are realized quite directly by adding, roughly, only book-keeping operations to avoid unnecessary steps. For any minimalist grammar G, these parsing models are guaranteed to accept all and only the elements of L(G). (In cases of non-sentences these parsing methods will not 'succeed', but will often detect non-trivial subconstituents, a presumably useful input to repair or learning strategies.) Humans, on the other hand, in the recognition of fluent speech, seem to use parsing methods that fail on certain kinds of fully grammatical structures; among the best-known examples are garden paths like *the horse raced past the barn fell* (Pritchett 1992), and highly ambiguous strings like *police police police police police police* (Barton et al. 1987: §3.4). It is an open question whether the human failures can be attributed to externally imposed limitations on mechanisms that can otherwise handle all constructions definable with Merge (cf. the chapter on processing in this volume, and the discussion of grammar-performance relations in (Berwick and Weinberg 1984).

Earlier formal studies of Government—Binding (GB) theory (Rogers 1994, 1999, Kracht 1995) concluded that it was a context-free grammar notation, up to indexing.[7] But those GB grammars required that each moved constituent c-command its trace, blocking 'remnant movement'. That requirement, sometimes called a 'proper binding condition' (PBC), is now generally regarded as too stringent (Abels 2007, Boeckx 2002, Hiraiwa 2003, Müller 2000, Müller 1996, Koopman and Szabolcsi 2000, Kayne 1994), and is not imposed here. Familiar remnant movement analyses include structures like these:

(10)

a. $[_{AP2}$How likely $[t_1$ to win]] is$_3$ John$_1$ t_3 t_2?
b. John $[_{VP2}$ reads $t_1]$ [no novels]$_1$ t_2.
c.

$[_{VP2}$	t_1	Gelesen]	hat	[das	Buch]$_1$	[keiner	t_2].
		read	Has	the	book	noone	

Notice that without the PBC, many of the common, simplistic assumptions about processing models will not work: processors cannot work left-to-right by putting moved elements into a memory store and then watching for the corresponding gap, since human languages not only allow constructions in which gaps precede the moved elements, but also ‘remnant movement’ structures like (10) where moved elements can contain moved elements (recursively, without bound).[8]

(p. 627) 27.3 Merge: conflated and persistent features

Minimalist grammars, as defined in the previous sections, partition the categories and the licensees into non-overlapping sets, but perhaps some properties of heads can enter into both selection and movement relations. For example, a verb might select a D, and a tense head might want that very same element to appear in its specifier because it is a D. The MGs above insist on having two features for these two different roles: category D can be selected, while only −D could be licensed by movement.

Inspired by Chomsky (1995b: §4.4.4) and Collins (1997: §5.2), let's (i) conflate the categories and licensees, using only features f and selectors/licensors =f, and (ii) specify that some subset of these features is persistent (or ‘interpretable’). (Non-persistent features are sometimes called ‘formal’.) If features are ordered and Merge steps do not erase persistent features, then everything following a persistent feature would be inaccessible. So we could remove the order, and assume that constituents have a set of features. That approach is common, but then when a head selects two constituents, what controls which constituent is selected first? To avoid that problem, let's keep ordered features but allow Merge to optionally erase persistent features, so that when they have played all their roles in a derivation they can be deleted. For this approach, we leave the previous definition of EM (6) unchanged; we modify the definition of IM (8) so that it is triggered by =x,

(11)
$$\mathrm{IM}\,(\mathrm{t}_1[=\mathrm{x}]) = \overset{>}{t_2^M \quad t_1}\ \{t_2[\mathrm{x}]^M \mapsto \varepsilon\} \quad \text{if SMC,}$$

and for persistent features we add variants of (6) and(11) that apply only to persistent features, leaving them unerased. (Persistent features are underlined.)

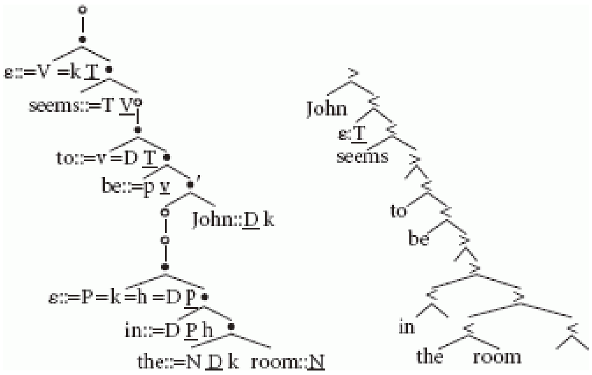
(12)
$$\mathrm{EM}'\,(t_1\,[=\mathrm{x}], t_2[\underline{\mathrm{x}}]) = \begin{cases} \overset{<}{t_1 \quad t_2[\underline{\mathrm{x}}]} & \text{if } |t_1| = 1 \\ \overset{>}{t_2[\underline{\mathrm{x}}] \quad t_1} & \text{otherwise,} \end{cases}$$
$$\mathrm{IM}'\,(t_1\,[=\mathrm{x}]) = \overset{>}{t_2[\underline{\mathrm{x}}]^M \quad t_1}\{t_2[\mathrm{x}]^M \mapsto \varepsilon\} \quad \text{if SMC,}$$

Let's call these grammars ‘conflated minimalist grammars’ (CMGs): Merge is now defined by (6), (11), and (12).

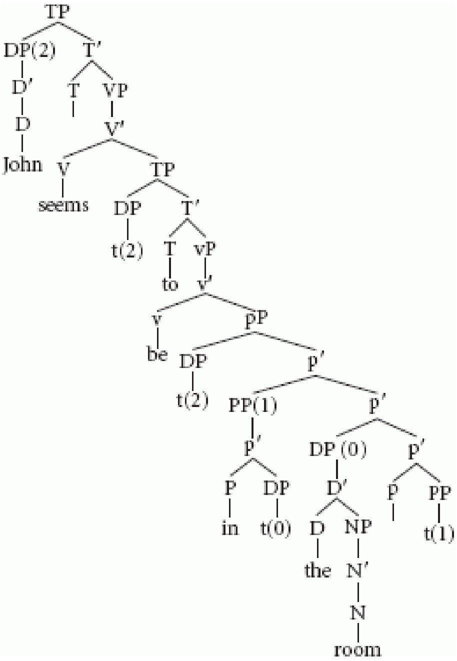
Example G2. In his discussion of persistent features, Collins (1997: 102) considers a structure that he gives schematically as follows:

(p. 628) **(13)** $[_{\mathrm{TP}}$ John$_i$ $[_{\mathrm{T'}}$ T$[_{\mathrm{VP}}$ seems $[_{\mathrm{TP}}$ t_i $[_{\mathrm{T'}}$ to be $[_{\mathrm{SC}}$ t_i in the room]]]]]

The following tiny CMG lexicon suffices to derive structures of this form. (Persistent features are underlined, and remnant movement triggered by feature h is used get the preposition into the right place. Cf. section 27.6.1 below on head movement.) The derivation shown below, left, yields the structure on the right:



The bare phrase structure reveals so little about the derivation, it can be useful to compute the corresponding redundant but more readable X-bar tree:



(p. 629) The MG finite partition property (9) still holds for CMGs: in useful CMG trees, there is a finite bound on the number of visible heads. And in fact, MGs and CMGs do not differ at all in the languages they define:

Theorem 2: *CMG≡MG.*

ε::=V =k *T*	seems::=T *V*	to::=v =D *T*
be::=p *v*	in::=D *P* h	ε:=P=k=h=D *p*
the::=*N D* k	room::*N*	John::*D* k.

A proof is sketched in Appendix A.1, below. This result obviously means that the question of whether humans use CMGs rather than MGs cannot be based simply on what expressions are in any particular language. In fact, it is not clear that an implementation of a CMG could be distinguished from an implementation of a MG: they might be different specifications of the very same implemented computation.[9]

Many other feature-checking and feature percolation schemes are proposed in the literature (and we will consider one more in section 27.4.2 below). Some are similar in expressive power and succinctness to the ones in MGs or CMGs; others are known to make the grammars as powerful as any computing device (Kobele 2005). Obviously, the real challenge is to see through the notational variants to find the most restrictive characterizations that can provide insightful, explanatory models of human language abilities.

27.4 Merge: Locality

27.4.1 Phases

It is possible that certain categories make everything in their complements invisible to syntactic operations, acting rather like the 'bounding nodes' of earlier work in the GB tradition. We could call such categories 'phases', after the similar proposals in the literature (Chomsky 2000a, 2001, Chomsky 2008a, Bošković 2003, Abels 2003). For example, the categories v and C could be phases in this sense. The following simple condition, which we will call PIC after the similar formulations of 'phase impenetrability conditions' in the literature, obtains the desired effect in MGs as soon as the phase feature becomes visible, which will always happen exactly at the point when the projection of the phase, including all movements to specifier positions (the 'edge' of the category), has been completed:

(p. 630) **(14)** (PIC) Merge cannot apply to a tree $t[f]$ if f is a phase and if the complement of the head of $t[f]$ has any syntactic features.

The PIC restricts both EM and IM, so no outstanding conditions in the comple- mentofanyphasecaneverappearinasuccessfulderivation.CallMGswiththis constraint 'phase-based minimalist grammars' (PMGs): they are (i) MGs as defined in sections 27.1 and 27.2, (ii) extended with a specification of which categories (if any) are phases, and (iii) with PIC. We can also consider 'phase-based conflated minimalist grammars' (PCMGs) by imposing PIC on the CMGs of section 27.3.

We can establish the following result, and easily adapt the standard MG parsing methods to the phase-based grammars:

Theorem 3: *PCMG≡PMG≡MG.*

A proof is sketched in Appendix A.2.

One of the intuitive ideas behind phase theory is that the material in the complements of phases is ready to be 'sent to the interfaces'—so that no syntactic operations can affect the interpretation or the linear order of pronounced elements inside a phase complement. Notice that, at least with regard to the pronounced sequences, we can establish this as a theorem about PMGs (and PCMGs) too: since no outstanding syntactic features are allowed in phase complements, it follows immediately that nothing can affect the linear order of phrases there. Chomsky apparently has some additional point in mind (e.g. Chomsky 2007: 16–17): perhaps phases could provide some analog of the finite partition property, so that, even without the assumption of SMC or anything like it, no search of unbounded structure would ever be needed to determine whether a derivational step can be taken (e.g. to find a matching head for IM). For example, suppose that the amount of syntactically visible material in a phase (not counting the material inside contained phases) were finitely bounded; that could be significant for recognition or any other algorithm that required discovering derivations. But phases as defined above provide no such guarantee.[10]

27.4.2 Relativized minimality

The SMC given in (7) above blocks derivations in which there are two *wh*-elements competing for the same position; intuitively, allowing either to move would mean that the other would not be checked by the closest available (p. 631) +*wh* position. In a natural grammar, this principle could block constructions like this:[11]

(15) What$_i$ do you wonder how$_j$ [to solve t_i t_j]

Rizzi's 'relativized minimality' suggests a natural generalization of the SMC, extending intervention beyond matching features to broader classes (Rizzi 2004a, 2001a, 1997). Not only can one *wh*-phrase not move across another, but also, for example, in (16) one subject cannot move across another, and in (17) one adverb cannot move across another (examples 16–19 from Rizzi 2001a):

(16) *John$_i$ seems that it is likely t_i to win

(17)

Rapidamente$_i$,	i	tecnici	hanno	(*probabilmente)	
rapidly,	the	technicians	have	probably	resolved
ti	il	problema	(Italian)		
	the	problema			

However, it seems that an adverb can sometimes be extracted across an adverb, e.g. if it is being moved to a stressed focus position,

(18)

RAPIDAMENTE$_i$,	i	tecnici	hanno	probabilmente	risolto
RAPIDLY,	the	technicians	have	probably	resolved
t_i	il	problem	(non	lentmente)	
	the	problem	not	slowly)	

A similar kind of 'selective island' effect is also suggested by Obenauer's (1994) examples:

(19)

a.

[Combien	de	livres]$_i$	a-t-il	beaucoup	consultés	t_i	
how-many	of	books	has-he	much	consulted		
							(French)

b. *Combien$_i$ a-t-il beaucoup consultés [t_i de livres]

Notice also that in (16), (17), and (19b), the intervening elements are blocking the movements because of features that will have been checked and deleted in an MG or CMG derivation at the time when the offending movement wants to take place. So in order to determine this kind of intervention, several changes in the MG formalism are needed.

We adapt MGs to capture a number of Rizzi's insights as follows. First, we conflate selectors with probes, and licensees with categories, distinguishing some subset of these as persistent, as in CMGs. Second, in order to obtain intervention effects like those suggested by (16) and (17), since checked features seem to be relevant, the Merge rule is modified so that, while the distinction between persistent and non-persistent features is maintained for checking relations, all features remain visible for intervention effects. Instead of marking progress on the requirements (p. 632) of lexical items by erasing features, we move a dot through the sequence. That is, lexical items will start with dot-initial feature list $\bullet\alpha_1\alpha_2 \ldots \cdot\alpha_n$; when α_1 is checked and becomes invisible to any other probe, we get the sequence $\alpha_1\bullet\alpha_2 \ldots \cdot\alpha_n$; and when α_1 is checked and persists (e.g. for the kind of cyclic movements mentioned in section 27.3), the feature sequence is left unchanged. The dot is similarly moved across later features, marking the unique feature that is visible to a probe/selector while leaving all the features available for the determination of blocking effects. In this system, the notation $t[f]$refers to a tree whose head has feature f immediately following the dot. Third, for any tree t, let *type* be a function mapping each basic feature f to features that will block movement of $t[f]$. And finally, potential interveners are defined in terms of c-command as follows.[12] For any subtree t_2 of tree t_1, let *cover* (t_2) be the set of features of heads t_3 such that $t_3{}^M$ c-commands $t_2{}^M$. With these definitions, we formulate this replacement for the SMC:

(20) (RMC) IM applies to t_1 [$= f$] only if (i) t_1[$= f$] has exactly one subtree t_2[f], and (ii) *cover* (t_2 [f]) ∩ type(f) = ∅.

Let's call grammars that are constrained in this way 'relativized minimalist grammars' (RMGs). In this kind of RMG it is possible to require that a =f movement of an adverb is blocked by another intervening adverb, while a different kind of movement triggered by a different feature =g can move the same adverb without being blocked. RMGs have the same expressive power as MGs, and standard parsing methods extend to them (Appendix A.3):

Theorem 4: *RMG≡MG.*

In recent work, Rizzi has proposed another, rather different restriction on movement. The reader will have noticed that in example G2, above, an 'abstract case' feature −k was checked in the tensed matrix clause. Rizzi observes that the subject requirement is not always associated with the case system in this way, and proposes a more general, semantically motivated account of why clauses need subjects and why subjects of tensed clauses tend not to be movable (Rizzi 2004b, Rizzi and Shlonsky 2007). Adapting his account slightly to the terms of the present framework:

(21) Certain designated positions are 'criterial' in the sense that they are dedicated to a particular interpretive property. These positions maybe identified as the specifiers created by +f features for f = q, top, foc,…

(p. 633) **(22)** (Criterial freezing) Nothing can be moved from a criterial position.

Rizzi shows how a range of subject—object asymmetries follow from the fact that there is a 'subject criterion' but not an 'object criterion'.[13] In the present framework, criterial freezing is achieved simply by making the criterial features non-persistent, and requiring (perhaps for semantic reasons) that these features always appear as the last feature of any head.[14]

27.4.3 Multiple movements and multiple Agree

The theory of Merge and its locality restrictions needs to allow for *wh*-in-situ and multiple *wh*-fronting (Rudin 1988, McDaniel 1989, Bošković 2007, 2002b). *Wh*-in-situ can be allowed simply by assuming that *wh*-phrases need not have formal features which must be checked by movement. Multiple *wh*-movement poses trickier problems for SMC (and also RMC and related constraints): they do not allow IM to apply when there are two *wh*-features. Several strategies might be worth pursuing. First, an SMC-like constraint might be tenable if we refine the *wh*-features (e.g. *wh*-nom, *wh*-acc, *wh*-manner) so that each *wh*-movement is triggered by a different *wh*-feature. Another strategy introduces a special 'absorption' or 'clustering' case of Merge which, upon encountering a second *wh*-phrase in a derivation, immediately merges those two phrases into a single one, leaving a single *wh*-feature (Gärtner and Michaelis 2007, Paperno 2008). This idea might fit with the special ordering ('tucking in') found in some multiple-*wh*-constructions, noted for example by Richards (1998). A third strategy relaxes SMC-like constraints to allow a more expressive formalism (perhaps similar to the '-SpIC -SMC' formalism mentioned by Gärtner and Michaelis 2005). It is not yet clear which approach can best fit the facts.

As noted in section 27.6.2 below, it is easy to extend MGs to allow 'feature movement', that is, instances of IM that check features in the usual way but which do not move anything. Operations of this sort have been proposed in analyses of agreement relations. But in that case, multiple agreement relations will pose the same puzzle as multiple movements.

(p. 634) 27.5 Merge: linearization

Merge is defined in section 27.2 so that complements are attached to the right of the selecting head, while specifiers are attached on the left. This underlying SVO order is stipulated in the definition of our Merge operation, but is conceivably the result of some other forces on linearization. We will consider that possibility, but first it is worth observing the unsurprising fact that all orders can obtained by movement. It is easy to establish that the MG definable languages are 'closed with respect to reversal'. That is, for any minimalist grammar G defining language L(G), the result of reversing every sentence of that language is a language L(G′), where G′ is another minimalist grammar. Is this a signal that the formalism is overly general? Perhaps not.

Greenberg's Universal 20 observes that certain orders of the basic elements of determiner phrases are common across languages, while others are extremely rare or unattested (Greenberg 1963). In a recent assessment of Greenberg's proposal, Cinque (2005) reports that only 14 of the possible 24 orderings of [Dem Num Adj N] are attested. These ordering facts are sometimes offered as evidence for the assumptions like the ones made in the definition of Merge in section 27.2, but if all orders can be derived by movement, it is natural to wonder whether all hope of explaining Universal 20 and related facts is lost.

While MGs are closed with respect to reversal, nevertheless some orders are easier to define than others. That is, some orders of elements require more steps in their derivation. In particular, suppose that head 1 selects head 2, head 2 selects head 3, and head 3 selects head 4. Then we can derive the order 1234 using EM only. Adding pairs of features (+f,−f) triggering movement to these heads, it turns out that we can derive many other orders, but not all of them. For example, we cannot derive the order 3214, as can be seen by trying all the possibilities. (Trying all the possibilities is quite tedious by hand, but is much easier when the process is automated.) Notice first that movement never changes the root category, so given our assumptions about selection, 1 must be the root of all derivable orders. If we just let head 1 trigger movement of head 3 to its specifier, we get the order 2341.Since the 4 is in the wrong place, we can try to move it first: 4 can move to the specifier of heads 1, 2, or 3, but in none of these positions can we separate it from 2 and 3. (Appendix A.4 shows MG grammars for derivable orders.)

Given the assumption that the selection order is 1234, and assuming no other heads are introduced, only 16 out of the 24 orders can be obtained by movement, and 14 of those 16 are the ones that Cinque reports as attested. It is hard to believe that this is an accident! So one might be tempted to assume that these linear asymmetries are coming from the underlying SVO order. But that assumption would be a mistake, as recently pointed out by Abels and Neeleman (2006). They point out that the patterns observed by Cinque are well accounted for even when there can be heads and specifiers on either side of the head.

(p. 635) One way to see this point is to consider another variant on MGs in which IM is unchanged, but the feature =X always triggers attachment on the right and X= triggers attachment on the left. That is:[15]

(23) $$\mathrm{EM}\,(t_1\,[\alpha],\, t_2[\mathrm{x}]) = \begin{cases} \overset{<}{\wedge} \\ t_1 \quad t_2 \quad \text{if } \alpha \text{ is =x} \\ \overset{>}{\wedge} \\ t_2 \quad t_1 \quad \text{if } \alpha \text{ is x=} \end{cases}$$

Let's call grammars defined by by (5), (8) and (23) 'directional MGs' (DMGs). Using the same proof strategy used for Theorems 2 and 3, it is easy to establish

Theorem 5: *DMG≡MG.*

In spite of this weak equivalence, the derivational complexities of various surface orders obtainable in DMGs and MGs can differ. Keeping the previous assumption that we have exactly four heads with the selection order 1234, DMGs derive the 8 of the 24 possible orders using EM only, and only 8 of the other orders are derivable. The following table compares the minimal derivational complexities of the various orders in MGs and DMGs with Cinque's assessment of the typological frequencies of each. To facilitate comparison, for Cinque's classification, we write 0 for 'unattested' orders, 1 for orders found in 'very few' languages, 2 for 'few', 3 for 'many', and 4 for 'very many'. Listing this classification in Figure 27.2, since all derivable orders are obtained with three or fewer licensees, we use 0 to indicate underivable orders and otherwise use the quantity $(4 - \ell)$ where ℓ is the minimum number of licensees needed to get the order. (See grammars in Appendix A.4.)

First, comparing the underivable and unattested orders, we see that MGs and DMGs are identically in good agreement with Cinque's results. And in addition, there is a tendency for rarer orders to be harder to derive. Given the nature of Cinque's typological classification and the way we have coded derivational complexity here, it is important to avoid reading too much into the quantitative details, but calculating Pearson correlation coefficients, we find a correlation of 0.62 between the MG ranks and Cinque's, and a correlation of 0.75 between the DMG ranks and Cinque's. So we see that DMGs fit the typological data without the assumption of a rigid SVO order.

order	Cinque	MG	DMG	order	Cinque	MG	DMG
1234	4	4	4	1324	0	0	0
1243	3	3	4	1342	1	3	4
1423	1	3	3	1432	3	2	4
4123	2	3	3	4132	1	2	3
2134	0	0	0	2314	0	0	0
2143	0	0	0	2341	1	3	4
2413	0	0	0	2431	2	2	4
4213	0	0	0	4231	2	2	4
3124	0	0	0	3214	0	0	0
3142	0	2	2	3241	0	1	2
3412	1	3	3	3421	1	2	4
4312	2	2	3	4321	4	1	4

Fig. 27.2 . Derivable and attested orders of four nominal heads: 1, 2, 3, 4

Up to this point, we have been coding linear order in the derived trees, but this is slightly redundant. Notice for example that in the simple MGs, both EM and IM put specifiers on the left. There is a missed generalization there, and we may find other simplifications of the grammar if we separate the determination of linear order (p. 636) from the calculation of hierarchical structure. A simple and elegant idea about this is explored by Kayne (1994) and has been much discussed. From our computational perspective, consider the grammars that result from retracting our assumption that the derived trees created by EM in (6) and IM in (8) are ordered. Then immediately, the two cases given in (6) collapse into one. We can consider what function properly maps the unordered, derived trees into linearly ordered structures. The theory of tree transducers (Kobele et al. 2007, Maneth 2004) provides a natural framework in which the computational properties of such functions can be studied.

27.6 More variations

27.6.1 head movement

The nature of head movement remains controversial, but many traditional instances of head movement can be captured with simple extensions of minimalist grammars (MGH), studied in (Stabler 1997, 2001, 2003, Michaelis 2002). A certain MG equivalent version of a ‘mirror’ theory inspired by Brody's proposals has also been carefully studied by Kobele (2002). These extensions too are all weakly equivalent to MGs. Intuitively, the effects of head movement, and of mirror theory, can be achieved by phrasal remnant movement. Many mysteries remain.

27.6.2 LF and PF movements

Many minimalist proposals consider the possibility that all constraints on syntax should come from the LF and PF interfaces, with the action of merge between these (p. 637) two interfaces kept as simple as possible. This leads some to ask: if IM is acting to check a feature, why does this ever involve movement of structure with phonetic material in it? This possibility was already familiar in analyses of ‘covert’ movement at least since Chomsky (1976). It is easy to add feature checking without overt movement of structure to MGs, a kind of covert ‘LF movement’, to MGs, as studied in the early work (Stabler 1997, Michaelis 1998). If some features are interpretable while others are not, as briefly mentioned in section 27.3 above, then there could also be processes that leave the interpretable features behind while moving phonetic features, ‘PF movement.’ Perhaps certain head movement phenomena and certain instances of scrambling should be treated as PF movements.

27.6.3 Adjunct Merge

The External Merge rule introduced in section 27.2 is sometimes called ‘argument merge’, and there are various proposals for another kind of operation for merging adjuncts (Lebeaux 1991, Chomsky 1993, Epstein et al. 1998, Fox 2003). These are usually proposed in order to allow a more natural account of the certain ‘reconstruction’ effects found with arguments but not with adjuncts. A proposal of this kind has been formally modeled by Gärtner and Michaelis (2008).

27.6.4 Sideward movement

The MG variants defined above all allow Merge to select freely from the lexicon. Some proposals compare derivations that involve exactly the same multiset of lexical elements.[16] Phases provide a different kind of domain, sometimes regarded as a ‘subarray’ of constituents (Chomsky 2000a, 2001, 2008a), from which the structure-building functions can select their arguments. On some conceptions, these intermediate workspaces admit a kind of ‘sideward movement’ (Starke 2001, Nunes 2001, Hornstein 1999, Boeckx and Hornstein 2004, Citko 2005) which moves an element from one tree to another which has not yet been merged or adjoined. Increasing the domain of Merge in this way seems to open the way to more unified accounts of various phenomena. Some preliminary studies of this kind of domain have been undertaken in Stabler (2006, 2007), again MG equivalent systems.

27.6.5 Copy and Delete

The MG variants considered so far have left nothing behind when something moves. In the implementation of standard parsing methods, this means simply **(p. 638)** that syntactic relations can be constructed from non-adjacent parts. There is no particular mystery in that; in all perceptual domains we sometimes recognize a unity among components that are (spatially or temporally) non-adjacent. And we have various kinds of discontinuity in artificial languages that we design for our own convenience; mutually dependent type declarations in many programming languages need not be adjacent, and are usually rejected or flagged if they are vacuous (Marsh and Partee 1984). But familiar arguments suggest that moved phrases sometimes seem to be interpreted as if they are in their original positions (Chomsky 1976, May 1977, Chomsky 1993), and sometimes we even seem to see all or part of the phonetic contents of the moved phrase in the original positions too (Kobele 2006, Bošković and Nunes 2007).

Suppose we keep the definition of EM unchanged from section 27.2, but identify a subset of features that trigger the use of this special additional case of IM, defined in terms of a function *g* from trees to trees:

(24)

$$\mathrm{IM}(t_1[+\mathrm{x}]) = \underset{t_2^M \quad t_1\{t_2[-\mathrm{x}]^M \mapsto g(t_2[-\mathrm{x}]^M)\}}{>} \qquad \text{if SMC}$$

Now we obtain different kinds of grammar depending on the function *g.* If *g* maps every tree to the empty tree ε, this is exactly the original IM. But now consider, for example, the function that *g* leaves the structure and phonetic contents of each tree untouched, removing only any outstanding syntactic features. Let's call these grammars ‘minimalist grammars with copying’ (MGC). MGCs have quite different computational properties from the previous MG variants, as discussed by Kobele (2006).[17] Again they define languages in an efficiently recognizable class that has been studied: so-called ‘parallel multiple context-free grammars’ (PMCFGs) (Seki et al. 1991):

Theorem 6: $CF \subset \boxed{MCFG \equiv MG} \subset \boxed{MGC \subseteq PMCFG} \subset CS.$

27.7 Next steps

The formal, computational studies of apparently diverse ideas in the minimalist program reveal some surprising properties and especially commonalities. (Cf. Gärtner and Michaelis 2005, a survey emphasizing some significant *differences* among minimalist proposals.) There will always be a big gap between new empirical **(p. 639)** speculations and what is well understood, but we can seek results that apply to large families of related proposals whenever possible, identifying common themes if only approximately. Further study will certainly illuminate important distinctions that have been missed here and bring deeper understanding of the fundamental mechanisms of human language.

Appendix 27.1. MG variants and MCFGs

A.1 CMG≡MG

To prove this claim from Theorem 2, we show (1) that CMG⊆MG and (2)MG⊆CMG.

To establish (1), it suffices to establish (3) CMG⊆MCFG since it is already known that (4) MCFG⊆MG (Michaelis

2001b, Harkema 2001a). In fact, the proof of (3) can be a very minor variation of the proof of MG⊆MCFG given by Michaelis, so we sketch the main ideas here and refer the reader to Michaelis's earlier proof (Michaelis 1998, 2001a) for full details.

(3) can be established in two steps. First we show (3a) every CMG using Merge is exactly equivalent to a grammar defined over tuples of categorized strings, and then (3b) we provide a simple recipe for converting the grammars over tuples of categorized strings into an exactly equivalent MCFG over tuples of strings. For (3a), a minor variation of the presentation of Stabler and Keenan (2003) suffices. The conversions between grammar formalisms needed for (3a–b) are very simple and easily automated (Guillaumin 2004).

For (3a), we define,

(25) vocabulary $\Sigma = \{$ every,some,student,...$\}$

types $T = \{::, \ :\}$ ('lexical' and 'derived', respectively)

syntactic features F of two kinds:

C, T, D, N, V, …, *wh*, case, … (basic features)
=C, =T, =D, =N, =V, …, =*wh*, =case, … (selectors/probes)

persistent features $= P \subseteq F$

chains $C = \Sigma^* \times T \times F^*$

expressions $E = C^*$

lexicon $Lex \subset \Sigma^* \times \{::\} \times F^*$, a finite set.

Then we define *merge* as the union of the following 7 functions, (3 for EM, 2 for IM, and then for persistent features: EM3′ and IM2′).

(26) For s, $t \in \Sigma^*$, for $\cdot \in \{:, ::\}$, γ, $\gamma' \in F^*$, $\delta \in F^+$, and where

(p. 640) $\alpha_1,\ldots,\alpha_k$, $\iota_1,\ldots,\iota_l$ $(0 \leq k, 1)$ are any chains, define:

$$\frac{s :: =f\gamma \qquad t \cdot f, \alpha_1, \ldots, \alpha_k}{st : \gamma, \alpha_1, \ldots, \alpha_k} \text{ EM1: lexical item selects a non-mover}$$

$$\frac{s : =f\gamma, \alpha_1, \ldots, \alpha_l \qquad t \cdot f, \iota_1, \ldots, \iota_k}{ts : \gamma, \alpha_1, \ldots, \alpha_l, \iota_1, \ldots, \iota_k} \text{ EM2: derived item selects a non-mover}$$

$$\frac{s \cdot =f\gamma, \alpha_1, \ldots, \alpha_l \qquad t \cdot f\delta, \iota_1, \ldots, \iota_k}{s : \gamma, \alpha_1, \ldots, \alpha_l, t : \delta, \iota_1, \ldots, \iota_k} \text{ EM3: any item selects a mover}$$

$$\frac{s : =f\gamma, \alpha_1, \ldots, \alpha_{i-1}, t : f, \alpha_{i+1}, \ldots, \alpha_k}{ts : \gamma, \alpha_1, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_k} \text{ IM1: final move of licensee}$$

$$\frac{s : =f\gamma, \alpha_1, \ldots, \alpha_{i-1}, t : f\delta, \alpha_{i+1}, \ldots, \alpha_k}{s : \gamma, \alpha_1, \ldots, \alpha_{i-1}, t : \delta, \alpha_{i+1}, \ldots, \alpha_k} \text{ IM2: non-final move of licensee}$$

where

(tuple-SMC) none of the chains $\alpha_1,\ldots,\alpha_{i-1}$, $\alpha_{i+1},\ldots,\alpha_k$ in IM1,IM2 or IM2′ has f as its first feature.

These first 5 functions apply regardless of whether the features involved are persistent. The functions for only persistent features (indicated by underlining) are similar, except the feature is not deleted. Since the undeleted feature is then available for Internal Merge, we have only the 'mover' cases:

$$\frac{s \cdot =f\gamma, \alpha_1, \ldots, \alpha_l \qquad t \cdot \underline{f}\gamma', \iota_1, \ldots, \iota_k}{s \cdot \gamma, \alpha_1, \ldots, \alpha_l, t : \underline{f}\gamma', \iota_1, \ldots, \iota_k} \text{ EM3}'\text{: any item selects a mover}$$

$$\frac{s : =f\gamma, \alpha_1, \ldots, \alpha_{i-1}, t : \underline{f}\gamma', \alpha_{i+1}, \ldots, \alpha_k}{s : \gamma, \alpha_1, \ldots, \alpha_{i-1}, t : \underline{f}\gamma', \alpha_{i+1}, \ldots, \alpha_k} \text{ IM2}'\text{: nonfinal move of licensee}$$

Define the structures S(G)=closure(Lex,{EM1,EM2,EM3,IM1,IM2,EM3′,IM2′}). The completed structures are those expressions $w \cdot C$, where C is the 'start' category and $\cdot \in \{:, ::\}$. The sentences L(G) = $\{w \mid w \cdot C \in S(G)$ for some $\cdot$ $\{:, ::\}\}$. Now it is easy to show by an induction on derivation depth that derivations from these grammars are isomorphic to the ones over trees, with the same lexical items and the same yields. For (3b) we convert these grammars to exactly equivalent MCFGs as in Michaelis's earlier proof: the tuples of categorized strings can be represented as tuple of strings with a single simple category, where the number of simple categories needed is finite by (9).

To complete the proof it remains only to show (2)MG⊆CMG. Given any MG, we remove any useless lexical items, and then produce a CMG simply by changing every +f in the MG to =f, and changing every −f to f (after renaming if necessary so that none of these new =f and f features are the same as ones already in the grammar). We let the set of persistent features $P = \emptyset$. These grammars will produce isomorphic derivations. Although the features are conflated in the CMG, since none of them are persistent, every step will delete a pair of features, and so every EM and IM step in the CMG will correspond to the same kind of step in the MG.

(p. 641) A.2 PCMG≡PMG≡MG

This claim from Theorem 3 can be proven with the same kind of strategy used above in A.1 and in Michaelis (1998). Here we consider only PCMG≡MG, since PMG≡MG can be established in an exactly analogous fashion. We show (1) that PCMG⊆MG and (2) MG⊆PCMG.

As in A.1, to establish (1), it suffices to show (3) PCMG⊆MCFG, which can be done in two steps: showing (3a) every PCMG using merge is exactly equivalent to a grammar defined over tuples of categorized strings, and then showing (3b) a simple recipe for converting the grammars over tuples of categorized strings into an exactly equivalent MCFG over tuples of strings.

Adapting step (3a) from A.1 for CMGs with phases, we adopt the definitions in (25), adding only a specification of a set of phases $Ph \subseteq F$. Merge is unchanged except that we impose the following condition on EM (i.e. on EM1, EM2, EM3, and EM3′):

(tuple-PIC) EM cannot apply if $f \in Ph$ and $k > 0$.

It is easy to show that this grammar on tuples of categorized strings is equivalent to the corresponding PCMG, and then (3b) can be established as in A.1 and in Michaelis (1998). The other direction, (2), is trivial, since a CMG is simply a PCMG where $Ph = \emptyset$.

A.3 RMG=MG

The results in the previous appendices A.1 and A.2 involve minor adjustments in the basic idea from Michaelis (1998), but this result requires a more substantial innovation. A complete presentation would also consider the extension of RMGs to head movement as an instance of internal merge (Rizzi 2001a, Roberts 2006), together with RMG parsing methods. Here we just sketch the main idea of the proof.

order	MG				DMG			
1234	1::=2 1	2::=3 2	3::=4 3	4::4	1::=2 1	2::=3 2	3::=4 3	4::4
1243	1::=2 1	2::=3 2	3::=4 +4 3	4::4 −4	1::=2 1	2::=3 2	3::4= 3	4::4
1423	1::=2 1	2::=3 +4 2	3::=4 3	4::4 −4	1::2= +4 1	2::=3 2	3::4= 3	4::4 −4
4123	1::=2 +4 1	2::=3 2	3::=4 3	4::4 −4	1::=2 +4 1	2::=3 2	3::=4 3	4::4 −4
1324	nd				nd			
1342	1::=2 1	2::=3 +3 2	3::=4 3 3	4::4	1::=2 1	2::3= 2	3::=4 3	4::4
1432	1::=2 1	2::=3 +3 2	3::=4 + 4 3 3	4::4 4	1::=2 1	2::3= 2	3::4= 3	4::4
4132	1::=2 +4 1	2::=3 +3 2	3::=4 3 3	4::4 4	1::=2 +4 1	2::3= 2	3::=4 3	4::4 4
2134	nd				nd			
2143	nd				nd			
2413	nd				nd			
4213	nd				nd			
2314	nd				nd			
2341	1::=2 1	2::=3 +3 2	3::=4 3 3	4::4	1::2= 1	2::=3 2	3::4= 3	4::4
2431	1::=2 +2 1	2::=3 2 2	3::=4 +4 3	4::4 4	1::2= 1	2::=3 2	3::4= 3	4::4
4231	1::=2 +2 1	2::=3 +3 2 2	3::=4 3	4::4 4	1::2= +4 1	2::=3 2	3::4= 3	4::4 4
3124	nd				nd			
3142	1::=2 +3 1	2::=3 +4 2	3::=4 3 3	4::4 4	1::=2 +3 1	2::=3 +4 2	3::=4 3 3	4::4 4
3412	1::=2 +3 1	2:::=3 2	3::=4 3 3	4::4	1::=2 +3 1	2::=3 2	3::=4 3 3	4::4
4312	1::=2 +3 1	2::=3 2	3::=4 +4 3 3	4::4 4	1::=2 +3 1	2::=3 2	3::4= 3 3	4::4
3214	nd				nd			
3241	1::=2 +4 +2 1	2::=3 +3 2 −2	3::=4 3 3	4::4 4	1::=2 +4 +2 1	2::3= 2 −2	3::=4 3	4::4 4
3421	1::=2 +2 1	2::=3 +3 2 −2	3::=4 3 3	4::4	1::2= 1	2::3= 2	3::=4 3	4::4
4321	1::=2 +2 1	2::=3 +3 2 −2	3::=4 +4 3 3	4::4 4	1::2= 1	2::3= 2	3::4= 3	4::4

Fig. 27.3 . Grammars for derivable orders of 4 heads

The proof is split in the usual way: (1) RMG⊆MG and (2)MG⊆RMG, where (1) is the challenging step. To establish

(1), it suffices to show (3) RMG⊆MCFG, which can be done in two steps: showing (3a) every RMG is exactly equivalent to some intermediate grammar G, and then showing (3b) a simple recipe for converting each such intermediate G into an exactly equivalent MCFG over tuples of strings. For previous results, the intermediate grammar G was a grammar over tuples of categorized strings (*of bounded size*), but for RMGs, a straightforward extension of that idea will not suffice. In RMGs, we need to keep track of both the active sequences of features and potential interveners. Notice that the interveners for each category can be represented as a subset of the finite set of features, and so this can be added to our categories without threatening the finite partition property. The problem is that when a remnant moves, the set of potential interveners for both the remnant and every moving part of the remnant changes, because the c-commanders of those elements will change. So the most natural proof strategy is to let the intermediate grammars be defined over trees, as RMGs are, but use trees for which it is evident that the finite partition property holds. This can be done because we can let the leaves of the tree be exactly the categorized strings that would have been used in a tuple-based grammar—and we know the length of these tuples is bounded—and then use a binary tree structure over these leaves to indicate hierarchical relations and potential interveners. So then in the MCFG the categories are not tuples of feature sequences but trees with k leaves, and since k is bounded and the number of binary trees and intervener specifications is also bounded, the number of categories in the MCFG is finitely bounded too, as required. **(p. 642)**

A.4 Derivable permutations of 1234 in MGs and DMGs

In Figure 27.3, we let the 'phonetic forms' of the heads 1, 2, 3, 4 be the same as their categories. Many orders can be obtained in multiple equally simple ways, only one of which is shown. We write 'nd' for 'not derivable'. (Obviously, all orders are derivable if we allow additional heads, as pointed out in the text.)

Notes:

Thanks to Chris Collins, Greg Kobele, and Jens Michaelis for help with this material.

(1) When Chomsky (2007: 23) stipulates that the probe/selector α remains the label, as in (1), he suggests that this is the 'simplest' idea, since then no other label needs to be sought. What does that mean? Notice that the label is presumably represented somehow, so the size of that representation will contribute to memory demands, and the nature of that representation will be relevant for ensuing computational steps. It is difficult to evaluate such matters in a principled and relevant way, since the time and space efficiency of an algorithm (i.e. how many steps are taken on each input, and how much memory is used) is determined relative to a computational architecture (what counts as a step; how is the memory accessed; can steps be taken in parallel)—see e.g. van Emde Boas (1990), Parberry (1996). Perusing any standard text on algorithms like (Cormen et al. 1991), one finds many surprises, and very many results that naive, informal intuition could not have led us to, so casual claims about complexity should be considered with appropriate care.

(2) Chomsky (1995b: 245) says that the set in (1) is simpler than the tree in (2), but it is not clear what the relevant notion of simplicity is. Perhaps the set is assumed simpler since it involves just four objects—α, $\{\alpha, \beta\}$, and $\{\alpha, \{\alpha\ \beta\}$—with a membership relation as defined by some standard set theory. Considering the tree on the other hand, we have three nodes, each labeled, and a dominance relation. Since these perspectives are so similar, at this point we will regard the tree and the set as notational variants, until we have found a reason to regard some distinguishing property of the notations as linguistically relevant.

(3) It is possible to assume that whatever requirements are imposed on the 'selection' relation are not part of the definition of Merge, but result from the action of other constraints. Obviously, this could provide a notational variant of the formalism given here, or could be quite different, depending on the nature of the constraints.

(4) Along with TAG and CCG languages, the MG definable languages are 'mildly context-sensitive' in the sense defined by Joshi (1985). This is discussed by Michaelis (1998) and Stabler (2004).

(5) As discussed by Salomaa (1973: §IV), Michaelis (2001a: §5.2), and Mateescu and Salomaa (1997: §3), a class of languages is a 'substitution-closed full abstract family of languages' just in case it is closed with respect to finite unions, finite products, Kleene star, arbitrary homomorphisms, inverse homomorphisms, and substitutions.

(6) The recognition problem for MCFGs or MGs can be solved ‘efficiently’ in the sense that the number of steps required can be bounded by a polynomial function of a natural measure of the input size, assuming that the steps are taken serially on a ‘random access machine’ of the kind mentioned by van Emde Boas (1990). Harkema presents a CKY-like algorithm for MG recognition that is guaranteed to need no more than $O(n^{4m+4})$ steps (Harkema 2000, 2001b), where n is the length of the input and m is a constant depending on the grammar (specifically, on the number of licensees that can occur in a derivation). Standard parsing algorithms like CKY and Earley's present many opportunities for parallelism, and so it is no surprise that much faster execution can be obtained when enough parallelism is available (Koulouris et al. 1998, Takashi et al. 1997, Hill and Wayne 1991).

(7) Some of the arguments in Ristad (1993) can be understood as showing that languages defined by GB syntax are more complex than context-free grammars, but these arguments include the indexing mechanisms of binding theory. Cf. the discussion of various sources of complexity in Barton et al. (1987).

(8) Although simplistic ‘slash-passing’ grammars are not easily extended to the MG class (or to the MGC class mentioned in Section 27.6.5 below), the ‘attribute grammar’ formalism of Bloem and Englefriet (2000) might be regarded as similar, and it can easily be extended to MGs, as shown by Michaelis et al. (2000).

(9) Familiar programs are implemented by ‘compiling’ them into codes that can be ‘directly’ executed by a computer, and it is reasonable to expect the neural implementation of our high-level linguistic performance models to be at least as involved. But the question of what should count as an implementation of a given program (particularly with ‘optimizing’ compilers) is not clear; addressing this question in the case of linguistic computations will certainly comprise part of the scientific problem. See e.g. Blass et al. (2008), Moschovakis (2001).

(10) Chesi (2007) notes that although phases have been argued to be of unbounded size (p.48, n.32), he can guarantee they are finite by stipulating a fixed finite bound on the number of arguments allowed in each phase and by assuming that the functional categories in each phase are not recursive. If these assumptions are warranted, they could of course be adopted here too, potentially giving us an alternative route to the finite partition property or something similar; the challenge is to defend those assumptions.

(11) We leave aside the argument—adjunct asymmetries in such movements, but consider them in Stabler (2010), and we postpone the discussion of multiple *wh*-extractions to Section 27.4.3 below.

(12) As usual, the dominance relation is reflexive (so in any tree, every node dominates itself), and in any tree with subtrees t_1, t_2, t_3, subtree t_1 *c-commands* subtree t_2 if and only if the root of t_2 is dominated by a sister of the root of t_1.

(13) Rizzi and Shlonsky observe that sometimes the subject can move, noting an example in Imbabura Quechua and other things, but also the simple English *Who came?* In the English question, they suggest, the ϕ-features of the finite clause and valued by *who* suffice to satisfy the subject criterion, so that *who* can move to the C domain: ‘So, Fin+Phi offers a kind of bypassing device, …allowing the thematic subject endowed with the wh- (or some other A′-) feature to move higher.’

(14) Abels (2007) observes that a ban on improper movement can also be enforced with restrictions on feature ordering.

(15) The DMGs introduced here generate more orders than allowed by the assumptions of Abels and Neeleman (Abels and Neeleman 2006, Abels 2007): Abels and Neeleman disallow movements that do not affect the noun, for reasons I do not understand. I think it is more interesting to compare all movements that involve only elements of the DP structure, as I do here, but the conclusions I mention here are the same in either case.

(16) Multisets are often called ‘numerations’ by linguists; computer scientists sometimes call them ‘heaps’ or ‘bags’. Multisets of elements of a set Lex can be formalized as functions from Lex into the natural numbers; each lexical item is mapped to the number of times it occurs.

(17) As discussed by Kobele (2006) and Stabler (2004), these languages are *not* ‘mildly context-sensitive’ in the sense defined by Joshi (1985). Cf. note 4 above.

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